

CMG Configuration Guide

CLOUD MOBILE GATEWAY

Release 25.7.R1

Final

DN1000134053

Issue 10-0

Nokia is committed to diversity and inclusion. We are continuously reviewing our customer documentation and consulting with standards bodies to ensure that terminology is inclusive and aligned with the industry. Our future customer documentation will be updated accordingly.

This document includes Nokia proprietary and confidential information, which may not be distributed or disclosed to any third parties without the prior written consent of Nokia.

This document is intended for use by Nokia's customers ("You"/"Your") in connection with a product purchased or licensed from any company within Nokia Group of Companies. Use this document as agreed. You agree to notify Nokia of any errors you may find in this document; however, should you elect to use this document for any purpose(s) for which it is not intended, You understand and warrant that any determinations You may make or actions You may take will be based upon Your independent judgment and analysis of the content of this document.

Nokia reserves the right to make changes to this document without notice. At all times, the controlling version is the one available on Nokia's site.

No part of this document may be modified.

NO WARRANTY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY OF AVAILABILITY, ACCURACY, RELIABILITY, TITLE, NON-INFRINGEMENT, MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, IS MADE IN RELATION TO THE CONTENT OF THIS DOCUMENT. IN NO EVENT WILL NOKIA BE LIABLE FOR ANY DAMAGES, INCLUDING BUT NOT LIMITED TO SPECIAL, DIRECT, INDIRECT, INCIDENTAL OR CONSEQUENTIAL OR ANY LOSSES, SUCH AS BUT NOT LIMITED TO LOSS OF PROFIT, REVENUE, BUSINESS INTERRUPTION, BUSINESS OPPORTUNITY OR DATA THAT MAY ARISE FROM THE USE OF THIS DOCUMENT OR THE INFORMATION IN IT, EVEN IN THE CASE OF ERRORS IN OR OMISSIONS FROM THIS DOCUMENT OR ITS CONTENT.

Copyright and trademark: Nokia is a registered trademark of Nokia Corporation. Other product names mentioned in this document may be trademarks of their respective owners.

©2025 Nokia.

Table of Contents

List of Figures.....	22
List of Tables.....	28
1 Getting started.....	34
2 Summary of changes.....	35
3 Mobile gateway features overview.....	38
3.1 4G EPC core network elements.....	38
3.2 3GPP reference points.....	41
3.3 Supported standalone and combination gateways.....	43
3.4 Control and User Plane separation support.....	44
3.5 RADIUS support.....	45
3.6 CDR formats.....	46
3.7 Offline charging standards.....	47
3.8 Online charging standards.....	47
3.8.1 Online charging standards summary.....	47
3.8.2 Deployment scenarios.....	49
3.8.3 Charging characteristics and charging profile.....	50
3.8.4 Bill shock prevention.....	50
3.8.5 Diameter messages.....	50
3.8.6 Re-authorization.....	50
3.8.7 Time-quota.....	51
3.8.8 Tariff change.....	51
3.8.9 Termination action and redirection.....	51
3.8.10 Failure handling.....	51
3.8.11 FUI support.....	52
3.8.12 CCR-I and CCR-E support.....	52
3.9 Gx interface standards.....	52
3.10 Policy and charging control.....	53
3.11 GPRS tunneling protocol.....	53
3.12 Application Assurance.....	53
3.13 Lawful Interception.....	55
3.14 IP reassembly support.....	56
3.15 L2TP/PPP regeneration support.....	56
3.16 VoLTE bearer preservation.....	56
3.17 Troubleshooting and analysis.....	56
3.18 SR OS services support.....	58
4 Cloud Mobile Gateway (CMG).....	59

4.1	CMG overview.....	59
4.1.1	CMG architecture.....	59
4.1.2	Inline LB and CP load distributor (LB-less) architecture options.....	61
4.2	CMG deployment options.....	62
4.2.1	CMG with GTP-U LB function.....	62
4.2.2	CMG without GTP-U LB function (LB-less).....	63
4.2.3	CMG in hybrid mode.....	64
4.3	CMG redundancy support.....	65
4.3.1	Intra-VNF redundancy (local HA) for CMG.....	65
4.3.1.1	MG-VM redundancy (1:1).....	66
4.3.1.2	MG-VM redundancy (N:K).....	68
4.3.1.2.1	Database as session storage.....	69
4.3.1.2.2	Deploying DB-VM with CMG.....	71
4.3.1.2.3	VM state management.....	72
4.3.2	Inter-VNF 1:1 geo-redundancy for CMG.....	73
4.4	CMG configuration.....	75
4.4.1	CMG instance configuration in LB mode.....	75
4.4.1.1	VM configuration.....	76
4.4.1.1.1	MG-VM configuration.....	76
4.4.1.1.2	LB-VM configuration.....	78
4.4.1.2	Redundancy group configuration for LB mode.....	79
4.4.1.2.1	1:1 redundancy group configuration.....	79
4.4.1.2.2	N:K redundancy group configuration.....	80
4.4.1.2.3	Non-redundant MG group configuration.....	81
4.4.1.3	Signaling interface configuration.....	82
4.4.1.4	APN configuration.....	85
4.4.1.5	SGi interface services configuration in LB mode.....	86
4.4.1.5.1	CGNAT configuration.....	87
4.4.1.5.2	IPSec/GRE configuration.....	92
4.4.1.6	Call failure rate configuration.....	93
4.4.2	CMG instance configuration in LB-less mode.....	94
4.4.2.1	VM configuration.....	96
4.4.2.1.1	MG-VM configuration.....	96
4.4.2.1.2	LB-VM configuration.....	97
4.4.2.2	Redundancy group configuration for LB-less mode.....	98
4.4.2.2.1	1:1 MG-VM redundancy group configuration.....	98
4.4.2.2.2	N:K group MG-VM redundancy configuration.....	99
4.4.2.2.3	Dynamic SAP for N:K MG-VM redundancy configuration.....	99
4.4.2.2.4	Non-redundant MG group configuration.....	100
4.4.2.3	Signaling interface configuration.....	100
4.4.2.4	MG group-specific options configuration.....	102
4.4.2.4.1	MG group-specific GTP-U configuration with 1:1 MG-VM redundancy.....	103
4.4.2.4.2	MG group-specific GTP-U configuration with N:K MG-VM redundancy.....	108
4.4.2.4.3	Signaling interface configuration for GTP-C redirection.....	110
4.4.2.4.4	MG group-specific APN configuration.....	114
4.4.2.5	Routing and redundancy configuration.....	117
4.4.2.5.1	Local prefix advertisement.....	118

4.4.2.5.2	Aggregate VPRN configuration for the MG groups VPRNs (GTP-U traffic).....	118
4.4.2.5.3	Aggregate VPRN configuration for the MG groups (SGi traffic).....	124
4.4.2.5.4	Egress route configuration.....	129
4.4.2.5.5	Redundancy configuration.....	131
4.4.2.6	SGi interface services configuration in LB-less mode.....	133
4.4.2.6.1	CGNAT configuration (1:1 MG-VM redundancy).....	134
4.4.2.6.2	CGNAT configuration (N:K MG-VM redundancy).....	138
4.4.2.6.3	IPSec/GRE configuration.....	141
4.4.3	CMG instance configuration in hybrid mode (LB and LB-less).....	141
4.5	Secure storage.....	143
5	Common gateway configuration.....	145
5.1	Gateway instance configuration.....	146
5.2	Gateway name configuration.....	146
5.3	Home PLMN ID configuration: MCC and MNC values.....	146
5.4	IP addresses configuration.....	146
5.4.1	Physical IP addresses (interface) configuration.....	146
5.4.2	System IP address (routing) configuration.....	147
5.4.3	Management IP address (network management) configuration.....	147
5.4.4	Mobile signaling IP addresses configuration.....	148
5.4.5	Router IP interfaces configuration.....	148
5.4.6	Recommendations for IP address configuration.....	149
5.4.7	Multiple IP addresses reservation from local IP address pool.....	150
5.5	VNF/CNF redundancy (geo-redundancy) configuration guidelines.....	150
5.5.1	VNF/CNF redundancy support.....	151
5.5.2	VNF/CNF redundancy redirection.....	153
5.5.2.1	Handling a partial network failure.....	153
5.5.2.2	Handling a manual switchover.....	153
5.5.3	ISSU minor and major release upgrade support.....	154
5.5.4	VNF/CNF redundancy configuration.....	154
5.5.4.1	Gateway interface configuration.....	155
5.5.4.2	Multi-chassis synchronization configuration.....	155
5.5.4.3	Routing policies configuration.....	157
5.5.4.4	Forcing a manual switchover.....	159
5.5.4.5	ICR UE-by-UE reconcile synchronization.....	160
5.5.4.6	ICR slave detection delay timer.....	162
5.5.4.7	ICR Forced Switchover based on Gx failover.....	163
5.5.5	Inter-VNF redundancy configuration for LB-less deployments.....	164
5.5.5.1	CMG configuration for ICR.....	164
5.5.5.2	MCS configuration.....	164
5.5.5.3	Routing policies configuration.....	169
5.6	Multiple access PDN connectivity.....	192
5.7	Enable multiple PDNs per APN.....	193
5.8	Ethernet PDU session configuration.....	195
5.9	APN configuration.....	197
5.9.1	APN configuration example.....	198
5.9.2	APN override using virtual APNs.....	200

5.9.2.1	APN override selection options.....	200
5.9.2.2	Virtual APN selection at the PDN/APN level.....	202
5.9.2.3	Unknown APN.....	203
5.9.2.4	Configuring APN override over Gx.....	204
5.9.2.5	APN override based on PCO.....	206
5.9.3	Load balancing over the SGi.....	207
5.9.4	APN-AMBR policing configuration.....	208
5.9.5	GBR/MBR calculation toward the RAN optimization.....	209
5.9.6	Corporate VPNs.....	209
5.10	SGW with LBO architecture for MEC.....	211
5.10.1	Configuring SGW with LBO Architecture for MEC.....	212
5.11	PGW selection for ePDG and TWAG.....	215
5.11.1	PGW selection overview.....	215
5.11.2	Configuring PGW selection for ePDG/TWAG.....	216
5.12	Use default profile values for GTP and Diameter signaling parameters.....	218
5.13	3GPP signaling configuration.....	218
5.13.1	S5 and Gx signaling using system IP interface configuration.....	219
5.13.2	Signaling configuration to use a separate IP interface.....	219
5.13.3	Gx interface.....	220
5.14	Diameter peer profiles configuration.....	220
5.14.1	DRA and Diameter server selection.....	221
5.14.2	Diameter TCP session and peer reset.....	223
5.14.3	Diameter TCP and SCTP connection reattempt.....	223
5.14.3.1	Fail-Retry functionality for TCP/SCTP transport.....	224
5.14.4	Diameter destination host and realm.....	225
5.14.5	Multiple destination realms for a Diameter peer (Gy interface) configuration.....	226
5.14.6	Multiple destination realms for a Diameter peer (Gx interface) configuration.....	227
5.15	Diameter interfaces overload control.....	227
5.16	Diameter overload indication conveyance.....	228
5.16.1	Capability announcement handling.....	228
5.16.2	Receiving overload report AVP.....	229
5.16.3	Overload report resulting actions.....	230
5.16.3.1	Overload report over the S6b interface.....	230
5.16.3.2	Overload report over the SWm interface.....	231
5.16.3.3	Overload report over the STa interface.....	233
5.16.3.4	Overload report over the Gx interface.....	233
5.17	Diameter origin host configuration.....	234
5.18	RADIUS interface.....	235
5.18.1	RADIUS interface configuration.....	236
5.18.2	Authentication (IP address allocation).....	236
5.18.2.1	Username and password-related parameters.....	239
5.18.3	Accounting.....	242
5.18.3.1	Duplicate accounting.....	243
5.18.3.1.1	Ignore Accounting Response.....	245
5.18.3.2	Interim accounting.....	245
5.18.3.3	Accounting ON and OFF messages.....	247
5.18.4	Load balancing.....	248

5.18.5 RADIUS disconnect.....	249
5.19 CDR configuration.....	251
5.20 P-CSCF selection.....	251
5.21 P-CSCF restoration.....	257
5.21.1 Initial attach and additional PDN connection — 3GPP access.....	257
5.21.2 Determining P-CSCF address to be used by the UE.....	258
5.21.3 P-CSCF failure detection upon terminating call.....	259
5.21.4 Non-3GPP access and terminating call failure.....	261
5.21.4.1 Basic restoration call flows.....	261
5.21.4.2 Extended restoration call flows.....	263
5.22 DNS interface profile.....	265
5.23 Gateway configuration example.....	266
5.24 Graceful shutdown and administrative session teardown.....	268
5.24.1 Session deletion priority.....	268
5.24.2 Gateway suspend state.....	269
5.24.3 Configuring the gateway shutdown.....	272
5.24.4 APN shutdown.....	272
5.24.4.1 Configurable back-off timer for APN shutdown.....	274
5.24.5 Commands for administrative session deletion.....	274
5.24.5.1 Administrative session deletion in CUPS deployments.....	275
5.24.5.2 Protection of dedicated bearers on clearing session.....	275
5.24.5.3 Other criteria for matching sessions to be deleted.....	275
5.24.5.4 Gateway and APN level clearing.....	276
5.24.5.5 Session clearing per APN, MG group, and PDN type.....	277
5.24.5.6 Session clearing per UP peer, APN, and IP prefix on the GW-C.....	278
5.24.5.7 Monitoring, overriding, and canceling ongoing session deletion.....	279
5.25 QCI values.....	280
5.25.1 Mission critical and PTT services.....	281
5.25.2 Real-time services.....	282
5.25.3 Operator-specific QCI to pre-release8 QoS parameter mapping.....	283
5.26 Wireless Priority Service on SGW-C/PGW-C/SMF over 5GC.....	284
6 PDN Gateway (PGW).....	287
6.1 PGW configuration example.....	287
6.2 PGW supported functionality.....	288
6.3 PGW with collocated 3GPP2 Home Agent support.....	288
6.4 Enterprise MIPv4 NEMO mobile router support over 4G and 3G networks.....	289
6.5 PGW 23.060 supported call flows.....	290
6.6 PGW 23.401 supported call flows.....	291
6.7 Alternative Gx/AAA call flow.....	292
6.8 Enhanced SGW restoration procedure.....	293
6.8.1 SGW failure with a UE in connected or idle mode.....	293
6.8.2 PGW detects SGW failure before Gx signaling.....	294
6.8.3 PGW detects SGW failure after Gx signaling.....	295
6.9 UE IP address allocation.....	296
6.9.1 Allocation of IP address types.....	296
6.9.2 UE IP address allocation methods.....	297

6.9.3	Controlling allocation of IP pool address to the end user.....	299
6.9.4	IP address pools configuration.....	301
6.9.5	Local IP address pools in VPRN configuration.....	304
6.9.6	IP address allocation configuration using a local DHCP server.....	307
6.9.6.1	DHCP server configuration for local IP address pools.....	307
6.9.6.2	Interface configuration in the SGi domain.....	308
6.9.6.3	Local user database configuration.....	308
6.9.6.4	DHCP client configuration.....	309
6.9.6.5	APN configuration.....	309
6.9.7	Monitoring UE IP address allocation from local pools.....	310
6.9.8	Monitoring connectivity/routability of IP addresses from IP pools.....	310
6.10	IP pool allocation mechanism.....	312
6.10.1	IP distribution to MG groups.....	312
6.10.2	Load-sharing of configured IP pools/prefixes.....	312
6.10.3	Micronet allocation.....	312
6.10.4	Round robin IP address allocation.....	313
6.10.5	IP address allocation from micronet.....	314
6.10.6	RADIUS IP address allocation in LB-less non-CUPS architecture.....	314
6.10.7	Centralized IP-pools functionality in LB-less deployments.....	315
6.11	NarrowBand-IoT SGi tunneling for non-IP PDNs.....	319
6.12	Mobile-to-All redirection.....	322
6.13	Diameter Routing Message Priority.....	322
6.14	Diameter DRA temporary overload support.....	324
6.15	eMPS-based IMS call priority services.....	325
6.15.1	eMPS IMS priority-initiated call over default bearer.....	326
6.15.2	eMPS IMS priority-initiated call with dedicated bearer.....	327
6.15.3	eMPS IMS termination and bearer deactivation.....	328
6.15.4	eMPS advance priority.....	329
6.15.4.1	DSCP marking of eMPS sessions.....	334
6.16	APN-level back-off time and load and overload control reporting.....	335
6.17	Time-conditioned APN-AMBR handling.....	336
6.18	Configuring the time-conditioned default EPS bearer QoS change handling.....	338
6.19	PGW support for ePDG selection using PCO.....	339
6.20	Session-setup timer.....	340
6.21	APN-AMBR control for roaming.....	341
6.22	DNS FQDN redirection.....	343
6.23	Delay tolerant network-initiated procedures.....	344
6.24	L2TP/PPP regeneration.....	345
6.24.1	L2TP PPP failure handling.....	346
6.24.2	IPCP DNS address assignment.....	347
6.24.3	L2TP PPP regeneration with the IMSI populated in ICRQ.....	350
6.24.4	L2TP PPP regeneration mobile router support.....	350
6.25	APN and serving PLMN rate control.....	351
6.26	SRVCC handover.....	353
6.27	PDN switching.....	353
6.27.1	Configuring PDN switching using IPSec and GRE tunnels.....	354
6.27.2	Configuring PDN switching using NAT.....	356

6.28	Capability of exposure over the T6c reference point.....	357
6.29	LTE multiple UE session over low orbit satellite connections.....	360
6.29.1	Multiple-UE sessions.....	361
6.30	APN block list.....	363
6.30.1	APN block list in CUPS.....	364
6.31	APN-based GTP-C PDN session attach rate limiting.....	365
6.32	Sending the back-off timer due to IP pool exhaustion.....	367
6.33	Configuring virtual APN selection based on PDN/PDU type.....	367
6.34	Configuring PGW support for 3GPP AAA server identifier.....	367
7	Serving Gateway (SGW).....	370
7.1	SGW instance configuration.....	370
7.1.1	SGW signaling configuration for using a single system IP interface.....	372
7.1.2	Configure each SGW signaling interface to use a separate IP interface.....	373
7.2	GTP reference point peers: (S1-u) eNodeB, (S11) MME, (S5) PGW.....	373
7.3	SGW system default profile values.....	374
7.4	SGW reference point path down behavior.....	377
7.5	Dual connectivity.....	378
7.6	CloT control plane optimization.....	379
7.7	Extended buffering.....	381
7.8	Network-triggered service restoration.....	382
7.8.1	Configuring an MME group for network-triggered service restoration.....	383
7.9	Configuring packet buffering time for PSM.....	384
8	Evolved Packet Data Gateway.....	385
8.1	Configuration requirements.....	385
8.2	IPSMG groups configuration.....	385
8.3	IKE and ESP security associations configuration.....	387
8.4	Enhanced certificate area support.....	389
8.5	Interface configuration using SWu reference points.....	390
8.6	Diameter reference point configuration using SWm.....	390
8.7	S2b reference point configuration.....	391
8.8	ePDG charging profile configuration.....	392
8.9	MOBIKE for ePDG configuration.....	393
9	Trusted Wireless Access Gateway.....	396
9.1	TWAG implementation.....	396
9.1.1	TWAG/ePDG combined gateway implementation.....	396
9.1.2	Supported features and functionality.....	397
9.1.3	Configuration requirements.....	399
9.2	TWAG open SSID access via captive portal.....	399
9.3	TWAG open SSID call flows.....	400
9.3.1	HTTP redirect with HTTP header enrichment.....	400
9.3.2	HTTP redirect with RADIUS accounting feed.....	401
9.3.3	Open SSID Stop Access.....	402

9.4	TWAG closed SSID access.....	403
9.4.1	S2a Mobility Over GTP (SaMOG) mode.....	403
9.4.2	Non-seamless WiFi Offload (NSWO) mode.....	405
9.5	TWAG closed SSID call flows.....	407
9.5.1	Initial SaMOG Attach (3GPP compliant Diameter STa).....	407
9.5.2	Initial SaMOG Attach (RADIUS-based).....	408
9.5.3	Initial NSWO Attach (3GPP compliant Diameter STa).....	410
9.5.4	Initial NSWO/Local break out Attach (RADIUS).....	410
9.5.5	Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - Diameter STa).....	410
9.5.6	Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - RADIUS).....	411
9.5.7	Inter WLAN AP handover to the same TWAG with PMK caching enabled on WLAN AP/WLC.....	411
9.5.8	MG TWAG-initiated Detach (SaMOG).....	412
9.5.9	MG TWAG-initiated Detach (NSWO).....	412
9.5.10	PGW-initiated Detach.....	413
9.5.11	STa-initiated Detach.....	413
9.5.12	RADIUS-initiated Detach.....	413
9.6	L2 VLAN as a connection method for WLAN APs.....	414
9.7	TWAG tracking per AP and location update.....	416
9.7.1	Closed SSID — subscriber location change based on EAP re-authentication.....	416
9.7.2	Closed SSID — subscriber location change based on receiving a RADIUS Accounting-Start message from the WLAN AP.....	417
9.7.3	Closed SSID — subscriber location change based on receiving a RADIUS Accounting-Interim message from the WLC.....	418
9.7.4	Closed and Open SSID — subscriber location change based on data trigger.....	419
9.7.5	Open SSID — subscriber location change based on DHCPv4 option 82.....	420
9.7.6	TWAG tracking configuration.....	421
9.8	TWAG NSWO with Gx support.....	422
9.9	TWAG configuration guidelines.....	423
9.9.1	SWw interfaces for TWAG sessions configuration.....	423
9.9.2	SWw tunnel profiles for TWAG sessions configuration.....	425
9.9.2.1	SWw tunnel profile configuration.....	426
9.9.2.2	DHCPv4 server configuration.....	427
9.9.2.3	SLAAC host profile configuration.....	428
9.9.3	TWAG for L2 VLAN access method configuration.....	428
9.9.4	APNs referenced in TWAG sessions configuration.....	430
9.9.5	Policy rule base configuration for APNs referenced in TWAG sessions.....	432
9.9.6	RADIUS AVP configuration for RADIUS accounting feed call flows.....	433
10	Traffic detection for SSG.....	435
10.1	TDF for SSG implementation overview.....	435
10.1.1	Features and functionality.....	435
10.1.2	Configuration requirements.....	437
10.2	Sd interface.....	437
10.2.1	Sd interface functions and features.....	437
10.2.2	Messages and AVPs.....	437
10.2.3	UE attach with Gx and Sd message call flows.....	438
10.2.4	ADC rules.....	438

10.2.5 Steering policy profiles.....	439
10.2.6 Event triggers and rate limiting.....	439
10.2.6.1 Event triggers.....	439
10.2.6.2 Event report indication procedures.....	440
10.2.6.3 Time of Day event triggers.....	440
10.2.6.4 Rate limiting.....	441
10.3 Gx interface.....	441
10.3.1 Gx interface in unsolicited mode.....	441
10.4 RADIUS support for fixed subscribers.....	442
10.5 Charging support.....	443
10.5.1 Offline charging.....	444
10.5.2 Online charging.....	445
10.5.3 Credit Control Failure Handling.....	445
10.5.4 TDF-CDR.....	446
10.6 SSG deployed as MP-TCP HAG.....	446
10.6.1 Subscriber context retrieval based on RADIUS CoA.....	447
10.6.2 MP-TCP HAG load balancer behavior in CMG.....	448
10.6.3 RADIUS-less HAG subscriber auto-create and DSL SPR RADIUS.....	449
10.6.3.1 RADIUS-less HAG auto-create model configuration.....	450
10.7 Auto-created subscribers for TCPO and AA features.....	451
10.8 Supported messages and AVPs.....	451
10.8.1 TDF information AVPs over the Gx interface.....	452
10.8.2 Diameter messages over the Sd interface.....	452
10.8.3 Supported Features AVP for application-based charging.....	453
10.8.3.1 TDF message command descriptions.....	453
10.8.3.2 Supported AVPs for the Sd interface.....	458
10.9 SSG TDF configuration guidelines.....	460
10.9.1 Sd interface configuration for the PDN.....	460
10.9.2 Diameter profile configuration for the Sd reference point.....	460
10.9.3 Downlink next-hop determination configuration.....	461
10.9.3.1 Next-hop address determination sequence.....	461
10.9.3.2 Next-hop address configuration for the VPRN IP pool.....	462
10.9.3.3 Next-hop address determination based on the PGW/GGSN address in the TSR.....	462
10.9.3.4 Default next-hop address configuration for the APN.....	463
10.9.4 Interface configuration to the PGW/GGSN for uplink traffic.....	464
10.9.5 Traffic redirection and steering profiles configuration.....	465
10.9.6 Steering for the SDF and APN configuration.....	465
10.9.7 Diameter signaling profile configuration for the PDN.....	465
10.9.8 Configuration and application of TDF charging profiles.....	466
10.9.9 Charging method configuration for policy rules.....	472
10.9.10 DPI and 5-tuple policy rule unit configuration for SSG TDF.....	475
10.9.11 Auto-create subscriber configuration for AA features.....	476
11 Traffic steering function.....	477
11.1 Service function chaining.....	477
11.2 SFC architecture.....	478
11.3 Traffic steering function support.....	479

11.3.1 SFC classifier with independent SFF.....	479
11.3.2 SFC classifier with integrated SFF.....	481
11.3.3 Networking and traffic flow.....	482
11.4 Steering policy profiles.....	483
11.4.1 Dynamic ADC rules for traffic steering.....	484
11.5 Traffic steering configuration guidelines.....	484
11.5.1 Traffic redirection and steering profile configuration.....	484
11.5.1.1 Traffic redirection and steering profile configuration for LB mode deployment.....	485
11.5.1.2 Traffic redirection and steering profile configuration for LB-less mode deployment.....	485
11.5.2 Steering profile configuration.....	486
11.5.3 Steering for the SDF and APN configuration.....	487
11.5.4 Incoming IP interface configuration for traffic from service chain.....	488
11.5.5 IP interface configuration for receiving uplink traffic from the mobile gateway on SSG.....	489
11.5.6 Traffic forwarding configuration for SFF function with CMG in LB-less mode.....	489
12 Charging configuration.....	492
12.1 Supported interfaces.....	492
12.2 Supported 3GPP release.....	492
12.3 OFCS summary.....	493
12.4 Home, visitor, and roamer charging profile configuration.....	494
12.5 Charging characteristics and charging profile.....	494
12.5.1 PDN/APN level charging profile selection.....	495
12.6 Bill Shock Prevention.....	495
12.7 Gy Rf synchronization events.....	496
12.8 CSG reporting.....	497
12.8.1 CSG Reporting Action IE handling.....	498
12.9 Location-based charging in the OFCS.....	498
12.10 Location-based charging in the OCS.....	500
12.11 User location change notification over Gn/Gp.....	501
12.12 Presence Reporting Area information reporting.....	502
12.12.1 Gx PRA functionality.....	502
12.12.2 S5 / S8 PRA functionality.....	503
12.12.3 Offline charging PRA functionality.....	504
12.12.4 Online charging PRA functionality.....	505
12.12.5 Sd-related PRA functionality.....	506
12.12.6 PRA configuration restriction and usage notes.....	506
12.12.7 PRA CLI commands.....	507
12.13 Multiple PRA reporting.....	507
12.13.1 Gx AVPs and messages.....	508
12.13.2 Support for multiple PRAs on Gy.....	511
12.13.3 GTP IEs and messages.....	512
12.13.4 Configuration restriction and usage notes.....	513
12.13.5 CLI commands.....	513
12.14 NarrowBand-IoT charging.....	513
12.15 Rating group level charging accuracy for delayed accounting.....	516
12.16 OAM switchover in case of lost access to storage nodes.....	517
12.17 Offline charging.....	518

12.17.1 CDR generation.....	518
12.17.1.1 Configurable CDR AVP options.....	518
12.17.1.2 Disabling event triggers on Gx.....	518
12.17.1.3 Minimum data threshold for CDR generation.....	519
12.17.1.4 Partial record trigger configuration.....	520
12.17.1.5 Generation of partial CDRs with zero volume.....	521
12.17.1.6 Generation of session CDRs with zero volume.....	522
12.17.1.7 Bearer metering in PGW CDRs.....	523
12.17.1.8 Directory structure configuration for local CDR storage.....	523
12.17.2 Offline charging record transfer.....	524
12.17.2.1 RF interface.....	524
12.17.2.1.1 Configure Rf reference-point, primary, and secondary peers.....	524
12.17.2.1.2 Configure the flash storage location.....	524
12.17.2.1.3 Include Release Identifier Extension field in ACR file and record header.....	525
12.17.2.2 Ga interface.....	525
12.17.2.2.1 GTP prime group configuration.....	525
12.17.2.2.2 Duplicate CDR prevention algorithm for Ga charging.....	526
12.17.2.2.3 CDR file transfer.....	527
12.17.2.2.4 Ga Interface (GTP') messages.....	527
12.17.2.3 Bp interface.....	529
12.17.3 CDR containers report to OCS.....	529
12.17.4 Secondary RAT usage data reporting for 5G NR (Option 3x).....	529
12.17.4.1 CDR parameter, IE, and AVP bindings for EPC offline charging.....	532
12.17.4.2 RAN Secondary RAT Usage Reports in GTP messages.....	532
12.17.4.3 List of RAN Secondary RAT Usage Reports CDR parameter.....	533
12.17.4.4 Secondary RAT Usage Reports in Diameter Rf charging.....	533
12.17.4.5 RAN Secondary RAT Usage Report AVP.....	533
12.17.4.6 CLI commands.....	534
12.18 Online charging.....	534
12.18.1 Online charging deployment scenarios.....	534
12.18.1.1 Non-roaming scenario: PGW/GGSN and OCS deployed in the H-PLMN.....	535
12.18.1.2 Roaming scenario: Home-routed (PGW/GGSN and OCS deployed in the H-PLMN).....	535
12.18.1.3 Roaming Scenario: Local-breakout (PGW/GGSN deployed in the V-PLMN and OCS deployed in the H-PLMN).....	536
12.18.1.4 OCS provisioning options.....	536
12.18.1.5 DCCA profile provisioning.....	537
12.18.2 Diameter messages for online charging.....	537
12.18.3 Gateway and OCS re-authorization.....	540
12.18.4 Time quota consumption.....	541
12.18.5 Support of tariff change.....	543
12.18.6 Termination action and redirection.....	543
12.18.6.1 HTTP traffic redirection sample configuration.....	543
12.18.6.2 UE URL append upon redirection.....	548
12.18.6.3 Termination action with Reporting Reason Final.....	548
12.18.6.4 Termination action with non-3GPP triggers.....	549
12.18.7 Failure handling.....	549
12.18.8 Credit Control Failure Handling.....	549
12.18.8.1 Failure Handling Continue options.....	553

12.18.9	Statistics collection for rating groups.....	554
12.18.10	Gy Assume Positive mode.....	555
12.18.10.1	Sample configuration.....	557
12.18.10.2	Gy Assume Positive use cases.....	558
12.18.10.2.1	CCR-I failure.....	558
12.18.10.2.2	CCR-U failure.....	559
12.18.10.2.3	CCR-T failure.....	560
12.18.10.2.4	CCR-U unknown session with Gy session restart.....	561
12.18.10.2.5	CCR-U session disconnect.....	563
12.18.10.2.6	CCR-I session disconnect.....	563
12.18.10.2.7	Variation on a CCR-I session disconnect.....	564
12.18.11	Signaling message rate limiting.....	565
12.18.12	Final Unit Indication.....	566
12.18.12.1	Multiple filter IDs with FUI Restrict and Redirect.....	567
12.18.12.2	Final Unit Action local policy for filter IDs.....	568
12.18.13	CCR event reporting.....	568
12.18.13.1	Session creation call flow.....	569
12.18.13.2	PLMN and time zone update call flow.....	570
12.18.13.3	eHRPD to LTE handover call flow.....	571
12.18.14	CCR-I on first rating group detection.....	572
12.18.15	Multiple Diameter peers with load balancing for Gx and Gy interfaces.....	573
12.18.16	OCS selection based on charging characteristics and IMSI/MSISDN range.....	573
12.18.17	Configurable event triggers.....	574
12.18.18	Gy terminate session on receiving FUA=Terminate for an MSCC failure code.....	574
12.18.19	Out-Of-Credit message with FUA=Terminate for Gy result code 4012.....	575
12.18.20	OCS reply on MSCC request for quota.....	575
12.18.21	Serializing CCR handling on Gy sessions with OCS.....	576
13	Policy configuration.....	578
13.1	Enabling release 13 policy and charging control CRs.....	578
13.2	Policy configuration using CLI batch capability.....	578
13.2.1	Begin, commit, and abort commands.....	579
13.2.2	Beginning and committing a policy configuration.....	579
13.2.3	Aborting a policy configuration.....	580
13.3	Policy modification without session shutdown.....	580
13.4	Gx interface configuration.....	582
13.4.1	Gx profile for APN-level timer and error handling.....	582
13.4.2	Gx synchronization after session or message loss.....	582
13.4.3	Gx profile based failure handling.....	584
13.4.4	VoLTE UE ULI support for billing.....	585
13.4.5	Override control for predefined and static rules.....	586
13.4.6	Called Station ID AVP.....	588
13.4.7	Diameter race condition handling for Gx and Sd interfaces.....	588
13.4.8	Gx Assume Positive.....	589
13.4.8.1	Gx Assume Positive configuration example.....	592
13.4.9	PCRF migration.....	594
13.4.10	Supported event triggers for the Gx interface.....	595

13.4.11 Gx access network information event trigger.....	596
13.4.12 PCC based on charging characteristics for Gx interface.....	596
13.4.13 Usage monitoring over Gx.....	597
13.4.13.1 Usage monitoring reporting accuracy enhancement.....	599
13.4.13.2 Monitoring key usage report event trigger override configuration.....	601
13.4.13.3 Service/application usage exclusion from IP-CAN/TDF session usage.....	601
13.4.14 PCC rule-level redirection configuration for the Gx interface.....	601
13.5 Filtering criteria configuration in policy rules.....	602
13.5.1 Configuring source address in policy rules filtering.....	602
13.5.2 Configuring port lists in policy rules filtering.....	603
13.5.3 Configuring IP address lists in policy rules filtering.....	604
13.6 Gateway-derived flow descriptions using DNS resolution.....	605
13.7 Policy rules precedence.....	608
13.8 Installing different rules with the same charging group.....	609
13.9 Sponsored data connectivity.....	609
13.10 Hybrid access bonding.....	610
13.10.1 Fixed access fill-out.....	611
13.11 Static predefined and local policies on PGW/GGSN.....	612
13.12 Dedicated bearer configuration options.....	613
13.12.1 Dedicated bearer idle timer per QCI.....	614
13.12.2 Support of multiple bearers diversion to AA.....	614
13.12.3 Dedicated bearer creation upon application detection.....	615
13.12.3.1 Dedicated bearer creation for ADC configuration.....	616
13.12.4 ADC extended to static PCC and VoIP optimization.....	617
13.12.4.1 Configuring static PCC rules for ADC.....	618
13.13 Protocol Configuration Options (PCO).....	620
13.14 AES encryption for HTTP redirection configuration.....	620
13.14.1 AES encryption using salt.....	622
13.15 Out of Credit and Reallocation of Credit event triggers.....	622
13.16 Locally-triggered default bearer modification with QoS update.....	623
13.17 QCI/ARP modification of default bearer.....	624
13.18 DSCP-based bearer binding.....	625
13.19 No QoS negotiation indication.....	626
13.20 RAN/NAS cause.....	627
13.21 AA application profile and ASO characteristics as part of pre-defined PCC rule.....	628
13.22 QoS policing per RG/SID.....	629
13.23 QoS policing per SDF.....	630
13.24 Bandwidth roundup for GBR and MBR.....	633
13.25 Enable/disable dynamic PCC per RAT type.....	634
13.26 Dynamic modification of AA start/stop notifications.....	636
13.27 High data rate support for 5G New Radio (option 3x).....	637
13.27.1 AVPs and information elements.....	638
13.27.2 CLI commands.....	640
13.27.3 QCI values.....	640
13.28 DNS snooping.....	641
13.28.1 Wildcard matching.....	641
13.28.2 Dynamic storage of returned IP addresses from DNS replies.....	642

13.28.3 Refresh of stored entries.....	643
13.28.4 Configuring DNS snooping.....	644
13.28.5 Forwarding DNS traffic to AA.....	645
13.29 Non-terminating policy rules.....	646
14 GPRS Tunneling Protocol.....	648
14.1 GTP configuration.....	648
14.2 GTP error indication messages.....	648
14.3 GTP retransmit timer configuration.....	650
14.4 SMS paging distinction.....	650
14.5 Pause of charging.....	651
14.6 GTP Diameter timestamp.....	652
14.7 GTP-C load and overload control.....	657
14.7.1 GTP load and overload configuration.....	659
14.7.2 PGW GTP-C node-level overload throttling.....	660
14.7.3 GTPv2-C APN load and overload control.....	665
14.7.3.1 APN load control.....	665
14.7.3.2 APN overload control.....	666
14.8 Signaling priority indication.....	667
14.9 GTP-C collision handling.....	668
15 IP fragmentation and reassembly.....	670
15.1 IPv4 reassembly for GTP-U tunnel ingress packets.....	670
15.2 PGW support for link MTU in PCO.....	671
15.3 IP reassembly MS-ISA.....	671
15.4 IP reassembly configuration.....	672
15.4.1 IP reassembly and IPSec on separate MS-ISA configuration.....	673
15.4.2 IP interface for IP reassembly configuration.....	673
15.4.3 IP reassembly and IPSec on the same MS-ISA configuration.....	674
15.4.4 IP reassembly ports, VLANs, and LAGs configuration.....	675
15.4.5 IP reassembly in CMG.....	677
15.4.6 Show command output.....	678
15.5 Diversion of fragmented traffic to IP reassembly MS-ISA.....	679
15.5.1 Applying filters to external interfaces.....	679
15.5.2 Configuring selective reassembly.....	681
15.6 TCP MSS configuration.....	681
15.6.1 TCP MSS static configuration.....	681
15.6.2 TCP MSS configuration for NAT.....	682
15.6.3 TCP MSS adjust configuration.....	682
16 IP tunnels.....	684
16.1 IP tunnels overview.....	684
16.1.1 Tunnel ISAs.....	686
16.1.1.1 Public tunnel SAPs.....	687
16.1.1.2 Private tunnel SAPs.....	688

16.1.1.3 IP interface configuration.....	688
16.1.1.4 GRE and IP-IP tunnel configuration.....	689
16.1.1.5 Tunnel configuration for ePDG.....	690
16.1.1.6 IP fragmentation and reassembly for IP tunnels.....	691
16.1.2 Operational conditions.....	693
16.1.3 QoS interactions.....	694
16.1.4 OAM interactions.....	694
16.1.5 Redundancy.....	695
16.1.6 Statistics collection.....	695
16.1.7 Security.....	695
16.1.7.1 GRE tunnel IPv4 multicast support.....	696
16.1.8 IKEv2.....	696
16.1.9 IKEv2 TS-List.....	697
16.1.10 SHA2 support.....	697
16.2 Using certificates for IPSec tunnel authentication.....	698
16.3 Trust-Author-Profile.....	698
16.4 Cert-profile.....	699
16.4.1 Cert-profile/Trust-anchor-profile versus Cert/Trust-anchor.....	700
16.5 Certificate Management Protocol version 2 (CMPv2).....	701
16.6 OCSP.....	701
16.7 Using CRL.....	702
16.8 Video wholesale example.....	703
16.9 IKEv2 remote-access tunnel.....	703
16.9.1 IKEv2 remote-access tunnel - Authentication.....	704
16.9.2 IKEv2 remote-access tunnel - EAP authentication.....	707
16.9.3 IPv6 over IPv4 IPSec tunnel.....	708
16.9.3.1 IPv6 over IPv4: Static LAN-to-LAN tunnel.....	708
16.9.3.2 IPv6 over IPv4: Dynamic LAN-to-LAN tunnel.....	709
16.9.3.3 IPv6 over IPv4: Remote-access tunnel.....	709
16.10 Configuring IPSec with CLI.....	710
16.10.1 Tunnel ISA provisioning.....	710
16.10.2 Tunnel group configuration.....	710
16.10.2.1 Tunnel group configuration for ePDG.....	711
16.10.3 Router interface configuration for IPSec.....	711
16.10.4 IPSec parameter configuration.....	712
16.10.5 IPSec in services configuration.....	712
16.10.6 X.509v3 certificate parameters configuration.....	713
16.10.6.1 Configuring the certificate enrollment.....	714
16.10.6.2 Configuring CA certificate/CRL import.....	714
16.10.7 MC-IPSec configuration.....	716
16.10.7.1 MIMP configuration.....	716
16.10.7.2 Multi-chassis synchronization configuration.....	717
16.10.7.3 Routing configuration for MC-IPSec.....	717
16.10.8 CMPv2 configuration and usage.....	719
16.10.9 OCSP configuration.....	719
16.10.10 IKEv2 remote-access tunnel configuration.....	720

17 Network Address Translation.....	724
17.1 Terminology.....	724
17.2 NAT overview.....	725
17.2.1 Principles of NAT.....	726
17.2.2 Application compatibility.....	726
17.3 Large Scale NAT.....	727
17.3.1 Port range blocks.....	727
17.3.1.1 Reserved ports and priority sessions.....	728
17.3.1.2 Prevent port block starvation.....	728
17.3.1.2.1 Dynamic port block starvation in LSN.....	729
17.3.1.2.2 Dynamic port block reservation.....	730
17.3.2 Timeouts.....	731
17.3.3 Watermarks.....	731
17.4 464XLAT support.....	732
17.5 NAT64 fragmentation.....	734
17.6 TCP MSS configuration for NAT.....	735
17.7 NAT logging.....	735
17.7.1 Syslog flow logging.....	736
17.7.2 Syslog/SNMP/local-file logging.....	739
17.7.2.1 Filtering LSN events to system memory.....	740
17.7.2.2 NAT logging to a local file.....	743
17.7.3 SNMP trap logging.....	743
17.7.3.1 NAT syslog.....	745
17.7.4 LSN NAT flow logging.....	746
17.7.4.1 Large scale NAT44 flow logging configuration example.....	747
17.7.5 NAT reporting EDRs.....	751
17.8 Logging of NAT binding records in CSV format configuration.....	752
17.9 Deterministic NAT.....	754
17.9.1 Supported deterministic NAT flavors.....	754
17.9.2 Number of subscribers per outside IP and per pool.....	755
17.9.3 Referencing a pool.....	755
17.9.4 Outside pool configuration.....	755
17.9.5 Rule mapping and the map command in deterministic LSN44.....	760
17.9.6 Hashing considerations in deterministic LSN44.....	763
17.9.7 Simultaneous support of dynamic and deterministic NAT.....	765
17.9.8 Selecting traffic for NAT.....	765
17.9.9 Inverse mappings.....	765
17.9.9.1 MIB approach.....	766
17.9.9.2 Offline approach to obtain deterministic mappings.....	766
17.9.10 Logging.....	768
17.10 Destination-based NAT (DNAT).....	768
17.10.1 Combination of SNAPT and DNAT.....	768
17.10.2 Forwarding model in DNAT.....	769
17.10.3 DNAT traffic selection via NAT classifier.....	770
17.10.4 DNAT configuration.....	770
17.10.4.1 DNAT traffic selection and destination IP address configuration.....	771
17.10.4.2 Micro-netting original source (inside) IP space in DNAT-only case.....	771

17.10.5 Destination NAT with port override.....	773
17.11 NAT optimizations for CMG.....	774
17.11.1 Processing NAT inline with LB-less architecture.....	774
17.11.2 Processing NAT inline on MG-VM with LB architecture.....	776
17.12 Enhanced statistics in NAT — Histogram.....	778
17.12.1 Configuration.....	780
17.13 Multiple NAT policies per inside routing context.....	782
17.13.1 Routing approach for NAT diversion.....	784
17.13.2 Filter-based approach.....	784
17.13.3 Default NAT policy.....	785
17.13.4 Scaling considerations.....	785
17.13.5 Multiple NAT policies and SPF configuration considerations.....	785
17.13.5.1 Multiple NAT policies and forwarding considerations.....	786
17.13.6 Logging.....	787
17.14 ISA feature interactions.....	789
17.14.1 MS-ISA use with service mirrors.....	789
17.14.2 LNS, Application Assurance and NAT.....	789
17.14.3 Subscriber-aware large scale NAT44.....	790
17.15 Support for SSG with CGNAT and firewall/AA functions.....	800
17.16 Configuring NAT with CLI.....	800
17.16.1 Provisioning a NAT ISA.....	800
17.16.2 ISA redundancy.....	801
17.16.3 Large scale NAT configuration.....	803
17.16.4 NAT configuration examples.....	805
18 Monitoring and troubleshooting tools.....	809
18.1 Call trace and packet capture.....	809
18.1.1 Capturing of SWu ESP drop reason.....	817
18.2 Network trace.....	819
18.2.1 Configuring a network trace profile.....	820
18.2.2 Enabling the network trace feature.....	821
18.3 Per Call Measurement Data.....	823
18.4 Key Performance Indicator and Key Capacity Indicator.....	824
18.4.1 KPI-KCI collection configuration.....	825
18.4.2 Statistics reporting.....	825
18.5 Prometheus metrics.....	825
18.6 Telemetry.....	826
18.7 GTP ping.....	826
18.8 UE ping.....	827
18.9 SNMP traps.....	827
Appendix A: Messages and information element reference.....	829
RADIUS interface messages.....	829
Access-Request.....	829
Access-Accept.....	835
Accounting-Request (Start).....	842

Accounting-Request (Stop).....	849
Accounting-Request (Interim-Update).....	856
Accounting-Request (Accounting-On).....	864
Accounting-Request (Accounting-Off).....	865
RADIUS CoA-Request.....	866
Disconnect-Request.....	868
Disconnect-Response (ACK).....	869
Disconnect-Response (NAK).....	870
GTPv2 interface messages.....	871
Path management messages (S11, S5, S8, S4, S2a, S2b).....	871
Tunnel management messages (S11, S5, S8, S4, S2a, S2b).....	872
Mobility management messages (S11, S5, S8, S4).....	915
CSFB messages (S11, S5, S8).....	916
PGW Restart (S11, S4).....	918
Gx interface messages.....	919
CC-Request.....	919
CC-Answer.....	924
Re-Auth-Request (RAR).....	928
Re-Auth-Answer (RAA).....	930
Gy interface messages.....	932
Credit Control Request (CCR).....	933
Credit Control Answer (CCA).....	942
Re-Auth-Request (RAR).....	946
Re-Auth-Answer (RAA).....	947
Abort-Session-Request (ASR).....	948
Abort-Session-Answer (ASA).....	949
TDF Credit Control Request message AVPs.....	950
TDF Credit Control Response message AVPs.....	956
Rf interface messages.....	958
Accounting-Request (ACR).....	959
Accounting-Answer (ACA).....	961
ePDG Rf interface attributes.....	962
SGW Rf interface attributes.....	975
PGW Rf interface attributes.....	993
TDF Rf interface attributes.....	1012
S6b interface messages.....	1012
Diameter AA-Request.....	1012
Diameter AA-Answer.....	1018
Diameter AS-Request.....	1024
Diameter Re-Auth-Request.....	1024
Diameter Re-Auth-Answer.....	1025
Diameter Session-Termination-Request.....	1026
Diameter Session-Termination-Answer.....	1027
Diameter Abort-Session-Request.....	1029
Diameter Abort-Session-Answer.....	1030
SWm interface messages.....	1031
Diameter EAP-Request.....	1032

Diameter EAP-Answer.....	1033
Diameter Re-Auth-Request.....	1035
Diameter Re-Auth-Answer.....	1036
Diameter AA-Request.....	1037
Diameter AA-Answer.....	1038
Diameter Session-Termination-Request.....	1039
Diameter Session-Termination-Answer.....	1040
Diameter Abort-Session-Request.....	1040
Diameter Abort-Session-Answer.....	1041
IKEv2 Notify messages.....	1042
Protocol Configuration Options (PCO) IE.....	1057
Gx interface Supported Features of Feature-List-ID 1.....	1062
Gx interface Supported Features of Feature-List-ID 2.....	1065
S6b Interface Supported-Features AVP.....	1066
Appendix B: Standards and protocol support.....	1067

List of Figures

Figure 1:	3GPP architecture for EPC core.....	39
Figure 2:	Mobile Network Deployment using PCEF with AA ISA.....	54
Figure 3:	EPC Bearer.....	55
Figure 4:	CMG VNFC architecture.....	60
Figure 5:	User plane LB architecture.....	61
Figure 6:	User plane load distributor architecture.....	62
Figure 7:	User plane LB architecture.....	63
Figure 8:	User plane LB-less architecture.....	64
Figure 9:	CMG in hybrid mode.....	65
Figure 10:	Intra-VNF 1:1 redundancy protection (CMG in LB mode).....	67
Figure 11:	Intra-VNF 1:1 redundancy protection (CMG in LB-less mode).....	68
Figure 12:	CMG deployment with DB-VMs.....	71
Figure 13:	Configured VM state in N:K redundancy.....	72
Figure 14:	Operational VM state in N:K redundancy.....	73
Figure 15:	Inter-VNF 1:1 (master/slave) geo-redundancy protection.....	74
Figure 16:	LB mode CMG network deployment.....	76
Figure 17:	Signaling interface configuration with common VPRN instance.....	83
Figure 18:	CMG with LB-less architecture.....	95
Figure 19:	MG group-specific network diagram for 1:1 MG-VM redundancy.....	103
Figure 20:	MG group-specific network diagram for N:K MG-VM redundancy.....	108
Figure 21:	GTP-C redirection for partial failure detection.....	111
Figure 22:	Routing and redundancy function with 1:1 MG-VM redundancy.....	117
Figure 23:	Routing and redundancy function with N:K MG-VM redundancy.....	118
Figure 24:	Network diagram of CMG in hybrid mode.....	142
Figure 25:	PGW/GGSN network architecture example.....	145
Figure 26:	PGW-C/SMF or PGW-U/UPF gateway VNF/CNF redundancy example.....	152
Figure 27:	SGW VNF/CNF redundancy example.....	152
Figure 28:	ePDG VNF redundancy example.....	152
Figure 29:	Multi-chassis synchronization configuration example.....	156
Figure 30:	MCS and BFD CMG configuration.....	164
Figure 31:	Routing policy configuration.....	169
Figure 32:	Policy statement configuration.....	170
Figure 33:	MG-VM configuration.....	174
Figure 34:	Export policy configuration.....	182
Figure 35:	Multiple PDN connections using a single access network in combined SGW/PGW (no MAPCON).....	192

Figure 36:	Multiple PDN connections using different access networks in combined SGW/PGW (MAPCON).....	193
Figure 37:	APN configuration example.....	198
Figure 38:	Virtual APN selection alternatives.....	201
Figure 39:	Virtual APN selection and suffixing at the PDN level.....	203
Figure 40:	Gx signaling for virtual APN selection.....	204
Figure 41:	Load balancing over the SGi.....	208
Figure 42:	IP pool overlapping based on L2TP.....	210
Figure 43:	All APNs steered over the same two LNS servers.....	211
Figure 44:	IP pool overlapping based on GRE tunnel.....	211
Figure 45:	SGW with LBO configuration flow.....	212
Figure 46:	E-UTRAN initial attach with RADIUS authentication and accounting.....	237
Figure 47:	PDP context activation with RADIUS authentication and accounting.....	238
Figure 48:	Detach with RADIUS accounting.....	243
Figure 49:	PDP context deactivation with RADIUS accounting.....	243
Figure 50:	Disconnect-ACK message call flow.....	250
Figure 51:	Disconnect-Request and Disconnect-NAK messages.....	250
Figure 52:	P-CSCF server pools.....	252
Figure 53:	P-CSCF server assignment with P-CSCF restoration support indication.....	258
Figure 54:	The PCRF indicates the chosen P-CSCF to the PGW.....	259
Figure 55:	Terminating call P-CSCF failure with P-CSCF restoration (MME-based).....	259
Figure 56:	Terminating call P-CSCF failure with P-CSCF restoration (PCRF-based).....	260
Figure 57:	Terminating call P-CSCF failure with P-CSCF restoration (Gx-based — Update).....	260
Figure 58:	Normal PDN activation — UE not indicating support for P-CSCF re-selection.....	261
Figure 59:	UE attach/tunnel establishment, Part 2 — PGW supports P-CSCF restoration.....	262
Figure 60:	Untrusted access P-CSCF restoration (S6b-based — Disconnect).....	262
Figure 61:	Untrusted access P-CSCF restoration (Gx-based — Disconnect).....	263
Figure 62:	UE attach/tunnel establishment, Part 1 — UE supports re-Selection.....	263
Figure 63:	UE attach/tunnel establishment, Part 2 — UE supports P-CSCF re-selection and PGW supports P-CSCF restoration.....	264
Figure 64:	Untrusted access P-CSCF restoration (S6b-based — Extended).....	264
Figure 65:	Untrusted access P-CSCF restoration (Gx-based — Extended).....	265
Figure 66:	PGW configuration example.....	287
Figure 67:	3GPP2 Home Agent.....	288
Figure 68:	NEMO-based mobile router support.....	290
Figure 69:	Default system call flow.....	292
Figure 70:	Alternative RADIUS call flow.....	293
Figure 71:	UE in idle mode.....	294
Figure 72:	UE in connected mode.....	294
Figure 73:	PGW detects SGW failure before Gx signaling.....	295

Figure 74:	PGW detects SGW failure after Gx signaling.....	296
Figure 75:	Injecting subscriber IP pools into RTM example.....	305
Figure 76:	Tunneling for non-IP PDNs.....	320
Figure 77:	Non-IP PDU tunneling.....	320
Figure 78:	Diameter DRA temporary overload support.....	325
Figure 79:	eMPS IMS priority-initiated call over default bearer.....	326
Figure 80:	eMPS IMS priority-initiated call with dedicated bearer.....	327
Figure 81:	eMPS IMS termination and bearer deactivation.....	328
Figure 82:	eMPS advance priority call flow.....	329
Figure 83:	PCRF-initiated dedicated bearer call flow with eMPS priority establishment.....	332
Figure 84:	PCRF-initiated dedicated bearer call flow without eMPS priority establishment.....	334
Figure 85:	ePDG selection using PCO.....	340
Figure 86:	Session-setup timer functionality.....	341
Figure 87:	PGW upgrades the APN-AMBR based on the local configuration.....	342
Figure 88:	Modify bearer command is received before the timer expires.....	342
Figure 89:	DNS FQDN redirect functionality.....	343
Figure 90:	Create/Update Bearer Request on hold.....	345
Figure 91:	PGW/GGSN support for PPP in the network.....	346
Figure 92:	DNS address assignment call flow.....	348
Figure 93:	IMSI/MSISDN in the ICRQ message.....	350
Figure 94:	L2TP PPP regeneration MR support.....	351
Figure 95:	Handover of PS to CS.....	353
Figure 96:	PDN switching using NAT (SNAT).....	354
Figure 97:	PDN switching using IPSec or GRE.....	354
Figure 98:	QoS modification by the SCEF for the PDN connection.....	358
Figure 99:	Remote office connection to HQ using LTE and low orbit satellites.....	360
Figure 100:	Low orbit satellites as the mobile backhaul for a remote eNodeB site.....	360
Figure 101:	Multiple transceiver installation for low orbit satellites.....	361
Figure 102:	SGi interface traffic routing in a multiple-UE session.....	363
Figure 103:	Charging function in a multiple-UE session.....	363
Figure 104:	PDN session attach rate limit (virtual APN serves all access types).....	366
Figure 105:	PDN session attach rate limit (virtual APN serves specific access types).....	366
Figure 106:	Dual connectivity.....	379
Figure 107:	S11-U used for short data transmission initiated from the UE.....	380
Figure 108:	Extended buffering.....	381
Figure 109:	Extended buffering with SGW relocation.....	382
Figure 110:	ePDG ISM-MG and ISA grouping.....	386
Figure 111:	ePDG and PGW-SGW two-card combination.....	387
Figure 112:	Open and closed SSID modes for TWAG implementation in CMG.....	396

Figure 113:	TWAG ePDG combined implementation and reference points.....	397
Figure 114:	TWAG Open SSID sccess call flow.....	399
Figure 115:	TWAG Open SSID user plane with EthoGRE tunnel.....	400
Figure 116:	Open SSID access call flow using HTTP redirect with HTTP header enrichment.....	401
Figure 117:	Open SSID access call flow using HTTP redirect with RADIUS accounting feed.....	402
Figure 118:	Open SSID Stop Access.....	403
Figure 119:	TWAG Closed SSID access in SaMOG mode.....	403
Figure 120:	TWAP function in TWAG.....	404
Figure 121:	High level call flow in SaMOG mode.....	405
Figure 122:	TWAG Closed SSID access in NSWO mode.....	406
Figure 123:	High level call flow in NSWO mode.....	406
Figure 124:	Initial SaMOG Attach using a 3GPP compliant Diameter STa call flow.....	407
Figure 125:	Initial SaMOG Attach based on RADIUS call flow.....	409
Figure 126:	Initial NSWO Attach using a 3GPP compliant Diameter STa call flow.....	410
Figure 127:	Initial NSWO/Local break out Attach based on RADIUS call flow.....	410
Figure 128:	Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - Diameter STa) call flow.....	411
Figure 129:	Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - RADIUS) call flow.....	411
Figure 130:	Inter WLAN AP handover to the same TWAG with caching enabled on WLAN AP/WLC call flow.....	412
Figure 131:	MG TWAG-Initiated Detach in SaMOG mode call flow.....	412
Figure 132:	MG TWAG-initiated Detach in NSWO mode call flow.....	413
Figure 133:	PGW-initiated Detach call flow.....	413
Figure 134:	PDN STa-initiated Detach call flow.....	413
Figure 135:	PDN RADIUS-initiated Detach call flow.....	414
Figure 136:	L2 VLAN connection of WLAN APs using VLAN tagging.....	414
Figure 137:	RADIUS traffic over Ethernet using EAP authentication.....	415
Figure 138:	RADIUS Accounting over Ethernet.....	415
Figure 139:	L2 aggregation in single or multiple sites.....	416
Figure 140:	Closed SSID UE location change based on EAP re-authentication.....	417
Figure 141:	Closed SSID UE location change based on receiving a RADIUS Accounting-Start message from the WLAN AP.....	418
Figure 142:	Closed SSID UE location change based on receiving a RADIUS Accounting-Interim message from the WLC.....	419
Figure 143:	Closed SSID UE location change based on data trigger.....	420
Figure 144:	Open SSID UE location change based on data trigger.....	420
Figure 145:	Open SSID UE location change based on DHCPv4 option 82.....	421
Figure 146:	WiFi Subscriber Attach (Closed SSID) — NSWO.....	423
Figure 147:	TDF reference architecture with policy and charging interfaces.....	435
Figure 148:	UE Attach with Gx and Sd message flows.....	438
Figure 149:	User traffic routing via SSG on Gx interface.....	442

Figure 150:	Subscriber Context Creation via SSG and Gx interface call flow.....	442
Figure 151:	Fixed subscriber context creation via RADIUS.....	443
Figure 152:	Fixed subscriber context creation via SSG and RADIUS interface call flow.....	443
Figure 153:	AA MP-TCP HAG deployment.....	447
Figure 154:	SSG HAG over RADIUS.....	448
Figure 155:	MP-TCP signaling.....	450
Figure 156:	Legacy deployment of service functions example.....	477
Figure 157:	Service function chaining example.....	478
Figure 158:	SFC architecture.....	479
Figure 159:	Mobile gateway as SFC classifier example.....	480
Figure 160:	SSG as SFC classifier example.....	481
Figure 161:	Mobile gateway with traffic steering functions example.....	482
Figure 162:	Network function as an SFC classifier: VLAN.....	483
Figure 163:	Network function as an SFC classifier with integrated SFF.....	483
Figure 164:	Uplink traffic forwarding for SFF with CMG in LB-less mode.....	490
Figure 165:	Downlink traffic forwarding for SFF with CMG in LB-less mode.....	490
Figure 166:	Duplicate CDR prevention algorithm.....	527
Figure 167:	NSA Option 3 family — Options 3, 3A, and 3X data flows.....	530
Figure 168:	Option 3X functionality.....	531
Figure 169:	Policy and charging architecture (non-roaming).....	535
Figure 170:	Policy and charging architecture (roaming home-routed).....	536
Figure 171:	Policy and charging architecture (roaming - LBO).....	536
Figure 172:	Time quota consumption example.....	542
Figure 173:	Time quota QHT example.....	543
Figure 174:	Gy Assume Positive response to CCR-I failure.....	559
Figure 175:	Gy Assume Positive response to CCR-U failure.....	560
Figure 176:	Gy Assume Positive response to CCR-T failure.....	561
Figure 177:	CCR-U unknown session with Gy Assume Positive session restart.....	562
Figure 178:	CCR-U session disconnect.....	563
Figure 179:	CCR-I session disconnect.....	564
Figure 180:	Variation of a CCR-I session disconnect.....	565
Figure 181:	Session creation for time zone and PLMN reporting.....	570
Figure 182:	PGW update for changes in UE time zone and PLMN.....	571
Figure 183:	eHRPD to LTE handover.....	572
Figure 184:	PCRF-initiated session recovery.....	583
Figure 185:	PGW-initiated session recovery.....	584
Figure 186:	Hybrid access bonding.....	611
Figure 187:	Locally trigger default bearer modification with QoS update.....	624
Figure 188:	No QoS negotiation indication.....	627

Figure 189:	GBR/MBR AMBR roundup process flow.....	634
Figure 190:	AAA detects failure based on origination time stamp.....	653
Figure 191:	AAA and PGW clean up after Maximum Wait Time.....	654
Figure 192:	PGW detects SGW delay failure based on origination time stamp.....	656
Figure 193:	PGW informing PCRF for dropped messages.....	659
Figure 194:	Tunnel ingress packet reassembly example.....	670
Figure 195:	Gateway IP reassembly flow example.....	672
Figure 196:	IP reassembly configuration with LAG utilization and load-balancing based on ingress GTP-UTEID.....	677
Figure 197:	7750 IPSec implementation architecture.....	684
Figure 198:	Example GRE over IPSec tunnel configuration.....	691
Figure 199:	IKEv2 TS-List.....	697
Figure 200:	Video wholesale configuration example.....	703
Figure 201:	Call Flow for psk-radius/cert-radius.....	705
Figure 202:	Typical call flow of EAP authentication.....	707
Figure 203:	Dynamic port block starvation in LSN.....	729
Figure 204:	IPv4 applications routed over IPv6 mobile networks.....	732
Figure 205:	Ping transmission from the IPv4 application to the IPv4 server example.....	733
Figure 206:	PLAT and DNS64 functionality.....	733
Figure 207:	Outside pool configuration.....	755
Figure 208:	Outside address ranges.....	760
Figure 209:	Deterministic LSN44 configuration example.....	764
Figure 210:	IP address/port translation modes.....	769
Figure 211:	DNAT provisioning model.....	773
Figure 212:	Destination NAT based on PCC rule.....	773
Figure 213:	NAT and MG grouping in LB-less architecture.....	774
Figure 214:	NAT configuration in LB-less architecture.....	775
Figure 215:	Active VM in local NAT in LB-less architecture.....	775
Figure 216:	Local loopback to peering router in LB-less architecture.....	776
Figure 217:	NAT and MG grouping in LB architecture.....	777
Figure 218:	NAT configuration in LB architecture.....	777
Figure 219:	Active VM in local NAT in LB architecture.....	778
Figure 220:	Local loopback to peering router in LB architecture.....	778
Figure 221:	Pool selection based on traffic destination.....	783
Figure 222:	NAT pool selection based on the inside source IP address.....	783
Figure 223:	SPF with multiple NAT policies.....	787
Figure 224:	Bypassing NAT policy rule.....	787
Figure 225:	Metadata inside data capture files.....	811

List of Tables

Table 1:	7750 SR services support.....	58
Table 2:	Vault configuration parameters.....	143
Table 3:	IP addressing recommendations.....	149
Table 4:	Node synchronization based on the UE-by-UE synchronization status.....	160
Table 5:	Node action based on the UE-by-UE synchronization status change.....	161
Table 6:	PGW/GGSN (S5/S8/Gn/Gp) behavior.....	193
Table 7:	SGW (S11) behavior.....	194
Table 8:	Supported Ethernet action - tag value combinations.....	197
Table 9:	Supported Ethernet action - tag value combinations.....	197
Table 10:	QCI-{FC, DSCP} mapping table.....	218
Table 11:	Sub-AVPs of the OC-OLR AVP.....	229
Table 12:	S6b interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload.....	231
Table 13:	SWm interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload.....	232
Table 14:	STa interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload.....	233
Table 15:	Gx interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload.....	234
Table 16:	RADIUS Access-Request message, GTP PCO/APCO information, and local configuration.....	239
Table 17:	Configuration examples for P-CSCF selection.....	253
Table 18:	QCI values defined by 3GPP standards.....	280
Table 19:	Feature-List-ID 1 features used in Gx.....	282
Table 20:	3GPP Standardized QCI Values.....	282
Table 21:	PGW 23.060 Supported Call Flows.....	290
Table 22:	PGW 23.401 supported call flows.....	291
Table 23:	Call flows and DRMP value in Diameter messages.....	323
Table 24:	S6b interface parameter values used in S6b eMPS bypass.....	331
Table 25:	Conditional-Policy-Information AVP support.....	336
Table 26:	Gx-specific Diameter AVPs.....	337
Table 27:	Message from the UE and response from the PGW/GGSN.....	348
Table 28:	Status code 200 parameters.....	358
Table 29:	GTP-C profile.....	374
Table 30:	Diameter profile.....	375
Table 31:	QCI-{FC, DSCP} mapping table.....	376
Table 32:	SGW reference point path down behavior.....	377
Table 33:	Diameter messages supported over the Sd interface.....	452
Table 34:	SSG TDF Sd AVPs.....	458

Table 35:	Supported charging interfaces.....	492
Table 36:	Change condition values populated at the SDC level.....	496
Table 37:	PRA information reporting AVPs in Gx messages.....	503
Table 38:	PRA information reporting IEs supported in GTP-C messages.....	504
Table 39:	PRA Information Reporting AVPs in Rf Messages Supported by SGW and PGW.....	504
Table 40:	PRA Information Reporting AVPs in Gy Messages Supported by PGW.....	505
Table 41:	PRA information reporting AVPs in Sd messages supported by SSG and TDF.....	506
Table 42:	Gx AVPs and messages for multiple PRA.....	508
Table 43:	PRA-Install AVP in messages supported by PGW.....	509
Table 44:	PRA-Remove AVP in messages supported by PGW.....	510
Table 45:	Event-Trigger AVP in messages supported by PGW.....	510
Table 46:	Supported Features of Feature-List-ID 1 used in Gx.....	511
Table 47:	Supported Features of Feature-List-ID 2 used in Gx.....	511
Table 48:	Event-Trigger AVP in messages supported by PGW.....	512
Table 49:	PRA information reporting IEs for multiple PRAs supported in GTP-C messages.....	513
Table 50:	RAN Secondary RAT Usage Report Bindings.....	532
Table 51:	GTP messages for RAN Secondary RAT Usage Reports.....	532
Table 52:	Secondary RAT Usage Report AVP.....	534
Table 53:	Error code handling.....	550
Table 54:	Trigger types for PLMN and time zone reporting.....	569
Table 55:	Gx profile CCFH action per call flow type.....	584
Table 56:	Supported sub-AVPs of the Override-Control AVP.....	587
Table 57:	Gx interface event triggers.....	595
Table 58:	Presence of usage monitoring AVPs in Gx messages.....	599
Table 59:	URL encryption for HTTP traffic redirection in CUPS and non-CUPS deployments.....	621
Table 60:	Examples for rounding upwards.....	634
Table 61:	Features of Feature-List-ID 2 used in Gx.....	638
Table 62:	Gx-specific Diameter AVPs.....	639
Table 63:	Gx re-used Diameter AVPs.....	639
Table 64:	Examples of configured and DNS-matched domain names.....	642
Table 65:	Message priority of network-initiated call flows.....	658
Table 66:	GTP-C overload condition and PGW throttling actions.....	661
Table 67:	APN LCI IE.....	666
Table 68:	APN OCI IE.....	667
Table 69:	Syslog message fields for NAT flow logging.....	736
Table 70:	Message fields.....	737
Table 71:	Flow creation and deletion template field descriptions.....	748
Table 72:	Contiguous number of subscribers.....	756
Table 73:	Preserving Det/Dyn port ratio with 2n subscribers.....	757

Table 74:	Fixed number of deterministic ports with 2n subscribers.....	758
Table 75:	Supported combinations of SNAPT and DNAT.....	769
Table 76:	Metadata acronyms.....	811
Table 77:	List of supported packet drop reasons.....	817
Table 78:	Generic Access-Request attributes.....	829
Table 79:	VSAs 26/10415 - 3GPP vendor-specific Access-Request attributes.....	831
Table 80:	Generic Access-Accept attributes.....	835
Table 81:	VSAs 26/6527 - Nokia vendor-specific Access-Accept attributes.....	838
Table 82:	VSAs 26/94 - Nokia vendor-specific Access-Accept attributes.....	840
Table 83:	VSAs 26/311 - Microsoft vendor-specific Access-Accept attributes.....	841
Table 84:	VSAs 26/10415 - 3GPP vendor-specific Access-Accept attributes.....	841
Table 85:	VSAs 26/2011 - Huawei vendor-specific Access-Accept attributes.....	842
Table 86:	Generic Accounting-Request (Start) attributes.....	842
Table 87:	VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Start) attributes.....	846
Table 88:	VSAs 26/6527 - Nokia vendor-specific Accounting-Request (Start) attributes.....	849
Table 89:	Generic Accounting-Request (Stop) attributes.....	849
Table 90:	VSAs 26/10415 - 3GPP-Vendor-Specific Accounting-Request (Stop) Attributes.....	853
Table 91:	VSAs 26/6527 - Nokia-Vendor-Specific Accounting-Request (Stop) Attributes.....	856
Table 92:	Generic Accounting-Request (Interim-Update) attributes.....	857
Table 93:	VSAs 26/10415 - 3GPP vendor-specific Accounting-Request (Interim-Update) attributes....	861
Table 94:	VSAs 26/6527 - Nokia vendor-specific Accounting-Request (Interim-Update) attributes....	864
Table 95:	Generic Accounting-Request (Accounting-On) attributes.....	864
Table 96:	Generic Accounting-Request (Accounting-Off) attributes.....	865
Table 97:	RADIUS CoA-Request attributes.....	867
Table 98:	VSAs 2626/6527/11 - Nokia vendor-specific RADIUS CoA-Request attributes.....	867
Table 99:	Generic Disconnect-Request attributes.....	868
Table 100:	VSAs 26/10415 - 3GPP vendor-specific Disconnect-Request attributes.....	869
Table 101:	Disconnect-Response (ACK) attributes.....	870
Table 102:	Disconnect-Response (NAK) attributes.....	870
Table 103:	Echo Request attributes.....	871
Table 104:	Echo Response attributes.....	872
Table 105:	Create Session Request.....	872
Table 106:	Bearer Context to be Created IE (for Create Session Request).....	876
Table 107:	Bearer Context to be Removed IE (for Create Session Request).....	877
Table 108:	Create Session Response.....	877
Table 109:	Bearer Context to be Created IE (for Create Session Response).....	880
Table 110:	Bearer Context to be Removed IE (for Create Session Response).....	881
Table 111:	Create Bearer Request.....	881
Table 112:	Bearer Context to be Created IE (for Create Bearer Request).....	883

Table 113:	Create Bearer Response.....	884
Table 114:	Bearer Context to be Created IE (for Create Bearer Response).....	885
Table 115:	Bearer Resource command.....	886
Table 116:	Bearer Resource Command Failure Indication.....	888
Table 117:	Modify Bearer Request.....	888
Table 118:	Bearer Context to be Modified IE (for Modify Bearer Request).....	891
Table 119:	Bearer Context Deleted IE (for Modify Bearer Request).....	891
Table 120:	Modify Bearer Response.....	891
Table 121:	Bearer Context to be Modified IE.....	893
Table 122:	Bearer Context Deleted (for Modify Bearer Response).....	894
Table 123:	Delete Session Request.....	894
Table 124:	Delete Session Response.....	895
Table 125:	Delete Bearer Request.....	896
Table 126:	Bearer Context within Delete Bearer Request IE.....	898
Table 127:	Delete Bearer Response.....	898
Table 128:	Bearer Context within Delete Bearer Response IE.....	899
Table 129:	Downlink Data Notification Request.....	900
Table 130:	Downlink Data Notification ACK.....	901
Table 131:	Downlink Data Notification Failure Indication.....	901
Table 132:	Delete Indirect Data Forwarding Tunnel Request.....	902
Table 133:	Delete Indirect Data Forwarding Tunnel Response.....	902
Table 134:	Modify Bearer command.....	902
Table 135:	Bearer Context IE within the commands.....	903
Table 136:	Modify Bearer Command Failure Indication.....	903
Table 137:	Update Bearer Request.....	904
Table 138:	Bearer Context IE to be Modified.....	905
Table 139:	Update Bearer Response.....	906
Table 140:	Bearer Context IE to be Modified.....	907
Table 141:	Delete Bearer Command.....	908
Table 142:	Bearer Context IE to be Deleted.....	909
Table 143:	Delete Bearer Failure Indication.....	909
Table 144:	Bearer Context IE to be Deleted.....	909
Table 145:	Create Indirect Data Forwarding Tunnel Request.....	909
Table 146:	Bearer Context IE within CIDFTR.....	910
Table 147:	Create Indirect Data Forwarding Tunnel Response.....	911
Table 148:	Bearer Context IE with CIDFTR.....	911
Table 149:	Release Access Bearer Request.....	912
Table 150:	Release Access Bearer Response.....	913
Table 151:	Modify Access Bearer Request.....	913

Table 152:	Bearer Context to be Modified IE.....	914
Table 153:	Bearer Context Deleted IE.....	914
Table 154:	Modify Access Bearer Response.....	914
Table 155:	Bearer Context to be Modified IE (for Modify Access Bearer Response).....	914
Table 156:	Bearer Context Deleted IE (for Modify Access Bearer Response).....	915
Table 157:	Change Notification Request IE.....	915
Table 158:	Change Notification Response IE.....	916
Table 159:	Suspend Notification.....	917
Table 160:	Suspend Acknowledge.....	917
Table 161:	Resume Notification.....	918
Table 162:	Resume Acknowledge.....	918
Table 163:	PGW Restart Notification.....	918
Table 164:	PGW Restart Notification Acknowledge.....	919
Table 165:	CC-Request attributes.....	919
Table 166:	CC-Answer attributes.....	925
Table 167:	VSAs 26/28458 - Nokia vendor-specific CC-Answer attributes.....	928
Table 168:	Re-Auth-Request attributes.....	928
Table 169:	Re-Auth-Answer attributes.....	931
Table 170:	Credit Control Request (CCR) attributes.....	933
Table 171:	Credit Control Answer (CCA) attributes.....	942
Table 172:	VSAs 26/6527 - Nokia vendor-specific CC-Answer attributes.....	946
Table 173:	Re-Auth-Request Gy attributes.....	946
Table 174:	Re-Auth-Answer Gy attributes.....	947
Table 175:	Abort-Session-Request Gy Attributes.....	948
Table 176:	Abort-Session-Answer Gy Attributes.....	949
Table 177:	TDF Debit / Reserve Units Request message attributes.....	950
Table 178:	TDF Debit / Reserve Units Response message attributes.....	957
Table 179:	Accounting-Request generic attributes.....	959
Table 180:	Accounting-Answer generic attributes.....	961
Table 181:	ePDG Rf attributes.....	963
Table 182:	SGW Rf attributes.....	976
Table 183:	PGW Rf attributes.....	994
Table 184:	Diameter AA-Request attributes.....	1012
Table 185:	Diameter AA-Request 26/10415 VSAs attributes.....	1015
Table 186:	Diameter AA-Answer attributes.....	1018
Table 187:	Diameter AA-Answer 26/10415 VSAs attributes.....	1022
Table 188:	Diameter AA-Answer 26/12951 VSAs attributes.....	1022
Table 189:	Diameter AA-Answer Self-Activation-Status Attributes.....	1023
Table 190:	Diameter AS-Request 26/12951 VSAs attributes.....	1024

Table 191:	Diameter Re-Auth-Request attributes.....	1024
Table 192:	Diameter Re-Auth-Answer attributes.....	1026
Table 193:	Diameter Session-Termination-Request attributes.....	1027
Table 194:	Diameter Session-Termination-Answer attributes.....	1028
Table 195:	Diameter Abort-Session-Request attributes.....	1029
Table 196:	Diameter Abort-Session-Answer attributes.....	1031
Table 197:	Diameter EAP-Request SWm attributes.....	1032
Table 198:	Diameter EAP-Answer SWm attributes.....	1034
Table 199:	Diameter Re-Auth-Request SWm attributes.....	1035
Table 200:	Diameter Re-Auth-Answer SWm attributes.....	1036
Table 201:	Diameter AA-Request SWm attributes.....	1037
Table 202:	Diameter AA-Answer SWm attributes.....	1038
Table 203:	Diameter Session-Termination-Request SWm attributes.....	1039
Table 204:	Diameter Session-Termination-Answer SWm attributes.....	1040
Table 205:	Diameter Abort-Session-Request SWm attributes.....	1041
Table 206:	Diameter Abort-Session-Answer SWm attributes.....	1042
Table 207:	ePDG to UE Notify message types.....	1042
Table 208:	UE to ePDGNotify message types.....	1043
Table 209:	SWm error mapping to UE IKEv2 error codes.....	1044
Table 210:	S2b error mapping to UE IKE error code.....	1047
Table 211:	Private Notify message - error types.....	1055
Table 212:	Protocol identifiers in configuration protocol options list of PCO IE.....	1058
Table 213:	Containers of additional parameter list, UE to Network direction.....	1058
Table 214:	Containers of additional parameter list, Network to UE direction.....	1060
Table 215:	Gx interface Supported Features of Feature-List-ID 1.....	1062
Table 216:	Gx interface Supported Features of Feature-List-ID 2.....	1065
Table 217:	S6b interface Supported-Features AVP.....	1066

1 Getting started

Get general information about this guide.

About this guide

This guide describes features and functionality provided by the Cloud Mobile Gateway (CMG). It describes how to provision, configure and implement related protocols, and manage the system.

This guide is organized into functional sections and provides concepts and descriptions of the implementation flow, as well as Command Line Interface (CLI) syntax and command usage.



NOTE:

- Configuration outputs shown in this guide are examples only; actual displays may differ depending on supported functionality and user configuration.
- This guide covers content for the release specified on the title page to the guide, and may also contain content that will be released in later maintenance loads. Refer to the applicable *7750 MG and CMG Release Notes* for information about features supported in each load of the Release software.
- The information in this guide is intended to be used in conjunction with the other guides in the CMG and the 7750 SR configuration guides. The 7750 SR guides describe SR OS service features that are supported by the CMG and are mostly used without modification. These features apply to all mobile gateway types, unless explicitly noted.
- MD-CLI commands for CMG are introduced in phases to each release. In the current release, this guide only includes classic CLI commands in, for example, feature descriptions and configuration examples. For more information about MD-CLI, refer to the *CMG and NRD MD-CLI Quick Reference Guide* and the *CMG and NRD MD-CLI User Guide*. For more information about the MD-CLI commands, refer to the *CMG MD-CLI Command Reference Guide*.
- For more information about the acronyms included in this guide, refer to the *Acronyms and Terms*.

Audience

This guide is intended for network administrators who are responsible for configuring routers. It is assumed that the network administrators have an understanding of networking principles and configurations.

2 Summary of changes

Changes between guide issues.

For more information about the CMG features and enhancements, refer to the *CMG Feature Overview*.

Changes between issue 9-0 (25.3.R1) and issue 10-0 (25.7.R1)

Feature or enhancement	Description	See
CMG realm limit of 256 realm ID per MG VM to be increased and dynamic flush of realm entries with no host in it (MG-4679)	Added a note about avoiding to combine a high number of Gx realms with a high number of PCRFs per realm	<i>Diameter destination host and realm</i>
—	Added a note about not combining different dedicated bearer rule types in the same rule base	<i>Dedicated bearer configuration options</i>
—	Added a note about the use of the debug>mobile>call-insight>all-ues command in live deployments	<i>Call trace and packet capture</i>

Changes between issue 8-0 (24.11.R1) and issue 9-0 (25.3.R1)

Feature or enhancement	Description	See
—	Editorial changes	—

Changes between issue 7-0 (24.7.R1) and issue 8-0 (24.11.R1)

Feature or enhancement	Description	See
4G and 5G WPS support on SGW/PGW/SMF based on preconfigured ARP and QCI_5QI values (MG-3045)	Added support for 4G and 5G WPS on SGW/PGW/SMF based on preconfigured ARP and QCI_5QI values	<i>Wireless Priority Service on SGW-C/PGW-C/SMF over 5GC</i>
ICR SWO based on Gx failover (MG-4223)	Added support for triggering an ICR FSWO when all Gx connections are detected as lost	<i>ICR Forced Switchover based on Gx failover</i>
IMSI in update request (MG-3942)	Added support for IMSI in update requests as per CR 1063 of 3GPP TS 29.060 and CR 1778 of 3GPP TS 29.274	<i>Modify Bearer Request</i>
—	Added a note for the case when PCRF sends the FQDN without port and transport information	<i>Multiple destination realms for a Diameter peer (Gy interface) configuration</i>
—	Added a note about defining the APN in “DNS query” format and updated the respective configuration example	<i>Configuring PGW selection for ePDG/TWAG</i>
—	7750 SR MG-specific content is removed in phases.	—

Changes between issue 6-0 (24.3.R1) and issue 7-0 (24.7.R1)

Feature or enhancement	Description	See
CMG support for network trace for non-SBI signaling messages and interfaces (MG-3499)	Added support for non-SBI interfaces in CMG and enhancements in vTAP solution.	Network trace Configuring a network trace profile Enabling the network trace feature
Domestic roaming in emergency (MG-4243)	Added support for domestic roaming for LTE Data, Voice over LTE, and Emergency Call	Gx interface Supported Features of Feature-List-ID 2
ePDG as CNF: Support for geo-redundancy (UE-by-UE sync) (MG-3889)	Added support for geo-redundancy in ePDG as CNF	ICR UE-by-UE reconcile synchronization
ePDG as VNF: ICR improvements – sync MG-ISM complete UE records for ePDG ICR (UE-by-UE) (MG-2655)	Added support for ICR UE-by-UE reconcile synchronization in ePDG as VNF	ICR UE-by-UE reconcile synchronization
Non-terminating policy rules (MG-2233)	Added support for marking predefined policy rules as Non-Terminating	Non-terminating policy rules

3 Mobile gateway features overview

Overview of the Cloud Mobile Gateway (CMG).

**NOTE:**

The terms "Mobile Gateway", "MG", and "gateway" refer to the following supported types of 7750 SR mobile gateways.

- Packet Data Gateway (PGW)
- Gateway GPRS Support Node (GGSN)
- Serving Gateway (SGW)
- evolved Packet Data Gateway (ePDG)
- Trusted Wireless Access Gateway (TWAG)
- Subscriber Services Gateway (SSG)
- 3GPP2 Home Agent (HA)

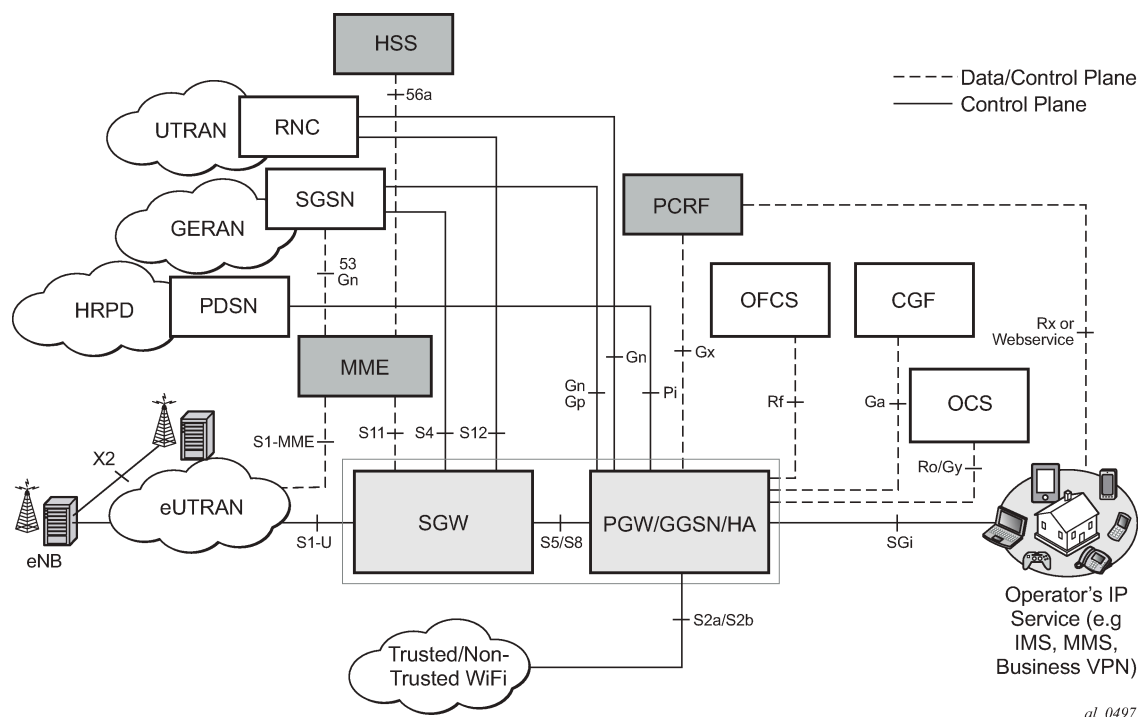
The terms "gateway" and "mobile gateway" also refer to the CMG product.

3.1 4G EPC core network elements

LTE is the name given to the fourth-generation (4G) networks. Today, third-generation (3G) networks exist in two modes: CDMA Rev 0/Rev A and UMTS. LTE has an evolution path for both architectures. 3GPP standard 23.401 defines the 3GPP evolution whereas standard 23.402 defines the non-3GPP evolution. The 7750 SR Mobile Gateway supports 23.401-based implementation compatible with Rel 13.9.0 (Dec 2016).

3GPP architecture for EPC core shows the Nokia Evolved Packet Solution covering 3G and 4G networks as well as trusted/non-trusted Wifi networks.

Figure 1: 3GPP architecture for EPC core



As seen in the architecture, the EPS contains the following network elements:

- UE — User Equipment
- Evolved UTRAN (E-UTRAN) eNodeB — supports the LTE air interface and includes functions for radio resource control, user plane ciphering, and Packet Data Convergence Protocol (PDCP)
- Mobility Management Entity (MME) — manages mobility, UE identities and security parameters
- Serving Gateway (SGW) — This is the node that terminates the interface toward E-UTRAN. For each UE associated with the EPS, at a given time, there is one SGW. SGW functions include:
 - the local mobility anchor point for inter-eNodeB handover
 - E-UTRAN idle mode downlink packet buffering and initiation of network triggered service request procedure
 - transport-level packet marking in the uplink and the downlink
 - UL and DL charging per UE, PDN, and QCI
- PDN Gateway (PGW) — This is the node that terminates the SGi interface toward the PDN. If a UE is accessing multiple PDNs, there may be more than one PGW for that UE. The PGW provides connectivity to the UE to external packet data networks by being the point of exit and entry of traffic for the UE. PGW functions include the following:
 - policy enforcement (gating and rate enforcement)
 - charging support
 - UE IP address allocation

- transport-level packet marking in the uplink and the downlink
 - UL and DL rate enforcement based on APN Aggregate Maximum Bit Rate (APN-AMBR)
 - Diameter Routing Message Priority (DRMP) support
 - SGi tunneling for non-IP PDNs
 - Enhanced Multimedia Priority Service (eMPS) calls with advance priority support
 - APN-level congestion control
 - The policy and charging rules function (PCRF) includes:
 - policy (QoS and gating) control
 - charging control
 - In a non-roaming scenario, there is only a single PCRF in the H-PLMN associated with one UE IP-CAN session. The PCRF terminates the Gx interfaces.
 - Offline Charging — The general charging architecture that LTE follows is defined in 3GPP TS 32.240.
 - primarily involves different Charging Trigger Functions (CTF) which act on a network element (SGW, PGW) to report accounting information to a Charging Collection Function (CCF)
-  **NOTE:**

Multiple OFCS instances are supported for Ga, including leCCF and C3S.
- Accounting information from LTE/EPC network elements is gathered so that the billing system can generate a consolidated record for each service.
 - Online Charging — The PGW/GGSN supports online charging functionality over the Diameter Gy/Ro interface per Release 11 December 2013 version of 3GPP standards.
 - 3GPP2 Home Agent functions include:
 - policy enforcement
 - charging support
 - UE IP address allocation
 - UE IP address preservation when the3 user re-attaches between LTE/eHRPD and HRPD
 - enterprise support for both regular UEs and mobile Routers (RFC5177)
 - IP-in-IP/GRE tunneling support

Related information

[Online charging standards summary](#)

The PGW/GGSN supports OCS functionality over the Diameter Gy/Ro interface per Release 11 December 2013 version of 3GPP standards.

3.2 3GPP reference points

The reference points used in the LTE-combined SGW/PGW environment.



NOTE:

3GPP version 13.6.0 December 2016 is supported for standards including S6b for TS 29.273 and S2a for 3GPP.

3GPP2 Pi	The interface between the PDSN and the public Internet.
3GPP2 RADIUS	The interface used in cdma2000 networks for AAA functionality.
Ga	Charging Interface between the gateway and the Offline Charging System (OFCS), based on the GTP-Prime protocol. Charging Data Records (CDRs) generated by the gateway are sent to the OFCS over Ga.
Gx	Provides transfer of (QoS) policy and charging rules from the PCRF to the Policy and Charging Enforcement Function (PCEF) in the PGW.
Gy/Ro	The reference point between the PGW and the Online Charging System (OCS) that provides flow-based online charging capabilities.
Gn/Gp	Provides inter-working between GERAN/UTRAN and E-UTRAN as specified by 3GPP standards. Toward the GERAN/UTRAN, the PGW appears as a GGSN toward the SGSN/RNC, supporting both non-direct and direct tunneling.
RADIUS	The RADIUS interface supported on the PGW/GGSN provides support for subscriber Authentication, Authorization, and Accounting capabilities based on RADIUS.
Rf	The charging interface between the gateway and OFCS (Offline Charging System), based on the Diameter protocol. Accounting Start/Stop/Interim records are sent to the OFCS over Rf.
S1-U	The reference point used between E-UTRAN and SGW for the per bearer user plane tunneling and inter eNodeB path switching during handover.

S11	Provides tunnel management between the SGW and the MME. It is used for attachment procedures as well as relocation.
S11-U	Provides a GTP-U interface between SGW and MME. It is used for Clot control plane optimization.
S12	Provides user plane tunneling when Direct Tunnel is established. It is based on the lu-u/Gn-u reference point using the GTP-U protocol as defined between SGSN and UTRAN or respectively between SGSN and GGSN.
SGi	The reference point between a PGW and the packet data network. The packet data network may be an operator external public or private packet data network or an intra-operator packet data network (to provision IMS services). The reference point corresponds to Gi for 3GPP accesses.
S2a	Provides the user plane with related control and mobility support between the following: • trusted non-3GPP IP access (3GPP2 - eHRPD (enhanced HRPD)) and the PGW acting as the Local Mobility Anchor (LMA) toward the HSGW (HRPD Serving Gateway) • TWAG (Trusted WLAN Access Gateway) and the PGW
S2b	Provides control and mobility support between the PGW/SGW and ePDG.
S4	This interface between the PGW and 2G/3G SGSNs is based on Gn/GTP (SGSN-GGSN). The user plane interface is based on GTP-U and the control plane is based on GTP-C.
S5/S8	Provides user plane tunneling and tunnel management between SGW and PGW. It is used for SGW relocation due to UE mobility and if the SGW needs to connect to a non-collocated PGW for the required PDN connectivity.
S6b	This reference point between a PGW and a 3GPP AAA server provides mobility-related authentication/authorization for access to trusted non-3GPP networks based on 3GPP TS 29.273 and TS 23.402.
SWm	This reference point is located between the 3GPP AAA Proxy (or Proxy/Server) and the ePDG and is used for AAA signaling. The SWm reference point is defined in 29.273 and uses RFC 4072 to carry EAP messages (Diameter-EAP-Request/Diameter-EAP-Answer). SWm also uses RFC 4005 (AA-Request/AA-Answer) for EAP-AKA fast re-authentication.

SWu

Supports handling of IPSec tunnels. SWu functionality includes UE-initiated tunnel establishment, user data packet transmission within the IPSec tunnel and tear down of the tunnel and support for fast update of IPSec tunnels during handover between two untrusted non-3GPP IP accesses.

SWw

Provides the link between the UE and a Wi-Fi access point.

Sx

This reference point is supported for communication between the control and user planes of the gateway (SGW, PGW/GGSN, combined SGW/PGW) for 4G packet core. The Sx interface utilizes the Packet Forwarding Control Protocol (PFCP) for control traffic and the GTP-U protocol for data traffic between the Control Plane (CP) and the User Plane (UP) function. The Sx interface consists of the following interfaces:

- Sxa — used for communication between the SGW-C and the SGW-U
- Sxb — used for communication between the PGW-C and the PGW-U
- Sxab — used for communication between the combined SGW-C/PGW-C and the combined SGW-U/PGW-U
- N4 — used for communication between the SMF and the UPF

Both the Sx CUPS (4G Control and User Plane Separation) and 5GC N4 interfaces utilize the PFCP protocol for control traffic and the GTP-U protocol for any data traffic between the CP and the UP function. The CP function applies to the control plane nodes (SGW-C, PGW-C, combined SGW-C/PGW-C, and SMF) and the UP function applies to the user plane nodes (SGW-U, PGW-U, combined SGW-U/PGW-U and UPF).

According to 3GPP specifications, a PFCP session refers to both the Sx and/or N4 sessions. PFCP associations describe the procedures to establish associations between EPC (SGW-C, PGW-C, SGW-U, PGW-U, and so on) and 5GC (SMF and UPF) nodes.

For more information, refer to the *CMG PFCP (Sx and N4) Interface Description* and the *CMG CUPS Configuration Guide*.

3.3 Supported standalone and combination gateways

The supported standalone and combination gateways for the CMG.

The following types of standalone gateways are supported in CMG.

PGW/GGSN

The Packet Data Gateway (PGW) has S5/S8 and optionally Gn/Gp configured, whereas the GGSN is a PGW with only Gn/Gp configured.

SGW

The Serving Gateway (SGW) node terminates the interface toward E-UTRAN. For each UE associated with the EPS, at a given time, there is one SGW.

ePDG

The evolved Packet Data Gateway (ePDG) provides secure access for a UE across untrusted non-3GPP access, usually assumed to be WiFi. Supported interfaces include SWm, SWu, and S2b.

TWAG

The Trusted Wireless Access Gateway (TWAG) is supported as a standalone gateway providing trusted wireless access, or as a combination gateway for ePDG and TWAG access, providing both trusted and untrusted WiFi access.

SSG TDF

The Subscriber Services Gateway (SSG) facilitates policy-based service steering including Traffic Detection Function (TDF) (as specified in 3GPP TS 29.212).

The following types of combined gateways are supported on single or separate Virtual Machines (VMs) for the CMG.

- A combined SGW/PGW is supported on a VM.
- A combined ePDG/SGW/PGW is supported with the ePDG on a VM and the SGW/PGW on a separate VM.
- A combined TWAG-ePDG is supported on a VM.
- A combined PGW-HA is supported on a VM.

**NOTE:**

Any combined gateway with a PGW can also be a GGSN.

3.4 Control and User Plane separation support

CMG supports separation of the CP (CP/SMF) and UP (UP/UPF) into independent network functions.

CMG support CP and UP separation per 3GPP-defined standard. For example, the PFCP signaling protocol is used for communication between CP and UP network elements in the Sx (4G CUPS) and N4 (5GC) interfaces.

CUPS provides flexibility for deployment of the mobile packet core and supports distribution of the UP functions closer to the user, while maintaining the CP function at a centralized location for ease of management.

For more information about the supported CUPS implementation, refer to the *CMG CUPS Configuration Guide* and the *CMG PFCP (Sx and N4) Interface Description*.

3.5 RADIUS support

The PGW/GGSN RADIUS interface provides a mechanism for performing authorization, authentication, and accounting (AAA) for subscriber Bearer/PDP contexts.

The PGW/GGSN RADIUS interface is based on the following standards:

- 3GPP TS 29.061: Interworking between the Public Land Mobile Network (PLMN) supporting packet based services and Packet Data Networks (PDN) (Release 13)
- IETF RFC 2865: Remote Authentication Dial In User Service (RADIUS)
- IETF RFC 2866: RADIUS Accounting
- IETF RFC 5176: Dynamic Authorization Extensions to RADIUS
- IETF RFC 3162: RADIUS and IPv6
- IETF RFC 2548: Microsoft Vendor-specific RADIUS Attributes
- IETF RFC 2868: RADIUS Attributes for Tunnel Protocol Support

The following are some of the RADIUS features supported on the PGW/GGSN:

IP address allocation

Can be configured for each APN on the PGW/GGSN with support for IPv4, IPv6, IPv4/IPv6 address allocation. If configured for RADIUS-based IP address allocation, the PGW/GGSN performs RADIUS authentication upon receiving a GTP Create PDP Context Request or Create Session Request.

Accounting

RADIUS accounting allows the RADIUS server to track volume and time usage for each PDN connection and support to assign a charging profile ID to the PDN connection. This may be useful, for example, when the APN is used for corporate access and also in wholesale configurations where the RADIUS server is owned by a separate ISP/network provider.

RADIUS accounting is also often used to inform upstream servers of the IP address assigned to a particular UE. For example, the operator may configure a set of WAP servers as a RADIUS server and therefore the PGW/GGSN will provide an association between the MSISDN and IP address.

The PGW also supports controlling Accounting ON and OFF messaging to work system or individual APN levels, and

selecting the load balancing procedure toward the RADIUS servers.

Periodic interim accounting is supported following configured intervals and triggered due to a RAT change event.

Virtual APN pre-authentication

The APN name provided by the UE is used by the MME/SGSN to select the PGW/GGSN. The Virtual APN feature allows the GGSN to override the GTP provided APN. The PGW/GGSN can use RADIUS pre-authentication so that a RADIUS server can provide the new APN.

RADIUS-initiated disconnect

The PGW/GGSN supports RADIUS-initiated disconnect per RFC 5176. This allows the RADIUS server to send a Disconnect-Message to release a particular PDN connection. Disconnect requests are accepted from any of the servers configured in the RADIUS group used for accounting by the APN.

RADIUS Framed-Route

The PGW/GGSN supports the RADIUS Framed-Route attribute in Access-Accept messages. This enables a RADIUS server to install a set of IPv4 static routes for networks behind a UE. These routes are injected into the PGW's routing protocols so that they can be reached through the UE. Each Framed-Route attribute (up to four per UE) contains a single IPv4 address prefix route that reflects an IPv4 address subnet that is reachable through the UE.

RADIUS Framed-IPv6-Route

The PGW/GGSN supports the RADIUS Framed-IPv6-Route attribute in Access-Accept messages. This enables a RADIUS server to install a set of IPv6 static routes for networks behind a UE. These routes are injected into the PGW's routing protocols so they can be reached through the UE. Each Framed-IPv6-Route attribute (up to four per UE) contains a single IPv6 address prefix route that reflects an IPv6 address subnet that is reachable through the UE.

Tunnel protocol

RADIUS-based tunneling allows the RADIUS server to dynamically select the tunneling protocol as well as the tunneling end points without static tunneling configuration. RADIUS-based tunneling is limited to IPv4 routes.

3.6 CDR formats

The gateway supports the following CDR formats from 3GPP TS 32.251.

- PGW-CDR formats including the presence category of the parameter in the event/CDR.; all supported fields are included by default; bearer-level charging for PGW-CDRs is supported
- SGW-CDR formats from including the presence of the field in the event/CDR

- ePDG IP CAN (ePDG-CDR) bearer messages including the presence category of the parameter in the event/CDR
- TDF-CDR for Application Based Charging for SSG; TDF-CDRs are supported as per 3GPP 32.251 / 32.298 and Diameter Rf as per 3GPP 32.251 / 32.299

For more information, refer to the *CMG CDR Description*.

3.7 Offline charging standards

The SGW and PGW both support Diameter Rf as per Release 11 3GPP standard for the EPC domain.

The following key offline charging system (OFCS) functionality is supported:

- Diameter ACR (start/interim/stop) and ACA commands
- Diameter over TCP and IPv6 or IPv4 transport
- Diameter charging sessions are set up from the ePDG, SGW, PGW, or TDF to the CDF on a per PDN session basis

Each IP-CAN session has a corresponding Diameter Rf charging session to the CDF.

- Each bearer within the PDN session has a corresponding traffic data volumes (TDV) container (ePDG, PGW, SGW) or service data container (SD) (TDF) for the Diameter charging session
- Charging-information addition triggers and partial record event triggers
- Default charging profiles for home subscribers, roamers (home-routed) as well as visiting (local break out) scenarios
- CCF connectivity outage and recovery
- Gy Rf synchronization events including Rf charging information addition triggers related to offline/online charging at the PGW

Related information

Offline charging

This key offline charging system (OFCS) functionality is supported by the CMG.

3.8 Online charging standards

Overview of the supported online charging system (OCS) standards and functionality

Related information

Online charging

The online charging functionality that is supported for the CMG.

3.8.1 Online charging standards summary

The PGW/GGSN supports OCS functionality over the Diameter Gy/Ro interface per Release 11 December 2013 version of 3GPP standards.

The relevant standards are as follows:

- 3GPP TS 32.251 v13.6.1 (Rel-13 June-2016): Telecommunication management; Charging management; Packet Switched (PS) domain charging
- 3GPP TS 32.295 v11.0.0 (Rel-11 April 2011): Telecommunication management; Charging management; Charging Data Record (CDR) transfer
- 3GPP TS 32.297 v11.0.0 (Rel-11 Sept 2012): Telecommunication management; Charging management; Charging Data Record (CDR) file format and transfer
- 3GPP TS 32.298 v11.8.0 (Rel-11 Dec 2013): Telecommunication management; Charging management; CDR Parameter Description
- 3GPP TS 32.299 v13.7.0 (Rel-13 Dec 2016): Telecommunication management; Charging management; Diameter charging application
- 3GPP TS 32.240 v13.2.0 (Rel-13 Dec 2016): Telecom Management.; Charging Management; Charging architecture and principles
- IETF RFC 4006
- IETF RFC 6733, *Diameter Base Protocol*; *Diameter Credit-Control Application*

For PGW/GGSN deployments that require support for Ga or Rf offline charging and Gy online charging interfaces per Release 11 December 2013 version of the 3GPP standards, the following are some key aspects of the functionality supported:

- Centralized Unit Determination with Centralized Rating is supported (OCS determines the Rating and units required for any service).
- Volume and Time quota monitoring is supported.
- Only Session Charging with Unit Reservation (SCUR) is supported. For LTE access (with S5-GTP), or for a GGSN with Gn/Gp connectivity to a SGSN, every bearer context/PDP context has a corresponding Diameter Credit Control session over Gy.
- Credit-Control is on a per Rating Group basis. For example, every Rating Group has an individual quota. There could be more than one Rating Group per PDP/Bearer Context. There is a Multiple-Services-Credit-Control AVP for each Rating-Group. All Service Data Flows (SDFs) belonging to the same Rating Group consume quota allocated for that Rating Group.



NOTE:

The SDF is an aggregate set of packet flows carried through the PCEF that matches a service data flow template.

- Reporting is on a per-Rating Group basis. All SDFs belonging to a particular Rating Group consume quota allocated for that Rating Group. Service-ID-level reporting is not supported. (However, service IDs belonging to a particular Rating Group are communicated as part of the MSCC).
- Requesting quota upon first packet detection for an SDF is the default behavior. The CCR-Initial message does not include any MSCC AVPs. Upon receiving a successful authorization from the OCS, the bearer is established. Subsequently, when the first packet corresponding to a service data flow ID is detected, a CCR-U is sent to the OCS with an MSCC AVP including the RSU and the Rating-Group/Service-ID.

Online charging standards caveats

The following aspects of the functionality are not supported:

- IEC (Immediate Event Charging) and ECUR (Event Charging with Unit Reservation)
- Credit Pooling
- Tight inter-working between offline and online charging
- Control of Offline Charging (Offline Charging AVP) by the OCS

3.8.2 Deployment scenarios

The following are examples of the online charging deployment scenarios:

- *Non-roaming scenario: PGW/GGSN and OCS deployed in the H-PLMN*
- *Roaming scenario: Home-routed (PGW/GGSN and OCS deployed in the H-PLMN)*
- *Roaming Scenario: Local-breakout (PGW/GGSN deployed in the V-PLMN and OCS deployed in the H-PLMN)*

For more information, see *Online charging deployment scenarios*.

Provisioning options

Some of the options for specifying the primary and secondary OCS addresses that a PGW interacts with include:

- Local provisioning on the PGW/GGSN at the gateway-level
Configuration context: `config>mobile>pdn>gy`
- Local provisioning on the PGW/GGSN at the APN-level. This overrides the configuration specified at the gateway-level.
Configuration context: `config>mobile>pdn>apn>charging`
- Local provisioning on the PGW/GGSN as part of the Charging Profile. This is the preferred option to support the Local Breakout scenario mentioned above. Note that the Charging Profile is picked using

the Octet 1 from the Charging Characteristics IE received from the HSS/MME at the time of IP-CAN session establishment.

Configuration context: `config>mobile>profile>charging>pgw-charging`

DCCA profile provisioning

The Diameter Credit Control Application (DCCA) profile associated with each Primary and Secondary OCS must be configured. The specified DCCA profiles are associated with the OCS peers (primary and secondary) during OCS provisioning.

The configuration context for a DCCA profile is: `config>mobile>profile>dcca`.

For information on supported DCCA profile configuration options, refer to the *CMG CLI Reference Guide*.

3.8.3 Charging characteristics and charging profile

The charging characteristics IE received from MME/HSS is used to pick a locally configured charging profile. A charging profile supports an option to enable or disable online charging.

Related information

Charging characteristics and charging profile

The charging characteristics (CC) IE received from MME/HSS is used to pick a locally configured charging profile. The selected charging profile is applicable for the IP-CAN session.

3.8.4 Bill shock prevention

Operators can control session data costs in real time by using the Bill Shock Prevention (BSP) feature. When the BSP feature is enabled, the roaming status is used to pick up a locally configured charging profile to be used for the IP-CAN session.

Related information

Bill Shock Prevention

The Bill Shock Prevention (BSP) feature enables operators to control session data costs in real time.

3.8.5 Diameter messages

The mobile gateway supports Diameter messages for online charging.

Related information

Diameter messages for online charging

3.8.6 Re-authorization

The mobile gateway supports re-authorization initiated either by the gateway (CTF) or by the OCS.

Related information

Gateway and OCS re-authorization

3.8.7 Time-quota

The mobile gateway supports time-quota consumption.

Related information

Time quota consumption

3.8.8 Tariff change

The Tariff-Time-Change AVP is used to determine the tariff switch time as described by RFC 4006, section 8.20. When this event is encountered, all current counts are captured and new counts are started. The counts are sent to the OCS in next CCR.

The client (gateway) reports usage which straddles the change time by sending two instances of the Used-Service-Unit AVP, one with Tariff-Change-Usage = 'unit-before-tariff-change' and one with 'unit-after-tariff-change'.

3.8.9 Termination action and redirection

The OCS can perform termination synchronously (if it has a CCR pending processing) by returning CCA with an appropriate failure code at the command level.

Alternatively, the OCS can send an ASR (Abort Session Request) message to terminate the Diameter Session. The gateway follows by sending a CCR (Terminate) to the OCS.

Final-Unit-Indication AVP in a CCA from the OCS at the Multiple Services Credit Control level (MSCC) is supported. Supported Final-Unit-Actions include TERMINATE and REDIRECT. When FUA = REDIRECT, OCS must send the redirect-server URL.

3.8.10 Failure handling

When the PGW determines a failure condition with the current OCS, the PGW uses the Credit-Control-Failure-Handling AVP (as defined in RFC 4006, sections 5.7 and 8.14) to determine what to do:

- This AVP can be received from the OCS in the CCA message or can be locally configured.
- The value received from the OCS always overrides any already existing value.

Failure Handling mechanisms according to the behavior described in Annex B of 3GPP TS 32.251 are supported.

The gateway also supports configuration of the credit control failure handling (CCFH) option.

Related information

Credit Control Failure Handling

Configuration of Credit Control Failure Handling (CCFH) is supported using the CLI command `cc-failover-handling` in the DCCA profile context.

3.8.11 FUI support

The mobile gateway supports the Final Unit Indication (FUI) functionality and configuration including FUI support at the MSCC level and command level, filter IDs with FUI Restrict and Redirect and FUA local policy for filter IDs.

Related information

Final Unit Indication

This section provides information about the supported FUI functionality and configuration including FUI support at the MSCC level and command level, filter IDs with FUI Restrict and Redirect and FUA local policy for filter IDs.

3.8.12 CCR-I and CCR-E support

The ability to configure sending a Credit Control Request Initial (CCR-I) message on first online rating group (RG) detection is provided. CCR Events (CCR-E) are used to report PLMN, time zone, and location changes to the OCS.

Related information

CCR event reporting

CCR Events (CCR-E) are used to report PLMN, time zone, and location changes to the OCS.

CCR-I on first rating group detection

The ability to configure sending a Credit Control Request Initial (CCR-I) message on first online rating group (RG) detection is provided.

3.9 Gx interface standards

The CMG Gx interface provides support for Release 11 Dec 2013 version of the 3GPP standards including the following:

- 3GPP TS 23.203 v13.10.0 (Rel-13 Dec 2016) - Policy and Charging Control Architecture
- 3GPP TS 29.212 v13.8.0 (Rel-13 Dec 2016) - Policy and Charging Control; Reference Points
- 3GPP TS 29.214 v11.11.0 (Rel-11 Dec 2013) - Policy and Charging Control over Rx Reference Point
- 3GPP TS 29.213 v13.8.0 (Rel-13 Dec 2016) - Policy and Charging Control Signaling Flows and QoS parameter mapping

The following specifies the PGW/GGSN standards compliance for Policy and Charging Control (PCC) features on the Gx interface and is in compliance with the 3GPP version of standards.

- Indication of Bearer Usage for IMS Signaling

- TAI and ECGI location change event triggers
- PGW/GGSN Gx enhancements for inter-operation with Gn/Gp SGSN
- Charging-correlation exchange handling updates
- Clarification of Flow-Description (IP-filter rule) restrictions
- AVP code/ M-bit setting updates
- HSS initiated subscribed QoS modification to be reported even when PCRF does not subscribe to those events
- Bi-directional filters support over Gx
- Supported Features AVP handling, as defined in Release 9
- Support for IMS Emergency Calls
- Support for Diameter race conditions handling

Related information

Gx interface configuration

The Gx interface policy and charging configuration options supported by the CMG.

3.10 Policy and charging control

The mobile gateway supports policy and charging control features for EPC nodes.

For CUPS-specific policy information, refer to the *CMG CUPS Configuration Guide* and to the *CMG PFCP (Sx and N4) Interface Description*.

Related information

Charging configuration

Policy configuration

Information and configuration guidelines for policy control implementation for EPC nodes.

3.11 GPRS tunneling protocol

The mobile gateway supports GPRS Tunneling Protocol (GTP) functionality and interface and signaling configuration.

Related information

GPRS Tunneling Protocol

This section describes information related to supported GPRS Tunneling Protocol (GTP) functionality.

3.12 Application Assurance

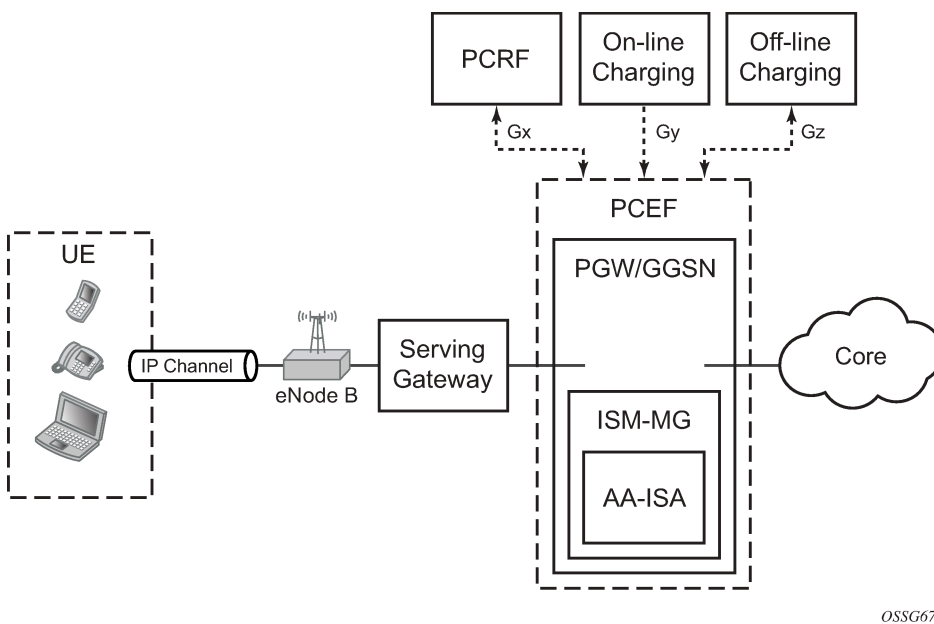
AA-ISA allows a mobile operator to identify applications being used by their subscriber when 5-tuple classification is insufficient.

The mobile gateway (PGW/GGSN) employs AA-ISA to serve as a deep packet inspection (DPI) function. This includes an extensive re-use of the AA signatures, classification and policy functions developed already for the wireline market (business and residential). AA-ISA in a PGW/GGSN node uses one of the ISM-MG MDA slots. Only traffic that requires application identifications/classifications will be diverted to the AA-ISA. In general, this results in default bearer traffic being diverted to the AA-ISA.

There are many policy actions that follow traffic identifications including stateful firewall, bandwidth policing, flow-count policing, remarking, TCP optimization, tethering detection, in-browser notifications, HTTP enrichment, and content-based charging.

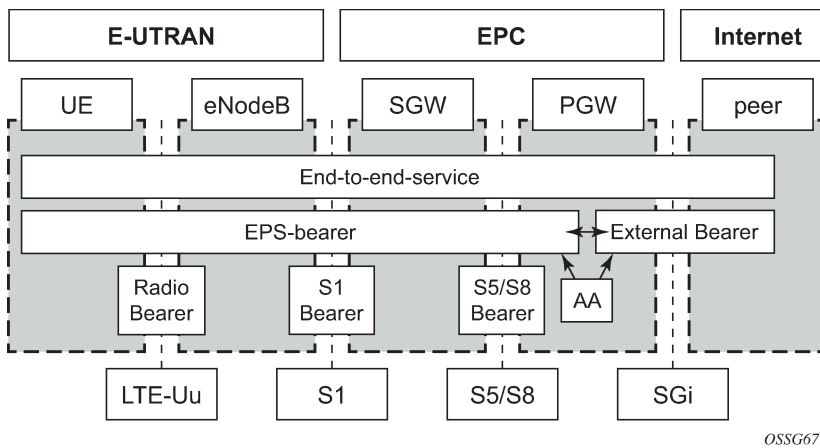
Mobile Network Deployment using PCEF with AA ISA shows a mobile network deployment using the Policy Enforcement function (PCEF) equipped with an AA-ISA. Within the PGW/GGSN, AA-ISA works in conjunction with ISM-MG. In this model, ISM-MG communicates with the AA-ISA for all dynamic configurations (such as creating subscribers). Refer to the *CMG Application Assurance Configuration Guide*.

Figure 2: Mobile Network Deployment using PCEF with AA ISA



The EPC traffic bearer is shown in *EPC Bearer*.

Figure 3: EPC Bearer



At the PGW, traffic is diverted to AA after tunnels are terminated (for example, tunnel encapsulation removed, such as GTP headers).

The default bearer (Internet/overtop video/audio) traffic is subjected to application assurance policies. Therefore, on the PGW, traffic to and from default bearer gets diverted into AA-ISA for further (>L4) analysis, classification and actions. Only traffic that requires >L4 analysis is diverted to AA-ISA by the ISM-MG.

3.13 Lawful Interception

The PGW/GGSN deployments include support for Lawful Interception per Release 11 Dec 2013 version of the 3GPP standards.

The following three lawful interception (LI) interfaces are supported:

X1 interface (CLI provisioning using SSH) Supports separate LI operator contexts and privileges. IMSI as a target identifier is supported, as well as IMEI (first 14 digits), MSISDN, and MAC. Target activation, target deactivation, target modification, and target retrieval (show command) operations are supported over the X1 interface.

X2 interface Based on TCP as the transport protocol with ASN.1-encoded Intercept Related Information (IRI) messages. PDP context activation, PDP context deactivation, start of interception with PDP context active and PDP context modification GGSN IRI events are supported per TS 33.106, TS 33.107 and TS 33.108. ASN.1 IRI messages comply with Annex B.9 of TS 33.108.

X3 interface (content of communication) Supports ULIC header (version 0) over UDP/IP as defined in 33.108, as well as ULICv1 over UDP and ULICv1 over TCP.

For more information, refer to the *CMG Lawful Interception Guide*.

3.14 IP reassembly support

The mobile gateway provides support for managing IP fragmentation and reassembly.

Related information

IP fragmentation and reassembly

This section provides information about IP fragmentation and reassembly.

3.15 L2TP/PPP regeneration support

The mobile gateway supports L2TP/PPP regeneration over LTE (S5/S8).

Related information

L2TP/PPP regeneration

The mobile gateway supports L2TP/PPP regeneration over LTE (S5/S8).

3.16 VoLTE bearer preservation

The mobile gateway supports reducing the impact of VoLTE sessions in case of PCRF failures. If the gateway detects that the PCRF has failed or restarted, VoLTE-related bearers are disconnected after the first packet on the VoLTE signaling bearer (the default bearer with QCI=5) has been received.

VoLTE bearers are preserved if there is no need for the PCRF to be contacted.

To enable this functionality, use the following command:

```
configure mobile pdn apn preserve-volte
```

To configure the time that the mobile gateway waits before determining that PCRF is not reachable, use the following command:

```
configure mobile pdn apn gx connection-failure-timer value
```



NOTE:

- If the connection failure timer is not configured, the default value is used.
- If the feature is not enabled, the connection failure timer has no effect.
- if the origin state identifier of the peer is increased the preserve-volte functionality is applied immediately, if enabled.

3.17 Troubleshooting and analysis

Call trace

The Mobile Gateway supports call trace and packet capture functionality. Call trace is a real-time troubleshooting and debugging tool that provides the ability to capture control and user plane messages from the mobile gateway.

Per Call Measurement Data

Per Call Measurement Data (PCMD) provides the ability to do performance analysis of mobile gateway functions on a per call session basis. Unlike KPI/KCI accounting that generates cumulative statistics over a specific time period, the PCMD feature supports analysis of each procedure at per call session granularity.

PCMD is a real-time diagnostics and troubleshooting tool that provides call and connection records for information such as call duration, quality, and disposition, and other important call events that occur during a call, such as call setup, failure, handover, and termination.

Refer also to the *CMG PCMD Guide*.

Performance and capacity indicators

Key Performance Indicator (KPI) and Key Capacity Indicator (KCI) counters are used for post processing of the system's behavior with respect to control plane signaling, bearer traffic pattern, and system resource usage during peak and non-peak hours.

Statistical data about the system behavior helps to understand the network activity and helps the operators to plan their network to handle the signaling, data bandwidth demands, and system resource needs.

GTP ping

The mobile gateway supports GTP ping functionality.

SNMP traps

The mobile gateway supports SNMP traps.

Related information

[Call trace and packet capture](#)

The call trace feature supports call trace and packet capture functionality.

[Per Call Measurement Data](#)

Per Call Measurement Data (PCMD) provides the ability to do performance analysis of mobile gateway functions on a per call session basis. Unlike KPI/KCI accounting that generates cumulative statistics over a specific time period, the PCMD feature supports analysis of each procedure at per call session granularity.

[Key Performance Indicator and Key Capacity Indicator](#)

Information about the Key Performance Indicator (KPI) and Key Capacity Indicator (KCI) counters that are used for post processing of the system behavior with respect to CP signaling, bearer traffic pattern, and system resource usage during peak and non-peak hours.

GTP ping

The reachability of GTP end-points, such as newly commissioned ENBs, MMEs and PGWs, can be checked using GTP ping commands.

SNMP traps

3.18 SR OS services support

Every CMG release is based on a 7750 SR release and therefore inherits many SR OS features. However, only a subset of 7750 SR features are explicitly verified within the CMG.

This section captures 7750 SR service features that are currently supported within the CMG, and which are mostly used without modification by the CMG. These features apply to all mobile gateway types, unless explicitly noted.

7750 SR services support lists supported 7750 SR services for the CMG.

Table 1: 7750 SR services support

BGP, BGPv6	IES	RADIUS, TACACS
BFD, BFD LAG	L2 services	Static MSS
BFD over (various) L2 connections	LACP	SDP GRE scaling
ECMP/LAG	LAG link profile	Static routing
FTP, SFTP	LDP	TEID-based hashing
Filters	MPLS, RSVP	VPLS
Route leaking	NAT	VPRN
GRE, IPSec	NTP	VRRP
GRE/IPSec reassembly	OSPF	
IS-IS	Routing policies	

4 Cloud Mobile Gateway (CMG)

The configuration guidelines specific to the Cloud Mobile Gateway (CMG). These guidelines are intended to be used after installation is completed as described in the *CMG Installation Guide*.

**NOTE:**

The terms "gateway" and "mobile gateway" also refer to the Cloud Mobile Gateway (CMG) product.

Related information

Common gateway configuration

Overview of the provisioning functionality that is common to all standalone and combined gateway types.

4.1 CMG overview

The CMG supports mobile gateway functionality that can be deployed on a generic computing infrastructure in a cloud environment. The CMG can support multiple gateway functions including PGW, GGSN, SGW, combined SGW/PGW/GGSN, TWAG, and ePDG.

This section provides an overview of the CMG architecture, deployment mode, and redundancy mode. For detailed installation and setup information, refer to the *CMG Installation Guide*.

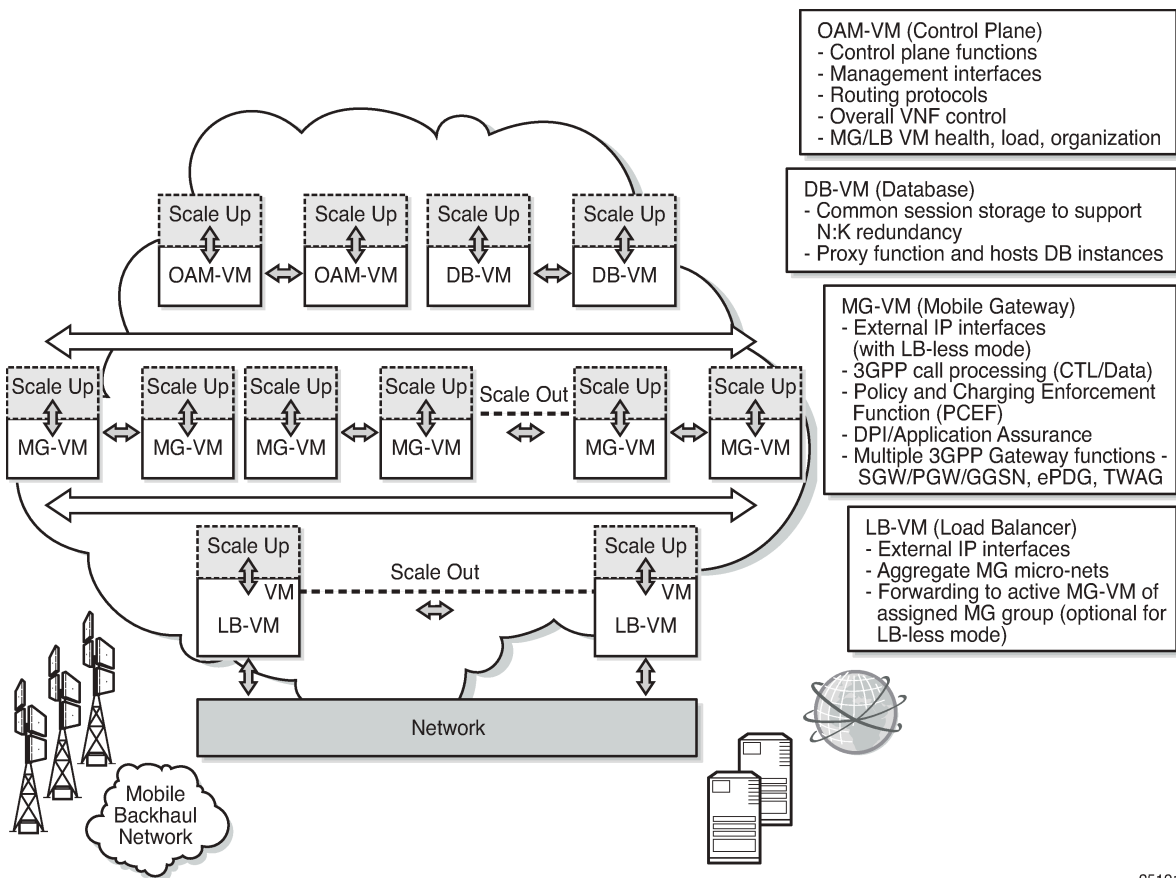
A CMG instance consists of multiple virtual machines (VMs) running on a generic computing infrastructure such as x86 servers. Each VM is dedicated to a specific set of functions that can be replicated across many similar VMs. A group of VMs is represented as a single instance of an application, where the VMs operate in sync with similar VMs of the group to support a network function. The ability to add multiple VMs for each function allows the CMG to scale horizontally and support a scaling range from a thousand to a million devices. The VM, within a CMG instance, is agnostic of other virtual machines present in the shared server environment.

4.1.1 CMG architecture

The CMG architecture comprises the following virtual network function components (VNFC), as shown in *CMG VNFC architecture*:

- Operations, Administration, and Maintenance (OAM-VM)
- Load Balancer VM (LB-VM)
- Mobile Gateway VM (MG-VM)

Figure 4: CMG VNFC architecture



25191

The CMG VNF components provide the following functions:

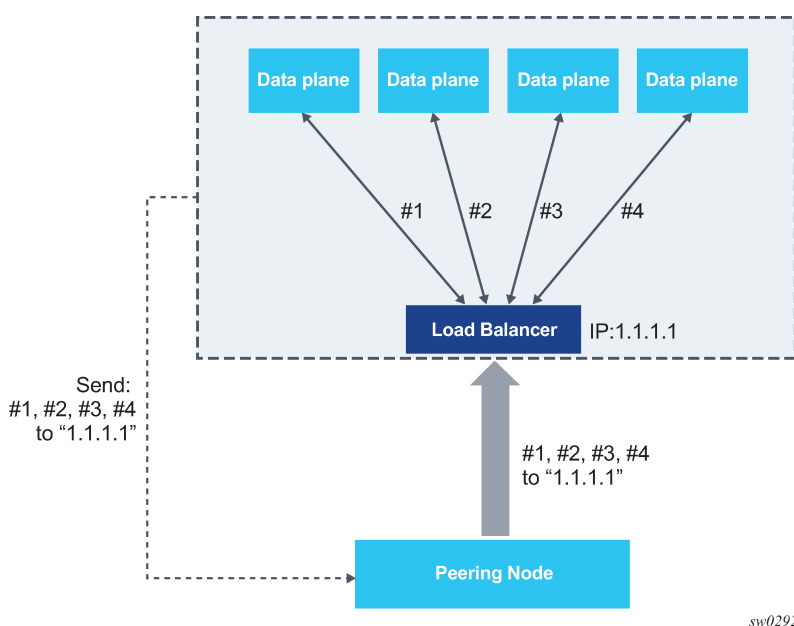
- OAM-VM supports CP functions including VNF and VNFC management, routing protocols, management interface (SNMP, TELNET, SSH, and CLI) for the configuration, KPI-KCI periodic XML report generation, and so on.
- LB-VM supports the following functions:
 - Provides network connectivity to the mobile gateway function and load distribution across the MG-VMs.
 - Forwards the GTP-C/GTP-U and UE addressed packets to the MG-VM.
 - Supports load balancing (LB) and load-balancing less (LB-less) modes of operation. When a CMG instance is configured to use the LB-VM for UP traffic (optional), it is identified as a CMG instance in LB mode. When a CMG instance is configured without the LB-VM for UP traffic, is identified as a CMG instance in LB-less mode.
 - Provides a single common IP address for network interface elements (MME, eNodeB, S/PGW, TWAG, and ePDG). Alternatively, each signaling and data plane interface can be configured with separate IP addresses. The ability to separate individual functions or merge multiple functions on individual interfaces allows maximum flexibility for various deployment cases of a CMG instance.
- The MG-VM supports the following functions:

- 3GPP call processing (Control and Data Plane), Policy and Charging Enforcement Function (PCEF), and Application Assurance (PCEF enhanced with ADC for application detection and control, and L7 service classification for policy charging control).
- 3GPP gateway functions such as SGW, PGW/GGSN, combined SGW/PGW/GGSN, ePDG, TWAG, and so on.
- Network connectivity for UP traffic for the CMG configured in LB-less mode.
- Supported service functions depending on the configurable personality of the MG-VM. As an SGW, the MG-VM provides bearer management services and the mobility anchor point. In an MG-VM that is configured as a PGW, the VM provides bearer management, IP anchor point support, lawful interception functionality, charging functionality, QoS enforcement, and advanced DPI functionality. When configured as an combined SGW/PGW, all the SGW and PGW functions are supported on a single MG-VM.
- External IP interfaces are configured on the MG-VM for a CMG instance deployed in LB-less mode.
- The DB-VM is the common session storage to support N:K redundancy. DB-VM supports proxy function and hosts DB instances.

4.1.2 Inline LB and CP load distributor (LB-less) architecture options

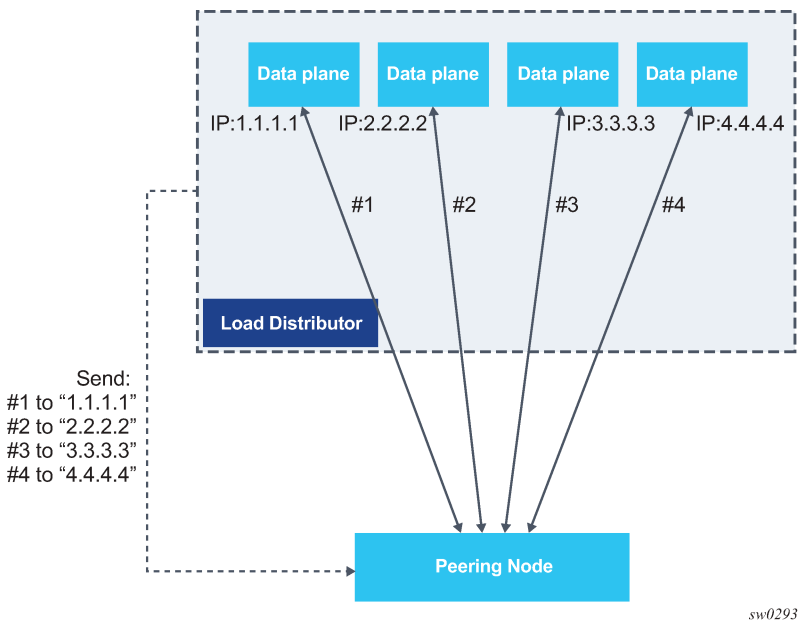
The LB for control and user plane traffic operates on a per-packet basis representing a set of session processing resources (for example, MG-VM). The LB is a receiving point of CP and UP traffic from the peering node. It receives and forwards traffic to the individual resource VMs in real-time, by associating each session with the MG-VM that is processing the session (for example, CP and UP traffic forwarding between the LB-VM and MG-VM on the CMG).

Figure 5: User plane LB architecture



The load distributor (LD) method for user traffic uses the CP signaling (for example, GTP-C) toward a peering node to distribute sessions across its own individual session processing resources (for example, MG-VM on the CMG). The load distributor keeps track of the utilization of the resources it is representing and assigns a new session to an anchor session processing unit (for example, MG-VM). The session distribution via CP signaling allows UP traffic to be distributed to a single session processing unit without the need of a load balancing function in the traffic path. Tunneling protocols such as GTP, IPSec, and so on, support the redirection method of load distribution, which uses a session between a load distributor and a session processing resource.

Figure 6: User plane load distributor architecture



4.2 CMG deployment options

The supported deployment options are the following:

- *CMG with GTP-U LB function*
- *CMG without GTP-U LB function (LB-less)*
- *CMG in hybrid mode*

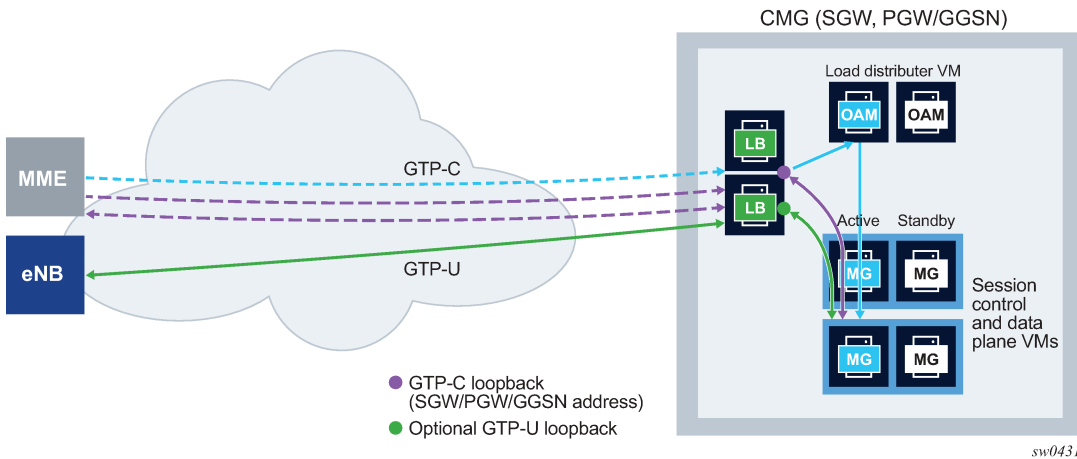
4.2.1 CMG with GTP-U LB function

CMG supports processing of user traffic received on an MG-VM through the LB-VM.

The CMG with a load balancer uses a common IP address for all the GTP-U sessions anchored across multiple MG-VMs present in the CMG instance. GTP-C signaling is used for setting up an individual GTP tunnel per PDN session across multiple MG-VMs with the use of a single IP address.

Multiple LB-VMs can be configured per CMG instance to allow for a scalable system. The peering router is configured with an ECMP path to the common IP address so that traffic is distributed across multiple LB-VMs. GTP-U UP traffic is received on the LB-VM and forwarded to the MG-VMs based on the identified session.

Figure 7: User plane LB architecture



The loopback IP address of the SGW, PGW, or combined SGW/PGW/GGSN is advertised to the peering nodes, and gateway selection (using DNS lookup) is performed based on the advertised single loopback address of the gateway. The peering node sends an initial GTP-C message (Create Session Request) to the loopback IP address that is configured to be received on the OAM-VM via the LB-VM. The CP traffic processing is identical on both LB and LB-less deployments of the CMG. That is, all CP traffic (GTP-C, Diameter, RADIUS, and so on) is received on a common loopback address of the gateway instance.

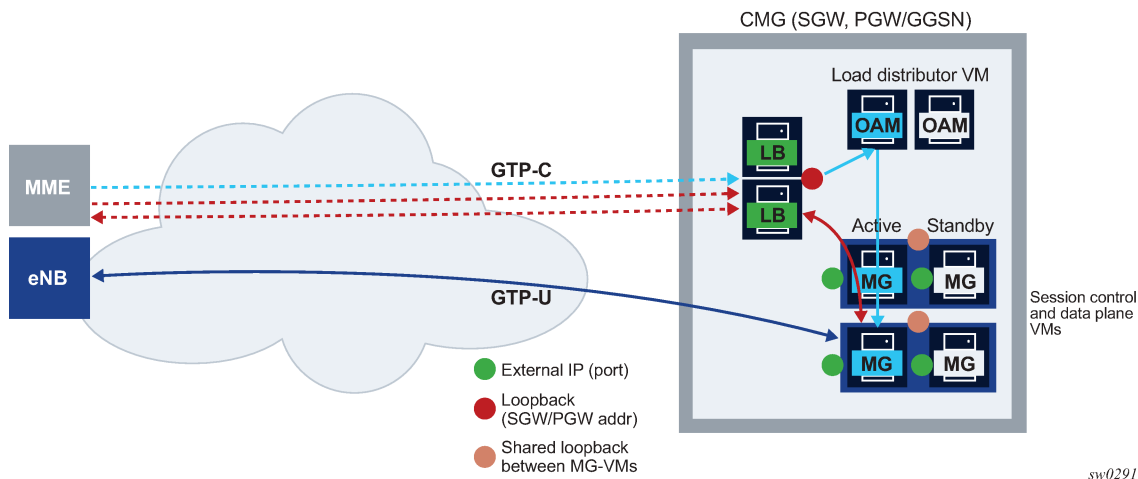
The OAM-VM receives the initial Create Session Request message on the SGW or combined SGW/PGW from the MME or on the PGW/GGSN from the SGW, SGSN, TWAG, or ePDG. The OAM-VM forwards the Create Session Request message to the anchor MG-VM, which responds with the common IP address as the FTEID of the UP. All control messages from the peering nodes are sent to the loopback IP address of the gateway.

4.2.2 CMG without GTP-U LB function (LB-less)

The CMG supports GTP-U traffic to be processed on an MG-VM without an LB-VM for UP traffic present in the system. The CMG without GTP-U load balancer (that is, LB-less mode) uses the GTP-U redirection method to distribute sessions on the MG-VM.

Each MG-VM is configured with one or more GTP-U interfaces (that is, I/O ports with IP addresses). GTP-C signaling is used to anchor the GTP-U traffic on a single MG-VM, to allow UP traffic to be sent and received directly from the MG-VM.

Figure 8: User plane LB-less architecture



The loopback IP address of the SGW, PGW, or combined SGW/PGW is advertised to the peering nodes and gateway selection (using DNS lookup) is performed based on the advertised single loopback address of the gateway. The peering node sends an initial GTP-C message (Create Session Request) to the loopback IP address that is configured to be received on the OAM-VM via the LB-VM (dedicated or integrated with MG-VM). The CP traffic processing is identical on both LB and LB-less deployments of the CMG. That is, all CP traffic (GTP-C, Diameter, RADIUS, and so on) is received on a common loopback address of the gateway instance.

The OAM-VM receives the initial Create Session Request message on the SGW or combined SGW/PGW from the MME or on the PGW/GGSN from the SGW, SGSN, TWAG or ePDG. The OAM-VM forwards the Create Session Request message to the anchor MG-VM, which responds with the local IP address as the FTEID of the UP. Subsequent control messages (that is, all non-zero GTP-C TEID messages) from the peering nodes are sent to the loopback IP address of the gateway.

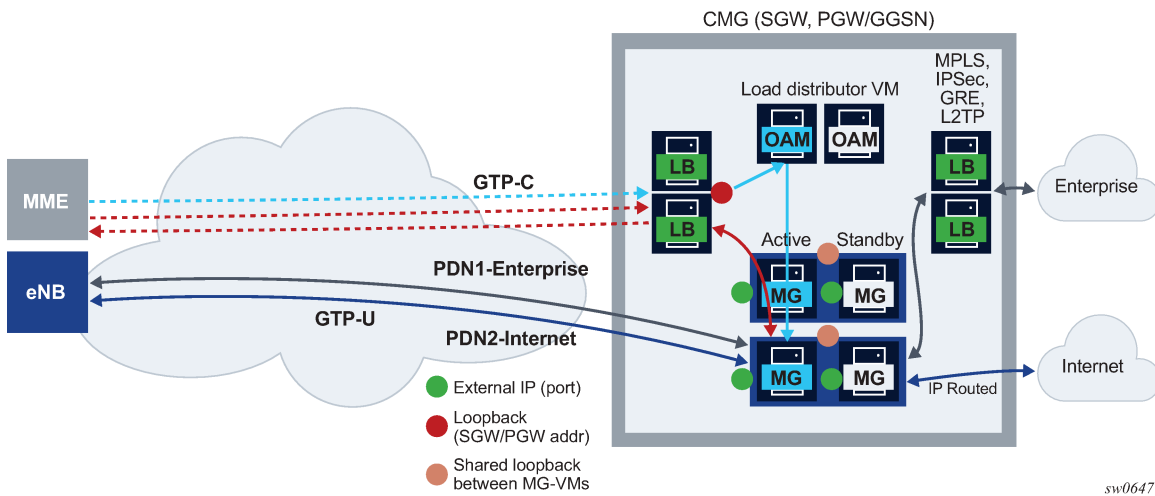
If a dedicated LB-VM is deployed for CP traffic, then the GTP-C traffic is received on the LB-VM, which then forwards it to the anchor MG-VM.

4.2.3 CMG in hybrid mode

The CMG supports hybrid mode which allows GTP-U traffic processing on an MG-VM with or without an LB-VM on the SGi interface.

User plane traffic on the SGi interface is supported either on the LB or the LB-less option per APN basis. For example, the Enterprise APN is configured to use the LB-VM option while the Internet APN is configured to avoid LB (that is, using the LB-less mode).

Figure 9: CMG in hybrid mode



Using the LB-VM on the SGi interface allows tunneling options like MPLS, IPSec, GRE, and L2TP with Enterprise PDN session by avoiding a tunnel per MG-VM basis and maintaining seamless connectivity from the gateway toward the enterprise network.

Avoiding the LB-VM on the SGi interface allows compute efficiency for high throughput Internet PDN sessions. Compute efficiency is maintained by avoiding inter-VM traffic with the LB-less mode.

4.3 CMG redundancy support

The CMG supports the following redundancy protection methods:

- intra-VNF 1:1 and N:K (CP and UP) redundancy (local HA)
- inter-VNF 1:1 redundancy (geo-redundancy)

4.3.1 Intra-VNF redundancy (local HA) for CMG

The OAM-VMs (VNF controller VMs) are configured as a 1:1 pair (active/standby) and the Load Balancer VMs (LB-VM) operate in an all-active mode.

In addition to routing protocols processing, the active OAM-VM also performs the operation and management function of the CMG instance. The active OAM-VM synchronizes its state with the standby OAM-VM to support redundancy of the OAM function. When a failure is detected on the active OAM-VM, the standby OAM-VM takes over the role of active processing and provides seamless failure of the previously active OAM-VM with no CP outage.

The active OAM-VM monitors all active MG-VMs and triggers a switchover to the alternate MG-VM in the event of an active MG-VM failure.

For the LB-VM, a peering router is configured so that traffic directed to an IP address configured on the gateway is reachable via all LB-VMs. The peering router must be configured with an ECMP path across all LB-VMs with failure detection using BFD or next-hop tracking. In case of an LB-VM failure, the peering

router detects it and redistributes traffic across the remaining active LB-VMs. This allows LB-VMs within a CMG instance to operate in an N:K redundancy model, where K is the number of LB-VMs that can fail simultaneously. The load balancer VMs (LB-VMs) must be dimensioned as per the N:K redundancy model.

VMs within a CMG instance can be distributed across separate zones within a data center to limit the impact of a zone-level failure. It is recommended that active and standby VMs are placed on a compute infrastructure in such a way so that they do not fail during a compute infrastructure level failure.

4.3.1.1 MG-VM redundancy (1:1)

In MG-VM redundancy model, the MG-VMs (call processing VMs) are configured in 1:1 (active/standby) pairs. That is, failure of the active MG-VM triggers a switchover to the paired standby MG-VM.

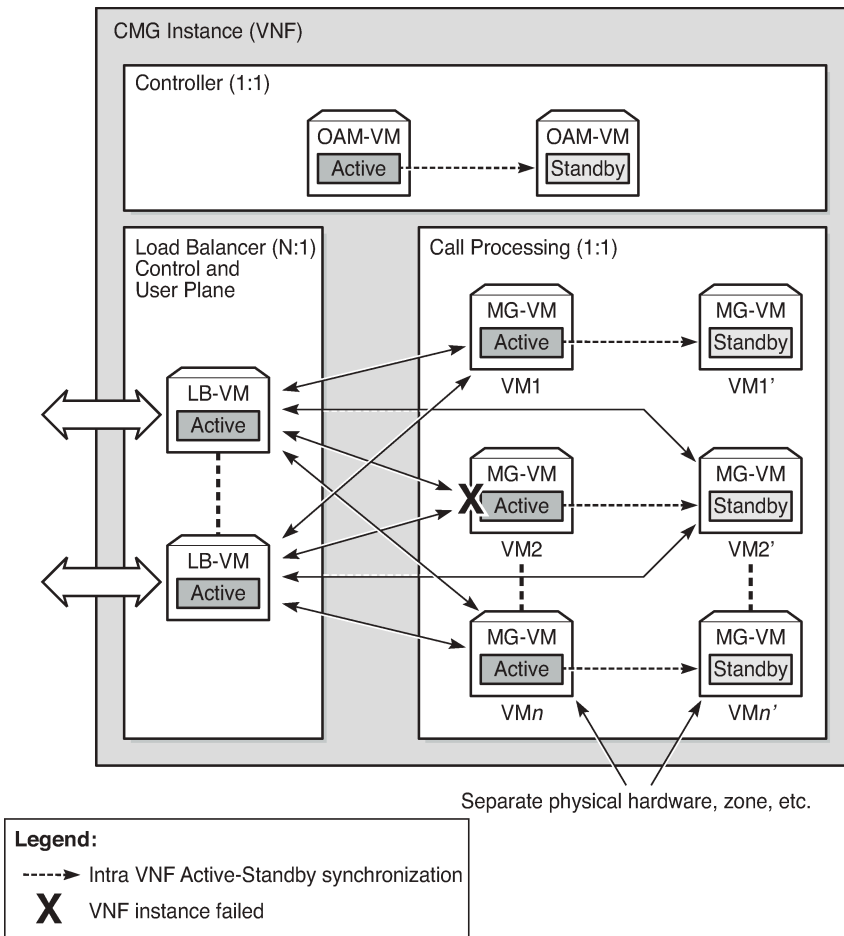
The active MG-VMs participate in call processing for subscriber sessions anchored on it. They also synchronize the subscriber's state to the paired standby MG-VM when a session event is triggered on the active MG-VM. A session event can be triggered due to signaling or UP traffic processing. The standby MG-VMs do not take part in call processing but maintain the subscriber state so that it can continue to support call processing of existing sessions with minimum outage upon failure of the active MG-VM.

CMG in LB mode

When the CMG is in LB mode, UP traffic is received on the LB-VM and traffic is forwarded to the anchor MG-VM identified for that session. To support intra-VNF 1:1 redundancy with CMG in LB mode, a failure of the active MG-VM is detected by the OAM-VM. The OAM-VM triggers a switchover to the standby MG-VM by programming each LB-VM to forward subsequent user traffic received on the LB-VM to the standby MG-VM. This supports seamless switchover of active and standby MG-VMs for peering nodes.

Intra-VNF 1:1 redundancy protection (CMG in LB mode) shows an example of CMG in LB mode that supports intra-VNF 1:1 (active/standby) redundancy protection.

Figure 10: Intra-VNF 1:1 redundancy protection (CMG in LB mode)

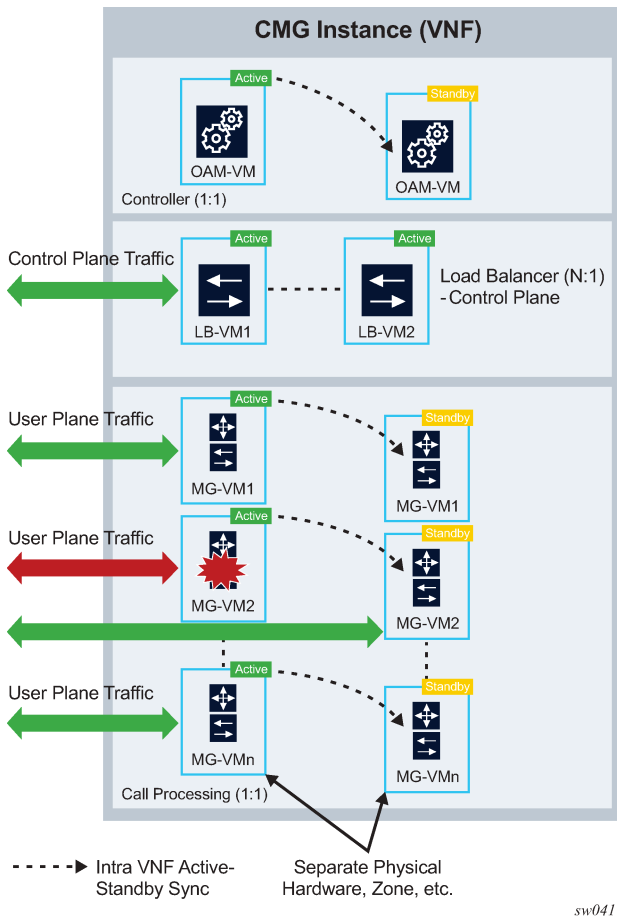


26326

CMG in LB-less mode

When the CMG is in LB-less mode, UP traffic is received directly on the MG-VM based on the GTP-U redirection method and the IP pool advertisement with the associated next-hop per MG-VM. The CMG supports intra-VNF 1:1 redundancy in LB-less mode. When the OAM-VM detects a failure of the active MG-VM, it triggers a switchover to the standby MG-VM by sending a routing update to the peering router. The peering router then forwards subsequent user traffic to the standby MG-VM. This supports seamless switchover of the active and standby MG-VMs for peering nodes, except for the immediate next-hop router.

Intra-VNF 1:1 redundancy protection (CMG in LB-less mode) shows an example of a CMG in LB-less mode that supports intra-VNF 1:1 (active/standby) redundancy protection.

Figure 11: Intra-VNF 1:1 redundancy protection (CMG in LB-less mode)

CMG in hybrid mode

When the CMG is deployed in hybrid mode (combined LB and LB-less) it supports the 1:1 (active/standby) redundancy method. The hybrid mode is applicable per APN on the SGi interface. Failure detection and convergence, due to failure of the MG-VM, is also implemented per APN basis. The MG-VM redundancy with hybrid mode is achieved by combining the redundancy methods of LB and LB-less modes.

For an APN configured in LB mode, the MG-VM failure is detected by the OAM-VM and it programs the LB-VM to update traffic to be forwarded to a new active MG-VM.

For an APN configured in LB-less mode, the MG-VM failure is detected by the OAM-VM and the BGP route update is sent to the peering router to update traffic to be forwarded to a new active MG-VM. This is achieved by changing the next-hop that is local to the new active MG-VM.

4.3.1.2 MG-VM redundancy (N:K)

CMG supports 1:1 redundancy of MG-VM with combined gateway (SGW, PGW/GGSN, ePDG, TWAG, and SSG/TDF), CUPS control plane (SGW-C, PGW-C/GGSN-C), and CUPS user plane (SGW-U, PGW-U/GGSN-U). With 1:1 redundancy of MG-VM, each active MG-VM has a paired standby MG-VM that is used when the active MG-VM fails.

If the active MG-VM is available, the standby does not perform any major function except for keeping the session state received from the active MG-VM. This means that, 50% of the compute resources (servers) are never used in such deployments which leads to inefficient utilization of compute resources.

During a VM or a server-level failure, only a limited set of resources are impacted and hence not all the standby resources are activated. It is only when a zone-level failure occurs that 1:1 redundancy is appropriate because 50% of the compute resources go out of service.

For deployments that do not require zone-level failure protection (significantly less than 50% resources will be impacted), the overhead of compute resources can be reduced by creating a shared pool of standby resources used for processing sessions that are impacted due to limited scope failure.

When a shared pool of standby resources is deployed and one of the resource is activated during failure, there is an inherent delay in convergence to program the impacted sessions on new resource. If there is a need for minimal delay in convergence, for example VoLTE or other real-time applications, it is recommended that 1:1 redundancy is deployed.

To save compute resources and support redundancy, CMG supports N:K (N active and K idle) redundancy. A K number of standby resources is deployed per CMG instance that gets activated upon failure of N primary resources (that is, N:K).

**NOTE:**

The CMG supports N:K redundancy with N active resources protected by K standby resources. That is, N active MG-VMs are protected by K standby MG-VMs. Typically, K depends on the scope of failure. For example, a VM-level failure is protected by K=1 and a server-level failure with a maximum of two VMs per server is protected by K=2.

4.3.1.2.1 Database as session storage

N:K redundancy requires the availability of the subscriber session context in a shared data store, as a backup copy, so it can be retrieved on the standby resource when the active resource fails.

In legacy systems, data was stored on the standby VM. With the availability of high performance in-memory databases, it is easier to use an off-the-shelf database to store the subscriber context and utilize the high availability principles of the database to simplify the application implementation. Using an in-memory database allows the application to use a single update to the database to support high availability of the subscriber context instead of managing multiple updates to local and geo-redundant nodes.

**NOTE:**

CMG supports N:K redundancy only for local redundancy. If N:K is deployed with geo-redundancy, the master and slave gateway instances maintain separate databases to store a backup of the session context.

To configure a cloud database, use the following commands:

```
configure mobile-gateway profile cloud-db profile-name
  description description-string
  server ipv4|ipv6 [port port]
    [no] shutdown
  exit
```

Multiple DB-VMs can be configured to allow for increased session scaling within the same VNF instance. Multiple cloud profiles can be configured containing any number of DB-VMs (up to 6) and they can be associated with a PDN afterward.

```
config>mobile>profile>cloud-db# info
-----
description      "prof1"
server 10.100.11.25 port 5678
    no shutdown
  exit

config>mobile>profile>cloud-db# info
-----
description      "prof2"
server 10.100.11.24 port 5678
    no shutdown
  exit
server 10.100.11.26 port 5678
    no shutdown
  exit
```

To associate a cloud database profile to use as an interface to an external cloud database, use the following commands:

```
configure mobile-gateway pdn gw-id
  cdbx cloud-db-profile profile-name
  interface [router router-instance] interface-name
```



NOTE:

The *router-instance* parameter can be either the base router instance or a VPRN service ID.

```
config>mobile>pdn>cdbx# info
-----
cloud-db-profile prof1
interface "dbNetwork"
```

The "dbNetwork" interface is the interface used for communicating with the DB proxy.

```
config>router# interface "dbNetwork"
```

```
config>router>if# info
```

```
-----
        address 10.100.11.1/24
        port 1/1/1:0
        no shutdown
-----
```

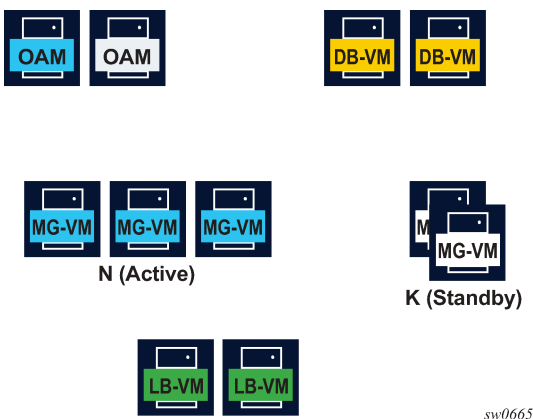
4.3.1.2.2 Deploying DB-VM with CMG

To support N:K redundancy, the DB-VM is added to the CMG instance and serves as the common session storage for all active MG-VMs.

During normal operation, each MG-VM stores its session state in the DB-VM, when the session is created and on any event that triggers change in the session's state.

When multiple DB-VMs are configured, active MG-VMs will be assigned to them based on their load status.

Figure 12: CMG deployment with DB-VMs



NOTE:

The number of DB-VMs required per CMG instance depends on the number of active MG-VMs and subscriber sessions to be stored in the database.

Besides hosting DB instances, the DB-VM also supports proxy function. The number of instances for the proxy function and DB instances depends on the session scale and call-model, as well as on the number of cores available per DB-VM.

**NOTE:**

To calculate the number of DB instances and proxy functions for deployment, contact your local Nokia representative.

4.3.1.2.3 VM state management

To support N:K redundancy, multiple MG groups must be configured and associated with a single resource pool. The number of MG groups associated with a resource pool, defines the number of MG-VMs that are in active state during normal operation.

A resource pool must be configured with a set of MG-VMs, that can operate in active or standby state. The difference of the MG groups associated with a resource pool and the number of MG-VMs associated with same resource pool, defines the K number (the number of MG-VMs that operate in standby state).

There is no configured state associated with an MG-VM, that is, an MG-VM is not marked as working or protect. Any MG-VM can operate in active or standby state. The priority of a VM to move between active and standby states is dictated by the availability of the MG-VM.

The following figures show an example of transition between active and standby state in MG-VMs.

Figure 13: Configured VM state in N:K redundancy

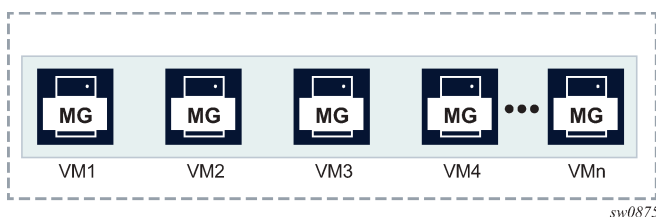
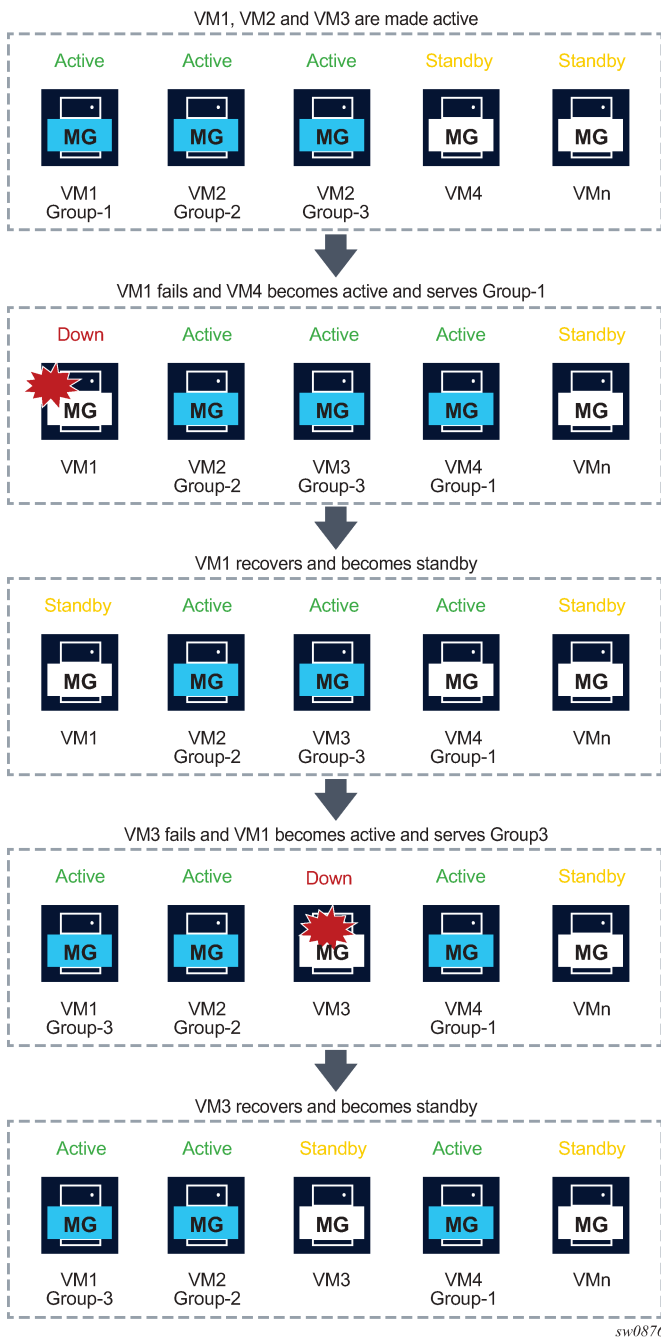


Figure 14: Operational VM state in N:K redundancy



4.3.2 Inter-VNF 1:1 geo-redundancy for CMG

The CMG supports inter-VNF 1:1 (master/slave) geo-redundancy protection by deploying CMG instances across locations geographically far from each other.

In this protection scheme, two independent CMG instances are configured as a 1:1 (master/slave) pair. The master CMG instance participates in call processing and data path forwarding. The master CMG instance synchronizes all bearer, SDF flow, and charging information to the slave CMG instance. The slave CMG

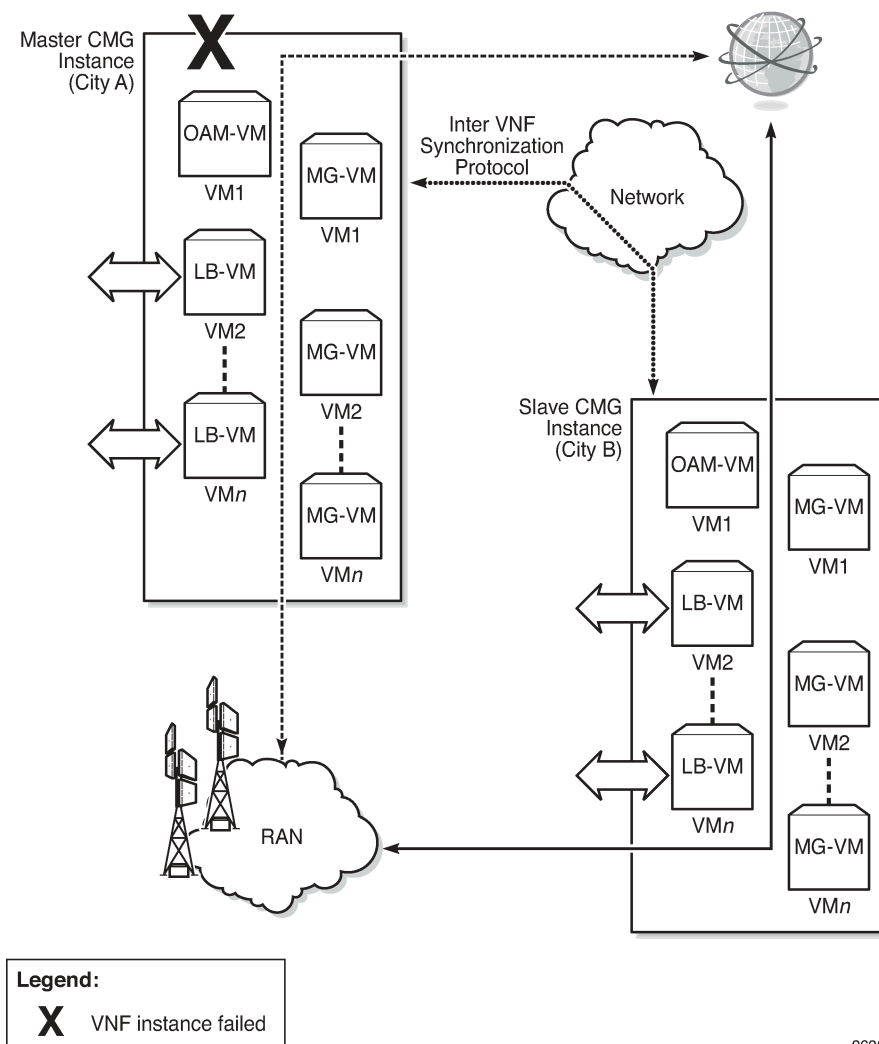
instance does not take part in call setup or data path forwarding. However, among other functions, bearer synchronization, SDF, and charging are available for hot slave service, thus assuring minimal outage.

Routing protocols are configured to advertise routes for locally configured 3GPP signaling interfaces (for example, GTP-C, GTP-U, and Diameter/RADIUS) and UE pools so that traffic is received on the master CMG instance (low metric toward the active and high metric toward the slave CMG instance).

Inter-VNF 1:1 redundancy (geo-redundancy) can be deployed independent of intra- VNF redundancy. For inter-VNF redundancy, the synchronization and election of the active and standby CMG use a unique Nokia inter-VNF synchronization protocol to ensure that bearer information is synchronized on a per MG-VM basis.

Inter-VNF 1:1 (master/slave) geo-redundancy protection shows an example of a CMG that supports inter-VNF 1:1 (master/slave) geo-redundancy protection across different locations.

Figure 15: Inter-VNF 1:1 (master/slave) geo-redundancy protection



26327

Related information

VNF/CNF redundancy (geo-redundancy) configuration guidelines

Description of VNF/CNF redundancy configuration for CMG.

4.4 CMG configuration

**NOTE:**

Any modification to the router/vprn-service interface does not reflect to the MG application, unless you restart the MG gateway.

4.4.1 CMG instance configuration in LB mode

The LB architecture of CMG provides a simple deployment option for gateway applications that do not require high throughput support, for example, SGW, PGW/GGSN for low throughput (Enterprise, M2M), ePDG, TWAG, and SSG. For very high throughput application such as Internet PDN, Nokia recommends deploying LB-less architecture.

By adding the LB-VM for user plane traffic, the gateway instance provides limited exposure of session processing resource (for example, MG-VM) to the external network. Redundancy method and routing configuration also becomes much simpler to configure and operate. In addition, SGi application such as MPLS, IPSec/GRE, L2TP can be supported seamlessly independent of the number of MG-VMs present in the system.

**NOTE:**

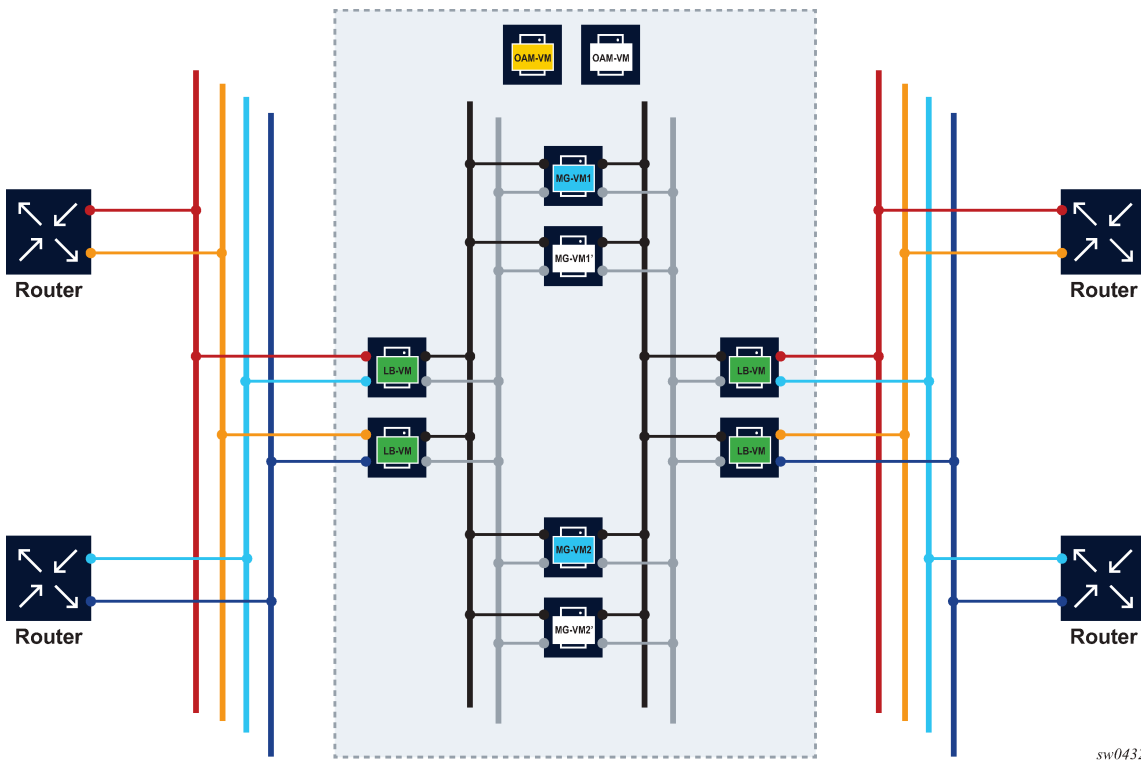
CMG in LB mode for combined mobile gateways supports SGW, PGW/GGSN, combined SGW/PGW/GGSN, ePDG, TWAG, and SSG.

CMG in LB mode for CUPS-based mobile gateways supports the following:

For CP	SGW-C and PGW-C/GGSN-C
For UP	SGW-U and PGW-U/GGSN-U

A CMG with LB mode architecture is deployed as shown in the network diagram of *LB mode CMG network deployment*. In this example, the CP and UP load balancing functions are provided on a dedicated pair of LB-VMs for core and SGi. It is not mandatory to deploy an individual LB-VM for each function. A single LB-VM can support both functions.

Figure 16: LB mode CMG network deployment



sw0432

Each LB-VM is configured with one or more IP port interfaces that allow network traffic to be received directly on the LB-VM, without any dependency on the number of MG-VMs present in the system. CP and UP traffic is received on the LB-VM and forwarded to respective MG-VMs based on session information received in the message.

Multiple LB-VMs can be configured per CMG instance. The peering router must be configured such that traffic is evenly sprayed across all network facing interfaces configured on the LB-VM.

The following are the high-level steps to configure an LB CMG instance:

1. *VM configuration*
2. *Redundancy group configuration for LB mode*
3. *Signaling interface configuration*
4. *APN configuration*
5. *SGi interface services configuration in LB mode*
6. *Call failure rate configuration*

4.4.1.1 VM configuration

The required configuration of the MG-VMs and LB-VMs for CMG in LB mode.

4.4.1.1.1 MG-VM configuration

The MG-VM is provisioned with the following functions:

Mobile Gateway (MG)	MDA 1
Application Assurance (AA)	MDA 2 (optional)
CGNAT (NAT)	MDA 4 (optional)



NOTE:

The AA and CGNAT functions are not required for CMG in LB mode with the following:

- SGW-U (UP function) only for CUPS architecture
- CP function for CUPS architecture (SGW-C, PGW-C, GGSN-C)
- SGW only for combined architecture
- ePDG only for combined architecture
- TWAG only for combined architecture

Example

This example shows the functions that must be configured on an MG-VM with LB mode.

```
config
  card 3
    card-type iom-v-mg
    mda 1
      mda-type isa-mg-v
      no shutdown
    exit
    mda 2
      mda-type isa2-aa-v
      no shutdown
    exit
    mda 4
      mda-type isa2-bb-v
      no shutdown
    exit
    no shutdown
  exit
```


If local HA is enabled, the second MG-VM is provisioned with the same set of functions. This is VM 4 (card 4) in the following example.

Example

```
config
  card 4
    card-type iom-v-mg
    mda 1
      mda-type isa-mg-v
      no shutdown
    exit
    mda 2
      mda-type isa2-aa-v
      no shutdown
    exit
    mda 4
      mda-type isa2-bb-v
      no shutdown
    exit
    no shutdown
  exit
```

4.4.1.1.2 LB-VM configuration

The LB-VM is provisioned with the following functions:

IO support	MDA 1
IP reassembly	MDA 2 (optional)
Tunnel (GRE/IPSec)	MDA 3 (optional)



NOTE:

IP reassembly or tunnel (GRE/IPSec) can be configured without IO support. It is recommended that IP reassembly and tunnel are configured with IO support for compute efficiency as it prevents inter-VM traffic.

Example

This example shows the functions that must be configured on an LB-VM with LB mode.

```
config
  card 1
    card-type iom-v
    mda 1
      mda-type m20-v
      no shutdown
    exit
    mda 2
      mda-type isa2-ip-reas-v
      no shutdown
    exit
    mda 3
      mda-type isa-tunnel-v
      no shutdown
    exit
    no shutdown
  exit
  card 2
    card-type iom-v-mg
    mda 1
      mda-type m20-v
      no shutdown
    exit
    mda 2
      mda-type isa2-ip-reas-v
      no shutdown
    exit
    mda 3
      mda-type isa-tunnel-v
      no shutdown
    exit
    no shutdown
  exit
```

4.4.1.2 Redundancy group configuration for LB mode

For CMG in LB mode support for 1:1 and N:K MG-VM redundancy in non-CUPS, CUPS, and 5GC deployments, refer to the *Usage Notes* section of the *7750 MG and CMG Release Notes*.

4.4.1.2.1 1:1 redundancy group configuration

Refer to the *Usage Notes* section of the *7750 MG and CMG Release Notes* for CMG deployments that support 1:1 MG-VM redundancy.

The following requirements apply when configuring the MG group for 1:1 redundancy:

- The gateway instance must be created before configuring the MG group.
- The number of MG groups must equal the number of active MG-VMs in the gateway instance.
- The redundancy type must be one-to-one and the gateway instance ID must be associated with the configured MG group.
- A pair of MG-VMs must be associated with the 1:1 redundant MG group, where the active MG-VM is configured as working and the standby MG-VM is configured as protect.

To configure the MG group for 1:1 redundancy, use the following commands:

```
configure mobile-gateway system group group-number
  gateway gw-id redundancy type
    card slot-number {working | protect}
```

```
configure mobile-gateway system group 1 gateway 1
  redundancy one-to-one
    card 1 working
    card 2 protect
```

4.4.1.2.2 N:K redundancy group configuration

For CMG deployments that support N:K MG-VM redundancy, refer to the *Usage Notes* section of the *7750 MG and CMG Release Notes*.

In an N:K redundant system, *N* is defined as the number of MG groups and *K* is the number of cards minus the number of MG groups.

The following requirements apply when configuring the MG group for N:K redundancy:

- The configuration is split between the MG group and the resource pool.
- The redundancy type and gateway ID are configured as part of the resource pool configuration, instead of the MG group.
- The resource pool must be configured with many-to-many redundancy and the gateway ID, prior to configuring the MG group.
- The cards are configured under the resource pool context.

**NOTE:**

A resource pool is used with the MG group for N:K redundancy only. For 1:1 redundancy, the MG group is still configured with working and protect MG-VMs per group, and the resource pool is not used.

- Multiple MG groups must be configured, the number of which must match the number of active MG-VMs that are expected to operate. For example, three active MG-VMs require three configured MG groups. The active and standby MG-VMs are associated with one resource pool, and each MG group is configured with a common resource pool. The active and standby MG-VMs are selected based on events in real-time; that is, the active and standby MG-VM is selected based on the status of each MG-VM. If there is a need for an MG-VM to become active, one of the standby MG-VMs is selected.

To configure the MG group for N:K redundancy, use the following commands:

```
configure mobile-gateway system
  resource-pool resource-pool-id redundancy redundancy-type gateway gw-id
    card slot-number
    card slot-number
  group group-number [resource-pool resource-pool-id]
```



NOTE:

In this example, five MG-VMs are configured in resource pool 1 and only three MG groups are configured; that is, N=3 and K=2 (three active and two standby MG-VMs).

```
configure mobile-gateway system
  resource-pool 1 redundancy many-to-many gateway 1
    card 1
    card 2
    card 3
    card 4
    card 5
  exit
  group 1 resource-pool 1
    no shutdown
  exit
  group 2 resource-pool 1
    no shutdown
  exit
  group 3 resource-pool 1
    no shutdown
  exit
```

4.4.1.2.3 Non-redundant MG group configuration

For CMG deployments that support configuration of non-redundant MG-VMs, refer to the *Usage Notes* section of the *7750 MG and CMG Release Notes*.

The following requirements apply when configuring the MG group for 1:1 redundancy:

- The gateway instance must be created before configuring the MG group.
- The number of MG groups must equal the number of active MG-VMs in the gateway instance.
- The redundancy type must be one-to-one and the gateway instance ID must be associated with the configured MG group.
- Only one MG-VM must be associated with a non-redundant MG group. The active MG-VM must be configured as working and no MG-VM is configured as protect.

**NOTE:**

The gateway instance must be created before configuring the MG group.

To configure a non-redundant MG group, use the following commands:

```
configure mobile-gateway system group group-number gateway gw-id  
card slot-number {working | protect}
```

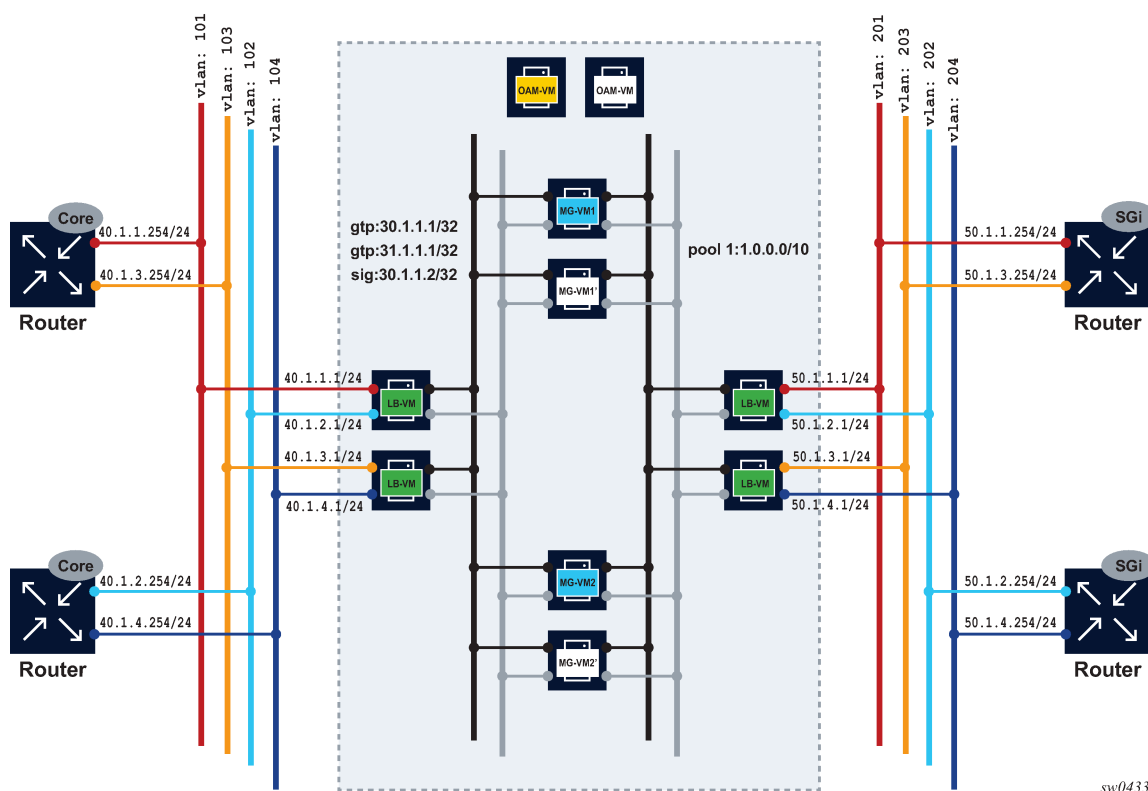
```
configure mobile-gateway system group 1 gateway 1  
card 1 working
```

4.4.1.3 Signaling interface configuration

The signaling interfaces (GTP-C, Diameter, RADIUS, and so on) can be configured in a common VPRN instance.

Signaling interface configuration with common VPRN instance shows an example of the signaling interface configuration using a common VPRN instance (VPRN 100).

Figure 17: Signaling interface configuration with common VPRN instance

**NOTE:**

There is no restriction when configuring the signaling interfaces across multiple VPRN instances.

Example

```
config service
  vprn 100 customer 1 create
    interface "gtp_lo1" create
      address 30.1.1.253/32
      loopback
    exit
    interface "gtp_lo2" create
      address 31.1.1.253/32
      loopback
    exit
    interface "sig_lo1" create
      address 30.1.1.252/32
      loopback
    exit
```

Example

This example shows GTP-C signaling reference point interfaces configured under the PGW. At least one GTP-C interface is required per reference point interface type per CMG instance.

```
config mobile pdn 1
  s11 "default"
    interface
      gtp-c router "vprn100" "gtp_lo1" s1-u "default"
    exit
  exit
  s5 "default"
    interface
      gtp-c router "vprn100" "gtp_lo1"
      gtp-u router "vprn100" "gtp_lo1"
    exit
  exit
  s8 "default"
    interface
      gtp-c router "vprn100" "gtp_lo2"
      gtp-u router "vprn100" "gtp_lo2"
    exit
  exit
  gn "default"
    interface
      gtp-c router "vprn100" "gtp_lo1"
      gtp-u router "vprn100" "gtp_lo1"
    exit
  exit
  gp "default"
    interface
      gtp-c router "vprn100" "gtp_lo2"
      gtp-u router "vprn100" "gtp_lo2"
    exit
  exit
```



NOTE:

Only signaling interfaces are configured on the CMG instance (LB mode) deployment as CP for CUPS, that is, the GTP-C interface only. GTP-U is configured on the UP function (SGW-U, PGW-U, GGSN-U).

```
config mobile pdn 1
  gx
    interface
      diameter router "vprn100" "sig_lo1"
```

```

    exit
    pcrf-selection
        primary-diameter-peer "Peer-Profile1"
        secondary-diameter-peer "Peer-Profile3"
    exit
exit
gy
    dcca-profile "DCCA-Prof1"
    interface
        diameter router "vprn100" "sig_lo1"
    exit
    primary-diameter-peer "Gy-Peer-Prof1"
    outage
        location cf1
            primary
        exit
        location cf2
    exit
    exit
exit
ga
    gtp-prime-group "gtp_server_group1"
    interface-gtp-prime router "vprn100" "sig_lo1"
exit

```

4.4.1.4 APN configuration

To configure traffic handling on the SGi interface for CMG in LB mode, each APN with associated routing instance requires a common VPRN across all MG groups.



NOTE:

Multiple APNs can share a VPRN instance.

The APN-specific configuration for SGi traffic is done under VPRN. The following is an example of the configuration steps shown in [Signaling interface configuration with common VPRN instance](#). IP interfaces within VPRN 201 are configured on LB-VM1 and LB-VM2 and IP pools are configured in VPRN 201.

Example

```

config service
    vprn 201 customer 1 create
        interface "lb1_port1_active" create
            address 10.1.1.1/8

```



```

        sap 1/1/1:201 create
        exit
    interface "lb1_port2_active" create
        address 10.1.2.1/8
        sap 1/1/2:202 create
        exit
    interface "lb2_port1 " create
        address 10.1.3.1/8
        sap 2/1/1:203 create exit
        exit
    interface "lb2_port2 " create
        address 10.1.4.1/8
        sap 2/1/2:204 create
        exit
    exit
    ip-local-pool "pool1"
        ipv4-prefix "1.0.0.0/10"
    exit
    static-route 1.0.0.0/10 black-hole

```

The VPRN associated with the APN (VPRN 201 in this example) must be configured under the APN.

```

config mobile pdn 1
    apn "apn.interntSgi.1.1.com"
    allow-emergency
    pco-option
        dns-server-ipv4
            primary 10.30.1.1
            secondary 10.30.1.2
        exit
    exit
    pcrf-selection
        dynamic-pcc fh-session disable fh-3xxx disable fh-4xxx disable
    exit
    ip-alloc-method local-pool
    pdn-type ipv4
    router "vprn201"
        ip-local-pool "pool1"
    exit
    authentication type diameter pri-dia-peer-profile "s6b_Peer-Profile1"
    sec-dia-peer-profile "s6b_Peer-Profile2" user-name imsi
    no shutdown
exit

```

4.4.1.5 SGi interface services configuration in LB mode

The following SGi interface services in LB mode can be configured:

- *CGNAT configuration*
- *IPSec/GRE configuration*

4.4.1.5.1 CGNAT configuration

In LB mode, the CGNAT function is enabled within the MG-VM by configuring a single NAT group using the local NAT option to apply local a NAT (that is, local CGNAT processing on the MG-VM).

The CGNAT processing on the MG-VM requires the addition of all the active and standby MG-VMs in a single NAT group. The active MDA limit option must be configured to be equal to the number of MG groups.

The NAT binding records (per port-block) can be exported to an external collector device by enabling RADIUS accounting policy.

This configuration example shows nat - radius as the RADIUS accounting policy.

```
config
  isa
    nat-group 1 create
      active-mda-limit 3
      mda 3/4
      mda 4/4
      mda 5/4
      mda 6/4
      mda 7/4
      mda 8/4
      local-nat
      radius-accounting-policy "nat-radius"
      no shutdown
    exit
```

The RADIUS accounting policy for NAT binding records can be configured as shown in the following example:

```
config
  aaa
    isa-radius-policy "nat-radius" create
      acct-include-attributes
      acct-delay-time
      acct-trigger-reason
      called-station-id
      calling-station-id
      circuit-id
```

```
class
dhcp-options
dhcp-vendor-class-id
frame-counters
framed-ip-addr
framed-ip-netmask
hardware-timestamp
inside-service-id
multi-session-id
nas-identifier
nas-ip-address
nas-port
nas-port-id
nas-port-type
nat-subscriber-string
octet-counters
outside-ip
outside-service-id
port-range-block
release-reason
remote-id
session-time
subscriber-data
subscriber-id
ue-creation-type
user-name
exit
servers

    access-algorithm round-robin

    retry 5

    router 3021

    source-address-range 192.168.2.14

    server 1 create

        accounting

        ip-address 192.168.1.3

        secret "vX.Y5EbmvdvLcJUG.PBN/gnKtsge7N2l1szbzIsLaks" hash2
```

```
        no shutdown

    exit

server 2 create

    accounting

    ip-address 192.168.1.4

    secret "vX.Y5EbmvdvlcJUG.PBN/gnKtsge7N2l1szbzIsLaks" hash2

    no shutdown

    exit

server 3 create

    accounting

    ip-address 192.168.1.5

    secret "vX.Y5EbmvdvlcJUG.PBN/gnKtsge7N2l1szbzIsLaks" hash2

    no shutdown

    exit

server 4 create

    accounting

    ip-address 192.168.1.6

    secret "vX.Y5EbmvdvlcJUG.PBN/gnKtsge7N2l1szbzIsLaks" hash2

    no shutdown

    exit

exit

exit
```

To enable the NAT function for an APN (per VPRN), a NAT policy must be configured on a VPRN associated with the APN (that is, inside the VPRN). The destination prefix must be configured to steer traffic toward the NAT for selective traffic based on the destination IP address. For example, IP address 0.0.0.0/0 will steer all traffic toward the NAT.

```
config service
vprn 201 customer 1 create
    nat
        inside
            nat-policy "nat-policy-1"
            destination-prefix 0.0.0.0/0
        exit
    exit
```

The NAT policy is configured with the IP address pool to be used from the public IP address space and associated with the VPRN configured for public IP addresses (that is, outside the VPRN).

```
config service
    nat
        nat-policy "nat-policy-1" create
        pool "public_pool_1" router 301
    exit
exit
```

The VPRN to be used for public IP addresses is configured with the external interfaces and also configured with the NAT outside policy.

```
vprn 301 name "301" customer 1 create
    description "SgiVprnOutside"
    vrf-export "expPlcy_sgiVprn"
    ecmp 10
    autonomous-system 1
    route-distinguisher 301:1
    mg-group 1 local-ecmp-egress
    interface "lb1_port1_active" create
        address 10.1.1.1/8
        sap 1/1/1:201 create
    exit
exit
interface "lb1_port2_active" create
    address 10.1.2.1/8
    sap 1/1/1:202 create
    exit
exit
interface "lb2_port1_active" create
    address 10.1.3.1/8
```

```

        sap 2/1/1:203 create
        exit
    exit
    interface "lb2_port2_active" create
        address 10.1.4.1/8
        sap 2/1/2:204 create
        exit
    exit
    nat
        outside
            pool "public_pool_1" nat-group 1 type large-scale create
            port-reservation blocks 70
            address-range 172.16.0.1 172.16.255.254 create
            exit
        no shutdown
        exit
    exit
exit

```

The route export policy must be configured on the outside VPRN and the prefix for the public IP addresses to be exported in route table.

```

config router policy-options
    prefix-list "prefix-vprn-301"
        prefix 172.16.0.0/16 exact
    exit
    community "sgi-mg" members "target:1:301"
        policy-statement "expPlcy_sgiVprn"
            entry 5
                from
                    protocol nat
                exit
                action accept
                    community add "sgi-mg"
                exit
            exit
        entry 10
            from
                prefix-list "prefix-vprn-301"
            exit
            action accept
                community add "sgi-mg"
            exit
        exit
    exit

```

4.4.1.5.2 IPSec/GRE configuration

To configure an IPSec/GRE tunnel, create a tunnel group with the VM hosting tunnel instance (tunnel instance is hosted on VM 1 in this example).

```
config
  isa
    tunnel-group 1 isa-scale-mode tunnel-limit-2k create
    multi-active
    mda 1/3
    no shutdown
  exit
exit
```

The VPRN associated with the APN must be configured with the GRE or IPSec tunnel interface.

```
config service
  vprn 201 name "201" customer 1 create
  description "sgi"
  route-distinguisher 201:201
  interface "local_lb1_port1" create
    address 10.1.1.1/8
    sap 1/1/1:201 create
  exit
exit
  interface "local_lb1_port2" create
    address 10.1.2.1/8
    sap 1/1/2:202 create
  exit
exit
  interface "local_lb2_port1" create
    address 10.1.3.1/8
    sap 2/1/1:203 create
  exit
exit
  interface "local_lb2_port2" create
    address 10.1.4.1/8
    sap 2/1/2:204 create
  exit
exit
  interface "private-gre1" tunnel create
    address 10.1.0.2/8
    sap tunnel-1.private:1 create
    ip-tunnel "gre_tun1" create
    dest-ip 10.1.0.1
```

```

        gre-header
        source 198.51.1.2
        remote-ip 192.168.1.2
        delivery-service 200
        no shutdown
    exit
exit
exit
static-route-entry 100.1.1.0/8
    next-hop 10.1.0.1
    no shutdown
exit
exit

```

The GRE or IPSec tunnel is routed over the public interface so a dedicated VPRN must be configured with public GRE or IPSec tunnel interface.

```

config service
    vprn 200 name "200" customer 1 create
    route-distinguisher 200:200
    interface "lb1_iprouted_port" create
        address 10.20.10.1/8
        sap 1/1/1:200 create
    exit
exit
interface "lb2_iprouted_port" create
    address 10.20.10.1/8
    sap 2/1/1:200 create
    exit
exit
interface "public_gre1" create
    address 192.168.1.1/16
    sap tunnel-1.public:1 create
    exit
exit
no shutdown
exit

```

4.4.1.6 Call failure rate configuration

The OAM can use a dedicated timer to monitor the call failure rate per MG group.

To configure the timer, use the following command.

```
configure mobile-gateway pdn mg-group-failure-rate call-failure-rate rate-  
percentage [monitor-time timer]
```

When the monitored call failure rate exceeds the configured `call-failure-rate` value and for the configured `monitor-time` period, the OAM instructs the MG group to perform a switchover. At the same time, an alarm is raised and is cleared once the switchover is completed.

**NOTE:**

The functionality is triggered after at least 10 failing calls.

After the switchover, the OAM waits for a configured number of minutes (configured using the `monitor-time` parameter) to get a report for the call failure rate. Whenever, within the configured number of minutes, the OAM receives this report, it starts monitoring the card. If the time expires and there is no call failure report, the OAM resets the logic for the MG group. If the OAM continues to detect the same call failure rate that exceeds the configured `call-failure-rate` value and for the configured `monitor-time` period, the OAM triggers an MG group suspend. At the same time an alarm is raised, and it is cleared once the switchover is completed.

When the MG group is on the suspend state, the OAM raises a new critical alarm to inform about the network condition.

After the switchover, if the OAM does not detect the same issue for the configured `monitor-time`, the alarms are cleared and the MG group is considered operational.

**NOTE:**

- This feature applies to PGW-C and SMF gateway types.
- The feature does not support scale-in setup.
- All available MG groups can be put on the suspend state.
- The suspended MG groups continue to receive signaling for already established sessions.

4.4.2 CMG instance configuration in LB-less mode

The LB-less architecture of CMG improves the throughput performance in the UP and enables the gateway instance to be deployed with minimum dependency on the cloud infrastructure.

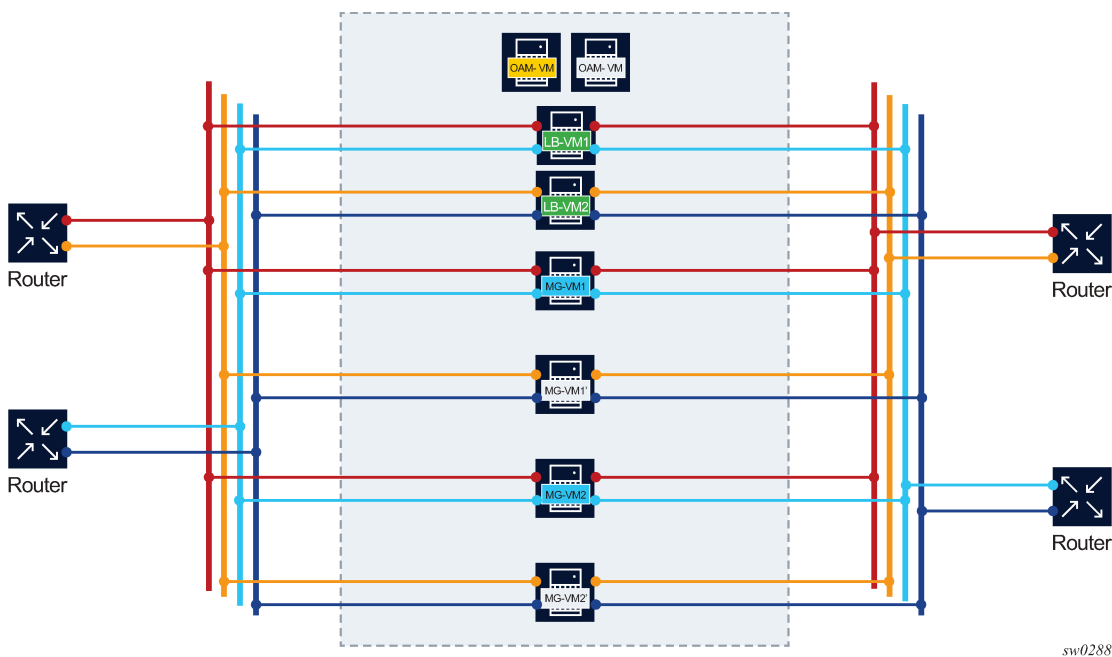
A cloud environment with limited I/O capacity on the LB-VM (due to virtual switch limitations or physical capacity) restricts the overall throughput capacity of the gateway, with the VM being able to support a much higher throughput. By removing the LB-VM from the path of the UP traffic, the gateway instance capacity scales better in infrastructures with limited I/O capacity achieving up to double the architecture with LB.

**NOTE:**

- CMG in LB-less mode only supports SGW, PGW/GGSN, and combined SGW/PGW/GGSN.
- CMG in LB-less mode for CUPS-based mobile gateways supports CUPS SGW-U, CUPS PGW-U/GGSN-U, and 5GC UPF.

A CMG with LB-less architecture is deployed as shown in the network diagram of *CMG with LB-less architecture*. In this example, CP load balancing function is provided on a dedicated pair of LB-VMs.

Figure 18: CMG with LB-less architecture



Each MG-VM is configured with one or more IP port interfaces that allow network traffic to be received directly on the MG-VM. Each MG-VM (or a pair of MG-VMs with local HA) is also configured with a GTP-U reference point interface that allows GTP-U traffic to be received directly on the MG-VM without a load balancing function present in the system.

**NOTE:**

Although the load balancing function for the CP traffic can be collocated with an MG-VM, it is recommended that a dedicated LB-VM is deployed for the CP load balancing function. When taking into account the LB-VM redundancy, then two LB-VMs must be configured to operate in active-active mode.

The following are the high-level steps to configure an LB-less CMG instance:

1. *VM configuration*
2. *Redundancy group configuration for LB-less mode*
3. *Signaling interface configuration*

4. *MG group-specific options configuration*
5. *Routing and redundancy configuration*
6. *SGi interface services configuration in LB-less mode*

4.4.2.1 VM configuration

This section describes the configuration of the MG-VMs and LB-VMs for CMG in LB-less mode.

4.4.2.1.1 MG-VM configuration

The MG-VM is provisioned with the following functions:

Mobile Gateway (MG)	MDA 1
Application Assurance (AA)	MDA 2 (optional)
I/O support	MDA 3
CGNAT (NAT)	MDA 4 (optional)



NOTE:

The AA and CGNAT functions are not required for CMG in LB-less mode with the following:

- SGW-U (UP function) only for CUPS architecture
- SGW only for combined architecture

This example shows the functions that need to be configured on an MG-VM to enable the LB-less mode.

```

config
  card 3
    card-type iom-v-mg
    mda 1
      mda-type isa-mg-v
      no shutdown
    exit
    mda 2
      mda-type isa2-aa-v
      no shutdown
    exit
    mda 3
      mda-type m20-v
  
```

```

        no shutdown
    exit
    mda 4
        mda-type isa2-bb-v
        no shutdown
    exit
    no shutdown
exit

```

If local HA is enabled, then the second MG-VM is provisioned with same set of functions, that is VM# 4 in the following example.

```

config
  card 4
    card-type iom-v-mg
    mda 1
      mda-type isa-mg-v
      no shutdown
    exit
    mda 2
      mda-type isa2-aa-v
      no shutdown
    exit
    mda 3
      mda-type m20-v
      no shutdown
    exit
    mda 4
      mda-type isa2-bb-v
      no shutdown
    exit
  no shutdown
exit

```

4.4.2.1.2 LB-VM configuration

The LB-VM is provisioned with the following functions:

I/O support	MDA 1
IP reassembly	MDA 2 (optional)
Tunnel (GRE/IPSec)	MDA 3 (optional)

This example shows the functions that need to be configured on an LB-VM with LB-less mode.

```
config
  card 1
    card-type iom-v-mg
    mda 1
      mda-type m20-v
      no shutdown
    exit
    mda 2
      mda-type isa2-ip-reas-v
      no shutdown
    exit
    mda 3
      mda-type isa-tunnel-v
      no shutdown
    exit
    no shutdown
  card 2
    card-type iom-v-mg
    mda 1
      mda-type m20-v
      no shutdown
    exit
    mda 2
      mda-type isa2-ip-reas-v
      no shutdown
    exit
    mda 3
      mda-type isa-tunnel-v
      no shutdown
    exit
    no shutdown
exit
```

4.4.2.2 Redundancy group configuration for LB-less mode

CMG in LB-less mode supports 1:1 and N:K MG-VM redundancy as described in the *Usage Notes* section of the *7750 MG and CMG Release Notes*.

This section describes configuration that is common for combined and CUPS (UP) deployments.

4.4.2.2.1 1:1 MG-VM redundancy group configuration

For CMG deployments that support 1:1 MG-VM redundancy, refer to the *Usage Notes* section of the 7750 MG and CMG Release Notes.

The 1:1 redundant MG group configuration of a CMG instance in LB-less mode is identical to the configuration of a CMG instance in LB mode.

Related information

[1:1 redundancy group configuration](#)

4.4.2.2.2 N:K group MG-VM redundancy configuration

For CMG deployments that support N:K MG-VM redundancy, refer to the *Usage Notes* section of the 7750 MG and CMG Release Notes.

In an N:K redundant system, *N* is defined as the number of MG groups and *K* is the number of cards minus the number of MG groups.

The MG group configuration of a CMG instance in LB-less mode is identical to the configuration of a CMG instance in LB mode.

Related information

[N:K redundancy group configuration](#)

4.4.2.2.3 Dynamic SAP for N:K MG-VM redundancy configuration

To support N:K redundancy with the LB-less architecture, a dynamic SAP is required for the VPRN interface, to enable failover from the active to the standby MG-VM.

The dynamic SAP decouples the L3 configuration from L2, such that a failure of the MG-VM is seamless at the L3 layer. For example, if the active MG-VM fails, the IP interface shifts from the physical port on the active MG-VM to the physical port on the standby MG-VM. Remapping of L3 to L2 is done using G-ARP on the standby MG-VM. As the IP interface shifts to the standby MG-VM, the peering nodes begin forwarding traffic to the standby MG-VM (new active), and session processing continues seamlessly to the peering nodes.

To configure the dynamic SAP including QoS policies and IP filters, use the following commands:

```
configure service vprn interface
  dynamic-sap dynamic-sap-id
    egress
      qos policy-id
      filter ip ip-filter-id
      filter ipv6 ipv6-filter-id
    ingress
      qos policy-id
```

```
filter ip ip-filter-id
filter ipv6 ipv6-filter-id
```

4.4.2.2.4 Non-redundant MG group configuration

For CMG deployments that support configuration of non-redundant MG-VMs, refer to the *Usage Notes* section of the *7750 MG and CMG Release Notes*.

The non-redundant MG group configuration of a CMG instance in LB-less mode is identical to the configuration for a non-redundant CMG instance in LB mode.

Related information

Non-redundant MG group configuration

4.4.2.3 Signaling interface configuration

The signaling interfaces (GTP-C, Diameter, RADIUS, and so on) are configured in a common VPRN instance (in this example VPRN 100).



NOTE:

As in the CMG deployment with LB, there is no restriction when configuring the signaling interfaces across multiple VPRN instances. The signaling interfaces on a CMG with LB-less mode deployment operate in the same way as in the CMG in LB mode.

```
config service
  vprn 100 customer 1 create
    interface "gtpc_lo1" create
      address 30.1.1.253/32
      loopback
    exit
    interface "gtpc_lo2" create
      address 31.1.1.253/32
      loopback
    exit
    interface "sig_lo1" create
      address 30.1.1.252/32
      loopback
    exit
```

This example shows GTP-C signaling reference point interfaces configured under the PGW. At least one GTP-C interface is required per reference point interface type per CMG instance.

**NOTE:**

Multiple GTP-C reference point interfaces per interface type are not supported.

```

config mobile pdn 1
s11 "default"
    interface
        gtp-c router "vprn100" "gtpc_lo1" s1-u "default"
    exit
exit
s5 "default"
    interface
        gtp-c router "vprn100" "gtpc_lo1"
        gtp-u router "vprn100" "gtpc_lo1"
    exit
exit
s8 "default"
    interface
        gtp-c router "vprn100" "gtpc_lo2"
        gtp-u router "vprn100" "gtpc_lo2"
    exit
exit
gn "default"
    interface
        gtp-c router "vprn100" "gtpc_lo1"
        gtp-u router "vprn100" "gtpc_lo1"
    exit
exit
gp "default"
    interface
        gtp-c router "vprn100" "gtpc_lo2"
        gtp-u router "vprn100" "gtpc_lo2"
    exit
exit

```

**NOTE:**

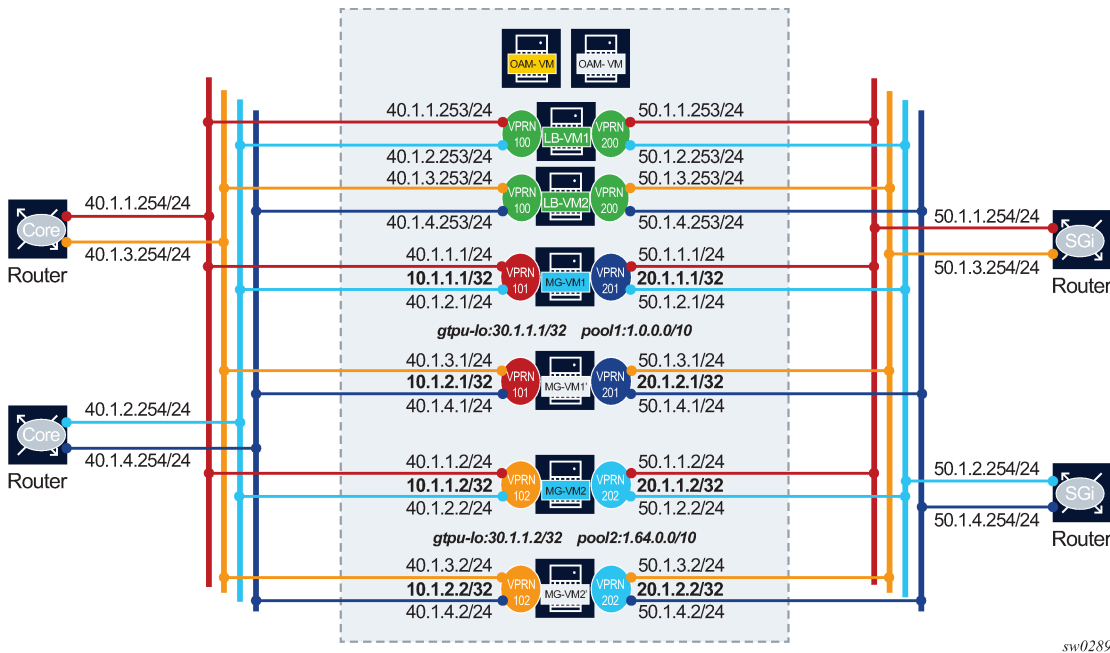
It is mandatory to configure a GTP-U interface for each reference point interface. This can be the GTP-C reference point interface, although it is not mandatory. For CMG deployment with LB-less mode, each MG-VM group must be assigned a unique GTP-U reference point interface. If no GTP-U reference point is configured for an MG group, the default GTP-U reference point (shown in the previous example)

will be used. Since this GTP-U IP address may not be configured on the MG group to which a session is anchored, this could lead to Inter-VM traffic.

```
config mobile pdn 1
  gx
    interface
      diameter router "vprn100" "sig_lo1"
    exit
    pcrf-selection
      primary-diameter-peer "Peer-Profile1"
      secondary-diameter-peer "Peer-Profile3"
    exit
  exit
  gy
    dcca-profile "DCCA-Prof1"
    interface
      diameter router "vprn100" "sig_lo1"
    exit
    primary-diameter-peer "Gy-Peer-Prof1"
    outage
      location cf1
        primary
        exit
      location cf2
    exit
  exit
  ga
    gtp-prime-group "gtp_server_group1"
    interface-gtp-prime router "vprn100" "sig_lo1"
  exit
```

4.4.2.4 MG group-specific options configuration

Figure 19: MG group-specific network diagram for 1:1 MG-VM redundancy



4.4.2.4.1 MG group-specific GTP-U configuration with 1:1 MG-VM redundancy

A CMG with LB-less mode requires each MG-VM group to host an independent VPRN for the GTP-U reference point interface.

In *MG group-specific network diagram for 1:1 MG-VM redundancy*, VPRN 101 is configured on MG-VM group 1 (includes MG-VM1 and MG-VM1'). VPRN 102 is configured on MG-VM group 2 (includes MG-VM2 and MG-VM2').

One or multiple IP interfaces are configured per MG-VM. For example, IP interfaces 40.1.1.1/24 and 40.1.2.1/24 are configured under VPRN 101 with ports on MG-VM1. IP interfaces 40.1.3.1/24 and 40.1.4.1/24 are configured under VPRN 101 with ports on MG-VM1'. The loopback IP address 30.1.1.1/32 in VPRN 101 is configured as one of the GTP-U reference point for MG group 1. Multiple GTP-U reference point interfaces can be configured for IP separation. For example, S8 and Gp interfaces configured with a dedicated loopback IP address (31.1.1.1/32) under VPRN 101 on MG group 1.

The MG group association with a VPRN must be enabled using the `mg-group` command.

This example shows VPRN 101 associated with MG group 1.

```
config service
  vprn 101 customer 1 create
    mg-group 1
    interface "grp1_gtpu_lo1" create
      address 30.1.1.1/32
      loopback
```

```

exit
interface "grp1_gtpu_lo2" create
    address 31.1.1.1/32
    loopback
exit
interface "grp1_port1_active" create
    address 40.1.1.1/24
    sap 3/3/1:101 create
exit
interface "grp1_port2_active" create
    address 40.1.2.1/24
    sap 3/3/2:102 create
    exit
exit
interface "grp1_port1_stdbby" create
    address 40.1.3.1/24
    sap 4/3/1:103 create
    exit
exit
interface "grp1_port2_stdbby" create
    address 40.1.4.1/24
    sap 4/3/2:104 create
    exit
exit

```

The GTP-U interfaces per MG group must be configured per reference point interface using the **group-specific** keyword. For example, S1-u, S5, Gn reference point interfaces are configured using grp1_gtpu_lo1 under VPRN 101 and **group-specific** association.

**NOTE:**

- The group-specific reference point can be created without shutting down the gateway.
- Only GTP-U is supported when using the **group-specific** keyword.

```

config mobile pdn 1
    s1-u "s1u_grp1" group-specific
        interface
            gtp-u router "vprn101" "grp1_gtpu_lo1"
        exit
    exit
    s5 "s5_grp1" group-specific
        interface
            gtp-u router "vprn101" "grp1_gtpu_lo1"
        exit
    exit

```

```

exit
s8 "s8_grp1" group-specific
    interface
        gtp-u router "vprn101" "grp1_gtpu_lo2"
    exit
exit
gn "gn_grp1" group-specific
    interface
        gtp-u router "vprn101" "grp1_gtpu_lo1"
    exit
exit
gp "gp_grp1" group-specific
    interface
        gtp-u router "vprn101" "grp1_gtpu_lo2"
    exit
exit

```

To associate group-specific GTP-U reference point interfaces to an MG group, an override reference point profile must be configured. Each MG group specific GTP-U reference point interface must be configured under the override profile.

```

config mobile pdn 1
    override-profile 1
        override ref-point-type s1u "default" with "s1u_grp1"
        override ref-point-type s5 "default" with "s5_grp1"
        override ref-point-type s8 "default" with "s8_grp1"
        override ref-point-type gn "default" with "gn_grp1"
        override ref-point-type gp "default" with "gp_grp1"
    exit

```

The override profile is associated with an MG group using the following MG group configuration with 1:1 MG-VM redundancy.

```

config mobile system
    group 1 redundancy one-to-one gateway 1
        override-profile 1
        card 3 working
        card 4 protect
        no shutdown
    exit

```

The above configuration steps must be repeated for each MG group when multiple MG groups are deployed per CMG instance.

**NOTE:**

CMG with LB-less mode supports overlapping IP address subnets to be configured across MG groups. A single IP address subnet across multiple MG groups enables flexible deployment of a CMG instance requiring minimum configuration on the peering routers connected to MG-VMs.

Since overlapping IP address subnets are allowed across MG groups, a subnet is repeated across MG-VMs. For example, in *MG group-specific network diagram for 1:1 MG-VM redundancy* the IP interfaces 40.1.1.2/24 and 40.1.2.2/24 are configured under VPRN 102 with ports on MG-VM2. IP interfaces 40.1.3.2/24 and 40.1.4.2/24 are configured under VPRN 102 with ports on MG-VM2'. These four subnet interfaces configured on MG group1 and MG group 2, are connected to four common broadcast networks configured on the peering router. This avoids creating new IP subnets during scale-out of MG groups. The VLAN consumption also remains constant, independent of the number of MG groups deployed per CMG instance.

This example provides configuration steps for MG group 2 shown in *MG group-specific network diagram for 1:1 MG-VM redundancy*.

```
config service vprn 102 customer 1 create
  mg-group 2
  interface "grp2_gtpu_lo1" create
    address 30.1.1.2/32
    loopback
  exit
  interface "grp2_gtpu_lo2" create
    address 31.1.1.2/32
    loopback
  exit
  interface "grp2_port1_active" create
    address 40.1.1.2/24
    sap 5/3/1:101 create
    exit
  exit
  interface "grp2_port2_active" create
    address 40.1.2.2/24
    sap 5/3/2:102 create
    exit
  exit
  interface "grp2_port1_stdbby" create
    address 40.1.3.2/24
    sap 6/3/1:103 create
    exit
  exit
  interface "grp2_port2_stdbby" create
    address 40.1.4.2/24
```

```
sap 6/3/2:104 create
exit
exit
```

In MG group 2, loopback 30.1.1.2/32 is configured under VPRN 102 as GTP-U reference point for S1-u, S5, and Gn interfaces. Loopback 31.1.1.2/32 is configured as a GTP-U reference point for S8 and Gp interfaces.

```
config mobile pdn 1
  s1-u "s1u_grp2" group-specific
    interface
      gtp-u router "vprn102" "grp2_gtpu_lo1"
    exit
  exit
  s5 "s5_grp2" group-specific
    interface
      gtp-u router "vprn102" "grp2_gtpu_lo1"
    exit
  exit
  s8 "s8_grp2" group-specific
    interface
      gtp-u router "vprn102" "grp2_gtpu_lo2"
    exit
  exit
  gn "gn_grp2" group-specific
    interface
      gtp-u router "vprn102" "grp2_gtpu_lo1"
    exit
  exit
  gp "gp_grp2" group-specific
    interface
      gtp-u router "vprn102" "grp2_gtpu_lo2"
    exit
  exit
```

Configure the override profile for MG group 2.

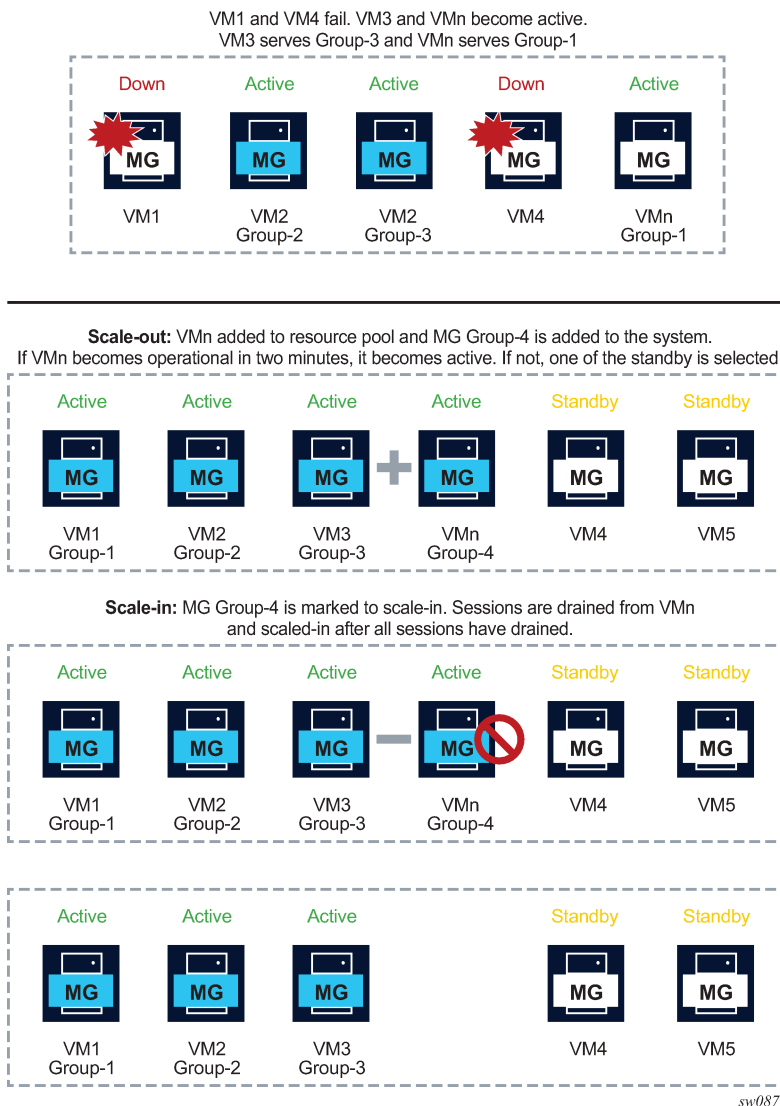
```
config mobile pdn 1
  override-profile 2
    override ref-point-type s1u "default" with "s1u_grp2"
    override ref-point-type s5 "default" with "s5_grp2"
    override ref-point-type s8 "default" with "s8_grp2"
    override ref-point-type gn "default" with "gn_grp2"
    override ref-point-type gp "default" with "gp_grp2"
  exit
```

Associate the override profile 2 with MG group 2.

```
config mobile system
  group 2 redundancy one-to-one gateway 1
    override-profile 2
    card 5 working
    card 6 protect
    no shutdown
exit
```

4.4.2.4.2 MG group-specific GTP-U configuration with N:K MG-VM redundancy

Figure 20: MG group-specific network diagram for N:K MG-VM redundancy



The override profile is associated with an MG group using the following MG group configuration with N:K MG-VM redundancy.

```
config mobile system
  group 1 gateway 1
    override-profile 1
    resource-pool 1
    no shutdown
  exit
```

With multiple active MG-VMs, the override profile is associated with each MG group using the following MG group configuration. A single resource pool is configured across all MG groups since the VMs are shared by multiple groups.

```
config mobile system
  group 2 gateway 1
    override-profile 2
    resource-pool 1
    no shutdown
  exit
```

With N:K MG-VM redundancy, the IP interfaces are configured as in 1:1 redundancy but without a physical port/SAP. There is no permanent association of an MG group to an MG-VM, so the physical port/SAP is selected when a specific MG-VM is becoming active.

One or multiple IP interfaces are configured per MG group. For example, IP interfaces 40.1.1.1/24 and 40.1.2.1/24 are configured under VPRN 101, with ports allocated dynamically on the MG-VM that will be active.

This example shows VPRN 101 associated with MG group 1.

```
config service
  vprn 101 customer 1 create
    mg-group 1
      interface "grp1_gtpu_lo1" create
        address 30.1.1.1/32
        loopback
      exit
      interface "grp1_gtpu_lo2" create
        address 31.1.1.1/32
        loopback
      exit
      interface "grp1_port1_active" create
        address 40.1.1.1/24
        dynamic-sap mg-group-1/3/1:101 create
      exit
    exit
```



```

    interface "grp1_port2_active" create
        address 40.1.2.1/24
        dynamic-sap mg-group-1/3/2:102 create
    exit
exit
exit

```

This example shows the configuration steps for MG group 2.

```

config service vprn 102 customer 1 create
    mg-group 2
    interface "grp2_gtpu_lo1" create
        address 30.1.1.2/32
        loopback
    exit
    interface "grp2_gtpu_lo2" create
        address 31.1.1.2/32
        loopback
    exit
    interface "grp2_port1_active" create
        address 40.1.1.2/24
        dynamic-sap mg-group-2/3/1:101 create
    exit
exit
    interface "grp2_port2_active" create
        address 40.1.2.2/24
        dynamic-sap mg-group-2/3/2:102 create
    exit
exit
exit

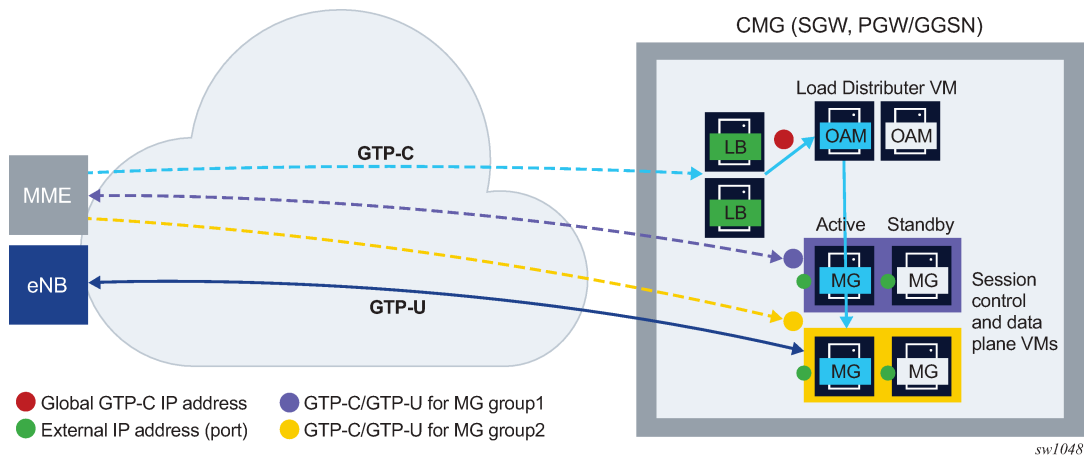
```

4.4.2.4.3 Signaling interface configuration for GTP-C redirection

This feature supports the GTP-C redirection method of signaling for GTP-based interfaces on the non-CUPS combined gateway CMG deployments in LB-less mode only.

The CMG supports a single IP address to represent all active MG-VMs within an instance. In case of multiple failures with local redundancy (for example, a double failure of both active and standby MG-VMs), it is possible that a standby MG-VM may not be available. As a result, the peering node cannot detect the failure of an MG resource, which could lead to extended clean-up time because the individual sessions must be deleted. To expedite the detection of failure of a subset of resources by a peering node, each MG-VM must communicate directly with the peering node. This is achieved using a GTP-C redirection method that responds directly from the MG-VM. In the case of a partial failure, the peering node detects the failure and cleans up the session without going through an individual session.

Figure 21: GTP-C redirection for partial failure detection



To enable GTP-C redirection-based signaling interfaces (S11, S5, S8, Gn, Gp, S2a, S2b, and S4/S12), the GTP-C interface is configured per MG group.

**NOTE:**

The GTP-C redirection method of signaling must be used with LB-less GTP-U configuration.

The following examples show GTP-C signaling reference point interfaces configured per MG group. The loopback interface for the GTP-C reference point (multiple for individual reference point type) is configured under the VPRN associated with the MG group. VPRN101 is associated with MG group 1 and VPRN102 is associated with MG group 2.

```
config service
  vprn 101 customer 1 create
    interface "grp1_gtp_lo1"
      create
        address 30.1.1.1/32
        loopback
      exit
    interface "grp1_gtp_lo2"
      create address
        31.1.1.1/32
      loopback
    exit
    interface "grp1_gtp_lo3"
      create address
        30.1.2.1/32
      loopback
    exit
  config service
```

```

vprn 102 customer 1 create
  interface "grp2_gtp_lo1"
  create
    address 30.1.1.2/32
    loopback
  exit
  interface "grp2_gtp_lo2"
  create address
    31.1.1.2/32
    loopback
  exit
  interface "grp2_gtp_lo3"
  create address
    30.1.2.2/32
    loopback
  exit

```

The GTP-C reference point for the interface that enables GTP-C redirection is configured with the "group-specific" flag as shown in the following example. A default interface per reference point interface type is also configured and is shared across all MG-VMs as the signaling reference point interface at the gateway level.

```

config mobile pdn 1
  s11 "default"
  interface
    gtp-c router "vprn100" "gtpc_lo1" s1-u "default"
  exit
exit
s11 "s11_grp1" group-specific
  interface
    gtp-c router "vprn101" "grp1_gtp_lo1" s1-u "default"
  exit
exit
s11 "s11_grp2" group-specific
  interface
    gtp-c router "vprn102" "grp2_gtp_lo1" s1-u "default"
  exit
exit

```

This example shows GTP-C signaling reference point interfaces configured per MG group and reference point interface type.

```

config mobile pdn 1
  s5 "default"
  interface

```

```
        gtp-c router "vprn100" "gtpc_lo1" s1-u "default"
    exit
exit
s5 "s5_grp1" group-specific
    interface
        gtp-c router "vprn101" "grp1_gtp_lo1"
    exit
exit
s5 "s5_grp2" group-specific
    interface
        gtp-c router "vprn102" "grp2_gtp_lo1"
    exit
exit
s8 "default"
    interface
        gtp-c router "vprn100" "gtpc_lo2"
    exit
exit
s8 "s8_grp1" group-specific
    interface
        gtp-c router "vprn101" "grp1_gtp_lo1"
    exit
exit
s8 "s8_grp2" group-specific
    interface
        gtp-c router "vprn102" "grp2_gtp_lo2"
    exit
exit
gn "default"
    interface
        gtp-c router "vprn100" "gtpc_lo3"
    exit
exit
gn "gn_grp1" group-specific
    interface
        gtp-c router "vprn101" "grp1_gtp_lo1"
    exit
exit
gn "gn_grp2" group-specific
    interface
        gtp-c router "vprn101" "grp2_gtp_lo1"
    exit
exit
```

To associate group-specific GTP reference point interfaces to an MG group, an override reference point profile must be configured as described in [MG group-specific options configuration](#). Each MG group specific GTP reference point interface must be configured under the override profile as shown in the following example.

```
config mobile pdn 1
  override-profile 1
    override ref-point-type s11 "default" with "s11_grp1"
    override ref-point-type s5 "default" with "s5_grp1"
    override ref-point-type s8 "default" with "s8_grp1"
    override ref-point-type gn "default" with "gn_grp1"
    override ref-point-type gp "default" with "gp_grp1"
  exit
```

This example shows the override profile for MG group 2 configuration.

```
config mobile pdn 1
  override-profile 2
    override ref-point-type s11 "default" with "s11_grp2"
    override ref-point-type s5 "default" with "s5_grp2"
    override ref-point-type s8 "default" with "s8_grp2"
    override ref-point-type gn "default" with "gn_grp2"
    override ref-point-type gp "default" with "gp_grp2"
  exit
```

4.4.2.4.4 MG group-specific APN configuration

To configure LB-less traffic handling on the SGi interface, each APN with a unique routing instance requires a dedicated VPRN per MG group. Each MG group anchors sessions from all APNs configured on the CMG instance and therefore each MG group requires as many VPRNs as the number of APNs.



NOTE:

Multiple APNs can share a VPRN instance per MG group, provided that they are sharing a routing instance. Each routing instance requires a dedicated VPRN for handling SGi traffic configured per MG group.

It is optional to use load balancing function on the SGi interface and independent of the load balancing function on the GTP-U interface.



NOTE:

The load balancing function for SGi traffic without deploying GTP-U load balancing function is not supported. It is mandatory to deploy the GTP-U load balancing function if a load balancing function is required for SGi traffic.

The load balancing function can be enabled per SGi VPRN basis. Some SGi VPRNs (that is, all APNs associated with this SGi VPRN) may operate in LB-less mode while other SGi VPRNs (all APNs associated with this specific SGi VPRN) may operate in LB mode.

The APN-specific configuration for SGi traffic is done under each VPRN per MG group basis. That is, IP interfaces within VPRN 201 are configured on MG-VM1 and MG-VM1' in MG group 1 and IP interfaces within VPRN 202 are configured on MG-VM2 and MG-VM2' in MG group 2.

The IP pools are configured in VPRN 201 for MG group 1 and VPRN 202 for MG group 2. The IP pool per MG group and static routes for local IP pool is configured as shown in the following example.

The following example provides configuration steps for MG group 1 (MG-VM1 and MG-VM1') shown in *MG group-specific network diagram for 1:1 MG-VM redundancy*.

```
config service
  vprn 201 customer 1 create
    mg-group 1
      interface "grp1_port1_active" create
        address 50.1.1.1/24
        sap 3/3/1:201 create
        exit
      exit
      interface "grp1_port2_active" create
        address 50.1.2.1/24
        sap 3/3/2:202 create
        exit
      exit
      interface "grp1_port1_stdbby" create
        address 50.1.3.1/24
        sap 4/3/1:203 create
        exit
      exit
      interface "grp1_port2_stdbby" create
        address 50.1.4.1/24
        sap 4/3/2:204 create
        exit
      exit
      ip-local-pool "pool1Grp1
        ipv4-prefix "1.0.0.0/10"
      exit
      static-route 1.0.0.0/10 black-hole
```

Example: MG group 2 (MG-VM2 and MG-VM2')

```

config service
  vprn 202 customer 1 create
    mg-group 2
    interface "grp2_port1_active" create
      address 50.1.1.2/24
      sap 5/3/1:201 create
    exit
  exit
  interface "grp2_port2_active" create
    address 50.1.2.2/24
    sap 5/3/2:202 create
  exit
  exit
  interface "grp2_port1_stdbby" create
    address 50.1.3.2/24
    sap 6/3/1:203 create
  exit
  exit
  interface "grp2_port2_stdbby" create
    address 50.1.4.2/24
    sap 6/3/2:204 create
  exit
  exit
  ip-local-pool "pool1Grp2"
    ipv4-prefix "1.64.0.0/10"
  exit
  static-route 1.64.0.0/10 black-hole

```

The list of VPRNs associated with an APN are configured under the APN. VPRN 201 (for MG group 1) and VPRN 202 (for MG group 2) are configured as shown in the following example. Each VPRN has a dedicated IP pool per MG group for dynamic UE IP address allocation, configured under the VPRN as shown in the previous example.

```

config mobile pdn 1
  apn "apn.interntSgiRedirect.1.1.com"
    allow-emergency
    pco-option
      dns-server-ipv4
        primary 10.30.1.1
        secondary 10.30.1.2
      exit
    exit
  pcrf-selection

```

```

dynamic-pcc fh-session disable fh-3xxx disable fh-4xxx disable
exit
ip-alloc-method local-pool
pdn-type ipv4
router "vprn201"
    ip-local-pool "pool1Grp1"
exit
router "vprn202"
    ip-local-pool "pool1Grp2"
exit
authentication type diameter pri-dia-peer-profile "s6b_Peer-Profile1"
sec-
    dia-peer-profile "s6b_Peer-Profile2" user-name imsi
    no shutdown
exit

```

4.4.2.5 Routing and redundancy configuration

Figure 22: Routing and redundancy function with 1:1 MG-VM redundancy

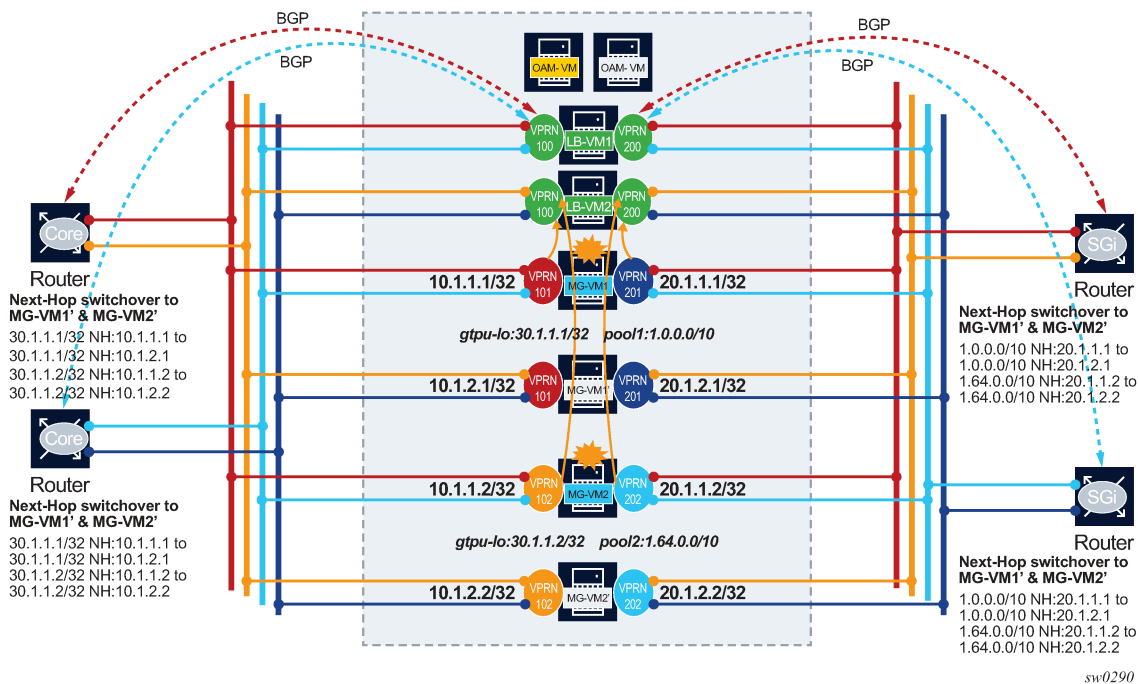
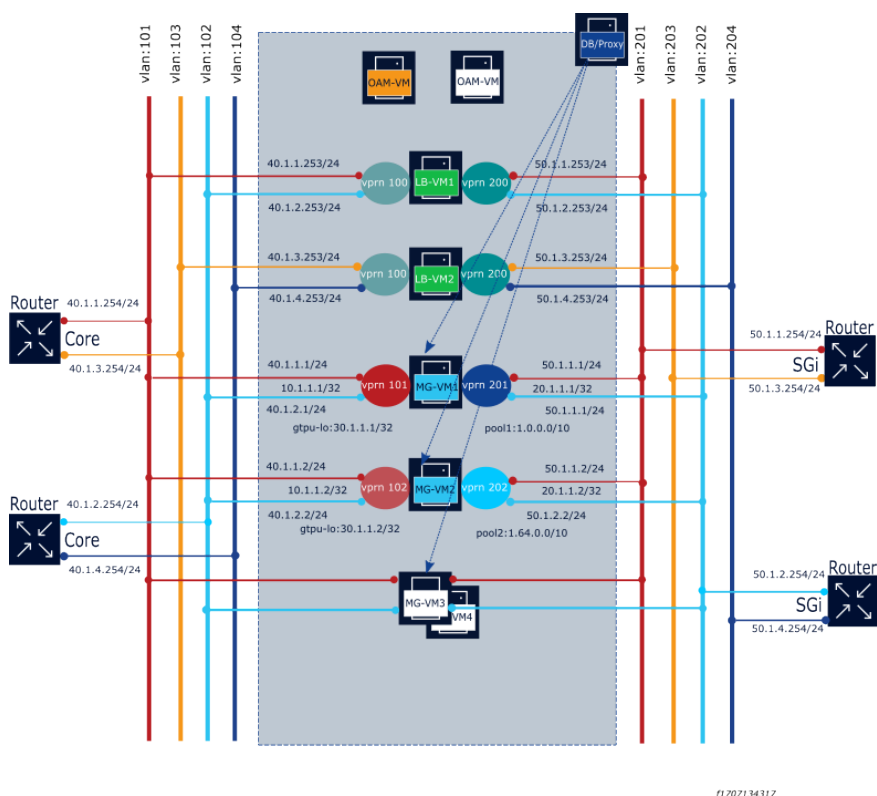


Figure 23: Routing and redundancy function with N:K MG-VM redundancy



NOTE:

Figure *Routing and redundancy function with N:K MG-VM redundancy* shows an example configuration with the subnets being re-used in the different VPRNs. Different subnets can also be used in different VPRNs.

4.4.2.5.1 Local prefix advertisement

The local routes for each MG group are aggregated in a single VPRN for GTP-U and APN VPRN (for example, VPRN 100 and VPRN 200 in *Routing and redundancy function with 1:1 MG-VM redundancy*) and advertised to peering routers via BGP. The routes are exported from a single VPRN per MG group and imported into an aggregate VPRN. Using route export/import into an aggregate VPRN supports running single BGP peering to each router, instead of running BGP peering from each VPRN per MG group.

NOTE:

LB-less mode supports both iBGP and eBPG peering.

4.4.2.5.2 Aggregate VPRN configuration for the MG groups VPRNs (GTP-U traffic)

For GTP-U traffic, the routes are exported from VPRN 101 and VPRN 102 with respective route targets as shown in [Routing and redundancy function with 1:1 MG-VM redundancy](#). The routes exported from VPRN 101 and 102 are imported into VPRN 100 using the list of route targets.

With 1:1 MG-VM redundancy, the route export policy requires the update of prefixes associated with the active and standby MG-VMs, based on the state of the MG-VM.

Example: MG group 1 — Route export policy for VPRN 101 (GTP-U traffic)

```
config router policy-options
  community "RefPt1_101" members "target:1:101"
  community "RefPt1_101_prtct" members "target:1:121"
  prefix-list "prefix_vprn101"
    prefix 30.1.1.1/32 exact
    prefix 31.1.1.1/32 exact
  exit
  policy-statement "vprn101_export"
    entry 10
      from
        prefix-list "prefix_vprn101"
        active-mg 3
      exit
      action accept
        community add "RefPt1_101"
      exit
    exit
    entry 15
      from
        prefix-list "prefix_vprn101"
        active-mg 4
      exit
      action accept
        community add "RefPt1_101_prtct"
      exit
    exit
  exit
```

In N:K MG-VM redundancy, the route export policy requires an update only of the prefixes that are associated with the active MG-VM.

Example: MG group 1 — Route export policy for VPRN 101 (GTP-U traffic)

```
config router policy-options
  community "RefPt1_101" members "target:1:101"
```

```
prefix-list "prefix_vprn101"
  prefix 30.1.1.1/32 exact
exit
policy-statement "vprn101_export"
  entry 10
    from
      prefix-list "prefix_vprn101"
    exit
    action accept
      community add "RefPt1_101"
    exit
  exit
```

Example: MG group 1 — Applying route export policy to VPRN 101

```
config service
  vprn 101 customer 1 create
  vrf-export "vprn101_export"
  autonomous-system 1
  route-distinguisher 101:1
```

Example: MG group 2 — Route export policy for VPRN 102 (GTP-U traffic)

```
config router policy-options
  community "RefPt1_102" members "target:1:102"
  prefix-list "prefix_vprn102"
    prefix 30.1.1.2/32 exact
  exit
  policy-statement "vprn102_export"
    entry 10
      from
        prefix-list "prefix_vprn102"
      exit
      action accept
        community add "RefPt1_102"
      exit
    exit
  exit
```

Example: MG group 2 — Applying route export policy to VPRN 102

```
config service
  vprn 102 customer 1 create
  vrf-export "vprn102_export"
```

```
autonomous-system 1
route-distinguisher 102:1
```

Configure an import policy for VPRN 100 to aggregate routes from individual VPRNs (for example, 101 and 102).

Example: With 1:1 MG-VM redundancy

```
config router policy-options
  community "RefPt1_101" members "target:1:101"
  community "RefPt1_102" members "target:1:102"
  community "RefPt1_101_prtct" members "target:1:121"
  community "RefPt1_102_prtct" members "target:1:122"
  policy-statement "vprn100_import"
    entry 10
      from
        community expression "[RefPt1_101] OR [RefPt1_101_prtct] OR
[RefPt1_102
] OR [RefPt1_102_prtct]"
      exit
      action accept
      exit
    exit
  exit
```

Example: With N:K MG-VM redundancy

```
config router policy-options
  community "RefPt1_101" members "target:1:101"
  community "RefPt1_102" members "target:1:102"
  policy-statement "vprn100_import"
    entry 10
      from
        community expression "[RefPt1_101] OR [RefPt1_102]"
      exit
      action accept
      exit
    exit
  exit
```

Example: Applying route export policy to VPRN 100

```
config service
vprn 100 customer 1 create
```

```

vrf-import "vprn100_import"
autonomous-system 1
route-distinguisher 100:1
interface "refpt_vprn_port1_wrk" create
    address 40.1.1.253/24
    sap 1/1/3:101 create
    exit
exit
interface "refpt_vprn_port2_wrk" create
    address 40.1.2.253/24
    sap 2/1/6:102 create
    exit
exit
interface "refpt_vprn_port1_prt" create
    address 40.1.3.253/24
    sap 1/1/3:103 create
    exit
exit
interface "refpt_vprn_port2_prt" create
    address 40.1.4.253/24
    sap 2/1/6:104 create
    exit
exit

```

The routes exported from a single VPRN per MG group are imported into an aggregate VPRN that forms a BGP peering and advertises routes to peering routers. The BGP session from VPRN 100 is formed toward each peering router and routes are advertised via single BGP peering. VPRN 100 is also used for configuring the signaling interfaces on the gateway as shown in the previous configuration example.

The BGP configuration within aggregate VPRN for GTP-U (for example, VPRN 100) will advertise the signaling interfaces and the GTP-U reference point interfaces to the peering router.

Configure the BGP route export policy to advertise routes to peering router.

```

config router policy-options
    policy-statement "exportVprn100Bgp"
        entry 10
            from
                community "RefPt1_101"
                family ipv4
            exit
            action accept
            next-hop 10.1.1.1
        exit
    exit
    entry 15
        from

```

```

        community "RefPt1_101_prtct"
        family ipv4
    exit
    action accept
        next-hop 10.1.2.1
    exit
exit
entry 20
    from
        community "RefPt1_102"
        family ipv4
    exit
    action accept
        next-hop 10.1.1.2
    exit
exit
entry 25
    from
        community "RefPt1_102_prtct"
        family ipv4
    exit
    action accept
        next-hop 10.1.2.2
    exit
exit
exit

```

The BGP route export policy is configured such that advertised routes are updated with the next-hop of the active MG-VM. Each MG-VM is configured with an associated loopback IP address. Upon failure of the MG-VM, the IP routes are updated to the peering router with the loopback IP address of new active MG-VM as the next-hop IP address.

Configure the BGP session from VPRN 100 to peering routers to advertise local routes.

```

config service
    vprn 100 customer 1 create
        interface "bgp_gtpu_loopback" create
            address 10.1.1.253/32
            loopback
        exit
    static-route 10.1.1.254/32 next-hop 40.1.1.254 bfd-enable
    static-route 10.1.1.254/32 next-hop 40.1.3.254 bfd-enable
    static-route 10.1.2.254/32 next-hop 40.1.2.254 bfd-enable
    static-route 10.1.2.254/32 next-hop 40.1.4.254 bfd-enable
    bgp
        family ipv4

```

```

min-route-advertisement 1
group "bgp_RefPt1"
  export "exportVprn100Bgp"
  peer-as 1
  neighbor 10.1.1.254
    local-address 10.1.1.253
    local-as 1
  exit
  neighbor 10.1.2.254
    local-address 10.1.1.253
    local-as 1
  exit
exit
no shutdown
exit
no shutdown
exit

```

where 10.1.1.254 is the loopback IP address on peering router 1 and 10.1.2.254 is the loopback IP address on peering router 2.



NOTE:

BFD must be enabled for each static route to track the path's status to the next-hop and to peering router. If BFD is not enabled then traffic will be dropped during link failure.

Related information

[Redundancy configuration](#)

4.4.2.5.3 Aggregate VPRN configuration for the MG groups (SGi traffic)

For SGi traffic, routes are exported from VPRN 201 and VPRN 202 with respective route targets as in the previous example. Routes exported from VPRN 201 and 202 are imported into VPRN 200 using the list of route targets. The BGP session from VPRN 200 is formed toward each peering router and the routes are advertised via single BGP peering.

Example: MG group 1 — Route export policy for VPRN 201 (SGi traffic)

```

config router policy-options
  community "interntSgi_201" members "target:1:201"
  community "interntSgi_201_prtct" members "target:1:221"
  prefix-list "prefix_vprn201"
    prefix 1.0.0.0/10 exact
  exit

```

```

policy-statement "vprn201_export"
  entry 10
    from
      prefix-list "prefix_vprn201"
      active-mg 3
    exit
    action accept
      community add "interntSgi_201"
    exit
  exit
  entry 15
    from
      prefix-list "prefix_vprn201"
      active-mg 4
    exit
    action accept
      community add "interntSgi_201_prtct"
    exit
  exit
exit

```

Example: Route export policy application to VPRN 201

```

config service
  vprn 201 customer 1 create
    vrf-export "vprn201_export"
    autonomous-system 1
    route-distinguisher 201:1

```

Example: MG group 2— Route export policy for VPRN 202 (SGi traffic)

```

config router policy-options
  community "interntSgi_202" members "target:1:202"
  community "interntSgi_202_prtct" members "target:1:222"
  prefix-list "prefix_vprn202"
    prefix 1.64.0.0/10 exact
  exit
  policy-statement "vprn201_export"
    entry 10
      from
        prefix-list "prefix_vprn202"
        active-mg 5
      exit
      action accept

```



```

        community add "interntSgi_202"
    exit
exit
entry 15
    from
        prefix-list "prefix_vprn202"
        active-mg 6
    exit
    action accept
        community add "interntSgi_202_prtct"
    exit
exit
exit

```

Example: Route export policy application to VPRN 202

```

config service
    vprn 202 customer 1 create
        vrf-export "vprn202_export"
        autonomous-system 1
        route-distinguisher 202:1

```

Example: Import policy for VPRN 200 to aggregate routes from individual VPRNs (for example, 201 and 202)

```

config router policy-options
    community "interntSgi_201" members "target:1:201"
    community "interntSgi_202" members "target:1:202"
    community "interntSgi_201_prtct" members "target:1:221"
    community "interntSgi_202_prtct" members "target:1:222"
    policy-statement "vprn200_import"
        entry 10
            from
                community expression "[interntSgi_201] OR [interntSgi_201_prtct]
OR [interntSgi_202] OR [interntSgi_202_prtct]"
            exit
            action accept
            exit
        exit
    exit

```

Example: Import policy application to VPRN 200

```

config service
  vprn 200 customer 1 create
    vrf-import "vprn200_import"
    autonomous-system 1
    route-distinguisher 200:1
    interface "bgp_wrk_interntSgi_port1" create
      address 50.1.1.253/24
      sap 2/1/3:201 create
      exit
    exit
    interface "bgp_wrk_interntSgi_port2" create
      address 50.1.2.253/24
      sap 2/1/6:202 create
      exit
    exit
    interface "bgp_prt_interntSgi_port1" create
      address 50.1.3.253/24
      sap 2/1/3:203 create
      exit
    exit
    interface "bgp_prt_interntSgi_port2" create
      address 50.1.4.253/24
      sap 2/1/6:204 create
      exit
    exit
  exit

```

The BGP configuration within the aggregate VPRN for SGi (for example, VPRN 200) will advertise the UE pool IP addresses to the peering router.

Example: BGP route export policy configuration to advertise routes to peering router

```

config router policy-options
  policy-statement "exportVprn200Bgp"
    entry 10
      from
        community "interntSgi_201"
        family ipv4
      exit
      action accept
        next-hop 20.1.1.1
      exit
    exit
  entry 15

```

```

        from
            community "interntSgi_201_prtct"
            family ipv4
        exit
        action accept
            next-hop 20.1.2.1
        exit
    exit
entry 20
    from
        community "interntSgi_202"
        family ipv4
    exit
    action accept
        next-hop 20.1.1.2
    exit
exit
entry 25
    from
        community "interntSgi_202_prtct"
        family ipv4
    exit
    action accept
        next-hop 20.1.2.2
    exit
exit
exit

```

The BGP route export policy is configured such that advertised routes are updated with the next-hop of the active MG-VM. Each MG-VM is configured with an associated loopback IP address. Upon failure of the MG-VM, the IP routes are updated to the peering router with the loopback IP address of new active MG-VM as the next-hop IP address.

Example: BGP session configuration from VPRN 200 to peering routers to advertise local routes

```

config service vprn 200
    interface "bgp_sgi_loopback" create
        address 20.1.1.253/32
        loopback
    exit
    static-route 20.1.1.254/32 next-hop 50.1.1.254 bfd_enable
    static-route 20.1.1.254/32 next-hop 50.1.3.254 bfd_enable
    static-route 20.1.2.254/32 next-hop 50.1.2.254 bfd_enable
    static-route 20.1.2.254/32 next-hop 50.1.4.254 bfd_enable

```

```

bgp
  family ipv4
  min-route-advertisement 1
  group "bgp_interntSgi"
    export "exportVprn200Bgp"
    peer-as 1
    neighbor 20.1.1.254
      local-address 20.1.1.253
      local-as 1
    exit
  neighbor 20.1.2.254
    local-address 20.1.1.253
    local-as 1
  exit
exit
no shutdown
exit
no shutdown
exit

```

where 20.1.1.254 is the loopback IP address on peering router 1 and 20.1.2.254 is the loopback IP address on peering router 2.

**NOTE:**

BFD must be enabled for each static route to track the path's status to the next-hop and to peering router. If BFD is not enabled then traffic will be dropped during link failure.

Related information

[Redundancy configuration](#)

4.4.2.5.4 Egress route configuration

Local routes from individual VPRNs must be configured in such a way so that the egressing traffic takes a local port avoiding the inter-VM traffic.

ECMP must be enabled to support multiple next-hops for a route. The number of ECMP next-hop depends on the number of the egress links to use from each MG-VM.

For example, to enable eight ECMP routes, use the following command:

```
config>router>ecmp 8
```

**NOTE:**

If the local HA is enabled, then the ECMP routes must be split across two MG-VMs. For example, if eight ECMP routes are configured per MG group, then four egress routes must be split across 2 MG-VMs to handle an equal amount of user plan traffic.

The **local-ecmp-egress** keyword must be enabled per MG group VPRN, with 1:1 MG-VM redundancy, to allow only local routes per MG-VM.

**NOTE:**

- The **local-ecmp-egress** keyword must be enabled per MG-group VPRN to allow only local routes per MG-VM.
- Not enabling the **local-ecmp-egress** keyword leads to inter-VM traffic and hence unoptimized traffic processing.

This example shows VPRN 101 configured with multiple static routes via each IP interface present in the MG group.

```
config service
  vprn 101 customer 1 create
    mg-group 1 local-ecmp-egress
      static-route 0.0.0.0/0 next-hop 40.1.1.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 40.1.2.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 40.1.3.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 40.1.4.254 bfd_enable
```

VPRN 102 configured with multiple static routes via each IP interface present on the system.

```
config service
  vprn 102 customer 1 create
    mg-group 2 local-ecmp-egress
      static-route 0.0.0.0/0 next-hop 40.1.1.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 40.1.2.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 40.1.3.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 40.1.4.254 bfd_enable
```

VPRN 201 configured with multiple static routes via each IP interface present on the system.

```
config service
  vprn 201 customer 1 create
    mg-group 1 local-ecmp-egress
      static-route 0.0.0.0/0 next-hop 50.1.1.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 50.1.2.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 50.1.3.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 50.1.4.254 bfd_enable
```

VPRN 202 configured with multiple static routes via each IP interface present on the system.

```
config service
  vprn 202 customer 1 create
    mg-group 2 local-ecmp-egress
      static-route 0.0.0.0/0 next-hop 50.1.1.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 50.1.2.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 50.1.3.254 bfd_enable
      static-route 0.0.0.0/0 next-hop 50.1.4.254 bfd_enable
```

4.4.2.5.5 Redundancy configuration

1:1 MG-VM redundancy

Each VPRN per MG group (GTP-U and SGi) is configured with a loopback address associated with an active and a standby MG-VM.

The following example shows VPRN 101 and VPRN 102 configured with 2 loopback addresses for active and standby. The peering router is configured with static routes in such way so that the first loopback IP address points to the active MG-VM and the second loopback IP address points to the standby MG-VM within a group.

The BGP routes are advertised so that the group-specific loopback IP address is advertised as a next-hop with the respective active MG-VM. For example, when an active MG-VM in a group is operational, then the BGP routes are advertised with the loopback IP address that is associated with the active MG-VM as next-hop so that the peering router forwards traffic to the active MG-VM. In case of failure of the active MG-VM, the BGP updates the routes so that the loopback address associated with the standby MG-VM is advertised as a next-hop for all routes, enabling the peering router to start forwarding traffic to the newly active MG-VM.

```
vprn 101 customer 1 create
  interface "grp1_lo1_active" create
    address 10.1.1.1/32
    loopback
  exit
  interface "grp1_lo1_stdby" create
    address 10.1.2.1/32
    loopback
  exit

vprn 102 customer 1 create
  interface "grp2_lo1_active" create
    address 10.1.1.2/32
    loopback
  exit
```

```
interface "grp2_lo1_stdbby" create
  address 10.1.2.2/32
  loopback
exit
```

The following example shows VPRN 201 and VPRN 202 configured with two loopback addresses for an active and a standby MG-VM.

```
vprn 201 customer 1 create
  interface "grp1_lo1_active" create
    address 20.1.1.1/32
    loopback
  exit
  interface "grp1_lo1_stdbby" create
    address 20.1.2.1/32
    loopback
  exit

vprn 202 customer 1 create
  interface "grp2_lo1_active" create
    address 20.1.1.2/32
    loopback
  exit
  interface "grp2_lo1_stdbby" create
    address 20.1.2.2/32
    loopback
  exit
```

**NOTE:**

BFD must be enabled for each static route to track the path's status to the next-hop and to peering router. If BFD is not enabled then traffic will be dropped during link failure.

BFD parameters must be configured under all IP interfaces to enable BFD with static routes configured on CMG, as shown in the following example.

```
config service
  vprn 101 customer 1 create
    interface "grp1_port1_active" create
      address 40.1.1.1/24
      bfd 100 receive 100 multiplier 3 echo-receive 100
      sap 3/3/1:101 create
    exit
```

N:K MG-VM redundancy

Each VPRN per MG group (GTP-U and SGi) is configured with a loopback address associated with an active MG-VM.

The following example shows VPRN 101 and VPRN 102 configured with one loopback address. The peering router is configured with static routes in such way so that the first loopback IP address points to the active MG-VM of the MG group.

The BGP routes are advertised with the group-specific loopback IP address that is associated with the active MG-VM as next-hop, so that the peering router forwards traffic to the active MG-VM. In case of failure of the active MG-VM, the BGP sends a route update as the selected standby MG-VM becomes the new active MG-VM, and enables the previously advertised loopback IP address as active.

```
vprn 101 customer 1 create
  interface "grp1_lo1_active" create
    address 10.1.1.1/32
    loopback
  exit

vprn 102 customer 1 create
  interface "grp2_lo1_active" create
    address 10.1.1.2/32
    loopback
  exit
```

The following example shows VPRN 201 and VPRN 202 configured with two loopback addresses for an active and a standby MG-VM.

```
vprn 201 customer 1 create
  interface "grp1_lo1_active" create
    address 20.1.1.1/32
    loopback
  exit

vprn 202 customer 1 create
  interface "grp2_lo1_active" create
    address 20.1.1.2/32
    loopback
  exit
```

4.4.2.6 SGi interface services configuration in LB-less mode

The following SGi interface services in LB-less mode can be configured:

- *CGNAT configuration (1:1 MG-VM redundancy)*

- *CGNAT configuration (N:K MG-VM redundancy)*
- *IPSec/GRE configuration*

4.4.2.6.1 CGNAT configuration (1:1 MG-VM redundancy)

In LB-less mode, the CGNAT function is enabled within the MG-VM by configuring a NAT group using the local NAT option per MG group.

The CGNAT with 1:1 MG-VM redundancy requires the addition of the active and standby MG-VM configured per MG group to be in a single NAT group. The active MDA limit option must be configured to be equal to 1.

The NAT binding records (per port-block) can be exported to an external collector device by enabling RADIUS accounting policy.

The following configuration example shows nat - radius as the RADIUS accounting policy used across all NAT groups.

```
config
  isa
    nat-group 1 create
      active-mda-limit 1
      mda 3/4
      mda 4/4
      local-nat
      radius-accounting-policy "nat-radius"
      no shutdown
    exit
    nat-group 2 create
      active-mda-limit 1
      mda 5/4
      mda 6/4
      local-nat
      radius-accounting-policy "nat-radius"
      no shutdown
    exit
```

To enable the NAT function in LB-less mode, a NAT policy must be configured on each VPRN associated with the APN (that is, inside the VPRN). The destination prefix must be configured to steer traffic toward the NAT for selective traffic based on the destination IP address. For example, IP address 0.0.0.0/0 will steer all traffic toward the NAT.

```
config service
  vprn 201 customer 1 create
    nat
      inside
        nat-policy "nat-policy-1"
        destination-prefix 0.0.0.0/0
```

```

        exit
    exit
exit
vprn 202 customer 1 create
    nat
        inside
            nat-policy "nat-policy-2"
            destination-prefix 0.0.0.0/0
        exit
    exit
exit

```

The NAT policy is configured with the IP address pool to be used from the public IP address space and associated with the VPRN configured for public IP addresses (that is, outside the VPRN).

```

config service
    nat
        nat-policy "nat-policy-1" create
            pool "public_pool_1" router 301
        exit
        nat-policy "nat-policy-2" create
            pool "public_pool_2" router 302
        exit
    exit

```

The VPRN to be used for public IP addresses is configured with the external interfaces and also configured with the NAT outside policy.

The following example shows the NAT outside the VPRN configured for MG group 1.

```

config service
    vprn 301 name "301" customer 1 create
        description "SgiVprnOutside_group1"
        vrf-export "expPlcy_sgiVprn_group1"
        ecmp 10
        autonomous-system 1
        route-distinguisher 301:1
        mg-group 1 local-ecmp-egress
        interface "grp1_port1_active" create
            address 10.1.1.1/8
            sap 3/3/1:201 create
        exit
    exit
    interface "grp1_port2_active" create
        address 10.1.2.1/8
        sap 3/3/1:202 create
    exit

```

```

        exit
    exit
    interface "grp1_port1_standby" create
        address 10.1.3.1/8
        sap 4/3/1:203 create
    exit
    exit
    interface "grp1_port2_standby" create
        address 10.1.4.1/8
        sap 4/3/2:204 create
    exit
    exit
    nat
        outside
            pool "public_pool_1" nat-group 1 type large-scale create
            port-reservation blocks 70
            address-range 172.16.0.1 172.16.255.254 create
        exit
        no shutdown
    exit
    exit
    exit

```

The following example shows the NAT outside the VPRN configured for MG group 2.

```

Config service
    vprn 302 name "302" customer 1 create
        description "SgiVprnOutside_group2"
        vrf-export "expPlcy_sgiVprn_group2"
        ecmp 10
        autonomous-system 1
        route-distinguisher 302:1
        mg-group 1 local-ecmp-egress
        interface "grp2_port1_active" create
            address 10.1.1.2/8
            sap 5/3/1:201 create
        exit
    exit
    interface "grp2_port2_active" create
        address 10.1.2.2/8
        sap 5/3/1:202 create
    exit
    exit
    interface "grp2_port1_standby" create
        address 10.1.3.2/8

```

```

        sap 6/3/1:203 create
        exit
    exit
    interface "grp2_port2_standby" create
        address 10.1.4.2/8
        sap 6/3/2:204 create
        exit
    exit
    nat
        outside
            pool "public_pool_2" nat-group 2 type large-scale create
            port-reservation blocks 70
            address-range 172.16.0.1 172.16.255.254 create
            exit
            no shutdown
            exit
        exit
    exit

```

The route export policy must be configured on the outside VPRN per MG group and the prefix for the public IP addresses to be exported in route table.

```

config router policy-options
    prefix-list "prefix-vprn-301"
        prefix 172.16.0.0/12 exact
    exit
    prefix-list "prefix-vprn-302"
        prefix 172.17.0.0/12 exact
    exit
    community "sgi-mgroup1" members "target:1:301"
    community "sgi-mgroup2" members "target:1:302"
    policy-statement "expPlcy_sgiVprn_group1"
        entry 5
            from
                protocol nat
            exit
            action accept
                community add "sgi-mgroup1"
            exit
        exit
        entry 10
            from
                prefix-list "prefix-vprn-301"
            exit
            action accept

```

```

        community add "sgi-mgroup1"
    exit
exit
policy-statement "expPlcy_sgiVprn_group2"
    entry 5
        from
            protocol nat
        exit
        action accept
            community add "sgi-mgroup2"
        exit
    exit
    entry 10
        from
            prefix-list "prefix-vprn-302"
        exit
        action accept
            community add "sgi-mgroup2"
        exit
    exit
exit

```

4.4.2.6.2 CGNAT configuration (N:K MG-VM redundancy)

With N:K MG-VM redundancy, the ISA for CGNAT function is anchored on the MG-VM that is selected to be active. There is no permanent association between the NAT group and the ISA, and therefore a dynamic ISA is configured per NAT.

Within the NAT group, the RADIUS accounting policy can be enabled for NAT binding records (per port-block) to be exported to an external collector device.

Similar to the dynamic SAP, in LB-less architecture N:K redundancy uses a dynamic MDA for failover from the active to the standby MG-VM, when local NAT is enabled on the MG-VM. The dynamic MDA decouples the NAT instance from the VM, such that failure of the active to the standby MG-VM is seamless to the overall system. For example, if the active MG-VM fails, the NAT instance shifts from the physical MG-VM that is associated with active, to the physical MG-VM that is associated with standby (new active).

The following configuration example shows nat - radius as the RADIUS accounting policy used across all NAT groups.

```

config
    isa
        nat-group 1 create
            active-mda-limit 1
            dynamic-mda mg-group-1/4
            local-nat
            radius-accounting-policy "nat-radius"

```

```

        no shutdown
    exit
    nat-group 2 create
        active-mda-limit 1
        dynamic-mda mg-group-2/4
        local-nat
        radius-accounting-policy "nat-radius"
        no shutdown
    exit

```

NAT policy and NAT inside the VPRN must be configured similar to 1:1 MG-VM redundancy. NAT outside the VPRN must be configured with dynamic SAP, as with SGi VPRN configured without CGNAT.

The following example shows the NAT outside the VPRN configured for MG group 1 with N:K MG-VM redundancy.

```

config service
    vprn 301 name "301" customer 1 create
        description "SgiVprnOutside_group1"
        vrf-export "expPlcy_sgiVprn_group1"
        ecmp 10
        autonomous-system 1
        route-distinguisher 301:1
        mg-group 1
        interface "grp1_port1_active" create
            address 10.1.1.1/8
            dynamic-sap mg-group-1/3/1:201
            create
            exit
        exit
        interface "grp1_port2_active" create
            address 10.1.2.1/8
            dynamic-sap mg-group-1/3/1:202 create
            exit
        exit
    nat
        outside
            pool "public_pool_1" nat-group 1 type large-scale create
            port-reservation blocks 70
            address-range 172.16.0.1 172.16.255.254 create
            exit
            no shutdown
            exit
        exit
    exit

```

The following example shows the NAT outside the VPRN configured for MG group 2.

```

config service
    vprn 302 name "302" customer 1 create
        description "SgiVprnOutside_group2"
        vrf-export "expPlcy_sgiVprn_group2"
        ecmp 10
        autonomous-system 1
        route-distinguisher 302:1
        mg-group 2
        interface "grp2_port1_active" create
            address 10.1.1.2/8
            dynamic-sap mg-group-2/3/1:201 create
            exit
        exit
        interface "grp2_port2_active" create
            address 10.1.2.2/8
            dynamic-sap mg-group-2/3/1:202 create
            exit
        exit
    nat
        outside
            pool "public_pool_2" nat-group 2 type large-scale create
            port-reservation blocks 70
            address-range 172.16.0.1 172.16.255.254 create
            exit
            no shutdown
            exit
        exit
    exit
exit

```

The route export policy must be configured on the outside VPRN per MG group and the prefix for the public IP addresses to be exported in the route table.

```

config router policy-options
    prefix-list "prefix-vprn-301"
        prefix 172.16.0.0/12 exact
    exit
    prefix-list "prefix-vprn-302"
        prefix 172.17.0.0/12 exact
    exit
    community "sgi-mgroup1" members "target:1:301"
    community "sgi-mgroup2" members "target:1:302"
    policy-statement "expPlcy_sgiVprn_group1"
        entry 5

```

```

        from
            protocol nat
        exit
        action accept
            community add "sgi-mgroup1"
        exit
    exit
    entry 10
        from
            prefix-list "prefix-vprn-301"
        exit
        action accept
            community add "sgi-mgroup1"
        exit
    exit
    policy-statement "expPlcy_sgiVprn_group2"
        entry 5
            from
                protocol nat
            exit
            action accept
                community add "sgi-mgroup2"
            exit
        exit
        entry 10
            from
                prefix-list "prefix-vprn-302"
            exit
            action accept
                community add "sgi-mgroup2"
            exit
        exit
    exit
exit

```

4.4.2.6.3 IPSec/GRE configuration

The IPSec/GRE tunnel configuration in LB-less mode is identical to the IPSec/GRE tunnel configuration in LB mode.

Related information

[IPSec/GRE configuration](#)

4.4.3 CMG instance configuration in hybrid mode (LB and LB-less)

CMG in hybrid mode provides the benefits of the LB and LB-less modes combined in a single gateway instance.

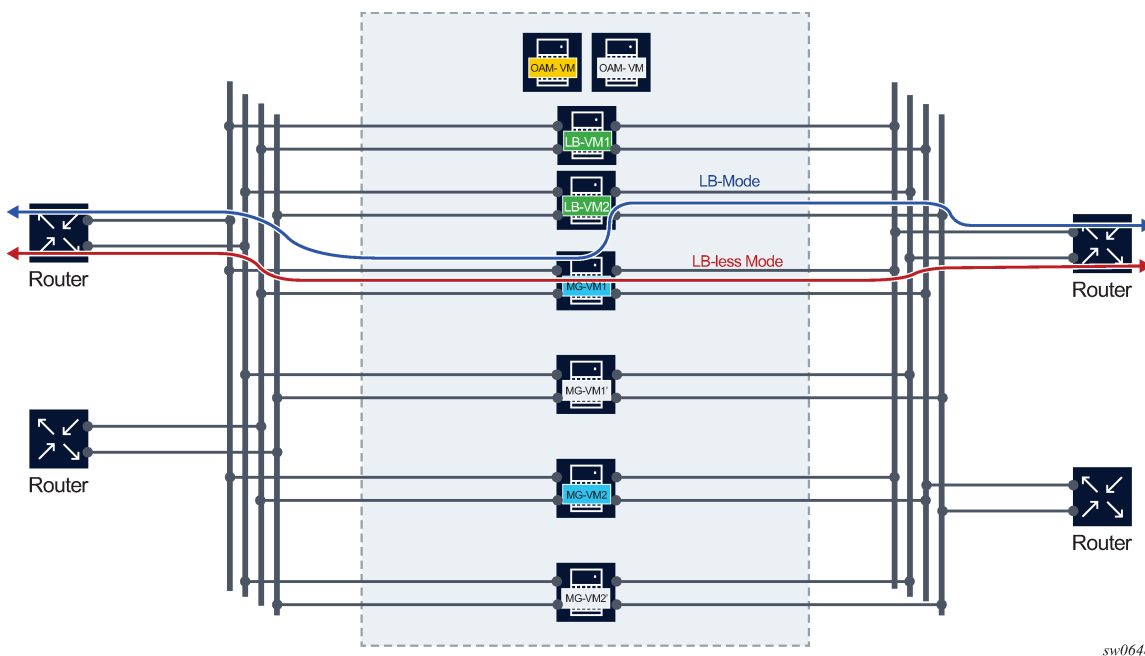
The UP between the RAN and the core network (all GTP-U interfaces) operates in LB-less mode for all PDN sessions. The UP toward the PDN network operates in LB or LB-less mode per APN basis.

CMG in hybrid mode supports the following deployments:

- non-CUPS combined (SGW, PGW/GGSN)
- CUPS user plane (SGW-U, PGW-U/GGSN-U)

A CMG instance deployed in hybrid mode combines the configuration required for LB-less and LB modes.

Figure 24: Network diagram of CMG in hybrid mode



sw0648

All signaling traffic is received/sent via the LB-VMs. The user plane traffic between the RAN and the core network is received/sent directly on the MG-VM. For an APN configured in LB-less mode, the user plane traffic from the PDN network is received/sent directly on the MG-VM. For an APN configured in LB mode, the user plane from the PDN network is received/sent via the LB-VM.

In this example, the control plane load balancing function is provided on a dedicated pair of LB-VMs. For simplicity, the user plane load balancing function is performed on a common pair of LB-VMs that are also used for control plane traffic.

To enable the gateway in LB-less mode, follow the steps in section [CMG instance configuration in LB-less mode](#). All steps must be followed, including the APN configuration, to enable the LB-less mode for a subset of APNs configured on the node.

To enable LB mode APNs, follow the steps in [APN configuration](#) of section [CMG instance configuration in LB mode](#).

4.5 Secure storage

CMG stores all configuration secrets in an encrypted form, following the latest encryption algorithms and strong encryption keys. The complete LI configuration is also kept in an encrypted form.

The "configuration secret" term includes passwords and passphrases used for authentication of users and authentication in various CP protocols and interfaces, keys for encryption of user plane traffic, CP information and sensitive information in various functions on the products, and any type of configuration information which needs to be protected.

These configuration secrets are configured in CLI with the [hash | hash2] optional parameter, using the following context, which indicates how encryption is done once the configuration is saved to disk.

```
configure system securitymanagement-interface classic-cli
  read-algorithm {hash | hash2 | all-hash}
  write-algorithm {hash | hash2 | all-hash}
```

For more information, refer to the *CMG CLI Reference Guide*.

The master encryption keys for the LI configuration and the configuration secrets are stored locally in the CMG instance by default. For additional security, CMG can be configured to connect to an external Vault service and store the master encryption keys to Vault. Vault is an open source tool widely used as the solution to manage and store secrets securely in networks.

When Vault is configured, CMG acts as a Vault client and after it authenticates to Vault, it offloads the secure storage of the master encryption keys, as described above, to Vault. Specifically, CMG supports storing the following keys to Vault:

- encryption key of all configuration secrets
- LI configuration encryption key
- admin authorized key for passwordless SSH connection of admin user

[Vault configuration parameters](#) describes the basic information that must be configured during the CMG deployment time for using Vault.

Table 2: Vault configuration parameters

Parameter	Description
vault-name	The service name, FQDN, or IP address of the external Vault server
vault-port	The port that Vault server uses

Table 2: Vault configuration parameters (continued)

Parameter	Description
vault-base-path	The absolute path for storage of secrets that the specific CMG instance will use

CMG supports integration with the Vault server version exposing the following APIs:

- kv version 1
- kv version 2

For more information about configuring a CMG instance to use Vault and how CMG authenticates to Vault, refer to the *CMG Installation Guide* and the *CMG CNF Installation Guide*.

5 Common gateway configuration

Overview of the provisioning functionality that is common to all standalone and combined gateway types.



NOTE:

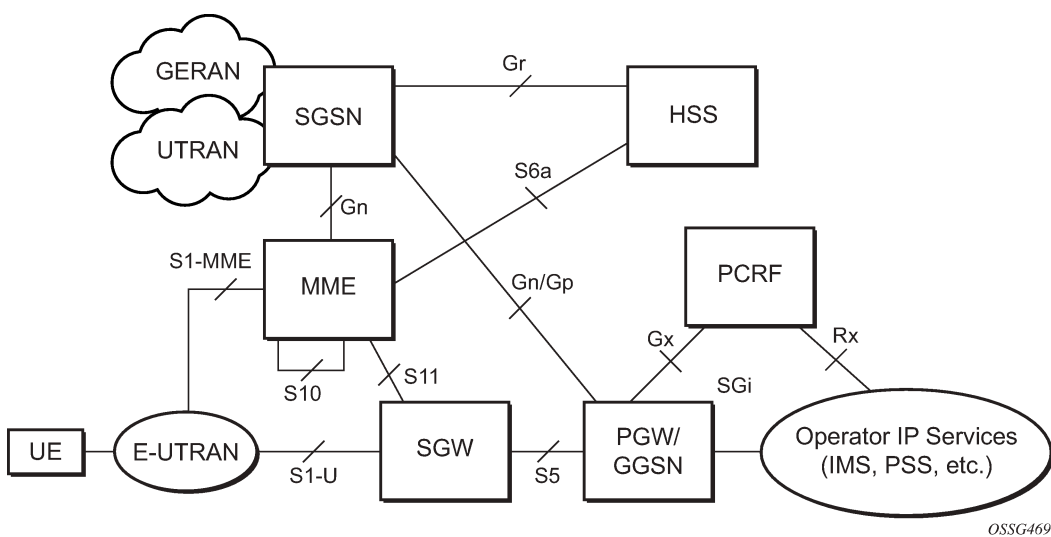
The terms "Mobile Gateway", "MG", and "gateway" refer to the following supported types of 7750 SR mobile gateways.

- Packet Data Gateway (PGW)
- Gateway GPRS Support Node (GGSN)
- Serving Gateway (SGW)
- evolved Packet Data Gateway (ePDG)
- Trusted Wireless Access Gateway (TWAG)
- Subscriber Services Gateway (SSG)
- 3GPP2 Home Agent

The terms "gateway" and "mobile gateway" also refer to the Cloud Mobile Gateway (CMG) product.

The gateway functionality allows 3G wireless operators to upgrade their current network to meet the bandwidth needs and also to seamlessly upgrade their network with a common platform as they migrate toward LTE networks.

Figure 25: PGW/GGSN network architecture example



Related information

[Charging configuration](#)

Policy configuration

Information and configuration guidelines for policy control implementation for EPC nodes.

5.1 Gateway instance configuration

One instance is supported per gateway. To configure the gateway instance, use the following command:

```
configure mobile-gateway pdn 1
```

5.2 Gateway name configuration

Configure the gateway with a name to uniquely identify it. The name includes the node ID in the gateway cluster and the group ID configured for the specified region.

The name is composed of the MCC and MNC value of the operator, the gateway type (SGW or PGW or ePDG or TWAG in this case), and the group and node IDs.

```
configure mobile-gateway pdn <gw-id> epc-node <MCC>.<MNC>.<SGW|PGW|ePDG|TWAG>.<Region String>.<Group Id>.<Node Id>
```

5.3 Home PLMN ID configuration: MCC and MNC values

Configure the Home PLMN ID of the network where the gateway network element is deployed. The MCC (three digits) and MNC (two or three digits) configured for the network element are used to compare with MCC and MNC of the IMSI to classify the users as homers, roamers, and visitors.

5.4 IP addresses configuration

The basic IP address provisioning guidelines.



NOTE:

For more information about UE IP address allocation for the PGW/GGSN, including configuration of local IP pools in VPRN and IP address allocation using a local DHCP server, see [UE IP address allocation](#).

5.4.1 Physical IP addresses (interface) configuration

An IP address for the physical I/O interface must be configured. The physical IP interface can be configured with IPv4, IPv6, or IPv4/IPv6 addresses.

Dual stack is supported; for example, IPv4 and IPv6 address configured on the same logical or physical interface.

Multiple I/O cards can be used as part of a LAG group for Layer 2 redundancy or individual I/O cards can be used as part of Layer 3 ECMP redundancy.

Separate VPRN services can be configured per Sx reference point to help operators partition the RAN domain, EPC core, and management domain. For example, S5 in VPRN-1, APNs mapped to VPRNs, and Ga in VPRN-3, can be configured when specifying the local interface along with VPRN instance.

5.4.2 System IP address (routing) configuration

The system IP address is used as router ID for the routing protocols (IGP, BGP) running on the system. This is normally the IP loopback address of the system, which is always up and can be reached through any of the physical interfaces configured on the system.

The system IP interface is configured with IPv4, IPv6 or IPv4v6 addresses.

```
configure router
  interface system
    address 10.20.1.1/32
    ipv6
      address 2005:: 1/128
    exit
  exit
```

5.4.3 Management IP address (network management) configuration

The management IP address is used for provisioning the MG chassis and for MG configuration using NFM-M.

The management IP address is normally the out-of-band management IP which is configured under the Boot Option File (BOF).

```
# show bof
=====
BOF (Memory)
=====
primary-image      cf3:\2.0.R5
primary-config     cf3:\test.cfg
address            138.120.189.56/24 active
static-route       128.0.0.0/2 next-hop 138.120.189.1
autonegotiate
duplex             full
speed              100
wait               5
persist            on
```

```
li-local-save
li-separate
console-speed 115200
=====
```

5.4.4 Mobile signaling IP addresses configuration

The IP interfaces used for mobile signaling protocols can be configured for each reference point, such as S5, S5-U, or Gx, based on the operator's network IP addressing schemes.

The source IP address of the signaling packet originated by the gateway is the IP interface configured for that respective reference point.

```
configure router
  interface "gx"
    address 110.112.134.111/32
    loopback
  exit
  interface "s5"
    address 10.112.14.1/32
    loopback
  exit
  interface "ga"
    address 10.112.24.1/32
    loopback
    ipv6
      address 3FFE::A70:1801/64
    exit
  exit
```

5.4.5 Router IP interfaces configuration

The following output shows a router IP interface configuration.

```
router
  interface "gx"
    address 10.10.8.1/24
    port 1/1/2
  exit
  interface "ip-10.10.6.3"
    address 10.10.6.3/24
    port 1/1/1
  exit
  interface "ip-10.30.1.3"
```

```

        address 10.30.1.3/32
        loopback
    exit
    interface "sgi"
        address 10.10.9.1/24
        port 1/1/3
        ipv6
            address 3FFE::A0A:901/112
        exit
    exit
    interface "system"
        address 10.20.1.3/32
        ipv6
            address 3FFE::A14:103/128

```

5.4.6 Recommendations for IP address configuration



NOTE:

When separate IP interfaces are used for gateway signaling IP interfaces configuration, you must check for each reference point whether a common IP must be configured to support all call flows.

Table 3: IP addressing recommendations

Configuration component	Interface name	IP address
SGW-PGW	S5	Customers can choose to share the same loopback across all logical interfaces (S5, S11, S1-U) or choose to use separate loopbacks for each logical interface Separate VPRN services can be configured per Sx reference-point to help operators partition the RAN domain, EPC core, management domain For example, S1-U, S11 in VPRN-1, S5/S5-U in VPRN-2, Rf in VPRN-3, which can be configured when configuring the local-interface along with VPRN instance
SGW-MME	S11	One IP loopback interface /32 or /64

Table 3: IP addressing recommendations (continued)

Configuration component	Interface name	IP address
SGW-MME	S11-U	One IP loopback interface /29 or /48 Must be configured if CloT control plane optimization support is required
SGW-eNodeB	S1-U	One IP loopback interface /29 or /48
SGW Management	OA&M	One EDN address per chassis Two Ethernet ports
Physical interface addresses	Routed interfaces	Two /30 addresses Two interface addresses for redundancy Two Ethernet ports Additional links can be added as needed

5.4.7 Multiple IP addresses reservation from local IP address pool

You can reserve up to three IPv4 and, or IPv6 addresses from a configured VPRN/IP local address pool. The reserved IP addresses are not allocated to any UE but they are used as client/loopback addresses for the RADIUS or the DHCP.

To reserve IP addresses, use the `config>profile>ip-profile` command. This command creates a list of reserved IP addresses from a VPRN/IP local address pool. The created IP profile name is then referred under the configured `config>service>vprn>ip-local-pools` command using the `reserved-ip-list` command with the IP profile name.

The reserved IP addresses can then be used for the RADIUS or the DHCP loopback under the same APN in the VPRN/base.

The ping command can use the `config>service>vprn>ip-local-pool>reserved-ip` command to reserve an IP address for the RADIUS or the DHCP loopback. In this case the `ip-profile` and the `reserved-ip-list` commands are not used.

5.5 VNF/CNF redundancy (geo-redundancy) configuration guidelines

Description of VNF/CNF redundancy configuration for CMG.

For more information about inter-VNF/CNF redundancy and intra-VNF/CNF redundancy for CMG, see [CMG redundancy support](#) section in the *Cloud Mobile Gateway (CMG)*.

**NOTE:**

The redundancy method does not change between VNF and CNF. This section is also applicable to CNF.

5.5.1 VNF/CNF redundancy support

Geo-redundancy allows a pair of gateways to be deployed in a redundant fashion.

This section describes geo-redundancy support for the mobile gateway platforms:

CMG Inter-VNF redundancy

CMG Inter-CNF redundancy

In case of node failure, the secondary node can continue to service UEs without invoking node failure recovery actions by the adjacent nodes, such as an SGSN, an SGW/PGW/GGSN, an OCS, a PCRF, a 3GPP AAA, an OFCS, an MME, a PGW-C/GGSN-C/SMF, or an ePDG.

The VNF/CNF redundancy function allows gateways to synchronize UE PDN sessions, bearer states, policies, and charging information so the secondary gateway can operate at full state upon switchover.

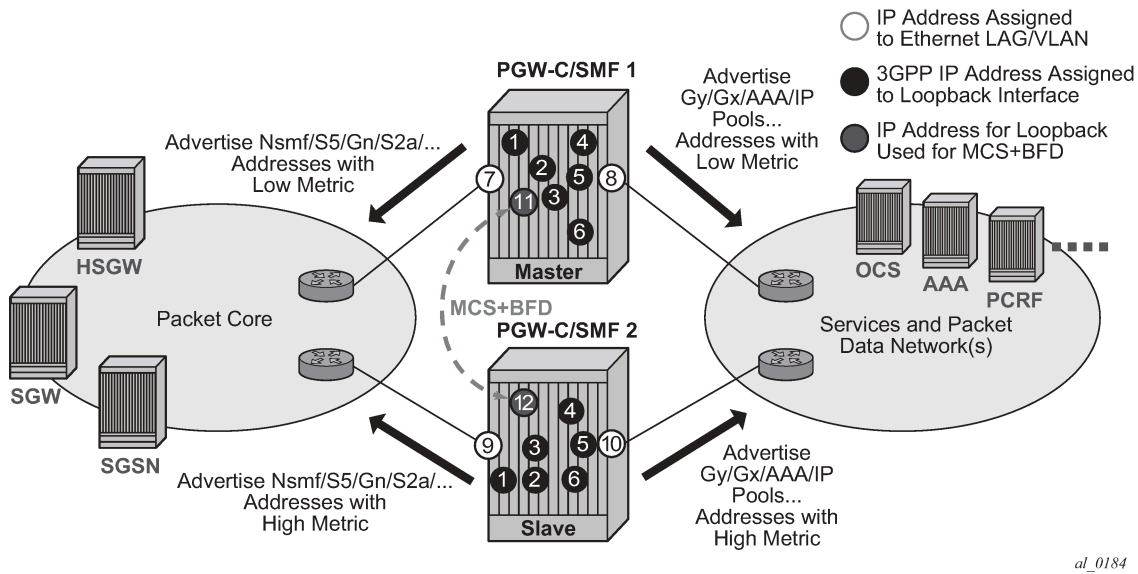
The administrative roles of the gateways are primary and secondary, and are provisioned by the operator. The operational role can be master or slave. Each gateway type can take the role of a master or slave; the VNF redundancy algorithm ensures that only one mobile gateway takes the master role at any given time. The VNF/CNF redundancy feature is integrated with routing protocols so that signaling and bearer traffic is attracted to the master gateway at all times.

Upon switchover, when a former slave gateway becomes a master, the new master gateway will modify its routing metric for the shared signaling and user plane IP interface addresses (such as Gn/Gp, S5/S8, N3/N9, AAA, Gy, and Gx) so that it attracts the corresponding traffic. In addition, the former master will modify its routing metrics so that it no longer attracts traffic.

[PGW-C/SMF or PGW-U/UPF gateway VNF/CNF redundancy example](#), [SGW VNF/CNF redundancy example](#), and [ePDG VNF redundancy example](#) show examples of PGW-C/SMF (control plane) or PGW-U/UPF (user plane), SGW, and ePDG VNF/CNF redundancy deployments.

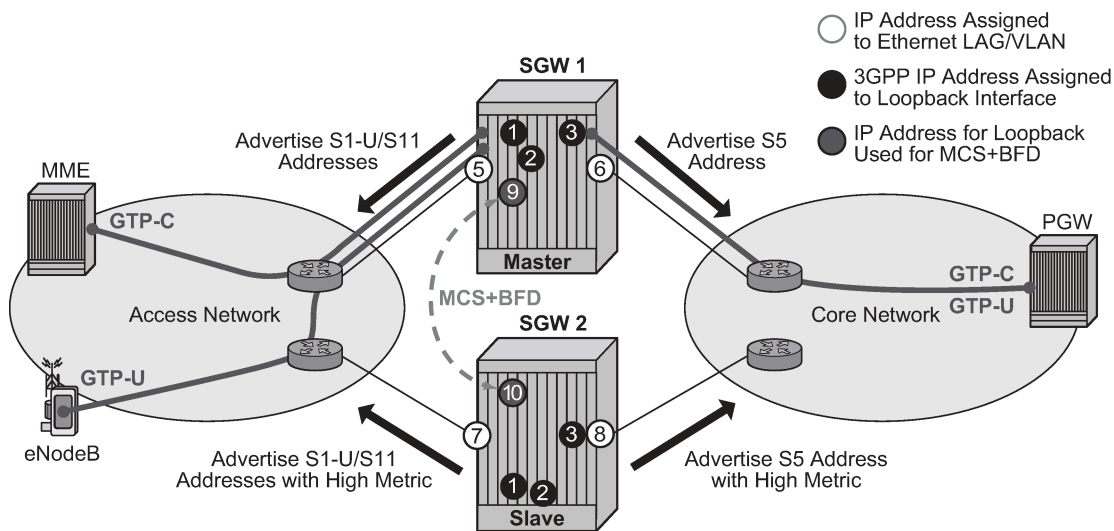
PGW-C/SMF is supported as inter-VNF/CNF only.

Figure 26: PGW-C/SMF or PGW-U/UPF gateway VNF/CNF redundancy example



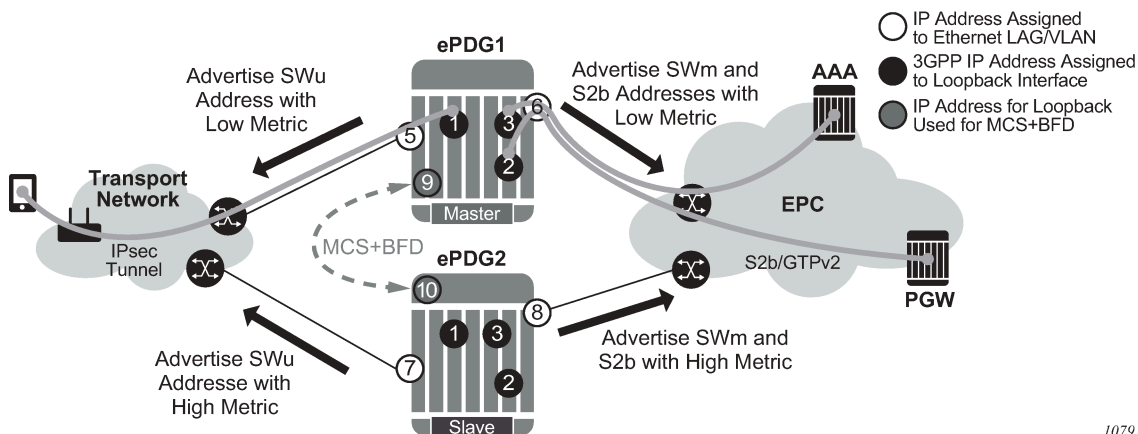
al_0184

Figure 27: SGW VNF/CNF redundancy example



al_0112

Figure 28: ePDG VNF redundancy example



1079

5.5.2 VNF/CNF redundancy redirection

VNF/CNF redundancy redirection enhances the gateway redundancy by improving the handling of possible partial network failures, further reducing data loss during a manual switchover due to route convergence delays.

VNF/CNF redundancy redirection introduces a tunneling mechanism between the mobile gateway pair with additional tunnel routes for the reference point interfaces over the MC mobile multi-chassis synchronization (MCS) interface. The tunnel used for redirection can be created with LDP, RSVP, or GRE, depending on the network between the nodes.

VNF/CNF redundancy redirection is supported on the reference point interfaces S1u, S5/S8, Gn/Gp, S11, S2a, S2b, Gx, Gy, s6b, Rf, SWm, and SWu.

5.5.2.1 Handling a partial network failure

Avoid a failover during a partial network failure.

When an VNF/CNF redundancy pair is running in a steady state, the mobile gateway in slave mode has its lower priority (high metric) routes advertised for the reference point interfaces; however, it also has additional tunnel routes active, creating a redirection tunnel from slave to master. The mobile gateway in master mode has its high priority (low metric) routes and attracts all traffic.

When a partial network failure occurs where some reference point traffic (for example, S5) ends up on the slave system due to a misconfiguration in the network or some other logical failure, the mobile gateway in slave mode receives that traffic only. However, with a redirection tunnel, that traffic is forwarded back to the current master. This allows the current master to continue to successfully handle the UE traffic without any interruption and does not require any immediate failover to be triggered. This gives you time to debug and correct the problem. The mobile gateway in slave mode also generates an SNMP trap indicating that traffic is running over the redirection tunnel.

The `config>router>if>mc-redirect-mode` command with the **epdg** keyword must be configured when the interface receiving traffic from the slave system is an ePDG.

5.5.2.2 Handling a manual switchover

To resolve packet drops due to route convergence delay, the manual switchover mechanism with VNF/CNF redundancy redirection also makes use of the MC-redirection tunnel mechanism, by creating a redirection tunnel from an old master to a new master for a limited time to bridge the route convergence delay.

A manual switchover minimizes data loss by synchronizing all UE state information. However, there is still the issue of route convergence delay in the network. The current manual switchover mechanism requires traffic and signaling routes to be withdrawn and re-advertised by a new master. This change in routing is subject to the route convergence delay in the network, causing packets to drop. In BGP networks, route convergence delays are in the order of tenths of seconds.

Like the current manual switchover mechanism, the old master (current master) will first synchronize UE state information to the new master (current slave) and the redirection tunnel is withdrawn. Once the

synchronization is successfully completed, the old master will not withdraw its reference point routes but instead will re-advertise routes for reference point interfaces with its slave metric (high metric) and active redirection tunnel route (slave (old master) to the new master). Any traffic that the old master still receives due to the route convergence delay is forwarded to the new master and any packet drops are avoided.

To avoid residual charging information loss between the VNF/CNF redundancy checkpoint and the switchover of traffic to a new master, immediate synchronization of statistics to the standby/slave for a planned switchover is supported. The mobile gateway tracks the statistics updated after the initial session state and check-points them to the paired resource (CMG MG-VM/pod). After the slave takes over the traffic and is in pre-master state, the remaining statistics will be check-pointed from the old master, which will be in pre-slave state.

5.5.3 ISSU minor and major release upgrade support

ISSU minor and major upgrade is supported for a broad set of Mobile Gateway releases, for operators who are using VNF/CNF redundancy.

Using the ISSU process, you can upgrade the standby/slave mobile gateways with a new major software release, and then perform a manual switchover, so the standby/slave mobile gateway becomes the active/master node, now running the new software release version. The operator can then upgrade the former active/master mobile gateway with the same new major software release. This is accomplished while continuing to handle all attached UE sessions without any impact to services.

For more information about the supported ISSU minor and major release upgrade process, refer to the applicable *7750 MG and CMG Release Notes*.

5.5.4 VNF/CNF redundancy configuration

The VNF/CNF redundancy configuration involves three steps:

1. *Gateway interface configuration*
2. *Multi-chassis synchronization configuration*
3. *Routing policies configuration*

The VNF/CNF redundancy pair must have the same configuration to match resources (CMG MG-VMs/pods). For VNF/CNF redundancy to work, both gateways in the pair must have the same configuration, including all CPMs and all ISM-MG cards of the same hardware version. Updates to the configuration after VNF/CNF redundancy is configured, must be made on the gateway running in slave mode.



NOTE:

- For the IP allocation by the SMF in CUPS, the network realm and the IP pool configuration for the up-peer-list profile must be the same between the gateways in the pair, and must be applied in the same order.

- For more information about the redundancy commands listed in the following sections, refer to the *7750 SR Basic System Configuration Guide*.
- For a configuration example of inter-VNF/CNF redundancy for LB-less deployments, see [Inter-VNF redundancy configuration for LB-less deployments](#).

5.5.4.1 Gateway interface configuration

To configure interfaces, follow the same steps as for standalone gateway interface configuration, with primary and secondary gateways S5, Gn, Gy, AAA, Nsmf and so on, configured identically using loopback interfaces.

For IP allocation, IP pools must be configured with static routes for each defined IP pool. These static routes are then used by the MCS to update route metrics in the event of a master/slave transition.

```
vprn 1 customer 1 create
:
:
    ip-local-pool "pool-1"
        hold-timer-sec 0
        ipv4-prefix "10.10.0.0/16"
    exit
    ip-local-pool "pool-2"
        hold-timer-sec 0
        ipv4-prefix "10.11.0.0/16"
        ipv4-prefix "10.12.0.0/16"
    exit
    static-route 10.10.0.0/16 black-hole
    static-route 10.11.0.0/16 black-hole
    static-route 10.12.0.0/16 black-hole
:
:
    no shutdown
exit
```

5.5.4.2 Multi-chassis synchronization configuration

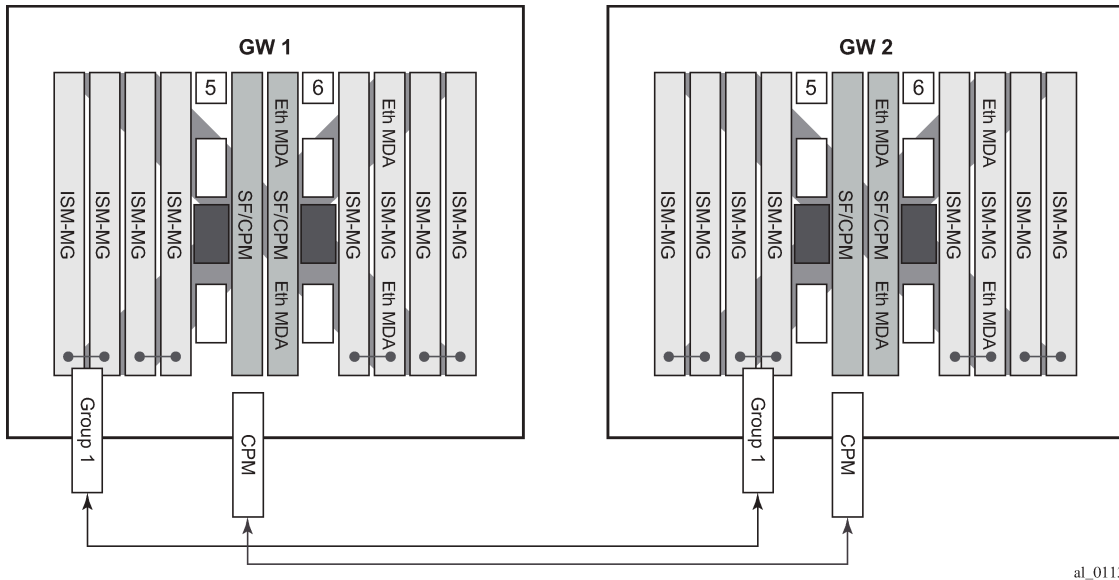
The multi-chassis synchronization (MCS) function elects a master gateway and synchronizes the UE state, policy, and charging information on the master gateway to the slave gateway.

The synchronization is done between the CPMs on the 7750 SR MG and OAM-VMs/pods on CMG. For 3GPP functions, synchronization is done between each of the ISM-MG groups on 7750 SR MG and MG-VMs/pods on CMG. The gateway must have the matching number of groups.

On the 7750 SR MG, physical position of the cards within the groups does not need to be the same between MG1 and MG2.

Similarly, the VM-ID of the MG-VM/pod for each redundancy group on paired gateway instances may not match.

Figure 29: Multi-chassis synchronization configuration example



The MCS function can be configured using IPv4 or IPv6 addresses, and must be configured in the global routing context. The MCS function uses an internal keepalive mechanism and can also be enabled with BFD for faster convergence.

Example: Primary gateway

```
multi-chassis
  peer 10.90.1.2 create
    description "MC peer 10.90.1.2"
    source-address 10.90.1.1
    mc-mobile
      mobile-gateway 1 role primary
      no shutdown
    exit
  exit
  no shutdown
exit
exit
```

Example: Secondary gateway

```
configure>redundancy>

multi-chassis
```

```

    peer 10.90.1.1 create
      description "MC peer 10.90.1.1"
      source-address 10.90.1.2
      mc-mobile
        mobile-gateway 1 role secondary
        no shutdown
      exit
    exit
  no shutdown
exit

```

In the case of a PGW, for MCS to track Gi/SGi interfaces, the track-mobile command needs to be enabled for each Gi/SGi interface under the config>service>vprn>interface context. All other traffic interfaces are automatically tracked.

```

vprn 2 customer 1 create
:
interface "SGi-Loopback" create
  address a.b.c.d/32
  loopback
  track-mobile
exit

```

5.5.4.3 Routing policies configuration

The sample configuration in this section uses a PGW as an example.

Before configuring the routing policies, the PGW must be up and the interfaces S5, Gn, Gi/SGi, and so on, and mc-mobile (MSC) must be configured. The PGW Inter-chassis/VNF redundancy uses 4-metric route policy configuration on all interfaces to ensure both nodes cannot become the master simultaneously:

Pri_m	Primary master metric (for example, 10)
Pri_s	Primary slave metric (for example, 100)
Sec_m	Secondary master metric (for example, 20)
Sec_s	Secondary slave metric (for example, 200)

The four metrics must be configured to satisfy:

```
Pri_m < Sec_m < Pri_s < Sec_s
```

The following example shows the configuration of the route policy for S5 using BGP Metric policy.

Example: Primary PGW

```
prefix-list "prefix-s5-icr"
  prefix <ip_address of s11> exact
exit
policy-statement "prefix-s5-icr"
  description "Policy From prefix-s5-icr To bgp"
  entry 10
  description "Entry 10 - From Prot. prefix-s5-icr To bgp"
    from
      prefix-list "prefix-s5-geored"
      state mobile-master
    exit
    to
      protocol bgp
    exit
    action accept
      metric set 10
    exit
  exit
  entry 20
    from
      prefix-list "prefix-s5-geored"
      state mobile-slave
    exit
    action accept
      metric set 100
    exit
  exit
  default-action reject
exit
```

Example: Secondary PGW

```
prefix-list "prefix-s5-icr"
  prefix <ip_address of s5> exact
exit
policy-statement "prefix-s5-geored"
  description "Policy From prefix-s5-icr To bgp"
  entry 10
  description "Entry 10 - From Prot. prefix-s5-icr To bgp"
    from
      prefix-list "prefix-s5-icr"
      state mobile-master
    exit
```

```
to
    protocol bgp
exit
action accept
    metric set 20
exit
exit
entry 20
    from
    prefix-list "prefix-s5-icr"
        state mobile-slave
    exit
    action accept
    metric set 200
    exit
exit
default-action reject
exit
```

5.5.4.4 Forcing a manual switchover

The manual switchover feature provides the ability to force a switchover, so the master gateway transitions to a slave role and its peer gateway transitions to master.

The manual switchover is not hitless, but synchronizes the complete charging information and state for each bearer, allowing the slave to transition with minimum disruption and data loss.

To force a switchover, use the VNF/CNF redundancy mobile switchover command:

```
admin redundancy mc-mobile-switchover mobile peer
```



CAUTION:

If the **now** option is specified, the switchover will be executed, but the new master node might have incomplete UE records. In some cases this may impact the service.



NOTE:

- If the **now** option is specified and the ICR is in an Up state, the manual switchover can be performed even when the Geo-Redundancy state is not hot.
- The **abort** option can be executed to abort a switchover only for an ongoing switchover that has been in progress for at least 5 minutes.

For the command description including parameter usage, see the `mc-mobile-switchover` command in *CMG CLI Reference Guide*.

5.5.4.5 ICR UE-by-UE reconcile synchronization

The ICR UE-by-UE reconcile synchronization functionality uses a database model of syncing the available UE data as one complete record, during a manual switchover (FSWO) or when a reconcile occurs due to lost and then restored MCS connectivity.

The UE-by-UE reconcile synchronization limits lock-steps and does not affect the processing of new attaches; therefore, ICR synchronization can be paced by the signaling module to use background CPU resources. If the CPU is busy, the updates go into the local DB tables and are synced when the CPU is available. This allows the master node to stay ahead of the slave node and does not stay in a lock-step state.

The ICR UE-by-UE reconcile synchronization from the ICR master to the ICR slave node is in UE steps and allows for monitoring the number of UEs that have been synchronized to the slave node by showing the current status of the synchronized UEs.



NOTE:

The ICR UE-by-UE reconcile synchronization should be enabled for all ICR-configured mobile gateway systems except for the HRPD Home Agent.

When enabled, UE-by-UE reconcile synchronization allows the execution of a manual switchover even if a resynchronization is in progress. Therefore, if for example, 90% of the UEs are synchronized, 90% of the UEs survive if a switchover is executed at that time.

To enable the ICR UE-by-UE reconcile synchronization, use the following command:

```
configure redundancy multi-chassis peer mc-mobile mc-complete-ue-sync
```

Node synchronization based on the UE-by-UE synchronization status describes the ICR master and slave nodes synchronization behavior, based on the status of the UE-by-UE reconcile synchronization feature.

Table 4: Node synchronization based on the UE-by-UE synchronization status

ICR gateway operational state	UE-by-UE functionality	Description
Master	Enabled	UE-by-UE synchronization
	Disabled	Synchronization based on the default (standard) functionality

Table 4: Node synchronization based on the UE-by-UE synchronization status (continued)

ICR gateway operational state	UE-by-UE functionality	Description
Slave	—	Synchronization based on how the master node is configured to perform the synchronization dependent on the communication over the MCS

The UE-by-UE reconcile synchronization feature can be configured on the fly, that is a shutdown of the gateway is not needed.

Node action based on the UE-by-UE synchronization status change describes the action of the ICR master and slave nodes when the status of the UE-by-UE reconcile synchronization feature changes.

Table 5: Node action based on the UE-by-UE synchronization status change

ICR gateway operational state	ICR gateway action
Master	When the UE-by-UE functionality status changes, it communicates to the slave node via the MCS that synchronization has changed (sync changed) and restarts synchronization according to the new configuration
Slave	When the UE-by-UE functionality status changes, it continues based on how the master node is currently syncing over the MCS When a sync changed is received over the MCS, it purges all current state information and is ready to resynchronize with the master node

To view more debugging information about the UE-by-UE reconcile synchronization, use the following command:

```
show redundancy multi-chassis mc-mobile peer ip-address
```

The following information is shown:

- status of the functionality (enabled or disabled)
- current status of the synchronization (0 to 100%)

- when a synchronization is in progress and the status displays less than 100%, the Geo-Redundancy State field indicates WARM to indicate that synchronization is in progress
- when the status displays 100%, the Geo-Redundancy State field indicates HOT
- an FSWO progress complete UE synchronization table to show the status of each synchronized MG group

Depending on the node that the `show>redundancy>multi-chassis>mc-mobile>peer` command is executed, it displays the following information:

Master node	Displays synchronization updates, for example, FSWO progress depending on the UE-by-UE reconcile synchronization feature status
Slave node	Displays synchronization information, for example, FSWO progress based on the configuration of the master node

5.5.4.6 ICR slave detection delay timer

The ICR slave traffic detection delay timer delays the start of traffic detection on the slave node for a configured period of time during an MCS connection loss between the ICR nodes. The timer can be configured on the fly and is activated in the next ICR MCS down event.



NOTE:

When the traffic detection mechanism is enabled, enabling the slave detection delay timer does not have any effect on the system.

To enable the ICR traffic detection delay timer, use the following command:

```
configure redundancy multi-chassis peer mc-mobile slave-traffic-delay-
timer delay-timer
```



NOTICE:

Before executing the command, make sure that a switchover is not in progress.

When the slave detection delay timer is enabled (that is, the ICR MCS is down), its duration cannot be changed; it can only be disabled or set its value to 0. This triggers the default traffic detection state mechanism.

When enabled, the slave traffic detection timer is activated on the ICR slave node as soon as the ICR MCS is lost (BFD detected lost) and the slave node performs the following:

- delays the start of traffic detection (the slave node stays as slave and continues to advertise slave metric for as long as the delay timer is running)

- if the `mc-redirect` command is enabled, continues to keep the MC redirect tunnel up and attempts to route any received traffic through it (regardless of whether traffic is successfully routed or not) for as long as the delay timer is running
- if the ICR MCS is back up before the delay timer expires, it stops the timer and continues as the slave node
- if the ICR MCS is still down when the delay timer expires, it starts the default traffic detection state mechanism

To view the status of the slave detection delay timer, use the following command:

```
show redundancy multi-chassis mc-mobile peer ip-address
```

5.5.4.7 ICR Forced Switchover based on Gx failover

Configure the CMG ICR (PGW, combined SGW/PGW, PGW-C, or combined SGW-C/PGW-C) to monitor all the Diameter Gx connections. When the MCS (mc-mobile) is up and all the Gx connections on the master CMG node are detected as lost, an ICR Forced Switchover (FSWO) is triggered.

When you enable the ICR FSWO based on the monitoring of all the Gx connections, the CMG uses the Diameter Gx path management and the ICR FSWO as follows:

- when the CMG ICR MCS is up, the Gx connections are monitored on the CMG in the master operational role
 - if all the Gx connections are detected as lost, a guard timer starts and when it expires and all the Gx connections are still lost, the CMG in the master operational role triggers an ICR FSWO to make the CMG in the slave operational role become the master
 - if all the Gx connections are detected again as up before the guard timer expires, the guard timer is cancelled and the CMG in the master operational role continues to monitor the Gx connections (no ICR FSWO is triggered)
- when the CMG ICR MCS is down, the Gx connection monitoring stops and the CMG ICR pair falls back to the current ICR traffic detection and failover handling

The configurable guard timer prevents an ICR FSWO due to a short Gx connectivity loss; for example, link flapping. Additionally, to prevent an ICR failover from happening immediately after the new master CMG detects all the Gx connections as lost, the new master CMG triggers another ICR FSWO only when Gx connections are detected as up and are again detected as lost.

To configure an ICR Forced Switchover based on Gx failover, use the following command:

```
config>redundancy>multi-chassis>peer>mc-mobile>diameter-connection-lost-  
detection>gx-failover guard-timer
```

To view the status of the ICR Forced Switchover based on Gx failover, use the following command:

```
show redundancy multi-chassis mc-mobile peer ip-address
```

5.5.5 Inter-VNF redundancy configuration for LB-less deployments

Configuring inter-VNF redundancy for LB-less mode deployments requires three steps (similar to LB mode deployments):

1. *CMG configuration for ICR*
2. *MCS configuration*
3. *Routing policies configuration*

5.5.5.1 CMG configuration for ICR

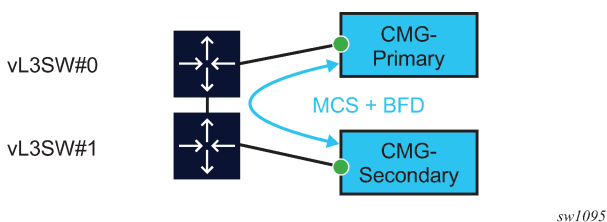
The secondary CMG configuration must meet the following rules (similar to LB mode deployments):

- the secondary CMG must have the same gateway configuration (combined SGW/PGW) as the primary (PGW) CMG
- the secondary CMG must have the same IP addresses for all reference point interfaces (S11, S5/S8, Gn/Gp, Gx, Gy, RADIUS, and so on) loopback addresses which can be either IPv4 or IPv6 addresses
- the secondary CMG must have the same number of configured MG-VMs with same capacity/scale as the primary CMG

5.5.5.2 MCS configuration

The MCS configuration must be configured in the global routing context using either IPv4 or IPv6 addresses. The MCS 'system' loopback and physical interface configuration with BFD parameters must be added.

Figure 30: MCS and BFD CMG configuration



System loopback and physical interface configuration

Example: Primary CMG configuration

```
config
  router Base
    interface "mc_int"
      address 10.10.50.1/24
```

```

        port 1/1/1:2
        no shutdown
    exit
    interface "system"
        address 10.20.1.3/32
        ipv6
            address 3ffe::a14:103/128
            bfd 100 receive 100 multiplier 3 echo-receive 100
        exit
        bfd 100 receive 100 multiplier 3 echo-receive 100
        no shutdown
    exit
exit
exit
exit

```

Example: Secondary CMG configuration

```

config
  router Base
    interface "mc_int"
      address 10.10.50.2/24
      port 1/1/5:2
      no shutdown
    exit
    interface "system"
      address 10.20.1.1/32
      ipv6
        address 3ffe::a14:101/128
        bfd 100 receive 100 multiplier 3 echo-receive 100
      exit
      bfd 100 receive 100 multiplier 3 echo-receive 100
      no shutdown
    exit
  exit
exit

```

For ICR ISSU, the following additional networking resources are required:

- new VLANs for connectivity between LB-VMs and peering router
- new subnets on LB-VMs and peering routers
- iBGP peering per router (eBGP peering is also supported)

The MSC iBGP configuration must be added to the base router.

MSC iBGP configuration

Example: Primary CMG configuration

```
config
  router Base
    autonomous-system 1
    policy-options
      prefix-list "prefix-system"
        prefix 10.20.1.3/32 exact
      exit
      policy-statement "prefix-system"
        entry 10
          from
            prefix-list "prefix-system"
          exit
          to
            protocol bgp
          exit
          action accept
        exit
      exit
    exit
  exit
  bgp
    connect-retry 2
    export "prefix-system"
    bfd-enable
    group "system"
      neighbor "vL3SW#0"
        min-route-advertisement 2
        peer-as 1
      exit
      neighbor "vL3SW#1"
        min-route-advertisement 2
        peer-as 1
      exit
    exit
    no shutdown
  exit
exit
exit
```

Example: Secondary CMG configuration

```

config
  router Base
    autonomous-system 1
    policy-options
      prefix-list "prefix-system"
        prefix 10.20.1.1/32 exact
      exit
    policy-statement "prefix-system"
      entry 10
        from
          prefix-list "prefix-system"
        exit
        to
          protocol bgp
        exit
        action accept
        exit
      exit
    exit
  exit
  bgp
    connect-retry 2
    export "prefix-system"
    bfd-enable
    group "system"
      neighbor "vL3SW#0"
        min-route-advertisement 2
        peer-as 1
      exit
      neighbor "vL3SW#1"
        min-route-advertisement 2
        peer-as 1
      exit
    exit
    no shutdown
  exit
exit
exit

```

The MSC ICR configuration must be added and the following parameters are specified:

- source and peer IP addresses for the MCS
- the primary role must be assigned to the primary CMG and the secondary role to the secondary CMG

- the synchronization method must be set using the `mc-complete-ue-sync` command
- the `mc-redirect` command must be disabled
- the `bfd-enable` command must be configured
- any other configuration must use the default settings

**NOTE:**

In the current release, the `mc-redirect` command is not supported in LB-less mode ICR deployments.

MSC ICR configuration

Example: Primary CMG configuration

```
config
  redundancy
    multi-chassis
      peer 10.20.1.1 create
        description "Mc peer 10.20.1.1"
        source-address 10.20.1.3
        mc-mobile
          bfd-enable interface system
          mc-complete-ue-sync
          mobile-gateway 1 role primary
          no shutdown
        exit
      exit
    no shutdown
  exit
exit
exit
exit
```

Example: Secondary CMG configuration

```
config
  redundancy
    multi-chassis
      peer 10.20.1.3 create
        description "Mc peer 10.20.1.3"
        source-address 10.20.1.1
        mc-mobile
          bfd-enable interface system
```

```

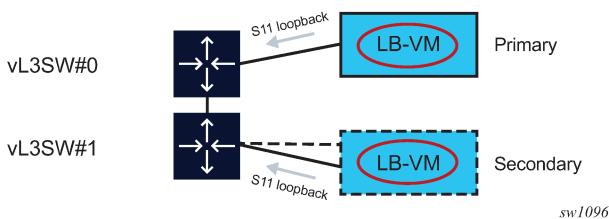
        mc-complete-ue-sync
        mobile-gateway 1 role secondary
        no shutdown
    exit
    exit
    shutdown
exit
exit
exit
exit
exit
exit

```

5.5.5.3 Routing policies configuration

The primary CMG has all the gateway signaling and data reference points configured as loopbacks with VPRNs and routing policies. The secondary CMG must be configured with the same loopback IP addresses and routing policies.

Figure 31: Routing policy configuration



Routing policy S11 interface configuration

Example: Primary CMG configuration

```

config>router>policy>
    prefix-list "S11_PdnTreePrefix"
    prefix 16.70.16.16/32 exact
    prefix 3ffe::1046:1010/128 exact
exit

```

Example: Secondary CMG configuration

```

config>router>policy>
    prefix-list "S11_PdnTreePrefix"
    prefix 16.70.16.16/32 exact
    prefix 3ffe::1046:1010/128 exact
exit

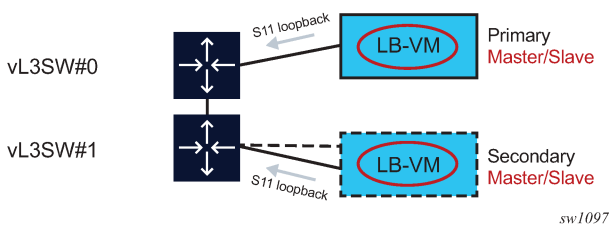
```

Both the primary and secondary CMGs must be configured with an additional policy statement for the "mobile-master" and "mobile-slave" states. The iBGP metric for each reference point prefix list for global/ residing on the LB-VM must also be configured.

**NOTE:**

eBGP using AS path prepend is also supported.

Figure 32: Policy statement configuration



Policy statement S11 interface configuration

Example: Primary CMG configuration

```
config>router>policy>
    policy-statement "S11_PdnTree"
        entry 10
            from
                prefix-list "S11_PdnTreePrefix"
                state mobile-master
            exit
            to
                protocol bgp
            exit
            action accept
                metric set 10
            exit
        exit
        entry 20
            from
                prefix-list "S11_PdnTreePrefix"
                state mobile-slave
            exit
            action accept
                metric set 30
            exit
        exit
```

```
        default-action drop
    exit
exit
```

Example: Secondary CMG configuration

```
config>router>policy>
    policy-statement "S11_PdnTree"
        entry 10
            from
                prefix-list "S11_PdnTreePrefix"
                state mobile-master
            exit
            to
                protocol bgp
            exit
            action accept
                metric set 20
            exit
        exit
        entry 20
            from
                prefix-list "S11_PdnTreePrefix"
                state mobile-slave
            exit
            action accept
                metric set 40
            exit
        exit
    default-action drop
    exit
exit
```

Both gateways must have the same VPRN network configuration for their interfaces for global/residing on the LB-VM.

VPRN S11 interface configuration

Example: Primary CMG configuration

```
configure
    service
        vprn 20071 name "20071" customer 1 create
        autonomous-system 73
```

```

route-distinguisher 20171:20171
interface "S11_PdnTreePGW1" create
    address 10.70.13.3/24
    bfd 4000 receive 4000 multiplier 3
    ipv6
        address 3ffe::a46:d03/120
    exit
    sap 1/1/4:70 create
    exit
exit
interface "S11_PdnTreeLoopback1" create
    address 16.70.16.16/32
    ipv6
        address 3ffe::1046:1010/128
    exit
    loopback
exit
exit
exit
exit

```

Example: Secondary CMG configuration

```

configure
service
    vprn 20071 name "20071" customer 1 create
    autonomous-system 73
    route-distinguisher 20071:20071
    interface "S11_PdnTreePGW1" create
        address 10.70.23.3/24
        bfd 4000 receive 4000 multiplier 3
        ipv6
            address 3ffe::a46:1703/120
        exit
        sap 2/1/3:70 create
        exit
    exit
    interface "S11_PdnTreeLoopback1" create
        address 16.70.16.16/32
        ipv6
            address 3ffe::1046:1010/128
        exit
        loopback
    exit
exit

```

```
    exit
exit
```

Both gateways must have the same BGP configuration in VPRN for their interfaces for global/residing on the LB-VM.

BGP VPRN S11 interface configuration

Example: Primary CMG configuration

```
configure
  service
    vprn 20071 name "20071" customer 1 create
      bgp
        export "S11_PdnTree"
        bfd-enable
        group "S11_PdnTreePGW1"
          neighbor "Vl3SW#0"
            min-route-advertisement 2
            peer-as 1
          exit
        neighbor "v13SW#1"
          min-route-advertisement 2
          peer-as 1
        exit
      exit
    exit
  exit
exit
```

Example: Secondary CMG configuration

```
configure
  service
    vprn 20071 name "20071" customer 1 create
      bgp
        export "S11_PdnTree"
        bfd-enable
        group "S11_PdnTreePGW1"
          neighbor "v13SW#0"
            min-route-advertisement 2
            peer-as 1
          exit
      exit
    exit
  exit
exit
```



```

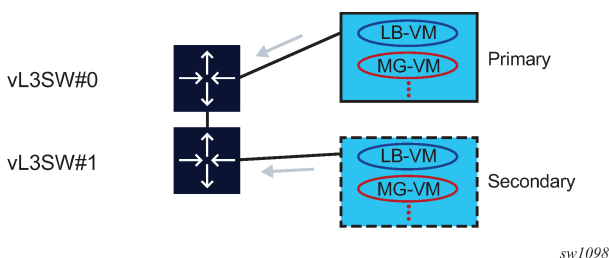
neighbor "v13SW#1"
    min-route-advertisement 2
    peer-as 1
exit
exit
exit
exit
exit
exit

```

The following configuration steps are needed for the MG-VMs:

- create a VPRN for each MG group (sub-VPRN) and one super VPRN to represent all these group-specific VPRNs
- create an interface for each port on the active and standby MG-VMs
- a new subnet is needed for each interface (this subnet can be reused for the next group's interface with a different host IP address)
- for the access side, create loopbacks for assignment to the reference points
- create IP address pools for the SGi side (each MG group requires its own separate IP address pool)
- add default static routes or advertise OSPF routes to the next hop interface of the router
- configure ECMP
- configure `mg-group` and `local-ecmp-egress` commands for each VPRN
- define the export policy to export the loopback/SGi IP address pools along with the active MG key and a community
- define an equivalent import policy for the super VPRN; this import policy covers all groups (matching the communities)
- the super VPRN has the BGP configured to the next hop routers

Figure 33: MG-VM configuration



Both gateways must have the same reference point interface configuration in the sub-VPRN (that resides on the MG-VMs).

Sub-VRPN reference point interface configuration

Example: Primary CMG configuration

```
config
  router
    policy-options
      community "RefPt1_101"
        members "target:1:101"
      exit
      community "RefPt1_101_prtct"
        members "target:1:121"
      exit
    exit
  exit
exit
```

Example: Secondary CMG configuration

```
config
  router
    policy-options
      community "RefPt1_101"
        members "target:1:101"
      exit
      community "RefPt1_101_prtct"
        members "target:1:121"
      exit
    exit
  exit
exit
```

Example: Primary CMG configuration

```
config
  router
    policy-options
      prefix-list "prefix_Grp1RefPt1"
        prefix 30.1.1.1/32 exact
        prefix 3ffe::1e01:101/128 exact
      exit
    exit
  exit
exit
```

```
exit
```

Example: Secondary CMG configuration

```
config
  router
    policy-options
      prefix-list "prefix_Grp1RefPt1"
        prefix 30.1.1.1/32 exact
        prefix 3ffe::1e01:101/128 exact
      exit
    exit
  exit
exit
```

Both the primary and secondary CMGs must be configured with an additional policy statement for the "mobile-master" and "mobile-slave" states. The iBGP metric for each reference point prefix list for global/residing on the MG-VM group must also be configured.



NOTE:

eBGP using AS path prepend is also supported.

Reference point interface configuration on MG-VM

Example: Primary CMG configuration

```
policy-statement "vprn101_export"
  entry 10
    from
      prefix-list "prefix_Grp1RefPt1"
      state mobile-master
      active-mg 3
    exit
  action accept
    community add "RefPt1_101"
    metric set 10
  exit
exit
entry 15
  from
    prefix-list "prefix_Grp1RefPt1"
    state mobile-master
```

```
        active-mg 4
    exit
    action accept
        community add "RefPt1_101_prtct"
        metric set 10
    exit
exit
entry 20
    from
        prefix-list "prefix_Grp1RefPt1"
        state mobile-slave
        active-mg 3
    exit
    action accept
        community add "RefPt1_101"
        metric set 30
    exit
exit
entry 25
    from
        prefix-list "prefix_Grp1RefPt1"
        state mobile-slave
        active-mg 4
    exit
    action accept
        community add "RefPt1_101_prtct"
        metric set 30
    exit
exit
exit
```

Example: Secondary CMG configuration

```
policy-statement "vprn101_export"
    entry 10
        from
            prefix-list "prefix_Grp1RefPt1"
            state mobile-master
            active-mg 3
        exit
        action accept
            community add "RefPt1_101"
            metric set 20
        exit
    exit
```

```

    entry 15
    from
        prefix-list "prefix_Grp1RefPt1"
        state mobile-master
        active-mg 4
    exit
    action accept
        community add "RefPt1_101_prtct"
        metric set 20
    exit
exit
entry 20
from
    prefix-list "prefix_Grp1RefPt1"
    state mobile-slave
    active-mg 3
exit
action accept
    community add "RefPt1_101"
    metric set 40
exit
exit
entry 25
from
    prefix-list "prefix_Grp1RefPt1"
    state mobile-slave
    active-mg 4
exit
action accept
    community add "RefPt1_101_prtct"
    metric set 40
exit
exit
exit

```

For MCS to track the Gi/SGi interfaces, the track-mobile command must be enabled for each Gi/SGi interface under the config>service>vprn>interface context. All other traffic interfaces are automatically tracked. Both the primary and secondary CMGs must have the same configuration.

Example: Primary CMG configuration

```

vprn 101 name "101" customer 1 create
    description "ThisIsTheSubVPRNForRefPt1InGrp1"
    vrf-export "vprn101_export"
    ecmp 5

```

```
autonomous-system 1
route-distinguisher 101:1
mg-group 1 local-ecmp-egress
interface "Grp1_RefPt1_loop1" create
    description "Loopback1ForRefPt1"
    address 30.1.1.1/32
    ipv6
        address 3ffe::1e01:101/128
    exit
    loopback
exit
interface "Grp1_RefPt1_BFD_Lo1" create
    description "BFDLoopback1ForRefPt1"
    address 40.40.1.1/24
    ipv6
        address 3ffe::2828:101/120
    exit
    loopback
exit
interface "Grp1_RefPt1_BFD_Lo2" create
    description "BFDLoopback2ForRefPt1"
    address 40.40.2.1/24
    ipv6
        address 3ffe::2828:201/120
    exit
    loopback
exit
```

Example: Secondary CMG configuration

```
vprn 101 name "101" customer 1 create
    description "ThisIsTheSubVPRNForRefPt1InGrp1"
    vrf-export "vprn101_export"
    ecmp 5
    autonomous-system 1
    route-distinguisher 101:1
    mg-group 1 local-ecmp-egress
    interface "Grp1_RefPt1_loop1" create
        description "Loopback1ForRefPt1"
        address 30.1.1.1/32
        ipv6
            address 3ffe::1e01:101/128
        exit
        loopback
    exit
```

```

interface "Grp1_RefPt1_BFD_Lo1" create
    description "BFDLoopback1ForRefPt1"
    address 40.40.1.51/24
    ipv6
        address 3ffe::2828:133/120
    exit
    loopback
exit
interface "Grp1_RefPt1_BFD_Lo2" create
    description "BFDLoopback2ForRefPt1"
    address 40.40.2.51/24
    ipv6
        address 3ffe::2828:233/120
    exit
    loopback
exit

```

Reference point interface configuration in sub-VRN (MG-VM)

Example: Primary CMG configuration

```

interface "Grp1_wrk_RefPt1_port1" create
    description "phyIntfOnWrkingCardGrp1Port1ForRefPt1"
    address 40.1.1.1/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:101/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 3/3/1:101 create
    exit
exit
interface "Grp1_wrk_RefPt1_port2" create
    description "phyIntfOnWrkingCardGrp1Port2ForRefPt1"
    address 40.1.2.1/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:201/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 3/3/2:102 create
    exit
exit
interface "Grp1_prt_RefPt1_port1" create

```

```

        description "phyIntfOnProtctCardGrp1Port1ForRefPt1"
        address 40.1.3.1/24
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
        ipv6
            address 3ffe::2801:301/120
            bfd 1000 receive 1000 multiplier 3 echo-receive 150
        exit
        sap 4/3/1:103 create
        exit
    exit
    interface "Grp1_prt_RefPt1_port2" create
        description "phyIntfOnProtctCardGrp1Port2ForRefPt1"
        address 40.1.4.1/24
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
        ipv6
            address 3ffe::2801:401/120
            bfd 1000 receive 1000 multiplier 3 echo-receive 150
        exit
        sap 4/3/2:104 create
        exit
    exit

```

Example: Secondary CMG configuration

```

interface "Grp1_wrk_RefPt1_port1" create
    description "phyIntfOnWrkingCardGrp1Port1ForRefPt1"
    address 40.1.1.51/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:133/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 3/3/1:101 create
    exit
exit
interface "Grp1_wrk_RefPt1_port2" create
    description "phyIntfOnWrkingCardGrp1Port2ForRefPt1"
    address 40.1.2.51/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:233/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 3/3/2:102 create
    exit

```



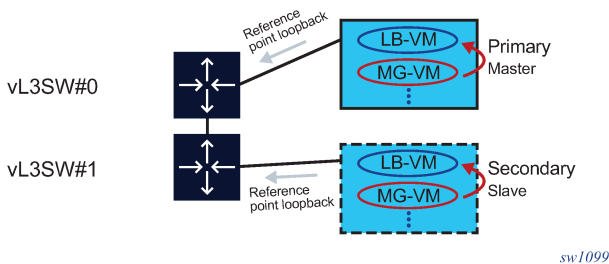
```

exit
interface "Grp1_prt_RefPt1_port1" create
  description "phyIntfOnProtctCardGrp1Port1ForRefPt1"
  address 40.1.3.51/24
  bfd 1000 receive 1000 multiplier 3 echo-receive 150
  ipv6
    address 3ffe::2801:333/120
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
  exit
  sap 4/3/1:103 create
  exit
exit
interface "Grp1_prt_RefPt1_port2" create
  description "phyIntfOnProtctCardGrp1Port2ForRefPt1"
  address 40.1.4.51/24
  bfd 1000 receive 1000 multiplier 3 echo-receive 150
  ipv6
    address 3ffe::2801:433/120
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
  exit
  sap 4/3/2:104 create
  exit
exit

```

BGP is not supported in the sub-VRPNs on the MG-VMs. Therefore, the local routes are exported to the super VRPN via an export policy, for example `vrpn101_export`. This configuration must be the same in both the primary and secondary CMGs.

Figure 34: Export policy configuration



Example: Primary CMG configuration

```

config
  service
    vrpn 101
      vrf-export "vrpn101_export"
    exit

```

```
exit
exit
```

Example: Secondary CMG configuration

```
config
  service
    vprn 101
      vrf-export "vprn101_export"
    exit
  exit
exit
```

The VRF import policy configuration for the super VPRN must be the same in both CMGs.

Example: Primary CMG configuration

```
config>router>policy>
  policy-statement "vprn10101_import"
    entry 10
      from
        community expression "[RefPt1_101] OR
[RefPt1_101_prtct]"
      exit
      action accept
      exit
    exit
  exit
```

Example: Secondary CMG configuration

```
config>router>policy>
  policy-statement "vprn10101_import"
    entry 10
      from
        community expression "[RefPt1_101] OR
[RefPt1_101_prtct]"
      exit
      action accept
      exit
    exit
  exit
```

The BGP export policy configuration for the super VPRN must be added to both CMGs.

Example: Primary CMG configuration

```
config>router>policy>
  policy-statement "exportVprn10101Bgp"
    entry 10
      from
        community "RefPt1_101"
        family ipv4
      exit
      action accept
        next-hop 40.40.1.1
      exit
    exit
  entry 15
    from
      community "RefPt1_101"
      family ipv6
    exit
    action accept
      next-hop 3ffe::2828:101
    exit
  exit
  entry 25
    from
      community "RefPt1_101 prtct"
      family ipv4
    exit
    action accept
      next-hop 40.40.2.1
    exit
  exit
  entry 30
    from
      community "RefPt1_101 prtct"
      family ipv6
    exit
    action accept
      next-hop 3ffe::2828:201
    exit
  exit
exit
```

Example: Secondary CMG configuration

```
config>router>policy>
```

```

policy-statement "exportVprn10101Bgp"
  entry 10
    from
      community "RefPt1_101"
      family ipv4
    exit
    action accept
      next-hop 40.40.1.51
    exit
  exit
  entry 15
    from
      community "RefPt1_101"
      family ipv6
    exit
    action accept
      next-hop 3ffe::2828:133
    exit
  exit
  entry 25
    from
      community "RefPt1_101 prtct"
      family ipv4
    exit
    action accept
      next-hop 40.40.2.51
    exit
  exit
  entry 30
    from
      community "RefPt1_101 prtct"
      family ipv6
    exit
    action accept
      next-hop 3ffe::2828:233
    exit
  exit
exit

```

The reference point interface configuration for the super VPRN must be available in both CMGs.

Example: Primary CMG configuration

```

config>service>
  vprn 10101 name "10101" customer 1 create

```

```
description "This_is_the_super_VPRN_for_RefPt1"
vrf-import "vprn10101_import"
autonomous-system 1
route-distinguisher 10101:1
interface "bgp_wrk_RefPt1_port1" create
    description "BGPIntfForRefPt1wrkport1"
    address 40.1.1.253/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:1fd/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 2/1/2:101 create
    exit
exit
interface "bgp_wrk_RefPt1_port2" create
    description "BGPIntfForRefPt1wrkport2"
    address 40.1.2.253/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:2fd/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 2/1/3:102 create
    exit
exit
interface "bgp_prt_RefPt1_port1" create
    description "BGPIntfForRefPt1prtport1"
    address 40.1.3.253/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:3fd/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 2/1/2:103 create
    exit
exit
interface "bgp_prt_RefPt1_port2" create
    description "BGPIntfForRefPt1prtport2"
    address 40.1.4.253/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:4fd/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
```

```
sap 2/1/3:104 create
exit
exit
```

Example: Secondary CMG configuration

```
config>service>
  vprn 10101 name "10101" customer 1 create
  description "This_is_the_super_VPRN_for_RefPt1"
  vrf-import "vprn10101_import"
  autonomous-system 1
  route-distinguisher 10101:1
  interface "bgp_wrk_RefPt1_port1" create
    description "BGPIntfForRefPt1wrkport1"
    address 40.1.1.252/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
      address 3ffe::2801:1fc/120
      bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 2/1/3:101 create
    exit
  exit
  interface "bgp_wrk_RefPt1_port2" create
    description "BGPIntfForRefPt1wrkport2"
    address 40.1.2.252/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
      address 3ffe::2801:2fc/120
      bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 2/1/6:102 create
    exit
  exit
  interface "bgp_prt_RefPt1_port1" create
    description "BGPIntfForRefPt1prtport1"
    address 40.1.3.252/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
      address 3ffe::2801:3fc/120
      bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 2/1/3:103 create
    exit
  exit
```

```

interface "bgp_prt_RefPt1_port2" create
    description "BGPIntfForRefPt1prtport2"
    address 40.1.4.252/24
    bfd 1000 receive 1000 multiplier 3 echo-receive 150
    ipv6
        address 3ffe::2801:4fc/120
        bfd 1000 receive 1000 multiplier 3 echo-receive 150
    exit
    sap 2/1/6:104 create
    exit
exit

```

The BGP configuration in the super VPRN must be available in both CMGs.

Example: Primary CMG configuration

```

#cont...
    bgp
        family ipv4 ipv6
        min-route-advertisement 1
        bfd-enable
        group "bgpToRtr"
            export "exportVprn10101Bgp"
            peer-as 1
            neighbor 40.1.1.254
                local-address 40.1.1.253
                local-as 1
            exit
            neighbor 40.1.2.254
                local-address 40.1.2.253
                local-as 1
            exit
            neighbor 40.1.3.254
                local-address 40.1.3.253
                local-as 1
            exit
            neighbor 40.1.4.254
                local-address 40.1.4.253
                local-as 1
            Exit
        exit
        no shutdown
    exit

```

Example: Secondary CMG configuration

```
#cont...
    bgp
    family ipv4 ipv6
    min-route-advertisement 1
    bfd-enable
    group "bgpToRtr"
        export "exportVprn10101Bgp"
        peer-as 1
        neighbor 40.1.1.254
            local-address 40.1.1.252
            local-as 1
        exit
        neighbor 40.1.2.254
            local-address 40.1.2.252
            local-as 1
        exit
        neighbor 40.1.3.254
            local-address 40.1.3.252
            local-as 1
        exit
        neighbor 40.1.4.254
            local-address 40.1.4.252
            local-as 1
        exit
    exit
    no shutdown
exit
```

The S11/S1U reference points configuration must be available in both CMGs.

Example: Primary CMG configuration

```
config>mobile>pdn
    s11 "default"
        interface
            gtp-c router "vprn20071" "S11_PdnTreeLoopback1" s1-u
        "default"
            gtp-u router "vprn20071" "S11_PdnTreeLoopback1"
        exit
        signaling
            gtp-c
            exit
        exit
```



```

exit
s11 "redir_s11_Grp1" group-specific
    interface
        gtp-c router "vprn101" "Grp1_RefPt1_loop1" s1-u "default"
        gtp-u router "vprn101" "Grp1_RefPt1_loop1"
    exit
exit
s1-u "default"
    interface
        gtp-u router "vprn20071" "S11_PdnTreeLoopback1"
    exit
    signaling
        gtp-u
            profile "S11_PdnTreeLoopback1Prof"
        exit
    exit
exit
s1-u "redir_s1u_Grp1" group-specific
    interface
        gtp-u router "vprn101" "Grp1_RefPt1_loop1"
    exit
exit

```

Example: Secondary CMG configuration

```

config>mobile>pdn
    s11 "default"
        interface
            gtp-c router "vprn20071" "S11_PdnTreeLoopback1" s1-u
"default"
            gtp-u router "vprn20071" "S11_PdnTreeLoopback1"
        exit
        signaling
            gtp-c
            exit
        exit
    exit
    s11 "redir_s11_Grp1" group-specific
        interface
            gtp-c router "vprn101" "Grp1_RefPt1_loop1" s1-u "default"
            gtp-u router "vprn101" "Grp1_RefPt1_loop1"
        exit
    exit
    s1-u "default"
        interface

```

```

        gtp-u router "vprn20071" "S11_PdnTreeLoopback1"
    exit
    signaling
        gtp-u
            profile "S11_PdnTreeLoopback1Prof"
        exit
    exit
exit
s1-u "redir_s1u_Grp1" group-specific
    interface
        gtp-u router "vprn101" "Grp1_RefPt1_loop1"
    exit
exit

```

The override profile configuration for the MG Group (under the config>mobile>pdn context) must be available in both CMGs.

Example: Primary CMG configuration

```

config>mobile>pdn
    override-profile 1
        description "Def_Prof for s11 For Grp 1"
        override ref-point-type s11 "default" with "redir_s11_Grp1"
        override ref-point-type s1u "default" with "redir_s1u_Grp1"
    exit

```

Example: Secondary CMG configuration

```

config>mobile>pdn
    override-profile 1
        description "Def_Prof for s11 For Grp 1"
        override ref-point-type s11 "default" with "redir_s11_Grp1"
        override ref-point-type s1u "default" with "redir_s1u_Grp1"
    exit

```

The override profile configuration of the MG group must be applied to both CMGs using the group command under the config>mobile>system context.

Example: Primary CMG configuration

```

config>mobile>
    system
        group 1 redundancy one-to-one gateway 1
            override-profile 1
            card 3 working

```

```

        card 4 protect
        no shutdown
    exit
exit

```

Example: Secondary CMG configuration

```

config>mobile>
  system
    group 1 redundancy one-to-one gateway 1
    override-profile 1
    card 3 working
    card 4 protect
    no shutdown
  exit
exit

```

5.6 Multiple access PDN connectivity

Multiple access PDN connectivity (MAPCON), as defined in 3GPP Release 10, is supported. With MAPCON, a UE can simultaneously request multiple PDN connections through different access networks. A typical use case is a UE selecting 3GPP access for one PDN connection and WiFi access for another.

Figure 35: Multiple PDN connections using a single access network in combined SGW/PGW (no MAPCON)

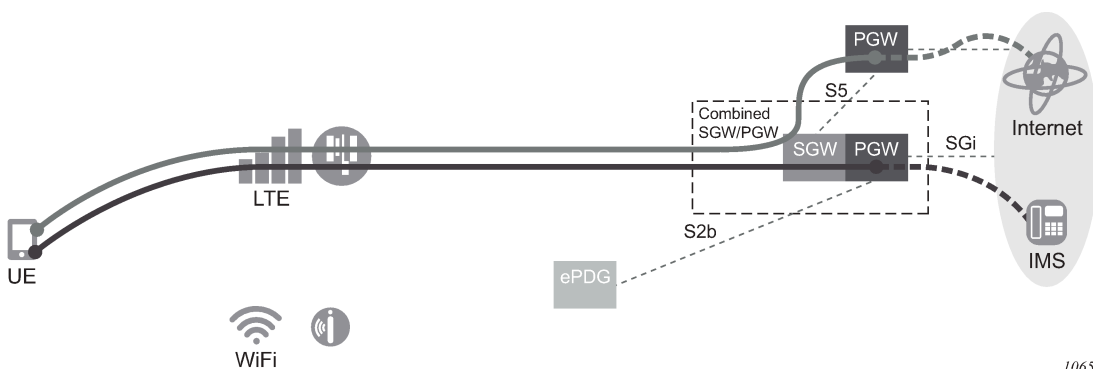
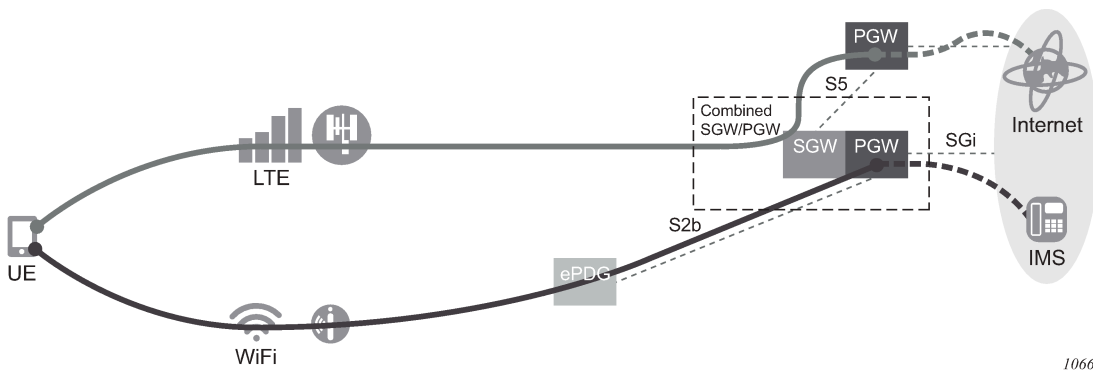


Figure 36: Multiple PDN connections using different access networks in combined SGW/PGW (MAPCON)

**NOTE:**

The MAPCON functionality is also supported in PGW and SGW functions.

To check whether a UE is using MAPCON, use the following show CLI PDN commands and note the reference point associated with each PDN connection in the detailed output:

```
show>mobile>pdn>pdn-context
```

```
show>mobile>pdn>pdn-context>imsi>detail
```

5.7 Enable multiple PDNs per APN

The option to enable multiple PDN sessions per APN for the gateway is supported. Multiple PDN sessions, independent of PDN type, can be set up per APN from each subscriber. The method used to identify unique PDN sessions is based on the EBI value. PDN sessions are identified by a unique pair of EBI and APN, which enables multiple PDN sessions of each PDN type to be established from each UE. The `config>mobile>pdn>allow-multiple-apn-pdn` CLI command enables multiple PDN sessions per APN.

Table 6: PGW/GGSN (S5/S8/Gn/Gp) behavior

Allow-multiple-PDN feature enabled	Existing PDN type	Second requested PDN type	Behavior
No	IPv4	IPv4	Reattach using second PDN
No	IPv4	IPv6	Reattach using second PDN

Table 6: PGW/GGSN (S5/S8/Gn/Gp) behavior (continued)

Allow-multiple-PDN feature enabled	Existing PDN type	Second requested PDN type	Behavior
No	IPv6	IPv4	Reattach using second PDN
No	IPv6	IPv6	Reattach using second PDN
No	IPv4v6	IPv4	Reattach using second PDN
No	IPv4v6	IPv6	Reattach using second PDN
No	IPv4v6	IPv4v6	Reattach using second PDN
Yes	IPv4	IPv4	Maintain both PDN
Yes	IPv4	IPv6	Maintain both PDN
Yes	IPv6	IPv4	Maintain both PDN
Yes	IPv6	IPv6	Maintain both PDN
Yes	IPv4v6	IPv4	Maintain both PDN
Yes	IPv4v6	IPv6	Maintain both PDN
Yes	IPv4v6	IPv4v6	Maintain both PDN

Table 7: SGW (S11) behavior

Allow-multiple-PDN feature enabled	Existing PDN type	Second requested PDN type	Behavior
No	IPv4	IPv4	Tear down UE
No	IPv4	IPv6	Tear down UE

Table 7: SGW (S11) behavior (continued)

Allow-multiple-PDN feature enabled	Existing PDN type	Second requested PDN type	Behavior
No	IPv6	IPv4	Tear down UE
No	IPv6	IPv6	Tear down UE
No	IPv4v6	IPv4	Tear down UE
No	IPv4v6	IPv6	Tear down UE
No	IPv4v6	IPv4v6	Tear down UE
Yes	IPv4	IPv4	Maintain both PDN
Yes	IPv4	IPv6	Maintain both PDN
Yes	IPv6	IPv4	Maintain both PDN
Yes	IPv6	IPv6	Maintain both PDN
Yes	IPv4v6	IPv4	Maintain both PDN
Yes	IPv4v6	IPv6	Maintain both PDN
Yes	IPv4v6	IPv4v6	Maintain both PDN

Related information[APN configuration](#)

5.8 Ethernet PDU session configuration

The configuration of the Ethernet PDU session type provides a PDU connectivity service that exchanges PDUs between the UE and the DN as L2 Ethernet frames to enable the support of industry IoT applications and FWA use cases using 5GS (5G core and 5G NR) as the underlying transport. A PDU session type in 5G can be IP (allows the transfer of L3 IP packets), Ethernet (allows the transfer of L2 Ethernet frames), or Unstructured.

The following limitations apply for the configuration of the Ethernet PDU session type.

- The UPF must be deployed on hardware level with Mellanox NICs.
- The UPF must be deployed as MG+ (LB-less).
- The spoof checking on the VF must be set to off at compute UPF node level.
- The trust checking on the VF must be set to on at compute UPF node level.

To configure a static route for internal forwarding in the VPRN of the N6 interface of the UPF, use the following command:

```
configure service vprn service-id static-route-entry 0.0.0.0/0 next-hop ip-int-name | ipv4-address | ipv6-address no shutdown
```

This affects both Ethernet and non-Ethernet traffic so that a separate VPRN must be used for the Ethernet traffic.

To enable the DL Ethernet frames detection and session matching on the UPF (based on C-VLAN and S-VLAN tagging) on a specific port, use the following command:

```
configure port ethernet eth-pdu-port
```

To configure the Ethernet PDN type on the SMF, use the following command:

```
configure mobile-gateway pdn apn pdn-type ethernet
```



NOTE:

If a virtual APN is used, the same configuration applies to the virtual APN.

To configure the Ethernet action and tag values on the UL frames, use the following command:

```
configure mobile-gateway pdn apn ethernet-action-up action action-type tag-value tag-value
```

To configure the Ethernet action and tag values on the DL frames, use the following command:

```
configure mobile-gateway pdn apn ethernet-action-dl action action-type [n3-c-vlan v-lan]
```

The supported Ethernet action and/or tag value combinations for the UL and DL frames are:

Table 8: Supported Ethernet action - tag value combinations

UL frames	DL frames
push tag-value n7-c-vlan	pop
push tag-value n7-s-vlan	pop
push-push tag-value n7-c-vlan-s-vlan	pop-pop

or:

Table 9: Supported Ethernet action - tag value combinations

UL frames	DL frames
swap tag-value n7-c-vlan	swap n3-c-vlan <i>vlan</i>
swap-push tag-value n7-c-vlan-s-vlan	swap-pop n3-c-vlan <i>vlan</i>

The Ethernet action and tag values must be configured under the APN on the SMF. If the swap and, or swap-pop actions are used, they must be configured on the UPF side and the default configuration must be used on the SMF side.

**NOTE:**

Only call-insight for internal traffic (MSM) (and not for ingress and egress) is supported.

5.9 APN configuration

This section provides information about Access Point Name (APN) configuration.

**NOTE:**

APN names containing uppercase and lowercase letters may be configured or received from external systems, but the system always handles them in a case-insensitive manner, as per 3GPP TS 23.003.

Related information

Policy configuration

Information and configuration guidelines for policy control implementation for EPC nodes.

5.9.1 APN configuration example

The APN consists of the following two parts:

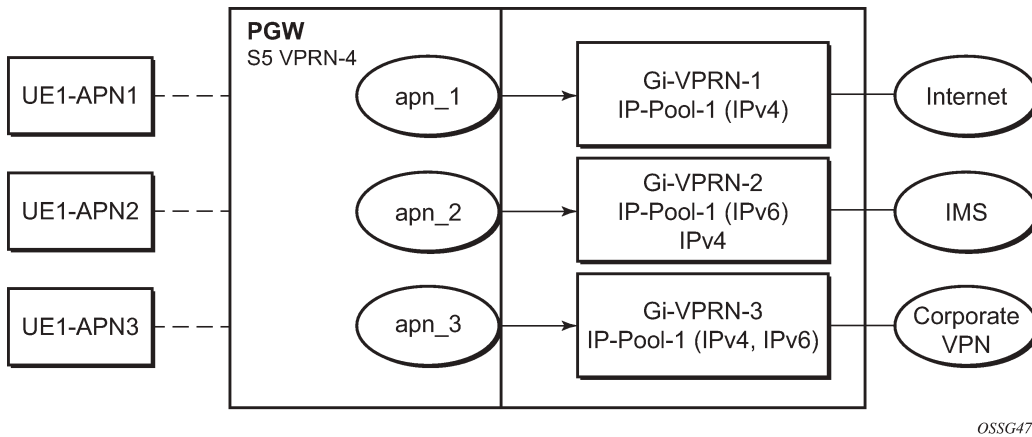
- APN network identifier (APN-NI); contains one or more labels
- APN operator identifier (APN-OI); uses the `mnc<MNC>.mcc<MCC>.gprs` format

The full APN includes both the APN-NI and APN-OI where the APN-OI is placed after APN-NI. The APN configuration can use both the full APN and APN-NI as APN name. If the PDP context is activated using GTPv1, the Create PDP Context request contains only the APN-NI, but CMG seeks to find a full APN with a matching APN-NI. If there are multiple matching full APNs, the first five or six digits of the IMSI are used to construct the APN-OI.

Each APN represents a service provided by an upstream service provider. APNs can be classified, for example, as consumer APNs such as Internet or IMS, and corporate APNs (using L2TP, IPSec, GRE-IP VPNs).

APN configuration example shows an APN configuration example with three APNs, each having own VPRN instance and IP pools in the PGW.

Figure 37: APN configuration example



NOTE:

This configuration is simplified to provide an overview. Refer to the following sections for more information about the various configuration options available under the APN.

```
#-----
echo "Mobile Gateway Configuration"
#-----
```

```
mobile-gateway
  pdn 1
    apn "apn_1"
    pco-option
      dns-server-ipv4
        primary 10.30.1.1
        secondary 10.30.1.2
      exit
    exit
    pdn-type ipv4
    router "Gi-VRN-1"
      ip-local-pool "IP-Pool-1"
        ipv4-prefix "1.2.3.0/24"
        exclusive
      exit
    exit
    selection-mode subscribed-apn
    no shutdown
  exit
  apn "apn_2"
    pco-option
      dns-server-ipv4
        primary 10.40.1.1
        secondary 10.40.1.2
      exit
    exit
    pdn-type ipv6
    router "Gi-VRN-2"
      ip-local-pool "IP-Pool-1"
        ipv6-prefix "ABCD::/48"
      exit
    exit
    selection-mode subscribed-apn
    no shutdown
  exit
  apn "apn_3"
    pco-option
      dns-server-ipv4
        primary 10.50.1.1
        secondary 10.50.1.2
      exit
    dns-server-ipv6
      primary "3FFE::B73:1801/64"
      secondary "3ffe::1f01:102"
    exit
```

```

exit
pdn-type ipv4 ipv6 ipv4v6
router "Gi-VRN-3"
    ip-local-pool "IP-Pool-1"
        ipv4-prefix "10.10.1.0/24"
        ipv6-prefix "4ffe:0:1::/48"
    exit
exit
no shutdown
exit

```

To view the APN configuration, use the following command:

```
show>mobile>pdn>apn
```

The output of this command includes the explicitly configured values for the output fields and the default values. This command can also be issued by users who do not have configuration context access.

To view APN-specific configuration, use one of the following commands:

```

show>mobile>pdn>apn>accounting
show>mobile>pdn>apn>authentication
show>mobile>pdn>apn>charging
show>mobile>pdn>apn>ip-alloc
show>mobile>pdn>apn>l2tp
show>mobile>pdn>apn>policy
show>mobile>pdn>apn>virtual-apn

```

5.9.2 APN override using virtual APNs

Virtual APNs allow the PGW/GGSN operator to override the APN specified by the UE, serving node, or HLR/HSS (referred to as the real APN) to consolidate or expand APNs.

When a UE attaches to the EPC packet core, it may provide an APN that indicates the network to which it desires to attach. Alternatively, the MME or SGSN may assign a different APN learned from the HSS during the authentication process. The APN is used by the MME or SGSN to determine the SGW/PGW/GGSN through which the UE is to be attached to the network. Ultimately, the attach request arrives at the PGW/GGSN in the form of a GTP Create Request message that contains an APN. The request may determine the VRN to which the UE's PDN connection should be bound and/or several configuration parameters (IP addressing, charging, DNS servers, and so on) that apply to the PDN connection.

5.9.2.1 APN override selection options

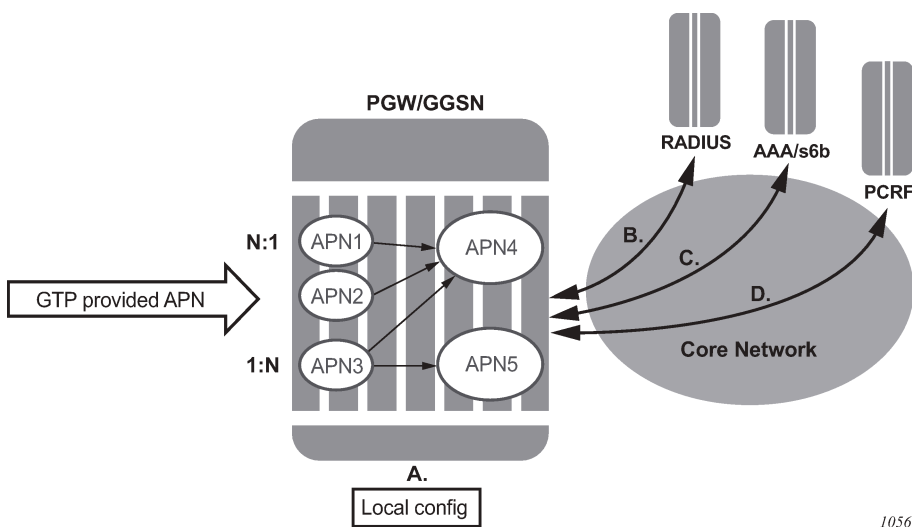
Using the virtual APN feature, you can override the APN that arrives in the GTP create message and assign the UE to an APN of choice.

The mapping can be done using the following options:

- Performed via specific RADIUS or Diameter pre-authentication messaging or S6b.
- Configured locally on the PGW/GGSN based on IMSI, MSISDN, IMEI, IMEISV TAC and SVN, roaming status, or charging characteristics (CC). The local configuration provides options to specify a static APN change for N:1 relation or to select the APN based on subscriber information (such as IMSI) 1:N relation.

There are also options to configure parameters on a per-interface basis to use either the real or virtual APN in the messages toward the peer network elements. Alternatively, it is possible to define a separate public APN string that is used toward external peers.

Figure 38: Virtual APN selection alternatives



It is possible to enable multiple virtual APN selection methods simultaneously for an APN. However, the real APN can only be overridden once. The following list provides the evaluation precedence if multiple methods are configured within the same APN.

1. Virtual APN (static 1:1 mapping), if configured, is always used first and any other virtual APN configurations are ignored.
2. Virtual APN list is evaluated second.
3. PCO-based override is evaluated third.
4. Virtual APN selection rules are evaluated fourth. If the rules match, then the virtual APN selection ends. If multiple rules match, the selection priority defines which match will be used.
5. Selection by pre-authentication, that is by external systems, is evaluated last.

In this example, the user ranges defined for the `virtual-apn-selection` command are evaluated first. If there is a match, the virtual APN is selected accordingly. If no rules match, the pre-auth command will

be executed. The pre-auth command can either return a new virtual APN or the same real APN value. In either case, the APN to be used must have all the mandatory configuration in place.

```
config>mobile>pdn>apn>
  virtual-apn-selection
    user-range "IMSIrange" type imsi 100000000000000 virtual-apn postpaid
    user-range "IMSIrange" type imsi 200000000000000 virtual-apn postpaid
    user-range "IMSIrange" type imsi 300000000000000 virtual-apn postpaid-m2m
  exit
  pre-auth type radius server-group "radius-servers" user-name msisdn
```

In this example, both the roaming-status and the cc parameters are used for the selection priority, with the former having priority. All roamers will use the real APN (if properly configured), visitors will match to the virtual APN AAA, and for home subscribers the APN is selected based on charging characteristics (cc) values.

```
config>mobile>pdn>apn>
  virtual-apn-selection
    selection-priority roaming-status cc imsi msisdn imei imeisv-tacsvn
    user-range roamer type roaming-status roaming virtual-apn xxx
    user-range visitor type roaming-status visiting virtual-apn aaa
    user-range home-1 type cc start-cc 256 virtual-apn bbb-1
    user-range home-2 type cc start-cc 512 end-cc 3840 virtual-apn bbb-2
    ...
    user-range home-n type cc start-cc 5860 end-cc n virtual-apn bbb-n
```

5.9.2.2 Virtual APN selection at the PDN/APN level

If you have multiple user ranges (for example, IMSI or MSISDN) that are applicable to several APNs, it is possible to use the virtual-apn-selection command at the PDN level to avoid configuring and maintaining each user range separately under each APN.

The virtual APN selection evaluation can be configured. By default, the virtual APN selection evaluation is made at the APN level if the virtual-apn-selection command is configured in the APN context (real APN in the Create Session Request). If not configured, the virtual APN selection evaluation is made at the PDN level. Virtual APN selection at the PDN level is evaluated first if the prefer-pdn command is enabled.

To prioritize the virtual APN selection evaluation at the PDN level, use the following command:

```
config>mobile>pdn>apn>virtual-apn-selection>prefer-pdn
```

This command only affects the real APN and is ignored for an APN used as a virtual APN for the new session. Only the append-string parameter is allowed for this purpose at the virtual-apn-selection command at the PDN level. Other parameters, such as pre-auth-type and virtual-apn do not support another APN override.

When the prefer-pdn command is enabled, evaluation of both PDN and APN level virtual APN selection is possible, and the PDN level evaluation is made first. There may be another APN level evaluation, if it is

configured for the selected APN. The selected APN can be a real APN, if no match at the PDN level is found, or a virtual APN if the `append-string` parameter is specified on the real APN upon the PDN level virtual APN selection.

Multiple match criteria can be used for the `virtual-apn-selection` command, and two criteria can be configured and evaluated at the same time. The criteria must have either one of the types **imsi**, **imei-tac**, or **msisdn** combined with **roaming-status** or **cc**.

The `selection-prio` command specifies the selected match in case of multiple matches. If more than one user range is defined that matches the same UE, the user range which has the highest number of matching criteria is selected. Prioritization of matching criteria is only needed when identical criteria are configured under the `config>mobile>pdn>virtual-apn-selection` and `config>mobile>pdn>apn>virtual-apn-selection` contexts.

If no match is found when evaluating the user ranges, a new virtual APN can be defined.

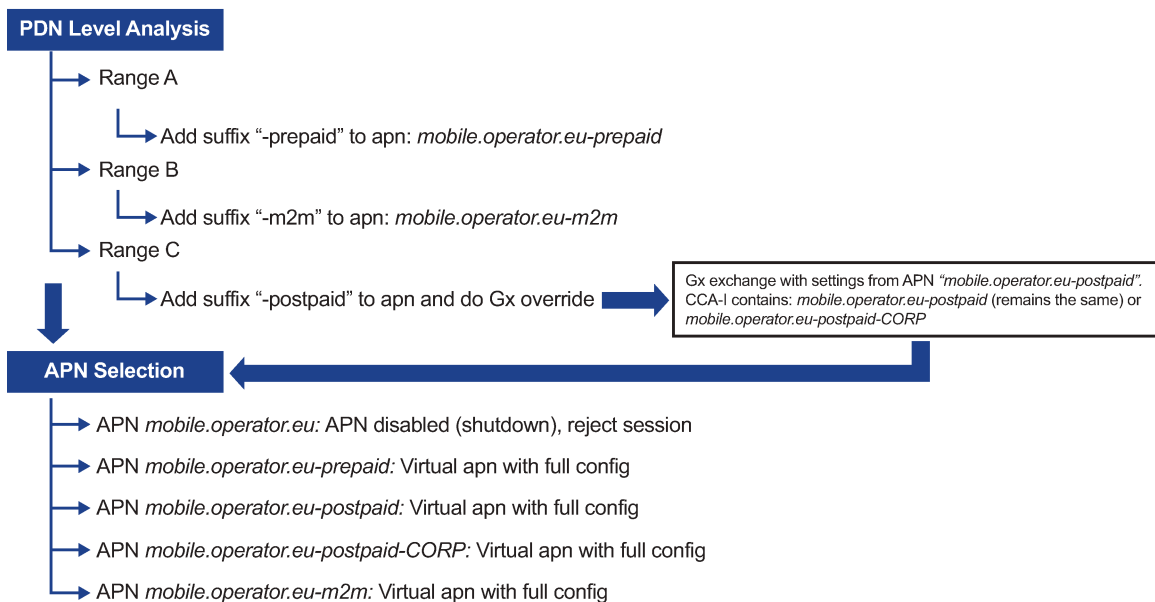
The `show>mobile>pdn>pdn-context>detail` command can be used to see the real APN, the virtual APN (if applicable), and the method that was used to select the virtual APN (if applicable).

Virtual APN selection and suffixing at the PDN level shows an example of virtual APN selection, where the virtual APN is always displayed here with lowercase letters, and suffixing at the PDN level.

The virtual APNs also have their own statistics. If a PDN connection is remapped to a virtual APN, the statistics for the real APN are not incremented.

Figure 39: Virtual APN selection and suffixing at the PDN level

-> Requested (real) APN: *mobile.operator.eu*



sw0012

5.9.2.3 Unknown APN

You can specify a special unknown APN. The unknown APN eases the configuration so that if an incoming APN value is not explicitly configured, the UE is associated with the unknown APN.

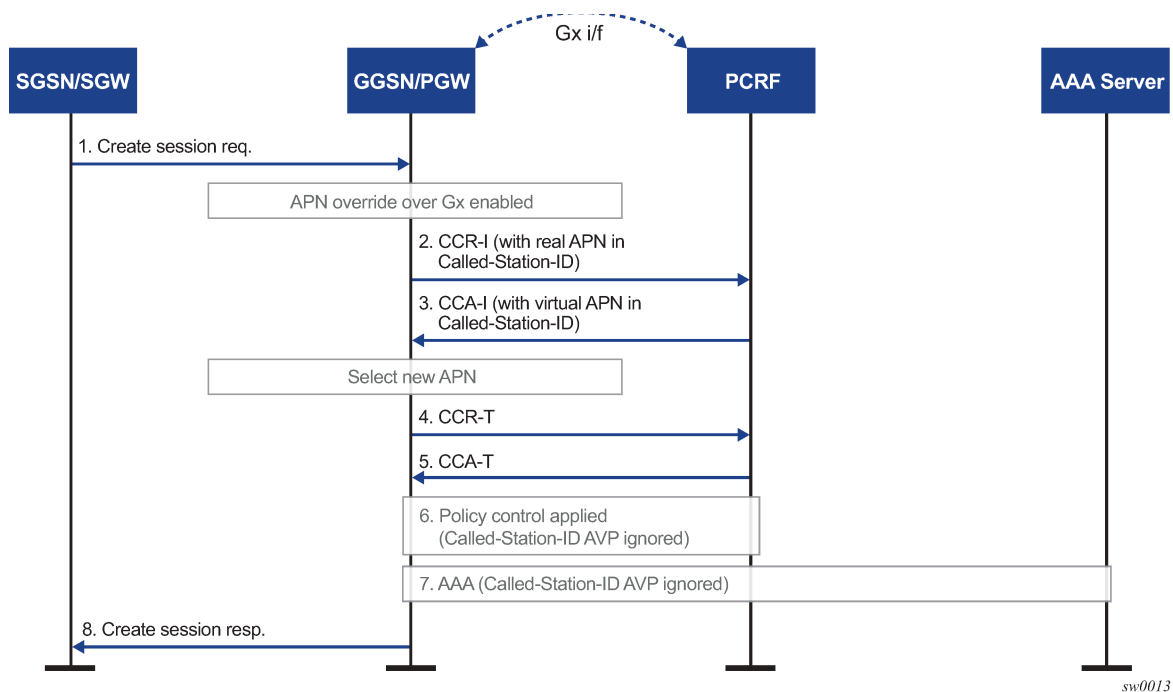
If unknown APN support is not enabled, each APN value that the operator needs to support must be fully configured. The virtual APN Selection field in the output of the `show>mobile>pd>pdn-context>detail` command shows the unknown APN as default.

5.9.2.4 Configuring APN override over Gx

Before you start

The PGW/GGSN supports a pre-authentication signaling phase prior to the actual APN selection to allow the PCRF to change the APN.

Figure 40: Gx signaling for virtual APN selection



The procedure for the APN override is as follows:

1. The SGSN/SGW/MME sends a Create PDP Context Request/Create Session Request message. The PGW checks if the APN override configurations exist.
2. If the Gx interface is enabled, the PGW sends a CCR-I message to the PCRF. At this stage there is no IP allocation, and therefore, the CCR-I message does not contain any of Framed-IP-Address or Framed-IPv6-Prefix AVPs.
3. The PCRF responds to the CCR message with a CCA-I message with the optional Called-Station-ID AVP included. This AVP has the value of the new (overriding) APN.
4. The PGW sends a CCR-T to the PCRF to close the existing PCRF session.
5. The PCRF responds to the CCR-T message with a CCA-T message.

6. The normal sequence follows. If the new APN is properly configured to use Gx, a new CCR-I, triggered by the new APN to create the new Gx session, is sent to the PCRF.
7. If the new APN contains configuration for AAA, the PGW will send an Access-Request message to the AAA server.
8. Session creation is completed.

A CCR-I will be sent solely for the purpose of requesting a new APN from the PCRF server. This gives the PCRF the flexibility to choose the APN based on more than just the incoming real APN. Only the Called-Station-Id retrieved from the CCA-I will be used for the new APN matching. The other attributes of the CCA-I message will be ignored and the session will be terminated because it is not required for anything else.

If the optional Called-Station-Id attribute is missing from the CCA-I, the create session will continue, if the real APN has sufficient configuration in place. If the real APN does not have sufficient configuration, the create session will be rejected and the Gx session will also be terminated.

**NOTE:**

The Called-Station-Id attribute is not contained in standard CCA messages, so the PCRF server implementation must be modified to support it (in the CCA-I).

The Called-Station-Id for Gx override must be valid, containing either the APN-NI value or the full APN value (APN-OI and APN-NI), as specified in section 9.1 of 3GPP 23.003. The PGW does not support a partial name for prefixing or suffixing instead of the original one from the UE.

For more information about the specific message attribute details, see [Gx interface messages](#).

The configuration and selection that the Diameter peer uses is determined in the same way as for a standard Gx reference point. This supports full reuse of the existing mechanisms for the Gx override feature.

The failover (failure response/timeout) logic for the later Gx communication works in the same way as for the CCR-I and CCR-T for the APN selection. If the Gx communication cannot be initiated or if it fails, the session creation will also fail.

**NOTE:**

This functionality is not currently available in N7 interface.

Procedure

1. Enable the APN override over the Gx interface.

Use one of the following commands.

Sub-steps

- a) At the PDN level.

```
configure mobile-gateway pdn gw-id virtual-apn-selection user-range user-range-name type user-range-type pre-auth-type type
```

Step example

```
configure mobile-gateway pdn 1 virtual-apn-selection user-range premiumUser1 type imsi 1000000000000001 2000000000000002 pre-auth-type diameter
```

- b) At the APN level.

```
configure mobile-gateway pdn gw-id apn {apn-name | unknown} virtual-apn-selection user-range user-range-name type user-range-type pre-auth-type type
```

Step example

```
configure mobile-gateway pdn 1 apn apn-name virtual-apn-selection user-range premiumUser1 type msisdn 1000000000000001 2000000000000002 pre-auth-type diameter
```

- c) In the pre-auth context, at the APN level.

```
configure mobile-gateway pdn gw-id apn {apn-name | unknown} pre-auth type
```

Step example

```
configure mobile-gateway pdn 1 apn apn1 pre-auth type diameter
```

2. Enable the dynamic policy control (PCC).

```
configure mobile-gateway pdn gw-id apn {apn-name | unknown} pcrf-selection dynamic-pcc
```

Step example

```
configure mobile-gateway pdn 1 apn apn1 pcrf-selection dynamic-pcc
```

5.9.2.5 APN override based on PCO

The PGW/GGSN supports GTP PCO-based APN override for both PAP and CHAP.

When the feature is enabled, and the Peer-Id in a PAP Authenticate-Request or Name in a CHAP response within the PCO contains a character @ followed by a string (APN_B), the indicated APN_B is used as a virtual APN. If the PCO does not contain a valid APN string (@APN_B), an explicit fallback (default) APN can be used. If the PCO does not contain a valid APN string and a fallback APN is not configured, the system reverts to using another virtual APN selection mechanism. If a virtual APN selection is not configured, then the system uses the real APN. The PCO-based APN override is applied after the static mapping (virtual APN and virtual APN list), but before the virtual APN selection. If the APN_B (or the explicit fallback APN) is not configured or the APN length exceeds the supported 80 characters, the attach will fail.

If both PAP and CHAP parameters are included in the PCO, only the CHAP parameters will be used.

The feature also supports using the username from the PCO without the @APN part in RADIUS messaging.

To enable PCO-based virtual APN selection, use the following command:

```
config>mobile>pdn>apn>virtual-apn-based-on-pco
```

To use the username from the PCO without the @APN part in RADIUS messaging, use the following commands:

```
config>mobile>pdn>apn
    accounting
    authentication
```

To configure a real or a virtual APN in RADIUS authentication or pre-authentication, use the following command:

```
config>mobile>pdn>apn>apn-network-id
```

To view the virtual APN selection method used, use the following commands:

```
show>mobile>pdn>pdn-context imsi imsi detail
```

```
show>mobile>pdn>pdn-context msisdn msisdn detail
```

To view APN and APN override statistics, use the following commands:

```
show>mobile>pdn>apn>statistics
```

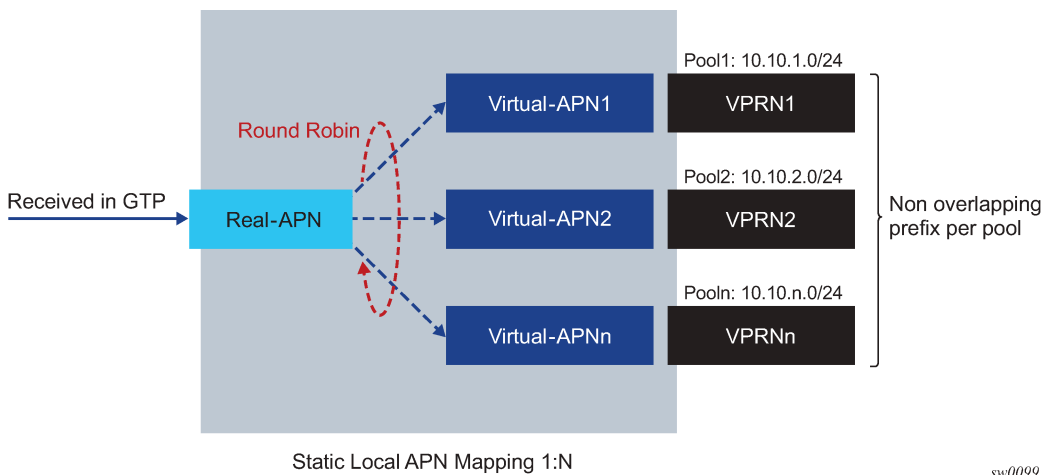
```
show>mobile>pdn>apn>apn-override-statistics
```

5.9.3 Load balancing over the SGi

This feature allows load balancing of PDN sessions and traffic over multiple VRFs/VPNs. PDN sessions that are setup for a real APN received in a GTP message can be distributed across multiple routing instances using real APN to virtual APN selection during attach.

Each virtual-APN is mapped to an individual VRF/VPN instance. A list of virtual APNs is statically configured and associated with a real APN that is received in GTP-C message on the PGW/GGSN. Each PDN session is anchored on one of the virtual APNs that are associated with the real APN. Distribution of PDN sessions across multiple virtual APNs is based on the round-robin method. All traffic for a PDN session is received/forwarded over the VRF/VPN that is associated with the virtual-APN selected during the round-robin distribution. This allows even distribution of traffic in case there is load balancing required across multiple instances of network functions (for example, DPI, Firewall, NAT, and so on) on the SGi interface.

Figure 41: Load balancing over the SGi



5.9.4 APN-AMBR policing configuration

Configure APN-AMBR uplink and downlink policing on a PGW/GGSN gateway. You can specify downlink only, uplink, and downlink, or none. By default, only downlink APN-AMBR policing is performed on the PGW/GGSN and none is used if no APN-AMBR policing is required (for example, policing is done downstream at a TDF).

To specify the type of AMBR policing to be enforced for an APN, use the following command:

```
config>mobile>pdn>apn>ambr-policing
```

The gateway obtains the APN-AMBR to be enforced for a particular subscriber through a specified HSS or local configuration under an APN context, a PCRF specified at the command level, or a PCRF specified through a predefined rule base.

This feature is supported on the ISM-MG-C. To manage the number of available policers for the ISM-MG-C, the operator has the ability to choose the policer distribution percentage between downlink only or uplink and downlink for the mobile gateway system.

To configure the total available policers between downlink only and uplink and downlink APN AMBR policing on the PGW/GGSN, use the following command:

```
config>mobile>system>policer-distribution
```

8.0.R3 AMBR policing and/or GBR bearer policing can also be disabled for an APN if needed; for example, for GBR bearers that do not require any policing (such as in a well-designed IMS network supporting VoLTE calls).

To disable AMBR policing and/or GBR bearer policing for the APN, use the following command:

```
config>mobile>pdn>apn>disable-gbr-policing-qci
```

The following example shows a sample configuration for the above CLI commands.

```
# configure mobile-gateway pdn 1 apn "ipd.nokia.com" ambr-policing ul-and-dl  
  
# configure mobile-gateway system policer-distribution 50
```

5.9.5 GBR/MBR calculation toward the RAN optimization

This feature provides optimization capability for a gateway to calculate the GBR/MBR toward the RAN. The implementation ensures that the gateway calculates only the exact value of bandwidth required, without losing radio resources.

The gateway normally calculates GBR/MBR based on active and installed rules for the bearer as defined in 3GPP/TS 29.212 Section 4.5.5.3. When a rule is installed and active, the gateway uses the GBR/MBR assigned in that rule to calculate the values toward the bearers. The gateway adds the GBR/MBR values from all active and installed rules, including flows for certain disabled rules. For example, for a call-on-hold scenario (call is put on hold and later adds a second call into conference), the subscriber only requires a predefined amount of bandwidth. However, GBR/MBR may still be calculated including rules that have the flow status disabled. As a result, more bandwidth than is required may be created on the RAN, which is considered to be an inefficient use of radio resources.

When the GBR/MBR optimization capability is enabled, the gateway uses a revised algorithm to calculate the GBR/MBR values, taking into account previously allocated RAN resources and disabled flows. The gateway also calculates the GBR/MBR such that at any given time, the allocated RAN resources do not drop below the previously allocated resources. The values of the GBR/MBR variables used in the algorithm are also updated.

The GBR/MBR optimization capability is disabled by default. To enable GBR/MBR optimization for the APN use the following command:

```
config>mobile>pdn>apn>optimize-gbr
```

5.9.6 Corporate VPNs

CMG supports corporate Virtual Private Networks (VPNs) using dedicated APNs and Virtual Private Routed Network (VPRN) services based on the 7750 SR routing support.

These services include:

- carrier grade routing capabilities, including full fledged IP-MPLS stack, widely deployed as MSE and PE router
- large scale IP-VPN support
- large scale routing and tunneling support
- high availability and fast convergence support
- VPN service management

The final APN used for a corporate subscriber can be decided by various methods; it can be configured in the UE, provided by HSS/MME to the PGW in the session creation, or even overridden in the PGW based on local IMSI analysis or on an external server (Gx, RADIUS, S6b), pointing to a dedicated virtual APN. The corporate VPN solution must be accompanied by an appropriate tunneling solution. In the PGW case, the most common tunneling solutions are based on L2TP, GRE, and IP-IP tunnels or IPSec.

The use of overlapping IP pool ranges between APNs that are under the same VPRN is also supported for both L2TP and GRE tunnels. The supported deployment models can either be one LNS/GRE server per APN or the same LNS servers for all APNs. This functionality applies for customers who prefer not to configure different APNs over different VPRNs (VRFs).

Figure 42: IP pool overlapping based on L2TP

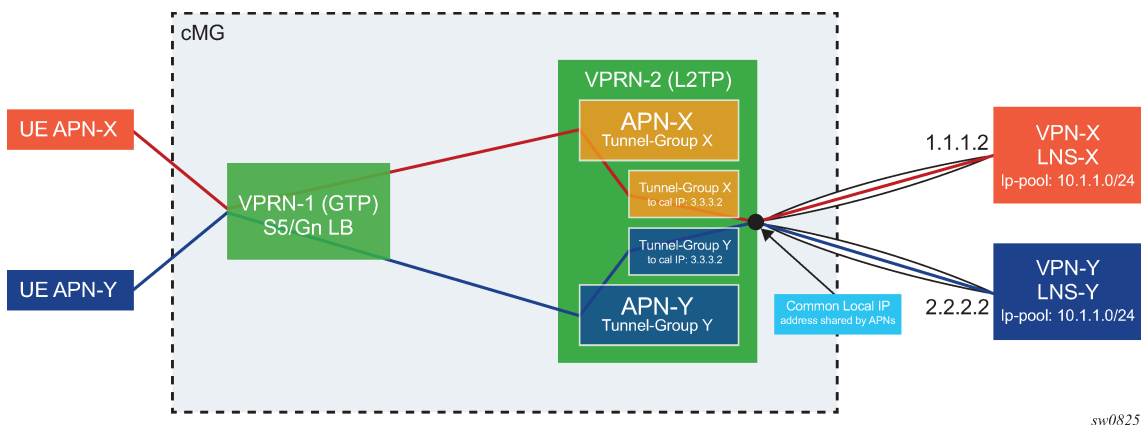
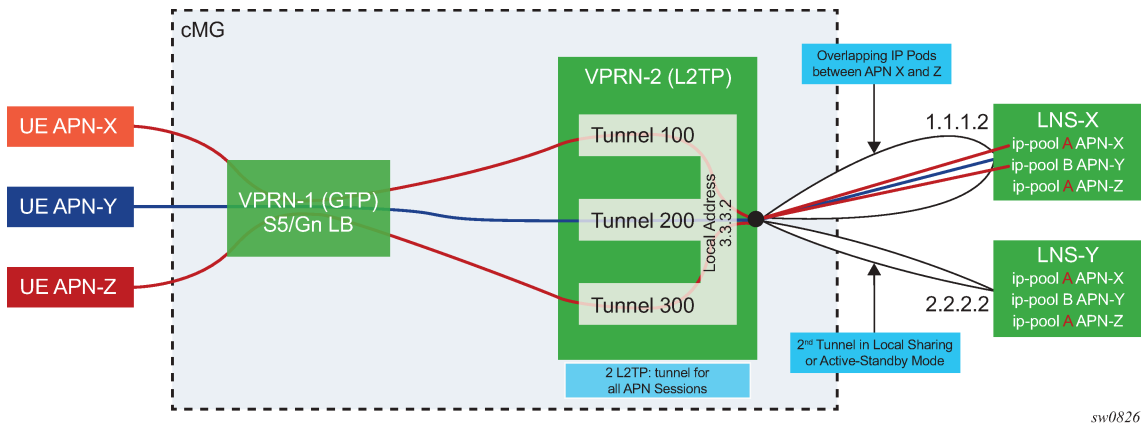
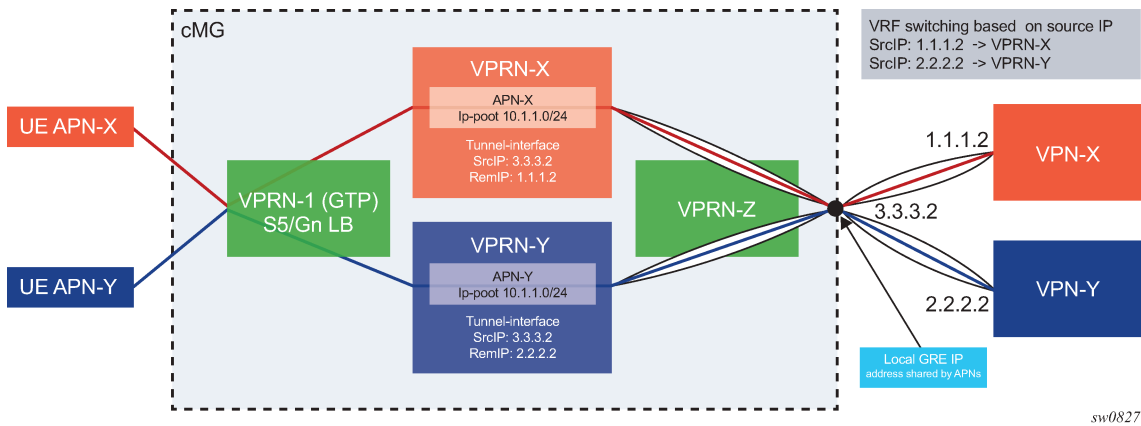


Figure 43: All APNs steered over the same two LNS servers



sw0826

Figure 44: IP pool overlapping based on GRE tunnel



sw0827

**NOTE:**

For more information about VPRNs and tunnels, see the 7750 SR OS documentation.

5.10 SGW with LBO architecture for MEC

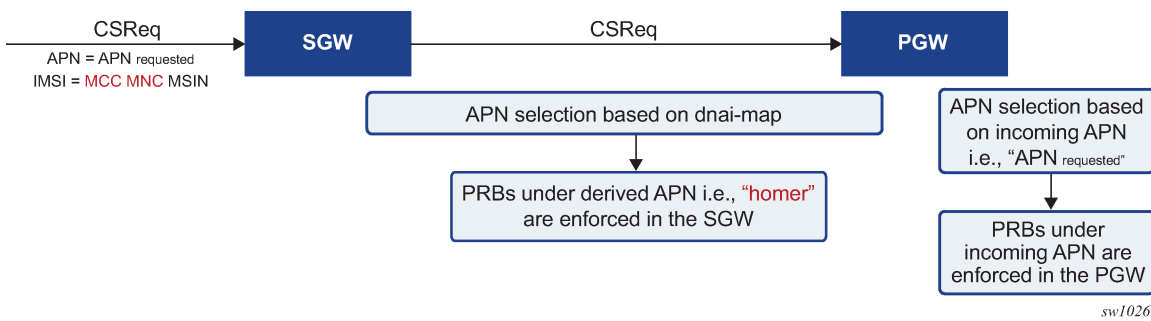
The SGW with LBO feature supports MEC in existing 4G networks and provides greater control over the granularity of traffic that requires steering. The feature allows subscribers to reach both the MEC and the operator's core applications in a selective manner over the same APN.

The MME performs the SGW selection as per 3GPP standards, based on the geographic location of the UE's TA, as provisioned in the operator's DNS. Traffic steering is based on the operator's chosen combination of policy rules, such as the APN, user ID, and the packet's 5-tuple. Other combinations are also possible and are mainly based on policy rules configuration. The traffic to be offloaded is steered over an SGi-LBO interface, which supports traffic separation through NAT and a level of security (firewall) that operators expect from a 3GPP compliant solution.

**NOTE:**

This feature is also supported in 4G CUPS deployments. For more information, refer to the *CMG CUPS Configuration Guide*.

Figure 45: SGW with LBO configuration flow



The following describes the SGW with LBO configuration flow shown in *SGW with LBO configuration flow*.

- All subscribers (roamers and homers) can use localized MEC services (for example, AR/VR).
- Entering the MEC area causes SGW relocation (a specific SGW covers specific TAs). The MME selects the SGW using the existing DNS/TA 3GPP procedure.
- Home subscribers attach to the SGW to offload traffic to MEC, using the home APN (APN "homer" in *SGW with LBO configuration flow*), which is different from the requested APN and is used only for configuring offloading policy rules. Traffic flows that do not require offloading use S5 toward the inbound PGW.
- Roaming subscribers attach to the SGW to offload traffic to the MEC using a roaming APN. The roaming APN is different from the requested APN and is used only for configuring offloading rules. Traffic flows that do not require offloading use S8 toward the outbound PGW.
- The APN configuration on the SGW uses policy rules to steer the traffic per APN. The APN is selected internally based on the DNAI mapping ID to APN mapping table, using the `dnai-map` command in the PDN context. This ensures that IMS traffic or other home-routed traffic is not affected. Only static rules are currently supported for traffic steering.
- S5/S8 bearer-forwarding traffic uses SGW CDRs, and LBO traffic uses PGW CDRs. This ensures that roaming partners are not affected by charging and gives users the flexibility to use MEC services even if a bill shock avoidance message is received.

5.10.1 Configuring SGW with LBO Architecture for MEC

Purpose

Perform the following steps to configure the SGW with LBO for MEC, to allow subscribers to reach both MEC and the operator's core applications in a selective manner over the same APN

Procedure

1. Configure a PLMN profile list for the mapping profile.

The list comprises one or more MCC+MNC values applicable to the operator.

```
configure mobile-gateway profile list plmn plmn-list-name
    mcc mcc-value mnc mnc-value
```

Step example

```
configure mobile-gateway profile list plmn homer-plmn-list
    mcc 311 mnc 486
    mnc 281 mnc 281
```

2. Configure policy rules for traffic steering per mapped APN.



NOTE:

The PRU must be configured before assigning it to a policy rule.

```
configure mobile-gateway profile policy-options policy-rule policy-rule-name
    policy-rule-unit policy-rule-unit-name qci qci arp arp precedence
    policy-rule-unit policy-rule-unit
        flow-description flow-description
            match
                remote-ip remote-ip
            exit
        exit
    flow-description flow-description
        match
            remote-ip remote-ip
        exit
    exit
exit
```

Step example

```
configure mobile-gateway profile policy-options policy-rule offload-traffic
    policy-rule-unit pru1 qci 9 arp 15 precedence 33005
    policy-rule-unit pru1
        flow-description 1
            match
                remote-ip 15.1.1.1/32
            exit
        exit
    exit
```



```

    flow-description 2
      match
        remote-ip ffe::f01:102/128
      exit
    exit
  exit
exit

```

3. Configure a mapped APN for the APN profile mapping.

```

configure mobile-gateway pdn gw-id apn apn-name
  router router-instance
  static-local-pcc
    begin
    dedicated-bearers
      default-bearer-qci qci qci arp arp
      policy-rule policy-rule-name
    exit
  exit
  commit
exit
no shutdown

```

Step example

```

configure mobile-gateway pdn 2 apn apn1
  router vprn30020
  static-local-pcc
    begin
    dedicated-bearers
      default-bearer-qci qci * arp *
      policy-rule offload-traffic
    exit
  exit
  commit
exit
no shutdown

```

4. Configure the DNAI map ID to APN mapping profile.

Configure the APN mapping profile using the DNAI map entry ID with any combination of matching criteria.

```

configure mobile-gateway pdn
  dnai-map entry-id {[plmn-list plmn-list-name] [incoming-apn-list apnlist]
  [mvno]} mapped-apn mapped-apn

```

The mapping may use any one or more of the PLMN list, incoming APN list and the **mvno** keyword (for MVNO subscribers). The DNAI map with the lowest *entry-id* has precedence over multiple matching entries. This allows the operator to choose the matching order by defining them in the preferred order of application. In the following examples, apn1 is selected because dnai-map 1 is preferred over dnai-map 2.

Step example

```
configure mobile-gateway pdn
  dnai-map 1 plmn-list homer-plmn-list mapped-apn apn1
```

Step example

```
configure mobile-gateway pdn
  dnai-map 2 incoming-apn-list apnlist1 mapped-apn apn2
```

5.11 PGW selection for ePDG and TWAG

The mobile gateway supports the ability to configure PGW selection algorithms for ePDG and TWAG configuration purposes.

The ePDG and TWAG select a PGW based on some combination of APN and PLMN ID (operator identifier). The ePDG/TWAG constructs an FQDN in a standard way and uses DNS to resolve the PGW, so as to enable choosing the closest PGW to the ePDG by matching a locally configured ePDG FQDN.

Additionally, the operator can configure a locally defined mapping of APN to a set of PGWs bypassing DNS, or configure a list of PGWs that are learned via DNS.

5.11.1 PGW selection overview

The ePDG/TWAG supports PGW selection at the APN level, using PGW FQDN matching against locally configured ePDG/TWAG FQDNs, statically configured PGW candidates, and dynamic (GTP-C) load information.

The ePDG/TWAG is able to locate the closest PGW and then statically configure it, by matching PGW FQDNs against locally configured ePDG/TWAG FQDNs. This is accomplished as defined in 29.303, using the right to left label matching method that is typically used by an MME to find co-located SGW/PGWs.

The ePDG/TWAG attempts to select the PGW that best matches the local FQDN, and then chooses between alternate PGWs, if the best-matched PGW is not reachable.

In the following example, the ePDG FQDN is specified as follows:

```
topon.eth0.epdg01.west.node.epc.mnc123.mcc123.3gppnetwork.org
```

Based on this FQDN setting, the ePDG matching algorithm selects the two west.node PGWs as best matches from the following list of resolved PGW candidates:

- `topon.eth0.pgww01.north.node.epc.mnc123.mcc123.3gppnetwork.org`
- `topon.eth0.pgww02.north.node.epc.mnc123.mcc123.3gppnetwork.org`
- `topon.eth0.pgww01.west.node.epc.mnc123.mcc123.3gppnetwork.org`
- `topon.eth0.pgww02.west.node.epc.mnc123.mcc123.3gppnetwork.org`

**NOTE:**

Best-match selection is used only for cases where the PGW has not already been selected. For example, it is used for an initial attach, but not for handovers from LTE to WLAN.

Additional information including priority and weight can be used to manage selection between the two possible PGW candidates. When topological matching is used, the system only considers results that have `topon` as the first label. All other results (for example, `topoff`) are ignored for topology matching, but are valid results when topological matching is disabled or does not provide an available PGW.

The ability to configure a static list of PGW identities for an APN can also be used. This is useful for several cases including the following:

- the DNS does not provide a PGW identity
- the PGW's provided by the DNS are unreachable
- the DNS is not configured

PGWs can be identified by the FQDN, IPv4 address, or IPv6 address. If an FQDN is specified, the system uses DNS to resolve an address. The configured APN is matched against the APN that would normally be resolved by DNS, which is the APN after modifications such as APN-OI replacement have occurred.

The ePDG/TWAG supports DNS selection and locally configured PGW selection simultaneously. In certain cases local configuration is used only as a backup, and in other cases it is used to override DNS resolution.

The priorities of dynamically learned PGW candidates (DNS) and those that result from a static APN mapping are not mixed. There is an absolute priority of static versus dynamic results and priority values between the two are not compared.

Load information is also used to select among multiple best match PGWs. However, if all best-match PGWs become overloaded, the ePDG/TWAG selects from non-best-match PGWs.

5.11.2 Configuring PGW selection for ePDG/TWAG

Purpose

CLI commands are used to configure the PGW selection entities, including the selection list profile, the FQDN information, topological matching capability, and so on.

A candidate list profile must first be configured for the PGW selection. The FQDN, IPv4 or IPv6 IP address may be specified for the list, as well as the priority, weight and S2a or S2b selection criteria.

Procedure

1. Configure the candidate list profile for the PGW selection.

```
configure>mobile>profile# list
pgw-selection-candidate-list list-name
    candidate {[fqdn string | ipv4 ip-address | ipv6 ipv6-address]}
    [priority num] [weight num] [ref-pt {[s2a | s2b]}]
```

Step example

This example shows a PGW candidate list profile configuration.

```
#config>mobile>profile>list# info
-----
      pgw-selection-candidate-list "pgwSelCandList1"
          candidate "topon.eth0.pgww01.east.node.eps.mnc123.mcc123.3gppnetwork.org"
priority 100 weight 100 ref-pt s2b
          candidate "topon.eth0.pgww02.east.node.eps.mnc123.mcc123.3gppnetwork.org"
priority 100 weight 100 ref-pt s2b
          candidate "topon.eth0.pgww03.west.node.eps.mnc123.mcc123.3gppnetwork.org"
priority 100 weight 100 ref-pt s2b
          candidate "topon.eth0.pgww04.west.node.eps.mnc123.mcc123.3gppnetwork.org"
priority 100 weight 100 ref-pt s2b
      exit
-----
```

2. Apply the ePDG/TWAG PGW selection to an APN.

```
configure>mobile>pdn# pgw-selection
    apn apn-name
    pgw-selection-candidate-list list-name
    topology-match-disable
    dns-results-preferred
    epdg-fqdn string
    twag-fqdn string
```



NOTE:

The APN must be defined in “DNS query” format.

Step example

This example shows an ePDG PGW selection configuration applied to an APN, including the PGW candidate selection list profile, topological matching disabled, and an ePDG FQDN reference.

```
#config>mobile>pdn>pgw-selection# info
-----
      apn "ims.apn.epc.mnc100.mcc200.3gppnetwork.org"
```

```

pgw-selection-candidate-list "pgwSelCandList1"
topology-match-disable
exit
epdg-fqdn "epdg01.west.node.epc.mnc123.mcc123.3gppnetwork.org"
-----

```

5.12 Use default profile values for GTP and Diameter signaling parameters

A gateway comes up with default profiles for GTP-C, GTP-U, and Diameter with default values used for signaling protocol parameters.

The default QCI-{FC, DSCP} mapping profile is specified in 3GPP standards.

The default profiles are not editable (the values cannot be modified). The operator can provide different respective profile values by configuring a new profile and associating the profile to a specific reference point or it can be applied globally.

Table 10: QCI-{FC, DSCP} mapping table

QCI (arp 1-15)	Forward class	Inprofile/Outprofile DSCP
1	nc	nc2
2	ef	ef
3	af41	af41/af42
4	h1	nc1
5	af21	af21/af22
6	af	af11/af12
7	af	af11/af12
8	l2	cs1
9	be	be

5.13 3GPP signaling configuration

The following 3GPP signaling configurations are supported:

- *S5 and Gx signaling using system IP interface configuration*
- *Signaling configuration to use a separate IP interface*
- *Gx interface*

5.13.1 S5 and Gx signaling using system IP interface configuration

The following shows an example configuration of a PGW instance to use common signaling profiles (Diameter) for all Diameter-based application.

The output also shows the configuration of a single system IP interface to be used as source interface for all the packets originated by the system.

```
# config>mobile>pdn# info
-----
      signaling
        diameter
          origin-host "diam.origRealm.com"
          origin-realm "origRealm.com"
          profile "diam-prof1"
        exit
      interface "system"
    exit
-----
```

5.13.2 Signaling configuration to use a separate IP interface

Based on operator requirements, separate IP interfaces can be configured for each of the reference-point.

The following output shows an example where Gx/S5 signaling separate IP interfaces.

```
#config>mobile#
-----
      config mobile-gateway pdn
        gx
          diameter-application 167772238
          interface
            diameter "Gx"
        exit
      s5
-----
```

```

        interface
            gtp-c "s5"
        exit
    exit
    no shutdown
exit all
-----

```

5.13.3 Gx interface

Gx is a reference-point for PGW interaction with PCRF. Gx is a Diameter-based application.

The application ID used for Gx application is 16777238.

5.14 Diameter peer profiles configuration

The Rf, Gx, Gy, and S6b interfaces are reference points for the gateway interaction with Diameter-based servers (Offline Charging, Online Charging, and so on). A Diameter profile must be configured with specific protocol timers or the default values are used.

A Diameter peer profile must also be configured with peer-specific parameters such as IP address, destination realm, and so on.

All Diameter interfaces on the gateway can be configured including Rf, Gx, Gy, and S6b. FQDN resolution to multiple IP addresses is supported for Rf interface only. Load balancing is supported for Gx, Gy, and Rf interfaces.

The following example shows a basic CLI configuration for the Diameter peer profile configuration.

```

configure mobile-gateway profile
    diameter-peer "Peer Profile 5"
        application-type rf
        destination-realm verizon_ccf5
        diameter-profile prof1
        interface "n10.10.7.82"
        peer 10.10.5.88 port 3878
        shutdown
    exit
    no load-balance
    transport tcp

```

The peer IP address can be an IP address or FQDN. If an FQDN is specified, then an IP address for the peer are obtained by a DNS a-record query.

An Rf Diameter peer within a Diameter profile may be modified without a gateway shutdown. In the case of Rf, for existing sessions, a change in the Diameter peer likely results in an incomplete record generated at

the leCCF. While the modification is being done on the primary Diameter peer, new sessions are sent to the secondary server (if configured). If a secondary server is not configured, new records are written to flash as per the Rf outage configuration settings.

For additional CLI configuration information, see the *Sig Rate Limit Profile* commands in the *CMG CLI Reference Guide*.

5.14.1 DRA and Diameter server selection

You can either perform the Diameter server selection locally by configuring a Diameter server in the Diameter profile used by the UE or use an external agent to do the selection. In this case, the selection is done by a Diameter Redirect Agent (DRA).

The Diameter peer profile configuration is used to configure Diameter servers and DRAs.

To configure a DRA in a Diameter peer profile, use the following command:

```
config>mobile>profile>diameter-peer>peer>enable-dra
```

The mobile gateway keeps an open connection to all the configured Diameter peers (DRAs and Diameter servers).

When a session is created, a CCR-I message is sent to the DRA as to any other Diameter server. The DRA does not process the request and performs a lookup for the Diameter Server that will handle the request. It replies with a CCA-I including the command to redirect the message to a Diameter server URI. The DRA sends the URI of the redirect server in the Redirect-Host AVP.

The mobile gateway node performs a local lookup, without DNS, to find the IP address of the redirect host. It searches under all Diameter peer profiles supported host lists to match the received URI. When a match is found, it uses the IP address configured for that Diameter peer as the Diameter server address. CCR-I is sent to this specific address.

To configure the supported host list for a Diameter peer profile, use the following command:

```
config>mobile>profile>diameter-peer>peer>supported-host
```



NOTE:

- The maximum number of hosts/DRAs that can be configured for a Diameter peer is 16. If a configured DRA is not available or does not reply, the CCR-I is sent to the next DRA in the list (this is not a failover). Round robin algorithm is used to balance the load on the DRAs.
- When the CCR-I is sent toward a DRA, it does not include a Destination-Host AVP. The Destination-Host AVP is included only in the CCR-I sent to the Diameter server.

The Redirect-Host-Usage AVP which is sent by the Diameter server defines how the provided URL is treated. For example, whether it is applied to all bearers for the same user, or it is cached in the client, and so on. The mobile gateway uses the value ALL_USER (6) of the Redirect-Host-Usage AVP. If a different option is received, it is treated as ALL_USER.

Example: Configuration of four DRAs for PCRF selection

```
configure mobile-gateway profile
    diameter-peer "dra"
        application-type gx
        destination-realm "nokia.com"
        diameter-profile "diam-prof1"
        load-balance
        peer 10.10.43.20 port 3868
            enable-dra
            no shutdown
        exit
        peer 10.10.43.21 port 3868
            enable-dra
            no shutdown
        exit
        peer 10.10.43.22 port 3868
            enable-dra
            no shutdown
        exit
        peer 10.10.43.23 port 3868
            enable-dra
            no shutdown
        exit
    exit
```

When a session is created, a CCR-I message is sent to a configured DRA. The DRA includes the URI of the selected PCRF in the CCA-I message. The URI is used to match one of the supported hosts from all the configured Diameter peers using the supported-host as a mapping key.

Example: Three PCRF servers

```
configure mobile-gateway profile
    diameter-peer "pcrf1"
        application-type gx
        destination-realm "nokia.com"
        diameter-profile "diam-prof1"
        peer 10.10.43.24 port 3868
            supported-host 1 "pcrf1.node.epc.mnc010.mcc440.nokia.com"
            no shutdown
        exit
    exit
    diameter-peer "pcrf2"
        application-type gx
        destination-realm "nokia.com"
```

```
diameter-profile "diam-prof1"
peer 10.10.43.25 port 3868
    supported-host 1 "pcrf2.node.epc.mnc010.mcc440.nokia.com"
    no shutdown
exit
exit
diameter-peer "pcrf3"
    application-type gx
    destination-realm "nokia.com"
    diameter-profile "diam-prof1"
    peer 10.10.43.26 port 3868
        supported-host 1 "pcrf3.node.epc.mnc010.mcc440.nokia.com"
        no shutdown
    exit
exit
```

For example, if the DRA replies with the Redirect-Host AVP value set to pcrf2.node.epc.mnc010.mcc440.nokia.com, the lookup results in matching with the PCRF server "pcrf2" and the CCR-I message is redirected to the IP of this server, which is 10.10.43.25.

5.14.2 Diameter TCP session and peer reset

This feature enables faster reset of a control plane TCP session on intra-VNF (MG-VM to MG-VM) failover.

The source TCP port can be configured to remain consistent between active and standby. When configured, a TCP session reset of the state session setup from the previously active port is triggered. Keeping the TCP port consistent between active and standby forces the peering node to enable a cleanup session from the previous active port and allows a new session from the now active port to setup immediately, without relying on the watchdog timer to detect the failure.

To enable keeping the same port as previous for a TCP session on an intra-VNF switchover, use the following command:

```
config>mobile>profile>diameter-peer>switchover-source-port-consistent
```

5.14.3 Diameter TCP and SCTP connection reattempt

The mobile gateway provides support for configuring a Tc timer which specifies the maximum amount of time the gateway waits before re-attempting to connect with a Diameter peer.

To configure the maximum time after which the node attempts to reconnect, in case of a transport connection failure, use the following command:

```
config>mobile>profile>diameter>connection-timer
```

**NOTE:**

If the Diameter peer deployment uses the STCP protocol, the reattempt behavior is impacted by both the connection timer and rto-init-max-timer of the STCP profile.

5.14.3.1 Fail-Retry functionality for TCP/SCTP transport

You can configure a fast mechanism to re-establish a connection with a Diameter peer, which specifies the number of attempts that will be made to re-establish the connection.

To configure the number of re-attempts made to establish a transport connection to a Diameter server, use the following command:

```
config>mobile>profile>diameter>fail-num-retries
```

If the command is not configured, the attempt to re-establish the connection is made after the default time period which is Tc seconds (30 seconds by default). When the functionality is enabled and the configured value is reached, the next attempt to reconnect is also made after the Tc time interval.

When the fail-num-retries command is enabled, the time after which the gateway re-attempts to establish connection with a Diameter server can also be configured.

To configure the time that the gateway re-attempts to establish connect, use the following command:

```
config>mobile>profile>diameter>fail-retry-interval
```

When no value is specified, the default value (0.5 seconds). The next attempts are made after the configuring the fail-retry-interval command.

**NOTE:**

- When the Diameter peer closes the connection due to a permanent failure, for example when the peer has terminated the connection stating that it does not need to communicate, the waiting time before trying to re-establish the connection can be configured using the perm-fail-retry-timeout command.
- This functionality applies when a Diameter connection is in closed state and does not affect the behavior prior to the connection being assumed as closed. For example, if a connection attempt message (either TCP SYN or SCTP INIT) is unanswered by the peer element, the retries that are made before the connection is moved to closed state, remain unaffected by this functionality and are not counted in the statistics for Connection attempts and Connection failures under the command show>mobile>pdn>path-mgmt-stats

This functionality applies to both TCP and SCTP protocols. For the TCP protocol, when enabled, the mobile gateway assumes that a connection is closed 5 seconds after the TCP SYN message has been sent without a reply from the peer. To assume an SCTP connection closed, the unanswered SCTP INIT message attempts follow the retry mechanism with the configured values set under the sctp-profile command.

5.14.4 Diameter destination host and realm

The PGW supports sending the origin realm received in the latest CCA, RAR, or ASR message from the OCS as the destination realm in subsequent messages to the OCS.

Similarly, the PGW supports sending the origin realm received in the latest CCA or RAR message from the PCRF as the destination realm in subsequent messages to the PCRF.

Upon failover or switchover to the secondary Diameter peer, the configured realm of the peer is populated as the destination realm in the message sent to the peer. This is the default behavior. Subsequent messages will populate the learned destination realm of the peer (for example, the origin realm in the CCA).

You can configure the PGW to enable sending the last known destination realm of a Diameter peer as the destination realm after a switchover to the secondary peer. A pre-condition for enabling this functionality is not to disable learning the destination realm; the CLI cross check applies to the same peer profile.

To configure the PGW to enable sending the last known destination realm of the Diameter peer as the destination realm after a switchover to the secondary peer, use the following command:

```
config>mobile>profile>diameter-peer>last-known-dest-realm-after-switchover
```

An alternative to using the learned destination realm of the peer is to disable sending the learned destination realm in subsequent messages.

To disable sending the learned destination realm in subsequent messages, use the following command:

```
config>mobile>profile>diameter-peer>disable-learnt-dest-realm
```

To view Diameter peer information, use the following command:

```
show>mobile>profile>diameter-peer
```



NOTE:

- On-the-fly changes to the `last-known-dest-realm-after-switchover` or `disable-learnt-dest-realm` command take effect on the next transaction for a session.
- The destination realm is learned only for a successful response from the peer (for example, for a successful CCA-I message).
- The destination realm is learned from the latest CCA, RAR, or ASR message from the peer.
- Enabling the `disable-learnt-dest-realm` command will disable dynamic learning of destination realms from then on. For existing sessions that already learned a destination realm, the realm that was last learned is used from then on.
- It is not recommended to combine a high number of Gx realms with a high number of PCRFs per realm. This can cause performance degradation in case of a sudden high rate of UE attaches that trigger a burst of Gx create session requests.

By default, the destination host AVP is populated with the last known value after a switchover to the secondary server for Gx, Gy, and S6b Diameter interfaces. This is required for deployments with a Diameter Routing Agent (DRA). However, deployments without a DRA may require that this AVP be clear so that the server accepts the CCR-U.

The operator can control the population of the destination host in the first message sent to the alternate server after a switchover occurs. This can be disabled for the Diameter peer profile using the following CLI command:

```
config>mobile>profile>diameter-peer>exclude-dest-host-after-switchover
```

5.14.5 Multiple destination realms for a Diameter peer (Gy interface) configuration

It is possible to configure the Gy interface to use multiple destination realms toward a selected Diameter peer, for example a DRA.

A table can be configured with entries that associate the host with a destination realm to be used toward the selected Diameter peer. One of the main purposes of this functionality is to allow a PCRF-specified Primary-Event-Charging-Function-Name FQDN (mapped in the configuration to the supported-host, port, and transport parameters) to use the associated destination realm toward the Diameter peer. In this scenario, the destination host can be excluded from the CCR-I for the PCRF-selected OCS server (via CLI configuration), because the DRA selects the appropriate OCS server host based on the destination realm.

This functionality also supports the selection of different destination realms served by a single DRA via local OCS selection, and provides the option to override the destination realm of the Gx-provided OCS address.

To configure the Diameter routing table for multiple destination realms support, use the following CLI commands.

```
config>mobile>pdn>gy#
dia-routing-table host-index supported-host fqdn port port transport {tcp |
sctp} realm dest-realm diameter-peer-group-list list-name
exclude-dest-host-pcrf-selected-server
```



NOTE:

If PCRF is expected to send the FQDN without port and transport information, you must configure the port as **3868** and the transport as **sctp** in the table to match the FQDN.

The no form of the `dia-routing-table` command can be used to delete the Diameter table, or the Diameter table entry (if the `host-index` is also specified).

To configure a destination realm that will override the destination realm configured for the Diameter peer profile, use the following DCCA profile commands, specifying the name of the destination realm configured in the `config>mobile>profile>diameter-peer` context.

```
config>mobile>profile># dcca profile-name  
destination-realm destination-realm-name
```

To view information for the Gy Diameter routing table, use the following command.

```
show mobile-gateway pdn ref-point-peers gateway gw-id gy dia-routing-table
```

5.14.6 Multiple destination realms for a Diameter peer (Gx interface) configuration

Multiple destination realms can be used per Diameter peer profile for an operator. This functionality is available through APN level configuration. This means that the operator can configure one or more destination realms in Gx interface per APN.

When destination realm configuration is used at APN level, destination realm that is configured in Diameter peer profile is ignored.

Separate APNs can be assigned to separate BWA regional operators and separate destination realms specific to those APNs. This allows for LTE core network sharing among different partners by assigning a different realm to each partner.

Each operator can use two DRAs in active/standby mode. All Gx messages are routed through a DRA to a specific PCRF according to the destination realm per APN that is included in the CCR message.

To configure a specific destination realm in Gx interface per APN, use the following command:

```
config>mobile>pdn>apn>destination-realm-gx
```

5.15 Diameter interfaces overload control

The Diameter interfaces overload control feature provides support for receiving and taking action on Diameter Experimental-Result-Code 5198 for the PGW.

If a Diameter server experiences overload or a DRA experiences overload against all of its Diameter servers, it may respond with the Experimental-Result-Code 5198 `DIAMETER_OVERLOAD_RETRY_NOT_ALLOWED_TO_ANY` toward the PGW.

If the PGW receives an Experimental-Result-Code 5198 during a call flow of an initial or update request, it takes the following actions:

- does not retry to a secondary DRA
- terminates the session with the GTP/PMIP cause code `USER_AUTHENTICATION_FAILED`

The session termination call flows always follow the normal procedure. The DRA is never expected to use the 5198 experimental result code in a Session Termination call flow with the PGW.

To view information about Diameter interface overload control codes, use the following commands:

```
show>mobile>pdn>ref-point-stats>s6b>failure-codes
```

```
show>mobile>pdn>ref-point-stats>gx>failure-codes
```

5.16 Diameter overload indication conveyance

Diameter Overload Indication Conveyance (DOIC) is supported over the S6b, Gx, SWm, and STa interfaces as per RFC 7683, 3GPP Release 13 TS23.275, and TS29.213.

As a DOIC node, the gateway operates as a reacting node and acts upon any Overload Reports (OLRs) received from the reporting node (AAA or PCRF). It then takes actions (throttling or diverting to an alternative peer) to perform the request in the OLR.

When the gateway operates as a reacting node, the DOIC has the following main functions:

- capability announcement handling
- action handling on the received OLR

The capability announcement and overload reporting are not generated in new Diameter messages. The DOIC information is piggybacked over existing Diameter messages by inserting new AVPs into existing Diameter requests and responses.

5.16.1 Capability announcement handling

The capability announcement handling is the ability of Diameter nodes to determine if other nodes in the path of a Diameter request support DOIC and is also referred to as DOIC Capability Announcement (DCA).

The DCA mechanism uses the OC-Supported-Features AVP (and optionally the OC-Feature-Vector AVP) to indicate the Diameter overload features that are supported.

The OC-Supported-Features AVP (AVP Code 621) is of type Grouped and is included in every Diameter request message a DOIC supporting node sends. It serves two purposes; to announce the node's support for DOIC and to contain the description of the supported DOIC features of the sending node.

The OC-Feature-Vector AVP (AVP Code 622) is of type Unsigned64 and contains a 64-bit flags field of the announced capabilities of a DOIC node. There are only two values specified:

- Value of zero (0x0000000000000000) — reserved
- OLR_DEFAULT_ALGO (0x0000000000000001) — default traffic abatement (loss) algorithm

When this flag is set by a DOIC reporting node, it means that the loss algorithm will be used for the requested overload abatement. As per standard, the OC-Feature-Vector AVP must include the OLR_DEFAULT_ALGO (0x0000000000000001).

**NOTE:**

The OLR_DEFAULT_ALGO flag is set in the OC-Feature-Vector AVP.

5.16.2 Receiving overload report AVP

The receiving overload report (OC-OLR) AVP (AVP Code 623) is of type Grouped and contains the information necessary to convey an overload report of an overload condition at the reporting node.

The application that the OC-OLR AVP refers to is identified by the Application-ID found in the Diameter message header. The host or the realm that the OC-OLR AVP is referring to is determined from the Origin-Host AVP and/or the Origin-Realm AVP found in the encapsulating Diameter command. The OC-OLR AVP is sent only from a reporting node.

The OC-OLR AVP consists of the following sub-AVPs:

- OC-Sequence-Number
- OC-Report-Type
- OC-Reduction-Percentage
- OC-Validity-Duration

The OC-OLR AVP message format is as follows:

```
OC-OLR ::= < AVP Header: 623 >
           < OC-Sequence-Number >
           < OC-Report-Type >
           [ OC-Reduction-Percentage ]
           [ OC-Validity-Duration ]
           * [ AVP ]
```

Table 11: Sub-AVPs of the OC-OLR AVP

AVP	Type	Description
OC-Sequence-Number (AVP Code 624)	Unsigned64	Used to identify if the OLR is for an existing overload condition or an update to an existing OLR when the gateway acts as a reactive node If the sequence number for the received OLR is:

Table 11: Sub-AVPs of the OC-OLR AVP (continued)

AVP	Type	Description
		• greater than the sequence number stored in the matching OCS entry, then a reacting node must update the matching OCS entry • less than or equal to the sequence number in the matching OCS entry, then a reacting node must silently ignore the received OLR; the matching OCS is not updated in this case
OC-Report-Type (AVP Code 626)	Enumerated	The following report types are supported on the gateway: • host-type OCS entry, that is, an overload action will be applied for any Destination-Host for which an OC-OLR is received A host-type is identified by the pair of Application-ID and the node's Diameter-Identity • realm-type OCS entry, that is, an overload action will be applied for the Destination-Realm for which an OC-OLR is received A realm-type is identified by the pair of Application-ID and realm.
OC-Reduction-Percentage (AVP Code 627)	Unsigned32	Indicates the percentage of the traffic that the sender is requested to reduce The range of values is 0 to 100
OC-Validity-Duration (AVP Code 625)	Unsigned32	Indicates, in seconds, the validity time of the overload report The number of seconds is measured after the reception of the first OC-OLR AVP with a given value for the OC-Sequence-Number AVP

5.16.3 Overload report resulting actions

This section describes the resulting actions upon receiving an overload report (OC-OLR AVP) over the different interfaces.

5.16.3.1 Overload report over the S6b interface

When an overload report is received over the S6b interface, the following actions are taken depending on the contents of the OC-OLR AVP.

When the OC-OLR AVP is received with the required traffic reduction and a realm OC-Report-Type AVP, the priority table shown in *S6b interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).



NOTE:

In case a secondary (alternative) peer is configured with the same realm, the PGW will attempt to verify/learn the OC-OLR AVP for the secondary peer.

When the OC-OLR AVP is received with the required traffic reduction, a host OC-Report-Type AVP, and the Diameter peer profile with a supported host list has a single or several hosts with all of them in overload, the priority table shown in *S6b interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).



NOTE:

If there are two or more hosts with variable amount of overload indication, the highest overload metric will be applied to the peer.

Table 12: S6b interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload

Reporting node	Reacting node	Message priority
AAA	PGW	• Priority 1 (always send, do not drop) 1. STR 2. AAR-U • Priority 2 If configured and not in overload, attempts an alternative peer first, then drops: 1. AAR-I

The PDN sessions known as emergency sessions are not throttled over the S6b interface.

5.16.3.2 Overload report over the SWm interface

When an overload report is received over the SWm interface, the following actions are taken depending on the contents of the OC-OLR AVP.

When the OC-OLR AVP is received with the required traffic reduction and a realm OC-Report-Type AVP, the priority table shown in *SWm interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).



NOTE:

In case a secondary (alternative) peer is configured with the same realm, the ePDG will attempt to verify/learn the OC-OLR AVP for the secondary peer.

When the OC-OLR AVP is received with the required traffic reduction, a host OC-Report-Type AVP, and the Diameter peer profile with a supported host list has a single or several hosts with all of them in overload, the priority table shown in *SWm interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).



NOTE:

If there are two or more hosts with variable amount of overload indication, the highest overload metric will be applied to the peer.

Table 13: SWm interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload

Reporting node	Reacting node	Message priority
AAA	ePDG	• Priority 1 (always send, do not drop) If configured, ePDG attempts an alternative peer: 1. STR 2. DER-U (with destination host populated) 3. AAR-U • Priority 2 If configured and not in overload, attempts an alternative peer first, then drops: 1. DER-I

The PDN sessions known as emergency sessions are not throttled over the S6b interface.

5.16.3.3 Overload report over the STa interface

When an overload report is received over the STa interface, the following actions are taken depending on the contents of the OC-OLR AVP.

When the OC-OLR AVP is received with the required traffic reduction and a realm OC-Report-Type AVP, the priority table shown in *STa interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).



NOTE:

In case a secondary (alternative) peer is configured with the same realm, the TWAG will attempt to verify/learn the OC-OLR AVP for the secondary peer.

When the OC-OLR AVP is received with the required traffic reduction, a host OC-Report-Type AVP, and the Diameter peer profile with a supported host list has a single or several hosts with all of them in overload, the priority table shown in *STa interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).



NOTE:

If there are two or more hosts with variable amount of overload indication, the highest overload metric will be applied to the peer.

Table 14: STa interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload

Reporting node	Reacting node	Message priority
AAA	TWAG	• Priority 1 (always send, do not drop) - If configured and not in overload, attempts an alternative peer: 1. STR 2. DER-U (with destination host populated) 3. AAR-U • Priority 2 - If configured and not in overload, attempts an alternative peer first, then drops: 1. DER-I

5.16.3.4 Overload report over the Gx interface

When an overload report is received over the Gx interface, the following actions are taken depending on the contents of the OC-OLR AVP.

When the OC-OLR AVP is received with the required traffic reduction and a realm OC-Report-Type AVP, the priority table shown in *Gx interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).

When the OC-OLR AVP is received with the required traffic reduction, a host OC-Report-Type AVP, and the Diameter peer profile with a supported host list has a single or several hosts with all of them in overload, the priority table shown in *Gx interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload* is used (the lowest priority is throttled first and the highest last).



NOTE:

If there are two or more hosts with variable amount of overload indication, the highest overload metric will be applied to the peer.

Table 15: Gx interface message priority for host and realm OC-Report-Type AVPs and all hosts in overload

Reporting node	Reacting node	Message priority
PCRF	PGW	- Priority 1 (always send, do not drop) - CCR-T - CCR-U - Priority 2 If configured and not in overload, attempts an alternative peer first, then drops: - CCR-I

5.17 Diameter origin host configuration

This feature provides the option to configure the gateway to send the Origin-Host Diameter AVP as a unique string per MG-VM in CMG (default configuration), or as a unique string per Diameter application running per MG-VM in CMG.

To configure the Origin-Host-AVP, use the following command:

```
config>mobile>pdn>signaling>diameter>origin-host name
```

In this case, the origin host name is pre-pended with the MG-VM ID that identifies the redundant pair. For example, if the origin host is 7750mg.nokia.com, then the Origin-Host-AVP will be sent with value mscp01.7750mg.nokia.com.

To override the default configuration, use the `origin-host` command in the following reference point contexts:

```
config>mobile>pdn>gx
config>mobile>pdn>gy
config>mobile>pdn>s6b
config>mobile>pdn>rf
config>mobile>pdn>sd
config>mobile>pdn>sta
config>mobile>pdn>sww
```

In a single ISM-MG redundant pair in the chassis, set a default value for the Diameter Origin-Host AVP and override this value for the Gx and S6b interfaces.

To specify 7750mg.nokia.com as the default value for the Origin-Host-AVP, use the following command:

```
config>mobile>pdn>signaling>diameter>origin-host 7750mg.nokia.com
```

To override the default value for the Gx interface, use the following command:

```
config>mobile>pdn>gx>origin-host gx-7750mg.nokia.com
```

To override the default value for the S6b interface, use the following command:

```
config>mobile>pdn>s6b>origin-host s6b-7750mg.nokia.com
```

Based on the above configuration, the Origin-Host-AVP will be sent with the following values (depending on the interface):

Gx	mscp01.gx-7750mg.nokia.com
S6b	mscp01.s6b-7750mg.nokia.com
All other reference points	mscp01.7750mg.nokia.com

5.18 RADIUS interface

This section provides information about the RADIUS authentication and accounting.



NOTE:

The RADIUS configuration and message formats should be the same for the PGW and the GGSN.

5.18.1 RADIUS interface configuration

The RADIUS source IP address is taken from the interface used for the RADIUS messaging. This interface can be one of the following (in descending priority order) depending on which have been configured in the system:

```
config>mobile>profile>radius-group>interface
```

```
config>mobile>pdn>radius>interface
```

```
config>router>interface "system"
```

5.18.2 Authentication (IP address allocation)

The IP address allocation method may be configured for each APN on the PGW/GGSN. One option is to use RADIUS authentication.

When RADIUS authentication is enabled, the PGW/GGSN performs RADIUS authentication upon a PDN connection activation (PGW) or for a primary PDP context activation (GGSN).

The PGW/GGSN only performs RADIUS authentication for the initial attach, additional PDN connection activation or primary PDP context activation. It does not perform authentication for dedicated bearer creation or for secondary PDP contexts.

Figure 46: E-UTRAN initial attach with RADIUS authentication and accounting

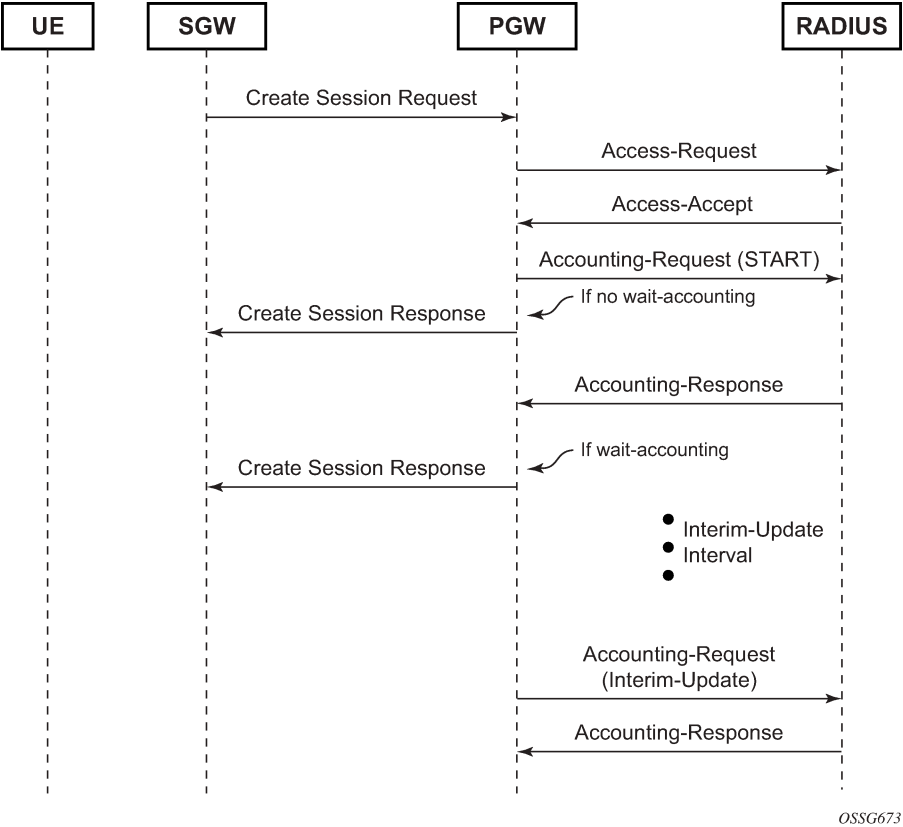
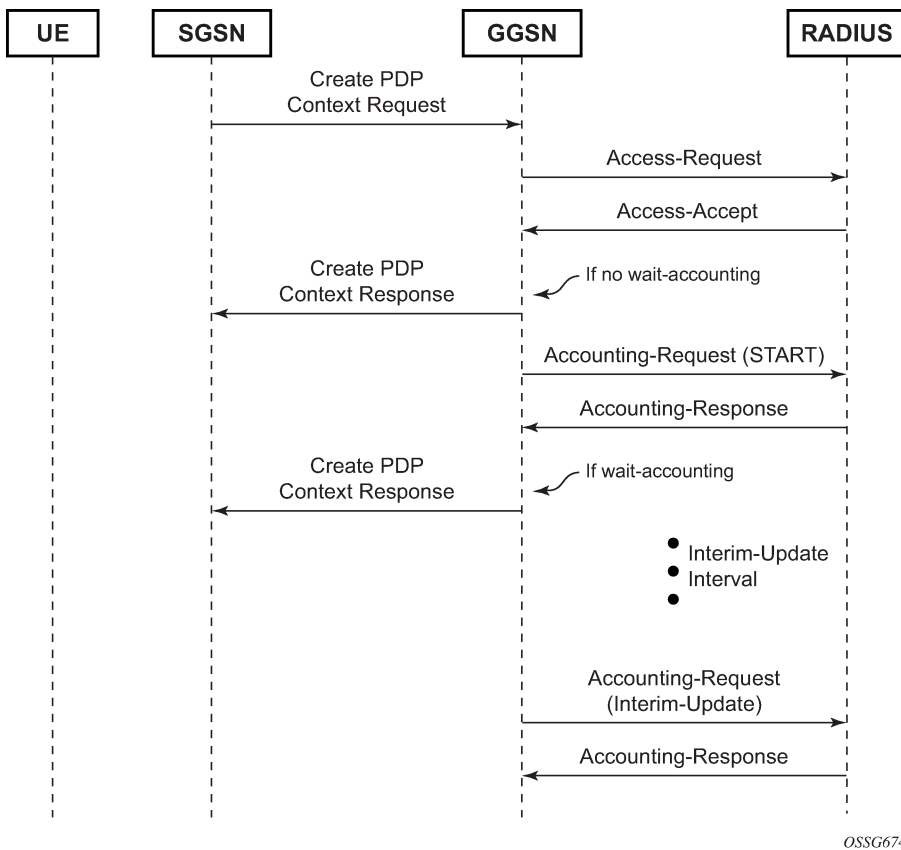


Figure 47: PDP context activation with RADIUS authentication and accounting



In addition to basic RADIUS allocation, the GGSN/PGW supports extended RADIUS server instructed local UE IP address allocation.

For more information, see the description of the `config>mobile>pdn>apn>ip-alloc-extended-radius-allocation` command.

There is also the possibility for the PDN connection activation to be triggered and established when RADIUS authentication is enabled even if the RADIUS server is unreachable. The session is established in a transparent mode, that is user authentication is skipped and IP address allocation is made from a local pool.

To enable the transparent mode functionality, use the following command:

```
configure mobile-gateway profile radius-group fh-session-no-response
{terminate | continue}
```

To configure the time and volume limits to control how long or the data volume threshold for which the PDN connection is allowed to exist in transparent mode, use the following commands:

```
configure mobile-gateway profile radius-group fh-time-limit time-limit
```

```
configure mobile-gateway profile radius-group fh-volume-limit volume-limit
```

5.18.2.1 Username and password-related parameters

The username can be the username received from the UE in Protocol Configuration Options (PCO) or Additional Protocol Configuration Options (APCO), during PDN connection activation. It can also be the MSISDN, IMSI, the APN, or the fixed value configured by the operator.

RADIUS Access-Request message, GTP PCO/APCO information, and local configuration describes how the attributes of the RADIUS Access-Request message are determined, based on the GTP PCO or APCO information and the local configuration of commands pre-auth and authentication in the config>mobile>pdn>apn context.

Table 16: RADIUS Access-Request message, GTP PCO/APCO information, and local configuration

No.	PAP or CHAP username included in GTP PCO/APCO	PAP password or CHAP challenge included in GTP PCO/APCO	MG username configuration	MG password configuration	MG PCO fallback configuration	Username in Access-Request message	PAP password or CHAP challenge in Access-Request message
1	N/A (does not affect)	N/A (does not affect)	imsi/msisdn/apn ¹	Not configured	N/A (does not affect)	UE imsi/msisdn/apn	Not included
2	N/A (does not affect)	N/A (does not affect)	imsi/msisdn/apn ¹	fixed or default	N/A (does not affect)	UE imsi/msisdn/apn	password
3	N/A (does not affect)	Yes (PAP or CHAP)	imsi-apn/msisdn-apn	N/A (does not affect)	N/A (does not affect)	UE imsi@apn/msisdn@apn	PAP password or CHAP challenge from GTP (A) PCO
4	N/A (does not affect)	No	imsi-apn/msisdn-apn	fixed or default	N/A (does not affect)	UE imsi@apn/msisdn@apn	password

¹ The APN is supported only in the authentication and not in the pre-authentication phase.

Table 16: RADIUS Access-Request message, GTP PCO/APCO information, and local configuration (continued)

No.	PAP or CHAP username included in GTP PCO/APCO	PAP password or CHAP challenge included in GTP PCO/APCO	MG username configuration	MG password configuration	MG PCO fallback configuration	Username in Access-Request message	PAP password or CHAP challenge in Access-Request message
5	N/A (yes or no)	N/A (yes or no)	fixed	N/A (does not affect ²)	N/A (does not affect)	fixed	fixed
6	Yes (PAP or CHAP)	Yes (PAP or CHAP)	pco	Not included or default	N/A (does not affect)	Username from GTP (A) PCO	PAP password or CHAP challenge from GTP (A) PCO
7	Yes (PAP)	Yes	pco	fixed	N/A (does not affect)	PAP username from GTP (A) PCO	password
8	Yes (PAP)	No	pco	Not configured	enabled	PAP username according to pco-fallback value	Not included
9	Yes (PAP)	No	pco	Not configured	disabled	PAP username from GTP (A) PCO	Not included

² Specifying value **fixed** for the user-name parameter in the authentication (and pre-auth) command means that a redirect will use both the username and password from the auth-fixed (or pre-auth-fixed) command. Therefore, if value **fixed** is specified, regardless of the setting of the password parameter in the authentication command itself would not be used.

Table 16: RADIUS Access-Request message, GTP PCO/APCO information, and local configuration (continued)

No.	PAP or CHAP username included in GTP PCO/APCO	PAP password or CHAP challenge included in GTP PCO/APCO	MG username configuration	MG password configuration	MG PCO fallback configuration	Username in Access-Request message	PAP password or CHAP challenge in Access-Request message
10	Yes (PAP)	No	pco	fixed or default	enabled	PAP username according to pco-fallback value	password
11	Yes (PAP)	No	pco	fixed or default	disabled	PAP username from GTP (A) PCO	password
12	Yes (CHAP)	No	pco	fixed or default	enabled	PAP username according to pco-fallback value	password
13	Yes (CHAP)	No	pco	Not configured	enabled	PAP username according to pco-fallback value	Not included
14	Yes (CHAP)	No	pco	N/A (does not affect)	disabled	Attach attempt is rejected: invalid IE	
15	No	N/A (does not affect)	pco	fixed or default	enabled	PAP username according to pco-fallback value	password

Table 16: RADIUS Access-Request message, GTP PCO/APCO information, and local configuration (continued)

No.	PAP or CHAP username included in GTP PCO/APCO	PAP password or CHAP challenge included in GTP PCO/APCO	MG username configuration	MG password configuration	MG PCO fallback configuration	Username in Access-Request message	PAP password or CHAP challenge in Access-Request message
16	No	N/A (does not affect)	pco	Not configured	enabled	PAP username according to pco-fallback value	Not included
17	No	N/A (does not affect)	pco	N/A (does not affect)	disabled	Attach attempt is rejected: invalid IE	
18	N/A (does not affect)	Yes (PAP or CHAP)	msisdn	pco_password	N/A (does not affect)	UE msisdn	PAP password or CHAP Challenge from GTP (A) PCO
19	N/A (does not affect)	No	msisdn	pco_password and default	N/A (does not affect)	UE msisdn	Default password

Similar support is provided for the user - range command in the following contexts:

```
configure mobile-gateway pdn virtual-apn-selection
```

```
configure mobile-gateway pdn apn virtual-apn-selection
```

5.18.3 Accounting

The PGW/GGSN can be configured to support RADIUS accounting which allows the RADIUS server to track volume and time usage for each PDN connection. This is useful when the APN is used for corporate

access and also in wholesale configurations where the RADIUS server is owned by another ISP/network provider.

Similar to RADIUS authentication, the PGW/GGSN sends RADIUS accounting start messages for the PDN connection and not for individual bearers or PDP contexts. Therefore the PGW/GGSN does not maintain separate RADIUS accounting sessions for dedicated bearers or secondary PDP contexts. The PGW/GGSN will report the volume RADIUS attributes as the sum of all bearers/PDP contexts belonging to the PDN connection.

Accounting stop messages are sent when all PDP contexts/bearers are deleted.

The figures below show some cases where the PDN connection is deleted. The PDN connection deletion may be initiated by the PGW/GGSN as well.

Figure 48: Detach with RADIUS accounting

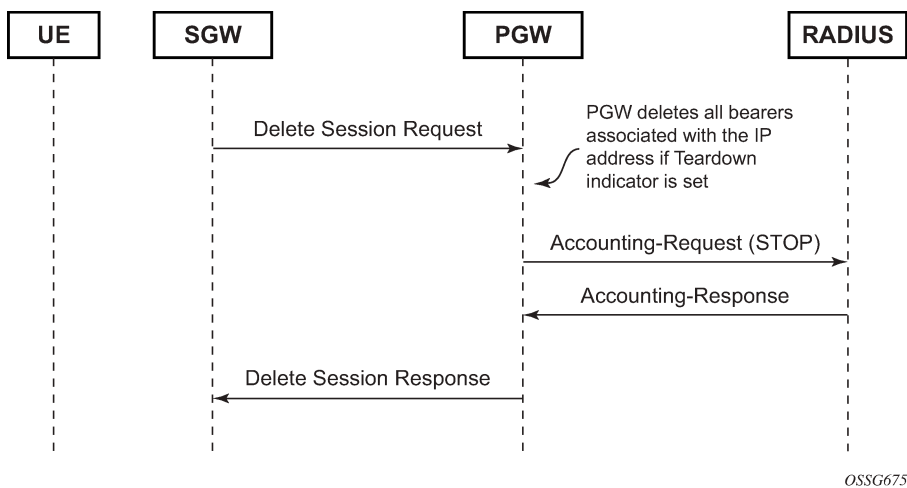
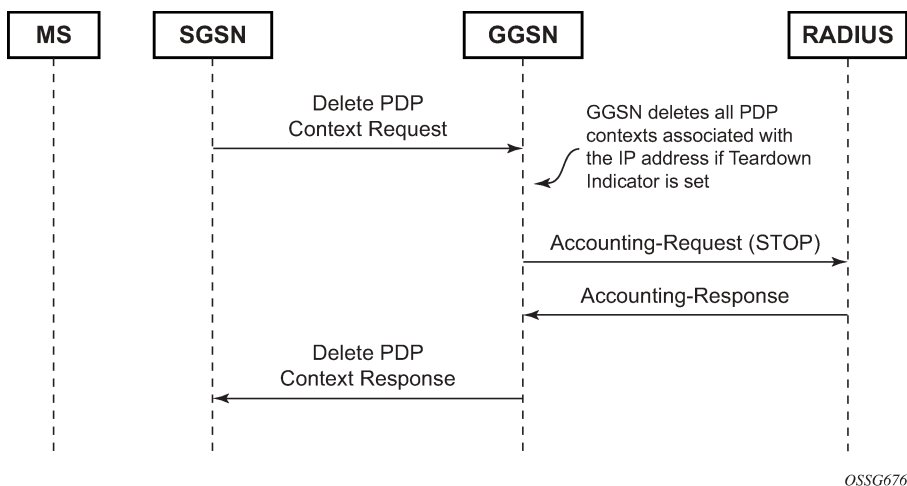


Figure 49: PDP context deactivation with RADIUS accounting



5.18.3.1 Duplicate accounting

Configure the PGW/GGSN to send a RADIUS accounting message to multiple servers simultaneously. The main purpose is to enable sending UE information to upstream servers.

This feature can be configured in cases where WAP servers or DPI devices may need to keep track of the IP addresses assigned to each UE along with other information (such as IMSI, charging ID, and so on) that is included in an accounting message.

To name a server group that is used for duplicate accounting, use the following command:

```
configure mobile-gateway pdn apn accounting duplicate-server-group
```

When duplicate RADIUS accounting groups are used, different AVP settings can also be configured to tune the message content according to the external server needs and capabilities.

To configure the RADIUS AVP setting, use the commands under the following context:

```
configure mobile-gateway profile radius-group radius-avp-options
```

This example shows a basic CLI configuration for duplicate accounting. The dup-group1 contains only one peer, while dup-group2 contains two peers. Each message is sent to both server groups, meaning to the peer in group 1, and to either of the peers in group 2, depending on load balancing (or to the higher priority peer if the two peers have different priority, not shown not in the example).

```
configure mobile-gateway profile
    radius-group "radius-servers"
        server-type both
        peer 1.2.3.4
            radius-profile "radius-profile-1"
            no shutdown
        exit
    radius-group "dup-group1"
        server-type acct
        peer 10.10.10.1
            radius-profile "radius-profile-WAP"
            no shutdown
        exit
    radius-group "dup-group2"
        server-type acct
        peer 20.10.10.1
            radius-profile "radius-profile-DPI"
            no shutdown
        exit
        peer 20.10.10.2
            radius-profile "radius-profile-DPI"
            no shutdown
        exit
configure mobile-gateway pdn 1 apn "Internet"
    accounting type radius user-name imsi server-group "radius-servers"
    duplicate-server-group "dup-group1"
```

```
duplicate-server-group "dup-group2"  
exit
```

5.18.3.1 Ignore Accounting Response

The PGW/GGSN can be configured to not wait for specific RADIUS servers to send an Accounting-Response.

To configure the PGW/GGSN not to wait for an Accounting-Response, use the following command:

```
config mobile-gateway profile radius-group ignore-acct-response
```

5.18.3.2 Interim accounting

The PGW/GGSN provides the following support for interim accounting:

- configurable, interval-based, periodic interim accounting messages
- when triggered due to a RAT change event
- when triggered due to a peer change or QoS change event
- when reaching a specified volume threshold
- for user location change events

To enable periodic interim accounting and configure the interval for sending interim-update messages to the accounting server, use the following command:

```
config>mobile>profile>radius-group>interim-update-interval
```

The Update PDP Context Request (Gn/Gp PGW and combined SGW/PGW), which results in a change of RAT, will trigger accounting messages to be sent to the AAA server. Similarly, a bearer modification (S5/S8 PGW and combined SGW/PGW), which results in a change of RAT, triggers accounting messages to be sent to the AAA server. Triggering an interim-accounting message due to a RAT change, re-initializes the periodic accounting timer as well as the volume threshold (if configured).

To enable interim accounting triggered by RAT change, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>rat-change-trigger
```

An Accounting-Request (INTERIM) message is sent when the primary PDP context/default EPS bearer is modified, and the SGSN/SGW or Quality of Service (QoS) is updated. The trigger for the PDP context/bearer modification can be the SGSN/SGW or the PCRF.

The QoS change proposed by the access network can be denied by the PCRF, so the QoS change is communicated to RADIUS only after the final QoS is accepted. Additionally, the PCRF bearer modification of the QoS will trigger accounting messages toward the AAA server. Here, the PCRF proposes a QoS which is

negotiated with the radio network (SGSN or MME). This can be downgraded and then sent to the PCRF. As in the previous case, the AAA will be informed of the agreed QoS (not the PCRF proposed QoS). In RADIUS, the default bearer QoS is sent to the AAA server.

The Interim accounting request for the primary PDP context/default bearer modification will be sent only if the modification changes value at least in one AVP compared to the previous accounting request.

To enable interim accounting triggered by a QoS change, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>qos-change-trigger
```

To enable interim accounting triggered by any location change event, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>user-loc-change
```

To enable interim accounting triggered by a peer change, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>peer-change-trigger
```

To enable interim accounting triggered by an eCGI change, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>ecgi-change
```

To enable interim accounting triggered by a TAI change, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>tai-change
```

To enable interim accounting triggered by a RAI change, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>rai-change
```

To enable interim accounting triggered by a CGI-SAI change, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>cgi-sai-change
```

The Accounting-Request (INTERIM) message may also be sent when a configured volume threshold is reached or exceeded. The volume threshold can be defined only based on the local configuration. By default, the volume threshold is not a reason for sending a RADIUS interim accounting request.

The volume counting is reset whenever a RADIUS INTERIM accounting request is sent due to any trigger. Additionally, the periodic accounting timer is reset if periodic interim accounting is enabled.

**NOTE:**

The local threshold is only a trigger; the RADIUS volume attributes always report cumulative counts including all octets since the PDP context/bearer activation.

To enable interim accounting when reaching a specified volume threshold, use the following command:

```
config>mobile>profile>charging>pgw-charging>radius-accounting>volume-limit-trigger
```



NOTE:

- Only a single interim accounting message is sent even if multiple triggers match due to a single event.
- Triggering an interim accounting message due to any reason re-initializes the periodic accounting timer and resets the volume counter.

To configure different AVP options for interim accounting messages, use the following command:

```
config>mobile>profile>radius-group>avp-options>acct-interim-avp-options
```

The interim AVP options apply to all the configured types of interim accounting messages.



NOTE:

The `acct-interim-avp-options` command must only be used when the `use-acct-interim-avp-options` command is configured under the `config>mobile>profile>radius-group>avp-options` context.

To view information about the RADIUS interim accounting configuration, use the following command:

```
show>mobile>profile>charging>pgw-charging
```

To view information about the RADIUS interim accounting for a PDN context, use the following command:

```
show>mobile>pdn>pdn-context>imsi>acct>radius
```

5.18.3.3 Accounting ON and OFF messages

The optional RADIUS Accounting ON and OFF are supported either as system or APN level messages. With the former approach, the PGW sends only one ON/OFF message per system, whereas in the latter, the PGW sends these messages for each APN from each node, for each relevant RADIUS accounting server.

With APN level messaging, the ON message is sent to all servers, despite any priority configurations. Issuing a shutdown command for the PDN clears the existing sessions in a controlled manner. The OFF message is sent per APN and per node when the last session has been removed.

To configure the RADIUS Accounting ON and OFF messaging to work on a per system or per APN level, use the following command:

```
config>mobile>pdn>radius>accounting-on-off-level
```

To configure RADIUS Accounting ON and OFF parameters, use the following commands:

```
config>mobile>profile>radius>  
    acct-on-off-initial-interval  
    acct-on-off-retry-interval
```

**NOTE:**

If the Called-Station-ID (APN) is excluded from the accounting messages, some other method must be used by the server for uniquely identifying the client and the APN as required.

To include the Called-Station-ID in the accounting messages, use the following command:

```
config>mobile>profile>radius-group>radius-avp-options>include-called-station-id-apn-in-on-off-messages
```

To further tune the behavior when using APN level messaging, use the commands in the following context:

```
config>mobile>pdn>apn>radius-accounting-on-off>  
    messages-sent  
    wait-for-ack  
    include-duplicate-servers
```

**NOTE:**

Public APN string is not supported with APN level ON and OFF messaging.

5.18.4 Load balancing

The default load distribution is based on a round-robin distribution between RADIUS servers of equal priority. The messages from the same bearer/PDP context are always sent to the same RADIUS server.

When alternative transaction-based load distribution is enabled, the server for all RADIUS authentication and accounting requests is selected using a round-robin distribution method; first the primary server is selected, then the secondary server, and then the primary server is selected again, and so forth. This means that server selection only depends on the previous RADIUS request. The authentication and accounting functions are handled separately from each other. This also applies when the same servers are used for both functions. The initial RADIUS Accounting-Request INTERIM and STOP may go to a different

server than the Accounting-Request START goes to. There is no guarantee that all RADIUS accounting requests related to the same bearer are sent to the same RADIUS server.

The transaction-based load balancing algorithm works per accounting session ID and not per system or per MSCP/MG VM. This means that the feature achieves statistical distribution as described, but the end result is not directly applicable to individual UEs.

The load distribution mode affects retries. By default, the retries are always sent to the same server as the initial request. The transaction-based load distribution applies to the retries as described in the previous paragraph, switching the destination server for each attempt.

**NOTE:**

When the server is changed, the request is removed from the transmission window of the previous server. Therefore, any response for that message will be dropped because the matching request is no longer in the transmission window.

The retransmission interval must be high enough to maximize the latency time between the RADIUS client and server. If no other servers are available, then server re-selection is not performed and retransmission is performed to the same server.

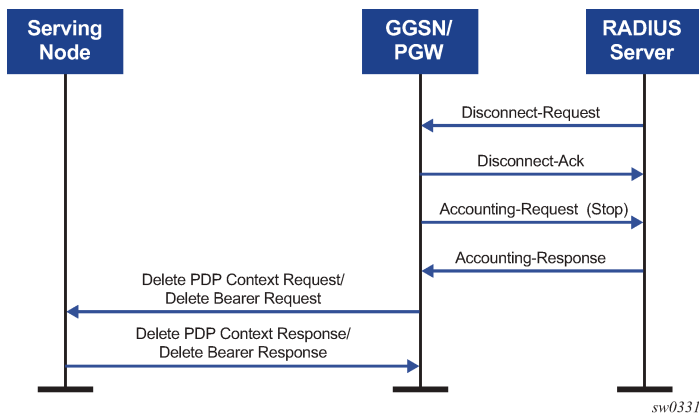
To configure the PGW/GGSN load balancing behavior for RADIUS authentication and accounting messages, use the following command:

```
config>mobile>profile>radius-group>transaction-based-load-balancing
```

5.18.5 RADIUS disconnect

The GGSN/PGW supports a RADIUS server-initiated Disconnect-Request message for the PDN connection (not for individual bearer or PDP context) termination triggered by external control servers. Disconnect requests are accepted from any of the servers configured in the RADIUS group used for accounting by the APN.

The Disconnect-ACK message is sent as a response to the Disconnect-Request message, if the disconnect succeeds. The associated PDN connection is deleted.

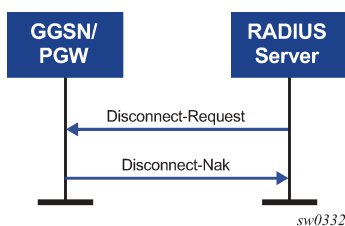
Figure 50: Disconnect-ACK message call flow

The system does not wait for a response for the delete request toward the serving node before responding to the RADIUS server with an ACK message.

A Disconnect-NAK message is sent as a response to a Disconnect-Request message when the GGSN/PGW cannot disconnect the PDN connection identified by the received Acct-Session-Id attribute in the Disconnect-Request message. A Disconnect-NAK message will be sent, for example, if there are no resources, the PDN connection does not exist, or the Disconnect-Request originated from a server that does not allow disconnect for the bearer for the bearer.

The Acct-Session-Id attribute is mandatory in the Disconnect-Request message and is used to find the target PDN connection. The additional attributes, for example, Framed-IP-Address or User-Name, must also match those used by the target PDN connection, otherwise a NAK is returned and the PDN connection is not removed.

The Error-Cause attribute indicates the reason for sending a NAK. If a Disconnect-Request is not accepted by the GGSN/PGW, the existing PDP connection is left active.

Figure 51: Disconnect-Request and Disconnect-NAK messages

If there are any Proxy-State attributes in the Disconnect-Request received from the RADIUS server, the GGSN/PGW includes those attributes in its response to the server.

CMG provides the option to configure some of the ACK/NAK AVP options using the commands under the following contexts:

```
config>mobile>profile>radius-group>avp-options>disconnect-ack-avp-options
```

```
config>mobile>profile>radius-group>avp-options>disconnect-nak-avp-options
```

The ACK/NAK AVP options are used only when the `use-disconnect-ack-avp-options` and `use-disconnect-nak-avp-options` commands are enabled under the `config>mobile>profile>radius-group>avp-options` context.

5.19 CDR configuration

CDR types are supported by the different gateway types. The CDR configuration is a PDN-level configuration.

```
config>mobile>pdn>charging
  cdr-type {pgwCdr|gCdr|eGCdr}
```

For more information about CDR, Rf messages, and field descriptions, refer to the *CMG CDR Description*.

Related information

CDR generation

MG allows generation of SGW-CDRs, PGW-CDRs, TDF-CDRs, and EPDG-CDRs.

Offline charging record transfer

The mobile gateway supports the Rf, Ga and Bp interfaces for transferring offline charging records.

Appendix A: Messages and information element reference

5.20 P-CSCF selection

A UE must register with a P-CSCF to make outbound IMS calls (UE-originated call) or to receive inbound IMS calls (UE-terminating call). The PGW provides the UE the P-CSCF addresses during the session creation. The (maximum) number of addresses to be sent to the UE is configured with parameter `max-allocated-pcscf-addresses`.

The PGW can learn the FQDN of the P-CSCF, via the S6b reference point, to discover a list of available P-CSCF addresses. The PGW can also be configured with a list of P-CSCF addresses or FQDN for dynamically resolving the addresses using DNS. This includes the ability to associate a weight and priority to each address so that the PGW allocates P-CSCF according to the priorities, and/or in a weighted round-robin fashion with the more highly weighted addresses being allocated more often.

The PGW can further select different P-CSCF groups using the UE's location, by configuring lists of Tracking Area Identities (TAI) and Location Area Identities (LAI) and associating a set of P-CSCF groups with each list.

To configure these lists, use the following command:

```
config>mobile>pdn>apn>pco-option>ue-location>tai-lai-list
```

To associate a set of P-CSCF groups with each list, use the following commands:

```
config>mobile>pdn>apn>pco-option>ue-location>tai-lai-list>pcscf-srvr-v4-group
```

```
config>mobile>pdn>apn>pco-option>ue-location>tai-lai-list>pcscf-srvr-v6-group
```

The local configuration of the P-CSCF addresses in the PGW can be done using the commands in context `config>mobile>pdn>apn>pco`, either by directly defining the server IP address (using commands `pcscf-srvr-ipv4` and `pcscf-srvr-ipv6`) or by linking to a separate P-CSCF group profile (using the `pcscf-srvr-v4-group` and `pcscf-srvr-v6-group` commands).

For example when prioritizing server groups, the operator can deploy multiple P-CSCF servers per location or data center, as well as additional (remote) ones for redundancy (for example, to protect against a site-level power outage).



NOTE:

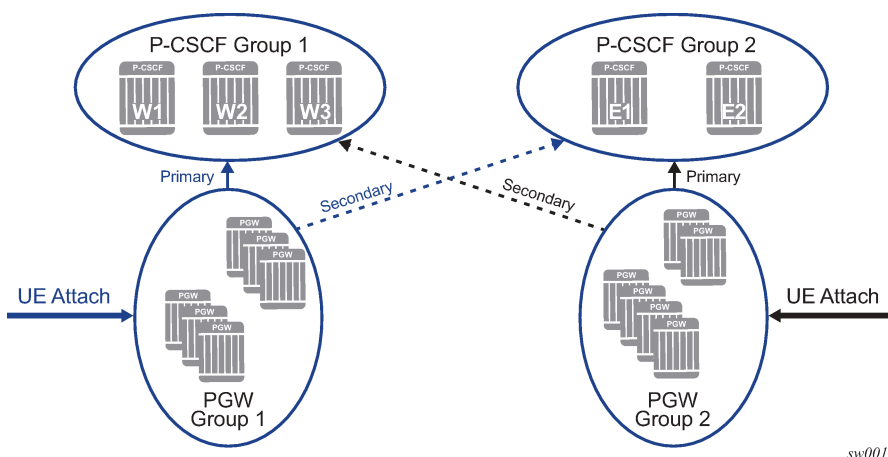
The modification of the P-CSCF group address type from IP address to FQDN or the opposite is different when using classic CLI or MD-CLI.

When using classic CLI, you have to remove first the IP address from the profile and then add the new FQDN to this profile. If this profile is linked to an APN, you must also unlink the APN from the profile, otherwise you cannot remove the last entry of the profile.

When using MD-CLI, the configured IP address is removed automatically and replaced by the FQDN without any notification.

P-CSCF server pools shows P-CSCF server pools located at West data center and East data center.

Figure 52: P-CSCF server pools



By linking different APNs to different groups, or creating multiple groups within one APN so that these groups are used with different priorities, it is possible to achieve load balancing between all the primary servers.

P-CSCF selection based on user range

You can configure the P-CSCF addresses sent to the UE per user range, based on the IMSI, MSISDN, or IMEI selection criteria. One P-CSCF IPv4 server group and one P-CSCF IPv6 server group is assigned to every user range, along with their optional secondary and tertiary groups. The P-CSCF server groups must be defined under the `config>mobile>profile` context, before associating them with a user - range.

To associate a PCSCF group with a user range, use the following command:

```
config>mobile>pdn>apn>pco>user - range
```

P-CSCF selection based on PCRF/PCF remote peer

You can configure the P-CSCF addresses sent to the UE based on the PCRF/PCF peer the PGW-C/SMF has communicated with. You can create one entry per APN and per Diameter peer or NF ID for these P-CSCF addresses. One `pcscf-srvr-v4-group` and/or one `pcscf-srvr-v6-group` is assigned to every Diameter peer or `nf-id`, along with their optional secondary and tertiary groups. The P-CSCF server groups must be defined under the `config>mobile>profile` context, before associating them with a Diameter peer or NF ID.

To configure an entry of P-CSCF addresses per APN and per specific Diameter peer/`nf-id`, use the following command:

```
config>mobile>pdn>pcf-pcrf-based-pcscf-allocation>entry
```

Configuration guidelines

The priority value (that is, the number of priority groups) dictates the allocation behavior. The main guideline is to allocate one address from each priority group which ensures that the system's current behavior is maintained. If there is only one server group, all addresses are taken from that group (and the weights are only applied to the first address). In this case, the priority value of the group does not apply.

[Configuration examples for P-CSCF selection](#) shows different configuration examples for the allocation of P-CSCF addresses. The allocation depends on the number of maximum P-CSCF addresses that are included in the PCO IE and the number of priority groups.

Table 17: Configuration examples for P-CSCF selection

Number of max. allocated P-CSCF addresses	Number of priority groups	P-CSCF address allocation
3	3	One address is taken from each group
3	2	Two addresses are taken from the higher priority

Table 17: Configuration examples for P-CSCF selection (continued)

Number of max. allocated P-CSCF addresses	Number of priority groups	P-CSCF address allocation
		group, and one from the lower priority group In case there is only one address in the higher priority group, two are taken from the lower priority group If this has only one address, only two addresses are finally sent to the UE
2	1	Both addresses are taken from the same group
2	3	Addresses are taken from the two highest priority groups

The mechanisms for the P-CSCF restoration are described in the [P-CSCF restoration](#) section. The restoration mechanisms use similar logic for the re-selection of the P-CSCF addresses as it is used for the initial server list selection.

Set the maximum number of P-CSCF addresses that will be included in the PCO IE using the following command:

```
config>mobile>pdn>pco-option>max-allocated-pcscf-addresses
```

Associate a P-CSCF server group to an APN and specify its priority using the following commands:

```
config>mobile>pdn>apn>pco>pcscf-srvr-v4-group
```

```
config>mobile>pdn>apn>pco>pcscf-srvr-v6-group
```

Verifying P-CSCF addresses

To list all the P-CSCF IPv4/IPv6 addresses allocated to the UE, use the following command:

```
show>mobile>pdn>pdn-context>imsi>detail
```



NOTE:

The P-CSCF server address information will only be shown when the P-CSCF server addresses are allocated to UE during the attach.

```
#show mobile-gateway pdn pdn-context imsi 110001234567000 detail
=====
PDN context detail
=====
IMSI                : 110001234567000
APN                 : nokia.com
Virtual APN         : N/A
Virtual APN sel     : -
Auth Status         : IMSI authenticated
IMEI                : 987654321012399
MSISDN              : 12345098761
PDN type            : IPv4                      LBI                : 5
Ref-point/Sig-pr*   : S5/GTPv2                  APN restriction     : 0
UL APN AMBR         : 5000 Kbps                  DL APN AMBR         : 5000 Kbps
Bearer contexts     : 1                        SDFs                : 0
Piggyback bearers requested : 0
Piggyback bearers creation successful : 0
Pri DNS IPv4 addr: 10.30.1.1                    Sec DNS IPv4 addr: 10.30.1.2
Pri DNS IPv6 addr: ::                            Sec DNS IPv6 addr: ::
Pri NBNS IPv4 ad*: 0.0.0.0                      Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: ::                            Sec NBNS IPv6 ad*: ::
Charging bearer * : Visiting

UE IPv4 address      : 5.127.255.0
S5 PGW Ctrl Teid    : 0x14500                    S5 SGW Ctrl Teid    : 0x10d00
S5 PGW V4 Ctl Adr: 16.16.16.16                  S5 SGW V4 Ctl Adr: 10.10.5.1
PCRF triggers       : N/A
Gx primary peer profile : Peer-Profile1

Gx secondary peer profile :

Gx peer profile in use : Peer-Profile1
```

```
Gx failover count      : 0

Gx peer origin         : Local configuration
PCRF hostname          :
PCRF address           : 10.10.43.2

PCRF port              : 3868

PCRF destination host  : primary-pcrf.nokia.com
Gx local address       : 16.16.16.19

PCRF enabled Gy Event Reporting :

Charging profile : 0                      Interim records *: 0
RF triggers      : 0

S6b primary peer profile : s6b_Peer-Profile1

S6b secondary peer profile :

S6b peer profile in use   : s6b_Peer-Profile1

S6b failover count       : 0

S6b peer hostname        :

S6b peer address         : 10.23.19.2

S6b peer port            : 3868

S6b peer dest host       : aaa.nokia.com
S6b peer origin          : diam.origRealm.com
S6b local address        : 16.16.16.25

Framed Routes           : 0
Redir Mob Ipv4 Addr     : ::
Redir Mob Ipv6 Addr     : ::
P-CSCF Ipv4 Addr        : 85.1.1.2
P-CSCF Ipv4 Addr        : 85.1.1.16
P-CSCF Ipv4 Addr        : 85.1.1.12
P-CSCF Ipv6 Addr        : 8570::1:1:2
P-CSCF Ipv6 Addr        : 8570::1:1:6
P-CSCF Ipv6 Addr        : 8570::1:1:1
L2TP Connection *: 0
```

```
Restoration Priority Level : 0
P-CSCF Reselection       : Y
Registered P-CSCF Addr   : ::
eMPS IMS priority session : No
Alloc ePDG svr Ipv4 Addr  : 0.0.0.0
Alloc ePDG svr Ipv6 Addr  : ::
Override Rating Group     : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
```

```
-----
Number of PDN contexts : 1
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

5.21 P-CSCF restoration

This section describes the P-CSCF restoration functionality.

When the PGW is informed of a failure that affects a UE's registered P-CSCF, it can update the list of P-CSCF addresses assigned to a UE, and send the new list of P-CSCF addresses to the UE so that it registers again to an available P-CSCF.

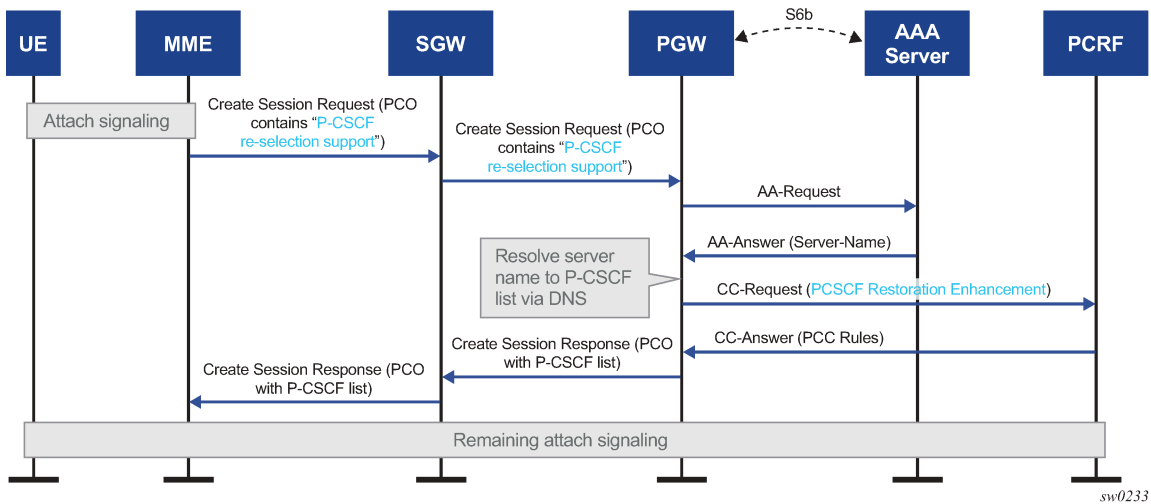
This feature provides support for the following:

- standardized and vendor-specific GTP restoration indication
- a standard mechanism whereby the PCRF identifies the P-CSCF server (used by UE) after the UE registration with VoLTE/IMS services
- 3GPP defined PCO option for the UE to indicate that it supports P-CSCF restoration
- non-3GPP accesses
- restoration indication from the MME (for LTE access)
- restoration indication from the AAA (for WLAN access)
- restoration indication from the PCRF (for LTE and WLAN access)

5.21.1 Initial attach and additional PDN connection — 3GPP access

The UE can indicate support for P-CSCF re-selection in the PCO. Therefore, the initial attach and additional PDN connection call flow must support this PCO attribute.

Figure 53: P-CSCF server assignment with P-CSCF restoration support indication



The PGW can be configured to indicate its support for P-CSCF restoration via the PCRF.

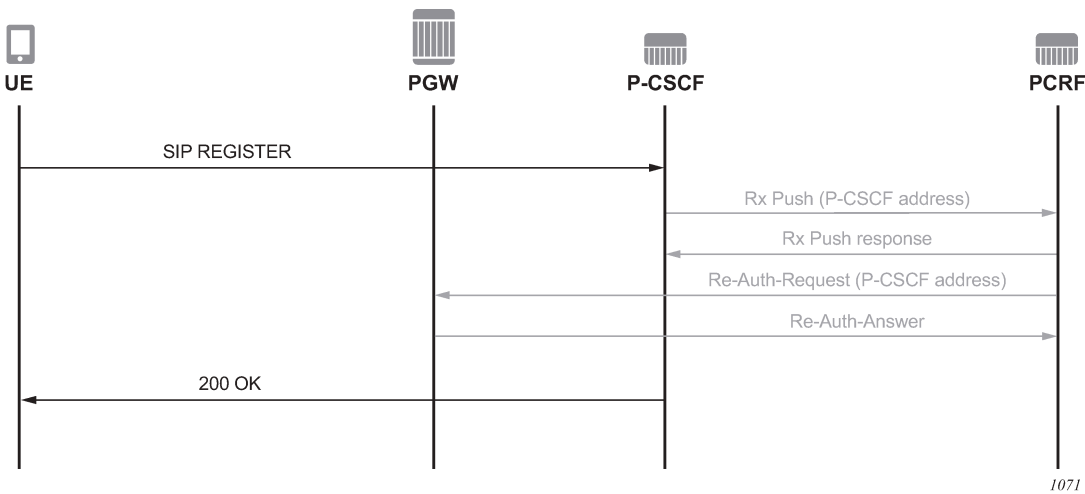
The PGW can obtain a P-CSCF server FQDN as part of the S6b exchange. Upon receiving an FQDN, the PGW resolves it to a list of P-CSCF addresses via DNS. If the FQDN had been resolved recently, the PGW uses the list in its DNS cache. The PGW performs its normal interaction with the PCRF. The PGW must provide a list of P-CSCF addresses in the Create Session Response.

Alternatively, the set of P-CSCF addresses can also be derived by configuring them via the PCO options.

5.21.2 Determining P-CSCF address to be used by the UE

The P-CSCF restoration method only informs the PGW that the P-CSCF for the UE's PDN connection has failed. The P-CSCF restoration method does not inform the PGW about which P-CSCF has failed; it assumes that the PGW already knows which P-CSCF is being used by the UE. If the PGW allocates multiple P-CSCF addresses to the UE during the attach procedure, it does not know which one the UE has used for IMS registration. 3GPP TS 23.380 describes how the PCRF can inform the PGW of the chosen P-CSCF after attach. Upon IMS registration by the UE, the P-CSCF will inform the PCRF that the UE has registered. The PCRF will then send a RAR to the PGW that contains the P-CSCF IP address.

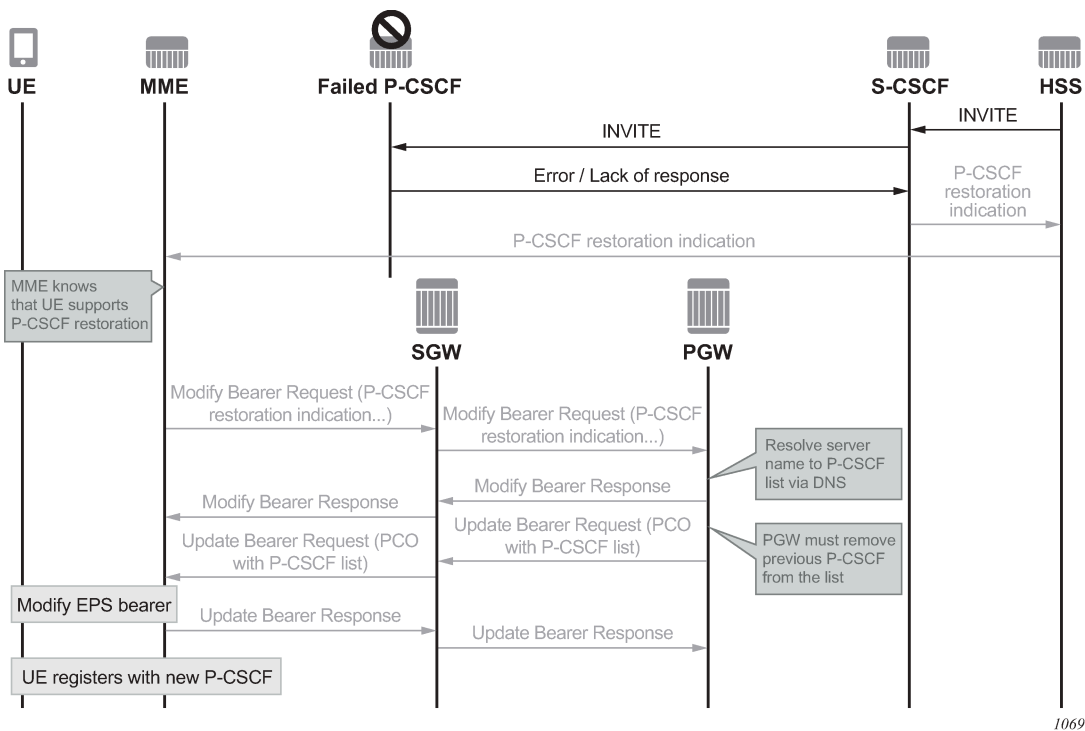
Figure 54: The PCRF indicates the chosen P-CSCF to the PGW



1071

5.21.3 P-CSCF failure detection upon terminating call

Figure 55: Terminating call P-CSCF failure with P-CSCF restoration (MME-based)



1069

1. The S-CSCF determines that the P-CSCF cannot handle the terminating call, either because it returned an error or because it failed to respond altogether.
2. The S-CSCF sends the P-CSCF restoration indication to the HSS.
3. The HSS sends the P-CSCF restoration indication to the MME.
4. The MME sends a Modify Bearer Request that includes the P-CSCF restoration indication.

5. The SGW forwards the MBR to the PGW.
 6. The PGW may choose to resolve a new list of P-CSCF addresses based on the FQDN previously received via S6b or via local configuration.
 7. The PGW creates a new list that does not include the registered P-CSCF address.
 8. The PGW sends an Update Bearer Request with a PCO containing the new list.
 9. The MME forwards the new list to the UE and the UE registers with a new P-CSCF.
- Alternatively, the PCRF may receive a P-CSCF restoration indication from the S-CSCF.

Figure 56: Terminating call P-CSCF failure with P-CSCF restoration (PCRF-based)

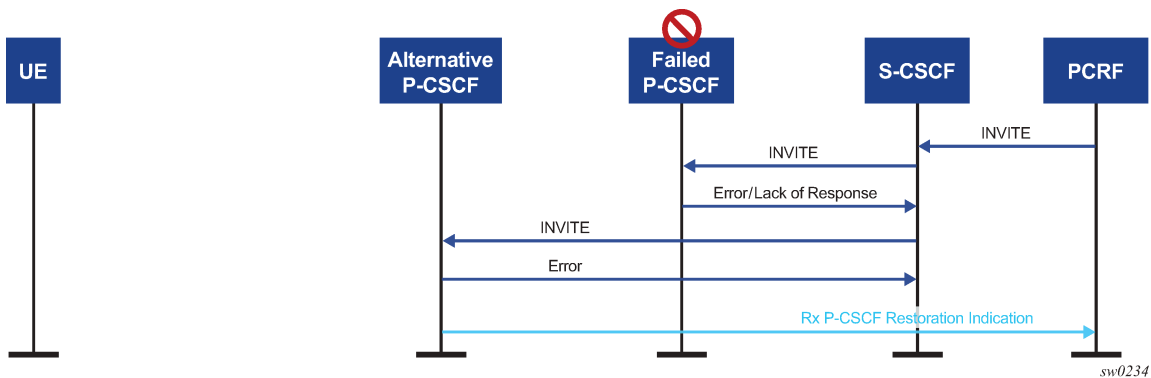
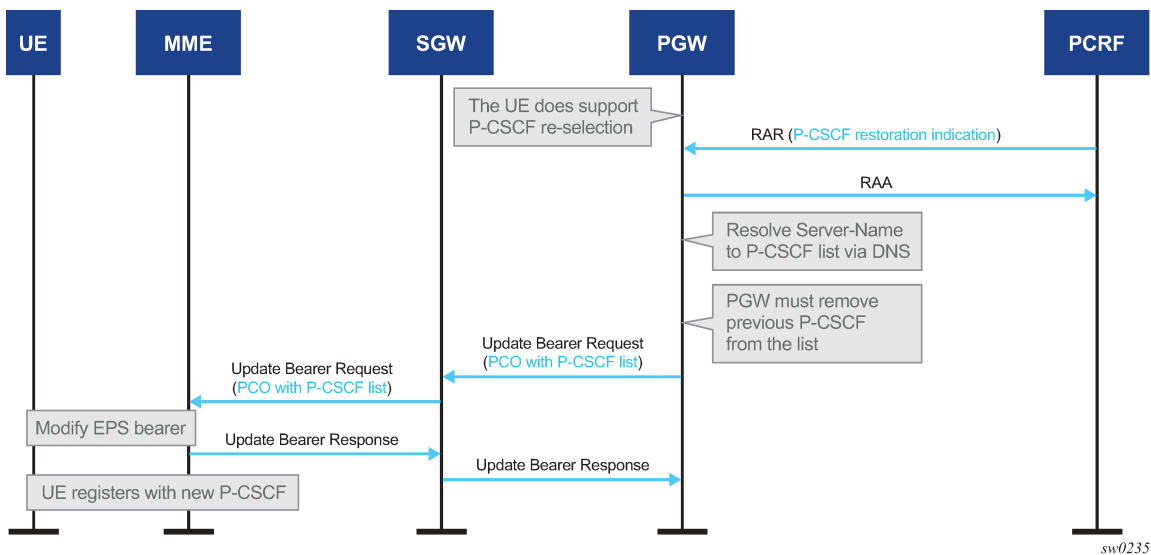


Figure 57: Terminating call P-CSCF failure with P-CSCF restoration (Gx-based — Update)



If the PGW indicates support for P-CSCF Restoration Enhancement, the PCRF will send the P-CSCF restoration indication via a ReAuth-Request message.

If the UE indicates support for P-CSCF re-selection, the PGW will then send an updated list of the P-CSCF addresses. If the UE does not indicate P-CSCF re-selection support, then the PGW can be configured to disconnect the PDN.

5.21.4 Non-3GPP access and terminating call failure

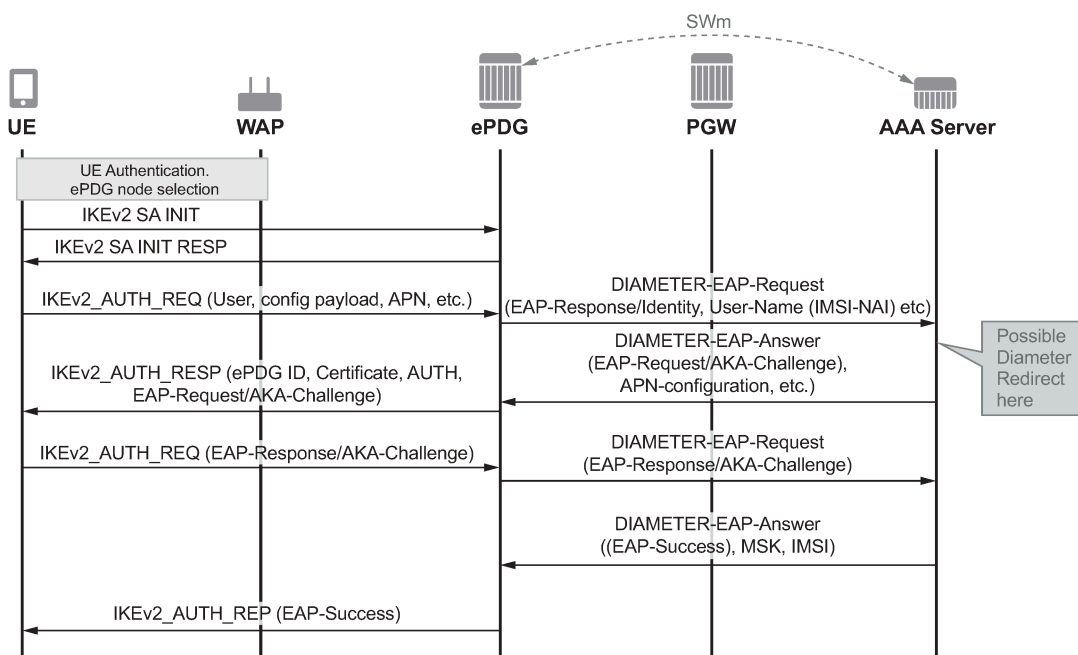
Although the 3GPP does not define a restoration option for non-3GPP access, it is assumed that the PGW is to support the HSS-based P-CSCF restoration mechanism, as there is no requirement for the PCRF-based P-CSCF restoration mechanism.

There are two methods of restoration defined by the 3GPP. The first method is a basic P-CSCF restoration whereby the PGW simply disconnects the PDN session and asks the UE to reactivate it. This is done if the UE has not indicated that it can support the optional extension. The second method is the optional extension, whereby the UE indicates support of restoration and the PGW sends an Update Bearer Request toward the ePDG with a new list of P-CSCF addresses.

5.21.4.1 Basic restoration call flows

Normal PDN activation — UE not indicating support for P-CSCF re-selection shows a PDN activation call flow in which the UE does not indicate support for P-CSCF restoration.

Figure 58: Normal PDN activation — UE not indicating support for P-CSCF re-selection

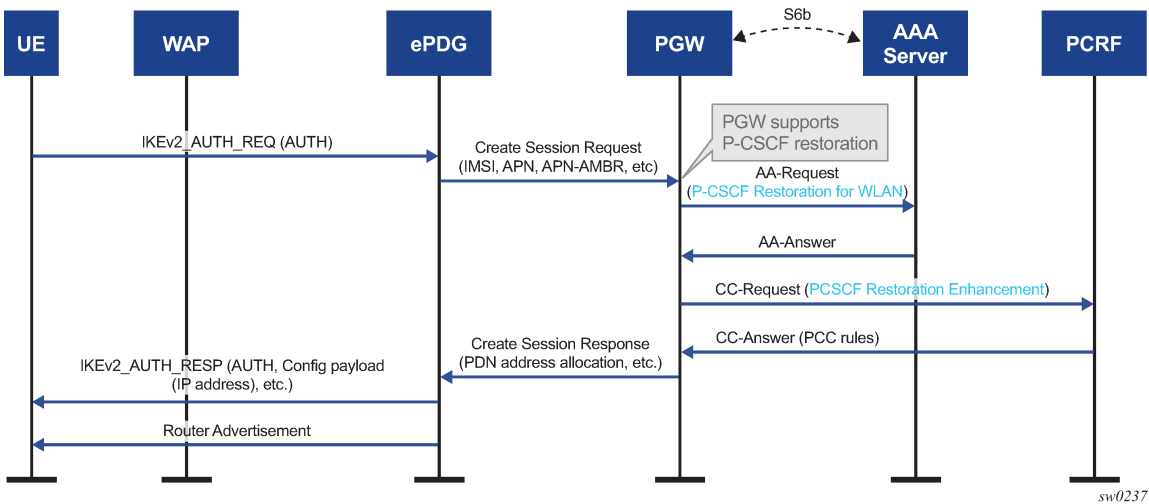


1073

The PGW can be configured to indicate support for P-CSCF restoration via both the S6b and Gx interfaces.

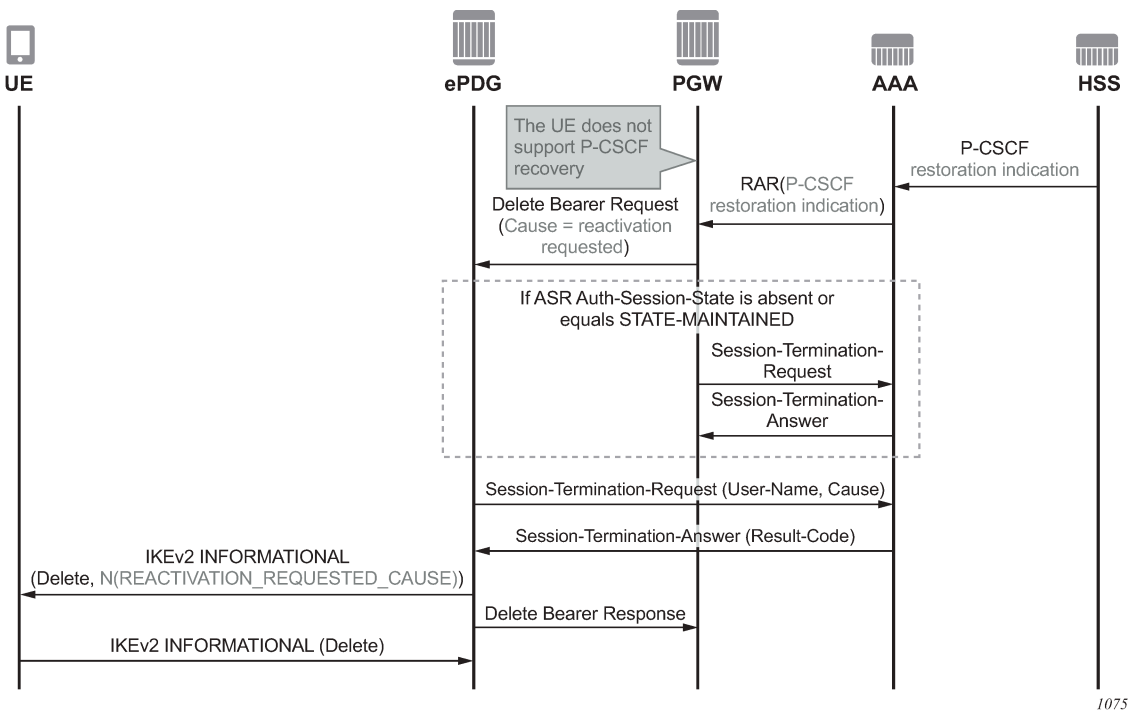
UE attach/tunnel establishment, Part 2 — PGW supports P-CSCF restoration shows PGW support for P-CSCF restoration through the S6b Supported-Features AVP.

Figure 59: UE attach/tunnel establishment, Part 2 — PGW supports P-CSCF restoration



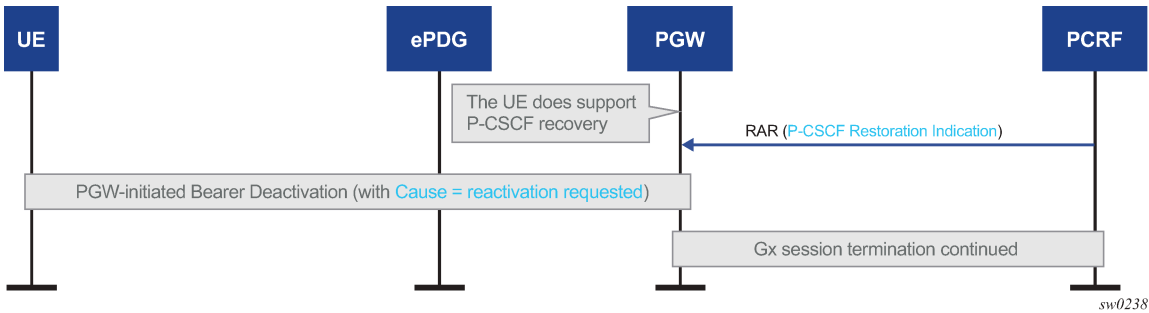
Untrusted access P-CSCF restoration (S6b-based — Disconnect) shows how the PGW receives a P-CSCF S6b-based restoration indication from the AAA in the RAR-flags AVP.

Figure 60: Untrusted access P-CSCF restoration (S6b-based — Disconnect)



Untrusted access P-CSCF restoration (Gx-based — Disconnect) shows how the PGW receives a P-CSCF Gx-based restoration indication from the AAA in the RAR-flags AVP.

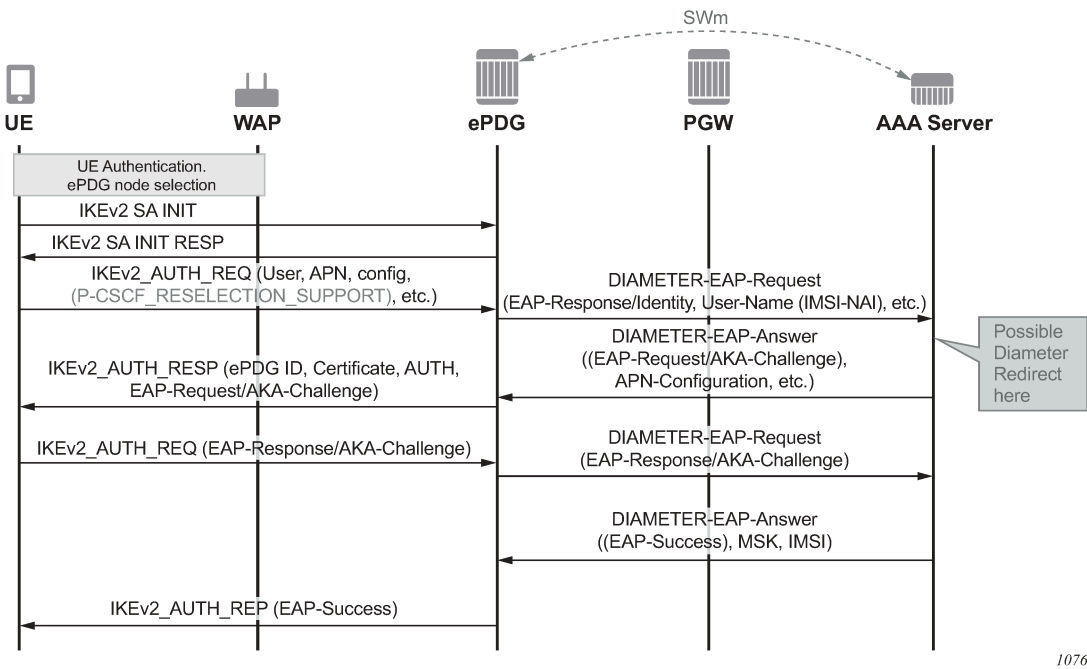
Figure 61: Untrusted access P-CSCF restoration (Gx-based — Disconnect)



5.21.4.2 Extended restoration call flows

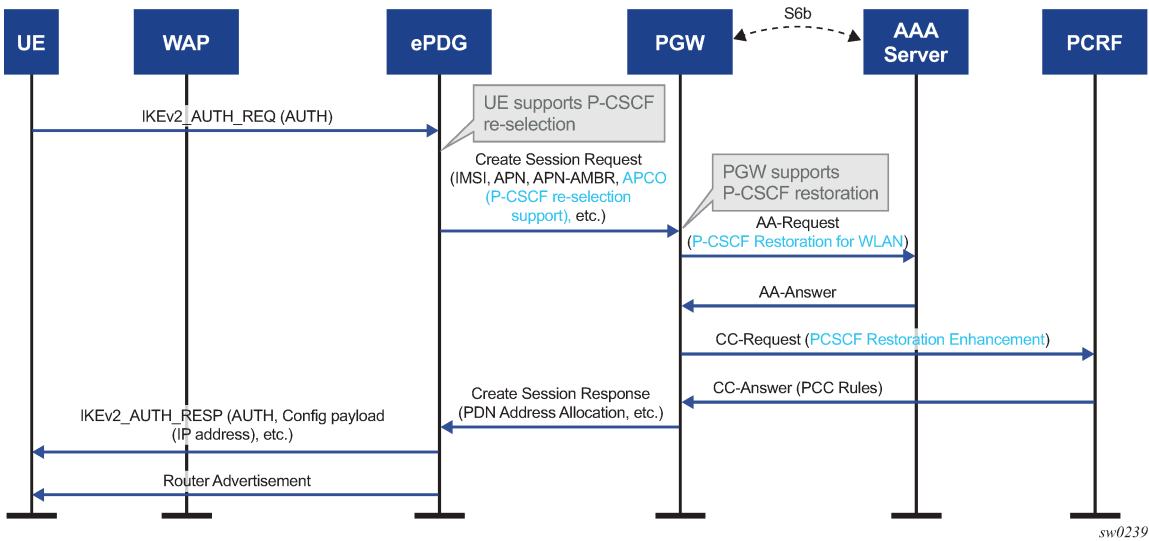
UE attach/tunnel establishment, Part 1 — UE supports re-Selection shows a PDN activation call flow in which the UE indicates support for P-CSCF restoration via a Notify message type.

Figure 62: UE attach/tunnel establishment, Part 1 — UE supports re-Selection



UE attach/tunnel establishment, Part 2 — UE supports P-CSCF re-selection and PGW supports P-CSCF restoration shows how the ePDG relays the UE's indication to the PGW via APCO, and if the PGW supports P-CSCF restoration, it indicates support to the AAA.

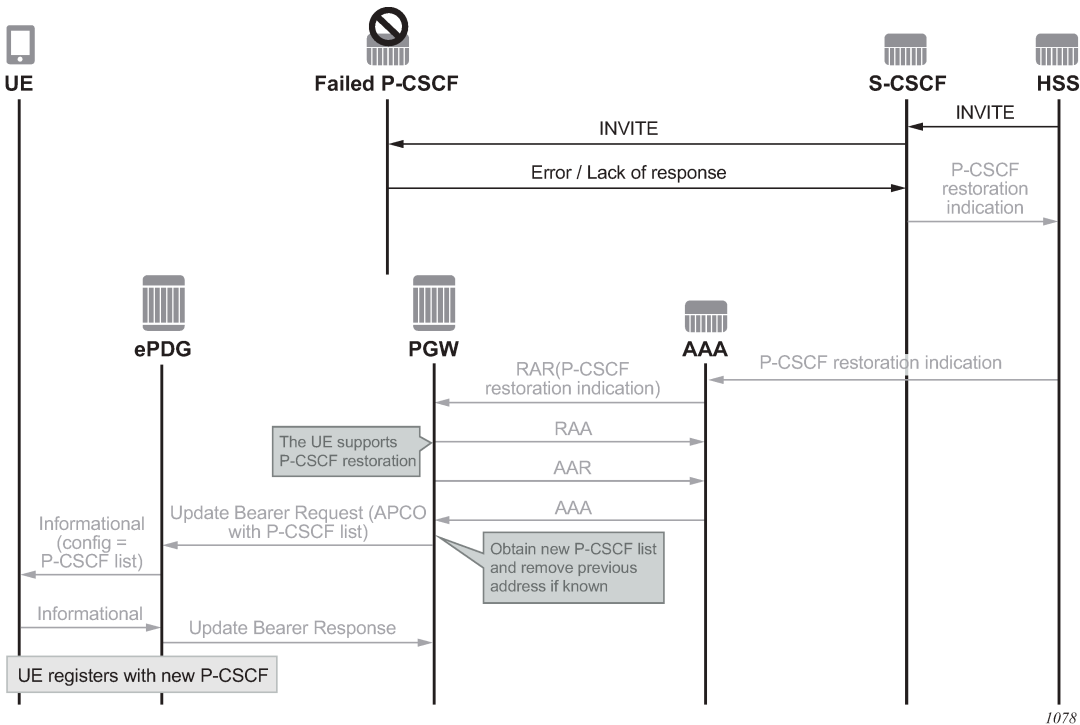
Figure 63: UE attach/tunnel establishment, Part 2 — UE supports P-CSCF re-selection and PGW supports P-CSCF restoration



The PGW can be configured to indicate support for P-CSCF restoration via both the S6b and Gx interfaces.

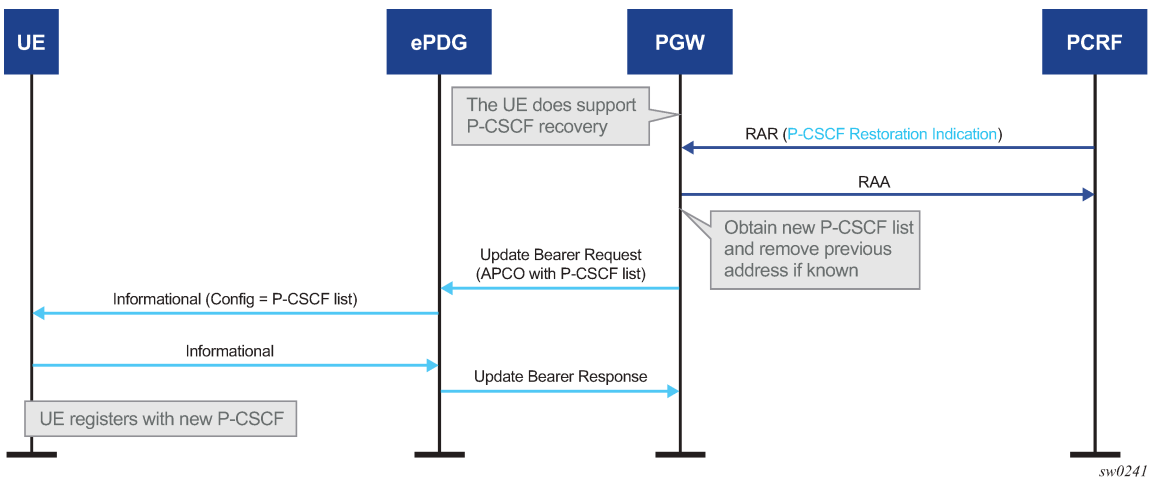
Untrusted access P-CSCF restoration (S6b-based — Extended) shows P-CSCF restoration support via the S6b interface.

Figure 64: Untrusted access P-CSCF restoration (S6b-based — Extended)



Untrusted access P-CSCF restoration (Gx-based — Extended) shows P-CSCF restoration support via the Gx interface.

Figure 65: Untrusted access P-CSCF restoration (Gx-based — Extended)



5.22 DNS interface profile

Use a DNS lookup profile for mobile gateway-specific CP sessions and allow flexible placement of the DNS server in any routing instance on the gateway.

A DNS lookup profile is configured with a list of DNS servers and parameters to use for DNS lookups and is configured per gateway instance.

DNS lookup for the following services is based on the configured DNS profile:

- Diameter peer FQDN resolution
- P-CSCF FQDN resolution
- PGW selection on ePDG using FQDN/NAPTR resolution

Services such as telnet, SSH, and FTP will continue to use the default DNS configuration in the base routing instance.

To configure a DNS profile for the mobile gateway, use the following commands:

```
config>mobile>profile>dns-profile
```

```
config>mobile>profile>dns-profile>ip-dscp
```

```
config>mobile>profile>dns-profile>ip-ttl
```

```
config>mobile>profile>dns-profile>message-retransmit
```

```
config>mobile>profile>dns-profile>primary-dns
```

```
config>mobile>profile>dns-profile>secondary-dns
```

```
config>mobile>profile>dns-profile>tertiary-dns
```

To configure a DNS profile to use for all DNS lookups on the control plane for specific reference point interfaces and to configure the source interface to use with the DNS profile for DNS lookups, use the following commands:

```
config>mobile>pdn>dns-client>dns-interface
```

```
config>mobile>pdn>dns-client>dns-profile
```

To view DNS profile information, use the following command:

```
show>mobile>profile>dns-profile
```

5.23 Gateway configuration example

The following output examples display basic gateway configuration including the following components. For additional configuration examples for specific gateways, see the related gateway sections.

- Physical Interface configuration {line cards, IP address}
- Logical IP Interface configuration {loopback IP interface}
- Routing Protocol configuration {static route, OSPF}
- Mobility gateway profiles, instance

```
#-----
echo "Card Configuration"
#-----
```

```

card 1
    card-type iom3-xp
    mda 1
        mda-type m20-1gb-xp-tx
    exit
exit
card 2
    card-type iom3-xp
    mda 1
        mda-type m20-1gb-xp-tx
    exit
exit
card 3
    card-type ism-mg-2pac-fp3
    mda 1
    exit
exit
#config>router# info
#-----
echo "IP Configuration"
#-----
    interface "s11"
        address 11.115.14.1/24
        port 2/1/4
    exit
    interface "s5"
        address 11.115.24.1/24
        port 1/1/4
        ipv6
            address 3FFE::B73:1801/64
        exit
    exit
    interface "system"
        address 11.20.1.15/32
        ipv6
            address EF15::1/128
        exit
#-----
echo "Static IPv6 Route Configuration"
#-----
    static-route 3FFE::B73:1802/128 next-hop 11.115.24.2
-----
#config# mobile-gateway
#config>mobile# info
-----

```

```

profile
    diameter "prof1"
        connection-timer 2
    exit
    diameter-peer "peer1"
        destination-realm "nokia.com"
        diameter-profile "prof1"
        peer 11.115.34.2
    exit
exit
system
    group 1 redundancy one-to-one gateway 1
        card 3 active
    exit
exit
pdn 1
    signaling
        diameter
            origin-host "diam.origRealm.com"
            origin-realm "origRealm.com"
            profile "prof1"
        exit
        interface "system"
    exit
    qci-policy "default"
    s11
        interface
            gtp-c "s11"
        exit
    exit
    s5
        interface
    exit
    exit
    no shutdown
exit

```

5.24 Graceful shutdown and administrative session teardown

Graceful shutdown means shutting down the services provided by the gateway with controlled session teardown. Limiting the rate of session teardowns helps to avoid causing disturbance to the surrounding network.

5.24.1 Session deletion priority

Four different priorities (critical, high, medium, or normal) can be assigned when several sessions are deleted, after a path failure.

The order of the session deletion priority is defined per QCI/ARP, by using the following configuration:

```
configure mobile-gateway pdn gw-id mass-deactivation qci qci-value [arp-pl arp-pl-value] session-delete-priority priority-value
```

With this configuration, the sessions with the highest priority are cleared first to re-establish the EU faster. For example, sessions are deleted according to the following priority:

- emergency calls as critical
- MPS calls as high
- IMS calls as medium
- other services like data services as normal

Sessions that match the QCI or QCI/ARP values are assigned with the defined priority value. The default session deletion priority is normal.

This configuration applies to:

- the PDN level clear command without filters (`clear mobile pdn bearer-context`)
- the APN level clear command without filters (`clear mobile pdn bearer-context apn apn1`)
- the clear commands at the APN level with group and PDN-type parameters (`clear mobile pdn bearer-context apn apn1 group1 pdn-type ipv4ipv6`)

This configuration also applies when the sessions are deleted due to path failure.



NOTE:

The above configuration does not apply to shutdown commands.

Related information

[Commands for administrative session deletion](#)

5.24.2 Gateway suspend state

A more graceful alternative for shutdown is to put the gateway in a suspended state and then gradually take the gateway out of service by clearing all the sessions with a `clear` command.

The suspended mode allows the gateway to serve the existing sessions and forward user traffic while the sessions are being deleted or other maintenance work is carried out.

The following command can be used to put the gateway in a suspended state.

```
configure mobile-gateway pdn suspend
```

**NOTE:**

The suspended mode is applied only in the control plane gateway if the control and user plane gateway are separate.

While in a suspended state, the gateway (PGW, SGW or combined SGW/PGW) does the following:

- keeps serving existing sessions for an uninterrupted end-user service. More specifically:
 - accepts all non-zero TEID Create Session Requests; for example, it allows creation of dedicated bearers or secondary PDP contexts
 - accepts new sessions for handover cases, that is, Create Session Requests with zero TEID and Handover Indication on S5, S2a, and S2b (applies to PGW only)
 - when in ePDG role, the gateway accepts a new attach in IKE for Handover, MOBIKE-related messages, AAA-initiated messages requiring further actions on IKE or GTP, and allows dedicated bearer creation
- rejects all other Create Session Requests; for example, a Create Session Request for a new UE is rejected. Handovers on the S11 interface are also always rejected (a different SGW can be selected by the MME).

While in a suspended state the ePDG does the following:

- rejects new Initial attach in the SWu, by sending an IKE_AUTH_Response that includes a Notify payload with a Private Notify Message Type "NETWORK_FAILURE"
- does not automatically tear down existing sessions. To continue with the graceful shutdown of the gateway, the operator needs to manually delete the session by using `clear>mobile>pdn>bearer-context` commands.

The suspended state is not supported by the TWAG, and SSG gateways and the suspend command does not have any effect on them.

While in a suspended state, the SMF does the following:

- keeps serving existing sessions for an uninterrupted end-user service. More specifically:
 - if an EPS or non-3GPP session has been created at the SMF and PGW-C, handovers and mobility to 5GS are accepted
- rejects new sessions

In the suspended state, the cause of the session rejection in the Create Session Response is by default set to 89 (Service Denied) for GTPv2 and 204 (System Failure) in GTPv1. Specifically for GTPv2, the session rejection cause for the suspended state, can be configured per interface as follows:

1. Configure the desired value in the GTP profile using the following command:

```
configure mobile-gateway profile gtp-profile gtv2-session-rejection-cause  
gw-suspend cause-code-value
```

2. Link the configured GTP profile to the specific interface; for example, s11 or s5.

In the suspended state, the SMF uses the following default values to reject new sessions:

- for the HTTP status code: 500 Internal Server Error
- for the application error: NF_FAILOVER
- for the N1 5GSM cause code: 8 (operator determined barring)

1. Configure the corresponding values in the N1 profile, using the following commands:

for the HTTP status code:

```
configure mobile-gateway profile n1-profile cause-code gw-suspend http-  
status-code http-status-code
```

for the application error:

```
configure mobile-gateway profile n1-profile cause-code gw-suspend  
application-error-code application-error-code
```

for the N1 5GSM cause code:

```
configure mobile-gateway profile n1-profile cause-code gw-suspend 5gsm-  
cause-code 5gsm-cause-code
```

2. Link the configured N1 profile to the used namf-comm-service instance.

If the V-SMF or the I-SMF is used, the V-SMF/I-SMF or H-SMF/SMF may be suspended. If the V-SMF/I-SMF is suspended, the used cause values are defined in the V-SMF/I-SMF and session rejection is not visible to the H-SMF/SMF. If the H-SMF/SMF rejects the session due to the suspension, the V-SMF/I-SMF forwards the N1 5GSM cause code received from the H-SMF/SMF. The HTTP status code and the application error received from the H-SMF/SMF are not forwarded to the AMF, because the used message towards the AMF is a request which does not include the HTTP status code or the application error.

In ICR deployments, the suspend state does not affect the operational role. A master continues to be the master after it is in the suspended state.

To view if the gateway is in the suspend state and for how long, use the following command:

```
show mobile-gateway system
```

Related information

[Commands for administrative session deletion](#)

5.24.3 Configuring the gateway shutdown

Before you start

A more drastic form of the gateway graceful shutdown can be initiated from CLI by using the shutdown command at this gateway instance:

```
configure mobile-gateway pdn shutdown
```

When started, a shutdown command runs to completion and it is not possible to cancel it. The command starts graceful deletion of the sessions on the gateway at a controlled rate. The session deletion rate is configurable under `config>mobile>pdn>session-deletion-rate`. By default, the rate is 512 session deletions per second per MSCP group.

The shutdown command puts the gateway instance in an administratively disabled state. In this state, the gateway does not accept new sessions, does not respond to path management echo requests, and switchover to standby card is not possible. The gateway forwards user traffic for sessions that are not yet deleted.

The shutdown command should be used only when the configured `session-deletion-rate` is high enough, depending on the number of the sessions served. The following method is recommended for clearing sessions from a gateway at any configured `session-deletion-rate`.



NOTE:

Some configurations on the PDN level can only be changed after shutting down the PDN; for example, when modifying a GTP profile that is referenced by a reference point.

After executing the shutdown command, the gateway will not reply to GTP echo requests. The peering network element will therefore consider the gateway dead and start deleting sessions.

Procedure

1. Put the gateway in suspend state.
2. Clear all the sessions.

Use the following command:

```
clear>mobile>pdn>bearer-context
```

5.24.4 APN shutdown

The APN shutdown can be initiated from CLI by using the shutdown command in the context of the specific APN:

```
configure mobile-gateway pdn apn shutdown
```

**NOTE:**

The APN shutdown is applied only in the control plane gateway if the control and user plane gateway are separate.

The shutdown command at the APN level affects only new sessions, meaning that the operational state of the APN related to the existing sessions remains up while the administrative state is down. The gateway continues to forward user traffic for the remaining sessions.

If the APN is in a shutdown state, the SMF uses the following default values to reject new sessions:

- for the HTTP status code: 403 Forbidden
- for the application error: DNN_NOT_SUPPORTED
- for the N1 5GSM cause code: 8 (operator determined barring)

1. Configure the corresponding values in the N1 profile, using the following commands:

for the HTTP status code:

```
configure mobile-gateway profile n1-profile cause-code apn-shutdown http-status-code http-status-code
```

for the application error:

```
configure mobile-gateway profile n1-profile cause-code apn-shutdown application-error-code application-error-code
```

for the N1 5GSM cause code:

```
configure mobile-gateway profile n1-profile cause-code apn-shutdown 5gsm-cause-code 5gsm-cause-code
```

2. Link the configured N1 profile to the used namf-comm-service instance.

If the V-SMF or I-SMF is used, the APN/DNN may be in a shutdown state in the H-SMF/SMF. If the H-SMF/SMF rejects the session due to the APN shutdown, the V-SMF/I-SMF forwards the N1 5GSM cause value received from the H-SMF/SMF. The HTTP status code and application error received from the H-SMF/SMF are not forwarded to the AMF, because the used message towards the AMF is a request which does not include HTTP status code or application error.

**NOTE:**

Some configurations on the APN level can only be changed after shutting down the APN.

5.24.4.1 Configurable back-off timer for APN shutdown

To configure the PGW to send a back-off timer value when rejecting new sessions on an APN that is in shutdown state, use the following command:

```
configure mobile-gateway pdn apn shutdown-back-off-timer timer-value
```

When this is configured on an APN and is in shutdown state, the PGW:

- rejects new create session requests with cause code 113 APN Congestion and includes the configured value of the back-off timer IE in the GTPv2 messages
- rejects new create PDP context requests with cause code 229 and includes the configured value of the back-off timer IE in the GTPv1 messages

5.24.5 Commands for administrative session deletion

The commands under the `clear>mobile>pdn>bearer-context` context can be used to perform the following:

- clear all the sessions from a gateway instance; for example, to continue the graceful gateway shutdown
- clear all the sessions from an APN; for example, to gracefully shutdown a specific APN on a PGW
- clear a group of sessions matching specific filtering criteria on a gateway instance or an APN
- clear a percentage of sessions from an APN; this administratively enforces a faster way of load balancing the sessions in more than one gateways serving the same APN.
- clear the session of a specific UE

Clearing all or a group of sessions at the gateway level is not supported on a standalone TWAG. Clearing all or a group of sessions at the APN level is not supported on ePDG and on standalone TWAG.

When clearing more than one session, the deletion rate is configured using the command `config>mobile>pdn>session-deletion-rate`. Alternatively, the operator can explicitly set the deletion rate using the `clear>mobile>pdn>session-deletion-rate` command. This rate is to be applied per MSCP group. The applied rate cannot exceed the configured rate at the PDN level. In general, the configured value or the value given explicitly in the `clear` command is treated by the system as the maximum rate to be applied, not as an enforced rate for the deletions. This is because there are other gateways mechanisms that take priority for defining the actual rate that is applied.

When clearing many sessions the `c clear` command may take longer to finish. The clearing of sessions will continue running on the new active card after an MSCP group switchover. For ICR deployments, the clearing will not continue after an ICR switchover on the new master chassis.

Related information

Other criteria for matching sessions to be deleted

5.24.5.1 Administrative session deletion in CUPS deployments

In CUPS deployment, the administrative massive session deletion must be initiated from the Control Plane gateway, so that all the relevant peers (User plane gateway, PCRF, Charging servers) are informed and are also tearing down their associated sessions. In 5GS, the administrative massive session deletion must be initiated from the SMF.



NOTE:

Executing clear commands on the user plane gateway result into local only session removal on the GW-U, because there is no message sent out to the GW-C to inform about the deletion. This results in a discrepancy between sessions on the GW-C and the GW-U.

5.24.5.2 Protection of dedicated bearers on clearing session

The SGW, PGW, and ePDG support the option to protect the sessions and the UEs having a dedicated bearer when a bearer-context `c clear` command is used to delete sessions.

The sessions with a dedicated bearer are left intact and will get deleted automatically, only after the dedicated bearer is deleted normally; for example, after the end of a VoLTE call. This option can be used to protect the ongoing VoLTE calls until they finish.

5.24.5.3 Other criteria for matching sessions to be deleted

The sessions that are selected for deletion can be filtered using the following criteria as parameters in the bearer-context `c clear` commands:

Idle-time

Selects sessions or UEs to be deleted based on the specified idle time. This can be used for example, to clear all sessions on an APN or all UEs on the gateway that are idle for more than 2 hours. Idle time filters are supported only when charging functionality is enabled on the gateway.

Connected time

Selects sessions or UEs to be deleted based on the connection time, for example, to clear all UEs on the gateway that were connected in the last hour.

UE location	Selects sessions or UEs to be deleted based on the UE location information. Currently, only the Tracking Area Code can be used as location selector.
Percentage	A quantitative criterion of how many sessions to delete from an MG group and an APN. When used together with the protect-dedicated-bearers option, the sessions without the dedicated bearers are deleted first. If the targeted number of sessions is not reached by deleting only the sessions without dedicated bearers, the deletion continues to the sessions with dedicated bearers, for example, by waiting the VoLTE call to end.
PCRF	Selects sessions or UEs to be deleted based on the PCRF used for the session.
P-CSCF	Selects sessions or UEs to be deleted based on the P-CSCF that is used for this session. If multiple P-CSCFs are pushed to the UE, the one that is actively used by the UE, is used to match the sessions to be deleted.

Use the following command to initiate session deletion for a selected group or all sessions on an entity (APN, gateway, MSCP group):

```
clear mobile-gateway pdn bearer-context
```

For the full list of parameters and usage descriptions for this command, see the bearer-context PDN clear commands in the *CMG CLI Reference Guide*.

Related information

Protection of dedicated bearers on clearing session

The SGW, PGW, and ePDG support the option to protect the sessions and the UEs having a dedicated bearer when a bearer-context clear command is used to delete sessions.

5.24.5.4 Gateway and APN level clearing

The bearer-context information may be cleared at the APN level using the APN parameter in the `clear>mobile>pdn>bearer-context` command. If APN is specified, the `clear` command is applied at the APN level. If not, the command is applied at the gateway level. At the APN level, the `clear` command runs over the sessions on the selected APN and matches the filters per session. For example, the idle time is applied per session and the matched sessions are cleared.

At the gateway level, the filters are applied per UE. If a filter matches a specific UE, all the sessions of that UE are cleared. For example, the `protect-dedicated-bearers` filter operates so that if a UE has a dedicated bearer at any session, all the UE sessions are protected until all its dedicated bearers are deactivated normally.

As an exception to the above rules, the **ue-clear** option is one option that can be used together with the **percentage** option at the APN level `clear` command. When the **ue-clear** is used, the APN level clear is done

until all the sessions of the selected UE are cleared. This is helpful, for example, when deleting sessions for load balancing purposes, in combined SGW/PGW deployments.

When the intention of a command is to clear all the sessions at the APN level, that is no filters are used, the APN must be in a shutdown state. When the intention of a command is to clear all the sessions at the gateway level, meaning that no filters are set, the gateway must be in suspended state.

**NOTE:**

By using the **protect-dedicated-bearers** filter without a **percentage** option, the intention is to clear all the sessions eventually. When using this filter the APN/gateway must be in a shutdown/suspended state respectively. In other cases, it is not necessary to shutdown/suspend the APN/gateway, although the use of the shutdown/suspend state is recommended.

5.24.5.5 Session clearing per APN, MG group, and PDN type

To delete sessions that belong to a specific APN and MG group and are established with a specific PDN type, use the following command:

```
clear mobile-gateway pdn bearer-context apn name group group pdn-type pdn-type
[session-deletion-rate session-deletion-rate]
```

To use the user plane activity as the trigger point for deletion, use the following command:

```
clear mobile-gateway pdn bearer-context apn name group group pdn-type pdn-type
data-packet-trigger
```

When the `data-packet-trigger` keyword is used, a session is deleted when user plane activity occurs, that is, a data packet is received on the PGW.

Parallel execution of session clearing per APN, per group, and PDN-type is supported for up to 16 parallel executions. To avoid a new session creation on a specific APN, MG-group, and PDN-type, the `block-list` must be configured.

To cancel a per APN, MG-Group and PDN-type ongoing deletions, use the following command:

```
clear mobile-gateway pdn bearer-context apn name group group pdn-type pdn-type
cancel
```

**NOTE:**

These commands are supported only in Cloud Mobile Gateway.

Related information

[APN block list](#)

Block entries under an APN are used to block new session creation on this APN on a specific MG-group and for specific PDN-Type.

Monitoring, overriding, and canceling ongoing session deletion

5.24.5.6 Session clearing per UP peer, APN, and IP prefix on the GW-C

To delete sessions on the GW-C that belong to a specific UP peer and APN and, optionally, use an IP address with a specific IP prefix, use the following command:

```
clear mobile-gateway pdn bearer context up-peer peer-ip apn apn-name [ipv4-
prefix ipv4-address/mask | ipv6-prefix ipv6-address/prefix-length]
```

This command triggers the PFCP Session Release requests for the selected sessions.

Parallel execution of session clearing per UP peer, per APN and per IP prefix is supported for up to 16 parallel executions. To avoid new session creations on a specific UP peer, APN and IP prefix, Nokia recommends before running the `clear` command, to configure the blocking for the specific peer, APN or IP prefix on the `up-peer-list`.

To cancel an ongoing session deletion triggered by the above CLI, use the following command:

```
clear mobile-gateway pdn bearer context up-peer peer-ip apn apn-name [ipv4-
prefix ipv4-address/mask | ipv6-prefix ipv6-address/prefix-length] cancel
```

To view the past and ongoing administrative deletions triggered, use the following command on the GW-C:

```
show mobile-gateway pdn clear-status
```

```
# show mobile-gateway pdn clear-status
```

```
=====
Mobile gateway clear status
=====
```

```
Gateway                               : 1
Suspended                             : No
Number of Sessions                     : 3547
Num of Sessions having a dedicated bearer : 827

Last administrative sessions clearing stats :
Started                                : 08/15/2017 02:39:28
Finished                               : 08/15/2017 02:40:33
Total Sessions to delete                : 650
Sessions deleted from APN                : 650
Sessions deleted                         : 1454
```

```
Ongoing Session Deletions
```

```
1.clearing on APN: apn1, Group: 1, PDN-type: ipv4, slow-loop
```

```
2.clearing on APN: apn2, Group: 2, PDN-type: ipv4, data-packet-trigger
3.clearing on APN: apn2, Group: 2, PDN-type: ipv4, data-packet-trigger
```

```
-----
no UP-Peer                APN                IP-prefix
-----
```

```
1.clearing on UP-Peer: 1.2.3.4, APN: apn4, IPv4-prefix: 10.10.10.1/255
2.clearing on UP-Peer: 1.2.3.5, APN: apn5, IPv6-prefix: 10:10::/128
-----
```

```
Number of gateways : 1
-----
```

```
Note:"Sessions deleted from APN" is only applicable when both "percentage" and
"apn"
are used in the clear command.
=====
```

Related information

APN block list in CUPS

In CUPS deployments, a block-list must be configured on both the GW-C and GW-U CMG instances. A block entry must be applied on the GW-C first, before it is applied on the GW-U, so that the blocked GW-U does not receive a new session request.

5.24.5.7 Monitoring, overriding, and canceling ongoing session deletion

To monitor the execution of an ongoing command, the operator can check the session counters statistics provided by the CLI show commands and by the KPI-KCI reports.

To monitor an ongoing session teardown, use the following command:

```
show mobile-gateway pdn clear-status
```

This show command provides statistics for the last clear command that was used. It also provides the list of ongoing session deletions per APN, per group, and per PDN-Type. For more information, see [Gateway and APN level clearing](#).

The list that defines ongoing command is used to cancel an ongoing clear command. For example:

```
# clear mobile pdn bearer-context apn apn1 group 1 pdn-type ipv4ipv6
# clear mobile pdn bearer-context apn apn2 group all pdn-type ipv6
# show mobile pdn clear-status
Ongoing Session Deletions:
1.clearing on apn: apn1, group: 1, pdn-type: ipv4ipv6
2.clearing on apn: apn2, group: all, pdn-type: ipv6
# clear mobile pdn bearer-context apn apn1 group 1 pdn-type ipv4ipv6 cancel
```

For any other commands and filters, canceling a clear command is not supported.

Running two or more different clear commands is supported for the per APN, per group, and per PDN-Type clearing. For any other clear commands and filters, running more than one commands at the same time, is not supported.

However, when a command is taking too long to finish, for example, `deletion-rate` is set to low value, or some VoLTE calls are too long when `protect-dedicated-bearers` is set, the command can be overridden by a subsequent command. For these cases, while there is an ongoing sessions deletion, the operator is allowed to:

- execute the command on the same entity, with the same filters and change the deletion rate and/or the **percentage** value
- execute the command on the same entity, without any filters, to select all the sessions of the entity for deletion
- execute the `clear` command at the gateway level for all sessions, or set the gateway in the shutdown state

5.25 QCI values

The EPS bearer traffic classification is performed according to QoS class identifier (QCI).

A QCI is a single scalar parameter which points to a set of QoS parameters with characteristics like Priority, Packet Delay Budget and Packet Error Loss Rate. Together, these characteristics describe the packet forwarding treatment that an Service Data Flow (SDF) aggregate receives edge-to-edge between the UE and the PCEF.

Table 18: QCI values defined by 3GPP standards

QCI value	Description
0	Reserved
1 to 4	GBR
5 to 9	non-GBR
10 to 64	Spare, for future use
65	GBR
66	
67	

Table 18: QCI values defined by 3GPP standards (continued)

QCI value	Description
68	Spare, for future use
69	non-GBR
70	
71 to 127	Spare, for future use
75	GBR
79	non-GBR
80	
82	GBR
83	
84	
85	
128 to 254	Operator specific
255	Reserved

5.25.1 Mission critical and PTT services

QCI values are available for mission critical and PTT (Push-to-Talk) services and allow you to better serve the needs of public safety services.

These QCI values can only be assigned upon request from the network. The UE and applications running on the UE are not allowed to request this QCI value.

The following QCI values are available for mission critical and PTT services:

- QCI = 65 (GBR)
- QCI = 66 (GBR)

- QCI = 69 (non-GBR)
- QCI = 70 (non-GBR)

Table 19: Feature-List-ID 1 features used in Gx

ID	Feature	Presence	Description
25	MissionCriticalQCIs	O	This feature indicates support for the Mission Critical QCI values 66, 67, 69 and 70 within the QoS-Class-Identifier AVP defined in clause 5.3.17

To configure a list of QCI values, use the following commands:

```
config>mobile>profile>list>qci-list
```

```
config>mobile>profile>list>qci-list>qci
```

5.25.2 Real-time services

3GPP TS 23.203 defines a set of standardized QCI values. 7750 SR MG considers QCI values 1, 2, 3, 5, and 7 as the key values for the network to support real-time quality of service for the end users.

Using dedicated QCI values helps to support simultaneous high speed LTE data and high quality voice and video calls and to ensure the expected Quality of Experience (QoE) levels for conversational and interactive services.

Table 20: 3GPP Standardized QCI Values

QCI	Resource type	Example services
1	GBR	Conversational voice
2		Conversational video (live streaming)
3		Real-time gaming
4		Non-conversational video (buffered streaming)

Table 20: 3GPP Standardized QCI Values (continued)

QCI	Resource type	Example services
5	Non-GBR	Video (buffered streaming), TCP-based (for example, email, chat, FTP, P2P file sharing, progressive video, and so on)
6		
7		Voice, video (live streaming), interactive gaming
8		Video (buffered streaming), TCP-based (for example, email, chat, FTP, P2P file sharing, progressive video, and so on)
9		

To provide an end to end solution, support for the respective QCIs is also required from the surrounding network elements (PCRF, MME, and eNodeB) and the UE for example for dedicated bearers, multiple APNs, and policies. The QCI value may also affect the transport QoS (IP DSCP), requiring support and consistent configurations on the transport equipment and links.

5.25.3 Operator-specific QCI to pre-release8 QoS parameter mapping

It is possible to configure a UMTS QoS policy with a mapping of operator-specific QCI values (128 to 254 as defined by 3GPP) to pre-release8 QoS parameters, for PCRF-specified QCI. This includes the traffic class, traffic handling priority, signaling-indication, SSD, and delay.

3GPP provides a recommended mapping of QCI values to pre-release8 QoS parameters in 3GPP TS 23.401 Annex E, however this does not include the operator-specific QCI values, which are considered to be the operators' responsibility.

An operator-specific QCI to pre-release8 QoS parameters mapping may be required, for example, when the PCRF assigns for certain subscribers, operator-specific QCI values that must be mapped to pre-release8 QoS for Gn sessions.

To allow multiple operator-specific QCI values to map to a specific traffic class, multiple entries are supported for each traffic class within the UMTS QoS policy profile; for example:

- QCI=128, TRAFFICCLASS=BACKGROUND, THP=NA, SSD=SPEECH, SI=TRUE, DELAY=4000
- QCI=129, TRAFFICCLASS=BACKGROUND, THP=NA, SSD=UNKNOWN, SI=FALSE, DELAY=2000
- QCI=150, TRAFFICCLASS=BACKGROUND, THP=NA, SSD=NA, SI=NA, DELAY=4000
- QCI=200, TRAFFICCLASS=INTERACTIVE, THP=HIGH, SSD=SPEECH, SI=NA, DELAY=4000



NOTE:

- The operator-specific QCI to pre-release8 QoS mapping functionality is supported for PCRF-specified QCI only.
- The operator-specific QoS profile is only used if the operator configures it; there is no default configuration.
- This functionality only considers the primary PDP QCI/ARP parameters and not the secondary PDP parameters; the gateway rejects secondary PDP Create Requests.
- Both the gateway and the PCRF must be configured to be aware of, and synchronized with, the operator-specific QCI values used.
- Show commands display the operator QCI configuration and statistics for the bearer that is applicable; for example, `show>mobile>profile>umts-qos-policy` and `show>pdn>bearer-context>bearer`.

Configuring an operator-specific QCI to pre-release8 QoS mapping

An operator-specific QCI to pre-release8 QoS parameter mapping is configured as part of a UMTS QoS policy, and is then applied at the PDN or APN level. The mapping includes the operator-specific QCI value and the traffic classification (conversational, streaming, interactive, or background). Optional attributes may also be configured including the traffic handling priority (high, normal, low, not available), signaling indication (true, false, not available), SSD (speech, unknown, not available), and delay (default is 0).

To configure an operator-specific QCI to pre-release8 QoS parameter mapping for a UMTS QoS policy, use the following commands:

```
config>mobile>profile# umts-qos-policy umts-qos-policy-name
operator-qci operator-qci traffic-class traffic-class traffic-
priority traffic-priority signaling-indication signaling-indication ssd ssd
delay delay
```

To apply a QoS policy with an operator-specific QCI to pre-release 8 QoS parameter mapping at the PDN or APN level, use the following commands:

```
config>mobile>pdn# umts-qos-policy umts-qos-policy-name
```

```
config>mobile>pdn>apn# umts-qos-policy umts-qos-policy-name
```

To view information about the an operator-specific QCI to pre-release8 QoS parameter mapping policy, use the following command:

```
show>mobile>profile# umts-qos-policy umts-qos-policy-name
```

Specifying the command with a policy name displays the details of that profile and without a policy name displays all existing profiles.

5.26 Wireless Priority Service on SGW-C/PGW-C/SMF over 5GC

The Wireless Priority Service on the combined SGW-C/PGW-C/SMF over 5GC applies a higher priority on the gateway to specific PDN or PDU sessions (for example, first responders who are required to have priority treatment as per government mandates) known as eMPS sessions (Multimedia Priority Sessions).

When you configure the WPS on the combined SGW-C/PGW-C/SMF, specific PDN or PDU sessions are elevated to an eMPS priority session based on the ARP values defined in the WPS profile. These ARP values are identified in the following call flows:

- (SGW-C/PGW-C) PDN session initial attach getting default Bearer QoS with an ARP value that matches a configured ARP value in the WPS profile
- (SMF) PDU session initial attach getting default QoSflow with an ARP value that matches a configured ARP value in the WPS profile
- (PGW/SMF) PCF Policy create/update with respond with default QoS bearer/flow with an ARP value that matches a configured ARP value in the WPS profile
- (SGW-C) Update Bearer Request/ Create Session Response default Bearer QoS with an ARP value that matches a configured ARP value in the WPS profile

A WPS-triggered eMPS priority PDN or PDU session gets priority and is not rejected, dropped or throttled due to any overload or congestion mitigation on the gateway.

Additionally, the WPS also enables you to configure the inclusion of the message-priority header in the subsequent GTPv2-C, SBI, and PFCP request and response messages for WPS-triggered eMPS priority PDN or PDU sessions. This informs adjacent NFs of the PDN or PDU sessions higher priority.

You can also configure a DSCP transport marking on the combined SGW-U/PGW-U/UPF for a WPS-triggered eMPS priority PDN or PDU session.

To configure a WPS profile on the CP function, use the following commands:

```
configure mobile-gateway profile wps-profile arp
configure mobile-gateway profile wps-profile arp message-priority
```

To enable the configured WPS profile, use the following command:

```
configure mobile-gateway pdn wps-profile
```

To configure a WPS profile with DSCP marking on the UP function, use the following command:

```
configure mobile-gateway profile wps-profile up-dscp-marking
```

To view the WPS configuration and eMPS statistics, use the following commands:

```
show mobile-gateway profile wps-profile
show mobile-gateway pdn apn statistics
show mobile-gateway pdn bearer-context
show mobile-gateway pdn call-flow-stats
```



```
show mobile-gateway pdn pdn-context imsi detail
```

6 PDN Gateway (PGW)

Configuration guidelines specific to the PGW.

The PGW is the node that terminates the SGi interface toward the PDN. If a UE is accessing multiple PDNs, there may be more than one PGW for that UE. The PGW provides connectivity to the UE to external packet data networks by being the point of exit and entry of traffic for the UE. PGW functions include:

- policy enforcement (gating and rate enforcement)
- charging support
- UE IP address allocation
- transport-level packet marking in the UL and the DL
- UL and DL rate enforcement based on APN Aggregate Maximum Bit Rate (APN-AMBR)

Related information

Common gateway configuration

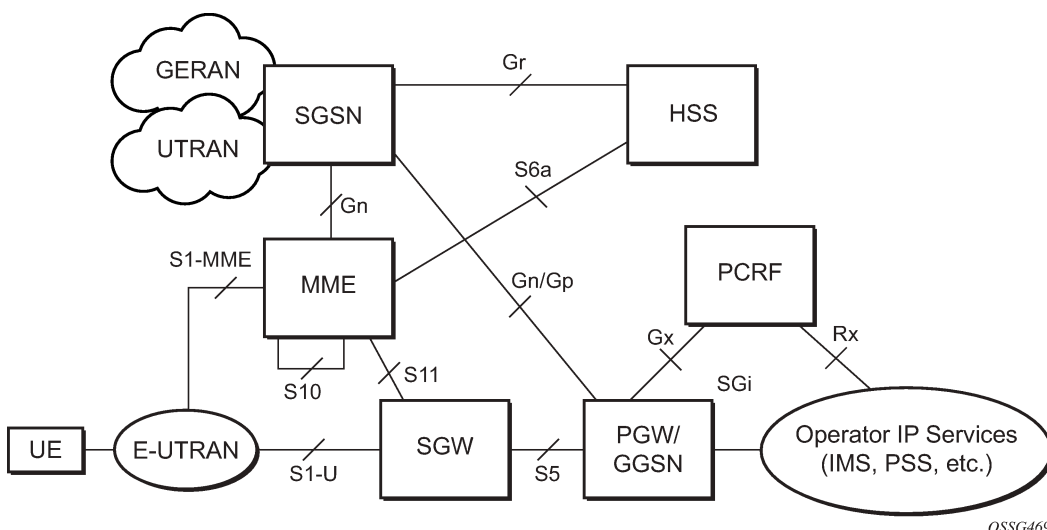
Overview of the provisioning functionality that is common to all standalone and combined gateway types.

6.1 PGW configuration example

The 7750 SR MG Gateway GPRS Support Node (GGSN) supports 3GPP Release 7 standard interworking with 3GPP Rel 7 SGSN in 3G Wireless networks for the creation of primary PDP contexts and secondary PDP contexts.

The PGW functionality allows 3G wireless operators to upgrade their current network to meet the bandwidth needs and also to seamlessly upgrade their network with a common platform as they migrate toward LTE networks.

Figure 66: PGW configuration example



6.2 PGW supported functionality

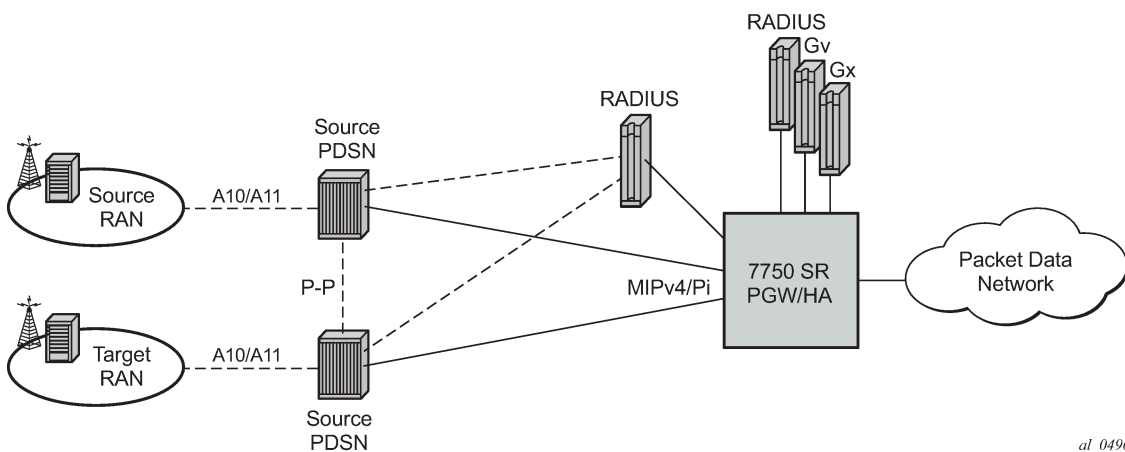
The PGW supports the following functionality:

- Gn/Gp interworking with pre-release 8 SGSN
- 3G to LTE, LTE to 3G interworking functionality
- Bypass SGSN for mobile data-traffic using direct tunnel to RNC over Gn interface
- The ability to generate CDRs, incorporates the CDF functionality and provide a Ga interface directly to a CGF for offline charging
- Four classes of QoS service
- Gi routing capabilities toward ISP, VPN, and IMS
- Periodic sending of unsolicited Router Advertisements toward the UE for IPv6 and IPv4V6 sessions.

6.3 PGW with collocated 3GPP2 Home Agent support

The combined PGW/GGSN and 3GPP2 Home Agent functionality allows 3G (CDMA Rev 0/Rev A) wireless operators to upgrade their current network to meet the bandwidth needs and also to seamlessly upgrade their network with a common platform as they migrate toward LTE networks.

Figure 67: 3GPP2 Home Agent



The 3GPP2 Home Agent supports the following:

- MIPv4 Standards support based on
 - 3GPP2 X.S0011-002-E (Mobile IP Access Services)
 - 3GPP2 X.S0011-003-E (Data Mobility and Resource Management)
 - 3GPP2 X.S0011-005-C (Accounting Services and 3GPP2 RADIUS VSAs)

- RFC 2002 / 3220 (IP Mobility Support)
- RFC 3024 (Reverse Tunneling for Mobile IP)
- RFC 3012, 4721 (Mobile IPv4 Challenge/Response Extensions)
- RFC 2794 (Mobile IP Network Access Identifier Extension for IPv4)
- RFC 2002, RFC 5944 (IP Mobility Support for IPv4)
- RFC 5177 (Network Mobility (NEMO) Extensions for Mobile IPv4)
- RFC 4784 (Verizon Wireless Dynamic Mobile IP Key Update)
- RFC 3543 (Registration Revocation in Mobile IPv4)
- RFC 3025 (Mobile IP Vendor/Organization Specific Extensions)
- Allow for UE IP address preservation when user re-attaches between LTE/eHRPD and HRPD.
- IP-in-IP / GRE tunneling support.
- RADIUS Standards support based on
 - 3GPP2 X.S0011-005-C (Accounting Services and 3GPP2 RADIUS VSAs)
 - RFC 2865 (Remote Authentication Dial In User Service (RADIUS))
 - RFC 4784 (Verizon Wireless Dynamic Mobile IP Key Update)
 - RFC 2866 (RADIUS Accounting)
 - RFC 2869 (RADIUS Extensions)
 - 3GPP TS 29.061
- 3GPP Gy and Gx reference point interfaces for online charging and policy control support.
- 3GPP S6b reference point for non-3GPP and 3GPP access is supported per version 11.6.0 July 2013
- SGi/Gi routing capabilities toward ISPs and VPNs.

6.4 Enterprise MIPv4 NEMO mobile router support over 4G and 3G networks

This feature provides support for Mobile IPv4 (MIPv4) NEMO-based Mobile Routers (CPEs) connection over LTE and eHRPD networks.

When enabled for an enterprise APN, the system includes support for an MIPv4 network Mobility (NEMO-HA) on the PGW platform to terminate MIPv4-based NEMO connections from Mobile Routers (MRs) that attach to a PDN connection using a default bearer (or optionally, dedicated bearers).

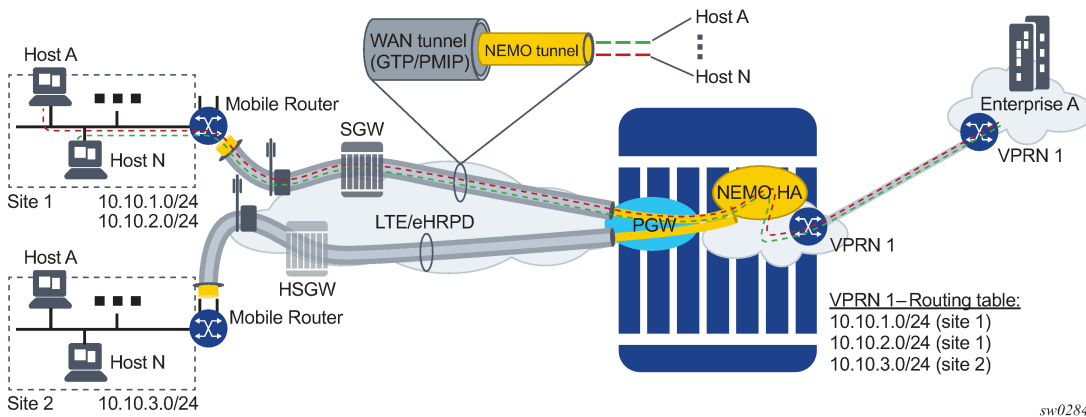
The MRs use a single PDN connection to the PGW. Thus, the MR first establishes a PDN connection to the PGW and when the connection is established, the MR dynamically negotiates with the NEMO HA the host

subnets that will be routable over the MR. The NEMO HA supports MRs using MIPv4 NEMO, based on RFC 5177, and Cisco vendor extensions carrying the host subnet LAN prefixes.

The NEMO functionality allows for bidirectional communication that is application-agnostic between hosts behind the MR and hides its mobility from the hosts on the WAN (3G/4G) network. Thus, the hosts on the subnets are fixed in relation to the MR and move homogeneously (as a whole).

NEMO-based mobile router support shows the NEMO-based MR connection over LTE and eHRPD networks.

Figure 68: NEMO-based mobile router support



NOTE:

The MR is not allowed to send any user data traffic (from itself or its connected hosts) directly over the WAN connection (LTE/eHRPD). Thus, the MR only uses the WAN connection directly for the MIP NEMO RRQ/RRP to setup the NEMO tunnel session. Once the NEMO session tunnel is established over the WAN connection, the MR must always send all user data traffic over the NEMO tunnel session on top of the WAN connection.

6.5 PGW 23.060 supported call flows

Table 21: PGW 23.060 Supported Call Flows

Call flow	23.060 Ref.
Location Management Procedures (A/Gb mode)	6.9.1
Location Management Procedures (Iu-mode)	6.9.2

Table 21: PGW 23.060 Supported Call Flows (continued)

Call flow	23.060 Ref.
Intersystem Change	6.13
PDP Context Activation Procedure	9.2.2.1, 9.2.2.1A
Secondary PDP Context Activation Procedure	9.2.2.1.1, 9.2.2.1.1A
Network Requested Secondary PDP Context Activation Procedure using Gn	9.2.2.3, 9.2.2.3A
Modification Procedures	9.2.3
Deactivation Procedures	9.2.4
RAB Release Procedure	12.7.2

6.6 PGW 23.401 supported call flows

PGW 23.401 supported call flows lists the PGW 23.401 supported call flows implementation compatible with Release 13.9.0 (December 2016).

Table 22: PGW 23.401 supported call flows

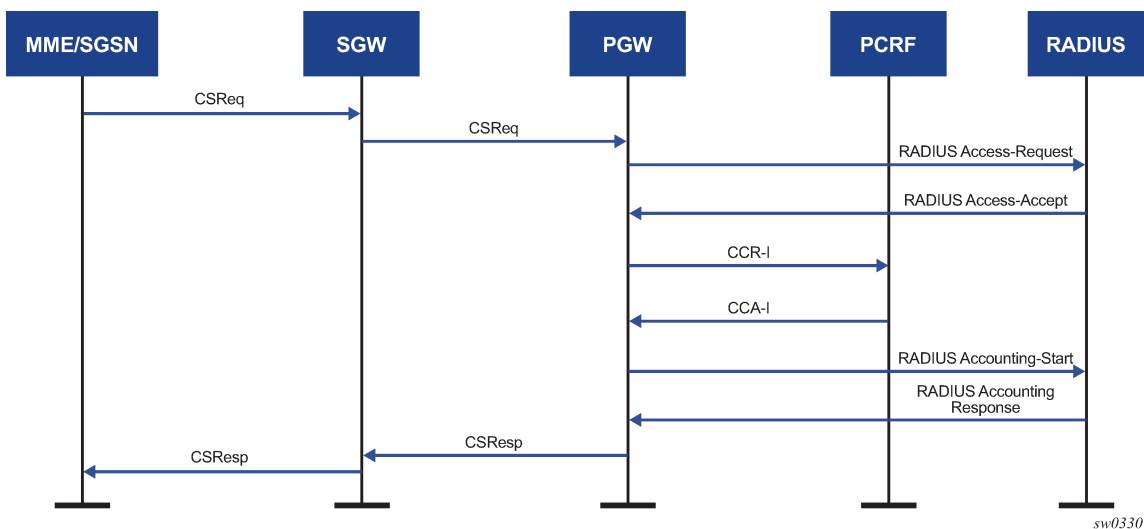
Call flow	23.401 Ref.
Initial Attach	5.3.2
Tracking Area Update Procedure	5.3.3
Detach Procedure	5.3.8
Dedicated Bearer Activation	5.4.1

Table 22: PGW 23.401 supported call flows (continued)

Call flow	23.401 Ref.
Bearer Modification with QoS Update	5.4.2
PGW-initiated Bearer Modification without QoS Update	5.4.3
PGW-initiated Bearer Deactivation	5.4.4
UE-requested Bearer Resource Modification	5.4.5
Handovers	5.5
UE-requested PDN Connectivity	5.10.2
UE- or MME-requested PDN Disconnection	5.10.3
Gn/Gp Inter-operation Procedures	D.3

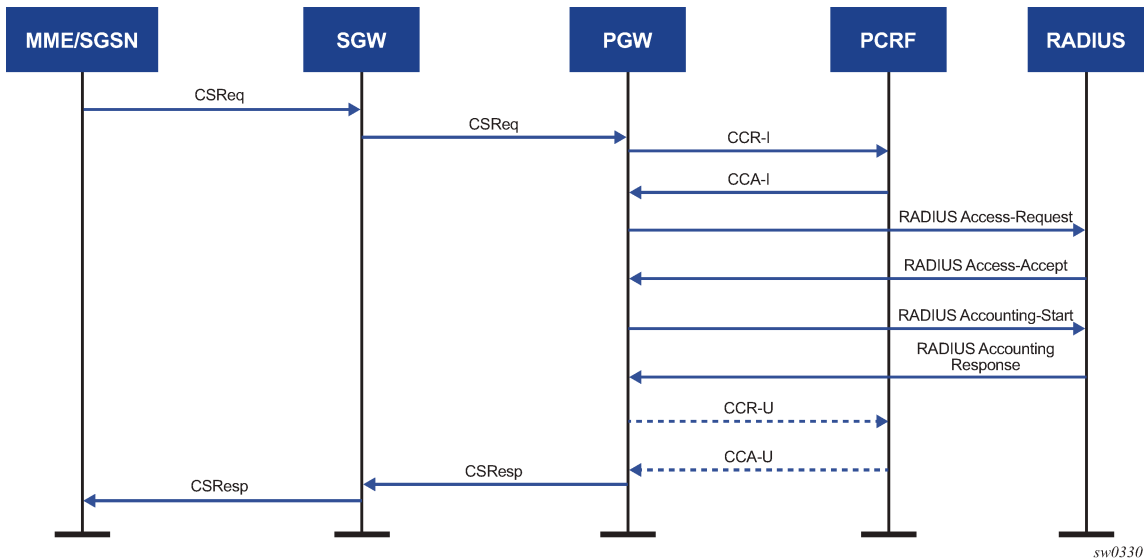
6.7 Alternative Gx/AAA call flow

Figure 69: Default system call flow



In the alternative Gx/RADIUS call flow, the PCRF (Gx) initial communication takes place before AAA (RADIUS). This also means that in a scenario where the UE IP address is allocated by the RADIUS server (or by the PGW as a fallback action), the allocated address needs to be communicated to the PCRF using a dedicated Gx transaction.

Figure 70: Alternative RADIUS call flow



To enable the alternative Gx/AAA call flow, use the following command:

```
config>mobile>pdn>gx-before-aaa
```

6.8 Enhanced SGW restoration procedure

This feature introduces support for the optional procedure in TS 23.007 called restoration of PDN connections after an SGW failure with MME-triggered SGW restoration.

After detecting an SGW failure, the MME attempts to restore the PDN connections eligible for restoration for all the UEs affected by the SGW failure. This procedure makes it possible to restore PDN connections affected by an SGW failure with or without a restart, and thus to resume delivery of downlink data toward the UE with less service interruption and signaling in the network.

6.8.1 SGW failure with a UE in connected or idle mode

If an SGW failure occurs when the UE is in connected or idle mode, the MME and PGW detect the failure either from a GTP-C message that has a restart counter incremented or due to a path failure.

The PGW and MME will maintain the PDN connections related to the SGW for the length of time configured in the enhanced-sgw-restoration command in the config>mobile>pdn>{s5 | s8}>signaling>gtp-c context. The MME will select a new SGW or the same SGW and pace the recovery

of the UEs. The PCRF, OCS, and OFCS message exchange may also happen, for example, if AN_GW_CHANGE trigger is armed or another event such as a time zone change is triggered.

Figure 71: UE in idle mode

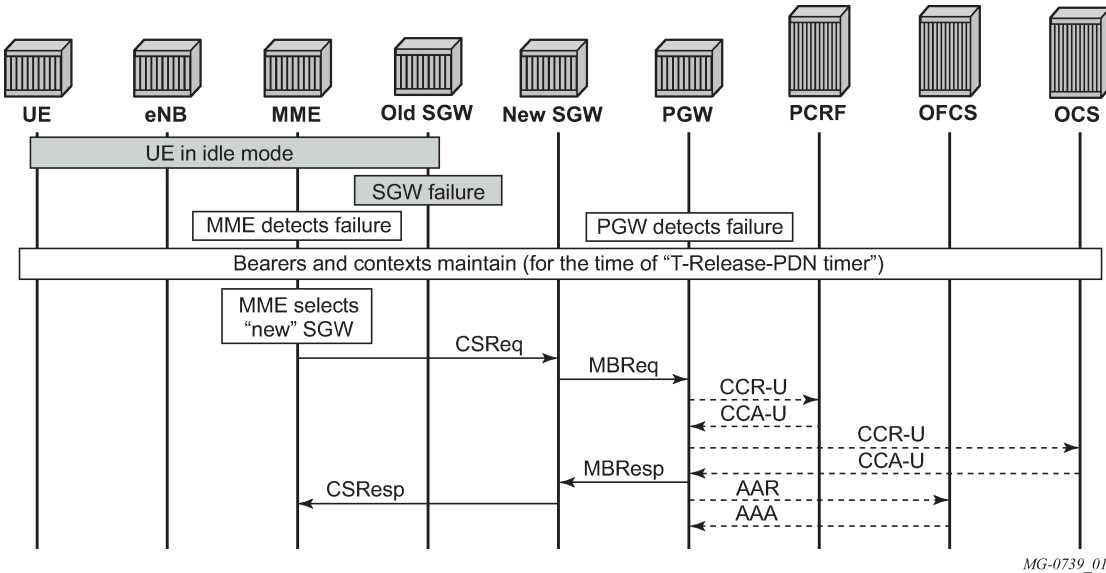
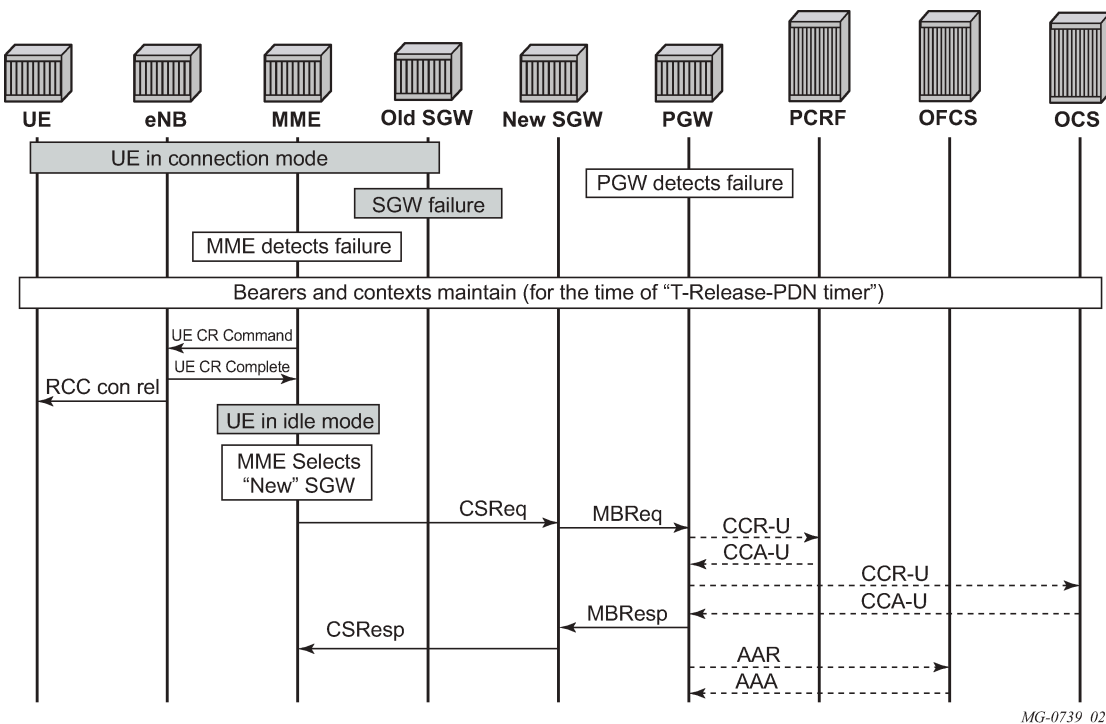


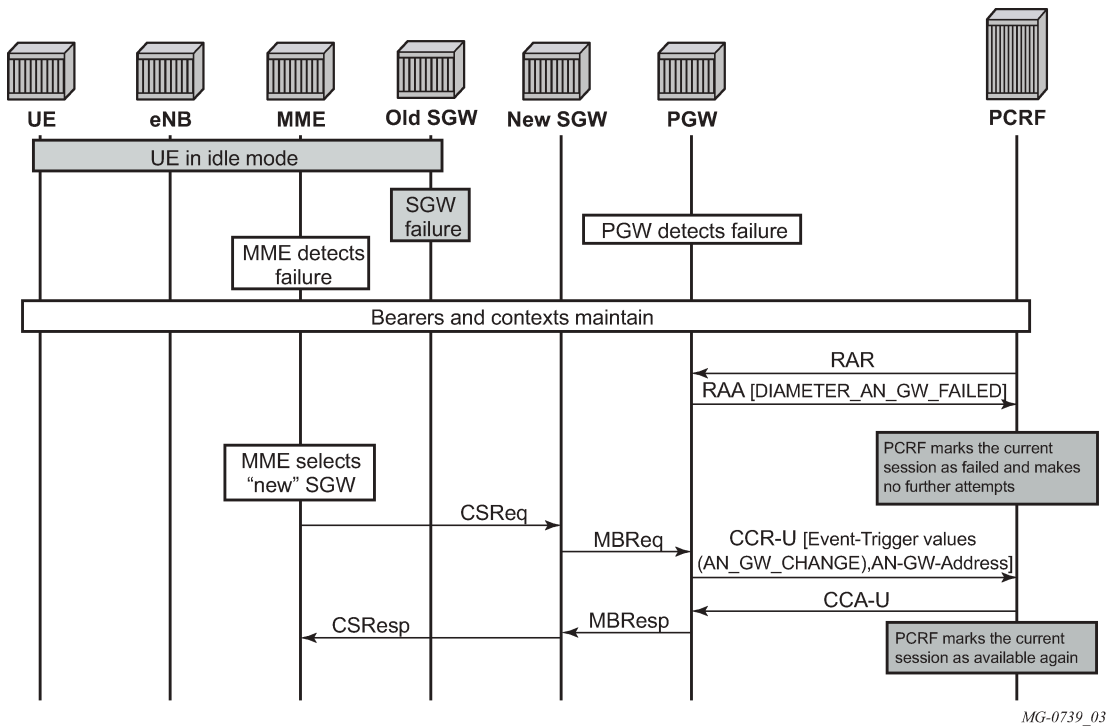
Figure 72: UE in connected mode



6.8.2 PGW detects SGW failure before Gx signaling

PGW detects SGW failure before Gx signaling shows the call flow when the PGW has already detected the SGW failure and is receiving Gx signaling requesting the rules to be modified or installed.

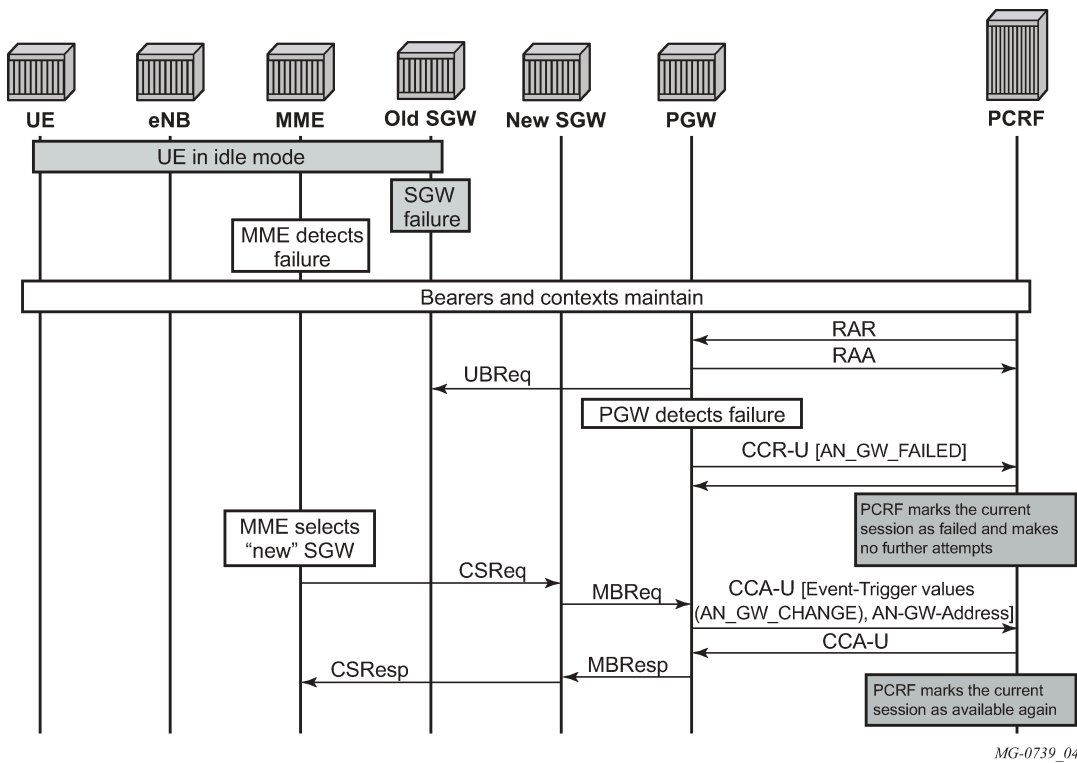
Figure 73: PGW detects SGW failure before Gx signaling



6.8.3 PGW detects SGW failure after Gx signaling

PGW detects SGW failure after Gx signaling shows the call flow when the PGW has not yet detected an SGW failure and receives the Gx signaling request with rules to be modify or installed.

Figure 74: PGW detects SGW failure after Gx signaling



6.9 UE IP address allocation

The purpose of User Equipment (UE) IP address allocation in PGW/GGSN is to define the IP address of the UE toward the Gi/SGi network. An IP address is allocated as part of UE attach.

6.9.1 Allocation of IP address types

The selection of the IP address pool from which the IP address is allocated, is based on the requested APN and the type of the requested address.

An IPv4 address is allocated when the UE requests for PDN/PDP type IPv4 or IPv4v6, and an IPv6 address (prefix as well as interface identifier) is allocated when the UE requests for either IPv6 or IPv4v6 PDN/PDP type.

When an IPv6 address is allocated to the UE, if the interface identifier value is 0 or undefined, the gateway changes the UE interface identifier to 1 before sending the allocated IPv6 address to the UE. Therefore, each UE can configure its own link-local IPv6 address as FE80::1. The gateway always configures its own link-local IPv6 address as FE80::2 ensuring a collision-free IPv6 link between the UE and the gateway. For IPv6 stateless address auto-configuration, the UE can choose any interface identifier to generate IPv6 addresses, other than link-local, without involving the network.

Certain IPv4 addresses, like subnet network address and subnet directed broadcast address, are not allocated to the end user when a dynamic allocation method is used. For IPv4 prefixes, as long as the prefix is minimally a /24, the first .0 address and the last .255 address are automatically suppressed. For TWAG

configurations, the last .254 address in the prefix are suppressed as in the default gateway. For example, if the prefix is 10.0.0.0/16, the reserved IP addresses are 10.0.0.0, 10.0.255.255, and 10.0.255.254 in the case of TWAG.

If the optional suppress-dotzero-broadcast-address flag is set, the system also suppresses all .0 and all .255 addresses. The following IP addresses would therefore not be allocated to the UEs from an IP pool configured with prefix 10.0.0.0/16:

10.0.0.0, 10.0.0.255

10.0.1.0, 10.0.1.255

...

10.0.255.0, 10.0.255.255

Prefixes smaller than /24 are only supported in limited scope.

**NOTE:**

The suppress-dotzero-broadcast-address will not take effect for IPv4 pools with masks from /25 to /29.

6.9.2 UE IP address allocation methods

PGW/GGSN supports the following UE IPv4 address and IPv6 prefix allocation methods:

Static address allocation

The IPv4 address, IPv6 prefix, or both (dual stack IPv4v6) are included in the incoming PDP context / EPS bearer creation request. This same address is always used by that specific UE, and has been received from HLR /HSS.

**NOTE:**

If the static IP address/prefix is not stored in the HSS/HLR subscription record, it may be configured on a per-user per-APN basis in the DHCP/RADIUS/L2TP server, and the PGW/GGSN can retrieve the IP address/prefix for the UE from that server. In this case, a static IP address/prefix is allocated by the same procedures as the dynamic IP address/prefix allocation.

Dynamic address allocation

The IPv4 address, IPv6 prefix, or both (dual stack IPv4v6) are allocated by dynamic methods, either from the local IP address pools or from the external systems.

IP address allocation options:

Dynamic IP assigned from local IP address pool

- The address pool is configured on the PGW/GGSN.
- The IP address is assigned by PGW/GGSN from the IP address pool based on local selection criteria.
- The IP pool is advertised on SGi on pool configuration. (For LB-less mode, separate ip-pools need to be configured per-MG-group).

AAA assigned pool

- The address pool is configured on the PGW/GGSN.
- The IP pool selection is done by the AAA server. The AAA determines the IP address pool for the UE and informs the PGW/GGSN.
- The IP address is assigned by the PGW/GGSN from the IP address pool selected by the AAA.
- The IP pool is advertised on SGi on pool configuration.

HSS-assigned static IP

- The IP address is assigned by the HSS arriving in the initial create request.
- No IP address pool is configured on the PGW/GGSN.
- Each IP address is allocated as a separate /32 for IPv4 or IP prefix for IPv6.
- The PGW/GGSN advertises assigned PDN session IP address on SGi on allocation.

AAA/DHCP assigned IP

- No IP address pool is configured on the PGW/GGSN.
- The AAA/DHCP server determines the IP address and informs the PGW/GGSN.
- Each IP address is allocated as a separate /32 for IPv4 or IP prefix for IPv6.
- The PGW/GGSN advertises assigned PDN session IP address on SGi on allocation.

AAA/DHCP/HSS assigned IP using remote-pool

- The IP address pool is configured in the APN VPRN as remote on the PGW/GGSN.
- The IP address is assigned by an external server and it is matching the IP address pool configured on PGW/GGSN.
- Avoids individual /32 for IPv4 and IP prefix for IPv6 by aggregating prefixes.

- The DHCP server may allocate an IP address for a finite lease time. The DHCP client in the PGW/GGSN renews the IP address before the lease expires.

Local IP address pool controlled by a separate local DHCP server

- A local DHCP server can be used to configure the IP address pools with prefixes containing less than 256 addresses.
- The IP address is assigned by the PGW/GGSN from the IP address pool based on local selection criteria.
- The IP pool is advertised on SGi on pool configuration.

L2TP

- The IP address negotiated during the L2TP/PPP IPCP negotiation is required to connect the endpoints of the tunnel, it is recorded by the gateway and sent to the UE in the initial create response.
- The gateway is not involved in the IP address allocation, and any IP allocation configuration in the APN is ignored.

The enhanced AAA allocation via the `config>mobile>pdn>apn>ip-alloc-allow-local` command is used to allow allocation of PDN type address from local pool when only one PDN address is provided by the AAA in response to the request for PDN type IPv4v6. In the case of HSS assigned static address, if only one PDN address is provided via GTP signaling, the missing PDN address should be allocated from the local pool.

Related information

MG group-specific APN configuration

To configure LB-less traffic handling on the SGi interface, each APN with a unique routing instance requires a dedicated VPRN per MG group. Each MG group anchors sessions from all APNs configured on the CMG instance and therefore each MG group requires as many VPRNs as the number of APNs.

6.9.3 Controlling allocation of IP pool address to the end user

Multiple IP address allocation methods can be configured on an APN. When more than one allocation method is configured, the allocation will be done based on the priority of the methods. For L2TP and DHCP, only that method is attempted before failing the attach. The AAA, HSS-static, and local pools, can all be configured with the following precedence order.

- If Layer 2 tunneling protocol (L2TP) is configured, use the IP address provided by the LNS or else the attach will fail.

The LNS is in full control for UE IP address allocation. Even in cases where a static IP address is included in the create session request, it is not used in further processing. In these cases the HSS-static IP address allocation method has to be enabled or else the attach will fail.

- If DHCP is configured, use the IP address provided by the DHCP server or else the attach will fail.

In case the static IP address is included in the create session request, it is relayed to the DHCP server in DHCPv4 Discover or DHCPv6 Solicit, which may or may not return this address to the PGW. This behavior is enabled irrespective of the configuration of the HSS-static IP address allocation method.

- If AAA is configured, try an AAA allocation. If this does not return an IP address, try the next option (HSS-static).
- If HSS-static is configured and available, use it. Otherwise try the next option (local-pool).
- If a local pool is configured, take the address from it. If local-pool does not return an IP address, the attach will fail.

The following command specifies the IP address allocation scheme in the APN context.

```
ip-alloc-method {local-pool | hss-static | aaa | dhcp-client | dhcp-relay}
[ {local-pool | hss-static | aaa | dhcp-client | dhcp-relay}... (up to 4 max) ]
```



NOTE:

This command configures the allowed allocation methods, not the relative priority.

It is also possible to separately configure the HLR/HSS-provided static IP address to be used for UE IP address allocation, if it is included in the GTP Create PDP Context Request/Create Session Request message. In this case, the APN will also allow dynamic IP address allocation methods which would normally be preferred over the static IP address.

To configure the UE IP address allocation methods to use the HLR/HSS-provided static IP address, use the following command:

```
configure mobile-gateway pdn apn ip-alloc-priority-order
```

When the **ip-alloc-priority-order** command is enabled, the static IP address/prefix provided in the CSR is used and, the behavior is as follows.

Provided static IP address is used without querying it from the DHCP server

The DHCP server query is only needed for UE IP address allocation, so in this case it is not needed and the whole sequence is skipped.

Provided static IP address is used without control by AAA (RADIUS, S6b)

The Access-Request messages are still sent because they are used for authentication as well as IP address allocation. The GGSN/PGW ignores any address that the AAA server (RADIUS, S6b) returns, if the static IP address of the corresponding type already exists. For RADIUS, the static IP address is additionally indicated to the server using the Framed-IP-Address attribute.

To view the configured UE IP address allocation method, use the following commands:

```
show mobile-gateway pdn apn statistics
```

```
show mobile-gateway pdn pdn-context imsi detail
```

Each APN/VPDN may support simultaneously dynamic address allocation from a local pool (a pool assigned by AAA) and static address allocation from a remote pool (IP address assigned by AAA).

```
show mobile-gateway pdn apn "ipd.nokia.com" statistics
```

```
=====
APN statistics summary
=====
```

```
Virtual APN      :
APN Admin State  : up                APN Oper State   : up
Router          : vprn1
PDN Type         : ipv4 ipv6 ipv4v6   Ip Alloc Method   : local hss
Ip Alloc Prio order: hss-static
Max-PDN-Connections: -              Min-PDN-Connections: -
...
...
-----
```

```
Number of cards : 1
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

```
show mobile-gateway pdn pdn-context imsi 123456789012301 detail
```

```
=====
PDN context detail
=====
```

```
IMSI : 123456789012301
APN : ipd.nokia.com
Virtual APN : N/A
...
...
UE IPv4 address : 100.0.0.1
UE IPv4 address source : AAA|HSS|Local Pool|DHCP|L2TP|unknown
UE IPv6 address : 4ffe:6400::15
UE IPV6 address source : AAA|HSS|Local Pool|DHCP|L2TP|unknown
...
...
```


6.9.4 IP address pools configuration

IP pools contain the IP addresses/prefixes and are linked to APNs via router contexts. Several APNs can refer to the same IP pools. Each VPRN may have multiple local IP pools, and each local IP pool may be comprised of multiple IPv4 and/or IPv6 prefixes.

An APN can be configured without IP pools. If the APN is configured under a VPRN, it will attempt to find an IP address from the default pools defined under the provisioned VPRN. In case no VPRN is configured under the APN (or the base router is explicitly configured), the IP pools are used from the base router.

If an IP address pool is configured as exclusive, only one APN or a TAI/LAI list is allowed to include that pool in its configuration. Furthermore, the hosts from an exclusive pool can only be allocated by the associated APN (the pool is not selected as one of the default pools).

The mobile gateway can allocate IP addresses configured in the APN or router context (pools/subnets) according to the round robin algorithm, which means that load balancing between the configured IP address pools/prefixes is applied (looping through all the available IP addresses). By doing so, an IP pool is used immediately after it is added to the system and thus can be tested for possible connectivity problems; for example, the subnet has not been exported via BGP or a filter (FW) is configured somewhere in the path.

To enable IP load balancing, use the following command:

```
configure router ip-pool-distribution
```

Grouping of IP address pools is also possible. When a specific IP address pool group is set through the S6b AAA attribute (framed-Pool or Framed-IPv6-Pool) only then those IP address pools belonging to that group are candidates. Load balancing is used between the selected IP address pools for allocating an IP address to a PDN connection. Configurable IP address pool high and low thresholds can be set. When these thresholds are configured, they indicate whether the specified threshold of an IP address pool usage is reached. When the configured thresholds are reached, alarms are generated.

To configure the thresholds for IP pool usage, use the following command:

```
configure router ip-local-pool pool-usage-threshold high-threshold high-threshold low-threshold low-threshold
```

There is also the option to configure whether added prefixes to the pool are forced to be used immediately. The next attach or PDN connection activation considers the added prefix as part of the available/candidate IP addresses to be used for IP address allocation in this case.

To configure the option that the added prefixes to a pool are used immediately in the next PDN connection activation, use the following command:

```
configure router ip-local-pool use-prefixes-immediately
```

The PGW/GGSN supports the configuration of a hold timer for a local IP address that causes the PGW/GGSN to hold an IP address for a period of time after a PDN connection is deleted. This means that the PGW/GGSN does not assign that IP address to any other PDN connection during this time. By default, the pool's hold timer is set to 180 seconds. Configuring it explicitly to value of 0 means that an IP address may

be assigned to another PDN connection immediately upon termination of PDN connection to which it had been assigned. By configuring a hold timer, the operator can avoid issues with a UE receiving traffic that had been intended for the UE that had previously used a reassigned IP address. However, while the hold timer is active, the PGW/GGSN does not assign the IP address to any PDN connection, even to the PDN connection that had just released it. The hold timer cannot be set for a remote pool.

There is also the possibility to modify the hold timer operation so that the PDN connection may be reassigned the held IP address if the UE attaches again before the hold timer expires. This hold for PDN feature appears more graceful toward the applications running over the PDN connection since they experience no change in IP address.

The following output is an example of an IP address pool. The router is provisioned with an IP pool named pool with two IPv4 prefixes suppressing the broadcast addresses, an IPv6 prefix, a hold timer interval of 300 seconds and provisioned as exclusive to APN.

```
configure
service vprn 1 customer 1 create
  ip-local-pool pool
    exclusive
    hold-timer-sec 300
    suppress-dotzero-broadcast-address
    ipv4-prefix "1.2.3.0/24"
    ipv4-prefix "1.2.4.0/24"
    ipv6-prefix "ABCD::/48"
  exit
exit
```

The PGW/GGSN can further select different pools using the UE's location, by configuring lists of Tracking Area Identities (TAI) and Location Area Identities (LAI) and associating a set of IP pools with each list. If the lists and address pools are configured so that address pools are not shared between different lists, this feature allows the operator a way to infer the location of the UE from its IP address. The PGW/GGSN does not automatically detach the UE's PDN connection if it moves to a location that belongs to a different list. This capability must be provided by the MME or SGSN.

The following example shows the basic configuration steps for TAI/LAI based IP allocation.

First, configure two TAI/LAI lists:

```
configure mobile-gateway profile tai-lai-list "List-IP-1"
config>mobile>profile>tai-lai-list# info
-----
      tai mcc 206 mnc 01 tac 0x63
      tai mcc 206 mnc 01 tac 0x64
-----

configure mobile-gateway profile tai-lai-list "List-IP-2"
config>mobile>profile>tai-lai-list# info
-----
      tai mcc 206 mnc 01 tac 0x65
-----
```

Assign these to an APN along with the appropriate local IP pool.

```
config>mobile>pdn>apn>router# tai-lai-list List-IP-1 ip-local-pool ipv4
config>mobile>pdn>apn>router# tai-lai-list List-IP-2 ip-local-pool bigpool
```

Ensure that a default local IP pool is available in case there is no TAI/LAI match.

```
config>mobile>pdn>apn# info
      router "vprn1"
        ip-local-pool "pool"
        tai-lai-list "List-IP-1"
          ip-local-pool "ipv4"
        exit
        tai-lai-list "List-IP-2"
          ip-local-pool "bigpool"
        exit
      exit
```

The individual IPv4/IPv6 prefixes in a local IP pool can be emptied by setting the block flag for that particular prefix. This feature is used to drain IP addresses from the prefix so the prefix will eventually no longer be in use. A prefix can be removed from an IP pool that is referenced in the APN configuration during runtime, as long as there are no allocated IP addresses associated with that prefix (used or held), and it is not the last prefix defined in the pool. The force flag will trigger the clearing of all sessions that run in the blocked IP prefixes

6.9.5 Local IP address pools in VPRN configuration

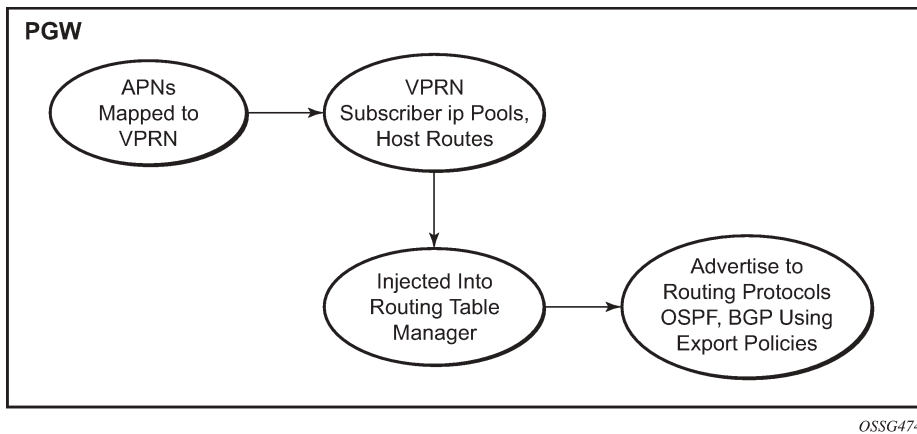
Configure injecting subscriber IP pools into RTM

Subscribers belonging to an APN are mapped to VPRN. The subscriber IP pool must be configured under VPRN for assigning addresses to UE.

Each subscriber IP pool can have IPv4 and/or IPv6 ranges defined.

The subscriber IP pool prefixes or aggregate prefixes have to be explicitly injected into the Routing Table Manager (RTM) via routing policies.

Figure 75: Injecting subscriber IP pools into RTM example



OSSG474

The subscriber IP pool prefixes are injected into the routing table with a new routing protocol type **Mobile-Subscriber Route**. The following are the types:

- The route type will be mobile-host for HSS-static subscribers and RADIUS or DHCP assigned addresses.
- For IP local pools, the addresses will appear in FIB as type internal-route (INT_ROUTE).

Export policies must be configured to advertise subscriber IP pool prefixes into respective routing protocols.

Advertising subscriber IP pools into routing protocols

Various routing protocols can be used to advertise externally the IP pools. The following example configuration shows how export policies can be configured in the OSPF context for advertisement as OSPF stub routes.

```

A:3ls-pgw>config>router>policy-options# info
-----
    prefix-list "pools"
        prefix 5.5.0.0/16 exact
    exit
    policy-statement "ippools"
        entry 10
            from
                prefix-list "pools"
            exit
            to
                protocol ospf
            exit
            action accept
        exit
    exit

```

```
exit
```

It is mandatory to introduce the IP pool prefixes to the routing system. This is achieved by configuring static black-hole routes for these prefixes, as shown in the following example.

```
A:3ld-pgw>configure service vprn 2 customer 1 create
A:3ld-pgw>config>service>vprn# info
```

```
-----
description "Network_Side"
...
ip-local-pool "pool"
    ipv4-prefix "5.5.0.0/16"
exit
static-route 5.5.0.0/16 black-hole
ospf 10.20.30.2
    export "ippools"
    area 0.0.0.0
        interface "system"
            hello-interval 5
            dead-interval 15
        exit
    interface "ip-10.10.6.3"
        hello-interval 5
        dead-interval 1
    exit
exit
exit
-----
```

The following example output shows the prefix in the neighbor router route-table.

```
A:3ls-SGW-2# show router 2 route-table protocol ospf
```

```
=====
Route Table (Service: 2)
=====
```

Dest Prefix	Type	Proto	Age	
Pref				
Next Hop[Interface Name]				Metric
5.5.0.0/16	Remote	OSPF	00h02m57s	
150				
10.10.6.3			1	
10.20.1.3/32	Remote	OSPF	02h04m29s	10

10.10.6.3	1000

No. of Routes: 2	
=====	

6.9.6 IP address allocation configuration using a local DHCP server

The gateway supports using a local DHCP server to configure IP address pools with prefixes containing less than 256 addresses. IP addresses allocated using the local DHCP server will be from the protocol Mobile Host.

6.9.6.1 DHCP server configuration for local IP address pools

The following restrictions apply when configuring IP address pools using a local DHCP server.

- The IP address pool in the DHCP server configuration cannot be configured as part of a local IP pool.
- The local DHCP server can only serve requests from the internal DHCP client.
- The local DHCP server configuration is not supported in an Inter-Chassis Redundancy setup.

This example shows the DHCP server configuration that can be used when configuring the local DHCP server IP address pools. The user database and pool are configured for the specified local DHCP server.

```

service
  vprn 2500 customer 1 create
  dhcp
    local-dhcp-server "dhcp-server-v4-2500" create
    user-db "dhcp-server-user-db"
    pool "dhcp-server-v4-pool" create
    options
      lease-time min 6
      lease-renew-time min 2 sec 24
      lease-rebind-time min 4 sec 2
    exit
    subnet 10.66.0.0/23 create
    options
      subnet-mask 255.255.254.0
      default-router 10.66.0.1
    exit
    address-range 10.66.0.0 10.66.0.127
    address-range 10.66.1.0 10.66.1.127
  exit
  subnet 10.152.0.0/23 create
  options
    subnet-mask 255.255.254.0

```

```

                                exit
                                address-range 10.152.0.0 10.152.0.127
                                address-range 10.152.1.0 10.152.1.127
                                exit
                            exit
                            no shutdown
                        exit
                    exit
                exit
            exit

```

6.9.6.2 Interface configuration in the SGi domain

This example shows configuration of the DHCP client interface, the DHCP server interface, and the VPRN physical interface that can be used when configuring the local DHCP server IP address pools.

```

service
    vprn 2500 customer 1 create
        autonomous-system 2500
        route-distinguisher 2500:2500
        interface "dhcpv4-phy-2500" create
            address 10.89.0.1/24
            sap 1/1/4:2500 create
            exit
        exit
        interface "dhcpv4-int-clnt-2500" create
            address 10.88.0.32/32
            loopback
        exit
        interface "dhcpv4-int-srvr-2500" create
            address 10.88.0.21/32
            local-dhcp-server "dhcp-server-v4-2500"
            loopback
        exit
        no shutdown
    exit

```

6.9.6.3 Local user database configuration

This example shows the configuration of the local user database that is referenced for the DHCP server configuration.

```

subscriber-mgmt
    local-user-db "dhcp-server-user-db" create

```

```

        ipoe
            match-list mac
            host "default" create
                address pool "dhcp-server-v4-pool"
                no shutdown
            exit
        exit
        no shutdown
    exit
exit

```

6.9.6.4 DHCP client configuration

This example shows the DHCP client configuration that can be used when configuring the local DHCP server IP address pools. A DHCP server group profile referencing the VPRN interface is configured for the mobile gateway context.

```

mobile-gateway
    profile
        dhcp-server-group "dhcpv4-int-grp-2500"
            interface router "vprn2500" "dhcpv4-int-clnt-2500"
            deadtime 300
            peer 10.88.0.21
            no shutdown
        exit
    exit
exit
exit

```

6.9.6.5 APN configuration

This example shows the mobile gateway APN configuration that can be used when configuring the local DHCP server IP address pools.

```

mobile-gateway
    pdn 1
        apn "ipd.2500.nokia.com"
        dhcpv4-server-group dhcpv4-int-grp-2500
        pco-option
            dns-server-ipv4
                primary 10.30.1.1
                secondary 10.30.1.2
            exit
        exit
        pcrf-selection

```



```

        dynamic-pcc fh-session disable fh-3xxx disable fh-4xxx disable
    exit
    ip-alloc-method dhcp-client
    pdn-type ipv4 ipv6 ipv4v6
    router "vprn2500"
    exit
    no shutdown
exit

```

6.9.7 Monitoring UE IP address allocation from local pools

KPI/KCIs

The following KPI/KCIs show the current utilization of IP pools:

- The KCIControlPlaneIpPool/VS.ipV4PoolUtilization and KCIControlPlaneIpPool/VS.ipV6PoolUtilization track system-wide utilization for each IP pool.
- The KCIControlPlaneVPRN/VS.ipUtilization provides an overview of all the IP pools within a VPRN.

Alarms

There are two dedicated alarms (notifications) for IP pool utilization.

- **tmnxMobGwPoolCapacityAlarmMinor**
Generated when either the IPv4 or IPv6 address pool usage reaches 85% of the pool size.
- **tmnxMobGwPoolCapacityAlarmMajor**
Generated when either the IPv4 or IPv6 address pool usage reaches 95% of the pool size.

Clearing of alarms is automatic when the utilization drops below the defined threshold.

CLI show commands

To monitor the IP pool utilization, use the following commands:

- `show mobile-gateway pdn apn statistics`
Contains information about IP addresses allocated from local pools for this APN, as well as the supported allocation methods.
- `show router ip-local-pool-stats`

Dedicated command for checking the IP pool utilization within a VPRN.

6.9.8 Monitoring connectivity/routability of IP addresses from IP pools

This feature utilizes an IP address from an IP address pool, of a specific APN, to check the connectivity between the PGW and a server on the SGi side for this APN.

The selected IP address is used as the source IP address of an ICMP Echo Request in a ping command. The UE connectivity from the PGW till an SGi destination is imitated. A reserved IP address from an IP address pool, used as the source IP address of the ping, is not allowed to be allocated to a UE.

To reserve an IP address 1.2.3.4 from mypool on vprn 2121 and run the ping test for the IPV4, use the following commands:

```
# tools perform router 2121 ip-local-pool "mypool" reserve-ipv4 1.2.3.4
# configure service vprn 2121 ip-local-pool "mypool" reserved-ipv4 1.2.3.4
# configure service vprn 2121 interface "ping-loopback" address 1.2.3.4
# ping ... source-address 1.2.3.4 vprn 2121 ... destination-address
```

To release the IP address back to the pool after the test, use the following command:

```
configure service vprn 2121 ip-local-pool mypool no reserved-ipv4
```

To reserve an IP address from mypool on vprn 2001 and run the ping test for the IPv6, use the following commands:

```
# tools perform router 2001 ip-local-pool "mypool" reserve-ipv6
INFO: CLI Successfully allocated IP address 3ffe:60:0:ff00::.
# configure service vprn 2001 ip-local-pool "mypool" reserved-ipv6
3ffe:60:0:ff00::
# configure service vprn 2001 interface "ping-loopback" ipv6 address
3ffe:60:0:ff00::1/128
# ping router 2001 ... destination-address source 3ffe:60:0:ff00::1
```

To release the IP address back to the pool after the test, use the following command:

```
configure service vprn 2001 ip-local-pool "mypool" no reserved-ipv6
```

The ping command can be used with an MG group-specific VPRN to check the connectivity of the active MG-VM toward an SGi server, in LB-Less deployments. The SGi interface link can be checked from all MG-groups and for all APNs (for example, that are on a different VPRN). This functionality applies to both IPv4 and IPv6 addresses.



NOTE:

Reserving an IP address from an IP address pool is only supported in the CMG platform and is not supported in CUPS deployments.

The same loopback interface, configured with a reserved IP address from a local IP address pool, is used to check the connectivity by executing a ping command from the SGi interface server side. In this case, the active MG-VM responds to the ping.

In case the local IP address is configured as remote, for example, due to an allocation done from RADIUS, and the locally reserved IP address is the one assigned by RADIUS for a new session, the new session request is rejected.

6.10 IP pool allocation mechanism

The IP pool allocation features the following functions:

- *IP distribution to MG groups*
- *Load-sharing of configured IP pools/prefixes*
- *Micronet allocation*
- *Round robin IP address allocation*
- *IP address allocation from micronet*
- *RADIUS IP address allocation in LB-less non-CUPS architecture*
- *Centralized IP-pools functionality in LB-less deployments*

6.10.1 IP distribution to MG groups

An MG group requests IP addresses from the CPM if there are sessions attached to the system. The CPM allocates addresses to an MG group in blocks of /24 (micronets).

MG groups can request more micronets from the CPM based on the call load. If all the addresses of the micronet are released and after the hold timer expires, the micronet returns back to the CPM. PDN/APN shutdown and reboot can also cause the release of the micronets back to the CPM.

6.10.2 Load-sharing of configured IP pools/prefixes

Multiple IP pools can be assigned to an APN. Also, load-sharing is performed among all IP pools configured per APN, and among all prefixes in the IP pool. Therefore, if there are three IP pools configured to an APN, the system chooses the least loaded IP pool for a micronet.

This is determined by calculating the total number of micronets for each IP pool and by checking how many are in use. The IP pool with the lowest percentage of micronets in use is chosen. When the IP pool is determined, the least loaded prefix is located, based on micronet usage count (lowest percentage of in-use micronets). Without load-sharing, the prefixes are chosen from lowest to highest.

6.10.3 Micronet allocation

When a prefix is selected, the micronets are chosen from tail, that is from the highest prefix to the lowest. For example, if the prefix is 10.0.0.0/16, the first micronet is always 10.0.255.0, then 10.0.254.0, and so on. When micronets return to the idle-list on the CPM they are always appended to the tail. So, they can get out of order depending on which order they returned to the tail. The 10.0.0.0 micronet is considered used only if the entire prefix is used up.

If 10.0.255.0, 10.0.254.0, 10.0.253.0 and 10.0.252.0 micronets are in use in the 10.0.0.0/16 prefix and the 10.0.255.0 micronet has no more IPs in use and hold timer has expired for all, the 10.0.255.0 micronet is released and placed in the idle-list. The idle-list now contains 10.0.0.0, 10.0.1.0, ..., 10.0.251.0 and 10.0.255.0. If the 10.0.252.0 micronet is also no longer in use, the idle-list becomes: 10.0.0.0, 10.0.1.0, ..., 10.0.251.0, 10.0.255.0 and 10.0.252.0. The idle-list is not reordered because micronets are just added to the tail. If a new micronet is required by an MG group, the 10.0.252.0 micronet is assigned taken from the tail. Therefore, IP addresses are reused (for example, 10.0.252.0) before new IP addresses used (for example 10.0.251.0).

6.10.4 Round robin IP address allocation

If round robin mechanism is enabled, the IP addresses that are released in MSCP, are placed in a standby list instead of being reused. If all the IP addresses in a micronet are in the standby list, the micronet is released back to the CPM to be used again later.

While processing a PDN connection activation, the oldest IP address on the standby list is used for the session, under the following conditions:

- there are no free IP addresses on the MG group
- the CPM does not have any idle micronets in the prefix
- no other pools/prefixes can be used, or
- the maximum number of micronets in the MG group is exhausted

Also, there is the case that one of the other MG groups has an IP address that has never been used, although it should be in the last micronet that is allocated to this specific MG group. Because it is not feasible to know this case in lack of awareness among MG groups, the operator cannot trigger a bounce. Therefore, instead of rejecting the session, the oldest IP address from the standby list on that MG group (not globally) is chosen.

CPM always prefers unallocated prefix micronets instead of released micronets from the idle-list. Therefore, it becomes more like a least recently used list.



NOTICE:

Nokia recommends not to use round-robin selection in CUPS deployments with the LB-less mode-based UP function. In the case where all UEs assigned to a specific IP address pool are detached, the internal algorithm still uses all available IP address pools in a load-sharing manner and does not load-balance those based on the occupancy of the IP address pool. Therefore, under this rare scenario,

session unbalancing across the UP MG groups may occur. When the rate of attaches and detaches amongst the configured IP address pools is approximately equal, the system remains balanced and the round-robin feature does not cause any issues.

6.10.5 IP address allocation from micronet

Within a single micronet, the IP allocation follows the FIFO logic. So when a micronet, for example 10.0.0.0/24, is sent to the MG group, the FIFO queue is 10.0.0.0, 10.0.0.1, 10.0.0.2, ..., 10.0.0.255. If the system needs to allocate 100 addresses, the FIFO queue is pointing at the IP address 10.0.0.100. Even if the 10.0.0.0 IP address is released back and its hold-timer expires, the 10.0.0.100 IP address, which is the head of the FIFO, is chosen.

6.10.6 RADIUS IP address allocation in LB-less non-CUPS architecture

In MG+ (LB-less) architecture, each MG-group in the CMG VNF is configured with its own IP pools. For RADIUS-based IP address allocation, the CMG VNF is able to use a unique loopback (LB) per MG-group requesting the IP addresses from the RADIUS server through the authentication procedure (Access-Request/Access-Accept) and the RADIUS server must be configured with unique IP pools per client (MG-Group LB address). The MG-group specific LB address is indicated to the server in the transport IP address of RADIUS messages, and also in the NAS-IP-Address AVP of the RADIUS message.

This LB address is also used as:

- A destination IP address of RADIUS Disconnect and CoA messages from the RADIUS server
- A source IP address of ping from the gateway to the RADIUS servers, as the customer can confirm connectivity (IPv4 only)
- Acct-On/Off messages that are also sent from the configured MG-group specific LB address

The `override-profile` command is used for overriding the CMG VNF RADIUS single LB address with MG-group-specific LB address. On-the-fly modifications (without MG-group or PDN shutdown) are supported for the `override-profile` entries.

Perform the following steps to configure on-the-fly changes for the `override-profile` entry.

Addition

1. Perform an APN shutdown.
2. Configure the RADIUS VPRN and interfaces to be used by the RADIUS group.
3. Add the override entry for the router and the interface to be used by the RADIUS group VPRN and interface.
4. Configure the RADIUS group with the VPRN and interface. The RADIUS group is now linked to the APN.
5. Activate (no shutdown) the APN.

Deletion

1. Perform an APN shutdown.
2. Delete the override entry (optional).
3. Delete the override-profile entry.

Modification of an override entry is achieved by deleting that entry, and adding the required one

1. Find the RADIUS groups with the VPRN and interface that are being modified in the override-profile entry.
2. Shutdown all the APNs using these RADIUS groups.
3. Delete the override-profile entry for the VPRN and interface.
4. Add the new override-profile entry for the VPRN and interface.
5. Activate (no shutdown) the APNs.

Restrictions

1. If the same RADIUS group is used by multiple APNs, then all the APNs using that RADIUS group need to be shutdown before adding the override entry.
2. Once a RADIUS group has an override mapping and is used by an APN, it is expected that all RADIUS groups used by the APNs will have override entries. If some RADIUS groups are allowed not to have override entries, the APN-based Acct-On/Off messages sent for each MG-group will look the same across MG-groups, which is not the required behavior.

6.10.7 Centralized IP-pools functionality in LB-less deployments

Using the centralized IP-pools functionality in LB-less deployments eliminates the requirement of splitting several IP-pools and subnets across the active and future active MG-groups (inside templates), and, therefore, reduces the administrator's effort and planning. This simplifies the IP allocation in a way that the dynamic scaling operations and configurations are simplified. In summary, the new functionality supports node-specific centralized IP-pools instead of MG group-specific IP-pools. While this centralized IP-pool functionality allows for easier integration with the elasticity operations, the legacy functionality is still allowed if required.

Prefixes for the centralized IP-pools functionality

For CMG and especially for dynamic scaling, the IP-pools per MG-group is not the optimized solution because all IP-pools and subnets are required to be pre-configured even though the MG-group is not active (scale-out). In the end, this leads to wasting of the IP addresses because scaling decisions are also made by other reasons besides the number of the active bearers and allocated IP addresses. As a result, when this

occurs, the pre-defined subnets may not be fully utilized. The centralized/shared IP-pools functionality is the optimum solution for CMG that is not affected by the scaling decision.

The prefixes of the centralized IP-pools are defined in a separate VPRN called the parent VPRN, and are referred by the MG group-specific VPRNs, called child VPRNs. The centralized IP-pool is split into fragments of a size configured with the `prefix-fragment-size` parameter that is configured on the parent VPRN and these fragments are then allocated to MG-groups. You must ensure that the configuration will not harm routing on the other side when you configure a low value for `prefix-fragment-size` parameter, or these prefixes are sparsely populated on the MG-groups if you configure a high value for `prefix-fragment-size` parameter. The following configuration parameters are blocked when the `prefix-fragment-size` or the `parent-router` parameters are configured:

- `exclusive`
- `group-name`
- `reserve-ip-list`
- `reserve-ipv4`
- `reserve-ipv6`
- `use-prefixes-immediately`
- `round-robin`
- `network-determined`

The maximum number of fragments that a prefix of a centralized IP-pool can be split into, is 256. You must configure the `prefix-fragment-size` parameter in a way that does not exceed this limit. For example, if you have an IPv4-prefix of 10.0.0.0/9 configured in an IP-pool, you must set the value for the `prefix-fragment-size` parameter to 17 or lower.

The `fragment-distribution-size` command (relevant to CUPS) must be configured for a network realm, for all IP-pools under the network realm so that the CP is aware of the number of the UPF MG-groups that are active. The CP sets immediately that many prefixes in use as configured in the `fragment-distribution-size`. For example, when configuring the `ipv4-prefix` command, three fragments of size 16 are set in use immediately (that is, 10.0.0.0/16, 10.1.0.0/16, and 10.2.0.0/16), and apply the round-robin (`ip-pool-distribution` command must be enabled) among the three fragments to achieve session load-balancing across the node MG-groups. If one of the three fragments becomes exhausted, (that is, more IP addresses are available), then, when a new micronet is requested from the CPM, the system allocates a new batch of three fragments, and applies the round-robin among the 3+3=6 fragments. Free fragments are sent back to CPM for reallocation. The number of free fragments available per IP-Pool is also available through KPI/KCI statistics reporting. The respective counter is `KCIControlPlaneIpPool.VS.freelpFragments` measured on parent VPRN on UP but also on CP VPRN.

SMF configuration

To configure the SMF, use the following command:

```
config
```

```

...
network-realm "nokia.com"
  fragment-distribution-size 3
  ip-local-pool "Super-pool-1"
    ipv4-prefix 10.0.0.0/10      prefix-fragment-size 16
    ipv6-prefix 3ffe::/44       prefix-fragment-size 48
  exit
exit

```

UPF configuration

To configure the UPF, use the following command:

```

config
  router
    ip-local-pool "Super-pool-1"
      remote
        ipv4-prefix 10.0.0.0/10      prefix-fragment-size 16
        ipv6-prefix 3ffe::/44       prefix-fragment-size 48
      exit
    exit
  service
    vprn 1
      parent-router base
      mg-group 1
      ip-local-pool "Super-pool-1"
        remote
      exit
    exit
    vprn 2
      parent-router base
      mg-group 2
      ip-local-pool "Super-pool-1"
        remote
      exit
    exit
    vprn 3
      parent-router base
      mg-group 3
      ip-local-pool "Super-pool-1"
        remote
      exit
    exit
  exit
exit

```


Non-CUPS combined SGW/PGW configuration

To configure the combined SGW/PGW for non-CUPs, use the following command:

```
config
mobile
pdn
apn
  router "vprn1"
exit
  router "vprn2"
exit
  router "vprn3"
exit
config
  router
    ip-local-pool "Super-pool-1"
      ipv4-prefix 10.0.0.0/10      prefix-fragment-size 16
      ipv6-prefix 3ffe::/44        prefix-fragment-size 48
    exit
  exit
  service
    vprn 1
      parent-router base
      mg-group 1
      ip-local-pool "Super-pool-1"
      exit
    exit
    vprn 2
      parent-router base
      mg-group 2
      ip-local-pool "Super-pool-1"
      exit
    exit
    vprn 3
      parent-router base
      mg-group 3
      ip-local-pool "Super-pool-1"
      exit
    exit
  exit
exit
```

Example

The existing block and force commands for specific fragments are also supported. To view the used fragments, use the show command, and then block or force the fragments required. The following example shows that the configured (centralized) prefix 6.0.128.0/17 has allocated the fragment 6.0.128.0/23 with 50% utilization, and that is then blocked to naturally drain and return to the CPM for reallocation.

```
show mobile-gateway pdn up-peer-list ip-pool-stats peer-ip 33.34.1.1
Network-realm    : nokia.com
Frag-Distro-Size: 2
-----
pool-name        : pool1v4
Apn-Exclusive    : disabled
Network-Determ   : disabled
Remote-Pool      : disabled
-----
Prefix
Block Force Used   Free   Held   Util%   Fragm Size
-----
6.0.128.0/17     No    No    0        0      0     0.00    23
6.0.128.0/23     No    No   254     254    0     50.00    0
-----
Total
Used   Free   Held   Util%
-----
Ipv4   254    254    0     50.00
Ipv6    0      0      0     0.00
-----
Reserv IP List   : -
-----
No IP Addresses configured
=====

/configure mobile-gateway profile pfcf up-peer-list peer 33.34.1.1 network-
realm nokia.com ip-pool pool1v4 ipv4-prefix 6.0.128.0/23 block
```



NOTE:

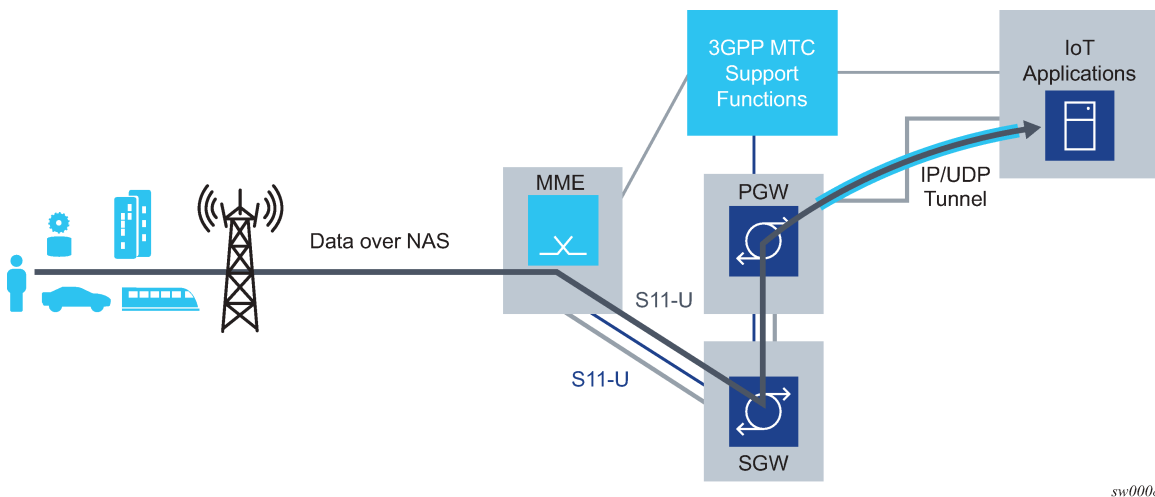
To block the entire prefix, the prefix-fragment-size must also be specified.

6.11 NarrowBand-IoT SGi tunneling for non-IP PDNs

Machine-to-Machine applications use non-IP data. The NB-IoT S-Gi tunneling for non-IP PDNs feature supports this functionality as described in 3GPP TS 23.720.

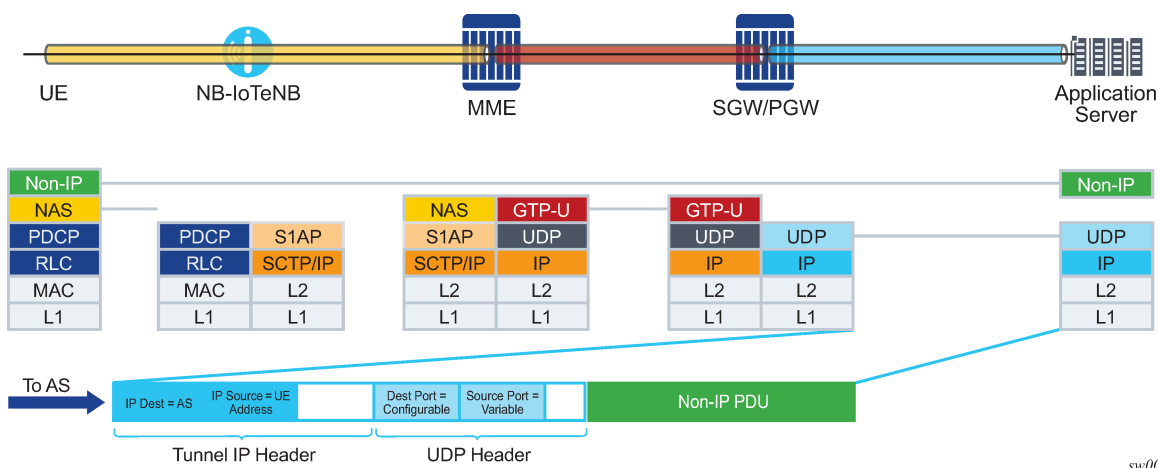
This feature is supported on the CMG.

Figure 76: Tunneling for non-IP PDNs



The EPC transports non-IP data between the UE and a set of application servers. This feature utilizes a method of tunneling non-IP data between the PGW and the application servers.

Figure 77: Non-IP PDU tunneling



The PGW supports the forwarding of non-IP packets to the application server. The SGi tunneling using UDP/IP is described in 3GPP 23.401 and is shown in *Non-IP PDU tunneling*.

The PGW allocates locally an IP address to the PDN connection even though it does not provide the IP address to the UE via the normal PDN address allocation.

Upon receiving a packet on the PDN connection (identified by the GTP-U TEID), the PGW encapsulates the packet in UDP and IP encapsulation where the:

- Source IP address is the IP address allocated to the PDN connection (the UE address)
- Destination address is the configured address of the application server
- UDP source port is the port chosen by the PGW
- UDP Destination port is the configured port to be used for non-IP encapsulation

Upon receiving a PDU back from the application server and provided that the UE source address is unique, the PGW will forward to the right DL TEID.

Use the following commands to configure the NB-IoT SGI tunneling feature.

To create a UDP tunnel group, use the following command:

```
configure mobile-gateway profile udp-tunnel-group
```

To associate an application server IP address and specify the destination UDP port for the tunnel, use the following commands:

```
configure mobile-gateway profile udp-tunnel-group peer
```

```
configure mobile-gateway profile udp-tunnel-group peer destination-port
```

To select non-IP for the PDN type, use the following command:

```
config mobile pdn apn pdn-type
```

**NOTE:**

When configuring an APN with non-IP PDN type, make sure that no other PDN type is selected for this APN.

To associate the non-IP PDN type with the UDP tunnel group previously created, use the following command:

```
configure mobile-gateway pdn apn non-ip-pdn udp-tunnel-group
```

The following commands have been updated to include the new non-IP PDN type:

```
show mobile-gateway pdn pdn-context
```

```
show mobile-gateway pdn pdn-context apn
```

```
show mobile-gateway pdn statistics
```

```
show mobile-gateway pdn bearer-context imsi detail
```

```
show mobile-gateway pdn apn statistics
```

```
show mobile-gateway pdn pdn-context imsi detail
```

To view information about the configured UDP tunnel groups, use the following command:

```
show mobile profile udp-tunnel-group
```

To view statistics for the tunnel groups toward application servers for non-IP PDN types, use the following command:

```
show mobile-gateway pdn application-server-stats udp-tunnel-group
```

6.12 Mobile-to-All redirection

This feature allows all upstream mobile traffic to be redirected to a core node by changing the IP address of the next hop.

The PGW/GGSN normally forwards traffic internally between two mobile subscribers that are attached to the same APN. Therefore, local mobile-to-mobile traffic may never traverse the core network. There may be cases when it is necessary to pass such traffic through a core node (such as a firewall or a Deep Packet Inspection node). In addition, there may be cases where all upstream traffic must be passed through a particular core node.

To configure the redirection per APN, use the following command:

```
configure mobile-gateway pdn apn redirect
```

6.13 Diameter Routing Message Priority

The PGW supports the Diameter Routing Message Priority (DRMP) AVP over the S6b interface. The PGW sends the DRMP AVP including a priority value for the eUTRAN/eHRPD call flow associated with the S6b Diameter request.

In S6b interface deployments where the Diameter Overload Indication Conveyance (DOIC) is only handled between the Diameter Routing Agents (DRAs) and the Diameter Servers, the DRMP AVP allows the PGW to add the relative priority of the access network call flow to the S6b Diameter request message to the DRA. The DRA can then factor that priority into routing, resource allocation, and overload abatement decisions.

The DRMP AVP format is based on RFC7944 and when enabled, it includes a configurable priority value that can be set for each of the network access call flows shown in *Call flows and DRMP value in Diameter messages*.

Table 23: Call flows and DRMP value in Diameter messages

Network access call flow	DRMP value in S6b Diameter Request message
EUTRAN initial attach IMSI/GUTI	The PGW includes the DRMP value in the S6b:AA-Request message to the 3GPP AAA
eHRPD initial attach	The PGW includes the DRMP AVP in the S6b:AA-Request message to the 3GPP AAA
WiFi initial attach	The PGW includes the DRMP AVP in the S6b:AA-Request message to the 3GPP AAA
EUTRAN-initiated detach	For any UE-, MME-, or HSS-initiated detach call flows, the PGW includes the DRMP AVP in the S6b:ST-Request message to the 3GPP AAA
eHRPD-initiated detach	For any UE- or HSGW-initiated detach call flows, the PGW includes the DRMP AVP in the S6b:ST-Request message to the 3GPP AAA
WiFi-initiated detach	The PGW includes the DRMP AVP in the S6b:ST-Request message to the 3GPP AAA
Re-Auth-initiated	For any AAA Re-Auth (RAR) call flow initiated by the AAA the PGW includes the DRMP AVP in the S6b:AA-Request message to the 3GPP AAA
PGW-initiated PDN Disconnect	The PGW includes the DRMP AVP in the ST-Request message to the 3GPP AAA. The PGW-initiated PDN disconnect also maps to cases where the PGW gets an ASR from the AAA to terminate the session and needs to send the S6b:ST-Request message back to the 3GPP AAA
eHRPD - LTE handover	The PGW includes the DRMP AVP in the S6b:AA-Request message to the 3GPP AAA

Table 23: Call flows and DRMP value in Diameter messages (continued)

Network access call flow	DRMP value in S6b Diameter Request message
LTE - WiFi handover	The PGW includes the DRMP AVP in the S6b:AA-Request message to the 3GPP AAA
eHRPD - WiFi handover	The PGW includes the DRMP AVP in the S6b:AA-Request message to the 3GPP AAA

To enable DRMP and configure the priority value for any of the above network access call flows, use the commands under the following context:

```
config>mobile>profile>s6b-profile>drmp
```

6.14 Diameter DRA temporary overload support

This feature provides support on the Gx interface for additional handling upon receiving a Diameter_Too_Busy (3004) message from a DRA that is in a temporary overload condition.

This includes deployments with a Primary and a Secondary DRA between the PGW and the PCRF (where the DRA does not support Diameter Overload Indication Conveyance) and when peer group lists are used even if Primary and Secondary DRA servers are not configured.

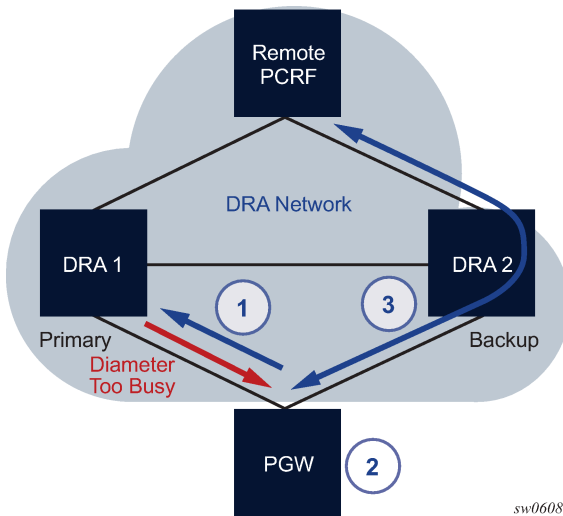
To enable the Diameter DRA overload handling based on the Diameter_Too_Busy message, use the following command:

```
configure mobile-gateway pdn gw-id gx dra-diameter-too-busy re-router-timer minute-value
```

To enable the Diameter DRA overload handling for a peer, use the following command:

```
configure mobile-gateway profile diameter-peer peer ip-addr | fqdn [port port] enable-dra
```

Figure 78: Diameter DRA temporary overload support



When this feature is enabled and the Primary DRA (DRA 1) responds with result code `Diameter_Too_Busy`, the PGW initiates the reroute timer and reissues the Gx request and any subsequent request messages to the Backup/Secondary DRA (DRA 2) for the duration of the reroute timer. When the timer expires, the PGW starts using the Primary DRA peer.

Any `Diameter_Too_Busy` messages received from the PCRF behind the DRA follows the configured Gx interface failure handling (continue, terminate, and so on) using the Origin-Host AVP to distinguish if the `Diameter_Too_Busy` message originates from the DRA or the PCRF.

In deployments with Primary and Secondary DRA servers, if both DRAs respond with a `Diameter_Too_Busy` message, the PGW terminates the session. If an RAR for an existing session is still issued via the Primary DRA, after a `Diameter_Too_Busy` message is received and the reroute timer is running, the PGW issues the RAA respond back to the Primary DRA and any subsequent requests for this session are sent to the Secondary/backup DRA.

6.15 eMPS-based IMS call priority services

The Enhanced Multimedia Priority Service (eMPS) allows for applying a higher priority in the network to an IMS-initiated voice/video call (for example, first responders who are required to have priority treatment as per government mandates).

An eMPS IMS-initiated call is marked with a higher priority in the network to minimize the risk of it getting dropped or throttled due to network overload. On the PGW, a UE will be known and treated as an eMPS priority PDN session once it is initiated by the PCRF over the Gx interface via the IMS system.

The PGW sees the UE as a regular UE during the initial attach and bearer setup and then proceeds with the normal default bearer set up. Once a UE with eMPS access initiates an IMS voice/video call via the IMS system, the IMS system will authorize it as an eMPS priority and inform the PCRF via the Cx/Rx interface. The PCRF will then initiate a bearer modification procedure to inform the PGW, based on the DRMP AVP value which is set to 0 and the updated QoS for the PDN session, with the default and the possible dedicated bearers.

The PCRF will then initiate a bearer modification procedure to inform the PGW, based on the DRMP AVP value which is set to 0 and the updated QoS for the PDN session with ARP values of 1, 2, or 3.

To view eMPS statistics, use the following commands:

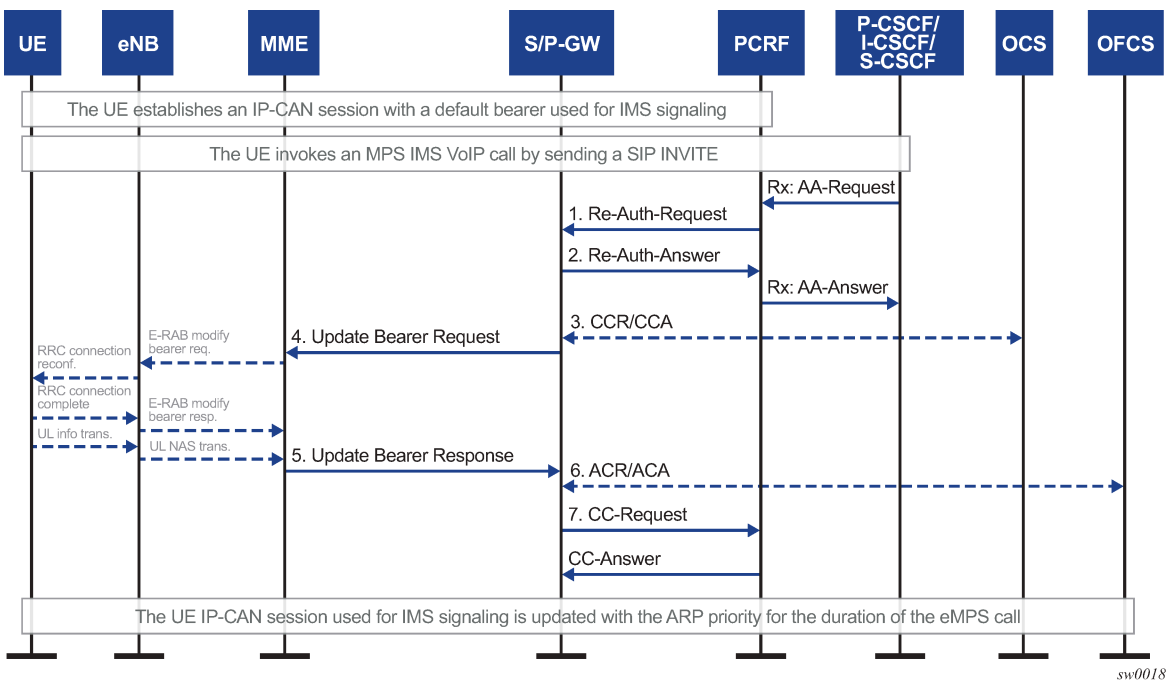
```
show mobile-gateway pdn apn statistics
```

```
show mobile-gateway pdn call-flow-stats
```

```
show mobile-gateway pdn pdn-context imsi detail
```

6.15.1 eMPS IMS priority-initiated call over default bearer

Figure 79: eMPS IMS priority-initiated call over default bearer



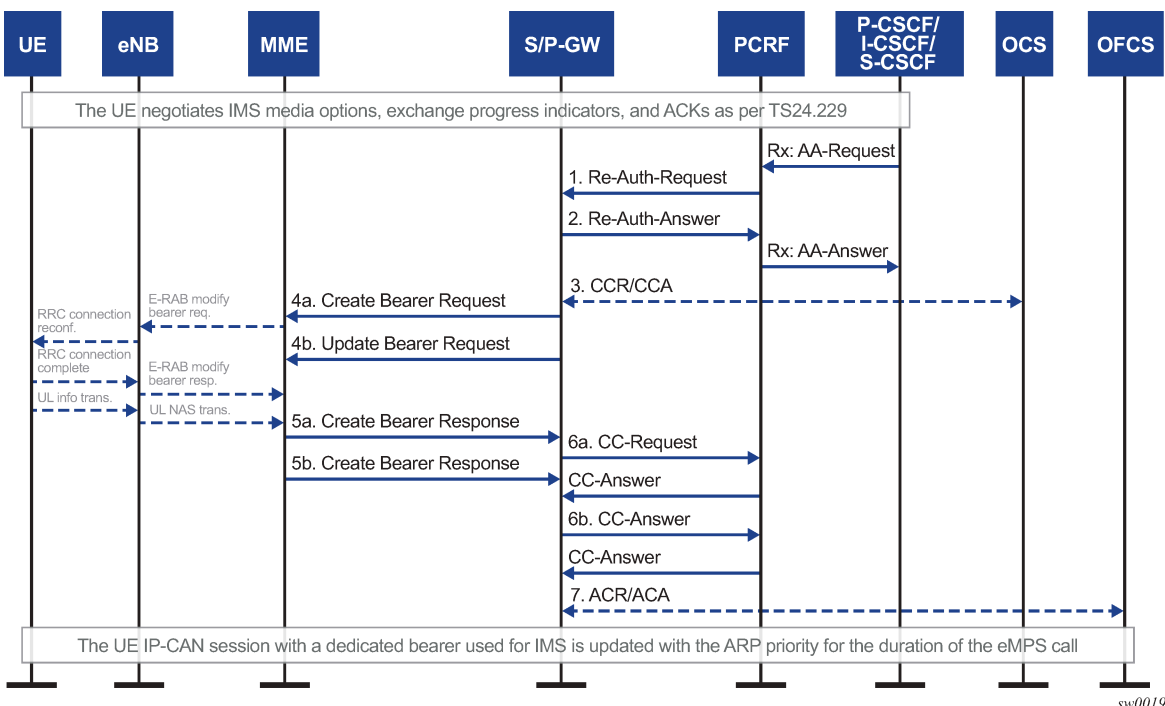
The following steps describe the call flow shown in *eMPS IMS priority-initiated call over default bearer*.

1. Based on the IMS P-CSCF Rx messaging, the PCRF initiates the authorization of a new flow by sending a Re-Auth-Request message to the PGW to update the default bearer ARP value change.
2. Based on the DRMP AVP value 0, the PGW tags the PDN session as an eMPS session and responds with RAA to the PCRF.
3. There are no special billing requirements for eMPS. Regular handling follows to determine if CCR-U should be sent.
4. Based on the RAR for the PCRF, the PGW uses the AVP information and the PCC-Rules with QoS and ARP parameters to initiate an update of the default bearer, and sends an Update Bearer Request.

5. The MME acknowledges the update bearer request by sending an Update Bearer Response.
For current support of the IMS-initiated eMPS default Bearer upgrade, the GTP-C MP-flag and message priority must be added to the Update Bearer Request.
6. The PGW sends a CC-Request (CCR) command to update the PCRF.
7. There are no special billing requirements for eMPS. Regular handling follows, if resource allocation notification was requested with Event-Trigger, to determine if ACR-Interim should be sent.

6.15.2 eMPS IMS priority-initiated call with dedicated bearer

Figure 80: eMPS IMS priority-initiated call with dedicated bearer



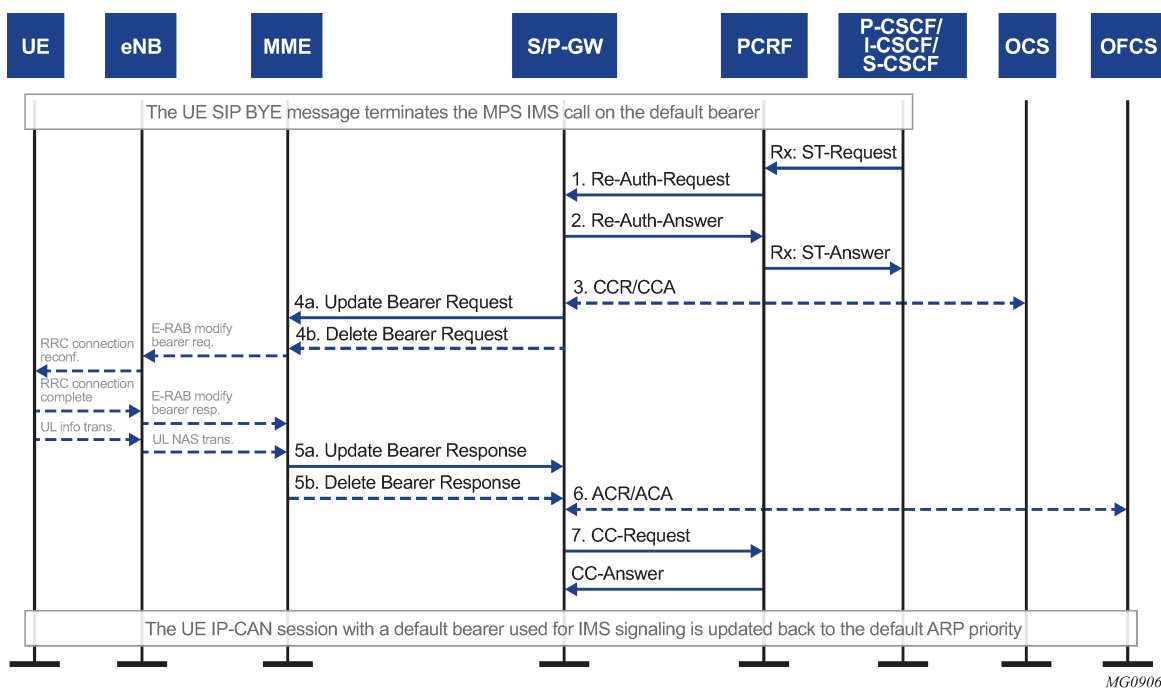
The following steps describe the call flow shown in *eMPS IMS priority-initiated call with dedicated bearer*.

1. Based on the IMS P-CSCF Rx messaging, the PCRF initiates the authorization of a new flow by sending a Re-Auth-Request message to the PGW to create a new dedicated bearer and update the default bearer ARP value.
2. Based on the DRMP AVP value 0, the PGW tags the PDN session as an eMPS session and responds with RAA to the PCRF.
3. There are no special billing requirements for eMPS. Regular handling follows to determine if CCR-U should be sent.
4. Based on the RAR from the PCRF, the PGW uses the AVP information and the PCC-Rules with the QoS and ARP parameters to initiate a new dedicated bearer and update the default bearer, and sends a Create Bearer Request and an Update Bearer Request.

5. The MME acknowledges the bearer activation and update bearer request by sending a Create Bearer Response and Update Bearer Response.
6. The PGW sends a CC-Request (CCR) command to update the PCRF, if resource allocation notification was requested with Event-Trigger.
7. There are no special billing requirements for eMPS. Regular handling follows to determine if ACR-Interim should be sent.

6.15.3 eMPS IMS termination and bearer deactivation

Figure 81: eMPS IMS termination and bearer deactivation



The following steps describe the call flow shown in *eMPS IMS termination and bearer deactivation*.

1. Based on the IMS CSCF Rx trigger, the PCRF initiates the bearer modification procedure by sending a RAR without the DRMP AVP and a Default-EPS-Bearer-QoS AVP with the original non-Priority ARP value.
2. The PGW removes the eMPS priority for the PDN session and responds with RAA to the PCRF.
3. There are no special billing requirements for eMPS. Regular handling follows to determine if CCR-U should be sent.
4. (a and b) Based on the RAR from the PCRF, the PGW uses the AVP information and the PCC-Rules with QoS and ARP parameters to update the default bearer and possibly remove any dedicated bearers that were created during the eMPS IMS calling.
5. (a and b) The MME acknowledges the bearer modification with Update Bearer Response and possibly Delete Bearer response.

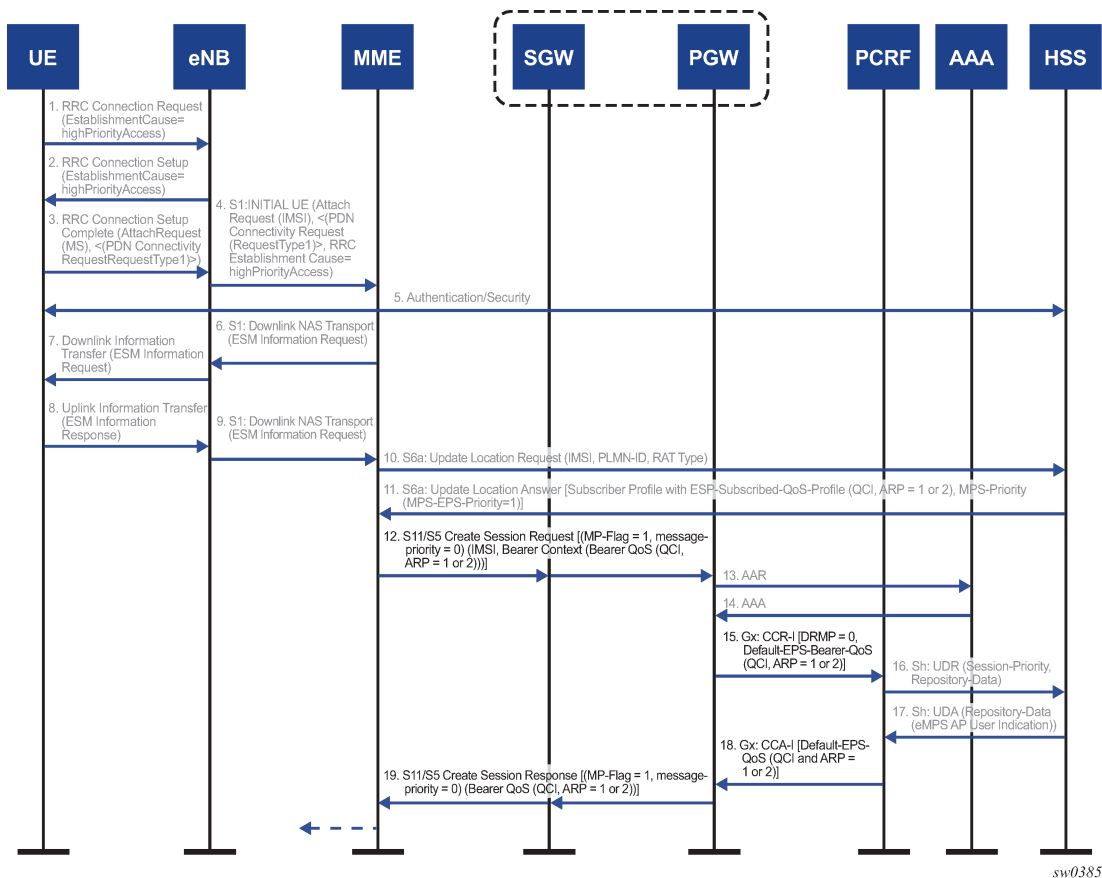
6. The PGW sends a CC-Request (CCR) command to update the PCRF.
7. There are no special billing requirements for eMPS. Regular handling follows to determine if ACR-Interim should be sent.

6.15.4 eMPS advance priority

The eMPS advance priority allows eMPS subscribers to attach to the network with the IMS PDN session default bearer that has been established with an eMPS priority for the duration of the PDN session.

The eMPS advance priority is initiated by the HSS/MME upon receiving an Establishment Cause with High Priority Access from the UE during the RRC connection establishment. The HSS will trigger the MME to initiate the eMPS priority PDN session to be established to the SGW/PGW.

Figure 82: eMPS advance priority call flow



The following steps describe the call flow shown in *eMPS advance priority call flow* (eMPS advance priority-related steps).

12. Based on the MPS-EPS-Priority set to 1 and ARP PL from HSS, the MME sends the Create Session Request message with the GTP-C header MP-flag set to 1 and the

priority set to 0. The QCI and ARP values are set to the values received from the HSS.

13. The PGW sends an AAR message (Session-ID, Origination-TimeStamp, Max-Wait-Time, MIP6-Agent-Info (PGW-FQDN), username (IMSI-NAI), APN, RAT-Type) to the 3GPP AAA to authorize the APN for the subscriber.

If the S6b interaction fails or timeouts, a local fallback can be configured to be used with the local PGW configuration for all the required parameters, including the P-CSCF and IP pool names, charging gateway host, charging group ID, session timeout, and idle timeout. It will then continue to message 15.

15. The PGW, based on the GTP-C header MP-flag set to 1 and the GTPv2-C priority field set to 0, issues a CCR-I message with the DRMP AVP set to 0 and the Default-EPS-Bearer-QoS AVP with the QCI and ARP values as received in the Create Session Request message.
18. For a successful eMPS call, the PCRF sets the ARP for the IMS APN to 1 or 2 and sends it to the PGW which prioritizes the message based on the received ARP.
19. The MME and the SGW prioritize the processing of the message based on the MP-flag and the GTPv2-C priority field in GTP-C header.

**NOTE:**

- The eMPS advance priority call flow is only supported for the LTE access type. No other access or handover to any other access type is supported.
- If the DRMP AVP is enabled on the S6b AAR, it will not indicate an eMPS call. It will be sent with a regular DRMP AVP value (that is, no change on the S6b interface).

The eMPS advance priority is supported with an S6b interface bypass in case the S6b connectivity fails when the authorization or re-authorization of an eMPS call flow is performed by the PGW.

The S6b interface bypass can be configured separately for the following (the local configuration will be used):

- no response from any of the available AAAs/DRAs
- all available AAAs/DRAs returning any of the following failure/error codes
 - 3xxx
 - 4xxx
 - 5xxx

The eMPS priority is enabled by default. To disable it, use the following command:

```
config>mobile>pdn>no ims-emps-priority
```

To configure the failure handling behavior for new sessions that fail to get a response from the AAA server, use the following commands:

```
config>mobile>profile>s6b-profile>fh-session-no-response
```

```
config>mobile>profile>s6b-profile>fh-time-limit
```

To configure the failure handling behavior based on the error code received, use the following commands:

```
config>mobile>profile>s6b-profile>message>
  result-code-3xxx
  result-code-4xxx
  result-code-5xxx
```

Table 24: S6b interface parameter values used in S6b eMPS bypass

S6b interface parameter	Description
Idle timeout	Uses the value configured in the <code>config>mobile>pdn>apn>idle-timeout</code> command
Session timeout	Uses the value configured in the <code>config>mobile>pdn>apn>session-timeout</code> or the <code>config>mobile>profile>s6b>fh-time-limit</code> command ³
Server name ⁴	Uses the value configured in the following commands: <pre>config>mobile>pdn>apn>pco> pcscf-srvr-ipv4 pcscf-srvr-ipv6 pcscf-srvr-v4-group</pre>

³ When the `fh-time-limit` command is configured and an eMPS S6b bypass scenario takes place either during an AAR-I or an ARR-U, the configured value of the `fh-time-limit` command will be applied as the session time for the S6b bypass session.

When the `fh-time-limit` command is not configured, the configured value of the `session-timeout` command will be used with its current behavior in case of an eMPS bypass scenario.

Therefore, regardless of the eMPS S6b bypass taking place or not, during AAR-I or AAR-U the session timeout started during the initial attach will continue to be used.

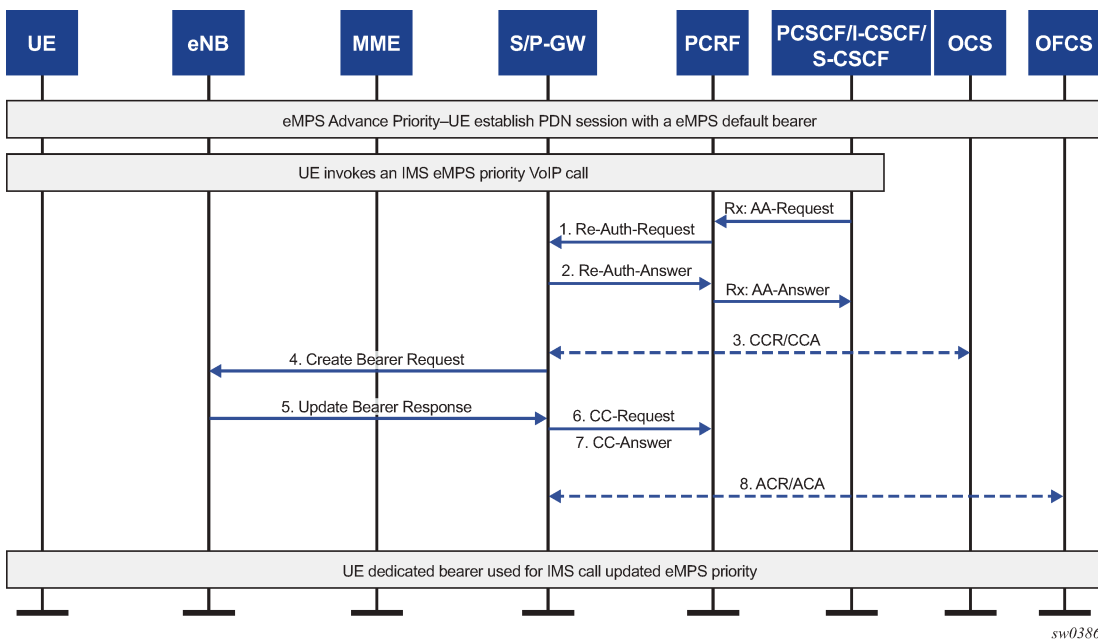
⁴ Used for the local fallback case.

Table 24: S6b interface parameter values used in S6b eMPS bypass (continued)

S6b interface parameter	Description
	pcscf-srvr-v6-group
Framed pool ⁴	Uses the value configured in the config>mobile>pdn>apn>ip-pool-name command
Framed IPv6 pool ⁴	Uses the values configured in the config>mobile>pdn>apn>router>ip-pool-name and the config>mobile>pdn>apn>ip-alloc-method commands

The eMPS advance priority also allows for dedicated bearers to be established (triggered via PCRF Gx:RAR). The dedicated bearer can be initiated with or without an eMPS priority.

Figure 83: PCRF-initiated dedicated bearer call flow with eMPS priority establishment



The following steps describe the call flow shown in *PCRF-initiated dedicated bearer call flow with eMPS priority establishment*.

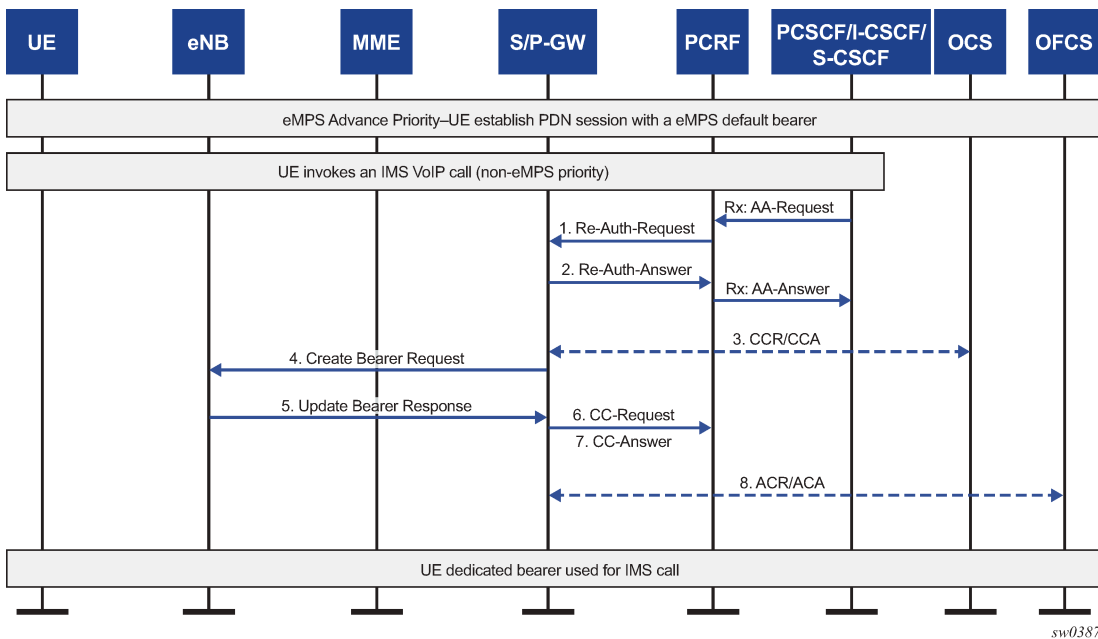
1. Based on the IMS P-CSCF Rx messaging, the PCRF initiates the authorization of a new flow by sending a Re-Auth-Request message to the PGW to create a new dedicated bearer with an ARP value of 1, 2, or 3 and the DRMP AVP set to 0.
2. The combined SGW/PGW responds with a RAR message to the PCRF.

3. There are no special billing requirements for eMPS; regular handling to determine if a CCR-U message should be sent follows.
4. The combined SGW/PGW stores the bearer context with the eMPS priority and issues a Create Bearer Request with the ARP value as received from the PCRF and the MP-flag in the GTPv2-C header set, and the message-priority set to 0.
5. The MME acknowledges the bearer activation and updates the bearer with eMPS priority and sends a Create Bearer Response (with the GTPv2-C header MP-flag set and the message-priority set to 0).
6. The PGW sends a CC-Request (CCR) command to update the PCRF.
7. There are no special billing requirements for eMPS; regular handling to determine if an ACR-Interim should be sent follows.

**NOTE:**

- For the eMPS advance priority, the default bearer is always kept with an eMPS priority. Therefore, the PCRF will not initiate any default bearer QoS update in any Gx:RAR message that would change the ARP value from 1 or 2. If the PCRF changes the default bearer to any other ARP value, the eMPS priority will be removed for the default bearer and any dedicated bearers for that PDN session.
- The PCRF may only send a Gx:RAR message to create/update a dedicated bearer with eMPS priority by including the DRMP AVP with a value of 0 and the ARP value is set to 1, 2, or 3.
 - If the PCRF sends a Gx:RAR message to create/update a dedicated bearer with the DRMP AVP set to 0 but an ARP value greater than 3, the dedicated bearer will not be upgraded with eMPS priority.
 - If the PCRF sends a Gx:RAR message to create/update a dedicated bearer with an ARP value of 1, 2, or 3 but the DRMP AVP is not set to 0, the dedicated bearer will not be upgraded with eMPS priority. If the dedicated bearer was in eMPS priority, it will be downgraded.

Figure 84: PCRF-initiated dedicated bearer call flow without eMPS priority establishment



The following steps describe the call flow shown in *PCRF-initiated dedicated bearer call flow without eMPS priority establishment*.

1. Based on the IMS P-CSCF Rx messaging, the PCRF initiates the authorization of a new flow by sending a Re-Auth-Request message to the PGW to create a new dedicated bearer with an ARP value change.
2. The combined SGW/PGW responds with a RAR message to the PCRF.
3. There are no special billing requirements for eMPS; regular handling to determine if a CCR-U message should be sent follows.
4. The combined SGW/PGW issues a Create Bearer Request with the ARP value and the GTPv2-C header with the MP-flag unset.
5. The MME acknowledges the bearer activation and updates the bearer request by sending a Create Bearer Response (with the GTPv2-C header MP-flag unset).
6. The PGW sends a CC-Request (CCR) command to update the PCRF.
7. There are no special billing requirements for eMPS; regular handling to determine if an ACR-Interim should be sent follows.

6.15.4.1 DSCP marking of eMPS sessions

The eMPS and advance priority support provide the option to configure a specific DSCP value for eMPS sessions. When a session is identified as an eMPS session, all packets of this session, over the GTP-C/ GTP-U and SGi (uplink) reference points, are marked with the configured eMPS-specific DSCP value.

For the SGW, PGW, and combined SGW/PGW, the DSCP marking is applied over the S1-U, S5-U, S5-C, S11, and SGi reference points.

To configure DSCP marking for eMPS sessions, use the following command:

```
configure mobile-gateway pdn gw-id ims-emps-priority cp-dscp-marking [value]  
up-uplink qci-policy-name up-downlink qci-policy-name
```

To configure the QCI values used when DSCP marking for eMPS sessions is enabled, use the following command:

```
configure mobile-gateway profile policy-options qci-policy qci qci-value
```

To configure the DSCP values used when DSCP marking for eMPS sessions is enabled, use the following command:

```
configure mobile-gateway profile policy-options qci-policy qci qci-value dscp  
{dscp-name | in dscp-name [out dscp-name] | out dscp-name [in dscp-name]}
```

For more information about the commands used for configuring DSCP marking for eMPS sessions, refer to the *CMG CLI Reference Guide*.

6.16 APN-level back-off time and load and overload control reporting

The PGW supports APN-level congestion control based on 3GPP TS 29.274 and 23.401.

Congestion control can be configured using the following two methods:

- PGW back-off time mechanism
- GTP-C APN-level load and overload control based on GTPv2-C

The PGW can be configured to use either the back-off time mechanism or the APN-level load and overload control mechanism for a specified APN.

To view the APN overload statistics, use the following command:

```
show mobile-gateway pdn apn-overload
```

Back-off Time mechanism

If back-off time is enabled and overload control is being performed, the PGW will reject every PDN connection request and include the PGW back-off time IE with the timer value configured for the specific APN that is in overload to the MME.

During the time the APN is in overload, the PGW will continue to reject any new PDN connection requests from the MME for this APN.

To enable the back-off time mechanism, the maximum number of PDN connections or maximum rate of PDN connection activations must be first configured using the following commands:

```
configure mobile-gateway pdn apn gtp-pdn-threshold max-activations
```

```
configure mobile-gateway pdn apn gtp-pdn-threshold max-connections
```

To configure the back-off time, use the following command:

```
configure mobile-gateway pdn apn overload-back-off-timer
```

6.17 Time-conditioned APN-AMBR handling

Time-conditioned APN-AMBR handling is supported on the Gx interface as per 3GPP PCC standards.

The PCRF has the ability to specify up to four instances of the Conditional-Policy-Information AVP, each of which includes an APN-AMBR and an Execution-Time. At the specified time, the PGW updates and enforces the corresponding APN-AMBR without interacting with the PCRF. The updated APN-AMBR is signaled toward the UE.

The following lists terminology specific for this feature.

Authorized APN-AMBR	The APN-AMBR for the total bandwidth usage of non-GBR QCI traffic at the APN.
Subsequent APN-AMBR	The APN-AMBR for the total bandwidth usage of non-GBR QCI traffic at the APN to be applied by the PCEF when the APN-AMBR change time is reached.
APN-AMBR change time	The time at which the PCEF will apply the Subsequent APN-AMBR for the total bandwidth usage of non-GBR QCI traffic at the APN.

Table 25: Conditional-Policy-Information AVP support

AVP \ Message	CCA(I)	CCA(U)	CCA(T)	RAR
0*4 [Conditional-Policy-Information]	√	√	N/A	√

Table 26: Gx-specific Diameter AVPs

Attribute name	AVP code	Clause defined	Value type (Note 2)	AVP flag rules (Note 1)				May encr	Acc type	Applicability
				Must	May	Should not	Must not			
Conditional-Policy-Information	2840	5.3.128	Grouped	V	P	–	M	Y	All	Both CondPolicyInfo
Execution-Time	2839	5.3.127	Time	V	P	–	M	Y	All	Both CondPolicyInfo

**NOTE:**

1. Both parameters (Subsequent APN-AMBR and APN-AMBR change time) are provided together.
2. The PCRF provides up to four instances of the parameters. If multiple instances are provided, the values of the APN-AMBR change time should not be too close to each other, to reduce the risk for a signaling overload. The PCEF will discard any previously received subsequent APN-AMBR instances, whenever the PCRF provides a new authorized APN-AMBR or subsequent APN-AMBR.
3. The PCRF can replace all instances that have been provided previously with a new instruction. A previously provided subsequent APN-AMBR or APN-AMBR change time parameter cannot be individually modified.
4. There is always an unconditional value for the APN-AM BR available at the PCEF. The initial value is received as subscribed APN-AMBR in an access-specific manner and the PCRF can overwrite it by providing an authorized APN-AMBR.
5. To reduce the risk of a signaling overload, the PCRF avoids simultaneous provisioning of authorized APN-AMBR for many UEs (for example, by spreading them over time).
6. The modification is made similarly to if the PCRF had modified the authorized APN-AMBR at that point in time.
7. A changed APN-AMBR will be communicated to the UE.

Support for time-conditioned APN-AMBR handling is indicated to the PCRF by setting the CondPolicyInfo feature bit within the Supported-Features AVP. This is populated based on the feature being enabled.

To enable the CondPolicyInfo feature bit, use the following command:

```
configure mobile-gateway profile gx supported-features conditional-policy-info
```

To view the configured conditional policy information, use the following command:

```
show mobile-gateway profile gx
```

To view the execution time and APN-AMBR for this feature, use the following command:

```
show mobile-gateway pdn pdn-context imsi gx-time-of-day-info
```

**NOTE:**

The `gx-time-of-day-info` command displays both APN-AMBR and QCI/ARP values, regardless of whether they are both installed. For example, if only QCI/ARP values are enabled, the APN-AMBR shows as 0 and vice versa.

Caveats

- The scope of this feature is limited to support for time-conditioned APN-AMBR handling (Supported-Features = CondPolicyInfo).
- This feature does not include support for conditional APN-AMBR handling (Supported-Features = ConditionalAPNPolicyInfo). Conditions include a list of applicable RAT types and/or a list of IP-CAN types.
- Time-conditioned APN policy information for the APN-AMBR change helps reduce the signaling load over the Gx interface. However, the APN-AMBR change needs to be communicated to the UE. Consequently a simultaneous change of the APN-AMBR for many UEs may introduce a signaling storm in the EPC (for example, over S5/S8/S2a/S2b). The PCRF can avoid this simultaneous change of the APN-AMBR, by, for example, spreading the time-conditioned change over time for many UEs.

6.18 Configuring the time-conditioned default EPS bearer QoS change handling

Time-conditioned default EPS bearer QoS change handling is supported on the Gx interface per 3GPP PCC standards. The feature is supported within the Conditional-Policy-Information AVP.

Purpose

This feature provides the PCRF with the ability to specify up to four instances of the Conditional-Policy-Information AVP, each of which includes APN-AMBR and/or Default-EPS-Bearer-QoS and an Execution-Time. At the specified time, the gateway updates and enforces the corresponding Default EPS Bearer QoS handling, without interacting with the PCRF. The updated default bearer QCI/ARP is signaled toward the UE.

This feature also supports the rule bound to default bearer feature as per 3GPP standards. The PCRF may indicate to the PCEF that a PCC rule is to be bound to the default bearer and remain on the default bearer.

The PCEF binds the indicated PCC rule to the default bearer, until the PCC rule is removed or until the PCRF modifies the PCC rule to set the Default-Bearer-Indication AVP to the value BIND_TO_APPLICABLE_BEARER.

**NOTE:**

The rule bound to default bearer feature must be enabled prior to enabling the Time-conditioned default EPS bearer QoS change feature.

Procedure

1. Enable the rule bound to default bearer option for the Supported Features AVP in the Gx profile.

```
configure mobile-gateway profile gx supported-features rule-bound-to-  
default-bearer
```

2. Enable the time-conditioned default EPS bearer QoS change option for the Supported Features AVP in the Gx profile.

```
configure mobile-gateway profile gx supported-features cond-policy-Info-  
default-qos
```

3. View the configuration and the time-conditioned default bearer QCI/ARP information.

```
show mobile-gateway profile gx
```

```
show mobile-gateway pdn-context imsi imsi gx-time-of-day-info
```

**NOTE:**

The `gx-time-of-day-info` command displays both APN-AMBR and QCI/ARP values, regardless of whether they are both installed. For example, if only QCI/ARP is enabled the APN-AMBR indicates 0 and vice versa.

6.19 PGW support for ePDG selection using PCO

The PGW (standalone or combined SGW/PGW) supports selection of the closest ePDG server when a UE attaches to LTE/eHRPD sessions. This functionality allows the UE to connect to the closest ePDG to the connected PGW, thus providing quicker connection time and fewer packet transfer e2e delays.

During the attach or mobility request, the UE can request the ePDG IPv6 and/or IPv4 address from the PGW, using the following PCO values:

- Operator Specific (FF01H — ePDG IPv6 Address Request)
- Operator Specific (FF02H — ePDG IPv4 Address Request)

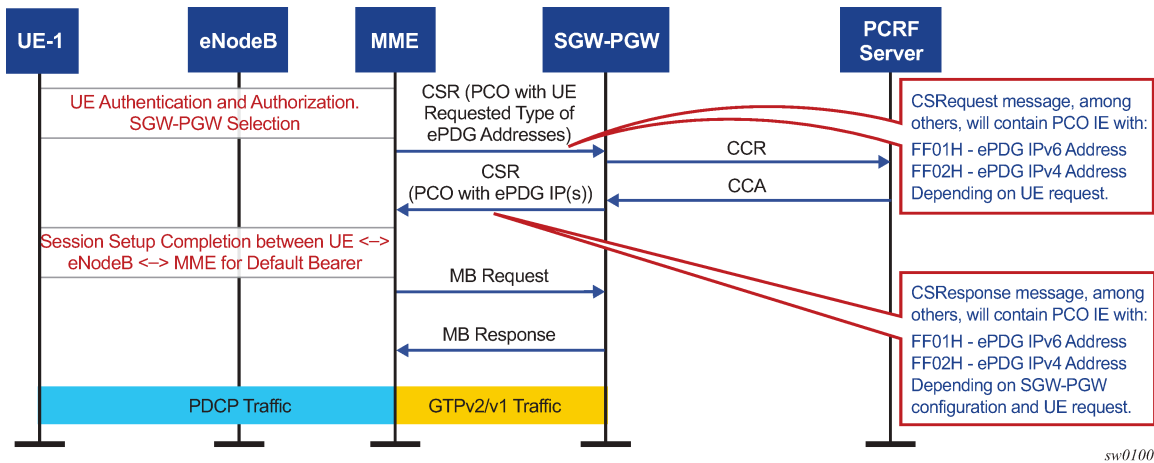
The PGW will offer the IPv4, IPv6, or IPv4 and IPv6 servers that are configured for the APN.



NOTE:

The PGW will send the ePDG servers in the CS response only if the **custom-1** option is specified for the operator PCO for the APN.

Figure 85: ePDG selection using PCO



To configure the ePDG server selection, use the following commands:

```
configure mobile-gateway pdn apn pco-option closest-v4-epdg-group
```

```
configure mobile-gateway pdn apn pco-option closest-v6-epdg-group
```



NOTE:

Server selection is based on the epdg-group profile list that has been specified for closest-v4-epdg-group and/or closest-v6-epdg-group.

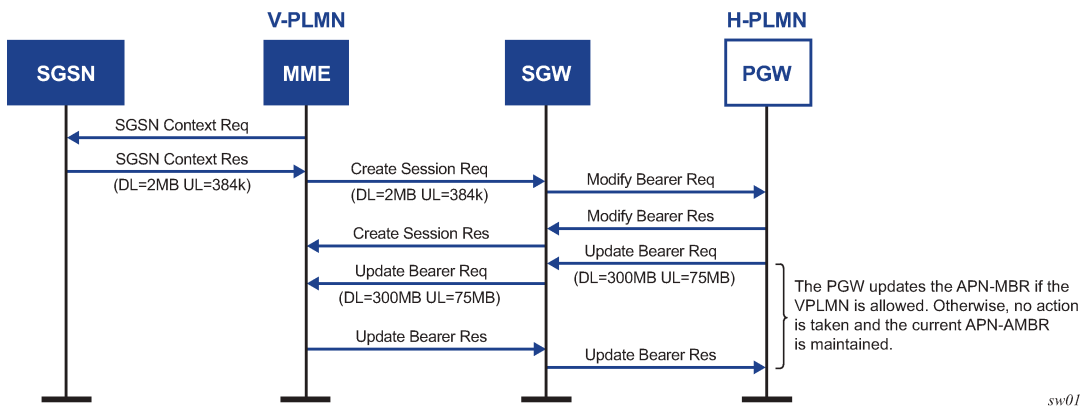
To see which ePDG servers, if any, are allocated to the UE, use the following command:

```
show mobile-gateway pdn pdn-context imsi detail
```

To obtain statistics about the number of UE sessions that requested ePDG servers and the number of UEs to which the PGW allocated ePDG servers, use the following command:

```
show mobile-gateway pdn apn statistics
```


Figure 87: PGW upgrades the APN-AMBR based on the local configuration

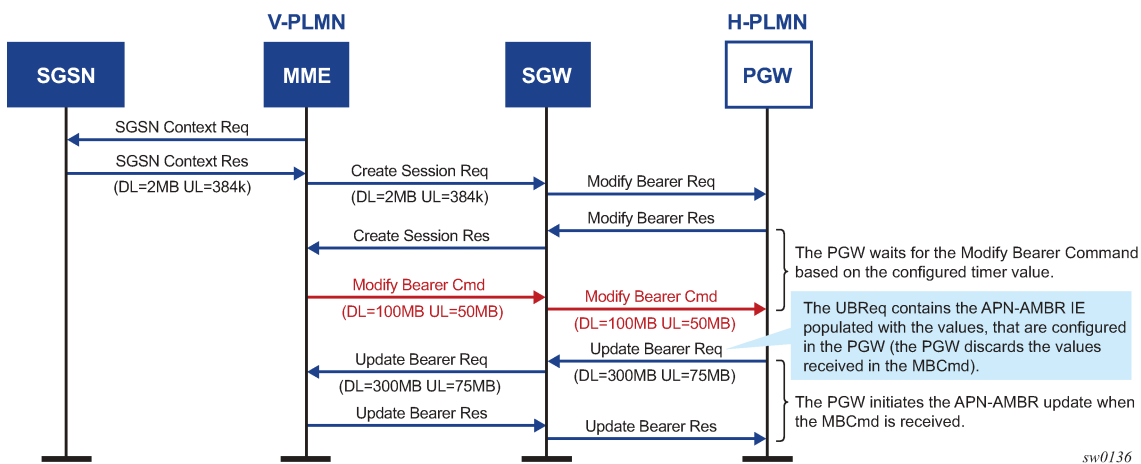


This functionality is needed in cases where the VPLMN SGSN downgrades the APN-AMBR because the bit rates are too high for UTRAN/GERAN, but the VPLMN MME fails to upgrade the APN-AMBR back to the original level when the UE moves back to the E-UTRAN. As a result, the subscriber will not get the optimal bit rates for the E-UTRAN. The local configuration allows the PGW to control the APN-AMBR in those APNs, where the PCRF is not deployed.

There is no configuration to downgrade the AMBR when the UE moves back to the UTRAN/GERAN, because the SGSN will downgrade the AMBR if the values are too high for the UTRAN/GERAN. The locally configured APN-AMBR is not applied during attach, because the MME gets the correct APN-AMBR from the HSS during attach.

The period of time that the PGW waits after a RAT change before it sends the Update Bearer Request for changing the APN-AMBR can also be configured. If the MME sends the Modify Bearer Command to the PGW before the timer expires, the PGW will update the APN-AMBR to the locally configured value in the reply to the Modify Bearer Command. This is applicable only when the UE is roaming, because the HPLMN cannot control how the VPLMN MME manages the APN-AMBR.

Figure 88: Modify bearer command is received before the timer expires



To configure the APN-AMBR control for roaming, use the following commands:

```
config>mobile>pdn>apn>roaming-handover
lte-qos-ambr
lte-qos-black-list
lte-qos-delay
```

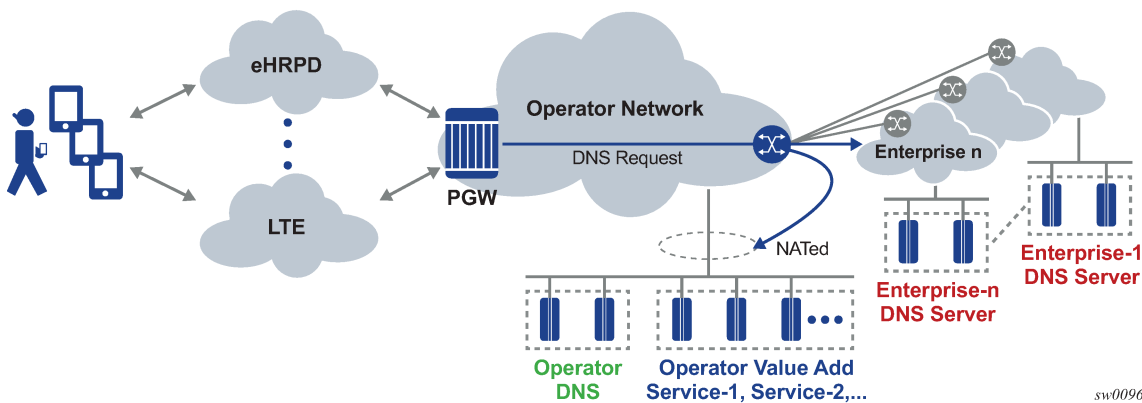
6.22 DNS FQDN redirection

DNS FQDN redirect is supported on the PGW. The PGW checks the DNS requests over any access network (HRPD, eHRPD, UMTS, or LTE) and matches it against a list of predefined FQDNs (configurable).

If there is a match, the DNS redirect function intercepts and sends the DNS request to a different (configurable) destination IP DNS server. If the request does not match any FQDN, the DNS redirect function forwards the DNS request to the original destination IP DNS server.

The DNS redirect functionality can be used in enterprise deployment scenarios, where an operator wants to offer additional value added services to enterprise users, regardless of the enterprise routing domain they are attached to.

Figure 89: DNS FQDN redirect functionality



To create FQDN lists and specify FQDNs within those lists, use the following commands:

```
configure mobile-gateway profile list fqdn-list
```

```
configure mobile-gateway profile list fqdn-list fqdn
```

To add the FQDN list as part of the flow description within a policy rule unit, use the following command:

```
configure mobile-gateway profile policy-options policy-rule-unit flow-
description match dns-fqdn-list
```

To configure the DNS redirect action value based on the flow description match of the FQDN, use the following command:

```
configure mobile-gateway profile policy-options action-rule-unit dns-redirect
```

To associate a DNS redirect server profile to an APN for FQDN DNS redirect, use the following command:

```
configure mobile-gateway pdn apn fqdn-dns-redirect-server
```

To view information regarding the DNS FQDN lists, the policy and action rule units, and the DNS redirect status, use the following show commands:

```
show mobile-gateway profile list fqdn-list
```

```
show mobile-gateway profile policy-options policy-rule-unit
```

```
show mobile-gateway profile policy-options action-rule-unit
```

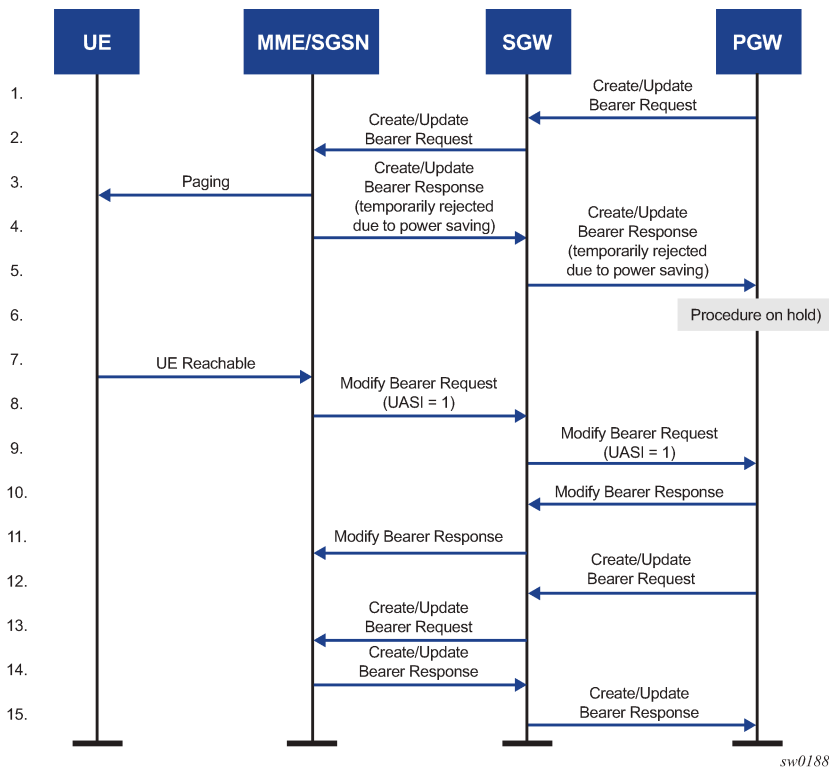
```
show mobile-gateway pdn apn dns-redirect status
```

6.23 Delay tolerant network-initiated procedures

High latency communication is supported for scenarios where paging takes longer because the UE is using Power Saving Mode (PSM) or extended idle mode Discontinuous Reception (eDRX).

Network-initiated procedure may be also delayed due to PSM or eDRX. An APN level configuration in the PGW can be used to determine if the network initiated procedures are delay tolerant and the maximum duration that the PGW will wait until the UE becomes reachable. The PGW will set the Delay Tolerant Connection Indication (DTCI) flag when the PDN connection is created. The MME/SGSN uses cause value UE is temporarily not reachable due to power saving (123) as a reply to the network initiated create or update procedure if paging takes longer than normal and the PGW has set a DTCI flag. The PGW puts the network initiate procedure on hold, until it receives the UE Available for Signalling Indication (UASI) flag from the MME/SGSN and the PGW will reattempt the procedure. The functionality is supported in GTPv2.

Figure 90: Create/Update Bearer Request on hold



If the network initiated procedure is about deleting a bearer, the MME/SGSN may delay sending the Delete Bearer Response until the UE becomes reachable.

Use the following command to enable support for delay tolerant network initiated procedures.

```
configure mobile-gateway pdn apn delay-tolerant-connection-suspension
```

6.24 L2TP/PPP regeneration

The mobile gateway supports L2TP/PPP regeneration over LTE (S5/S8).

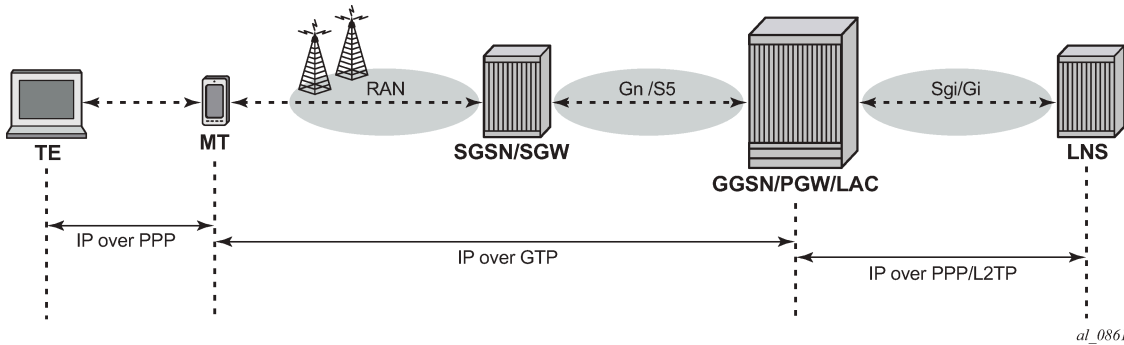
L2TP tunneling is used to serve corporate VPN services, to provide different tunneling options to connect upstream ISPs on the SGi/Gi interface. The PGW/GGSN in this case functions as an L2TP Access Concentrator (LAC), to tunnel traffic to the third-party router acting as an L2TP Network Server (LNS).

The PGW/GGSN configured as a LAC enables communication with the LNS, for the establishment of Virtual Private Network (VPN) tunnels between the operator and a subscriber's corporate network. L2TP is often used in VPN networks because it provides more control over authentication and IP address assignment. An operator may do a first level of authentication, however PPP is used to exchange user name and password and IPCP is used to request an address.

To support PPP negotiation between the PGW/GGSN and the corporation, an L2TP tunnel must be set up in the PGW/GGSN running a LAC service. With VPN access based on PPP regeneration, the PGW/GGSN

supports PPP in two different areas of the network, with two different sets of PPP endpoints as shown in *PGW/GGSN support for PPP in the network*.

Figure 91: PGW/GGSN support for PPP in the network



In the first section in *PGW/GGSN support for PPP in the network*, IP over PPP is established between the TE and MT (over the R-Interface). From there IP packets are transported over GTPv1 (Gn/Gp) or GTPv2 (S5) to the PGW/GGSN, through the RAN and SGSN/SGW, using IP PDP Type (there is no PPP between the mobile device and the PGW/GGSN). From the PGW/GGSN on the Sgi/Gi interface to the LNS, IP over PPP over an L2TP tunnel to the enterprise network LNS is supported. The second set of PPP endpoints occurs between the PGW/GGSN (LAC) and the LNS, with the PGW/GGSN also acting as the PPP client for the mobile device.



NOTE:

Alternative options can be used instead of the first PPP section. For example, an operator may use dialer software on the mobile device, which does not require the first section.

6.24.1 L2TP PPP failure handling

PPP keepalives

PPP/LCP Echo Requests/Replies can be exchanged between the PGW/GGSN LAC and the LNS to ensure that session-level connectivity between the LAC and the LNS remains established. The PGW/GGSN LAC never initiates a PPP/LCP Echo Request toward the LNS. When the PGW/GGSN LAC receives a PPP/LCP Echo Request from the LNS, it responds with a PPP/LCP Echo Reply message.

L2TP Hello messages

L2TP Hello messages and ZLB replies can be exchanged between the PGW/GGSN LAC and the LNS to ensure that the tunnel connection remains established. The PGW/GGSN LAC initiates a L2TP Hello message toward the LNS if it has not received one from the LNS within the configurable hello-interval parameter (under the l2tp, group, or tunnel contexts). The PGW/GGSN LAC waits for a ZLB response from the LNS and if it does not receive it, it attempts as many times as the max-retries-estab parameter and according to the intervals described in *L2TP retransmissions*. If the PGW/SGSN LAC never receives

the ZLB message from the LNS after all retry attempts, it terminates the tunnel by sending a Stop Control Connection Notification (StopCCN) message. When the PGW/GGSN LAC receives a Hello message from the LNS, it responds with a ZLB message.

L2TP retransmissions

The PGW/GGSN LAC implements a method to retransmit a control message if no response/acknowledgement is received on time.

The implementation is according to RFC 2661 and it is described as follows.

The first retransmission of the message happens after 1 second and each subsequent retransmission employs an exponential back-off interval (that is, the second retransmission after 2 seconds, the third retransmission after 4 seconds, and so on). There is an 8 seconds cap upon the maximum interval between the retransmissions. The retransmission count can be configured using the `max-retries-not-estab` (when the L2TP session is not established) and the `max-retries-estab` (when the L2TP session is established) commands in the `l2tp`, `group`, or `tunnel` contexts. For more information about these commands, refer to the *CMG CLI Reference Guide*.

PPP retransmissions

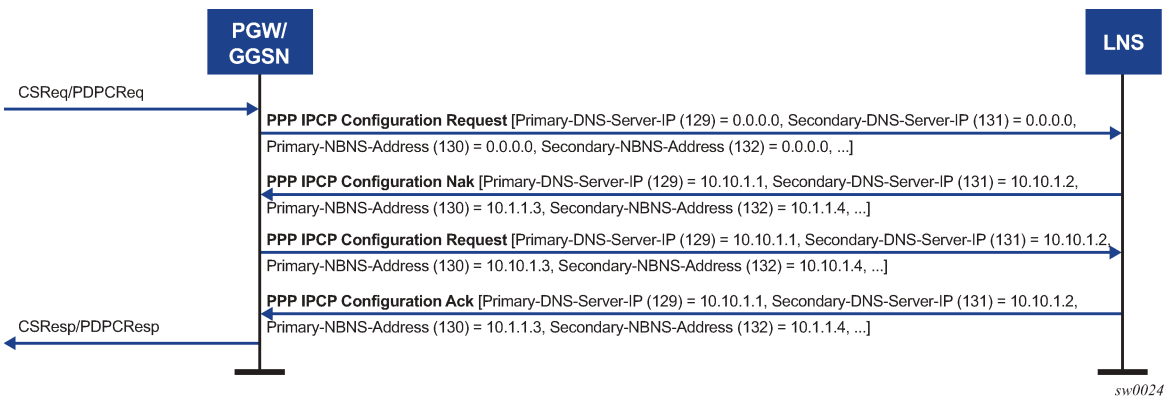
The PGW/GGSN LAC retransmits a PPP request message if no acknowledgement is received within 1.5 seconds. A second retransmission of the message happens after 2 seconds and a third (and last) retransmission happens after 4 seconds. If a response is not received after the final retry attempt (that is, within 4 seconds), the PGW/GGSN LAC initiates a session termination action. Hence, the PPP keepalive total timeout interval is approximately 11 seconds. Currently, the PPP retransmission timer and the PPP retransmission count are not configurable.

6.24.2 IPCP DNS address assignment

The DNS address assignment via PPP IPCP for L2TP/PPP Regeneration is based on the RFC1877, which defines the method for negotiating with the remote peer, the address of the DNS and NBNS servers to be used on the local end of the link.

The local peer requests an invalid server address and the remote peer specifies the address by NAKing this option and returning the IP address of a valid DNS server.

Figure 92: DNS address assignment call flow

**NOTE:**

This functionality does not change the current behavior over the GTP-C S5/Gn interface.

Message from the UE and response from the PGW/GGSN below shows the end-to-end behavior depending on what is received from the UE, and the corresponding PGW/GGSN response.

Table 27: Message from the UE and response from the PGW/GGSN

CSReq/PDP context Req from UE	Response from PGW/GGSN
PCO-IPCP container with the Primary/Secondary DNS and Primary/Secondary NBNS addresses set to 0.0.0.0	Assigned DNS and NBNS addresses (from LNS, RADIUS or local) back to the CSResp/PDP Context Resp PCO-IPCP container
PCO-DNS container with Primary/Secondary DNS with any address	Assigned DNS addresses (from LNS, RADIUS or local) back to the CSResp/PDP Context Resp PCO-DNS container
PCO-IPCP container with Primary/Secondary DNS addresses that do not match the assigned DNS addresses (from LNS, RADIUS or local)	PCO-IPCP container with the DNS addresses rejected in the CSResp/PDP Context Resp
PCO-DNS and PCO-IPCP containers where the PCO-IPCP has Primary/Secondary DNS addresses that do not match the assigned DNS addresses (from LNS, RADIUS or local)	PCO-IPCP container with DNS addresses from the CSReq/PDP Context Req rejected in the CSResp/PDP Context Resp PCO-DNS container with assigned DNS addresses (from LNS, RADIUS or local) in the CSResp/PDP Context Resp

Table 27: Message from the UE and response from the PGW/GGSN (continued)

CSReq/PDP context Req from UE	Response from PGW/GGSN
PCO-DNS and PCO-IPCP containers where the PCO-IPCP has Primary/Secondary DNS addresses that do not match the assigned DNS addresses (LNS, RADIUS or local) and the Primary/Secondary NBNS addresses set to 0.0.0.0	PCO-IPCP container with DNS addresses from the CSReq/PDP Context Req rejected in the CSResp/PDP Context Resp Does not include any NBNS addresses in the PCO-IPCP container in the CSResp/PDP Context Resp PCO-DNS container with assigned DNS addresses (LNS, RADIUS or local) in the CSResp/PDP Context Resp
PCO-DNS and PCO-IPCP containers where the PCO-IPCP has Primary/Secondary DNS addresses that do not match the assigned DNS addresses (LNS, RADIUS or local) and Primary/Secondary NBNS addresses that match	PCO-IPCP container with DNS addresses from the CSReq/PDP Context Req rejected in the CSResp/PDP Context Resp Does not include any NBNS addresses in the PCO-IPCP container in the CSResp/PDP Context Resp PCO-DNS container with assigned DNS addresses (LNS, RADIUS or local) in the CSResp/PDP Context Resp
PCO-DNS and PCO-IPCP containers where the PCO-IPCP has Primary/Secondary DNS addresses that do not match the assigned DNS addresses (LNS, RADIUS or local) and Primary/Secondary NBNS addresses that do not match	PCO-IPCP container with DNS and NBNS addresses from the CSReq/PDP Context Req rejected in the CSResp/PDP Context Resp PCO-DNS container with assigned DNS addresses (LNS, RADIUS or local) in the CSResp/PDP Context Resp

**NOTE:**

If the UE does not include the PCO DNS/NBNS, the IPCP DNS/NBNS option to the LNS is not sent.

The priority for selecting the DNS/NMBS server is the following:

1. via PPP IPCP
2. via RADIUS
3. via local configuration

To configure the DNS and NBNS address configuration option, use the following commands:

```
configure mobile-gateway pdn apn ppp-regeneration exclude-dns-server-option
```

```
configure mobile-gateway pdn apn ppp-regeneration exclude-nbns-server-option
```

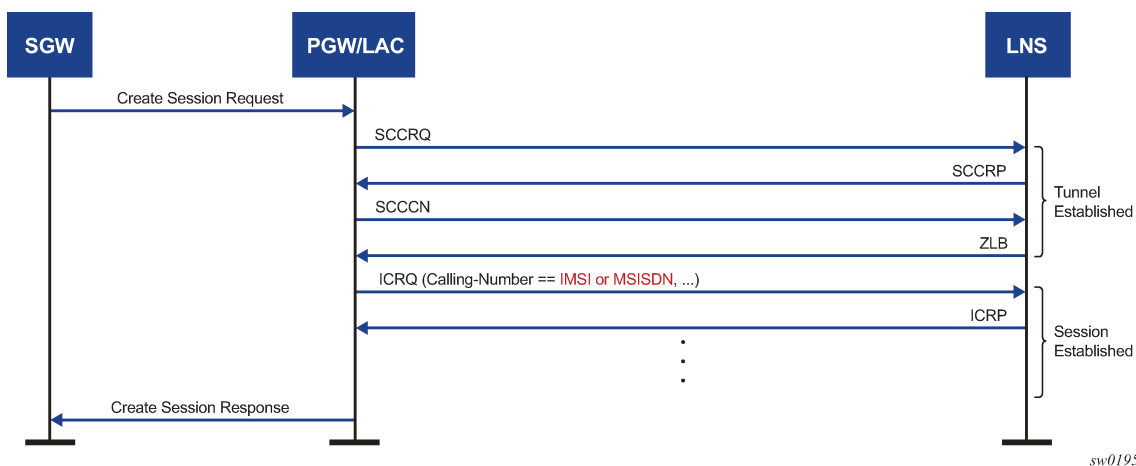
To view the status of the PPP IPCP negotiation information for the DNS and the NBNS, use the following command:

```
show mobile-gateway pdn apn
```

6.24.3 L2TP PPP regeneration with the IMSI populated in ICRQ

This feature provides the option to send the IMSI or the MSISDN in the Calling-Number AVP of the ICRQ message during the L2TP session establishment.

Figure 93: IMSI/MSISDN in the ICRQ message



The value of the Calling-Number is encoded as a text string. The value that is configured to be included in the ICRQ message is independent of the value that is configured to be sent for PAP/CHAP username in the L2TP ICCN and PPP messages. For example, the ICRQ message may be configured to contain the IMSI in the Calling-Number AVP, while the PAP authentication in L2TP ICCN may be configured to include the MSISDN.

To configure the ICRQ message enhancement, use the following command:

```
configure mobile-gateway pdn apn ppp-regeneration icrq-calling-number-type
```

To view information about the ICRQ message enhancement configuration, use the following command:

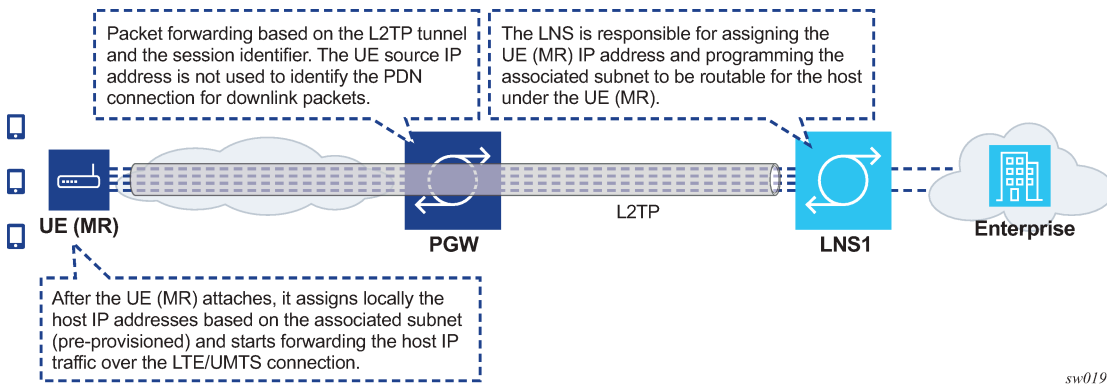
```
show mobile-gateway pdn apn statistics
```

6.24.4 L2TP PPP regeneration mobile router support

This feature provides an option to disable anti-spoofing for APNs that have L2TP/PPP regeneration enabled.

This feature supports Mobile Router (MR) CPE access using multiple IP addresses (hosts) over the UE (GTP, L2TP, LNS session uplink) and multiple destination IP addresses over the same downlink session. The LNS handles the routing northbound of the L2TP tunnel traffic.

Figure 94: L2TP PPP regeneration MR support



To configure a UE to act as an MR, use the following command:

```
configure mobile-gateway pdn apn ppp-regeneration mobile-router
```

To view whether a UE is configured to act as an MR, use the following command:

```
show mobile-gateway pdn apn statistics
```

6.25 APN and serving PLMN rate control

This feature adds support for serving PLMN and APN rate control. Both rate control features are relevant for any use case where the amount of traffic is very limited, for example, for IoT devices which generate very few data.

The APN rate control is managed by the PGW. It applies to the S1-U, S4-U/S12, and S11-U interfaces. The APN rate control defines the number of packets the UE is allowed to send or receive in the APN. If the UE has multiple bearers in the same APN, the rate limit applies to all the bearers in that APN. The APN rate limit is defined as 1 to 16777215 packets per time unit; the time unit is either minute, hour, day or week. The UE is responsible for excluding the MO exception data from the uplink rate control and informing the PGW through the MME to use those data for charging. Exception data is used by CIoT (Cellular Internet of Things) monitoring devices to report exceptional events. For example, a smoke detector may send an exception report when smoke is detected.

The APN rate control is configured in the PGW. The APN rate limits will be defined according to the APN level configuration, where the rate limit is applied to all UEs connected to the APN. If the rate limit is defined in

the APN configuration, and the UE has indicated that it supports the APN rate control, the PGW will pass the rate limit to the UE in the PCO or ePCO.

To enable the APN rate control in the PGW, use the following command:

```
configure mobile-gateway profile policy-options apn-rate-control ul-rate-control-in-pgw
```

To configure the uplink rate limit for the APN rate control, use the following command:

```
configure mobile-gateway profile policy-options apn-rate-control ul-rate-limit ul-rate-limit
```

Since exception reports may be dropped by the PGW due to the UL policy enforcement (although the UE sends them), a higher threshold, specific for exception reports, has been introduced which the PGW has to consider for the UL traffic policy.

In particular, when the PDN connection is created, the UE needs to advertise the support of additional APN rate control for exception data to the network in the PCO/ePCO of the Create Session Request message. If the UE has advertised the support of additional APN rate control for exception data to the network, the network (PGW) provides an additional rate for exception data reporting to the UE through the PCO/ePCO in the Create Session Response message. It is the number of the additional allowed exception reports per unit time.

To configure the additional UL rate limit for exception data, use the following command:

```
configure mobile-gateway profile policy-options apn-rate-control additional-ul-rate-limit-for-exception-data
```

The serving PLMN rate control allows the MME to restrict the maximum rate for user plane traffic. The UE will enforce the rate limit for UL traffic and the PGW will enforce the rate limit for DL traffic. The configured rate limit in the MME is specified as number of packets per deci hour (6 minutes). The rate limit is defined within the serving PLMN Rate Control IE in the Create Session Request or Modify Bearer Request.

The serving PLMN rate control can be enabled only if the PDN connection may use only S11-U for the user plane. The MME uses the Control Plane Only PDN Connection Indication (CPOPCI) to inform the PGW that the user plane may use only S11-U. The CPOPCI flag can be set to one (1) only when the PDN connection is created, because the status cannot be changed later. The CPOPCI flag will be ignored if it is received after the PDN connection has been created.

The UE and PGW will enforce the rate limit for traffic.

The APN rate control does not depend on the CIOT flag in the Node Features IE of Echo messages. The APN rate control can be enabled even if the CIOT flag is not set.

Use following commands to configure the APN rate control:

```
configure mobile-gateway profile policy-options apn-rate-control
```

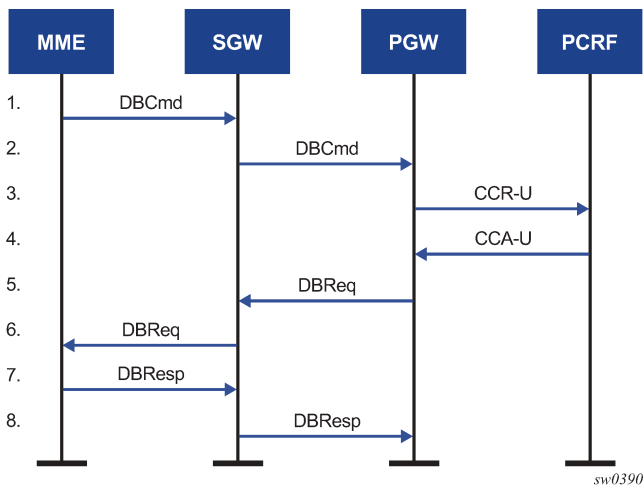
**NOTE:**

The CLI batch commands `begin` and `commit` must be used when creating or editing APN rate control. Enter the `begin` command to enter the editing mode. The changes in the editing mode will not have impact to existing sessions until the `commit` command is entered.

6.26 SRVCC handover

Handover of PS to CS shows the Single Radio Voice Call Continuity (SRVCC) handover of the PS to the CS as specified in 3GPP TS 23.216.

Figure 95: Handover of PS to CS



1. The MME may set the VB (Voice Bearer) bit in the Bearer Flags IE of Delete Bearer Command. It is used to indicate that the dedicated bearer is a voice bearer, but the call continues in the CS network.
2. If the PGW received the VB bit, the PGW uses the PS_TO_CS_HANDOVER in Rule-Failure-Code AVP. This way, the PCRF can inform the P-CSCF that the related voice bearer was terminated because the voice call was moved to the CS.
3. The PGW sends the CCR-U to inform the PCRF about the dedicated bearer termination.

6.27 PDN switching

PDN switching supports VRF traffic steering using nexthop redirect rules to apply NAT (SNAT) or IPSec/GRE tunneling per PCC rule.

For example, a PCC rule matching the flow will be redirected toward the NAT or IPSec/GRE tunnel for a PDN session while the remaining traffic is routed based on the destination address. PDN switching also allows traffic to flow out of an alternate VPRN to the SGi VPRN.

Figure 96: PDN switching using NAT (SNAT)

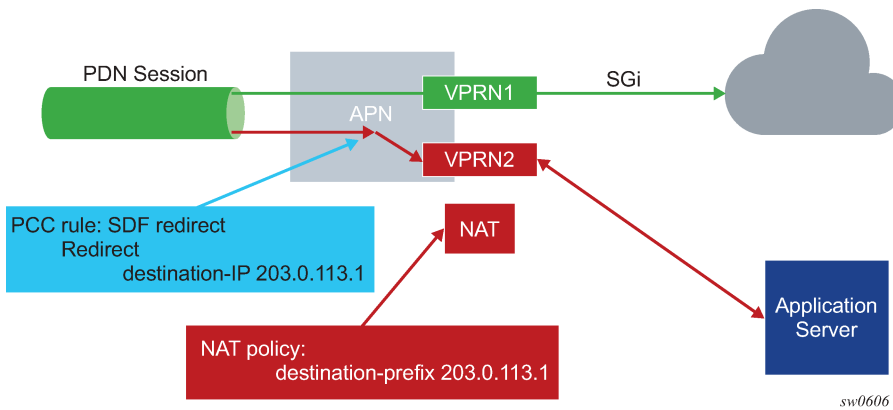
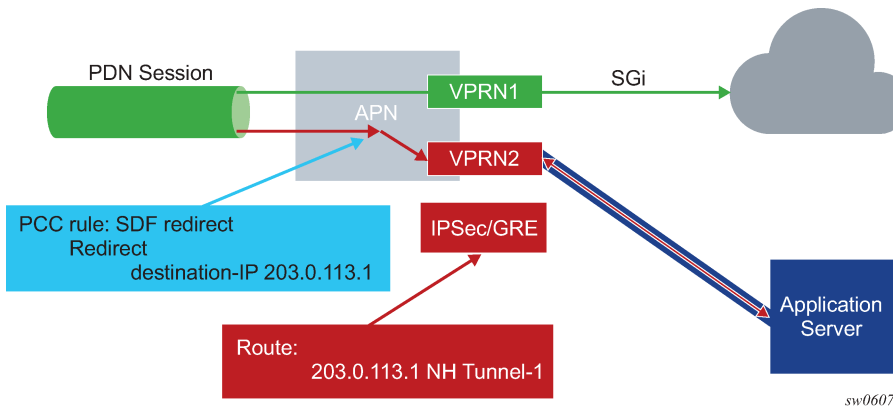


Figure 97: PDN switching using IPsec or GRE



6.27.1 Configuring PDN switching using IPsec and GRE tunnels

Purpose

The following is an example of how to configure PDN switching using IPsec and GRE tunnels.



NOTE:

The IP address that must be used in a traffic redirect profile as well as in NAT policy is a type of Martian IP address.

Procedure

1. Configure IPsec/GRE (private/public VPRNs). For information about IP tunnels and configuring IPsec with CLI, refer to the *CMG CLI Reference Guide*.
2. Create a traffic redirect profile.

**NOTE:**

The VPRN routers specified are the private VPRN routers for GRE and IPsec.

Step example

```
policy-options
begin
  traffic-redirect-profile "GRE_redirect"
    router "vprn121"
    traffic-redirect-target 1
      uplink-next-hop-ipv4 10.84.85.201
    exit
  exit
  traffic-redirect-profile "IPsec_redirect"
    router "vprn121"
    traffic-redirect-target 1
      uplink-next-hop-ipv4 10.84.85.202
    exit
  exit
exit
```

3. Link the traffic redirect profile to a policy rule unit through a flow gate status context.

Step example

```
policy-rule-unit "tcp_tunnel_202"
  flow-description 1
    match
      protocol 6
      remote-ip 10.84.85.202/32
      remote-port eq 80
    exit
  exit
  flow-gate-status traffic-redirect "IPsec_redirect"
exit
policy-rule-unit "tcp_tunnel_201"
  flow-description 1
    match
      protocol 6
      remote-ip 10.84.85.201/32
      remote-port eq 443
    exit
  exit
exit
```

```
flow-gate-status traffic-redirect "GRE_redirect"
```

4. Create policy rules for each policy rule unit; see *Static predefined and local policies on PGW/GGSN*.

6.27.2 Configuring PDN switching using NAT

Purpose

The following is an example of how to configure PDN switching using NAT.

Procedure

1. Create a traffic redirect profile.

Step example

```
begin
  traffic-redirect-profile "1"
    router "vprn121"
    traffic-redirect-target 1
    uplink-next-hop-ipv4 203.0.113.1
  exit
exit
```



NOTE:

The IP address specified in this step for the uplink next hop is the same as the destination IP address configured for the NAT policy in step 3.

2. Link the traffic redirect profile to a policy rule unit through a flow gate status context.

Step example

```
policy-rule-unit nat_tunnel_52
  flow-description 1
    match
      protocol 6
      remote-ip 10.85.60.52/32
    exit
  exit
  flow-gate-status traffic-redirect "1"
exit
```

**NOTE:**

The value "1" specified for `flow-gate-status traffic-redirect` is the traffic redirect profile configured in step 1.

3. Configure the NAT policy with a destination prefix that uses the same IP address as the traffic redirect profile.

Step example

```
nat
inside
  nat-policy nat_dest_203
    destination-prefix 203.0.113.1/32
  exit
```

**NOTE:**

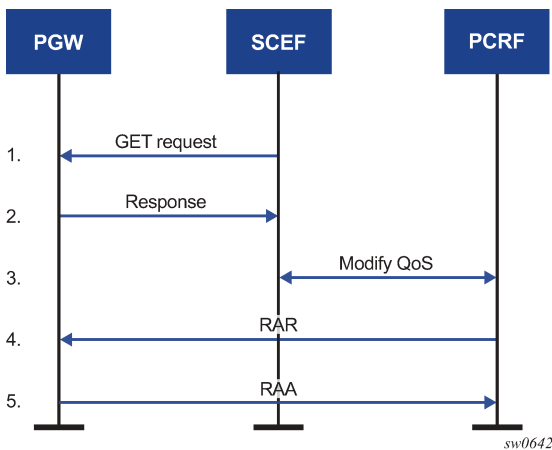
The IP address specified in this step as the destination prefix is the same as the IP address configured for the traffic redirect profile in step 1.

6.28 Capability of exposure over the T6c reference point

The PGW supports the capability of the exposure function based on the custom HTTP T6c reference point. It supports the Binding Support Function (BSF) discovery service for the EPC, where the Service Capability Exchange Function (SCEF) can discover the PCRF of the PDN connection.

The SCEF sends an HTTP GET request to the PGW over the T6c interface. The request contains the UE IP address and the APN. If the PGW finds the matching PDN connection, it returns to the realm and the host identity of the PCRF, which controls the PDN connection. The SCEF can open the Rx session to the PCRF as an AF and request, for example, QoS modifications via PCRF.

Figure 98: QoS modification by the SCEF for the PDN connection



The T6c interface is based on HTTP1.1 (HTTP 1.0 is not supported) and the transport protocol is TCP over IPv4 or IPv6 address.

The following parameters may be present in the GET request:

- IPv4 address of the UE
- IPv6 address of the UE APN (APN-NI of the PDN connection)
- APN (APN-NI of the PDN connection)

The value must be the real APN, if the virtual APNs are already used.

The following example shows a valid GET request:

```
GET/epcpc/PoS/v1/queryPCRF?IPv4=x.x.x.x&APN=xxx
HTTP/1.1
Host: localhost
```

If a unique matching PDN connection is found, status code 200 is used in the response and *Status code 200 parameters* lists the parameters in the payload.

Table 28: Status code 200 parameters

Parameter	Type	Presence	Description
PCRFHost	String	M	The host name of the PCRF (Destination-Host in Gx)
PCEFHost	String	M	The host name of the PCEF (Origin-Host in Gx)
MSISDN	String	M	The MSISDN of the matching session

Table 28: Status code 200 parameters (continued)

Parameter	Type	Presence	Description								
			If the MSISDN is undefined, the dummy MSISDN with a 15 digits-set to zero is used (as specified in 3GPP 23.003)								
RATType	Integer	O	The RAT type of the session. The values are the same as in the Gx interface. The typical values are: <table><tr><td>1000</td><td>UTRAN</td></tr><tr><td>1001</td><td>GERAN</td></tr><tr><td>1004</td><td>E-UTRAN</td></tr><tr><td>1005</td><td>E-UTRAN NB-IoT</td></tr></table>	1000	UTRAN	1001	GERAN	1004	E-UTRAN	1005	E-UTRAN NB-IoT
1000	UTRAN										
1001	GERAN										
1004	E-UTRAN										
1005	E-UTRAN NB-IoT										
APN	String	O	The APN-NI of the session (same as in the request)								
IPv4	String	O	The IPv4 address of the session								
IPv6	String	O	The IPv6 address of the session								

To enable and configure the T6c interface, use the following commands:

```
configure mobile-gateway pdn t6c
  interface
  http-port
  uri
  t6c-profile
```

To configure the presence of optional parameters in the response, use the following commands:

```
configure mobile-gateway profile t6c-profile
  include-rat-type
  include-apn
  include-ip
```

If no matching session is found, status code 404 appears in the response. Status codes 400 and 500 may also appear to indicate an error.

6.29 LTE multiple UE session over low orbit satellite connections

The combined SGW/PGW supports LTE over low orbit satellite technology to provide Internet and enterprise-access services as well as backhaul for mobile operators.

Figure 99: Remote office connection to HQ using LTE and low orbit satellites

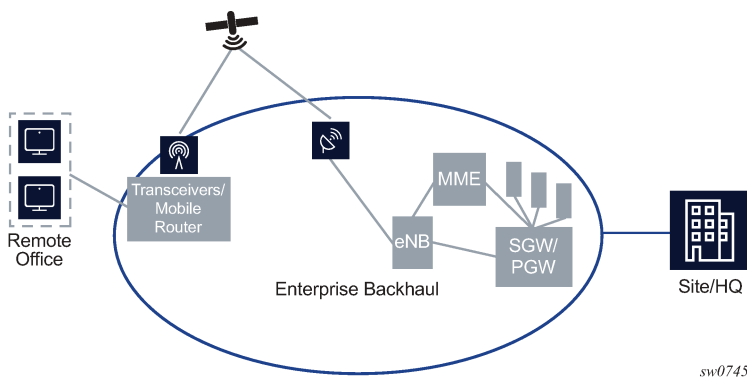
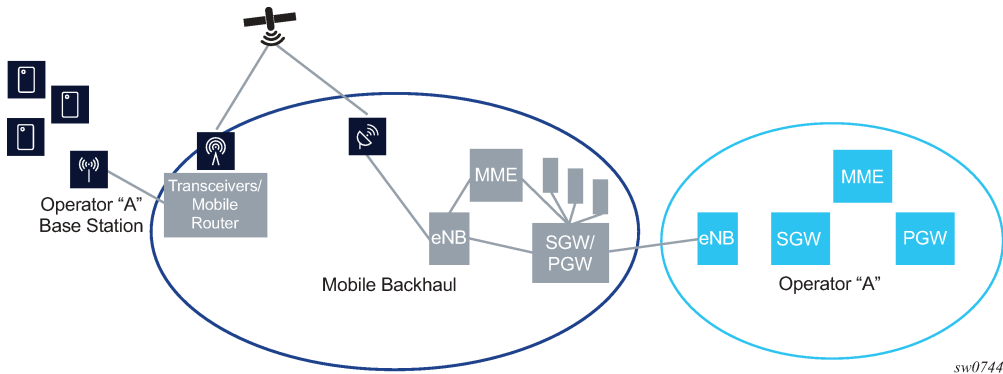


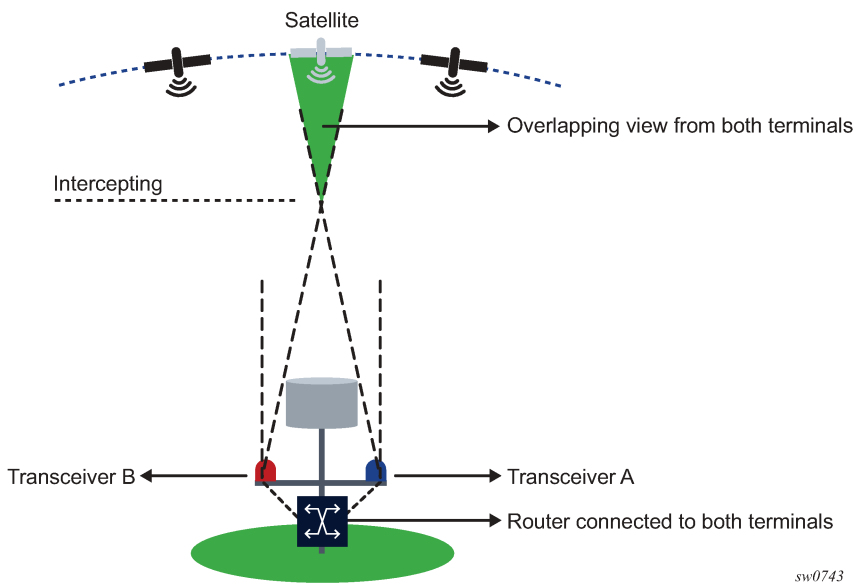
Figure 100: Low orbit satellites as the mobile backhaul for a remote eNodeB site



The low orbit trajectory satellites move overhead across the sky at significant speed, disappearing for short periods of time. The satellite receiver Line Of Sight (LOS) may also be blocked by nearby obstructions, further impacting the access to the satellite.

To overcome this, the mobile router/CPE uses multiple transceivers (up to three) to address the loss of LOS by placing the transceivers around the obstruction. Each of these transceivers occasionally experiences loss of LOS, but at any given point, at least one transceiver has a clear LOS. These transceivers run LTE on top and the combined SGW/PGW perceives them as UEs.

Figure 101: Multiple transceiver installation for low orbit satellites



For each mobile router/transceiver site, the combined SGW/PGW handles up to 3 transceivers, attaching each of them as a UE (multiple UE session) with its own LTE PDN session to the PGW. However, only one UE, at any given time, is used for uplink and downlink data traffic.

6.29.1 Multiple-UE sessions

Support for sessions with multiple UEs on the combined SGW/PGW is implemented by leveraging the **RADIUS Framed-Route** attribute, the **bonding** functionality, and a **multiple UE session mode** configured on the PDN and APN levels, allowing individual UEs to become part of a multiple-UE session with the MR and its transceivers/UEs.

To configure multiple-UE sessions on the PDN level, use the following command:

```
configure mobile-gateway pdn gw-id allow-multi-ue-session
```



NOTE:

To enable multiple UE session support, the `config>mobile>pdn>allow-duplicate-msisdn` command must be enabled.

To configure multiple-UE sessions on the APN level, use the following command:

```
configure mobile-gateway pdn gw-id apn {apn-name | unknown} multi-ue-bonding
```



NOTE:

For call trace debugging of multiple-UE sessions, a common identifier (MSISDN) must be used to get the expected output.

To configure the time that the uplink traffic detection waits before selecting another UE in the multiple-UE session after a new UE has just been detected and selected, use the following command:

```
configure mobile-gateway pdn gw-id apn {apn-name | unknown} hybrid-access  
link-switch mode uplink-traffic hysteresis time
```

When enabling this functionality, each UE attaches using the standard network LTE attach, but the UEs (up to three) that belong to the same multiple UE session have the same MSISDN but different IMSI. In the multiple UE session, the RADIUS server and the Gx interface only see the IMSI of the first UE (out of the three) to attach to the LTE. For the Gy interface, there is a Gy session for each bearer.

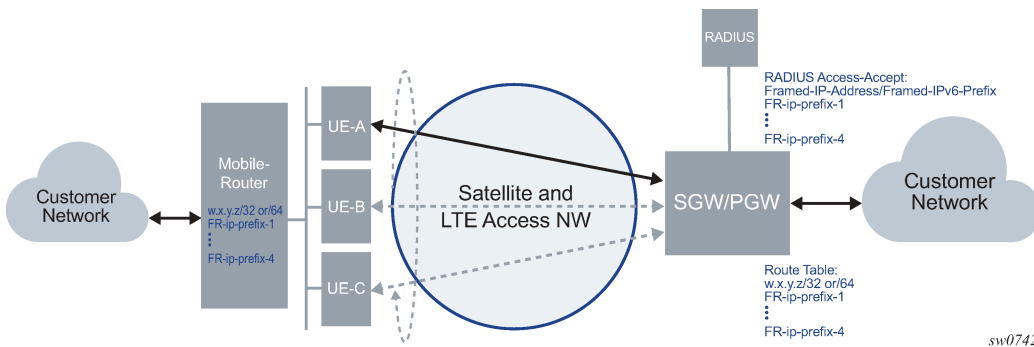
The multiple UE session APN and the use of the bonding functionality means that only the first UE's Create Session Request triggers a RADIUS Access-Request message to receive the assigned Framed-Route attributes from the RADIUS server. The other UEs are locally programmed with the same Framed-Route attributes. The RADIUS server is able to identify any of the UEs that belong to the same multiple UE session (based on the MSISDN) and assign the same Framed-Route attributes for the UE that attaches first. The Framed-Route attribute in the RADIUS Access-Accept message is coded as per the RFC865 (IPv4)/RFC3162 (IPv6) and contains the IP prefixes that must be reachable behind the UEs. The PGW routes these Framed-Route attributes out on the SGi interface. The same Framed-Route attributes are also pre-provisioned on the mobile router behind the UEs.

The RADIUS server also sends the Framed-IP-Address/Framed-IPv6-Prefix attributes with the IPv4 address/IPv6 prefix that is sent in the PAA IE. This means that all UEs (up to three) receive the same IP address in the PAA IE. However, the UE does not use the PAA IE address to send any traffic. All traffic is routed from the mobile router via the UEs based on the pre-provisioned routes (the same routes as in the Framed-Route attributes on the PGW).

The PGW routes uplink traffic to the UEs on the SGi interface. As shown in [SGi interface traffic routing in a multiple-UE session](#), the PGW forwards downlink traffic with destination IP addresses including any of the routes w.x.y.z/32, /64, or ip-prefix-1 to ip-prefix-4, to the currently selected next hop, that is, UE-A, UE-B, or UE-C. The selected UE will forward the traffic on to the customer network.

When the next hop UE for downlink traffic is selected, the PGW supports traffic sensing on the data plane. Therefore, the PGW detects and determines which UE is currently receiving uplink traffic from the PGW and programs it (using bonding functionality) as the next hop for all downlink traffic for the destination IP of any of the Framed-Route attributes. This is based on the low orbit satellites and the frequent change in UEs due to LOS and on that only the active UE (over the satellites) can receive traffic.

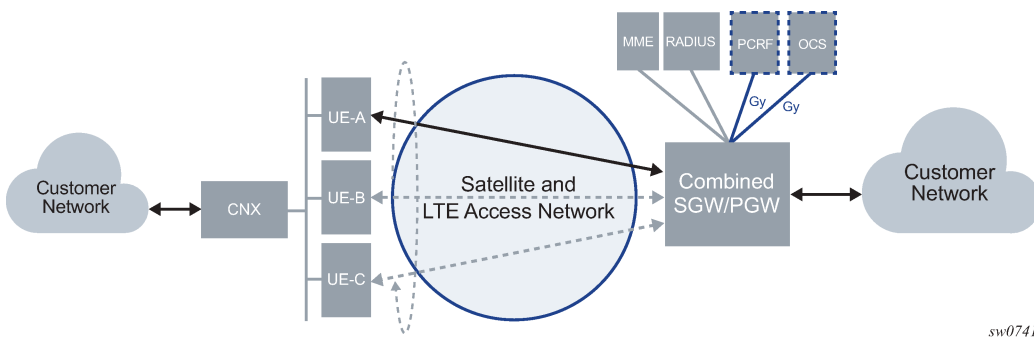
Figure 102: SGi interface traffic routing in a multiple-UE session



For the Gx interface, only one UE (the first that attaches) triggers the CCR-I and receives the CCA-I, with the PCC rules also applicable to the rest of the UEs (locally programmed to use the same PCC rules). The PCC rules are received as static rule base/names from the PCRF if there is one locally configured static PCC rule.

Charging is supported, based on the Gy interface, using bearer-level charging with a Gy session for each bearer. For example, for three UEs with two dedicated bearers and a default bearer, there are nine bearers, that is, nine Gy interface sessions. The MSISDN is the same in all sessions, but the IMSI is different (based on the attached UE).

Figure 103: Charging function in a multiple-UE session



6.30 APN block list

Block entries under an APN are used to block new session creation on this APN on a specific MG-group and for specific PDN-Type.

To add block list entries, use the following command:

```
configure mobile-gateway pdn apn block-list entry entry-id group group pdn-type pdn-type
```

When an incoming create session request comes in the PGW for an APN and this APN contains a block-list configuration, the gateway load balancing algorithm tries to find a group that is not blocked for the requested PDN-Type. If this group is not found, the session is rejected.

The PGW allows the creation of the session when the requested PDN-Type is dual stack and either IPv4 or IPv6 is blocked on a specific APN and MG-group and the `pdn-type-conversion` command is enabled.

If the incoming create session request is dual stack, any of the IPv4 or IPv6 is blocked in some groups and the `pdn-type-conversion` is enabled, the PGW load balancer selects the less loaded MG group that can serve this session either as single stack (in case of IPv4 or IPv6 blocked) or as dual stack (no blocking).

Existing sessions on a blocked APN, group, and PDN-type are still served (mobility events, dedicated bearer activation, and so on) but charging is not performed for these sessions. User plane traffic of sessions on blocked resources is dropped.

**NOTE:**

It is possible to see the quota request/response messages on the Gy interface. These are used to allocate quotas from the OCS, but the session is not charged. Charging is performed only when traffic is unblocked and charging events are created.

To initiate session deletion for existing sessions on a blocked resource, see [Session clearing per APN, MG group, and PDN type](#).

6.30.1 APN block list in CUPS

In CUPS deployments, a `block-list` must be configured on both the GW-C and GW-U CMG instances. A block entry must be applied on the GW-C first, before it is applied on the GW-U, so that the blocked GW-U does not receive a new session request.

On the GW-C, block commands under an APN per UP peer and per `ip-pool` prefix are used to block a new session creation on a UP peer for this APN and/or prefix.

To add block list entries, use the following commands:

```
configure mobile-gateway profile pfcg up-peer-list peer apn [no] block
```

```
configure mobile-gateway profile pfcg up-peer-list peer network-realm ip-pool  
  ipv4-prefix | ipv6-prefix prefix/length [block [force]]
```

When a new session request is received in the PGW-C for an APN, a blocked UP peer as a whole or the specific IP prefix is excluded from the selection for this session.

On the GW-U, block entries under an APN are used to block new session creation on this APN on a specific MG-group.

To add block list entries, use the following command:

```
configure mobile-gateway pdn apn block-list entry entry-id group group pdn-  
type pdn-type
```

When a session request is received in the PGW-U for an APN and this APN contains a `block-list` configuration, the PFCP session is rejected in case the MG-group for the incoming session, as dictated from the UE IP address, is blocked. The PFCP rejection cause is set to 75 - No Resources Available in this case.

The GW-C, upon receiving the rejection cause 75 in the PFCP Session Establishment response, rejects the new session request back to the MME.

Existing sessions on a blocked resource (APN and group on the GW-U, UP peer, APN and IP prefix on the GW-C) are still served (mobility events, dedicated bearer activation, and so on). User plane traffic of existing sessions which are on a blocked resource on the GW-U is dropped and not charged.

**NOTE:**

Similarly as in the non-CUPS deployments, it is possible to see the Session Report Request/Response (initiated from GW-U) on the Sx interface, the quota request/response messages on the Gy interface and the Session Modification Request / Response (initiated from GW-C) on the Sx interface. These are used to allocate quotas from the OCS, but the session is not charged. Charging is performed only when traffic is unblocked and charging events are created.

To initiate session deletion for existing sessions on a blocked resource, see [Session clearing per UP peer, APN, and IP prefix on the GW-C](#).

6.31 APN-based GTP-C PDN session attach rate limiting

The APN-based GTP-C PDN session attach rate limiting functionality configures a threshold for the maximum number of GTP-C PDN attach requests per second.

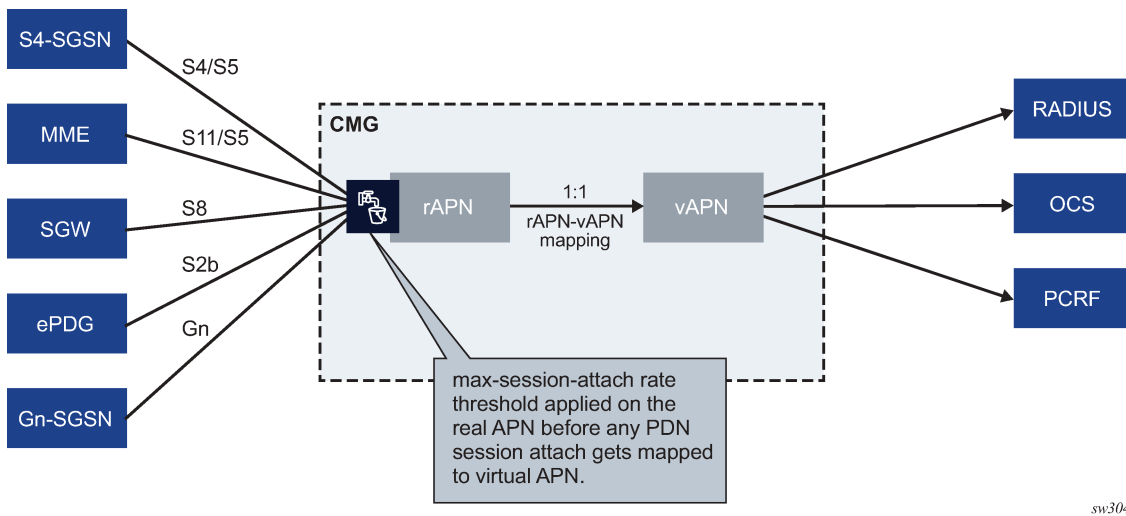
The threshold limit applies to the new PDN (GTPv2) and PDP (GTPv1) session attach requests per second that are received for the APN. The PDN attach requests that are above the configured threshold are rejected with the GTP-C cause code #73 (No resources available).

For example, if the threshold value is set to 1000, new session requests up to 1000 PDN attaches are allowed and processed. The session requests above the 1000 threshold are rejected with the GTP-C cause code #73 (No resources available).

The APN-based GTP-C PDN session attach rate limiting support is applied on the real APN to all new PDN session attaches, before any mapping to a virtual APN occurs. This allows for indirect attach rate limiting support for a virtual APN.

In cases where all access types are served by a virtual APN, the use of 1:1 mapping between the real APN and the virtual APN is used, as shown in [PDN session attach rate limit \(virtual APN serves all access types\)](#). The maximum PDN session attach rate threshold is applied on the real APN before any mapping to the virtual APN and therefore, it also provides throttling support to the virtual APN.

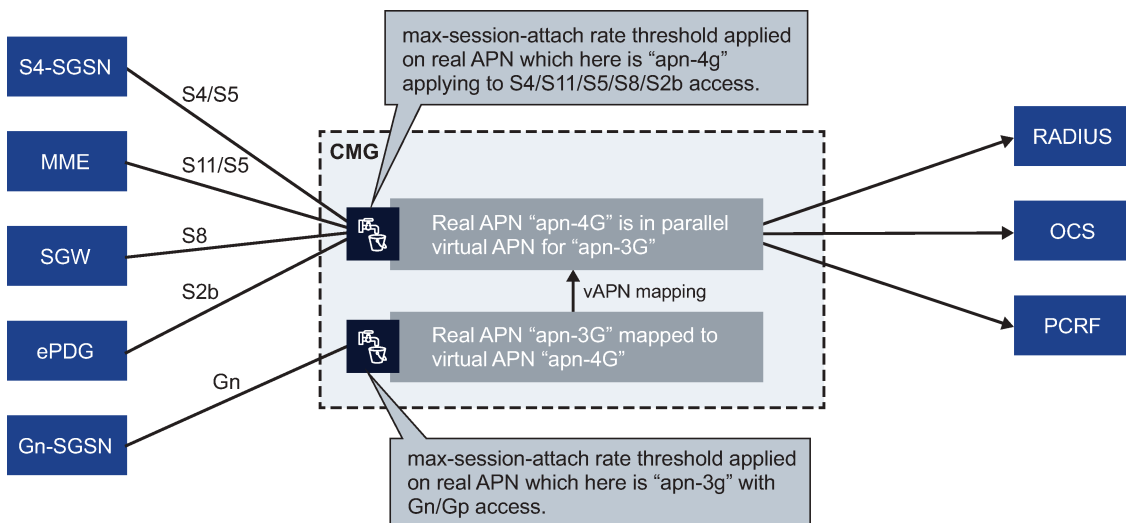
Figure 104: PDN session attach rate limit (virtual APN serves all access types)



sw3047

In other cases, the APN serves as a real APN for certain access types and as a virtual APN for other access types, as shown in *PDN session attach rate limit (virtual APN serves specific access types)*. In this scenario, the maximum PDN session attach rate is still configured on the real APN and the threshold is applied before any mapping to the virtual APN occurs. Therefore, throttling support is also provided to the virtual APN.

Figure 105: PDN session attach rate limit (virtual APN serves specific access types)



sw3048

The APN-based GTP-C PDN session attach rate limiting support can configure whether the emergency attaches are throttled or not. It can also enable the generation of an SNMP alarm when the threshold is reached and cleared.

To enable the APN-based GTP-C PDN session attach rate limiting, use the following command:

```
configure mobile-gateway pdn gw-id apn apn-name max-session-attach max-attach-limit [alarm] [include-emergency]
```

6.32 Sending the back-off timer due to IP pool exhaustion

This feature introduces the option to signal BACK OFF TIMER (EPC Timer GTPv2 IE) back to the MME when there is an IP pool exhaustion use case in the real APN.

Procedure

1. Configure the back-off timer when there is an IP pool exhaustion.

```
configure mobile-gateway pdn apn ip-pool-exhaust-back-off-timer
```

2. View the IP pool exhaustion back-off timer.

```
show mobile-gateway pdn apn
```

6.33 Configuring virtual APN selection based on PDN/PDU type

This feature introduces the option, for CMG to select a virtual APN based on PDN/PDU type value.

Procedure

1. Configure the virtual APN based on PDN.

```
configure mobile-gateway pdn virtual-apn-selection
```

2. Configure the virtual APN based on APN.

```
configure mobile-gateway apn pdn virtual-apn-selection selection-prio
```

```
configure mobile-gateway apn pdn virtual-apn-selection user-range
```

3. View the virtual APN.

```
show mobile-gateway pdn apn
```

6.34 Configuring PGW support for 3GPP AAA server identifier

Configure the PGW to select, on the S6b interface, the same AAA server as the one selected by ePDG/TWAG on the SWm/STA respectively for the subscriber authentication and authorization. The ePDG provides the AAA server information to the PGW.

Procedure

1. Configure the S6b node identifier.

```
configure mobile-gateway pdn gw-id s6b node-identifier-list list-name
```

Step example

```
configure mobile-gateway pdn 1 s6b node-identifier-list node-list1
```

2. View the S6b reference point statistics.

```
show mobile-gateway pdn ref-point-stats s6b
```

3. Configure the AAA host realm list.

```
configure mobile-gateway profile list aaa-host-realm list-name
```

Step example

```
configure mobile-gateway profile list aaa-host-realm aaa-list1
```

4. Configure the ID of the AAA host realm list.

```
configure mobile-gateway profile list id id realm realm-string host host-string diameter-peer-group-list group-list-name
```

Step example

```
configure mobile-gateway profile list id 2 realm realm2 host host1 diameter-peer-group-list diameter-group-list2
```

5. View the AAA host realm list.

```
show mobile-gateway profile list aaa-host-realm list-name
```

Step example

```
show mobile-gateway profile list aaa-host-realm list1
```

Step result

```
# show mobile-gateway profile list aaa-host-realm <set-1>
=====
AAA Hosts Realms
Profile name : set-1
```

```
Entry Id Host Realm diameter-peer-group-list
1 aaa-test.realm.com realm.com peer-group-aaa-1

# show mobile-gateway profile list aaa-host-realm <set-1>
=====
AAA Hosts Realms
Profile name : set-1
Entry Id Host Realm diameter-peer-group-list
1 aaa-test.realm.com realm.com peer-group-aaa-1
```

7 Serving Gateway (SGW)

Configuration guidelines specific to the SGW.

The SGW is the node that terminates the interface toward E-UTRAN. For each UE associated with the EPS, at a given time, there is one SGW.

SGW functions include:

- local mobility anchor point for inter-eNodeB handover
- E-UTRAN idle mode downlink packet buffering and initiation of network triggered service request procedure
- transport-level packet marking in the uplink and the downlink
- UL and DL charging per UE, PDN, and QCI

Related information

Common gateway configuration

Overview of the provisioning functionality that is common to all standalone and combined gateway types.

7.1 SGW instance configuration

A serving mobile gateway instance must be configured to use the respective global profile; that is, GTP-c or Diameter profiles, which can be applied for all the signaling interfaces or configured specifically to the reference-points.

```
A:3ls-SGW-2# configure mobile-gateway
A:3ls-SGW-2>config>mobile# info
-----
    profile
      diameter "diam-prof2"
      exit
      diameter "diameter_profile1"
        connection-timer 5
      exit
      diameter-peer "loadb" ==> this peer is for load balance, load balance
        on per PDN level
        application-type rf
        destination-realm "nokia.com"
        diameter-profile "diameter_profile1"
        load-balance
        peer 10.11.7.105 port 3868
          no shutdown
        exit
        peer 10.11.7.106 port 3868
```

```

        no shutdown
    exit
    peer 10.11.7.107 port 3868
        no shutdown
    exit
exit
diameter-peer "cdfPeer1"
    application-type rf
    destination-realm "nokia.com"
    diameter-profile "diameter_profile1"
    peer 10.11.7.101 port 3868
        no shutdown
    exit
exit
charging
    sgw-charging 1 ==> the profile id used to match the
        charging characteristics , if there is CC in the CSReq , it has to
        be matched to one of the charging profiles, if does not match, the
        call will be rejected
        off-line
            enable-offline
            pri-cdf-dia-peer "loadb"
            sec-cdf-dia-peer "cdfPeer2"
            charging-info-trigger
                qos-change
                tariff-time-change
                tariff-time start 10:30 end 15:30
                user-loc-change
            exit
            partial-record-closure-trigger
                max-change-condition 2
                ms-time-zone-change
                rat-change
                time-limit 55
                volume-limit 1
            exit
        exit
    exit
    sgw-charging 2
        off-line
            enable-offline
            pri-cdf-dia-peer "cdfPeer2"
            charging-info-trigger
                qos-change
                tariff-time-change
                tariff-time start 10:30 end 15:30
                user-loc-change
            exit
            partial-record-closure-trigger
                max-change-condition 2

```

```

        ms-time-zone-change
        rat-change
        time-limit 3
        volume-limit 1
    exit
    exit
    exit
    exit
pdn 1
    charging
        charging-prof home 1 ====> this is default charging profile,
        used only if there is no CC in CSReq...
    exit
    signaling
        diameter
            origin-host "diam.origRealm.com"
            origin-realm "origRealm.com"
            profile "default"
        exit
        gtp-c
            profile "s5"
        exit
        interface "system"
    exit
    qci-policy "test"
    rf
        acct-interim-interval 5
        acct-level pdn-level
        primary-diameter-peer "cdfPeer3" ====> This will be used when
        the CC matches the charging profile, but no peers specified in the
        charging profile is in service
        secondary-diameter-peer "cdfPeer4"
    exit
    no shutdown
exit

```

7.1.1 SGW signaling configuration for using a single system IP interface

The following example shows an SGW instance that is configured to use common signaling profiles (Diameter) for all Diameter-based applications.

The example also shows how to configure a single system IP interface to be used as the source interface for all the SGW signaling packets originated by the system.

```

configure mobile-gateway pdn 1
    signaling
        diameter
            origin-host "diam.origRealm.com"

```

```

        origin-realm "origRealm.com"
        profile "prof1"
    exit
    interface "system"
    exit
exit

```

7.1.2 Configure each SGW signaling interface to use a separate IP interface

Based on operator requirements, separate IP interfaces can be configured for each SGW reference point, or separate VPRNs can be configured for each SGW reference point. The following displays an example where separate S11, S5, and Rf signaling IP interfaces are used for SGW signaling IP interfaces configuration.



NOTE:

When separate IP interfaces are used for SGW signaling IP interfaces configuration, you must check for each reference point whether a common IP must be configured to support all call flows.

```

configure mobile-gateway pdn 1
    rf
        interface
            diameter "Rf"
        exit
    s11
        interface
            gtp-c "s11"
        exit
    exit
    s5
        interface
            gtp-c "s5"
        exit
    exit
    no shutdown
exit all

```

7.2 GTP reference point peers: (S1-u) eNodeB, (S11) MME, (S5) PGW

The SGW IP address is statically configured in MME. MME peer configuration in SGW is not required.

The eNodeB and PGW peer IP addresses are conveyed to SGW from MME during the UE Attach signaling procedure.

7.3 SGW system default profile values

The SGW system provides default profiles for GTP and Diameter with default values used in the signaling protocol parameters.

For the default QCI-{FC, DSCP} behavior, SGW preserves and propagates the DSCP values received in UL directions from eNodeB toward PGW and in DL directions from PGW toward eNodeB.

Provisioned QCI policies support overriding the DSCP values received in UL and DL directions.

The default profiles are not editable, therefore, the values cannot be modified. The operator can fine tune respective profile values per their needs by configuring a new named profile and then associating that profile to the respective reference-point or it can be applied globally.

- *GTP-C profile*
- *Diameter profile*
- *QCI-{FC, DSCP} mapping table*

Table 29: GTP-C profile

Profile	Configurable	Default
gtp-c default	No	cannot modify
[no] msg-retransmit time-out	0 to 30 seconds	5
retry-count	1 to 8	3
[no] ip-dscp	0 to 63	56
[no] ip-ttl	1 to 255	255
[no] keep-alive time-out	0, 60 — 180 seconds	60
retry-count	1 to 8	3
t3-response	1 to 8	3

Table 30: Diameter profile

Profile	Configurable	Default
Diameter default	No	cannot modify
[no] connection-timer	0-->180 seconds	30
[no] dpr-timeout	1 to 3600 seconds	1800
[no] ip-dscp	0 to 63	56
[no] ip-ttl	1 to 255	255
[no] perm-fail-retry-timeout	0 to 1440 minutes	30
[no] dns-refresh-interval	0 to 24 hours	6
[no] transaction-timer	1 to 180 seconds	5
[no] watchdog-timer	1 to 180 seconds	30
Exit	—	—
Diameter profile name	32 chars max	none
[no] connection-timer	0-->180 seconds	30
[no] dpr-timeout	1 to 3600 seconds	1800
[no] ip-dscp	0 to 63	56
[no] ip-ttl	1 to 255	255
[no] retry-count	1 to 8	3

Table 30: Diameter profile (continued)

Profile	Configurable	Default
[no] transaction-timer	1 to 180 seconds	5
[no] watchdog-timer	1 to 180 seconds	30
Exit	—	—
diameter-peer profile-name	32 chars max	none
[no] destination-realm name	80 chars max	none
[no] diameter-profile	32 chars max	none
[no] load-balance	load-balance/no load-balance	no load-balance
[no] peer	peer ip and port	none
[no] transport	transport tcp/no transport	no transport
application-type	gx, rf	none
Exit	—	—

Table 31: QCI-{FC, DSCP} mapping table

QCI (arp 1-15)	Forward class	Inprofile/Outprofile DSCP
1	nc	nc2
2	ef	ef

Table 31: QCI-{FC, DSCP} mapping table (continued)

QCI (arp 1-15)	Forward class	Inprofile/Outprofile DSCP
3	af41	af41/af42
4	h1	nc1
5	af21	af21/af22
6	af	af11/af12
7	af	af11/af12
8	l2	cs1
9	be	be

7.4 SGW reference point path down behavior

Table 32: SGW reference point path down behavior

Reference point	Behavior				
S5/S11/S4/ S2a/S2b GTPv2 Peer path down (missed keep alive)	A GTPv2 path is up when it has one or more UE attaches on the reference point. This is indicated in the Path Mgmt State output field of the command <code>show>mobile>pdn>ref-point-peers</code> Periodic keep alive (GTPv2 Echo messages) are sent when the GTPv2 path comes up The sending of keep alive messages is based on the attributes configured using the command <code>config>mobile>profile>gtp</code> The default keep alive attributes are: <table> <tr> <td>interval</td><td>60 seconds</td></tr> <tr> <td>retry counts</td><td>4 seconds</td></tr> </table>	interval	60 seconds	retry counts	4 seconds
interval	60 seconds				
retry counts	4 seconds				

Table 32: SGW reference point path down behavior (continued)

Reference point	Behavior
	timeout 5 seconds Using the default values, the gateway will send periodic messages at 60 second intervals. If a GTP-C Echo Response is not received within the 5 second timeout period, the gateway retries four times. If the gateway hears a GTP-C Echo Response from the remote end, it aborts the retry procedure and restarts the keep alive interval timer for this peer. If no response is received before the end of the retry timeout sequence, the gateway declares the path to be down. Example with default keep alive settings 1. A peer comes up due to a UE attach on a reference point. 2. The keep alive interval timer is started for this reference point. 3. The keep alive interval issues a GTP-C Echo message at 60 seconds. 4. Assuming that the GTP-C Echo Response is not reaching the gateway, the retry timeout sequence is as follows: • first retry at 65 seconds • second retry at 70 seconds • third retry at 75 seconds • fourth retry at 80 seconds 5. The fourth retry is declared as timed-out (failed) at 85 seconds, after waiting the default five seconds.
S11 GTPv2 Peer restart notified	Bearers context resources associated with that S11 peer are released and delete events are sent to other EPC core network elements, such as PGW
S1-U/S12/S11-U GTP-U Peer path down (missed keep-alives)	SNMP trap is sent on S1 peer down event and bearer context resources are not released

7.5 Dual connectivity

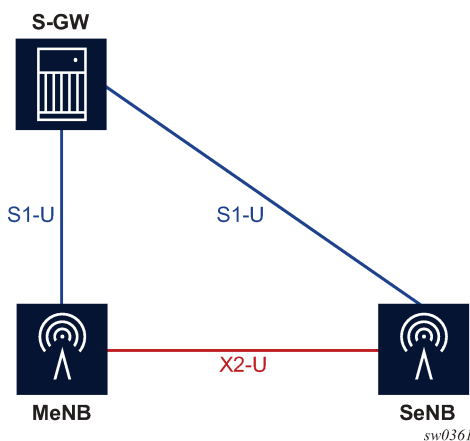
Dual connectivity allows two eNodeBs to serve a UE in the connected state.

There are three bearer types:

Master Cell Group (MCG) bearer	the S1-U interface is terminated in the Master eNodeB (MeNodeB)
Secondary Cell Group (SCG) bearer	the S1-U interface is terminated in the Secondary eNodeB (SeNodeB)
Split bearer	the S1-U interface is terminated in the MeNodeB, but the MeNodeB may offload traffic to the SeNodeB via the X2 interface

If there are only MCG bearers or split bearers, then S1-U interface is terminated only in the MeNodeB and dual connectivity is transparent to the SGW. If there are SCG bearers, dual connectivity is visible to the SGW and some of the bearers of the UE are served by the SeNodeB.

Figure 106: Dual connectivity



The MME may use the E-UTRAN-initiated RAB modification procedure to move the bearers of the UE between the MeNodeB and the SeNodeB. The SGW will send the End Marker when the user plane is switched to the new eNodeB.

The SGW will reject requests where the secondary eNodeB is visible to the SGW.

To enable dual connectivity, use the following command:

```
configure mobile-gateway pdn dual-connectivity
```

7.6 CloT control plane optimization

Cellular Internet of Things (CloT) control plane optimization supports passing of user plane data via the MME by encapsulating the user plane packets in NAS. This eliminates the need to establish radio bearers for short data transactions.

The usage of S11-U is also visible in offline charging data produced by the SGW. The MME may use the CIOT flag of the Node Features IE to determine whether the SGW supports CIOT control plane optimization or any other related CIOT features.

To include the CIOT flag in Node Features IE of Echo Request/Response, use the following command:

```
config>mobile>profile>gtp>gtpv2-node-features>include-ciot
```

To enable the S11 GTP-U interface (S11-U), use the following command:

```
config>mobile>pdn>s11>interface>gtp-u
```

To include S11-U related information into charging, use the following commands:

- ```
config>mobile>profile>charging>cdr-avp-options-profile>cp-ciot-eps-optimisation-indicator
```

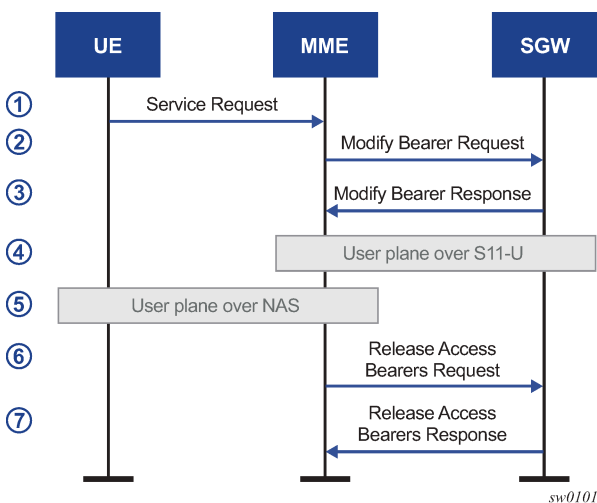
It specifies if the Cp Ciot EPS Optimisation Indicator is included or not in CDRs.

- ```
config>mobile>profile>charging>sgw-charging>offline>charging-info-trigger>change-in-user-plane
```

It specifies whether change in user plane to UE trigger is enabled.

S11-U used for short data transmission initiated from the UE shows how S11-U is used for short data transmission initiated from UE. MME sends its S11-U F-TEID in Modify Bearer Request and the SGW its S11-U F-TEID in Modify Bearer Response. After data transmission is completed, the S11-U tunnel is released. S11-U may be activated also during attach, TAU, PDN connection activation and network triggered service request.

Figure 107: S11-U used for short data transmission initiated from the UE



**NOTE:**

It is possible to configure a different IP address for the S11-U and S1-U reference points, but this should be done only if all the MME nodes in the local PLMN support this specific deployment.

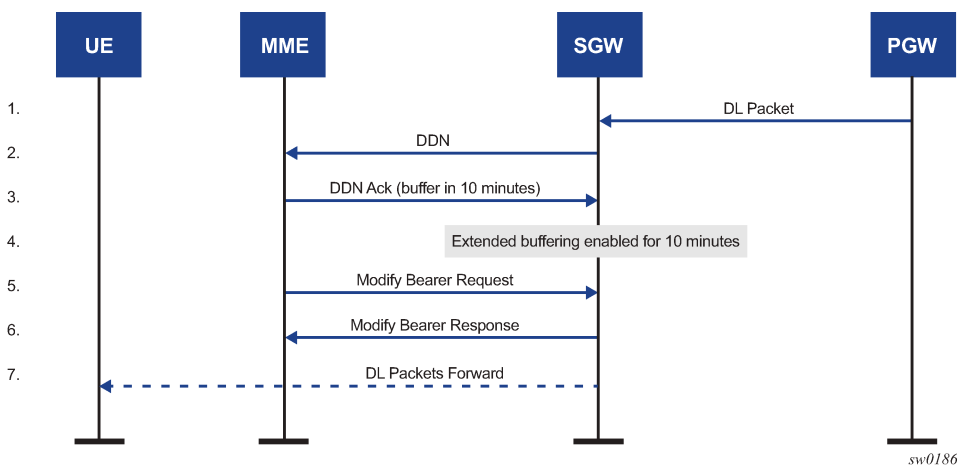
7.7 Extended buffering

High latency communication is supported for the scenarios where paging will take longer because the UE is using the Power Saving Mode (PSM) or the extended idle mode Discontinuous Reception (eDRX). This functionality prevents loss of user-plane packets, reduces paging attempts, and avoids failures of network initiated procedures.

If the paging is triggered by the downlink traffic, the MME may activate extended buffering in the SGW. The DL Buffering Duration IE of the Downlink Data Notification Acknowledge message will indicate how long the SGW should buffer the downlink packets. Local configuration can be used to define the maximum duration for buffering the downlink packets. The MME may also include the DL Buffering Suggested Packet Count IE in the DDN Ack to indicate how much buffering might be required.

Extended buffering shows the scenario where extended buffering is enabled for 10 minutes and the MME restores the user plane before the time limit is reached.

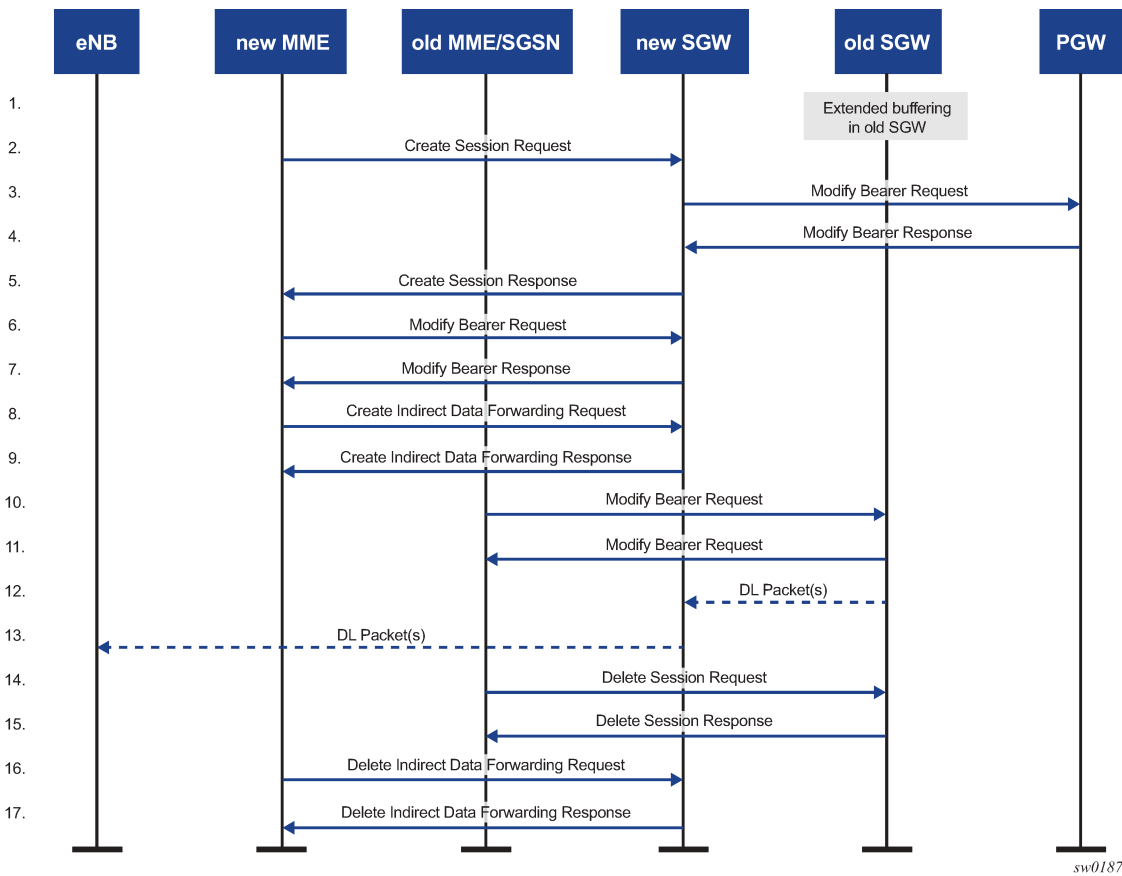
Figure 108: Extended buffering



Extended buffering may last several minutes and it is possible that SGW relocation occurs during extended buffering. The MME will then setup an indirect data forwarding path, through which the source SGW will send the buffered packets to the target SGW, which will then forward the packets to the eNodeB (S1-U), S4-SGSN (S4-U) or the MME (S11-U).

Extended buffering with SGW relocation shows signaling related data forwarding during TAU with SGW relocation.

Figure 109: Extended buffering with SGW relocation



NOTE:

The PGW pause of charging feature should not be enabled with High Latency Communication devices.

To configure the maximum duration for extended buffering in the S11 and S4 reference points, use the following commands:

```
configure mobile-gateway pdn s11 signaling gtp-c ddn-extended-buffering-time
```

```
configure mobile-gateway pdn s4 signaling gtp-c ddn-extended-buffering-time
```

7.8 Network-triggered service restoration

The SGW supports the standard network-triggered service restoration procedure defined in 3GPP TS 23.007, to allow UEs to re-attach quickly to the network after an MME/SGSN restart.

The configurable MME/SGSN pool for MME/SGSN failure handling, allow the operator to define the MME/SGSN groups for the network triggered service restoration procedure. If the MME/SGSN fails and is listed in one of the configured MME/SGSN groups, the SGW may send a DL Data Notification message to any of the

MMEs/SGSNs of the same MME/SGSN group as part of the network service restoration. The legacy method for using private extension in the Echo Request/Response can also be used to define MME pools. The new configuration must be used if the MME does not support the private extension of the legacy method.

If an optional QCI list is included in these commands, only the session with the matching QCI is eligible for restoration.

To configure the length of the sessions after MME/SGSN failure, use the following commands:

```
configure mobile-gateway pdn gw-id s11 signaling gtp-c restoration-  
time restoration-time [qci qci-value]
```

```
configure mobile-gateway pdn gw-id s4 signaling gtp-c restoration-  
time restoration-time [qci qci-value]
```

7.8.1 Configuring an MME group for network-triggered service restoration

Procedure

1. Create an MME group.

```
configure mobile-gateway profile mme-group MME-group-id
```

Step example

```
configure mobile-gateway profile mme-group 1
```

2. Configure the MME nodes.

Configure the MME nodes which may be used for network triggered service restoration.

```
configure mobile-gateway profile mme-group MME-group-id peer ipv4|ipv6
```

Step example

```
configure mobile-gateway profile mme-group 1 peer 192.168.10.10  
configure mobile-gateway profile mme-group 1 peer 192.168.10.20
```

3. Enable the MME group for the S11 reference point.

```
configure mobile-gateway pdn gw-id s11 sig-ref-point-name mme-group MME-  
group-id
```

Step example

```
configure mobile-gateway pdn 1 s11 s11_int mme-group 1
```

7.9 Configuring packet buffering time for PSM

This feature adds support for the extension of the CMG `ddn-extended-buffering-time` command by increasing the maximum value of the DDN extended buffering time to 24 hours. This enhancement allows more advanced and flexible IoT devices integration.

Procedure

1. Configure the S4 DDN extended buffering time.

```
configure mobile-gateway pdn s4 signaling gtp-c ddn-extended-buffering-time
```

2. Configure the S11 DDN extended buffering time.

Configure the MME nodes which may be used for network triggered service restoration.

```
configure mobile-gateway pdn s11 signaling gtp-c ddn-extended-buffering-time
```

8 Evolved Packet Data Gateway

The configuration guidelines and examples for configuring the ePDG. This information is intended to be used in conjunction with the information in the *CMG ePDG Guide*, which provides more information and complete feature descriptions for ePDG.

Related information

Common gateway configuration

Overview of the provisioning functionality that is common to all standalone and combined gateway types.

8.1 Configuration requirements

The following configurations are required for the ePDG:

- IOM-XP3-B/C
- ISM-MG redundancy



NOTE:

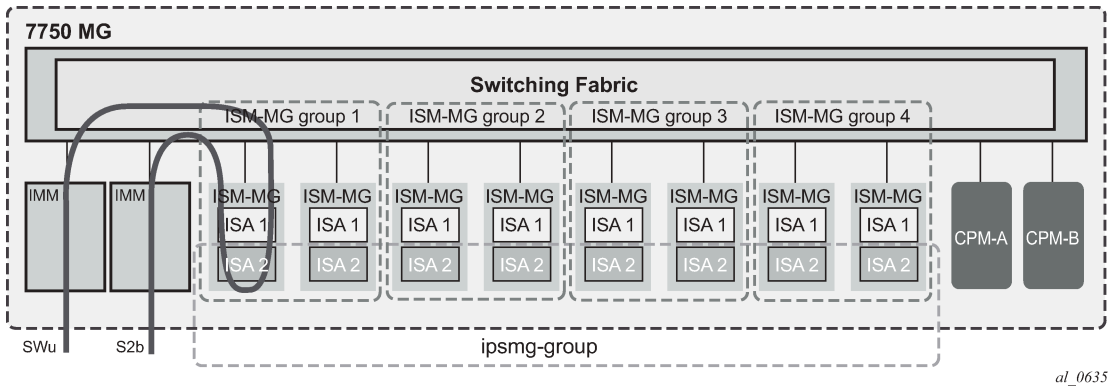
- The ISM-MG also supports the ePDG as a unique application.
- When using the ePDG as a combination gateway with PGW/SGW/GGSN, the S2b interface of the ePDG and the PGW should be on the same VPRN.

8.2 IPSMG groups configuration

The ePDG can be implemented as a standalone gateway or as a combination of the ePDG and PGW/SGW running on the same chassis.

To configure the ePDG, the second ISA in an ISM-MG must be used for IPSec. The MDA type must be configured as `isa2-ipsmg`. A group configuration is used, similar to the Application Assurance groups, where all `isa2-ipsmg` MDA types will then be assigned to a single group. However, the IPSMG redundancy will be inherited by the ISM-MG group on which the ISA is operating.

Figure 110: ePDG ISM-MG and ISA grouping



First, configure the second ISA as an IPSMG MDA type. In this example, card slot 2 is being configured.

```
config card slot 2 card-type ism-mg-2pac-fp3
config card mda 1 mda-type isa-mg
config card mda 2 mda-type isa2-ipsmg
```

Next, add MDA 2 to an IPSMG group. An IPSMG group refers to an Integrated Services Adapter (ISA) Internet Protocol Security (IPSec) Mobile Gateway (MG) group.

```
config isa ipsmg-group create 1
config isa ipsmg-group 1 mda 2/2
```

The ISA IPSMG group commands are described in the *7750 SR Mobile Gateway Multi-Service Integrated Services Adapter Guide*.

Associate the ipsmg-group with the PDN gateway ID:

```
config mobile-gateway pdn 5 ipsmg-group 1
config isa ipsmg-group 1 mda 2/2
```

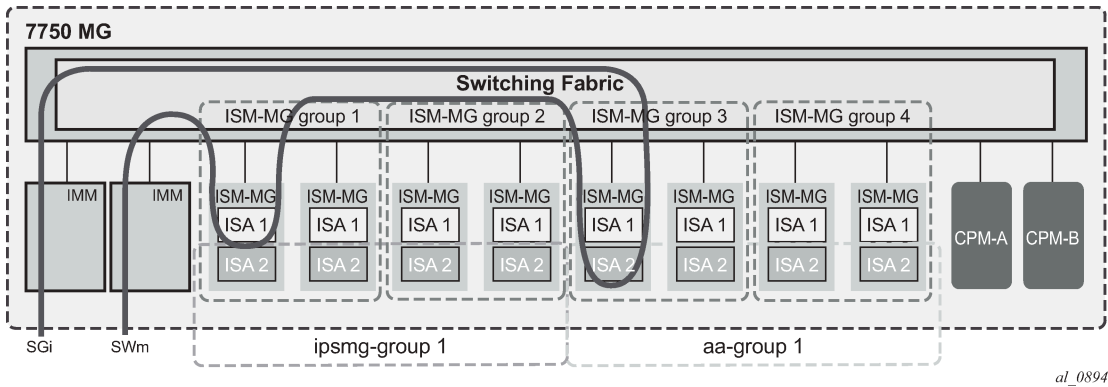


NOTE:

The ePDG can also be implemented in combination with PGW and SGW gateways on the same chassis, using a two-card combination scenario. This is done by defining ISM-MG groups as usual for the PGW-SGW, but only including a subset of the second ISAs in the ipsmg-group, so that the second ISA is free to support AA.

In *ePDG and PGW-SGW two-card combination*, groups 1 and 2 are defined to be part of the ePDG and groups 3 and 4 are part of the PGW. In this scenario, the second ISA for the PGW is free to support Application Assurance.

Figure 111: ePDG and PGW-SGW two-card combination

**NOTE:**

When using the ePDG as a combination gateway with PGW/SGW/GGSN, the S2b interface of the ePDG and the PGW must be on the same VPRN.

The ePDG can also be implemented as a combination gateway with TWAG, providing both trusted and untrusted WiFi access.

Related information*Trusted Wireless Access Gateway*

This section describes the implementation of the Trusted Wireless Access Gateway (TWAG) supported as a standalone gateway or as a combined gateway for both ePDG and TWAG access.

8.3 IKE and ESP security associations configuration

An IPSec profile is used to configure the IKE and ESP security associations. For ePDG configurations, the IPSec profile must be applied to each SWu interface in the PDN.

The IPSec profile has two main sections configured using the `ike - sa` and `esp - sa` commands. For each section, up to ten algorithm suites (proposals) can be configured to allow different algorithm suites that the ePDG accepts from a peer. IKE or ESP chooses the first matching proposal, using a match order of 1- 2- 3....- 10. The proposals must therefore be configured in decreasing order of preference, with the most-preferred suite as proposal 1, the second-preferred suite as proposal 2, and the least-preferred suite as proposal 10. Proposal 1 can also be thought of as the "preferred" proposal, proposal 2 to 9 as the "acceptable" proposal, and proposal 10 as the "last resort" proposal.

If none of the proposals offered by the peer (mobile device) match any of the proposals configured on the ePDG, the ePDG sends a NO_PROPOSAL_CHOSEN error notification back to the peer.

Note that the IKE and ESP profiles support encryption including DES, 3DES, AES128, AES192 and AES256. The ePDG also supports Diffie-Hellman groups 1, 2, 5, 14, 15 for the IKE-SA and ESP-SA.

The commands in the following context configures ePDG encryption:

```
configure mobile-gateway profile ipsec esp-sa proposal-entry encryption
```

```
configure mobile-gateway profile ipsec ike-sa proposal-entry encryption
```

The command in the following context configures the Diffie-Hellman group:

```
configure mobile-gateway profile ipsec esp-sa proposal-entry dh-group
```

In addition, the UE can convey the WiFi AP MAC address via its IKE identity. The ePDG can send the MAC address over S2b and the PGW can send it via Gx. The UE can also send its IMEI via a custom operator IKE configuration attribute to the ePDG, which forwards it via SWm to the AAA. The ePDG understands custom operator IKE configuration attributes provided for the UE to request and receive lists of P-CSCF addresses. The PGW is able to receive and send them via custom operator attribute numbers.

This example shows an IPSec profile with three proposal entries configured for IKE and ESP.

```
esp-sa
  lifetime 85
  replay-window 32
  nat-traversal-dynamic-updates
  proposal-entry 1
    auth sha256
    encryption aes256
  exit
  proposal-entry 2
    auth sha384
    dh-group 2
    encryption aes192
  exit
  proposal-entry 3
    auth md5
    dh-group 15
    encryption 3des
  exit
exit
ike-sa
  cert "ftp://*: *@135.243.192.247//usr/global/tools/lte/certs/
  ca2048Epdg310012Cert.der"
  dpd-interval 130
  lifetime 115
  local-identity "epdg.epc.mnc012.mcc310.pub.3gppnetwork.org"
  mobike check-return-routability
  nat-traversal-dynamic-updates
  private-key "cf1:ca2048Epdg310012Key.der"
```

```

proposal-entry 1
    auth md5
    encryption 3des
exit
proposal-entry 2
    auth sha256
    dh-group 15
exit
proposal-entry 3
    auth sha512
    dh-group 1
    encryption aes256
exit
exit

```

Dynamic updates for NAT traversal can only be enabled for ESP SA if the replay window option is enabled. In addition, the replay window cannot be disabled while NAT traversal is enabled. For more information about command usage, refer to the IPsec Profile Commands section in the *CMG CLI Reference Guide*.

8.4 Enhanced certificate area support

The ePDG certificate area offers ability to create multiple certificate profiles with different ePDG certificates or certificate chains per profile. Certificate validation is also supported via CRL and OCSP.

To import certificates and create CA profiles, use the following 7750 SR OS commands:

```

admin certificate
    crl-update ca ca-profile-name
    import type {cert|key|crl} input url-string output filename format

```

```

configure system security pki ca-profile
    cert-file filename
    crl-file filename
    ocsp
        responder-url url-string
        service service-id
    revocation-check {crl|crl-optional}

```

```

configure system security pki
    certificate-expiration-warning hours [repeat repeat-hours]
    crl-expiration-warning hours [repeat repeat-hours]

```


To view the CA profile, use the following command 7750 SR OS commands:

```
show certificate ca-profile
```

For more information and command descriptions, refer to *7450 ESS, 7750 SR, and VSR Multiservice Integrated Service Adapter and Extended Services Appliance Guide* and *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

To enable support for creating certificate entry profiles, use the following commands:

```
configure mobile-gateway profile ipsec-profile ike-sa cert-entry cert-entry-id
  epdg-cert epdg-cert-name
  priv-key epdg-priv-key
  send-chain number-of-chains
  shutdown
  url-base-address url
```

To view the IPsec profile, use the following command:

```
show mobile-gateway profile ipsec-profile profile-name
```

8.5 Interface configuration using SWu reference points

The ePDG uses multiple SWu reference points to name the interfaces, IPSec profiles, and IPSMG groups, similar to other reference points including S5, S11, and so on.

The SWu reference points are configured in the PDN context, as shown in the following example.

```
config>mobile>pdn>swu# info detail
-----
  interface "swu"
  ipsec-profile "epdgProfile"
-----
```

8.6 Diameter reference point configuration using SWm

SWm is used to configure Diameter-based reference points, similar to the configuration of other Diameter-based reference points. Configuration elements include the peer group, SWm profile, transaction timer, retry count, and Diameter interface.

The application type must be set to SWm for the Diameter peer profile, as shown in the following example.

```
configure
  mobile-gateway
    profile
```

```

        diameter-peer "swm_Peer-Profile1"
            application-type swm
            destination-realm "nokia.com"
            diameter-profile "s6b_diam-prof1"
            peer 10.10.68.2 port 3868
            no shutdown
        exit
    exit
exit
exit
exit
exit

```

The SWm peer group and related options can then be configured in the PDN context, as shown in the following output example.

```

#config>mobile>pdn>swm#
-----
    interface
        diameter router "vprn20031" "sgw_swm_geored_lo"
    exit
    no retry-count
    transaction-timer 3
    peer-group 1 primary-diameter-peer "swm_Peer-Profile1" secondary-
diameter-peer "swm_Peer-Profile2"
-----

```

8.7 S2b reference point configuration

S2b is the interface interconnecting the ePDG and PGW and is used for CP and UP tunneling. S2b is also used for both initial attach and handover attach session establishment and continuation. S2b is a GTP protocol based interface. Different parameters for GTP-C and GTP-U are configurable per S2b interface.

Two interfaces can be created under two different VPRNs, as shown in the following example.

```

/show router 20032 interface
s2b_epdg_lo16          Up          Up/Down    VPRN    loopback
10.10.52.9/32          n/a
/show router 20033 interface
sgw_s5_geored_lo      Up          Up/Down    VPRN    loopback
10.10.53.1/32         n/a

```

Multiple S2b interfaces can be configured, as shown in the following example.

```

/configure mobile-gateway pdn 2
s2b "default"

```

```

        interface
            gtp-c router "vprn20033" "sgw_s5_geored_lo"
            gtp-u router "vprn20033" "sgw_s5_geored_lo"
        exit
    exit
s2b "epdg16"
    interface
        gtp-c router "vprn20032" "s2b_epdg_lo16"
        gtp-u router "vprn20032" "s2b_epdg_lo16"
    exit
exit

```

8.8 ePDG charging profile configuration

Charging profiles are configured in the mobile-gateway charging profile context and are applied in the PDN context.

The following basic example shows ePDG charging profile configuration with the QoS trigger and volume limit enabled for offline charging.

```

config>mobile>profile>charging>epdg>$ info detail
-----
        no description
        off-line
            enable-offline
            no pri-cdf-dia-peer
            no sec-cdf-dia-peer
            charging-info-trigger
                qos-change
                no tariff-time-change
                no tariff-time
            exit
        partial-record-closure-trigger
            no max-change-condition
            no time-limit
            volume-limit 1000
        exit
    exit

```

The following example shows ePDG charging profile 1 assigned as the default in the PDN charging context.

```

#config>mobile>pdn>charging$ info
-----
        charging-prof home 1
        charging-prof visiting 1

```

charging-prof roaming 1

8.9 MOBIKE for ePDG configuration

For MOBIKE configuration, the ePDG supports the following options:

- dynamic updates for source IP addresses for IPsec ESP SA and IKE SA
Dynamic updates for NAT traversal can only be enabled for ESP SA when the replay window is also enabled.
- ability to specify check return routability MOBIKE to enable the UE to change the IP address for its end of the IKE SA if it moves to a new network access point, without requiring the SA to be re-established
- ability to include or exclude the UE local IP address and UE UDP port for BBAI
- ability to configure the ePDG to send a modify bearer request if the UE local IP address or UE UDP port changes, for the BBAI function.

```
configure>mobile-gateway# profile
ipsec-profile
  esp-sa
    nat-traversal-dynamic-updates
  ike-sa
    mobike [check-return-routability]
    nat-traversal-dynamic-updates
```

```
configure>mobile-gateway# profile
policy-options
  begin
  gx-avp-options <profile-number>
    exclude-ue-local-ip-address
    exclude-ue-udp-port
  commit
```

```
configure>mobile-gateway# pdn
send-mbr-for-bbai
```

For an IPsec configuration example, see [IKE and ESP security associations configuration](#).

The following is an example of a basic ePDG configuration with BBAI enabled.



NOTE:

When using the ePDG as a combination gateway with PGW/SGW/GGSN, the S2b interface of the ePDG and the PGW must be on the same VPRN.

```
#config>mobile>pdn# info
-----
    send-mbr-for-bbai
    home-plmn-list "basicPlmn"
    ipsmg-group "1"
    charging
        charging-prof home 1
    exit
    signaling
        diameter
            origin-host "diam.origRealm.com"
            origin-realm "origRealm.com"
            profile "s6b_diam-prof1"
        exit
    exit
    kpi-kci
    exit
    dns-client
    exit
    pco-option
    exit
    s2b "epdg2"
        interface
            gtp-c router "vprn20033" "s2b_epdg_lo2"
            gtp-u router "vprn20033" "s2b_epdg_lo2"
        exit
        match
            apn-list "list1"
        exit
    exit
    swm
        interface
            diameter router "vprn20031" "sgw_swm_geored_lo"
        exit
        transaction-timer 3
        peer-group 1 primary-diameter-peer "swm_Peer-Profile1" secondary-
diameter-peer "swm_Peer-Profile2"
    exit
    swu "epdg"
        interface router "vprn20031" "sgw_swu_geored_lo"
        ipsec-profile "epdgProfile"
    exit
    no shutdown
-----
```

For more information about MOBIKE implementation for the ePDG, refer to the *CMG ePDG Guide*. For MOBIKE CLI command descriptions, refer to the *IPSec Profile Commands* section in the *CMG CLI Reference Guide*.

Related information

IKE and ESP security associations configuration

An IPSec profile is used to configure the IKE and ESP security associations. For ePDG configurations, the IPSec profile must be applied to each SWu interface in the PDN.

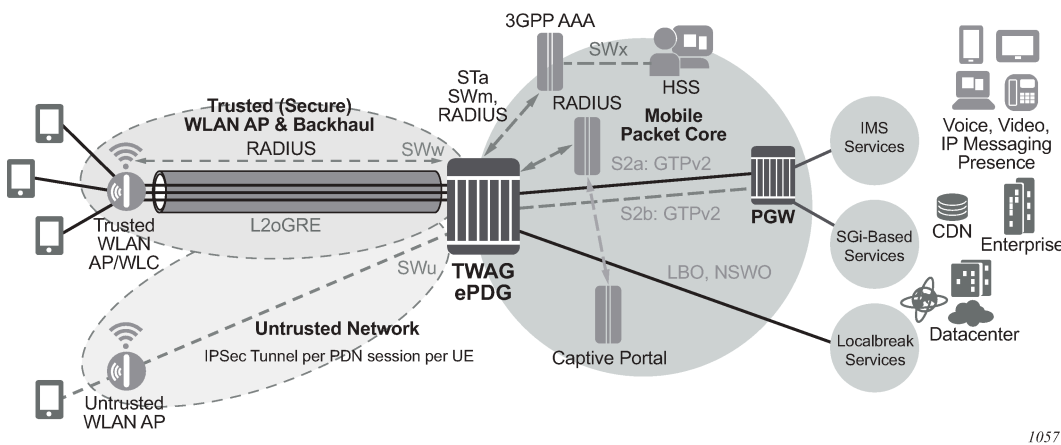
9 Trusted Wireless Access Gateway

This section describes the implementation of the Trusted Wireless Access Gateway (TWAG) supported as a standalone gateway or as a combined gateway for both ePDG and TWAG access.

9.1 TWAG implementation

The Nokia TWAG implementation supports both the Open SSID mode (captive portal) for device authentication, and Closed SSID mode (EAP based SIM, AKA, AKA', TLS and so on) for user authentication. As a result, all flavors of trusted WiFi architectures (for example, homespot, hotspot, small cell and so on) are supported.

Figure 112: Open and closed SSID modes for TWAG implementation in CMG



1057

In Open SSID mode, the device is not authenticated by the WiFi access point. Instead, the traffic is directed to the TWAG which performs HTTP redirection to a web portal for device authentication. If the authentication is successful, then the web portal back end is able to communicate with the TWAG to allow access to the device.

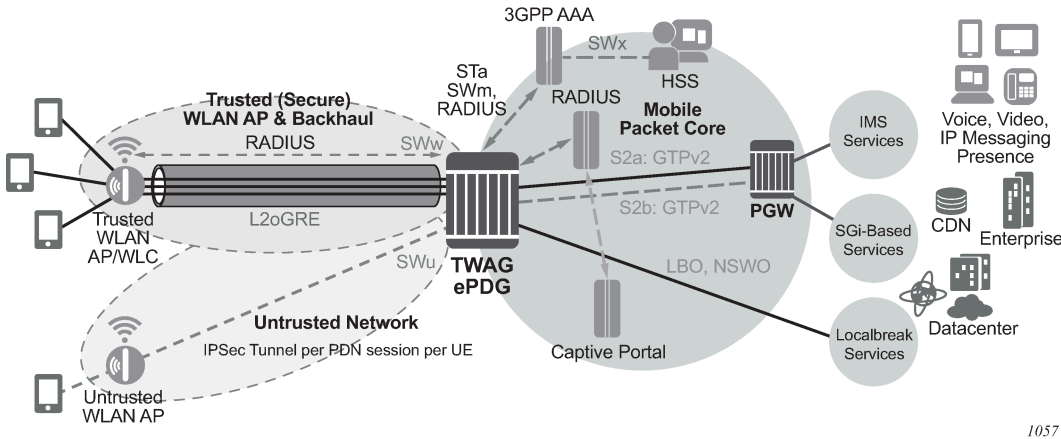
In Closed SSID mode, the device is authenticated by the WiFi access point and the AAA authentication is performed via the TWAG by proxying AAA messages between the WiFi network and the 3GPP AAA server. Furthermore, the 3GPP integration provides subscriber access to the Evolved Packet Core (EPC) over a Trusted WiFi infrastructure so that they can access the same services via WiFi or 3GPP access. Authentication to the WiFi Access network and the EPC is performed in a single step using the 3GPP AAA server (and HSS).

9.1.1 TWAG/ePDG combined gateway implementation

Trusted WiFi Access Gateway (TWAG) is supported as a standalone or as a combined gateway for both ePDG and TWAG access, providing both trusted and untrusted WiFi access on the same ISM-MG-C and chassis.

The combined ePDG and TWAG run as independent sessions, thus an IMSI based UE is attached either via ePDG or TWAG, but not to both at the same time.

Figure 113: TWAG ePDG combined implementation and reference points



1057

9.1.2 Supported features and functionality

The TWAG/ePDG supports the following functionality, as per 3GPP specifications, Rel 11, Dec. 2013 versions:

- single system for both TWAG-based UE access and ePDG-based UE access
- NFM-M support
- TWAG for Open SSID access (captive portal-based access)
 - AAA features
 - RADIUS accounting
 - RADIUS CoA
 - WLAN AP connectivity
 - L2oGRE (soft GRE) IPv4
 - fragmentation and reassembly
 - UE connectivity
 - DHCPv4 support for host IPv4 assignment
 - SLAAC host support for IPv6 assignment
 - HTTP redirect support with or without HTTP header enrichment for web portal authentication
 - session timeout management
- TWAG for Closed SSID access (EAP-based access, SaMOG and NSWO)

- AAA features
 - support for RADIUS facing WLAN AP
 - RADIUS accounting
 - translation proxy functions between RADIUS and DIAMETER
 - EAP SIM based support (AKA', AKA, and SIM)
 - EAP non-SIM based support with RADIUS (local breakout)
- WLAN AP connectivity
 - L2oGRE (soft GRE) IPv4 and IPv6
 - fragmentation and reassembly
- UE connectivity
 - SaMOG (S2a Mobility Over GTP)
 - NSWO (Non-Seamless WiFi Offload) - local break out
 - DHCPv4 support for client IPv4 assignment
 - SLAACv6 support for client IPv6 assignment
 - SLAAC host support for IPv6 assignment
- Interface support
 - SWw - WiFi facing L2oGRE and RADIUS
 - STa (Diameter) / RADIUS - AAA facing - (IPv4/IPv6)
 - S2a (GTPv2) - PGW facing - (IPv4/IPv6)
- offline charging based on RADIUS accounting
- RADIUS Accounting-Interim trigger when the subscriber's location is updated
- WLAN AP connectivity via EthoGRE (soft GRE tunneling)
- fragmentation/reassembly support
- call trace
- lawful interception including support for PGW connected to a TWAG over S2a
- per-SSID or per-VLAN (qtag) access configuration supporting different action functionality per SSID or VLAN ID from the WLAN AP
- inter-WLAN AP mobility
- HTTP redirect to captive portal with or without HTTP header enrichment
- SR MG DPI/Application Assurance support

Related information

Evolved Packet Data Gateway

The configuration guidelines and examples for configuring the ePDG. This information is intended to be used in conjunction with the information in the CMG ePDG Guide, which provides more information and complete feature descriptions for ePDG.

Appendix B: Standards and protocol support

9.1.3 Configuration requirements

TWAG or TWAG/ePDG combined gateway configurations are supported.

The following is required for the TWAG or combined TWAG/ePDG on the 7750 SR MG platform:

- ISM-MG-C
- FP3-based IMM
- ISM-MG redundancy



NOTE:

The ISM-MG also supports ePDG as a unique application.

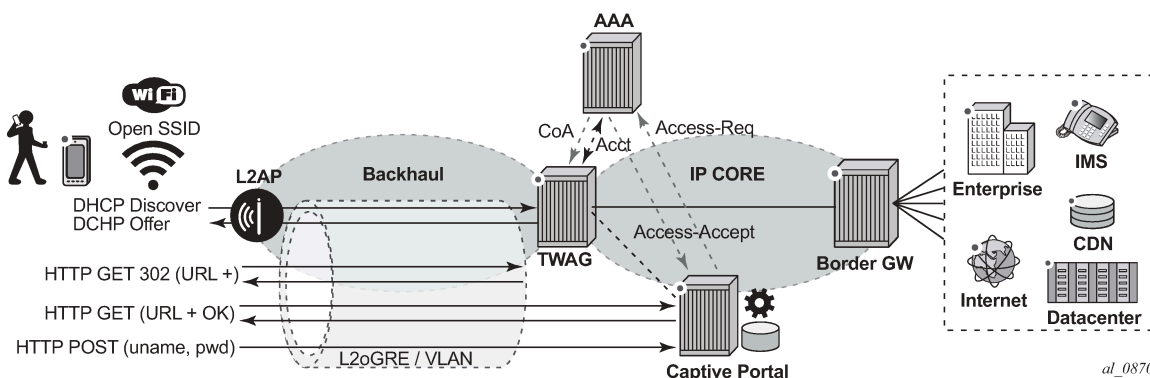
9.2 TWAG open SSID access via captive portal

TWAG open SSID access is supported via a captive portal. Open SSID is WiFi access without using 802.11i/802.1x SIM authentication-based exchange and encryption, and use of EAP-AKA' as defined in 3GPP.

User access is provided using portal-based authentication, by redirecting user HTTP traffic to a web portal, where the users are prompted for credentials, one-time passwords, and so on.

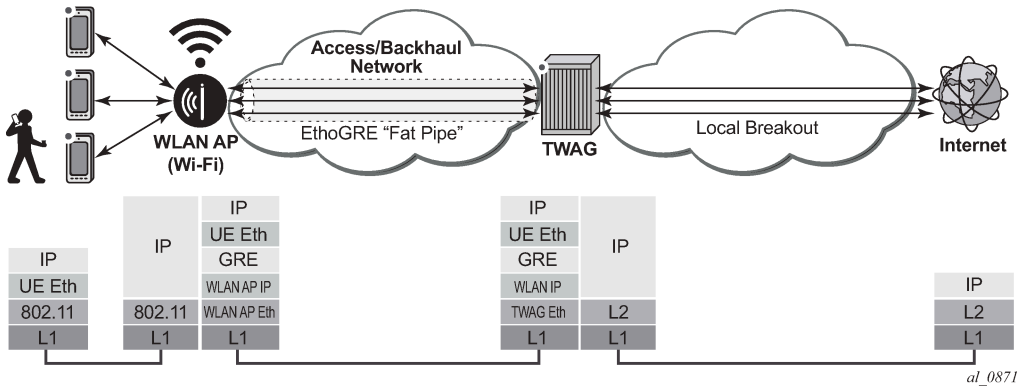
Several variations can be used for relaying the required user and TWAG system information to the portal. The gateway supports both HTTP header enrichment and RADIUS accounting feed options.

Figure 114: TWAG Open SSID access call flow



In the Open SSID model, the TWAG is connected with the WLAN APs using EthoGRE tunnels (soft GRE tunneling). Any traffic from the UE is L2 bridged by the APs to the TWAG.

Figure 115: TWAG Open SSID user plane with EthoGRE tunnel



Related information

TWAG open SSID call flows

The supported open SSID TWAG call flows.

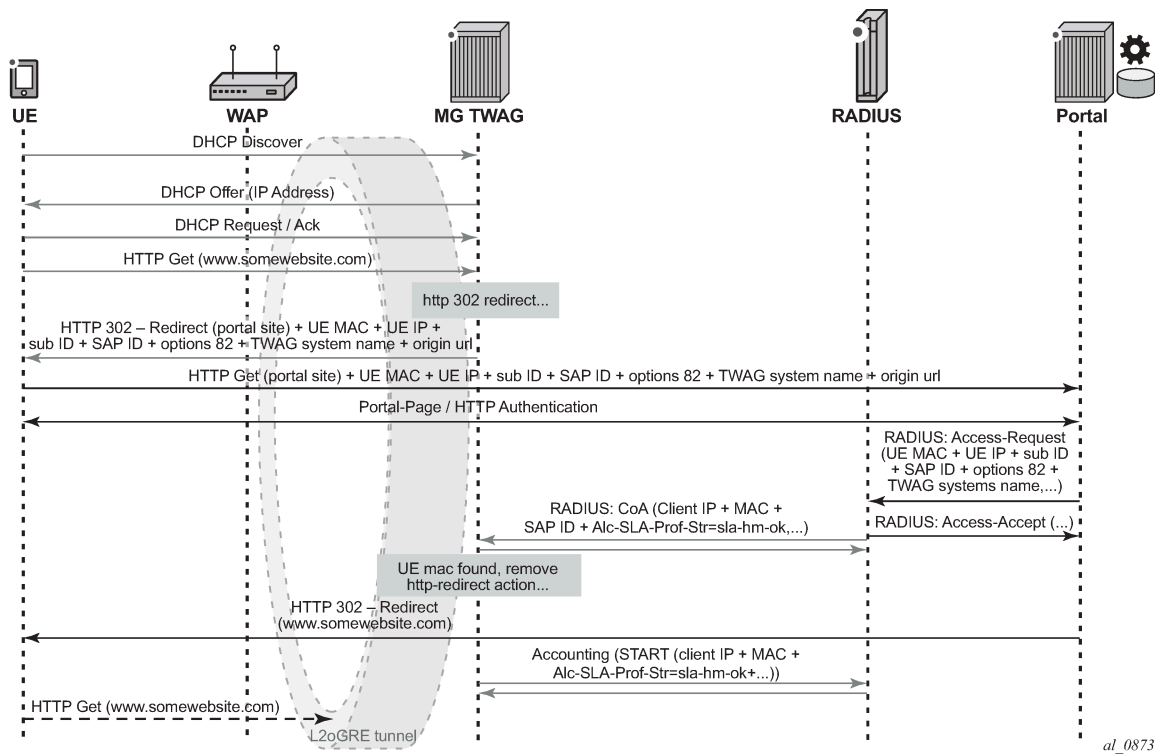
9.3 TWAG open SSID call flows

The supported open SSID TWAG call flows.

9.3.1 HTTP redirect with HTTP header enrichment

The TWAG supports the Open SSID access call flow process using HTTP redirect with HTTP header enrichment.

Figure 116: Open SSID access call flow using HTTP redirect with HTTP header enrichment

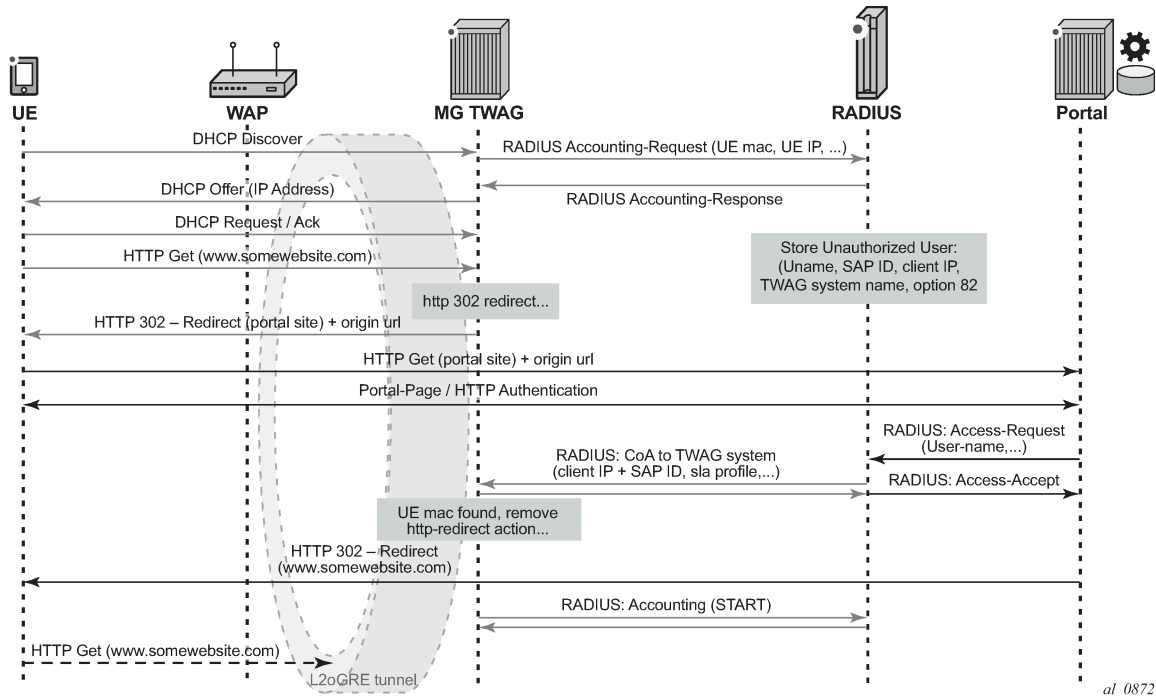


The following steps describe the call flow shown in *Open SSID access call flow using HTTP redirect with HTTP header enrichment*:

1. The UE discovers WiFi coverage and issues a DHCP Discovery requesting an IP address.
2. The TWAG receives the DHCP Discovery over the soft GRE tunnel, dynamically establishes a GRE tunnel (if one does not exist already), and does a UE MAC lookup. If the UE is identified as new, the TWAG assigns an IP address and applies ACL filter rules and HTTP redirect actions. The TWAG then responds with the DHCP Offer and the UE responds with a DHCP Acknowledgment.
3. The UE attempts access using an HTTP client.
4. The TWAG receives the UE HTTP request over the GRE tunnel. With the redirect action applied for the UE, the TWAG responds back with the HTTP 302 redirect, including the portal URL and HTTP header enrichment. The information about the UE and TWAG that the portal and RADIUS server need is appended. This includes information that the RADIUS server uses in the RADIUS CoA after portal-UE authentication is successful.
5. The authentication/authorization might involve multiple HTTP round-trip messages between the UE and captive portal, which the TWAG ACL filter rules pass through transparently.
6. When UE authentication is successful, the captive portal provides the UE and CoA access information to the RADIUS server.
7. The TWAG removes the ACL filter rules and the redirect action to enable UE access.

9.3.2 HTTP redirect with RADIUS accounting feed

Figure 117: Open SSID access call flow using HTTP redirect with RADIUS accounting feed



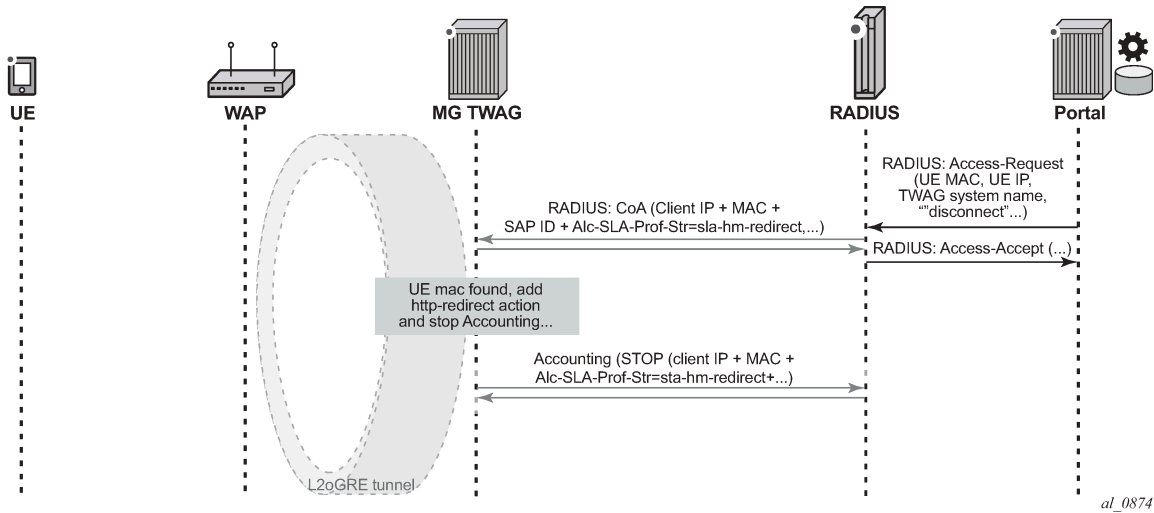
The following describes the supported call flow:

1. The UE discovers WiFi coverage and issues a DHCP Discovery requesting an IP address.
2. The TWAG receives the DHCP Discovery over the soft GRE tunnel, dynamically establishes a GRE tunnel (if one does not exist already), and does a UE MAC lookup. If the UE is identified as new, the TWAG assigns an IP address and applies ACL filter rules and HTTP redirect actions. The TWAG then responds with the DHCP Offer and the UE responds with a DHCP Acknowledgment.
3. The TWAG issues a RADIUS account message to the AAA server with the UE information required to handle the portal authentication and CoA after authentication is successful. This message includes User-Name, Calling-Station-Id, Framed-IP-Address, Called-Station-Id, and so on.
4. The UE tries to get access using an HTTP client.
5. The TWAG receives the HTTP request over the GRE tunnel and, with redirect action installed, responds back to the UE with the HTTP 302 redirect to the portal.
6. The authentication/authorization may involve multiple HTTP round-trip messages between the UE and captive portal, which the TWAG ACL filter rules pass through transparently.
7. When the portal authentication is successful, the RADIUS server issues a CoA, which the TWAG uses to identify the UE, remove the ACL filters and the HTTP redirect action, and start accounting.

9.3.3 Open SSID Stop Access

Depending on the captive portal functionality, there may be a request to stop access to a UE. In this case, the captive portal can communicate with the RADIUS server and the server issues a new CoA to the TWAG to stop access and reapply a redirect action for the UE.

Figure 118: Open SSID Stop Access



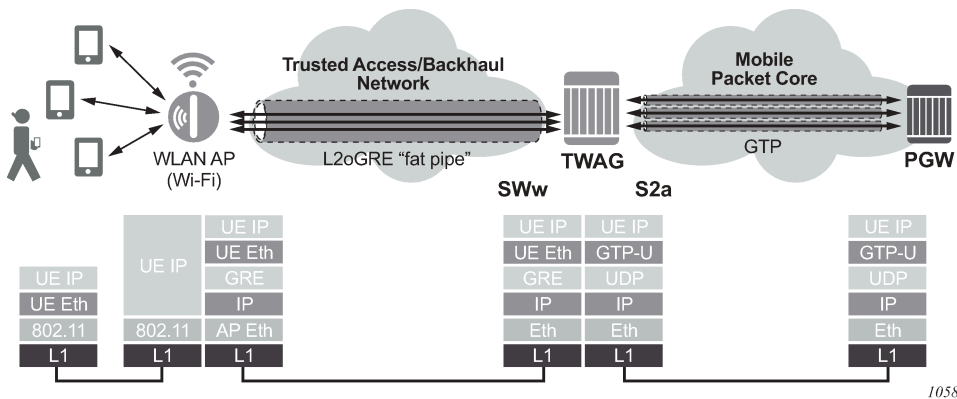
9.4 TWAG closed SSID access

The supported closed SSID TWAG access modes.

9.4.1 S2a Mobility Over GTP (SaMOG) mode

SaMOG provides access to the EPC for a 3GPP IMSI based UE by using a UE-AP-TWAG point-to-point link/tunnel and an S2a GTP tunnel between TWAG and PGW.

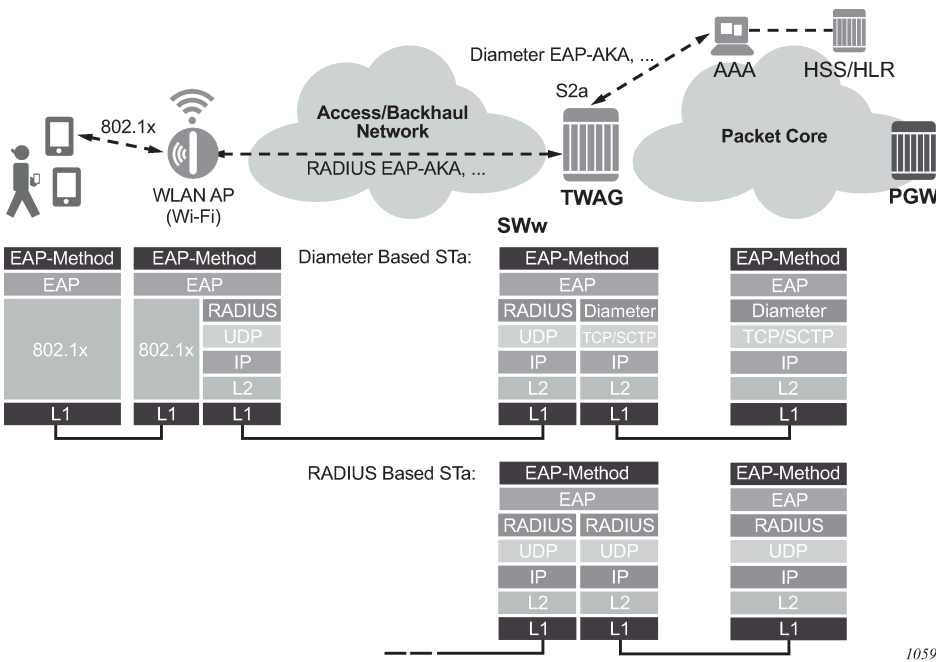
Figure 119: TWAG Closed SSID access in SaMOG mode



In the control plane, the UE access is based on EAP authentication. The MG TWAG allows the use of multiple EAP SIM-based authentication methods, such as EAP-SIM, EAP-AKA or EAP-AKA'.

The MG TWAG supports the Trusted WLAN AAA Proxy (TWAP) function as defined in 3GPP. The TWAP function supports both Diameter (3GPP STa) and RADIUS toward the AAA server. Due to the limited functionality on the Wireless LAN (WLAN) Access Points (APs) and Wireless LAN Controllers (WLCs), RADIUS is always used between the WLAN APs/WLCs and the TWAG.

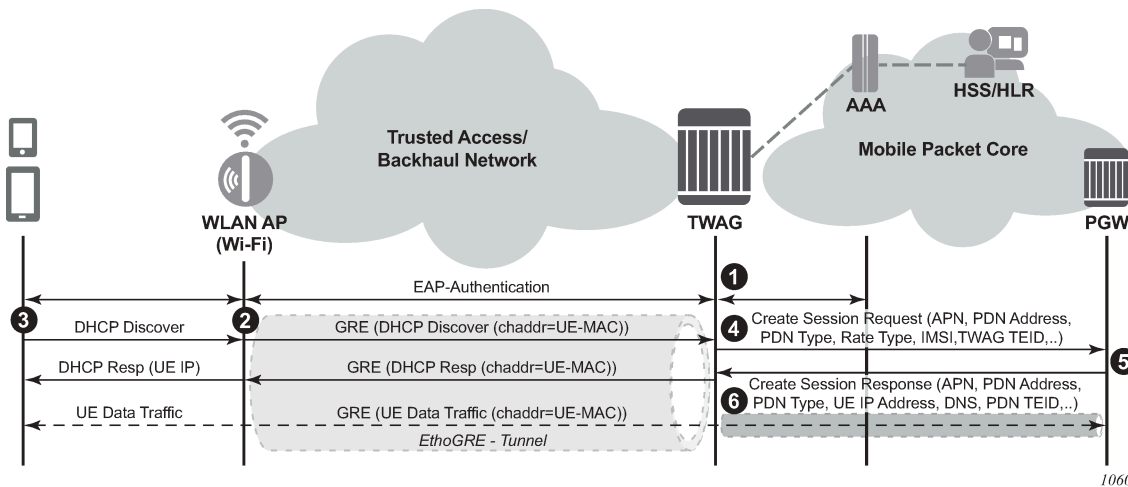
Figure 120: TWAP function in TWAG



The 802.1X exchange takes place over the WiFi interface, which on WLAN AP/WLCs translates into RADIUS EAP authentication toward the AAA server via the TWAP function of the MG TWAG. Depending on the use of 3GPP AAA (STa - Diameter) server or RADIUS AAA server, the TWAP function will act as a Diameter translation proxy or as a RADIUS proxy respectively.

Once the EAP-based authentications have been successfully completed, the GTP-C message exchange over the S2a between the MG TWAG and the PGW is completed to finalize the UE attach to the EPC over trusted WiFi.

Figure 121: High level call flow in SaMOG mode



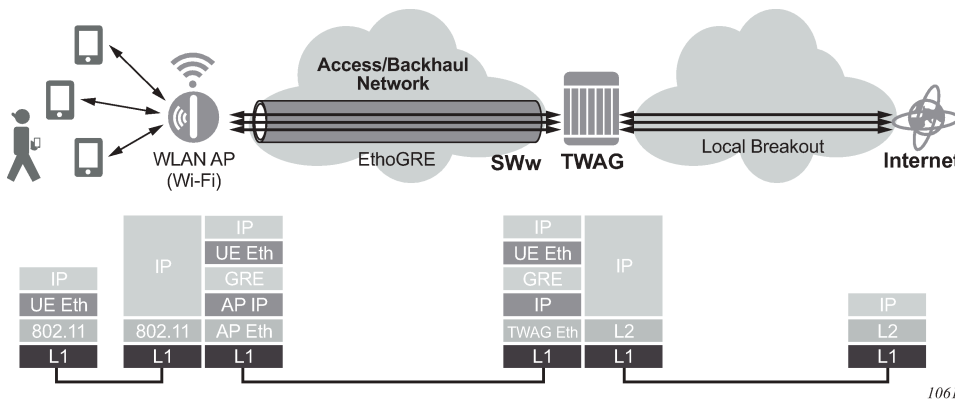
The following steps describe the call flow shown in *High level call flow in SaMOG mode*:

1. The AAA in the final EAP success, also includes the UE subscription (Default APN and PGW selection) and UE access information (S2a, PDN Type, IMSI and so on).
2. With EAP successfully received, the WLAN AP starts bridging any UE data packets over the L2oGRE tunnel.
3. After the 802.1x/EAP is successfully received, the UE requests the IP address.
4. When receiving the UE data packets on the L2oGRE tunnel, the TWAG automatically creates a GRE tunnel context using the cached UE subscription information and initiates a Create Session Request on S2a to PGW.
5. The PGW sets up a GTP-U default bearer and assigns the UE IP address and DNS in the Create Session Response.
6. TWAG responds with the UE IP address and sets up GTP-U default bearer to the PGW and association with GRE tunnel context.

9.4.2 Non-seamless WiFi Offload (NSWO) mode

NSWO is also referred to as EAP Authentication-based local offload on the TWAG. Similarly to the SaMOG mode, the access between the UE-AP-TWAG uses the same point-to-point link/tunneling, but the UE data traffic is then directly offloaded from the TWAG.

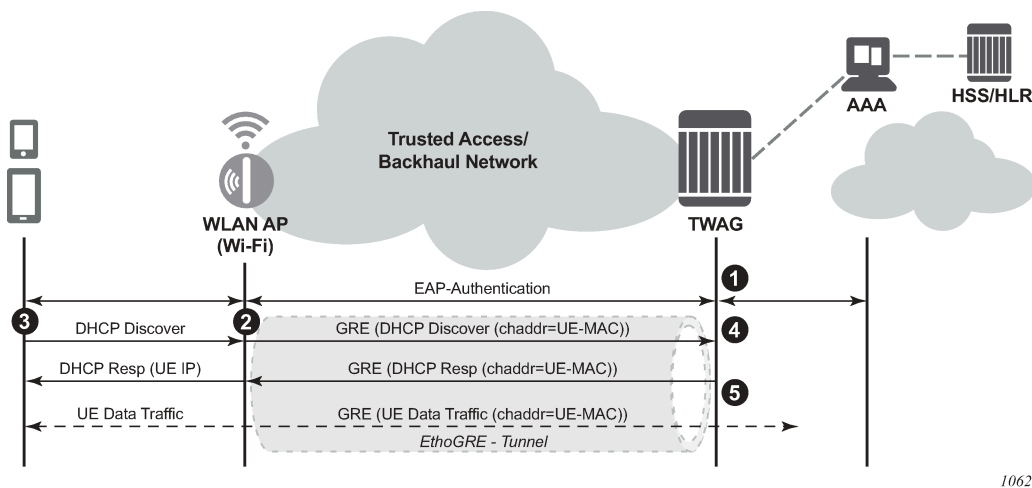
Figure 122: TWAG Closed SSID access in NSWO mode



In the control plane, the UE access is based on EAP authentication. The MG TWAG allows the use of multiple EAP-based authentication methods, such as SIM-based (NSWO) EAP authentication (EAP-SIM, EAP-AKA, and EAP-AKA') as well as non SIM-based EAP authentication (EAP-TLS, PEAP-MSCHAPv2, and so on).

The MG TWAG supports the Trusted WLAN AAA Proxy (TWAP) function as defined in 3GPP. The TWAP function supports both Diameter (3GPP STa) and RADIUS toward the AAA server. Due to the limited functionality on the Wireless LAN (WLAN) Access Points (APs) and Wireless LAN Controllers (WLCs), RADIUS is always used between the WLAN APs/WLCs and the TWAG.

Figure 123: High level call flow in NSWO mode



The following steps describe the call flow shown in *High level call flow in NSWO mode*:

1. The AAA in the final EAP success, also includes the UE subscription and UE access information (NSWO, PDN Type and so on).
2. With EAP successfully received, the WLAN AP starts bridging any UE data packets over the L2oGRE tunnel.
3. After the 802.1x/EAP is successfully received, the UE requests the IP address.

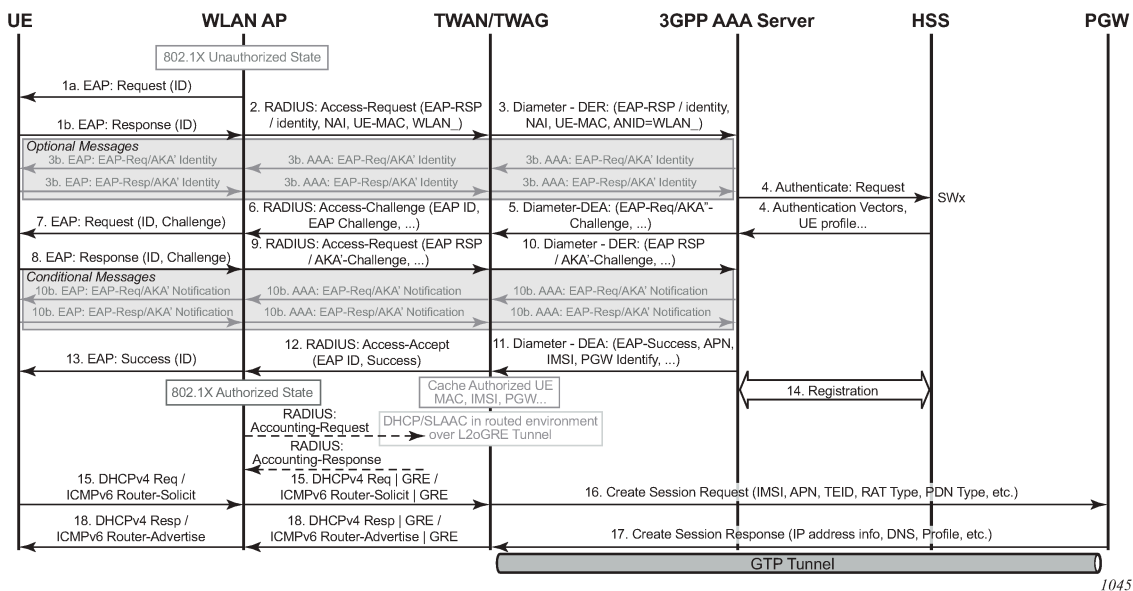
- The WLAN gateway automatically creates a GRE tunnel context using the cached UE subscription information.
- TWAG responds with the UE IP address from the local pool and sets up the UE GRE tunnel context (UE MAC address) to be routed locally.

9.5 TWAG closed SSID call flows

The supported closed SSID TWAG call flows.

9.5.1 Initial SaMOG Attach (3GPP compliant Diameter STa)

Figure 124: Initial SaMOG Attach using a 3GPP compliant Diameter STa call flow



The following steps describe the call flow shown in *Initial SaMOG Attach using a 3GPP compliant Diameter STa call flow*:

- The UE and WLAN AP start the 802.1X EAP Response/Request Identity message exchange.
- The WLAN AP encapsulates the EAP Response/Identity message into the EAP message attribute in the RADIUS Access-Request message and send it via the TWAG.
- The RADIUS/Diameter proxy translator on the TWAG translates and proxies the RADIUS EAP message into a Diameter-EAP-Request.
- The 3GPP AAA Server receives the Diameter-EAP-Request that contains the EAP Response/Identity message, the subscriber identity and the access type over the STa interface. The 3GPP AAA Server identifies the subscriber and communicates with the HSS.

5. Multiple round trips of RADIUS/Diameter EAP authentication message exchanges take place between the WLAN AP and the AAA via the TWAG.
6. Multiple round trips of RADIUS Access-Challenge (EPA ID, EAP Challenge) message exchanges between the TWAG and the WLAN AP.
7. Multiple round trips of EAP Request (EAP ID, EAP Challenge) message exchanges between the WLAN AP and the UE.
8. Multiple round trips of EAP Response (EAP ID, EAP Challenge) message exchanges between the UE and the WLAN AP.
9. Multiple round trips of RADIUS Access-Request (EAP ID, EAP Challenge) message exchanges between the WLAN AP and the TWAG.
10. Multiple round trips of Diameter-EAP-Request (EPA Request, AKA' Challenge) message exchanges between the TWAG and the AAA.
11. The 3GPP AAA Server sends the EAP Success message to the WLAN AP. The 3GPP AAA Server also includes a Pair wise Master Key (PMK).
12. The PGW address, UE subscription information with APN list/default APN, IMSI, PDN-Type, and so on are also included. The TWAG caches and associates those with the UE MAC.
13. The WLAN AP stores the PMK keying information and issues an 802.1x EAP Success message exchange with the UE and allows the UE traffic being L2 bridged over the L2oGRE tunnel.

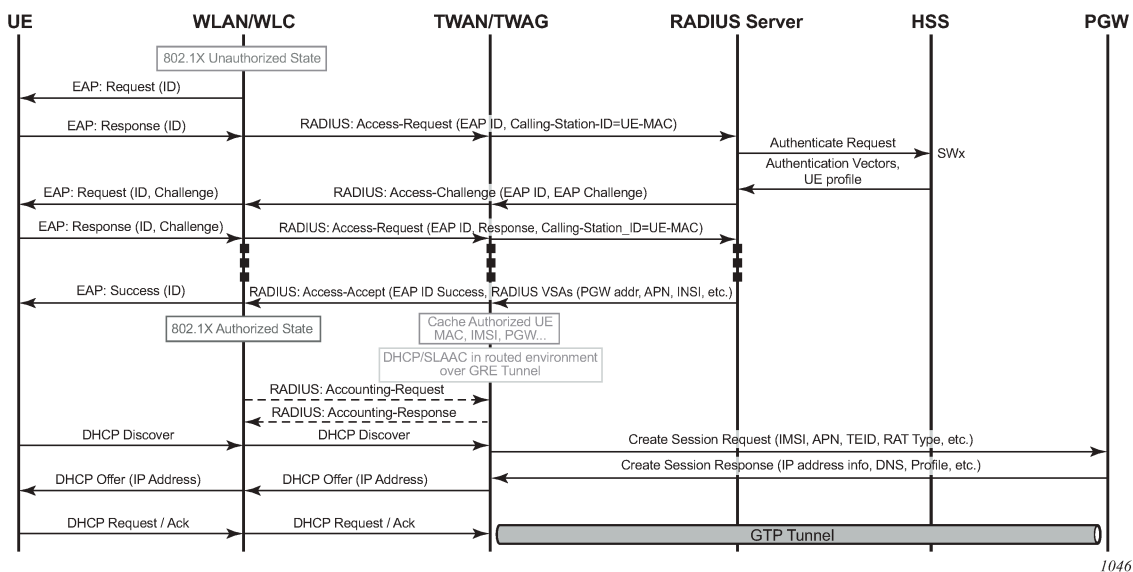
**NOTE:**

The WLAN AP may send a RADIUS Accounting-Request/Start message to inform the TWAG that the UE has now authorized access.

14. The 3GPP AAA server registers to the HSS and the AAA server stores access session information related to the subscriber including the access network identity.
15. The UE sends a DHCPv4 Discover/Request message that the WLAN AP will be L2 bridged over the GRE tunnel to the TWAG.
16. The TWAG selects the PGW based on the PGW Identity (received as part of the reply from the 3GPP AAA EAP Success message) and sends a Create Session Request to the PGW.
17. The PGW returns a Create Session Response message, including the IP address and DNS addresses allocated for the UE, and PDN connection with the default bearer is established.
18. The TWAG returns a DHCPv4 response (with the IP address and DNS addresses from PGW) back to the UE over the L2oGRE tunnel.

9.5.2 Initial SaMOG Attach (RADIUS-based)

Figure 125: Initial SaMOG Attach based on RADIUS call flow



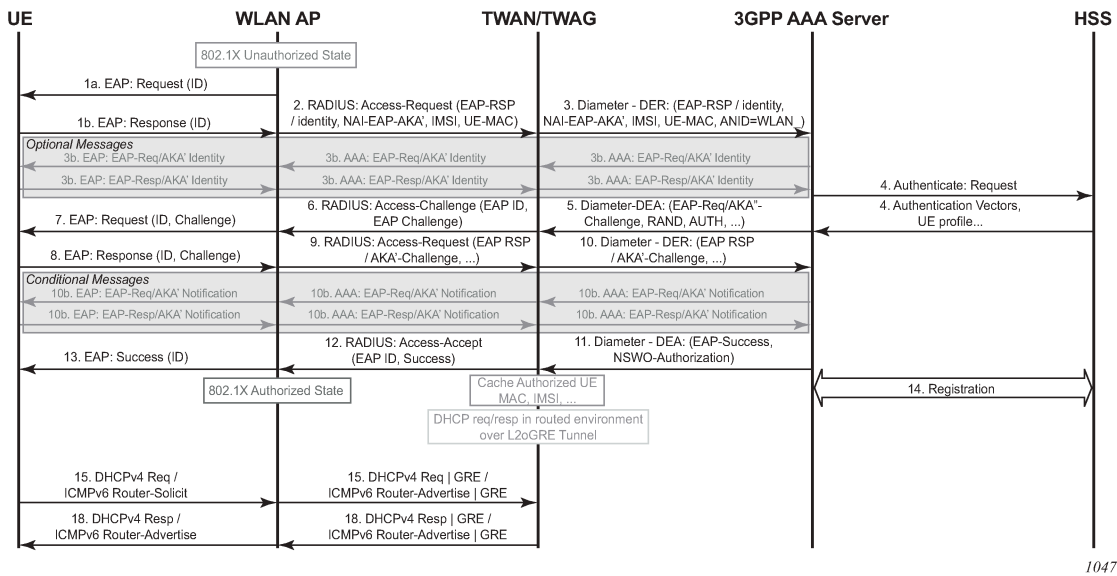
The following steps describe the call flow shown in *Initial SaMOG Attach based on RADIUS call flow*:

1. The UE and WLAN AP start the 802.1X EAP Response/Request Identity message exchange.
2. The WLAN AP encapsulates the EAP Response/Identity message into the EAP message attribute in the RADIUS Access-Request message and sends it via the TWAG.
3. The TWAG proxies the RADIUS Access-Request message to the RADIUS server and RADIUS may communicate with HSS.
4. Multiple round trips of RADIUS EAP authentication message exchanges take place between the WLAN AP and the AAA, via the TWAG as RADIUS proxy.
5. The 3GPP AAA Server sends the EAP Success message to the WLAN AP. The 3GPP AAA Server also includes a Pair wise Master Key (PMK). The WLAN AP stores the keying material to be used in communication with the authenticated UE and allows UE traffic being L2 bridged over the L2oGRE tunnel.

The PGW address, UE subscription information with APN list/default APN, IMSI, PDN-Type, and so on are also included.
6. After being successfully authenticated, the UE sends a DHCPv4 Discover/Request message that it will be L2 bridged over the GRE tunnel to the TWAG.
7. TWAG selects the PGW and sends a Create Session Request message to the PGW.
8. The PGW returns a Create Session Response including the IP address and DNS addresses allocated for the UE, and PDN connection with the default bearer is established.
9. The TWAG returns a DHCPv4 response (with the IP address and DNS addresses from PGW) back to the UE over the L2oGRE tunnel.

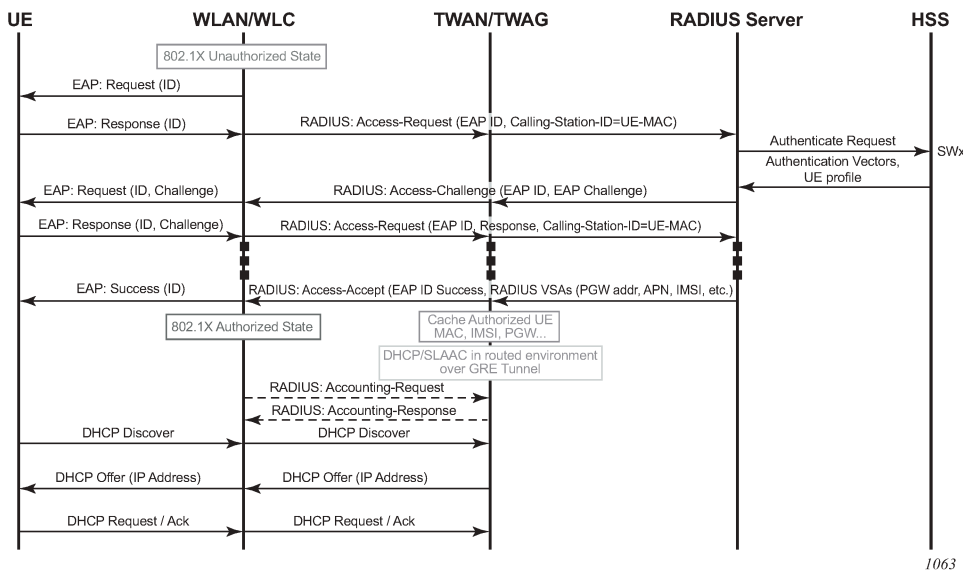
9.5.3 Initial NSW0 Attach (3GPP compliant Diameter STa)

Figure 126: Initial NSW0 Attach using a 3GPP compliant Diameter STa call flow



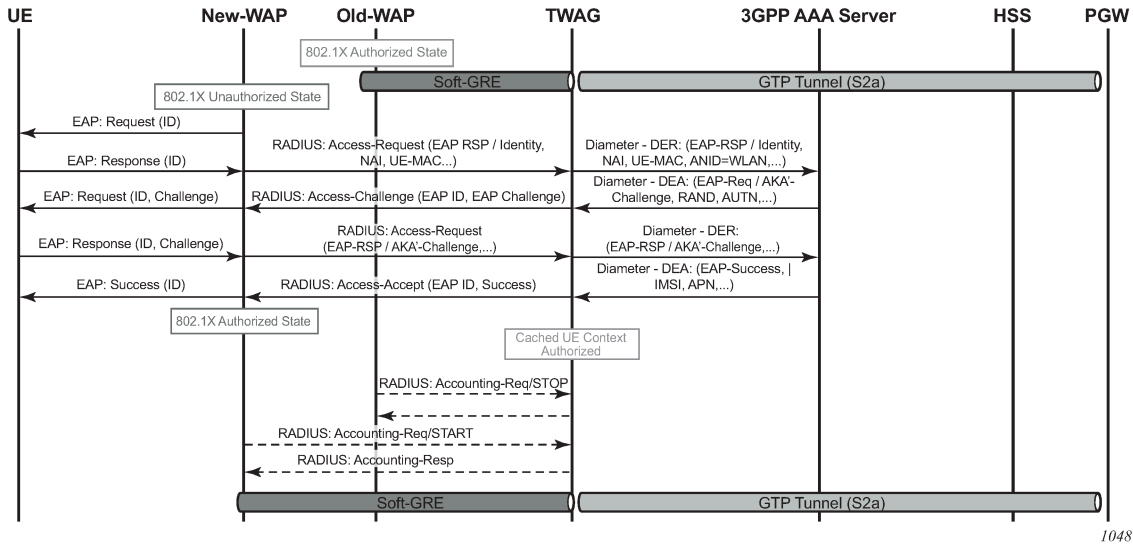
9.5.4 Initial NSW0/Local break out Attach (RADIUS)

Figure 127: Initial NSW0/Local break out Attach based on RADIUS call flow



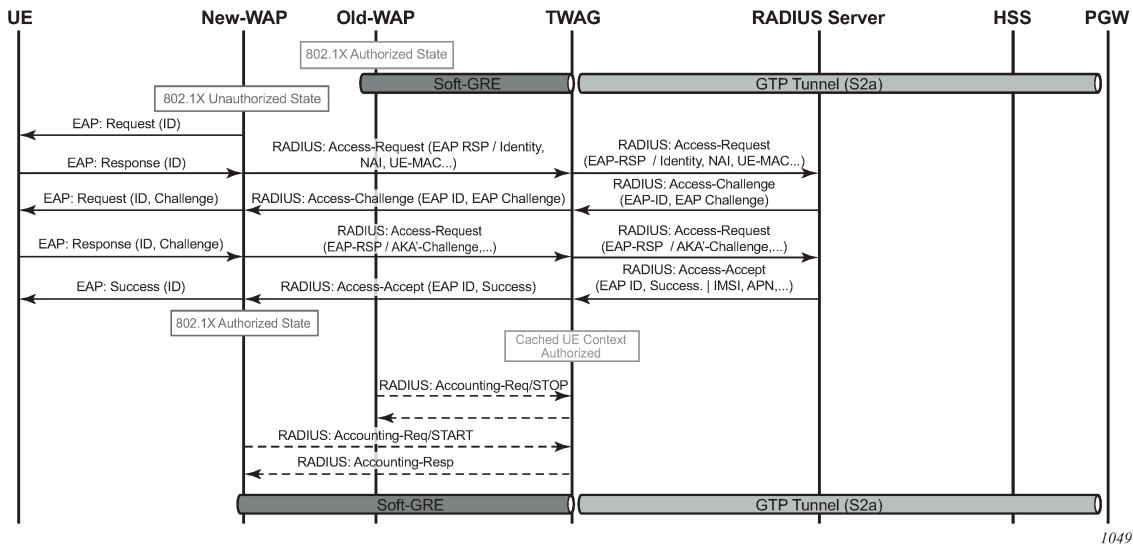
9.5.5 Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - Diameter STa)

Figure 128: Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - Diameter STa) call flow



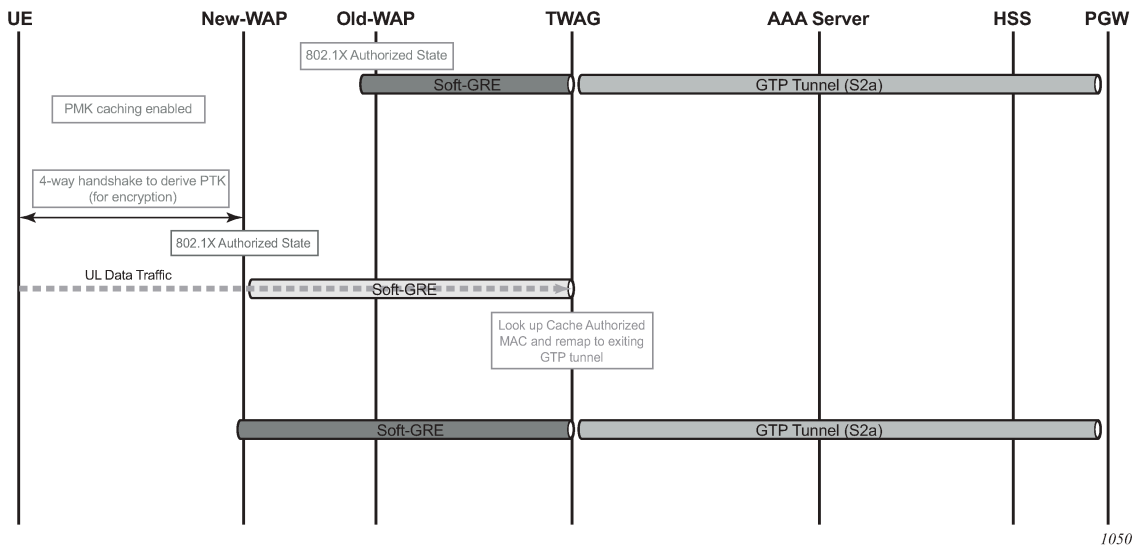
9.5.6 Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - RADIUS)

Figure 129: Inter WLAN AP handover to the same TWAG (EPA RE-AUTH - RADIUS) call flow



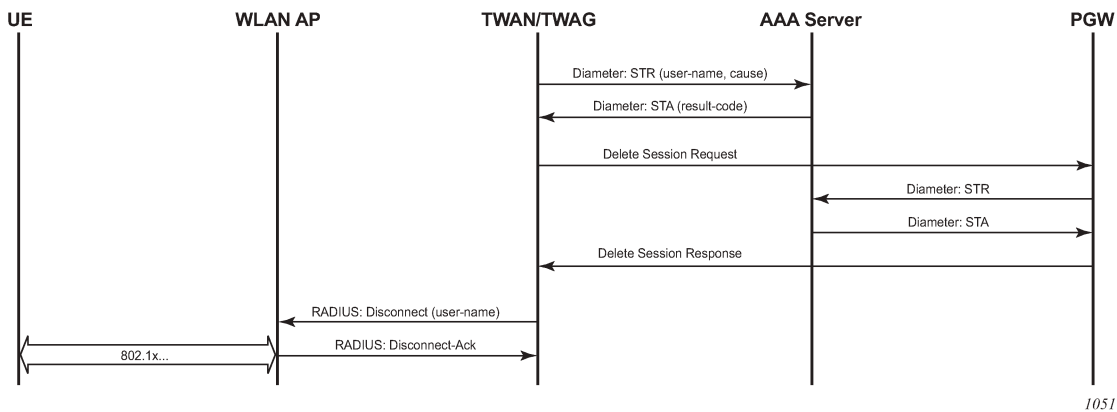
9.5.7 Inter WLAN AP handover to the same TWAG with PMK caching enabled on WLAN AP/WLC

Figure 130: Inter WLAN AP handover to the same TWAG with caching enabled on WLAN AP/WLC call flow



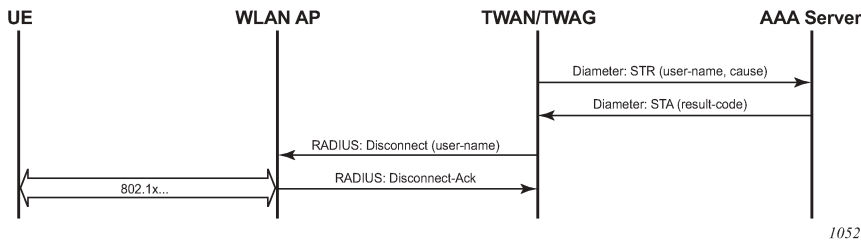
9.5.8 MG TWAG-initiated Detach (SaMOG)

Figure 131: MG TWAG-Initiated Detach in SaMOG mode call flow



9.5.9 MG TWAG-initiated Detach (NSWO)

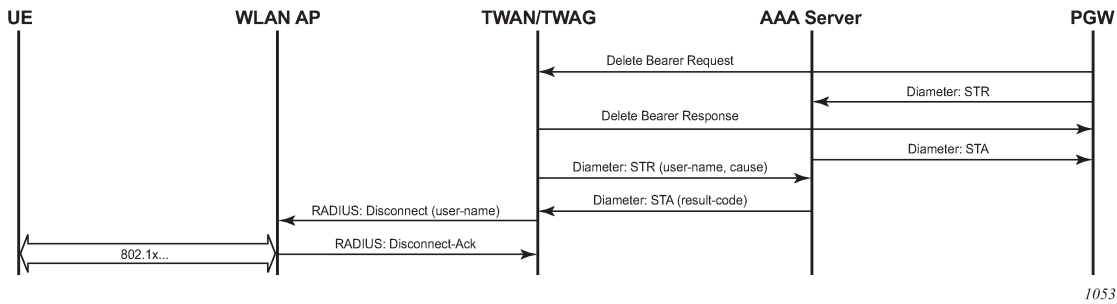
Figure 132: MG TWAG-initiated Detach in NSWO mode call flow



1052

9.5.10 PGW-initiated Detach

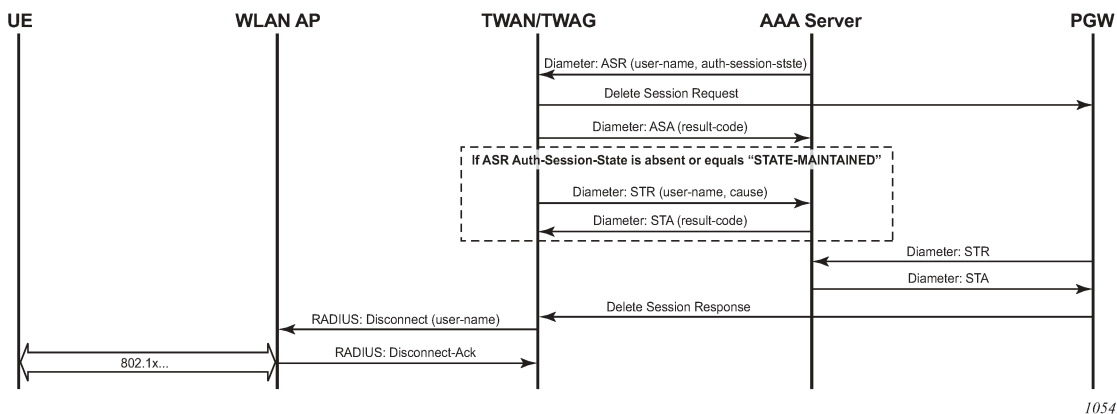
Figure 133: PGW-initiated Detach call flow



1053

9.5.11 STa-initiated Detach

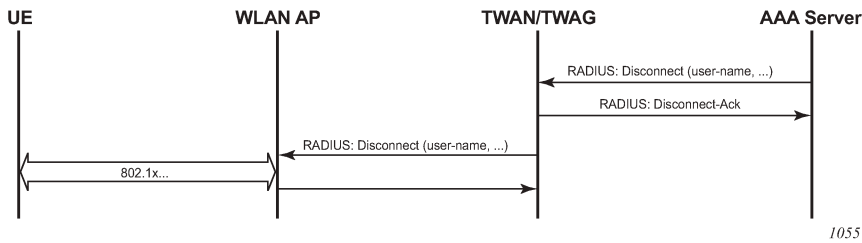
Figure 134: PDN STa-initiated Detach call flow



1054

9.5.12 RADIUS-initiated Detach

Figure 135: PDN RADIUS-initiated Detach call flow

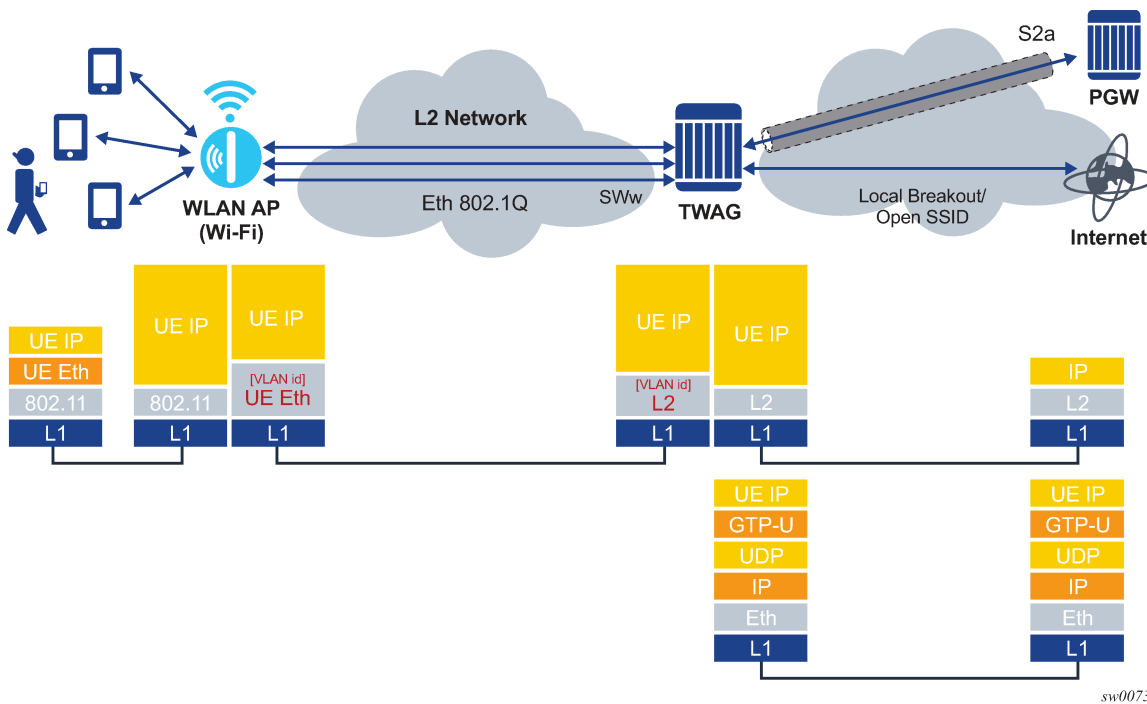


9.6 L2 VLAN as a connection method for WLAN APs

The VLAN access method allows the connection of WLAN APs and/or a Wireless LAN Controller (WLC) directly on Ethernet to a TWAG using 802.1q VLAN tagging.

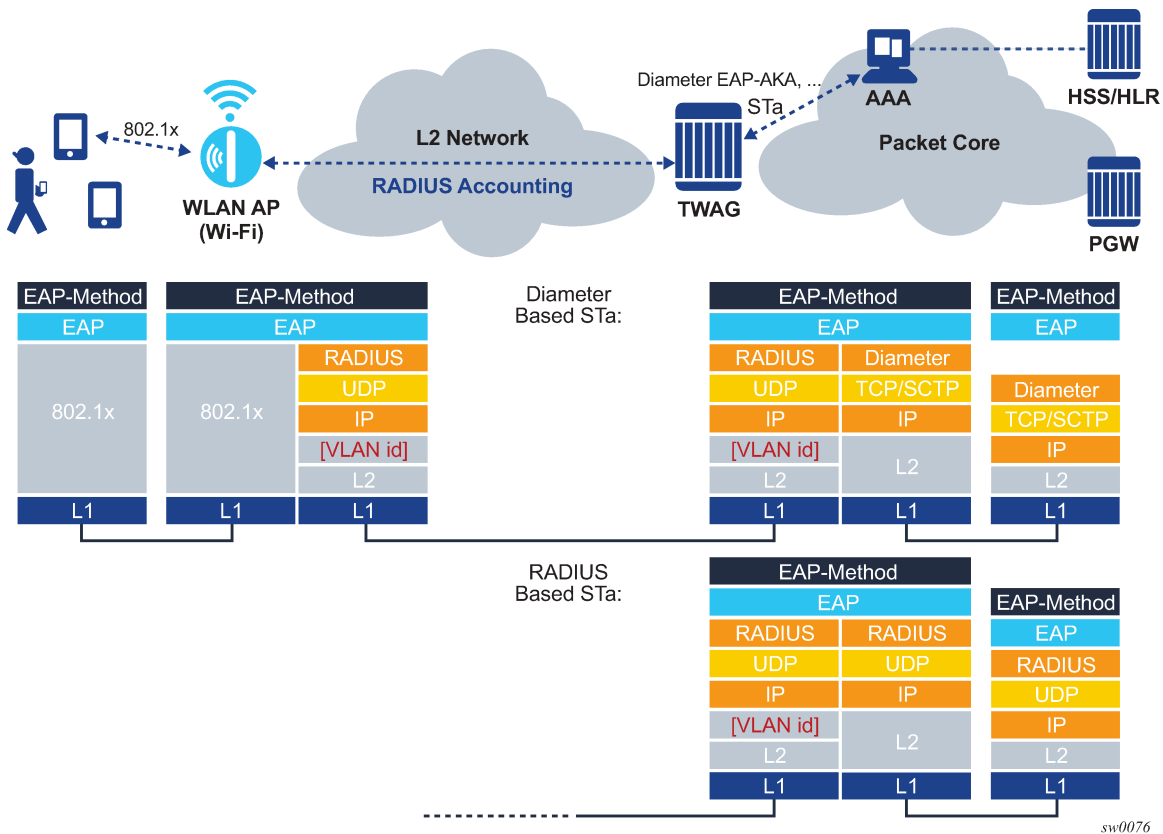
WLAN APs and the TWAG are connected directly (L2 aggregated) over L2 (Ethernet). The WLAN AP/WLC inserts the 802.1q VLAN ID, based on the SSID and L2-bridge of the UE packet to the TWAG.

Figure 136: L2 VLAN connection of WLAN APs using VLAN tagging



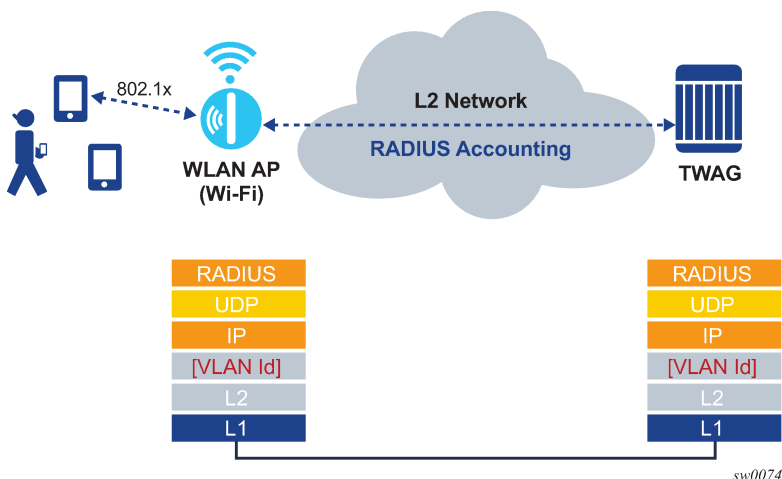
In the case of closed SSID, RADIUS traffic can also be configured to be sent over the same Ethernet connection using a different VLAN ID.

Figure 137: RADIUS traffic over Ethernet using EAP authentication



WLAN APs can be configured to send RADIUS Accounting START/STOP messages for inter-WLAN AP handover as the UE moves from one AP to another.

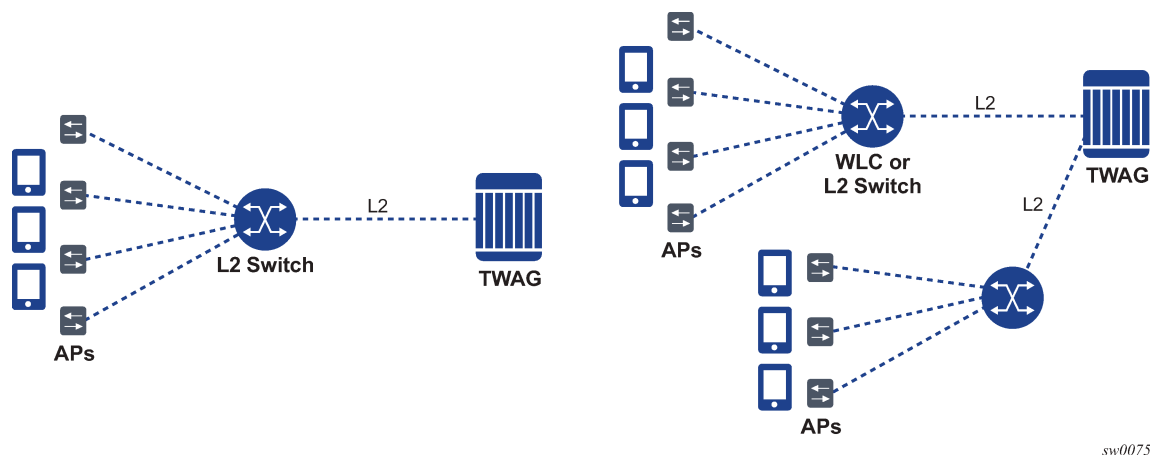
Figure 138: RADIUS Accounting over Ethernet



The VLAN connection mode is typically implemented using L2 aggregation. One use case is a single SSID represented with a VLAN ID and the L2 aggregated APs via an L2 switch/WLC at one site, as shown in the left part of *L2 aggregation in single or multiple sites*. Another case is when there are more than one sites at

different locations, with L2 aggregation via a L2 switch/WLC at each site using two SSIDs (one for each site) represented by two VLAN IDs as shown in the right part of *L2 aggregation in single or multiple sites*.

Figure 139: L2 aggregation in single or multiple sites



To enable L2 access for TWAG, use the following command:

```
config>service>vpn>enable-l2-access
```

9.7 TWAG tracking per AP and location update

This feature introduces RADIUS Accounting-Interim trigger update to the AAA server, when the TWAG detects a change in the UE's WLAN location due to mobility. This allows operators to use business logic in the AAA server, based on the location of Trusted WiFi subscribers.

In Closed SSID access mode, the TWAG detects and reports the UE WLAN location change based on the following events:

- RADIUS Access-Request from a new AP
- RADIUS Accounting-Start from a new AP
- RADIUS Accounting-Interim from a WLC
- data-triggered mobility

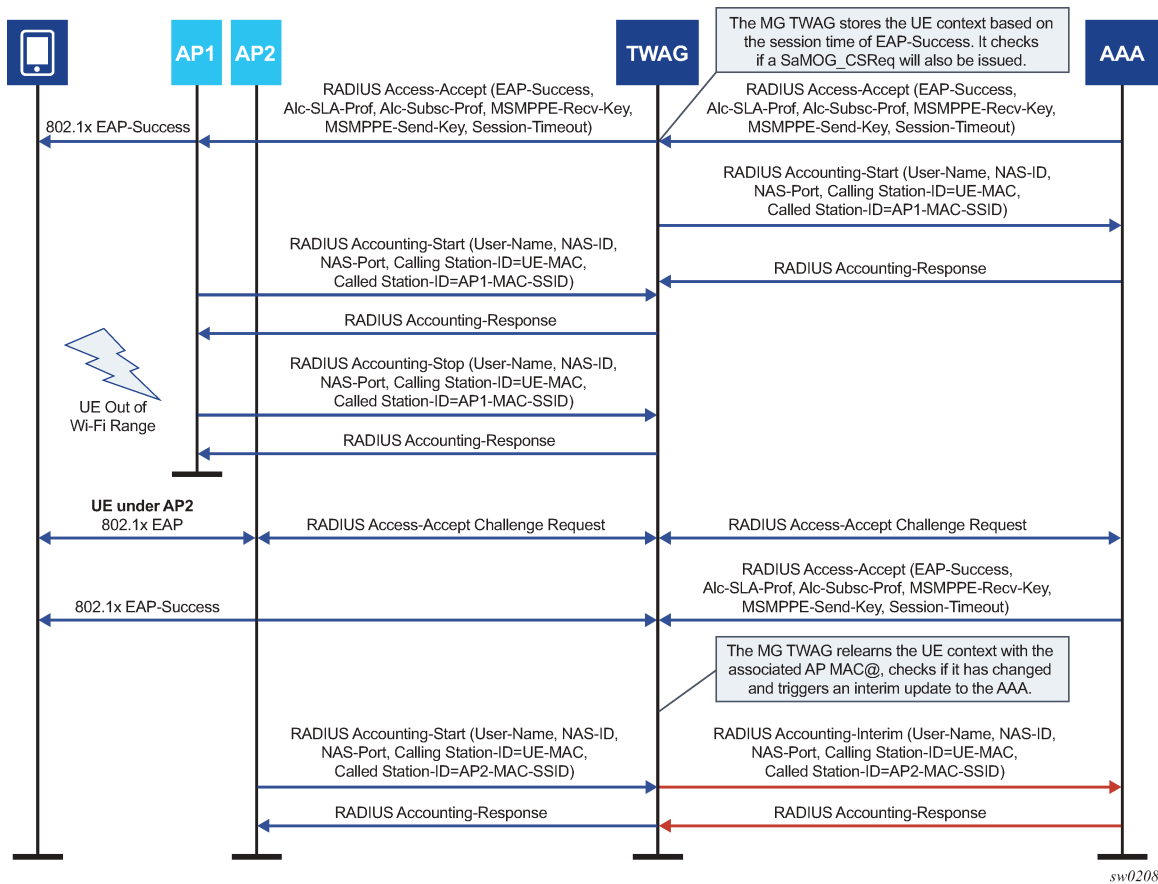
In Open SSID access mode, the TWAG detects and reports the UE WLAN location change based on the following events:

- data-triggered mobility
- DHCP option 82

9.7.1 Closed SSID — subscriber location change based on EAP re-authentication

When a UE gets re-authenticated during the EAP authentication, the TWAG checks the Called-Station-Id (that contains the AP MAC@) in the RADIUS Access-Request message and compares it to the stored Called-Station-Id of the associated UE context. If the Called-Station-Id is different, then a triggered Interim-Accounting update message is sent to the accounting server including the Called-Station-Id == MAC@ of the new WLAN AP.

Figure 140: Closed SSID UE location change based on EAP re-authentication

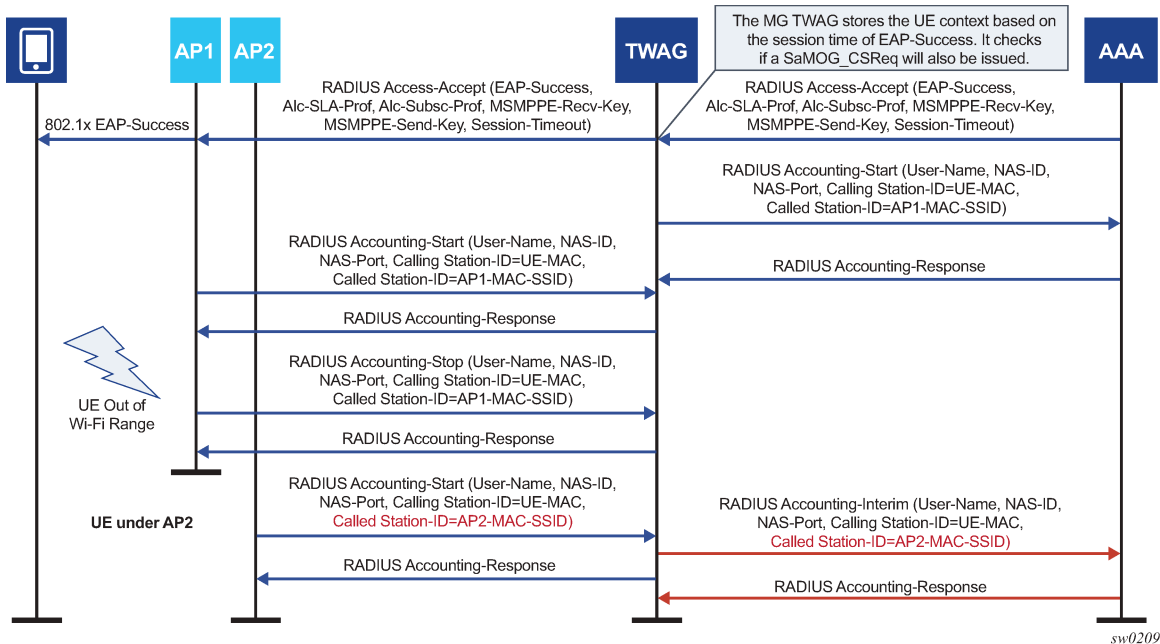


9.7.2 Closed SSID — subscriber location change based on receiving a RADIUS Accounting-Start message from the WLAN AP

The TWAG receives RADIUS traffic directly for the WLAN APs and the WLAN APs are configured to send RADIUS Accounting-Start to the TWAG, when a UE is connected and has authorized access.

The TWAG checks the Called-Station-Id (that contains the AP MAC@) in the Accounting-Start message and compares it to the stored Called-Station-Id in the associated UE context. If the Called-Station-Id is different, then a triggered Interim-Accounting update message is sent to the accounting server including the Called-Station-Id == MAC@ of the new WLAN AP.

Figure 141: Closed SSID UE location change based on receiving a RADIUS Accounting-Start message from the WLAN AP

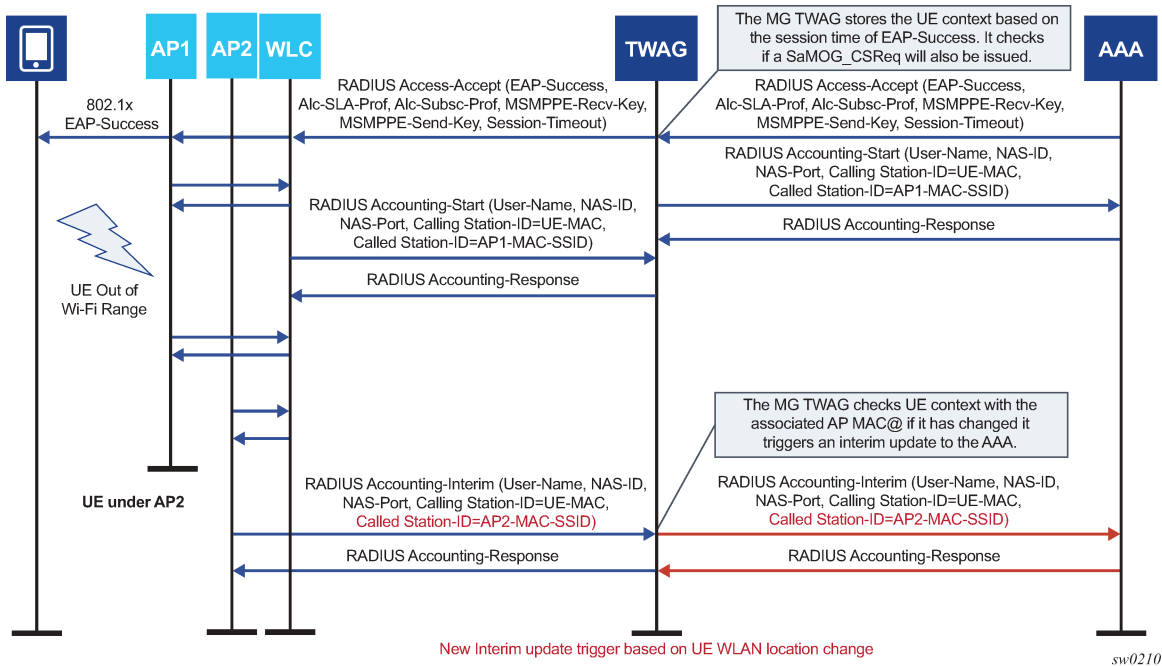


9.7.3 Closed SSID — subscriber location change based on receiving a RADIUS Accounting-Interim message from the WLC

When the TWAG receives RADIUS traffic via the WLC, the WLC may be configured to send RADIUS Accounting-Interim update messages to the TWAG on behalf of the WLAN APs.

The TWAG checks the Called-Station-Id (containing the AP MAC@) in the Accounting-Start or Accounting-Interim message and compares it against the stored Called-Station-Id in the associated UE-context. If the Called-Station-Id is different, then a triggered Accounting-Interim update message will be sent to the accounting server, including the Called-Station-ID == MAC@ of the new WLAN AP.

Figure 142: Closed SSID UE location change based on receiving a RADIUS Accounting-Interim message from the WLC



9.7.4 Closed and Open SSID — subscriber location change based on data trigger

In the case of Soft-GRE tunneling that is based on WAN IP@ of the AP tunnel end-point in the received packet data, the TWAG compares the Called-Station-Id (containing the AP MAC@) to the stored WAN IP@ in the associated UE-context. If the WAN IP@ of the AP tunnel end-point is different, then a triggered Accounting-Interim update message is sent to the accounting server, including the Called-Station-ID == MAC@ of the new WLAN AP.

Figure 143: Closed SSID UE location change based on data trigger

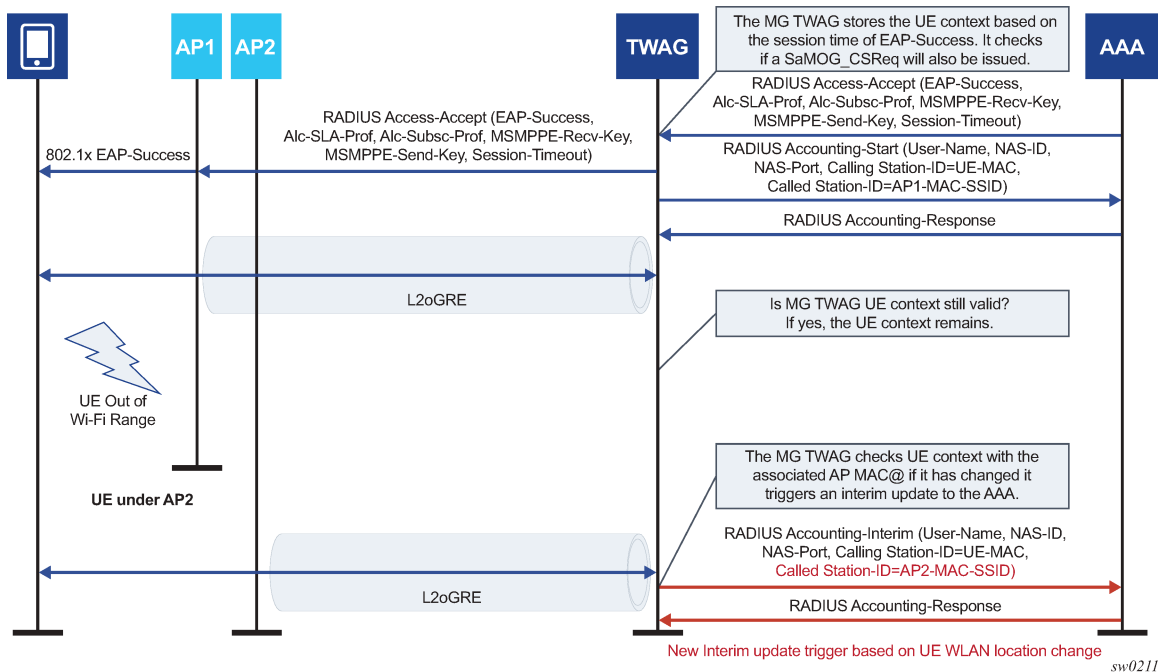
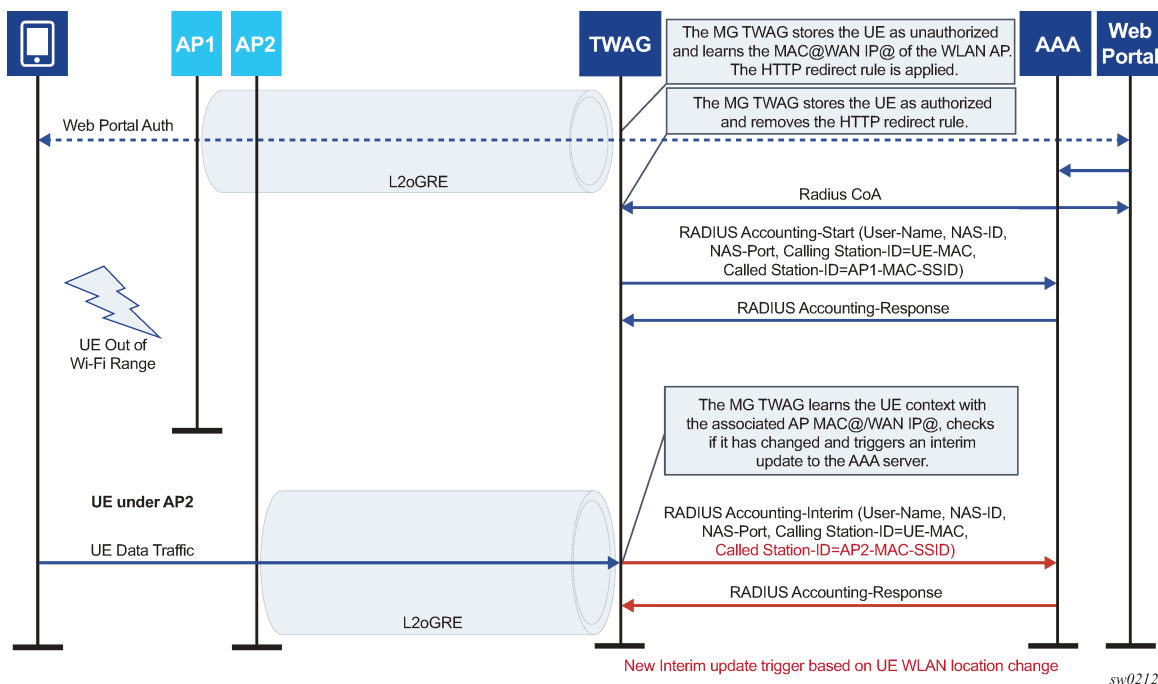


Figure 144: Open SSID UE location change based on data trigger

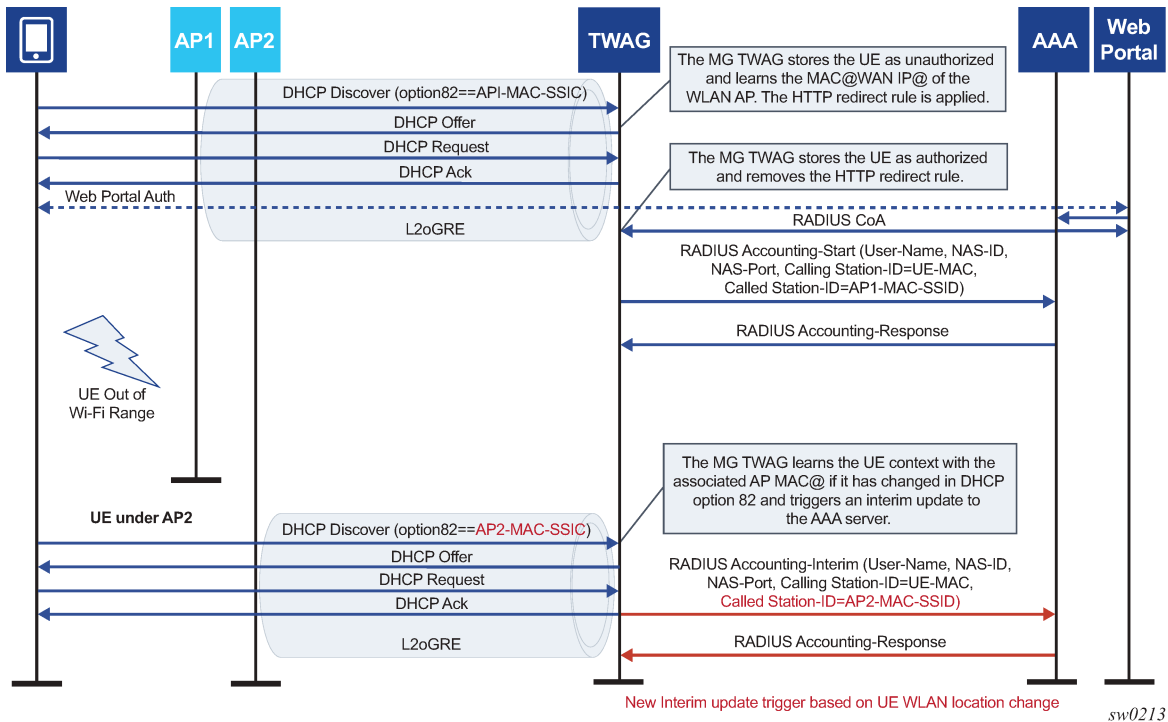


9.7.5 Open SSID — subscriber location change based on DHCPv4 option 82

The MAC@ of the AP in the received DHCP option 82, is also used to compare it against the current stored MAC@ of the AP in the associated UE-context. If the WAN IP@ of the AP is different, then a triggered

Accounting-Interim update message is sent to the accounting server, including the Called-Station-ID == MAC@ of the new WLAN AP.

Figure 145: Open SSID UE location change based on DHCPv4 option 82



9.7.6 TWAG tracking configuration

To enable the triggering of RADIUS Accounting-Interim updates to the AAA server when the TWAG detects a change in the UE's WLAN AP location, use the following command:

```
configure mobile-gateway profile charging twag-charging radius-accounting
wlan-ap-location-change
```



NOTE:

The TWAG charging profile must be referenced under the `config>mobile>pdn>charging>charging-prof-home` context.

If the WLAN APs are configured to send the RADIUS accounting requests to inform the TWAG of the UE being attached to it and for the TWAG to act on Accounting Start and Accounting Interim to detect UE location change, you must also enable the following command:

```
configure mobile-gateway profile sww-radius-profile acct-triggered-mobility
```


To enable the TWAG to use the RADIUS accounting requests sent by the WLAN APs to check if the UE's WLAN location has changed and program the remote next-hop based on re-authentication or accounting from WLAN AP or WLC, use the following command:

```
configure mobile-gateway profile sww-radius-profile fast-inter-ap-handover
```

To enable sending the ARP over GRE to verify the WLAN AP MAC@ which must be enabled in case Soft-GRE and RADIUS Accounting-Interim location trigger updates are enabled to the AAA server when the TWAG detects a UE's WLAN AP location changes due to data-based mobility, use the following command:

```
config mobile-gateway profile sww-tunnel-profile ap-mac-discovery
```

To view TWAG charging information, use the following command:

```
show mobile-gateway profile charging twag-charging
```

9.8 TWAG NSWO with Gx support

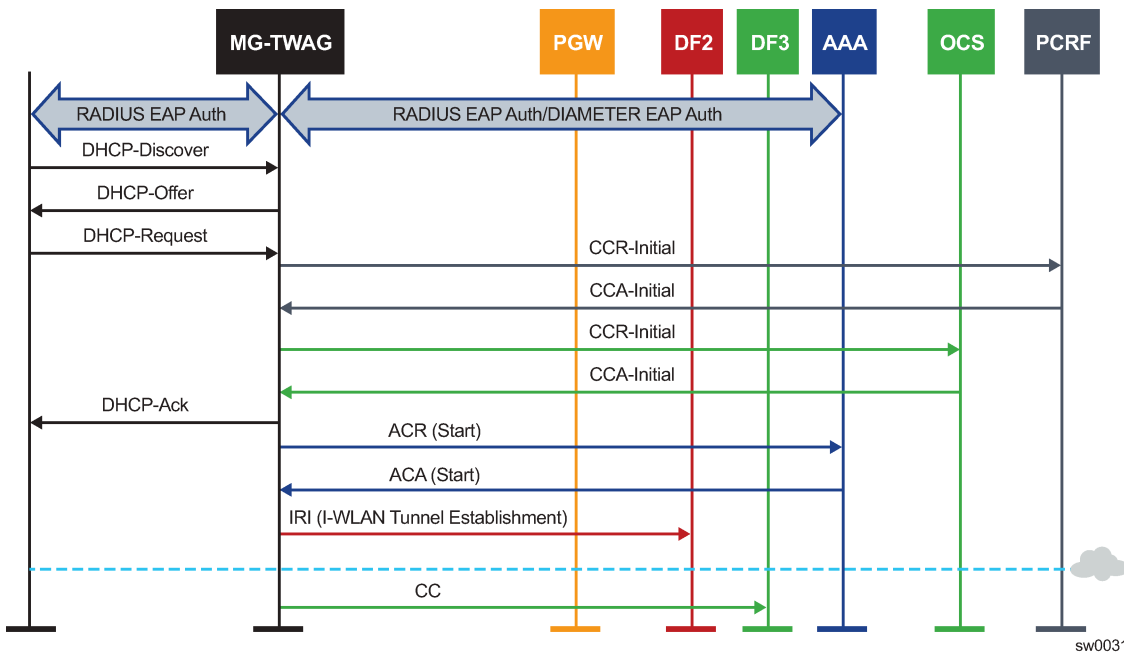
Gx interface support is provided on TWAG for non-seamless Wireless Offload (NSWO) scenarios.



NOTE:

- TWAG Gx Interface support is currently only available for closed SSID - NSWO.
- Gx interface support on TWAG is not provided for closed SSID with S2a mobility Over GTP (SaMOG).
- PCRF policy takes precedence over Authorization, Authentication, and Accounting (AAA); for example, APN-AMBR policing takes precedence over AAA.
- Gx sessions exist as long as the subscriber IP is valid and terminates when the subscriber IP session ends.
- IP-CAN-Type = non-3GPP-EPC
- Negotiation of the Bearer Control Mode (BCM) does not apply.
- PCRF subscription to changes of IP-CAN type, RAT type, or Access Network Information does not apply.

Figure 146: WiFi Subscriber Attach (Closed SSID) — NSW0

**NOTE:**

Online and offline charging functionality is planned for a later release.

The following commands are used to configure TWAG NSW0 with Gx support:

```
configure mobile-gateway profile policy-options gx-avp-options exclude-an-trusted
```

```
configure mobile-gateway profile policy-options gx-avp-options exclude-twan-identifier
```

9.9 TWAG configuration guidelines

The steps and commands required to configure open and closed TWAG implementations.

9.9.1 SWw interfaces for TWAG sessions configuration

The SWw reference point is used to configure router interfaces for TWAG SSID implementation. The SWw reference point can be used to configure up to 16 SWw interfaces per TWAG. An SWw tunnel profile must be configured and tied to the signaling interface for the SWw reference point.

Open SSID TWAG implementation

Use the following syntax to configure a SWw interface for Open SSID TWAG implementation.

```
configure>mobile-gateway# pdn
sww ref-point-name
interface
sww-coa [router router-instance] interface-name port port secret secret [hash|hash2]
sww-tunnel [router router-instance] interface-name [connect-method connect-method]
signaling
sww-tunnel-profile profile-name
```

For additional configuration information, see the SWm, SWu, and SWw PDN commands in the *CMG CLI Reference Guide*.

Example: SWw interface configuration for open SSID TWAG implementation

```
#config>mobile>pdn>sww# info
-----
interface
  sww-coa router "vprn20005" "radius-client" port 3799 secret
  "r.5/.HP5LvWN98QLbervXGh1GskgswGg1k2StsbdBNs" hash2
  sww-tunnel router "vprn3003" "AP-endpoint" connect-method gre-tunnel
exit
signaling
  sww-tunnel-profile "swwTunnel_1"
exit
-----
```

Closed SSID TWAG implementation

Use the following syntax to configure a SWw interface for Closed SSID TWAG implementation.

```
configure>mobile-gateway# pdn
sww ref-point-name
interface
sww-auth [router router-instance] interface-name port port secret secret [hash|
hash2]
sww-rad-disconnect secret secret [hash|hash2]
sww-tunnel [router router-instance] interface-name [connect-method connect-method]
signaling
sww-tunnel-profile profile-name
sww-radius-profile profile-name
```

For additional configuration information, see the SWm, SWu, and SWw PDN commands in the *CMG CLI Reference Guide*.

Example: SWw interface configuration for closed SSID TWAG implementation

```
#config>mobile>pdn>sww# info
-----
interface
  sww-auth router "vprn3003" "listening_radius_server" port 1812 secret
  ".KDuRv8aknHpZpSN0r8FP5yn90B7WQNC" hash2
  sww-rad-disconnect secret ".KDuRv8aknHpZpSN0r8FP5yn90B7WQNC" hash2
  sww-tunnel router "vprn3003" "AP-endpoint2" connect-method gre-tunnel
exit
signaling
  sww-tunnel-profile "swwTunnel_2"
  sww-radius-profile "swwTunnel_2"
exit
-----
```

SWw RADIUS profile configuration

Example: SWw RADIUS profile configuration

```
#config>mobile>profile>sww-radius-profile# info detail
-----
no description
no disconnect-port
no retry-count
no retry-timeout
disconnect-optional-attributes
  no exclude-user-name
  no exclude-framed-ip-address
exit
access-accept-optional-attributes
  no include-alc-wlan-apn-name
  no include-alc-subsc-id-str
  no include-alc-mgw-interface-type
  no include-alc-msisdn
  no include-3gpp-pdp-type
  no include-3gpp-ggsn-address
  no include-3gpp-gprs-negotiated-qos-profile
  no include-3gpp-charging-characteristics
  no include-3gpp-imsi
  no include-3gpp-msisdn
exit
-----
```

9.9.2 SWw tunnel profiles for TWAG sessions configuration

An SWw tunnel profile must be configured and tied to the signaling interface for the SWw reference point.

9.9.2.1 SWw tunnel profile configuration

The SWw tunnel profile specifies the DHCPv4 server profile, a VLAN qtag-to-APN mapping, an SSID-to-APN mapping, the default access type, a default APN, and a SLAAC host profile. For Closed SSID TWAG implementation, the UE cache time can also be specified.

The TWAG RADIUS accounting feed option must be either enabled or disabled for all the APNs associated with the profile. This determines the method used to relay user and TWAG information to the portal. HTTP redirect with header enrichment or RADIUS accounting feed may be used.

A policy rule base must also be configured and referenced under the APNs for the TWAG session.

Open SSID TWAG implementation

Use the following syntax to configure an SWw tunnel profile for Open SSID TWAG implementation.

```
configure>mobile-gateway# profile
sww-tunnel-profile
  default-apn name [portal|eap]
  description
  default-access-type [portal|eap]
  dhcpv4-server profile
  slaac-host profile
  qtag value apn name [portal|eap]
  ssid name apn name [portal|eap]
```

The following example shows an SWw tunnel profile configuration that can be tied to the SWw signaling interface. For more information about SWw profile configuration, refer to the section *SWw Profile Commands* in the *CMG CLI Reference Guide*.

```
#config>mobile>profile>sww-tunnel-profile# info
-----
  default-apn portal "ipd.nokia.1.1.com"
  description "NOKIA WIFI"
  default-access-type portal
  dhcpv4-server "dhcpProf_1"
  slaac-host "slaac1_prof"
  qtag 101 apn "ipd.nokia.2.1.com" portal
  qtag 102 apn "ipd.nokia.3.1.com" portal
  ssid "nok.wifi1.hotspot.com" apn "ipd.nokia.2.1.com" portal
  ssid "nok.wifi2.hotspot.com" apn "ipd.nokia.3.1.com" portal
-----
```

Closed SSID TWAG implementation

Use the following syntax to configure an SWw tunnel profile for Closed SSID TWAG implementation.

```
configure>mobile-gateway# profile
sww-tunnel-profile
  default-apn name [portal|eap]
  description
  default-access-type [portal|eap]
  dhcpv4-server profile
  slaac-host profile
  ue-cache-time seconds
  qtag value apn name [portal|eap]
  ssid name apn name [portal|eap]
```

The following example shows an SWw tunnel profile configuration that can be tied to the SWw signaling interface. For more information about SWw profile configuration, refer to the section *SWw Profile Commands* in the *CMG CLI Reference Guide*.

```
#config>mobile>profile>sww-tunnel-profile# info
-----
  default-apn eap "ipd.nokia.1.1.com"
  description "NOKIA WIFI2"
  default-access-type eap
  dhcpv4-server "dhcpProf_2"
  slaac-host "slaac2_prof"
  ue-cache-time 3600
  qtag 101 apn "ipd.nokia.3.1.com" eap
  qtag 102 apn "ipd.nokia.4.1.com" eap
  ssid "nok.wifi3.hotspot.com" apn "ipd.nokia.3.1.com" eap
  ssid "nok.wifi4.hotspot.com" apn "ipd.nokia.4.1.com" eap
  exit
-----
```

Related information

Policy rule base configuration for APNs referenced in TWAG sessions

A policy rule base must be referenced under APNs configured for the SWw tunnel profile for TWAG sessions.

APNs referenced in TWAG sessions configuration

You can use HTTP redirect with header enrichment or RADIUS accounting to send the AAA the values to be used in the COA from the RADIUS server.

9.9.2.2 DHCPv4 server configuration

This example shows a DHCPv4 server profile configuration.

```
#config>mobile>profile>dhcpv4-server# info detail
-----
no description
no hold-time
no ignore-rapid-commit
lease-time hrs 1 mins 0 secs 0
exit
-----
```

9.9.2.3 SLAAC host profile configuration

This example shows a SLAAC host profile configuration.

```
#config>mobile>profile>slaac-host# info detail
-----
no description
router-advertisements
  no force-unicast-ip
  no include-rdnss
  no max-rtr-adv-interval
  no min-rtr-adv-interval
  no mtu
  no preferred-lifetime
  no rdnss-lifetime
  no reachable-time
  no retransmit-time
  no router-lifetime
  no valid-lifetime
exit
-----
```

9.9.3 TWAG for L2 VLAN access method configuration

You can connect WLAN APs and a TWAG directly (L2 aggregated) over L2 (Ethernet). The configuration is identical to the GRE connection method but L2 access must be specified in the SWw tunnel configuration under VPRN and an SWw reference point must be created.



NOTE:

The following must be taking into account when configuring a TWAG for L2 VLAN access:

- RADIUS and SWw tunnel must be in different VPRNs
- only different VLAN tags are allowed under a VPRN

For additional configuration information, refer to the sections SWm, SWu and SWw, PDN and Profile commands in the *CMG CLI Reference Guide*.

Example: RADIUS VPRN configuration

```
# configure service vprn 4002
#config>service>vprn# info
-----
description "Default Description For VPRN ID 4002"
route-distinguisher 4002:4002
interface "AP-radius1" create
    address 201.3.1.1/24
    sap 1/1/3:201 create
        description "AP/radius-terminating interface"
    exit
exit
service-name "XYZ Vprn 4002"
no shutdown
```

Example: SWw tunnel VPRN configuration

```
#config>service# vprn 4003
#config>service>vprn# info
-----
description "Default Description For VPRN ID 4003"
route-distinguisher 4003:4003
interface "AP-endpoint1" create
    address 201.3.3.1/24
    sap 1/1/3:101 create
        description "AP1-terminating interface"
    exit
exit
interface "AP-endpoint2" create
    address 201.3.4.1/24
    sap 1/1/5:102 create
        description "AP2-terminating interface"
    exit
exit
interface "AP-endpoint5" create
    address 201.3.5.1/24
    sap 1/1/3:103 create
        description "AP5-terminating interface"
    exit
exit
```



```

interface "AP-endpoint6" create
    address 201.3.6.1/24
    sap 1/1/3:104 create
        description "AP6-terminating interface"
    exit
exit
interface "AP-endpoint7" create
    address 201.3.7.1/24
    sap 1/1/5:105 create
        description "AP7-terminating interface"
    exit
exit
interface "AP-endpoint8" create
    address 201.3.8.1/24
    sap 1/1/5:106 create
        description "AP8-terminating interface"
    exit
exit
interface "sww-tunnel-1" create
    address 9.9.9.20/32
    loopback
exit
enable-l2-access
service-name "XYZ Vprn 4003"
no shutdown

```

Example: SWw reference point configuration

```
# configure mobile-gateway pdn 1 sww "swwrefpt1"
```

Example: SWw interface configuration

```

#config>mobile>pdn>sww# info
-----
interface
    sww-coa router "vprn20005" "radius-client" port 1812 secret "r.5/
.HP5LvWn98QLbervXGh1GskgswGg1k2StsbdBNs" hash2
    sww-tunnel router "vprn4003" "AP-endpoint2" connect-method l2-access
exit
signaling
    sww-tunnel-profile "swwTunnel_1"
exit
-----

```

9.9.4 APNs referenced in TWAG sessions configuration

You can use HTTP redirect with header enrichment or RADIUS accounting to send the AAA the values to be used in the COA from the RADIUS server.

This option is configured in the CLI by enabling or disabling the TWAG RADIUS accounting feed option for all APNs associated with the SWw tunnel profile that is to be applied to the SWw interface.

```
configure>mobile-gateway# pdn
apn
    twag-radius-acct-feed
```

If twag-radius-acct-feed is disabled, HTTP header enrichment is used instead of RADIUS accounting feed. If twag-radius-acct-feed is enabled, associated RADIUS AVP options should also be configured.

In addition, a policy rule base must be configured for all the APNs referenced in the TWAG sessions.

The following APNs must be configured for the SWw tunnel profile:

- APN for qtag mapping
- APN for SSID mapping
- default APN
- unknown APN

The SWw tunnel profile uses the configured default APN only if there are no qtag-to-APN or SSID-to-APN matches. An APN with the name "unknown" must also be configured with all the correct policies. The TWAG will revert to the "unknown" APN if there are no qtag-to-APN or SSID-to-APN matches and the default APN is not configured. If an "unknown" APN is not configured, discover packets will be dropped.

Example: APN with TWAG RADIUS accounting feed configured

```
#config>mobile>pdn>apn# info
-----
    pco-option
        pcscf-srvr-ipv4 10.40.1.1
        dns-server-ipv4
            primary 10.30.1.1
            secondary 10.30.1.2
        exit
        nbns-srvr-ipv4
            primary 10.30.1.1
            secondary 10.30.1.2
        exit
    exit
    pdn-type ipv4 ipv6 ipv4v6
    router "vprn1"
        ip-local-pool "pool1.1.0"
    exit
    accounting type radius user-name imsi server-group "RADIUS Group 1"
```

```

exit
static-predefined-pcc
    default-policy-rule-base "redirectRuleBase"
exit
twag-radius-acct-feed
no shutdown

```

Related information

RADIUS AVP configuration for RADIUS accounting feed call flows

Policy rule base configuration for APNs referenced in TWAG sessions

A policy rule base must be referenced under APNs configured for the SWw tunnel profile for TWAG sessions.

9.9.5 Policy rule base configuration for APNs referenced in TWAG sessions

A policy rule base must be referenced under APNs configured for the SWw tunnel profile for TWAG sessions.

The following is an example of a policy configuration for call flows with HTTP header enrichment. The policy rule base configured for HTTP header enrichment is referenced under the APN for the TWAG session to be in redirect until the user is authenticated.

```

#config>mobile>profile>policy-options#
begin
policy-rule-unit "Re_dns"
    flow-description 1
        match
            protocol 17
            remote-port eq 53
        exit
    exit
exit
policy-rule-unit "redirect_portal"
    flow-description 1
        match
            remote-ip 15.1.1.30/32
        exit
    exit
exit
policy-rule-unit "redirect_all_http"
    flow-description 1
        match
            protocol 6
            remote-port eq 80
        exit
    exit
    flow-gate-status http-redirect "http://15.1.1.30:80/nokia.
html?mac=$MAC&ip=$IP&sub=$SUB&sap=$SAP&aca=wg1-
bi0514&orig_url=$URL&rid=$RID&cid=$CID"
exit

```

```

charging-rule-unit "CRU_RG1"
    rating-group 1
    charging-method online
exit
charging-rule-unit "CRU_RG50"
    rating-group 50
    charging-method online
exit
charging-rule-unit "CRU_RG9999"
    rating-group 9999
    charging-method online
exit
policy-rule "allow_dns" policy-rule-unit "Re_dns" charging-rule-
unit "CRU_RG9999" qci * arp * precedence 101
policy-rule "allow_portal" policy-rule-unit "redirect_portal" charging-
rule-unit "CRU_RG1" qci * arp * precedence 82
policy-rule "redirect_all_http" policy-rule-
unit "redirect_all_http" charging-rule
unit "CRU_RG50" qci * arp * precedence 100

policy-rule-base "redirectRuleBase"
    policy-rule "allow_dns"
    policy-rule "allow_portal"
    policy-rule "redirect_all_http"
exit
commit

```

For call flows with RADIUS accounting feed enabled, a similar policy rule base can be used. The redirect URL in this case does not need the enrichment parameters.

**NOTE:**

Policy rule bases on the TWAG should preferably be configured with wildcard QCI and ARP values.

The following example shows how the policy rule base configured for HTTP header enrichment is referenced under the APN for the TWAG session to be in redirect until the user is authenticated.

```

#config>mobile>pdn>apn#
    static-predefined-pcc
        default-policy-rule-base "redirectRuleBase"
    exit

```

9.9.6 RADIUS AVP configuration for RADIUS accounting feed call flows

RADIUS AVP options must be configured if the `twag-radius-acct-feed` is enabled for the APNs used by the SWw tunnel profile.

```
configure>mobile-gateway# profile
radius-group radius-group-name
...
radius-avp-options
  include-nas-port-id
  include-alc-subsc-id-str
  include-alc-subsc-prof-str
  include-alc-sla-prof-str
```

10 Traffic detection for SSG

Overview of the supported Traffic Detection Function (TDF) for the Subscriber Services Gateway (SSG).

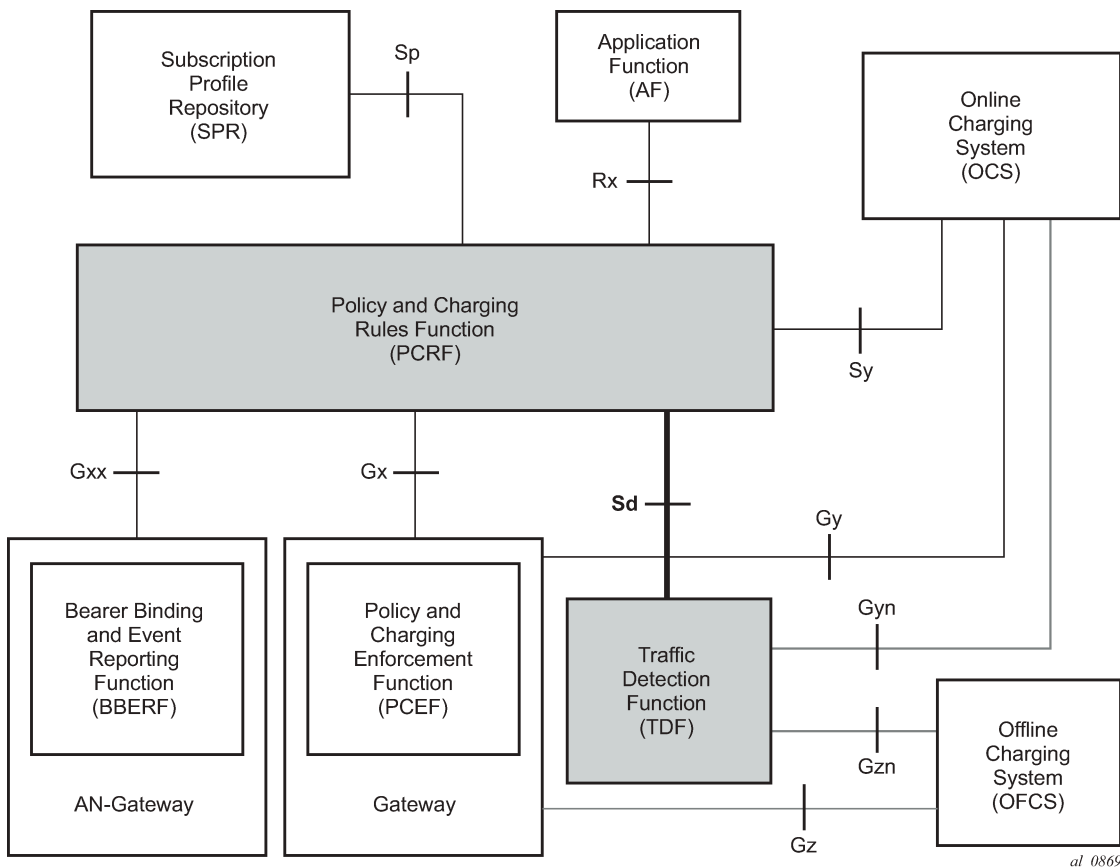
10.1 TDF for SSG implementation overview

Overview of the TDF for the SSG supported for the 7750 SR MG.

The SSG facilitates policy-based service steering including TDF functionality (as specified in 3GPP TS 29.212). The following interfaces are associated with the TDF:

- Sd interface to the PCRF
- Gyn interface to the OCS
- Gzn interface to the OFCS

Figure 147: TDF reference architecture with policy and charging interfaces



10.1.1 Features and functionality

The SSG implementation supports the following features and functionality for policy-based service steering including TDF:

- Sd interface between the PCRF and TDF as specified in 3GPP TS 29.212, including support for solicited mode of operation
- ABC (Application Based Charging) in the Supported Features AVP sent to the PCRF in the TSA
- upon TDF session establishment, the SSG builds the context, may associate a steering policy, and pins the session on an ISM-MG
- activation and deactivation of predefined ADC rules and rule bases
- optional service function chaining (SFC) steering function that can be enabled using gateway-level steering profiles and policies, where the usual hierarchy of SDF-level profiles override the gateway level
- event triggers over the Sd interface
- event report triggers over the Sd interface
- redirect information AVP within an ADC rule for redirecting UL traffic for an application
- PCRF-specified offline and online charging addresses over the Sd interface
- sending the TDF information AVP to the PCRF in a CCR-I message
- configurable option to ignore the received charging characteristics (in a TSR) and apply a default charging profile, and the ability to configure a default charging profile on a per-APN basis for the following:
 - visiting case (the subscriber belongs to a different PLMN)
 - roaming case (home-routed traffic)
 - home subscriber case (the subscriber belongs to the same PLMN as the TDF)
 - unconditionally (always ignored)
- charging information addition triggers
- partial record closure triggers
- ACR message AVPs
- online charging per TDF session and per application (Application Based Charging)
- CCR update to the OCS for session-level triggers including ULI change, CSG information change, change of UE presence in presence reporting area
- CCR message AVPs
- Credit Control Failure handling
- TDF-CDR for registering offline and online charged failures

10.1.2 Configuration requirements

The following configurations are required for TDF for the SSG:

- IOM-XP3-B/C
- ISM-MG redundancy

10.2 Sd interface

The functionality supported by the Sd interface.

10.2.1 Sd interface functions and features

The Sd reference point provides the interface between the PCRF and the SSG TDF.

The Sd interface supports the following functionality:

- establishment and termination of sessions between the PCRF and TDF
- enabling the PCRF to have dynamic control over application detection and control behavior at a TDF
- solicited and unsolicited application reporting
- delivery of IP-CAN session-specific parameters between the PCRF and TDF
- provisioning of ADC rules for the purpose of application detection, enforcement, and charging at the TDF
- start and stop notification of a detected application and transfer of flow descriptions from the TDF to PCRF

10.2.2 Messages and AVPs

The SSG TDF feature supports online charging functionality over the Diameter Gy/Ro interface and offline charging over the Diameter Rf interface.

The SSG indicates support for Application Based Charging (ABC) in the supported features AVP sent to the PCRF in the TSA. The base functionality for the Sd reference point is defined by 3GPP Release 11. Optional features are an extension to this functionality.

Related information

Supported messages and AVPs

The Diameter messages and AVPs supported over the Sd interface for the SSG TDF.

Appendix A: Messages and information element reference

SSG TDF configuration guidelines

The steps and commands required to configure SSG TDF.

10.2.3 UE attach with Gx and Sd message call flows

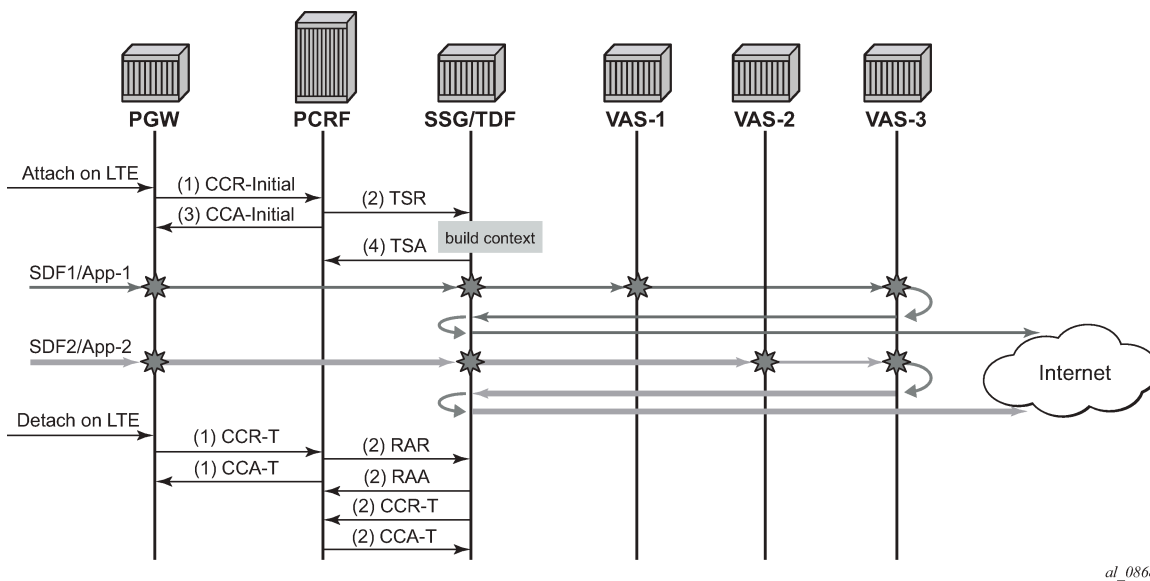
UE Attach with Gx and Sd message flows shows a basic call flow for a UE attach with Gx and Sd message flows.

In this call flow, upon establishing a TDF session, the SSG builds the context, associates a steering policy, and pins the session on an ISM-MG. A TDF-Session-Request message provides the following context parameters:

- Subscription-Id
- Framed-IP-Address
- SessCalled-Station-Id (APN and so on)

The Origin-Host message sent in the TSA indicates the Origin-Host of the ISM-MG to which the session is pinned and facilitates subsequent messages for the session being sent directly to the ISM-MG. The CPM distributes and load-balances the sessions on the ISM-MG cards.

Figure 148: UE Attach with Gx and Sd message flows



al_0868

10.2.4 ADC rules

The SSG supports activation and deactivation of predefined ADC rules and rule bases, and installation, modification, and removal of dynamic ADC rules.

Predefined rules are configured in the TDF and can be activated or deactivated by the PCRF referencing the rule name. Similarly, predefined rule bases are configured in the TDF and can be activated or deactivated by the PCRF referencing the rule base name. For overlapping rules, the rule with the highest presence is applied for enforcement, charging, and so on.

The following fields are applicable for a predefined ADC rule:

- flow description (AA charging group (TDF application identifier, IP-tuple))
- rating group
- service identifier
- reporting level
- charging method (online, offline, both, or none)
- metering method (time-based, volume-based, or both)
- monitory key
- direction
- flow gate status
- QoS uplink and downlink bit-rate
- mute notification

The SSG also supports installation, modification, and removal of dynamic ADC rules. When dynamic and predefined ADC rules have the same precedence, the dynamic ADC rule takes precedence.

An application's UL traffic can also be redirected using the Redirect Information AVP within an ADC rule. Redirection can be enabled or disabled using the Redirect-Support AVP. The PCRF-provided redirect address takes precedence over the locally configured address; the steering profiles apply to the redirected traffic.

10.2.5 Steering policy profiles

Steering policy profiles can be configured and applied at the APN or SD levels.

Related information

Traffic steering function

10.2.6 Event triggers and rate limiting

The event triggers, rate limiting, and usage monitoring features that are supported over the Sd interface.

10.2.6.1 Event triggers

The SSG supports the event triggers over the Sd interface. The PCRF can subscribe to these triggers and the TDF reports them to the PCRF when they occur.

The following event triggers are supported:

- NO_EVENT_TRIGGERS
- OUT_OF_CREDIT

- REALLOCATION_OF_CREDIT
- REVALIDATION_TIMEOUT
- USAGE_REPORT
- APPLICATION_START
- APPLICATION_STOP
- CREDIT_MANAGEMENT_SESSION_FAILURE

10.2.6.2 Event report indication procedures

The SSG supports event report indication procedures over the Sd interface. The TDF subscribes to these and the PCRF notifies the TDF when the PPGW tells the PCRF that these events have occurred.

Unlike the PGW, the TDF has no direct GTP connectivity to the packet core and therefore notification of these events is received indirectly via the PCRF (upon TDF subscription):

- SGSN_CHANGE
- RAT_CHANGE
- PLMN_CHANGE
- IP_CAN_CHANGE
- RAI_CHANGE
- ULI_CHANGE
- NO_EVENT_TRIGGERS
- AN_GW_CHANGE
- UE_TIMEZONE_CHANGE
- TAI_CHANGE
- ECGI_CHANGE
- USER_CSG_INFO_CHANGE
- USER_CSG_HYBRID_SUB_INFO_CHANGE
- USER_CSG_HYBRID_UNSUB_INFO_CHANGE
- CHANGE_OF_UE_PRESENCE_IN_PAR

10.2.6.3 Time of Day event triggers

Time of Day (ToD) procedures are supported over the Sd interface. This includes support for the following event triggers, which can also facilitate ToD related steering policy use cases:

- REVALIDATION_TIMEOUT

The PCRF subscribes to this event trigger in a TSR, CCA, or RAR message, along with revalidation time. Upon timer expiration, the TDF sends a CCR-U to request ADC rules. This can be disabled by the REVALIDATION_TIMEOUT Event Trigger.

- Rule-Activation-Time and Rule-Deactivation-Time

Part of an ADC Rule Install and apply to all rules and rule bases within that ADC Rule Install.

10.2.6.4 Rate limiting

The SSG supports rate limiting of bandwidth for sessions and applications, using the QoS information AVP as listed below.

- The maximum requested bandwidth UL and DL options within the QoS information AVP for the command level are used to configure rate limiting at the session level.
- The maximum requested bandwidth UL and DL options within the QoS information AVP for the ADC rule are used to specify rate limiting at the application level.

10.3 Gx interface

The functionality supported by the Gx interface.

10.3.1 Gx interface in unsolicited mode

The Gx interface in unsolicited mode is also supported on the SSG and can be used for creating a subscriber context for a mobile subscriber.

To use the Gx unsolicited mode, all user traffic must be routed via the SSG using routing support on the platform. Once user traffic transits via the SSG node, a data trigger is initiated with the source IP address that was previously unseen. The data trigger initiates a Gx message toward the PCRF with the IP address of the subscriber as a key to lookup subscriber context. A response from the PCRF provides the SSG with the subscriber information, such as IMSI, APN, and PCC rules. The SSG uses the response from the PCRF to create a subscriber context and applicable policies.

Figure 149: User traffic routing via SSG on Gx interface

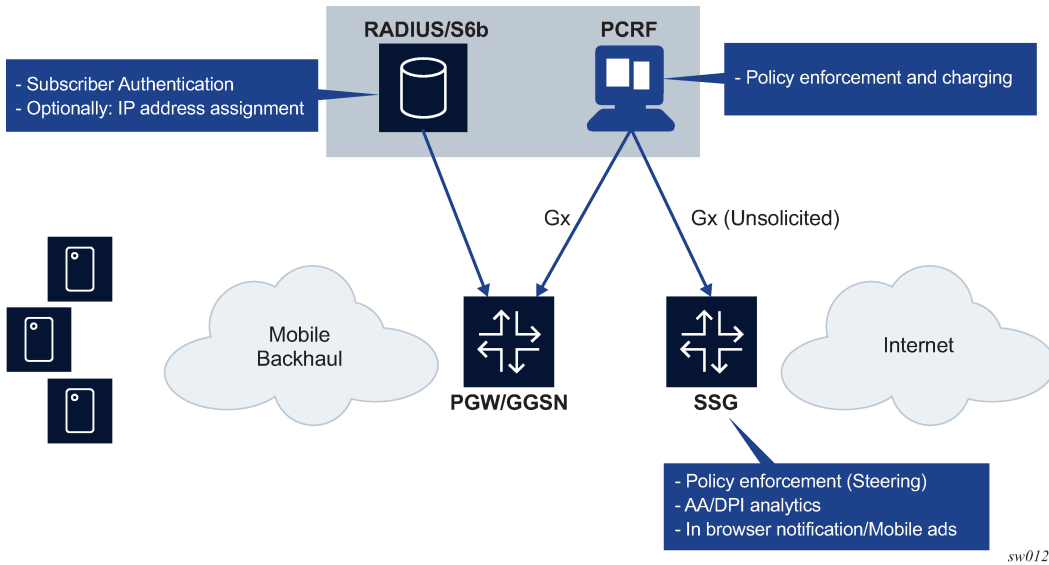
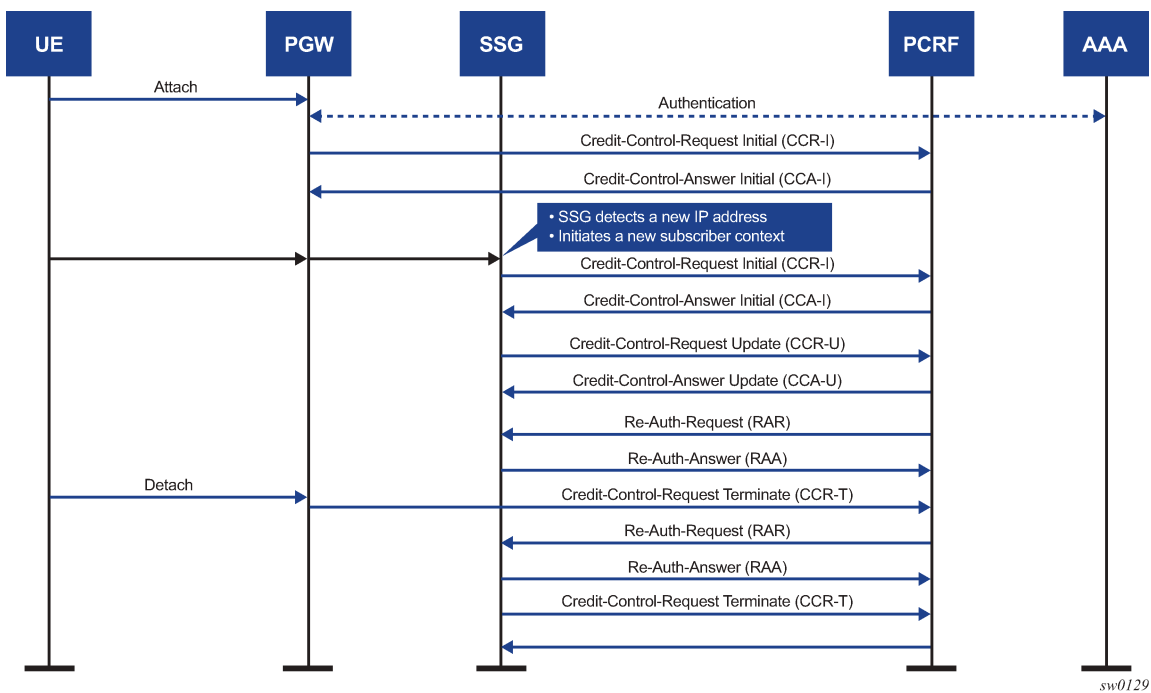


Figure 150: Subscriber Context Creation via SSG and Gx interface call flow



10.4 RADIUS support for fixed subscribers

In addition to SSG support for mobile subscribers, support for fixed subscribers is also available.

Subscriber context creation for a fixed UE is supported using the RADIUS interface. All user traffic is routed via the SSG using routing support. Once the user traffic transits via the SSG node, a data trigger is initiated with the source IP address that was previously unseen. A data trigger initiates a RADIUS message toward the AAA with the IP address of the subscriber as a key to lookup the subscriber context. A response from

the AAA in the form of a RADIUS CoA message provides the SSG with the subscriber information, such as the subscriber ID stored in the Subscriber Profile Repository (SPR), PCC rules, and so on. The SSG uses a response from the AAA to create a subscriber context and applicable policies.

Figure 151: Fixed subscriber context creation via RADIUS

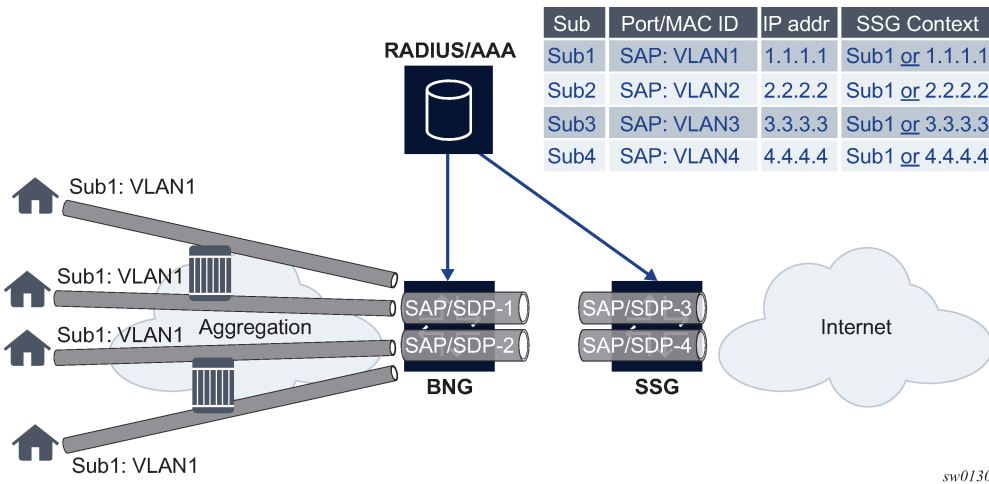
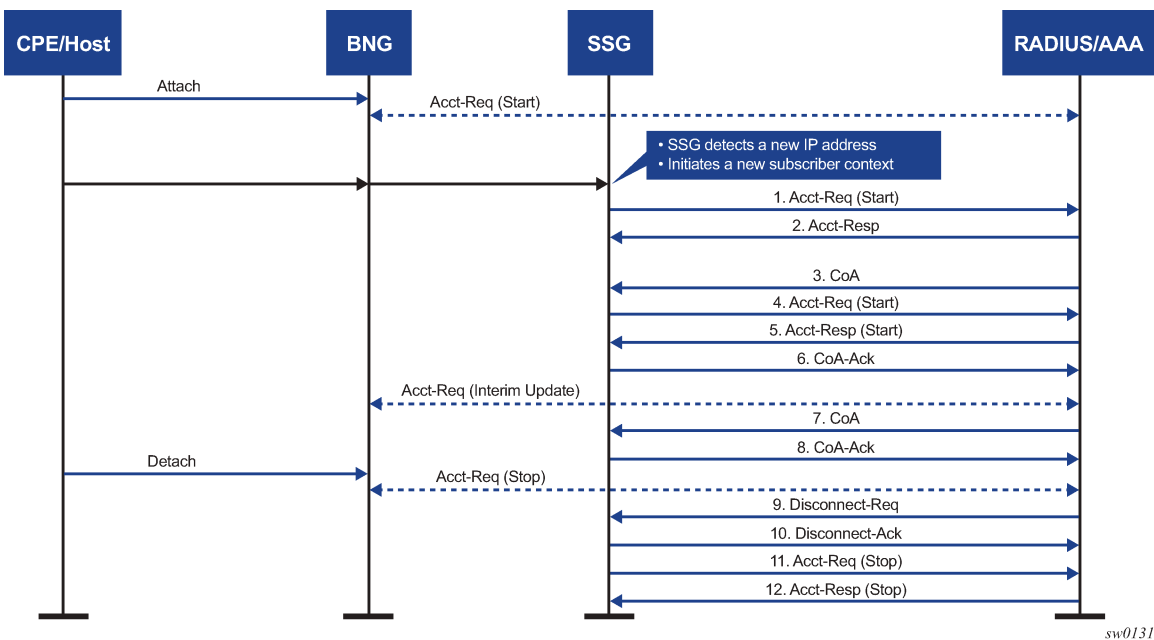


Figure 152: Fixed subscriber context creation via SSG and RADIUS interface call flow



10.5 Charging support

The SSG supports PCRF-specified default charging methods, using offline and online AVPs at the command level. A PCRF-specified default charging method overrides locally configured charging methods. The default charging method also applies to every ADC rule (instantiated for the session) where another charging method is omitted. The charging characteristics received in a TSR are used to select a charging profile to

apply for the session. PCRF-specified offline and online charging addresses are also supported over the Sd interface. PCRF-specified addresses take precedence over locally configured addresses.

The TDF Information AVP includes the TDF-Destination-REALM and TDF-Destination-Host AVPs, or the TDF-IP-Address AVP. If the TDF Information AVP is pre-provisioned in the PCRF and the value is also received in a CC-Request from the PCEF, the value received in the CC-Request takes precedence over the pre-provisioned value.

Related information

Messages and AVPs

The SSG TDF feature supports online charging functionality over the Diameter Gy/Ro interface and offline charging over the Diameter Rf interface.

Appendix A: Messages and information element reference

10.5.1 Offline charging

The SSG TDF functionality supports the Gzn interface to the OFCS. Diameter Rf and GTP'-based Ga are supported.

OFCS charging is supported at a flow-based charging (FBC) level. For offline charging over a Gzn interface, Application Based Charging is supported. Charging information for application traffic is categorized by rating group or a combination of rating group and service ID.

A configurable option is supported to ignore received charging characteristics (in a TSR) and apply a default charging profile in the following cases. The default charging profile for these cases is configurable on a per-APN basis.

- Visiting case (subscriber belongs to a different PLMN)
- Roaming case (home-routed traffic)
- Home Subscriber case (subscriber belongs to the same PLMN as the TDF)
- Unconditional (always ignored)

The following additional charging information triggers are supported for the SSG TDF and are applicable to the Ga and Rf interfaces:

- serving node changes (SGSN/AGW)
- user location change
- user CSG information change
- change of UE presence in PRA
- tariff time change
- application time limit expiry
- application volume limit expiry
- application unit limit expiry

- termination of application traffic

The following partial record closure triggers are supported for the SSG and are applicable to the Ga and Rf interfaces:

- volume limit (data)
- time limit (duration)
- MS time zone change
- PLMN change
- RAT change
- maximum number of charging condition changes
- management intervention

10.5.2 Online charging

The SSG TDF functionality supports the Gyn interface to the OCS. For online charging over the Gyn interface, Application Based Charging is supported. Online charging is supported at the rating-group level. Charging information for application traffic is categorized by rating group.

For each TDF session subject to online charging, the TDF initiates one credit management session toward the OCS. Credit control for online charging is on a per-rating-group basis. Independent credit control for individual services or applications is achieved by assigning a unique rating group in the corresponding ADC rule.

Volume-based and time-based charging models are supported. The TDF supports the following rating-group level trigger conditions:

- threshold reached
- QHT (Quota Holding Time) expiration
- quota exhaustion
- validity time reached
- forced reauthorization (RAR from OCS)

The TDF supports the following session-level trigger conditions:

- ULI change
- CSG information change
- tariff time change

10.5.3 Credit Control Failure Handling

The SSG supports the following actions for CCFH:

Terminate

This is the default behavior. The TDF indicates to the PCRF that the charging session was terminated, and the OCS does not allow the service to continue. The PCRF determines whether to continue or terminate the associated TDF session.

Retry and terminate

If the OCS and TDF support failover procedures and there is an alternate OCS available, the TDF attempts a failover to an alternative OCS. Otherwise, the TDF indicates to the PCRF that the charging session was terminated and the OCS does not allow the service to continue. The PCRF determines whether to continue or terminate the associated TDF session.

Continue

If the OCS and TDF support failover procedures and there is an alternate OCS available, the TDF will attempt a failover to an alternative OCS. Otherwise, the TDF indicates to the PCRF that the charging session was terminated and the OCS allows the service to continue. The PCRF determines whether to continue or terminate the associated TDF session.

If the configured session-restore retry timer (`session-restore-retry-timer` command) expires and the OCS does not respond to the CCR sent by the gateway, then the session continues (as a result of the CCFH=Continue that the OCS provided previously) until the failure-handling session timer (`fh-session-continue timer` command) expires.

If a TDF session is both offline and online charged, the SSG registers the failure situation in the TDF-CDR. If the session is only online charged, the execution of the CCFH continue action results in a TDF-CDR being opened.

10.5.4 TDF-CDR

The SSG supports generating a TDF-CDR for Application Based Charging. TDF-CDRs are supported as per 3GPP 32.251 / 32.298. Diameter Rf is supported as per 3GPP 32.251 / 32.299.

For a description of the TDF-CDR formats, refer to the *CMG CDR Description*.

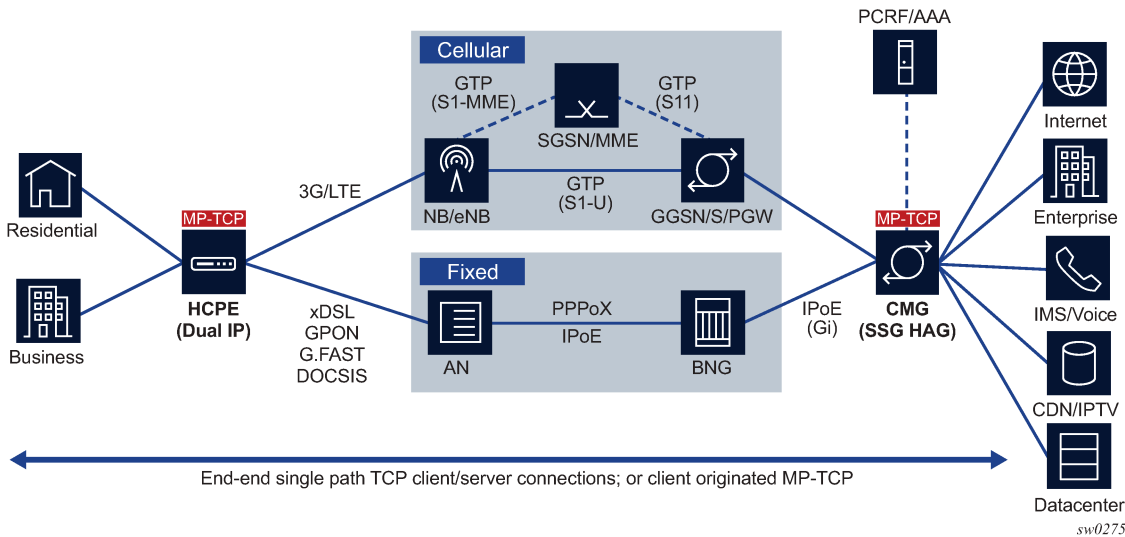
10.6 SSG deployed as MP-TCP HAG

The SSG Multi-Path TCP (MP-TCP) HAG solution allows MP-TCP subscribers to use a TCP connection over two simultaneous access paths (fixed and mobile).

The HAG functionality is implemented in Application Assurance (AA) while the CMG diverts TCP traffic to AA. Diverting TCP traffic to AA can be done using policy rules. This provides subscribers with increased bandwidth (by multiplexing two transport networks) and coverage because the end-to-end session remains active as long as any access path is available.

AA MP-TCP HAG deployment shows an AA MP-TCP hybrid access gateway deployment. For more information about AA support of MP-TCP HAG, refer to the AA MP-TCP HAG section in the *CMG Application Assurance Configuration Guide*.

Figure 153: AA MP-TCP HAG deployment



To configure subscriber access over fixed and mobile paths, for the specified RADIUS or Gx interface, use the following command:

```
configure service vprn ip-local-pool transit-sub
```

To configure a subscriber mapping profile and address, use the following command:

```
configure mobile-gateway profile subscriber-mapping address
```

To configure a subscriber mapping ID and TCP-IP profile for an APN, use the following commands:

```
configure mobile-gateway pdn apn subscriber-mapping-profile
```

The KPI/KCIs for this feature are implemented in AA. CMG does not implement KPI/KCIs for this feature.



NOTE:

- The Gx and Sd interfaces are not supported for retrieving the subscriber's context.
- Only IPv4 or IPv6 addresses can be mapped in the subscriber mapping table. Combination of IPv4 with IPv6 address type is not supported.

10.6.1 Subscriber context retrieval based on RADIUS CoA

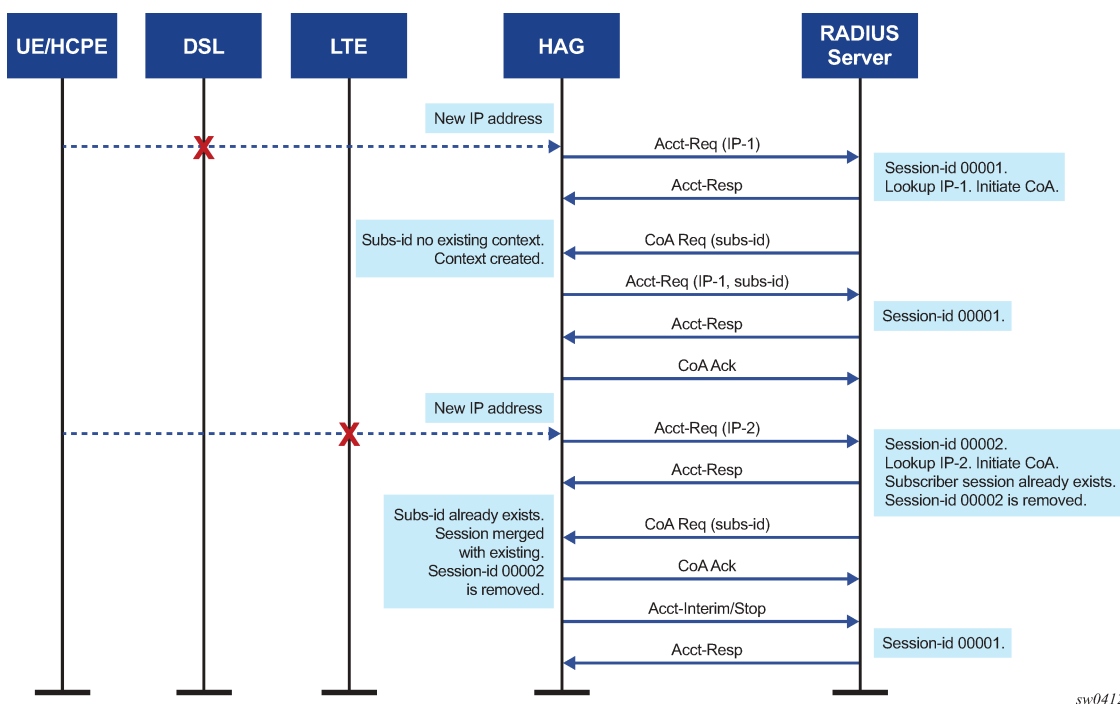
One of the options available for the SSG HAG to retrieve subscriber information is based on RADIUS.

With this approach, when user traffic is detected from a source IP address (DSL or LTE) that does not have associated subscriber context, the SSG HAG triggers a procedure to retrieve the subscriber context (such as subscriber identifier, policies).

The SSG HAG sends an Accounting Request message. The RADIUS server responds and sends a Change-of-Authorization message (CoA) to transfer the subscriber information to the SSG HAG.

The same procedure is triggered for both access paths. When the subscriber ID provided by RADIUS matches, then the LTE and DSL paths are merged. RADIUS accounting is performed for the merged session.

Figure 154: SSG HAG over RADIUS



sw0412

10.6.2 MP-TCP HAG load balancer behavior in CMG

Traffic distribution across different MG-VMs is based on micronets (/24 IPv4 or /56 IPv6) which are provisioned by Load Balancers (LBs).

The SSG/HAG allows configuration of IP pools which are split into micronets by the OAM-VM. When the Load Balancer VM (LB-VM) detects traffic from an unassigned micronet, the OAM-VM will assign it to the MG-VM that is less loaded.

The MP-TCP HAG receives two separate IP flows, one for the DSL and one for the LTE access path. These two flows must be directed to the same MG-VM so that they are merged. Because the IP addresses or ranges corresponding to these flows are independent, the micronet scheme is modified to enable the MP-TCP functionality.

The following scenario for an MP-TCP subscriber sending traffic over 2 access paths describes this behavior.

The MP-TCP subscriber with two IP addresses, IP_{DSL} and IP_{LTE} , starts sending traffic over the DSL access path. The LBs in the HAG receive the IP_{DSL} packet and check if there is any micronet forwarding rule installed. If there is not any rule installed, the OAM-VM assigns the new micronet to an MG-VM. When traffic is redirected to this specific MG-VM, the IP_{DSL} associated with it is unknown to the MG-VM and the subscriber ID information is retrieved through the RADIUS CoA message.

When the HCPE function element sets up a second TCP flow over LTE, the HAG receives packets that have an IP_{LTE} assigned. Another MG-VM is assigned to this IP_{LTE} flow based on the same micronet forwarding rules. As a result, the two IP_{DSL} and IP_{LTE} flows can be redirected to different MG-VMs.

But, the subscriber ID information that is retrieved through the RADIUS CoA message for the IP_{LTE} flow is the same with the subscriber ID information associated with the IP_{DSL} flow. So, the new session ID created for the IP_{LTE} flow is deleted and the two flows are combined in the same session where the LTE flow is reassigned to the MG-VM of the DSL flow.

The LBs are then updated with a /32 forwarding rule for the IP_{LTE} .



NOTE:

All the LBs in a CMG appliance are active. The entire UE-to-MG-VM mapping table is forwarded to all the LB-VMs within an appliance. This ensures that upon failure of a LB the remaining LBs can take over the traffic and redirect it to the correct MG-VM instance. No session state is lost on the MG-VMs, thus ensuring service continuity. On the network side, traffic redirection to a new LB relies on the existing routing protocols (for example, BGP and ECMP).

Related information

[Cloud Mobile Gateway \(CMG\)](#)

The configuration guidelines specific to the Cloud Mobile Gateway (CMG). These guidelines are intended to be used after installation is completed as described in the CMG Installation Guide.

10.6.3 RADIUS-less HAG subscriber auto-create and DSL SPR RADIUS

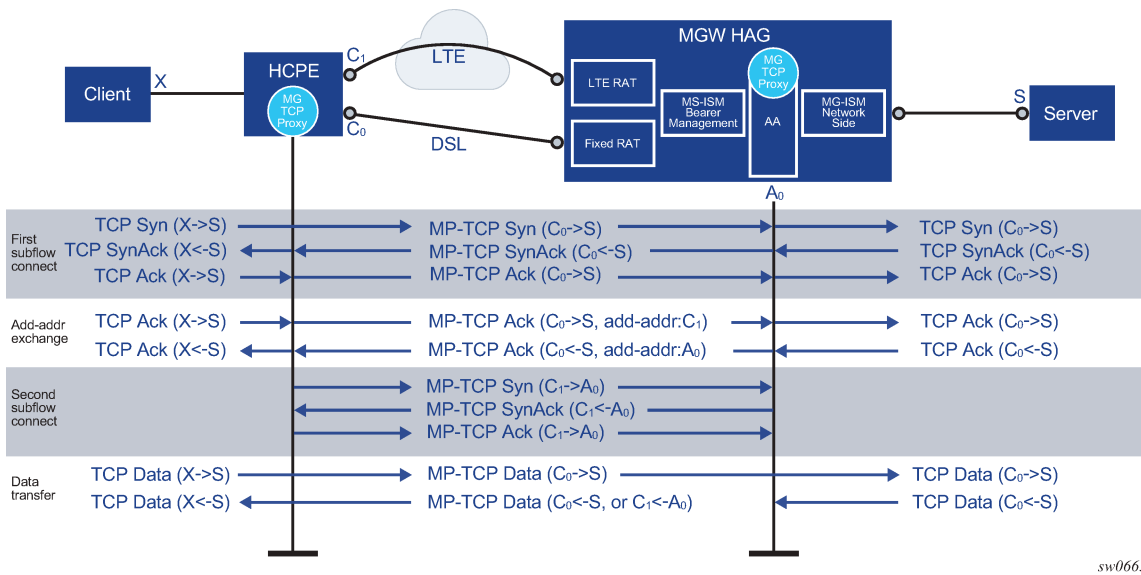
For the converged Subscriber Policy Repository (SPR) model, you must implement a common policy domain (converged SPR AAA) with the subscriber IP address and policy state aligned between both the fixed (DSL) and mobile (LTE) networks.

Service providers usually manage fixed and mobile networks independently and want to avoid implementing a common policy domain only for managing the MP-TCP services.

The RADIUS-less auto-create model supports auto-creation of the subscriber context including automatic creation of the subscriber ID based on the primary path IP address, and auto-learning of the secondary path IP address from the MP-TCP ADD_ADDR option (defined in RFC 6824). This deployment model is useful where differentiated sub-policy and charging is not required. The HAG auto-creates the primary path subscriber ID and relies on HCPE provided MP-TCP options to determine the IP address for the secondary path.

An alternate deployment model uses the SPR RADIUS for the DSL path and the MP-TCP provided ADD_ADDR option to create the secondary LTE path.

Figure 155: MP-TCP signaling

**NOTE:**

The diagram use of ADD_ADDR is only for reference. RFC allows to send ADD_ADDR in any message. Due to the length of ADD_ADDR it is usually sent in a separate ACK as described in RFC 6824. However the exchange of ADD_ADDR in both directions is a requirement for the successful setup of the primary path.

In the RADIUS-less HAG auto-create model, a new mobile gateway subscriber context is auto-created when the first packet is sent from an HCPE IP address C₀ for which the mobile gateway has no existing subscriber state. The second HCPE IP address C₁ is added to the same mobile gateway subscriber context when the ADD_ADDR for C₁ is received by the HAG. The subscriber context is removed after inactivity timeout from both paths.

In the DSL SPR RADIUS model, when the first packet is sent from an HCPE IP address C₀ for which the mobile gateway has no existing subscriber state, the SPR is contacted by a mobile gateway request for subscriber context for this DSL IP address. The SPR responds with a subscriber identifier and an optional subscriber policy. The second HCPE IP address C₁ is automatically added to the same mobile gateway subscriber context when the ADD-ADDR for C₁ is received by the HAG. The subscriber context is removed after inactivity timeout from both paths, or when instructed by the SPR.

Related information**MP-TCP HAG load balancer behavior in CMG**

Traffic distribution across different MG-VMs is based on micronets (/24 IPv4 or /56 IPv6) which are provisioned by Load Balancers (LBs).

10.6.3.1 RADIUS-less HAG auto-create model configuration

For the RADIUS-less HAG auto-create model, the transit-sub protocol must be configured as **auto-create** for the DSL IP pool and as **mptcp** for the LTE IP pool.

For the DSL SPR RADIUS model, the transit-sub protocol for the DSL IP pool must be configured as **radius**, and as **mptcp** for the LTE IP pool.

For more information about configuring the MP-TCP HAG, refer to the AA MP-TCP HAG section in the *7450 ESS, 7750 SR, and VSR Multiservice Integrated Service Adapter and Extended Services Appliance Guide*.

```
config>service>vprn>ip-local-pool
ip-local-pool "mptcp-primary"
                                remote
                                next-hop-gw-ipv4 "31.1.1.2"
                                transit-sub fixed protocol auto-create
                                ipv4-prefix "10.0.0.0/16"
                                exit
ip-local-pool "mptcp-secondary"
                                remote
                                next-hop-gw-ipv4 "41.1.1.2"
                                transit-sub mobile protocol mptcp
                                ipv4-prefix "20.0.0.0/16"
                                exit
```

Enabling of ip-bonding-assist on the AA side.

```
config>app-assure>group>mp-tcp-prxy># info
-----
bonding-assist
-----
```

10.7 Auto-created subscribers for TCPO and AA features

Subscribers can be created automatically based on unique IP addresses per PDN sessions.

Analytics for inline applications such as TCPO and application rate limiting do not require per-subscriber profile. Instead a common policy can be applied across all subscriber sessions. Because per subscriber policies are not required, the overhead of subscriber creation using data triggered (Gx or RADIUS) or via Sd interface can be avoided by automatically creating subscribers.

To automatically create subscriber contexts, use the **auto-create** protocol option of the transit-sub command.

10.8 Supported messages and AVPs

The Diameter messages and AVPs supported over the Sd interface for the SSG TDF.

Related information

[Appendix A: Messages and information element reference](#)

10.8.1 TDF information AVPs over the Gx interface

The SSG supports sending TDF Information AVPs to the PCRF in a CCR-I message.

The TDF Information AVP is either received over the Gx interface within the initial CC-Request received from the PCEF or pre-provisioned at the PCRF. The TDF Information AVP includes either the TDF-Destination-Realm and TDF-Destination-Host AVPs or the TDF-IP-Address AVP. If the TDF Information AVP is pre-provisioned in the PCRF and the value is also received in a CC-Request from the PCEF, the value received in the CC-Request takes precedence over the pre-provisioned value.

10.8.2 Diameter messages over the Sd interface

The SSG TDF feature supports online charging functionality over the Diameter Gy/Ro interface and offline charging over the Diameter Rf interface.

Diameter messages supported over the Sd interface describes the types of Diameter messages that are supported over the Sd interface. For solicited mode, the PCRF initiates the Sd session with a TDF-Session-Request message and the SSG responds with a TDF-Session-Answer message.

For solicited mode, only CC-Request (CCR) Update and Terminate messages are valid.

Table 33: Diameter messages supported over the Sd interface

Message	From/To	Description
TDF-Session-Request	PCRF to TDF	Establishes a TDF session
TDF-Session-Answer	TDF to PCRF	Response to TSR
CCR-Update	TDF to PCRF	For solicited mode, only CCR-Update and CCR-Terminate are applicable
CCA-Update	PCRF to TDF	—
CCR-Terminate	TDF to PCRF	For solicited mode, only CCR-Update and CCR-Terminate are applicable

Table 33: Diameter messages supported over the Sd interface (continued)

Message	From/To	Description
CCA-Terminate	PCRF to TDF	—
RAR	PCRF to TDF	PCRF can trigger session termination using Session-Release-Cause in RAR
RAA	TDF to PCRF	—
CCR-Initial	TDF to PCRF	Unsolicited mode only

Related information

SSG TDF configuration guidelines

The steps and commands required to configure SSG TDF.

Appendix A: Messages and information element reference

10.8.3 Supported Features AVP for application-based charging

The SSG indicates support for Application Based Charging (ABC) in the Supported Features AVP sent to the PCRF in the TSA.

The base functionality for the Sd reference point is defined by 3GPP Release 11. Any optional feature is an extension to this functionality. When optional features are included, the "M" bit is cleared for the Supported Features AVP.

10.8.3.1 TDF message command descriptions

The TDF-Session-Request (TSR) command, indicated by the Command-Code field set to 8388637 and the "R" bit set in the Command Flags field, is sent by the PCRF to the TDF to establish the TDF session and to provision the ADC rules. It may also include the requested event triggers.

The following shows the supported message format:

```
<TS-Request> ::= < Diameter Header: 8388637, REQ, PXY >
    < Session-Id >
    { Vendor-Specific-Application-Id }
    { Origin-Host }
    { Origin-Realm }
    { Destination-Realm }
    [ Destination-Host ]
    [ Origin-State-Id ]
    [ OC-Supported-Features ]
```



```

    *[ Subscription-Id ]
    *[ Supported-Features ]
    [ Framed-IP-Address ]
    [ Framed-Ipv6-Prefix ]
    [ IP-CAN-Type ]
    [ RAT-Type ]
    [ AN-Trusted ]
    [ User-Equipment-Info ]
    [ QoS-Information ]
0*2[ AN-GW-Address ]
    [ 3GPP-SGSN-Address ]
    [ 3GPP-SGSN-Ipv6-Address ]
    [ 3GPP-GGSN-Address ]
    [ 3GPP-GGSN-Ipv6-Address ]
        [ 3GPP-Selection-Mode ]
    [ Dynamic-Address-Flag ]
    [ Dynamic-Address-Flag-Extension ]
    [ PDN-Connection-Charging-ID ]
    [ 3GPP-SGSN-MCC-MNC ]
    [ RAI ]
    [ 3GPP-User-Location-Info ]
    [ Fixed-User-Location-Info ]
    [ User-CSG-Information ]
    [ TWAN-Identifier ]
    [ 3GPP-MS-TimeZone ]
    [ 3GPP-Charging-Characteristics ]
    [ Called-Station-Id ]
    [ Charging-Information ]
    [ Online ]
    [ Offline ]
    *[ ADC-Rule-Install ]
    [ Revalidation-Time ]
    *[ Usage-Monitoring-Information ]
    *[ CSG-Information-Reporting ]
    *[ Event-Trigger ]
    [ Presence-Reporting-Area-Information ]
    [ Logical-Access-ID ]
[ Physical-Access-ID ]
[ 3GPP2-BSID ]
    *[ Proxy-Info ]
    *[ Route-Record ]
    *[ AVP ]

```

The TDF-Session-Answer (TSA) command, indicated by the Command-Code field set to 8388637 and the "R" bit cleared in the Command Flags field, is sent by the TDF to the PCRF in response to the TS-Request command.

The following shows the supported message format:

```
<TS-Answer> ::= < Diameter Header: 8388637, PXY >
    < Session-Id >
    { Vendor-Specific-Application-Id }
    { Origin-Host }
    { Origin-Realm }
    [ Result-Code ]
    [ Experimental-Result ]
    [ Origin-State-Id ]
    [ OC-Supported-Features ]
    [ OC-OLR ]
    *[ Supported-Features ]
    *[ ADC-Rule-Report ]
    [ Event-Report-Indication ]
    [ Error-Message ]
    [ Error-Reporting-Host ]
    *[ Failed-AVP ]
    *[ Proxy-Info ]
    *[ Route-Record ]
    *[ AVP ]
```

The CC-Request (CCR) command, indicated by the Command-Code field set to 272 and the "R" bit set in the Command Flags field, is sent by the TDF to the PCRF to request ADC rules or to inform the PCRF about the application detection. It is also sent to the PCRF when the TDF session terminates, following the receipt of the corresponding RAR command from the PCRF.

The following shows the supported message format:

```
<CC-Request> ::= < Diameter Header: 272, REQ, PXY >
    < Session-Id >
    { Auth-Application-Id }
    { Origin-Host }
    { Origin-Realm }
    { Destination-Realm }
    [ OC-Supported-Features ]
    { CC-Request-Type }
    { CC-Request-Number }
    [ Credit-Management-Status ]
    [ Destination-Host ]
    [ Origin-State-Id ]
    [ Framed-IP-Address ]
    [ Framed-Ipv6-Prefix ]
```

```

    * [ ADC-Rule-Report ]
    * [ Application-Detection-Information ]
    * [ Event-Trigger ]
    [ Event-Report-Indication ]
    * [ Usage-Monitoring-Information ]
    * [ Proxy-Info ]
    * [ Route-Record ]
    * [ Supported-Features ]
    * [ AVP ]

```

The CC-Answer (CCA) command, indicated by the Command-Code field set to 272 and the "R" bit cleared in the Command Flags field, is sent by the PCRF to the TDF in response to the CCR command. It is used to provision ADC rules and event triggers for the TDF session and to acknowledge the report of the application traffic start and stop.

The following shows the supported message format:

```

<CC-Answer> ::= < Diameter Header: 272, PXY >
    < Session-Id >
    { Auth-Application-Id }
    { Origin-Host }
    { Origin-Realm }
    [ Result-Code ]
    [ Experimental-Result ]
    { CC-Request-Type }
    { CC-Request-Number }
    [ OC-Supported-Features ]
    [ OC-OLR ]
    [ Framed-Ipv6-Prefix ]
    [ QoS-Information ]
    * [ Event-Trigger ]
    [ Event-Report-Indication ]
    [ Origin-State-Id ]
    * [ Redirect-Host ]
    [ Redirect-Host-Usage ]
    [ Redirect-Max-Cache-Time ]
    * [ ADC-Rule-Remove ]
    * [ ADC-Rule-Install ]
    [ Revalidation-Time ]
    * [ Usage-Monitoring-Information ]
    [ Error-Message ]
    [ Error-Reporting-Host ]
    * [ Failed-AVP ]
    * [ Proxy-Info ]
    * [ Route-Record ]
    * [ Supported-Features ]

```

```
*[ AVP ]
```

The Re-Auth-Request (RAR) command, indicated by the Command-Code field set to 258 and the "R" bit set in the Command Flags field, is sent by the PCRF to the TDF to provision ADC rules using the PUSH procedure for solicited application reporting. It is also used to provision event triggers, to report event report indications for the TDF session for solicited application reporting, and to request the TDF session termination for both solicited and unsolicited application reporting.

The following shows the supported message format:

```
<RA-Request> ::= < Diameter Header: 258, REQ, PXY >
    < Session-Id >
    { Auth-Application-Id }
    { Origin-Host }
    { Origin-Realm }
    { Destination-Realm }
    { Destination-Host }
    { Re-Auth-Request-Type }
    [ Session-Release-Cause ]
    [ Origin-State-Id ]
    [ OC-Supported-Features ]
    [ QoS-Information ]
    *[ Event-Trigger ]
    [ Event-Report-Indication ]
    *[ ADC-Rule-Remove ]
    *[ ADC-Rule-Install ]
    [ Revalidation-Time ]
    *[ Usage-Monitoring-Information ]
    *[ Proxy-Info ]
    *[ Route-Record ]
    *[ AVP]
```

The Re-Auth-Answer (RAA) command, indicated by the Command-Code field set to 258 and the "R" bit cleared in the Command Flags field, is sent by the TDF to the PCRF in response to the RAR command.

The following shows the supported message format:

```
<RA-Answer> ::= < Diameter Header: 258, PXY >
    < Session-Id >
    { Origin-Host }
    { Origin-Realm }
    [ Result-Code ]
    [ Experimental-Result ]
    [ Origin-State-Id ]
    [ OC-Supported-Features ]
    [ OC-OLR ]
    *[ ADC-Rule-Report ]
```

```
[ Event-Report-Indication ]
[ Error-Message ]
[ Error-Reporting-Host ]
*[ Failed-AVP ]
*[ Proxy-Info ]
*[ AVP ]
```

10.8.3.2 Supported AVPs for the Sd interface

The Sd-specific AVPs and rule formats supported by the SSG TDF.

Table 34: SSG TDF Sd AVPs

Attribute name	AVP code	Clause defined	Value type ¹	AVP flag rules ²				May encr	Acc. type
				Must	May	Should Not	Must Not		
ADC-Rule-Base-Name	1095	5b.3.4	UTF8String	M,V	P	–	–	Y	All
ADC-Rule-Definition	1094	5b.3.3	Grouped	M,V	P	–	–	Y	All
ADC-Rule-Install	1092	5b.3.1	Grouped	M,V	P	–	–	Y	All
ADC-Rule-Name	1096	5b.3.5	OctetString	M,V	P	–	–	Y	All
ADC-Rule-Remove	1093	5b.3.2	Grouped	M,V	P	–	–	Y	All
ADC-Rule-Report	1097	5b.3.6	Grouped	M,V	P	–	–	Y	All

Notes:

1. The value types are defined in RFC 6733.
2. The AVP header bit denoted as “M” indicates whether support of the AVP is required. The AVP header bit denoted as “V” indicates whether the optional Vendor-ID field is present in the AVP header. The

AVP header bit denoted as “P” indicates the need for encryption for end-to-end security. For further details, see the RFC 6733.

The following shows the supported format for the ADC-Rule-Install AVP.

```
< AVP Header: 1092 >
*[ ADC-Rule-Definition ]
*[ ADC-Rule-Name ]
*[ ADC-Rule-Base-Name ]
[ Rule-Activation-Time ]
[ Rule-Deactivation-Time ]
*[ AVP ]
```

The following shows the supported format for the ADC-Rule-Definition AVP.

```
< AVP Header: 1094 >
{ ADC-Rule-Name }
[ TDF-Application-Identifier ]
[ Service-Identifier ]
[ Rating-Group ]
[ Reporting-Level ]
[ Online ]
[ Offline ]
[ Metering-Method ]
[ Precedence ]
[ Flow-Status ]
[ QoS-Information ]
[ Monitoring-Key ]
[ Redirect-Information ]
[ Mute-Notification ]
*[ AVP ]
```

The following shows the supported format for the ADC-Rule-Remove AVP

```
< AVP Header: 1093 >
*[ ADC-Rule-Name ]
*[ ADC-Rule-Base-Name ]
*[ AVP ]
```

The following shows the supported format for the ADC-Rule-Report AVP.

```
< AVP Header: 1097 >
*[ ADC-Rule-Name ]
*[ ADC-Rule-Base-Name ]
[ PCC-Rule-Status ]
[ Rule-Failure-Code ]
```

```
[ Final-Unit-Indication ]
*[ AVP ]
```

10.9 SSG TDF configuration guidelines

The steps and commands required to configure SSG TDF.

10.9.1 Sd interface configuration for the PDN

The Sd interface must be configured for the PDN. Parameters to be configured include the primary and secondary Diameter interface, the peer group, retry count, and transaction timer as shown in the following CLI syntax.

```
configure>mobile-gateway# pdn
sd
    diameter-redirect
    interface
        diameter [router router-instance] interface-name
        multi-homing
            diameter [router router-instance] secondary-peer-group peer-group-id
            primary-pcrf profile-name [secondary-pcrf profile-name]
            retry-count value
            transaction-timer value
```

Example: Sd interface configuration on the PDN

```
#config>mobile>pdn>sd# info
-----
        interface
            diameter router "vprn20002" "sd-lo"
        exit
        peer-group 1 primary-pcrf "temp-gx-for-sd" secondary-pcrf "temp-gx-for-sd-secondary"
-----
```

10.9.2 Diameter profile configuration for the Sd reference point

The Diameter peer specifies peer information for Diameter applications. For the TDF Sd reference point, the Diameter peer application type is configured as Sd. The router interface and peer information must also be specified.

Use the following CLI syntax to configure the parameters for a Diameter peer profile.

```
configure>mobile-gateway# profile
diameter-peer
  application-type
  destination-realm
  diameter-profile
  interface router
  peer
    supported-host
    no shutdown
```

Example: Diameter peer profile configuration for the Sd reference point

```
#config>mobile>profile# diameter-peer "temp-gx-for-sd"
#config>mobile>profile>diameter-peer# info
-----
      application-type sd
      destination-realm "nokia.com"
      diameter-profile "diam-prof1"
      peer 20.0.0.2 port 3868
          no shutdown
      exit
-----
```

10.9.3 Downlink next-hop determination configuration

Configure the downlink next-hop determination for user traffic between the PGW and SFC.

10.9.3.1 Next-hop address determination sequence

For the SSG TDF implementation, the source IP packet processing can be configured for downlink user traffic received from the SFC and for uplink user traffic received from the PGW/GGSN. For the downlink traffic, the next-hop determination sequence must be specified.

If the SSG TDF is implemented together with the SSG/steering function, the interface between a standalone SSG node (or PGW/GGSN combined with SSG) and the SFC can be configured using the steering command with the `dl-from-sfc` keyword to enable packet processing for downlink user traffic received from the SFC.

The downlink next hop toward the PGW is determined based on the VRPN IP pool configuration, matching the AVP received in the TSR message to the address list profile configured for the APN or the default APN configuration.

The following is the sequence in which the next-hop information is determined.

1. Check for the next-hop address in the IP local pool configuration; see [Next-hop address configuration for the VPRN IP pool](#).
2. Determine if a 3GPP-GGSN-Address/3GPP-GGSN-Ipv6-Address AVP was received in the TSR message. If the AVP is present, check for the next-hop address configured for the APN steering option by matching the PGW/GGSN address to the configured address list profile; see [Next-hop address determination based on the PGW/GGSN address in the TSR](#).
3. If a valid next-hop address is not obtained by step 1 or 2, try to use the default next-hop address configured for that APN; see [Default next-hop address configuration for the APN](#).

If the next-hop information cannot be determined by these steps, the call fails.

Related information

[Traffic steering function](#)

10.9.3.2 Next-hop address configuration for the VPRN IP pool

The downlink next hop toward the PGW can be determined based on the VPRN IP pool configuration. The IP pool configuration is used to specify the next-hop information for the interface.

Use the following CLI syntax to configure the VPRN IP pool.

```
configure>service# vprn
ip-local-pool
  next-hop-gw-ipv4 ip-address
  next-hop-gw-ipv6 ip-address
  ipv4-prefix ipv4-address/mask [block]
  ipv6-prefix ipv4-address/mask [block]
```

Example: VPRN IP local pool configuration

```
#config>service>vprn# ip-local-pool "sd-pool.1.1"
#config>service>vprn>pool# info
-----
                remote
  next-hop-gw-ipv4 "31.1.1.2"
  next-hop-gw-ipv6 "3ffe::1f01:102"
  ipv4-prefix "1.0.0.0/16"
  ipv6-prefix "4ffe:0:1::/48"
-----
```

10.9.3.3 Next-hop address determination based on the PGW/GGSN address in the TSR

The next-hop address can be determined based on the 3GPP-GGSN-Address/3GPP-GGSN-Ipv6-Address AVP that is received in the TSR message. If the AVP is present, the next-hop address configured for the APN steering option is checked by matching the PGW/GGSN address to the configured address list profile.

Use the following CLI syntax to configure an address list profile.

```
configure>mobile-gateway# profile
addr-list addr-list-name
      ip-address ip-address/prefix-length
```

Example: Address list profile configuration

```
# configure mobile-gateway profile addr-list "ggsn-sig-addr-list"

#config>mobile>profile>addr-list# info
-----
                ip-address 10.10.5.1/32
                ip-address 3ffe::a0a:501/128
-----
```

The address list profile can be specified for the gateway signaling address for the APN steering function; use the CLI syntax below.

```
configure>mobile-gateway# pdn
apn
      steering
            gw-signaling-addr addr-list-name
```

Example: Address list profile configured for the APN steering function

```
# configure mobile-gateway pdn 1 apn "ipd.nokia.1.1.com" steering gw-
signaling-addr "ggsn-sig-addr-list"

#config>mobile>pdn>apn>steering>gw-signaling-addr# info
-----
                gw-addr-ipv4 31.1.1.3
                gw-addr-ipv6 3ffe::1f01:103
-----
```

10.9.3.4 Default next-hop address configuration for the APN

A default next-hop address can be configured for the APN.

This address will be used if the next hop cannot be determined based on the IP pool configuration for the VPRN or the PGW/GGSN address in the TSR.

```
configure>mobile-gateway# pdn
apn
    steering
        default-addr-ipv4
        default-gw-addr-ipv6
```

Example: A configured default next-hop address for an APN

```
#config>mobile>pdn>apn>steering# info
-----
                default-gw-addr-ipv4 31.1.1.4
                default-gw-addr-ipv6 3ffe::1f01:104
-----
```

10.9.4 Interface configuration to the PGW/GGSN for uplink traffic

The interface between a standalone SSG and PGW/GGSN can be configured for the SSG TDF.

The steering command with the **ul-from-gw** keyword enables the source IP-based packet processing for uplink user traffic received from the PGW/GGSN. User traffic received from the PGW/GGSN is processed on the anchor ISM-MG card. Source IP-based packet processing allows locating of the anchor ISM-MG card on the IOM.

If the SSG TDF is implemented together with the SSG/steering function, the interface between a standalone SSG node (or PGW/GGSN combined with SSG) and the SFC can be configured using the steering command with the **dl-from-sfc** keyword to enable packet processing for downlink user traffic received from the SFC.

Use the following CLI syntax to configure an interface to enable packet processing for downlink user traffic received from the SFC.

```
configure>service# vprn
interface
    steering [dl-from-sfc]
```

Example: Interface configuration with source IP-based packet processing for uplink user traffic from the specified PGW/GGSN

```
interface "interface_Connected_Pgw" create
```

```
address 31.1.1.1/24
bfd 4000 receive 4000 multiplier 3
ipv6
    address 3ffe::1f01:101/120
exit
sap 1/1/4:1 create
    description "sap-1-31.1.1.1"
exit
steering ul-from-gw
exit
```

Related information

Traffic steering function

10.9.5 Traffic redirection and steering profiles configuration

A traffic redirection profile configures the uplink and downlink next-hop information that is used within a steering profile for the SSG TDF configuration. The traffic redirection profile must be defined in the policy options profile context, prior to association with a steering profile

Related information

Traffic steering function

10.9.6 Steering for the SDF and APN configuration

A steering profile policy can be configured and associated within a policy rule unit for SDF-level steering. It can also be associated at the APN level as a default steering profile for an APN.

When a profile is enabled for the SDF level, it overrides a profile specified at the APN level. If no profile is specified at the SDF level, the one specified at the APN level is inherited.

Related information

Traffic steering function

10.9.7 Diameter signaling profile configuration for the PDN

Example: Signaling configuration for the Diameter host, realm, and profile for the PDN

```
configure>mobile-gateway# pdn
signaling
    diameter
        origin-host
        origin-realm
```

profile

```
#config>mobile>pdn>signaling# info
-----
        diameter
            origin-host "diam.origRealm.com"
            origin-realm "origRealm.com"
            profile "diam-prof1"
        exit
-----
```

10.9.8 Configuration and application of TDF charging profiles

The SSG TDF supports online charging at the rating-group level. Charging information for application traffic is categorized by rating group.

Offline charging is supported at a Flow-based Charging (FBC) level. Charging information for application traffic is categorized by rating group or a combination of rating group and service ID.

Charging profiles can be configured in the mobile gateway charging profile context, and then applied for the APN or the EPC node. Use the following CLI syntax to configure a charging profile.

```
configure>mobile-gateway# profile
charging
  off-line
    charging-info-trigger
      [no] serving-node-change
      tariff-time start time end time
      [no] tariff-time-change
      [no] user-loc-change
    enable-offline [session-level | fbc | both]
    partial-record-closure-trigger
      max-change-condition max-change-condition
      no max-change-condition
      [no] ms-time-zone-change
      [no] plmn-change
      [no] rat-change
      time-limit time-limit
      no time-limit
      volume-limit volume-limit
      no volume-limit
    pri-cdf-dia-peer profile-name
    no pri-cdf-dia-peer
    sec-cdf-dia-peer profile-name
    no sec-cdf-dia-peer
```

```

        no shutdown
    on-line
        dcca-profile profile-name
        no dcca-profile
        enable-online [session-level | rg-level]
        no enable-online
        pri-ocs-dia-peer profile-name
        no pri-ocs-dia-peer
        sec-ocs-dia-peer profile-name
        no sec-ocs-dia-peer

```

Example: TDF charging profile configurations

```

charging
  tdf-charging 0
    on-line
      enable-online rg-level
      pri-ocs-dia-peer "Gy-Peer-Prof15"
      sec-ocs-dia-peer "Gy-Peer-Prof16"
    exit
  exit
  tdf-charging 1
    off-line
      enable-offline fbc
    exit
    on-line
      dcca-profile "DCCA-Prof1"
      enable-online rg-level
      pri-ocs-dia-peer "Gy-Peer-Prof1"
      sec-ocs-dia-peer "Gy-Peer-Prof2"
    exit
  exit
  tdf-charging 2
    on-line
      enable-online rg-level
      pri-ocs-dia-peer "Gy-Peer-Prof3"
      sec-ocs-dia-peer "Gy-Peer-Prof4"
    exit
  exit
  tdf-charging 3
    on-line
      enable-online rg-level
      pri-ocs-dia-peer "Gy-Peer-Prof5"
      sec-ocs-dia-peer "Gy-Peer-Prof6"
    exit

```

```
exit
tdf-charging 4
  on-line
    enable-online rg-level
    pri-ocs-dia-peer "Gy-Peer-Prof7"
    sec-ocs-dia-peer "Gy-Peer-Prof8"
  exit
exit
tdf-charging 5
  on-line
    enable-online rg-level
    pri-ocs-dia-peer "Gy-Peer-Prof9"
    sec-ocs-dia-peer "Gy-Peer-Prof10"
  exit
exit
tdf-charging 6
  on-line
    enable-online rg-level
    pri-ocs-dia-peer "Gy-Peer-Prof11"
    sec-ocs-dia-peer "Gy-Peer-Prof12"
  exit
exit
tdf-charging 7
  on-line
    enable-online rg-level
    pri-ocs-dia-peer "Gy-Peer-Prof13"
    sec-ocs-dia-peer "Gy-Peer-Prof14"
  exit
exit
tdf-charging 10
  description "Online only"
  on-line
    dcca-profile "DCCA-Prof1"
    enable-online rg-level
    pri-ocs-dia-peer "Gy-Peer-Prof1"
  exit
exit
tdf-charging 11
  description "Offline only"
  off-line
    enable-offline fbc
  exit
exit
tdf-charging 12
  description "Offline only with CIT"
```

```
        off-line
            enable-offline fbc
            charging-info-trigger
                serving-node-change
                user-loc-change
            exit
        exit
    exit
    tdf-charging 13
        description "Offline only with PRCT"
        off-line
            enable-offline fbc
            partial-record-closure-trigger
                max-change-condition 4
                ms-time-zone-change
                plmn-change
                rat-change
                time-limit 4
                volume-limit 6000
            exit
        exit
    exit
    tdf-charging 14
        description "Offline only with CIT and PRCT"
        off-line
            enable-offline fbc
            charging-info-trigger
                serving-node-change
                user-loc-change
            exit
            partial-record-closure-trigger
                max-change-condition 4
                ms-time-zone-change
                plmn-change
                rat-change
                time-limit 4
                volume-limit 6000
            exit
        exit
    exit
    gy-avp-options 1
    exit
exit
```


**NOTE:**

These configurations are applied in the APN configuration example.

Example: Charging profiles configured for APNs or for the EPC nodes

```
pdn 1
  aa-group "1:1" def-app-profile "Profile1"
  apn "ipd.nokia.1.1.com"
    aggr-rate-dl 6000
    aggr-rate-ul 6000
    pcrf-selection
      dynamic-pcc fh-session disable fh-3xxx disable fh-4xxx disable
    exit
  router "vprn1"
  exit
  svc-steering
    default-steering-profile ssg-profile
  exit
  no shutdown
exit
apn "ipd.nokia.1.2.com"
  aggr-rate-dl 6000
  aggr-rate-ul 6000
  charging
    charging-prof home 10
    charging-prof visiting 10
    charging-prof roaming 10
  exit
  router "vprn2"
  exit
  svc-steering
    default-steering-profile ssg-profile2
  exit
  no shutdown
exit
apn "ipd.nokia.1.3.com"
  aggr-rate-dl 6000
  aggr-rate-ul 6000
  charging
    charging-prof home 11
    charging-prof visiting 11
    charging-prof roaming 11
  exit
```

```
router "vprn3"
exit
svc-steering
    default-steering-profile ssg-profile3
exit
no shutdown
exit
apn "ipd.nokia.1.4.com"
aggr-rate-dl 6000
aggr-rate-ul 6000
charging
    charging-prof home 14
    charging-prof visiting 14
    charging-prof roaming 14
exit
router "vprn4"
exit
svc-steering
    default-steering-profile ssg-profile4
exit
no shutdown
exit
apn "ipd.nokia.1.5.com"
aggr-rate-dl 6000
aggr-rate-ul 6000
charging
    charging-prof home 15
    charging-prof visiting 15
    charging-prof roaming 15
exit
router "vprn5"
exit
svc-steering
    default-steering-profile ssg-profile5
exit
no shutdown
exit
epc-node "310.012.pgw.region1.3.4"
home-plmn-list "basicPlmn"
charging
    charging-prof home 1
    charging-prof visiting 1
    charging-prof roaming 1
    node-id "NOK-NODE"
exit
```

```

signaling
  diameter
    origin-host "diam.origRealm.com"
    origin-realm "origRealm.com"
    profile "diam-prof1"
  exit
exit

```

10.9.9 Charging method configuration for policy rules

The **charging rules policy** option supports the configuration of charging methods to support different charging scenarios for the SDF.

Charging scenarios include online charging, offline charging, both online and offline charging, and no charging. The **none** option supports scenarios when an SDF is not to be charged (offline and online charging is disabled). The rating group and monitoring key information can also be specified.

The configured charging rules can be associated with policy rules for the steering profile. Use the following CLI syntax to configure a charging rule unit.

```

configure>mobile-gateway# profile
policy-options
  begin
  charging-rule-unit
  charging method
  monitoring-key
  rating-group
commit

```

Example: Charging rule unit configurations

```

charging-rule-unit "onLine1100"
  rating-group 100
  charging-method both
  monitoring-key "23"
exit
charging-rule-unit "onLine1101"
  rating-group 101
  charging-method both
  monitoring-key "24"
exit
charging-rule-unit "onLine1102"
  rating-group 102
  charging-method both
  monitoring-key "25"

```

```
exit
charging-rule-unit "onLine1103"
    rating-group 103
    charging-method both
    monitoring-key "26"
exit
charging-rule-unit "onLine1104"
    rating-group 104
    charging-method both
    monitoring-key "27"
exit
charging-rule-unit "vrg100-1100"
    rating-group 100
    service-identifier 1100
    charging-method both
    reporting-level service-id
exit
charging-rule-unit "vrg100-1110"
    rating-group 100
    service-identifier 1110
    charging-method both
    reporting-level service-id
exit
charging-rule-unit "vrg101-1101"
    rating-group 101
    service-identifier 1101
    charging-method both
    reporting-level service-id
exit
charging-rule-unit "vrg101-1111"
    rating-group 101
    service-identifier 1111
    charging-method both
    reporting-level service-id
exit
charging-rule-unit "vrg102-1102"
    rating-group 102
    service-identifier 1102
    charging-method both
    reporting-level service-id
exit
charging-rule-unit "vrg102-1112"
    rating-group 102
    service-identifier 1112
    charging-method both
```

```
reporting-level service-id
```

Example: Policy rules referencing the charging rule units

```
policy-rule "onLine1Any5" policy-rule-unit "onLine1AnyAny" charging-rule-unit
  "onLine15" qci 9 arp 1 precedence 4000
policy-rule "onLine1100-100" policy-rule-unit "onLine1100" charging-rule-unit
  "onLine1100" qci 9 arp 1 precedence 3999
policy-rule "onLine1101-101" policy-rule-unit "onLine1101" charging-rule-unit
  "onLine1101" qci 9 arp 1 precedence 3998
policy-rule "onLine1102-102" policy-rule-unit "onLine1102" charging-rule-unit
  "onLine1102" qci 9 arp 1 precedence 3997
policy-rule "onLine1103-103" policy-rule-unit "onLine1103" charging-rule-unit
  "onLine1103" qci 9 arp 1 precedence 3996
policy-rule "onLine1104-104" policy-rule-unit "onLine1104" charging-rule-unit
  "onLine1104" qci 9 arp 1 precedence 3995
policy-rule "vrg100-1100-onLine1100" policy-rule-unit "onLine1100" charging-
rule-unit "vrg100-1100" qci 9 arp 1 precedence 1001
policy-rule "vrg100-1110-onLine1101" policy-rule-unit "onLine1101" charging-
rule-unit "vrg100-1110" qci 9 arp 1 precedence 1000
policy-rule "vrg101-1101-onLine1102" policy-rule-unit "onLine1102" charging-
rule-unit "vrg101-1101" qci 9 arp 1 precedence 1011
policy-rule "vrg101-1111-onLine1103" policy-rule-unit "onLine1103" charging-
rule-unit "vrg101-1111" qci 9 arp 1 precedence 1010
policy-rule "vrg102-1102-onLine1104" policy-rule-unit "onLine1104" charging-
rule-unit "vrg102-1102" qci 9 arp 1 precedence 1021
policy-rule "vrg102-1112-onLine1105" policy-rule-unit "onLine1105" charging-
rule-unit "vrg102-1112" qci 9 arp 1 precedence 1020
  policy-rule-base "onLine1"
    policy-rule "onLine1Any5"
    policy-rule "onLine1100-100"
    policy-rule "onLine1101-101"
    policy-rule "onLine1102-102"
    policy-rule "onLine1103-103"
    policy-rule "onLine1104-104"
  exit
  policy-rule-base "virtualRG"
    policy-rule "vrg100-1100-onLine1100"
    policy-rule "vrg100-1110-onLine1101"
    policy-rule "vrg101-1101-onLine1102"
    policy-rule "vrg101-1111-onLine1103"
    policy-rule "vrg102-1102-onLine1104"
    policy-rule "vrg102-1112-onLine1105"
```

**NOTE:**

This application is also shown in *DPI and 5-tuple policy rule unit configuration for SSG TDF*.

10.9.10 DPI and 5-tuple policy rule unit configuration for SSG TDF

The following examples show an Application Assurance group policy and 5-tuple policy rule configuration and the supporting mobile gateway configuration.

Example: Application Assurance configuration

```
#config>app-assure>group>policy# charging-group "googleCG"
#config>app-assure>group>policy>charge-grp# info
-----
                notify-start-stop flow-based
-----
#config>app-assure>group>policy>app-filter entry 33 create
                expression 1 http-host eq "*.google.*"
                application "google"
                no shutdown
                exit
-----
```

Example: 5-tuple policy rule configuration for the mobile gateway

```
#config>mobile>profile>policy-options#
    begin
    policy-rule-unit "anyany"
        flow-description 1
        exit
    exit

#config>mobile>profile>policy-options>policy-rule-unit# info
    no mute-notification
    flow-description 1
        match
            aa-charging-group "googleCG"
        exit
    exit

#config>mobile>profile>policy-options>policy-rule "google" policy-rule-unit
"googlePru" charging-rule-unit "CRU4" qci * arp * precedence 1400
```

```
#config>mobile>profile>policy-options>policy-rule "anyany" policy-rule-unit
  "anyany" charging-rule-unit "CRU0" qci * arp * precedence 2400
#config>mobile>profile>policy-options>policy-rule-base "anyanyNoSteering"
  policy-rule "anyany"
  policy-rule "google"
  exit
commit
```

10.9.11 Auto-create subscriber configuration for AA features

To configure auto-create subscribers for AA features and TCP-O, use the following syntax.

Use the **auto-create** keyword when configuring transit subscribers.

```
configure router ip-local-pool
```

```
configure service vprn ip-local-pool transit-sub {fixed | mobile} protocol {gx
| radius | auto-create}
```

Use the following syntax to configure the PDN APN settings for this feature.

```
configure mobile-gateway pdn 1 apn apn_name
transit-sub
data-triggered
attach-response-wait-timer 10
attach-retry-hold-timer 30
exit
```

Use the following syntax to map the UE IP to the subscriber ID.

```
configure mobile-gateway profile subscriber-
mapping subscriber_mapping_profile_name address ip subscriber-id subscriber_id
```

Use the following command to configure the subscriber mapping profile.

```
configure mobile-gateway pdn 1 apn apn_name
subscriber-mapping-profile subscriber_mapping_profile_name
```

11 Traffic steering function

Overview of Nokia implementation of the traffic steering function for the gateway (PGW/GGSN, PGW-U/GGSN-U) and the Subscriber Services Gateway (SSG).

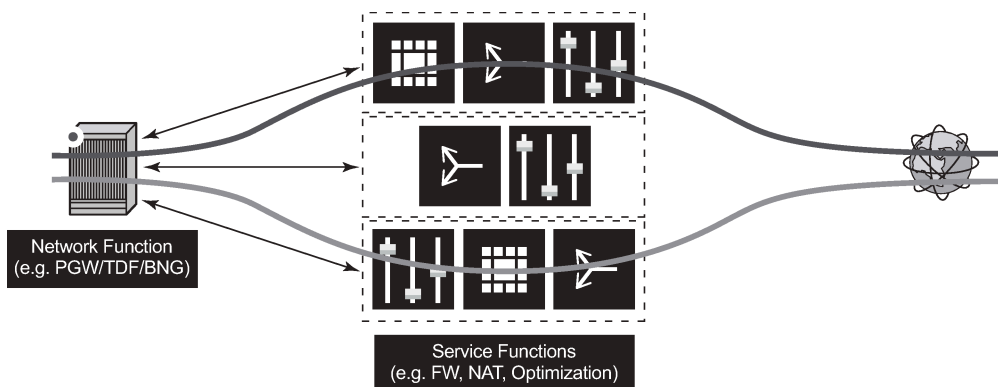
11.1 Service function chaining

The Service Function Chaining (SFC) architecture decouples the classification and steering functions.

Legacy methods used to deploy value-added service functions in packet core have been inflexible and restrictive. Applying third party value-added services for functions that are supported on gateway requires classification of traffic, followed by steering of traffic using applicable service functions. In legacy systems, classification and steering functions are coupled and both are performed on a network function node. Service functions are statically connected in sequence. Classification is performed on a network function node and traffic is redirected to the first service function node to apply a set of service functions in a specific sequence. Information for an applicable set and sequence of service functions is not carried in the packet, so each set of service functions and sequences requires a unique instance.

Because the network is statically configured to connect service functions, it is not possible to change applicable service functions dynamically. To make applicable service functions dynamic, reclassification can be used at every service function. However, this adds to overhead and does not use network resources efficiently.

Figure 156: Legacy deployment of service functions example



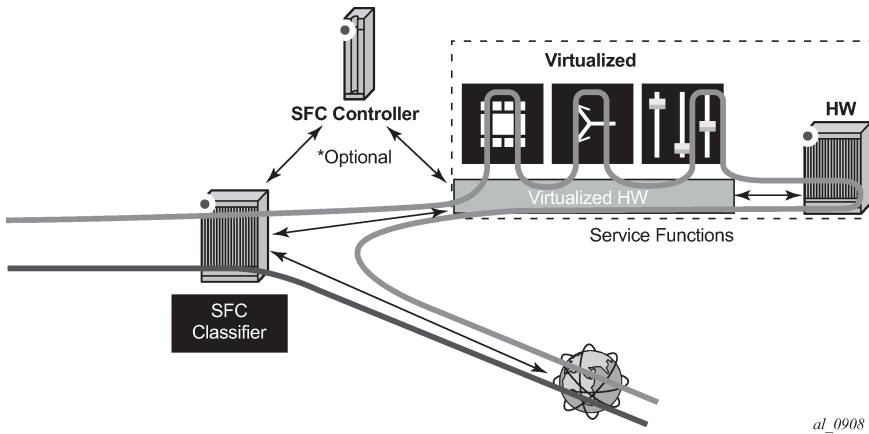
al_0875

With the Service Function Chaining (SFC) architecture, classification and steering functions are decoupled. The network function is not aware of applicable service functions. It is only aware of classification rules and service chain identifiers. Each service chain identifier maps to an applicable set and sequence of service functions. Service functions are connected via a networking layer (SDN) that can dynamically map each packet to a specific set and sequence of service functions based on a predefined service chain.

Traffic forwarding rules (service chain identifier and applicable service functions) are defined by the SFC controller and distributed to the network forwarding layer to steer traffic using a specific set of service

functions in a predefined sequence. Network functions embed a service chain identifier in each packet to help the networking layer steer traffic using applicable service functions.

Figure 157: Service function chaining example



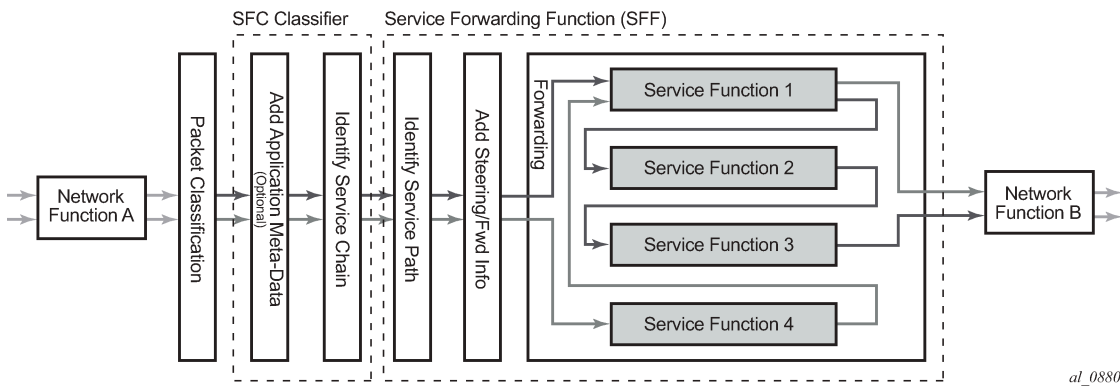
11.2 SFC architecture

SFC involves packet classification, such as a flow in a video stream, or SFC classification such as optionally adding application meta-data in each packet (for example, Subscriber IMSI-1).

The SFF identifies the service path for each user flow. A service path is a sequence of service functions (for example, traffic forwarded to the Threat Management function, followed by the Analytics function). SFC can also identify and apply a service chain, which is a sequence of service functions (for example, SC1: Threat Management > Analytics).

With independent SFC classifier and SFF functions, the traffic steering policies are enforced by the SDN SFC controller that is also managing the SFF function instances. For L2-based forwarding, the MAC address of each service function is available to the SDN SFC controller. The SDN SFC controller programs the SFF with the service chain forwarding information (that is, based on the source MAC address, forward to the destination MAC address). The SFF can add steering information in each packet (for example, the packet of user flow1 is embedded with service chain 1, the packet of user flow2 is embedded with service chain 2, and so on). It also performs forwarding through service functions based on steering information embedded in the traffic packet header.

Figure 158: SFC architecture



al_0880

11.3 Traffic steering function support

The traffic steering functions supported are the following:

- *SFC classifier with independent SFF*
- *SFC classifier with integrated SFF*
- *Networking and traffic flow*

11.3.1 SFC classifier with independent SFF

With this method of deployment, the SFC classifier identifies traffic to be steered via service chain and forwards traffic to an independent SFF.

Traffic to be steered via a specific service chain is mapped to a service chain identifier. SFF steers traffic via service function instances and sends traffic back to the SFC classifier for further action.



NOTE:

Only VLAN ID is supported as service chain identifier.

Mobile gateway as SFC classifier

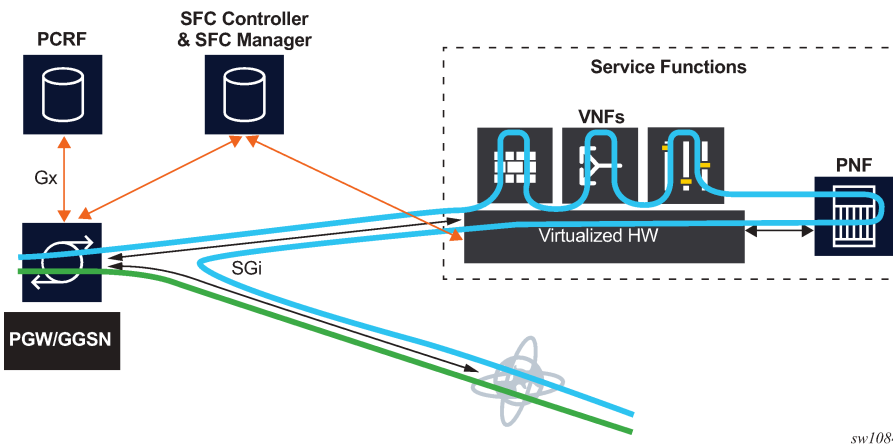
The mobile gateway (PGW/GGSN, PGW-U/GGSN-U, UPF) performs packet classification for the SFC. Upon packet classification and identification of an applicable service chain, the gateway forwards traffic to the node that is performing the SFF.

The SFC steering policy (service chain identifier and next hop) is defined on the gateway. The match criteria and the applicable steering policy are enabled via local policy configuration or by PCRF. SFC forwarding rules (service chain identifier and mapped service chain path) are defined by the SFC controller and distributed to the network forwarding layer to steer traffic using a specific set of service functions in the predefined sequence. When matching traffic is classified on the gateway, the service chain identifier is embedded in the packet to help the networking layer steer traffic via the applicable SFC.

**NOTE:**

Nokia recommends consulting a Nokia representative about the available support for CMG deployment as SFC classifier.

Figure 159: Mobile gateway as SFC classifier example



SSG as SFC classifier

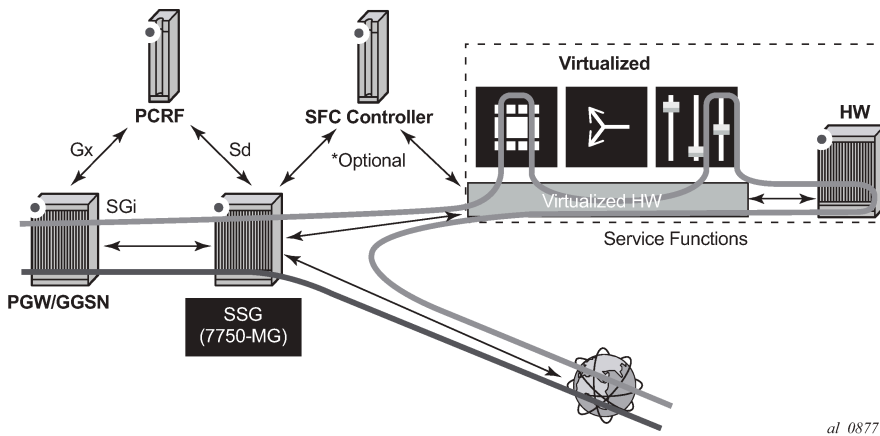
The SSG performs packet classification for the SFC while traffic is forwarded between gateway and external network via SSG. Traffic forwarding is setup such that all traffic is received by SSG using route advertisement for UE addresses that are configured on gateway.

Upon packet classification and identification of an applicable service chain, the SSG forwards traffic to the node that is performing the SFF. The SFC steering policy (service chain identifier and next hop) is defined on the SSG. The match criteria and the applicable steering policy are enabled via the Sd interface. SFC forwarding rules (service chain identifier and mapped service chain path) are defined by the SFC controller and distributed to the network forwarding layer to steer traffic using a specific set of service functions in the predefined sequence. When matching traffic is classified on the SSG, the service chain identifier is embedded in the packet to help the networking layer steer traffic via the applicable SFC.

**NOTE:**

Nokia recommends consulting a representative about the available support for CMG deployment as SFC classifier.

Figure 160: SSG as SFC classifier example



11.3.2 SFC classifier with integrated SFF

With this method of deployment, the SFC classifier identifies traffic to be steered via mapped service chain.

On classification and mapping of traffic flow to a service chain, the SFC classifier steers traffic via service function instances. As the SFC classifier is also acting as SFF, traffic between each service function instance is received back on the SFC classifier/SFF for further action as SFF.



NOTE:

Only VLAN ID is supported for forwarding between service function instances.

Mobile gateway as SFC classifier and SFF

The gateway (PGW/GGSN, PGW-U/GGSN-U, UPF) performs packet classification for the SFC. Upon packet classification and identification of an applicable service chain, the gateway forwards traffic to the first service function instance as it is also performing the SFF.

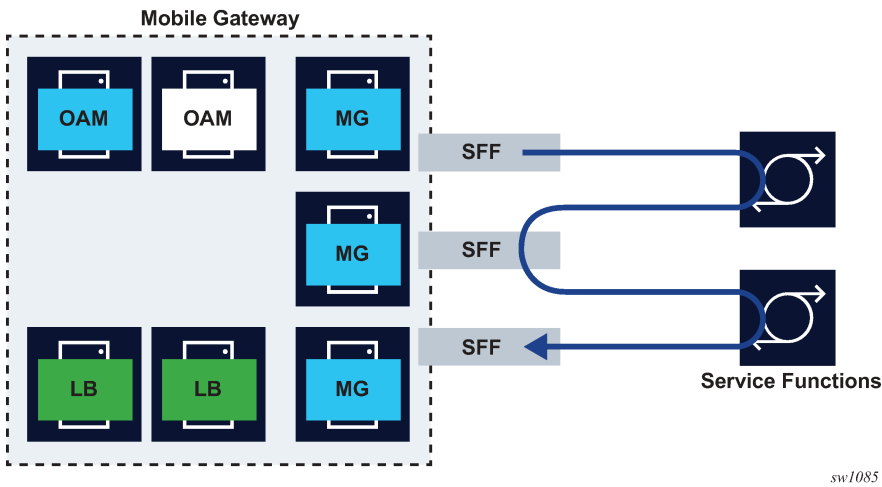
The SFC steering policy (service chain identifier and next hop) is defined on the gateway. In addition, network must be configured such that traffic is forwarded back to the gateway for further steering of traffic via the next service function instance in chain.



NOTE:

Gateway as SFC classifier with integrated SFF function is supported with CMG operating in LB-less mode only.

Figure 161: Mobile gateway with traffic steering functions example



SSG as SFC classifier and SFF

The SSG performs packet classification for the SFC while traffic is forwarded between gateway and external network via SSG. Traffic forwarding is setup so that all traffic is received by SSG using route advertisement for UE addresses that are configured on gateway.

The SSG performs packet classification for the SFC. Upon packet classification and identification of an applicable service chain, the SSG forwards traffic to the first service function instance as it is also performing the Service Forwarding Function (SFF). The SFC steering policy (service chain identifier and next hop) is defined on the SSG. In addition, network must be configured such that traffic is forwarded back to the SSG for further steering of traffic via the next service function instance in chain.



NOTE:

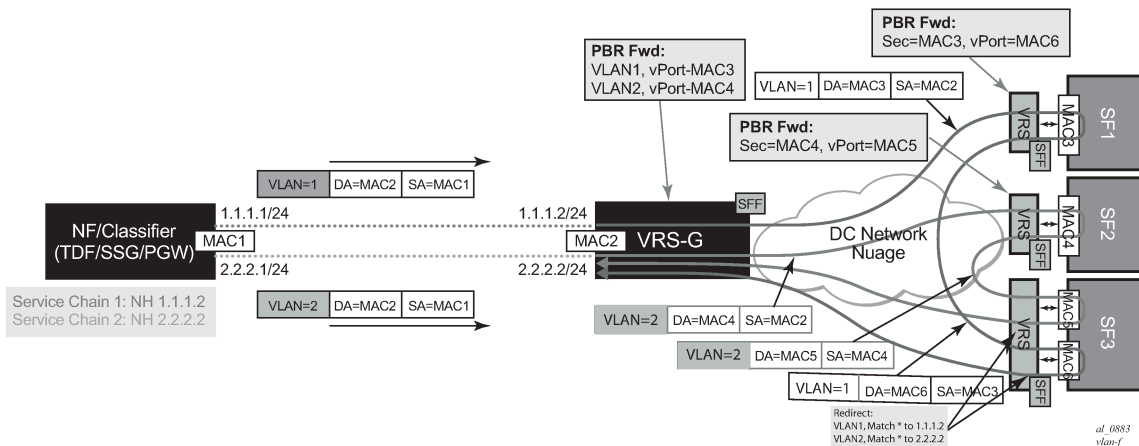
SSG as SFC classifier with integrated SFF function is not supported.

11.3.3 Networking and traffic flow

Traffic flow with independent SFF

Classification criteria is provided by PCC rules. The steering profile uses the SFC identifier for SFC path selection and steering. The steering profile is configured on gateway or SSG. Traffic is steered to the SDN layer via a PNF/VNF switch (for example, Nuage VRS-G). Traffic is redirected to the VRS-G using the steering profile defined on the SSG.

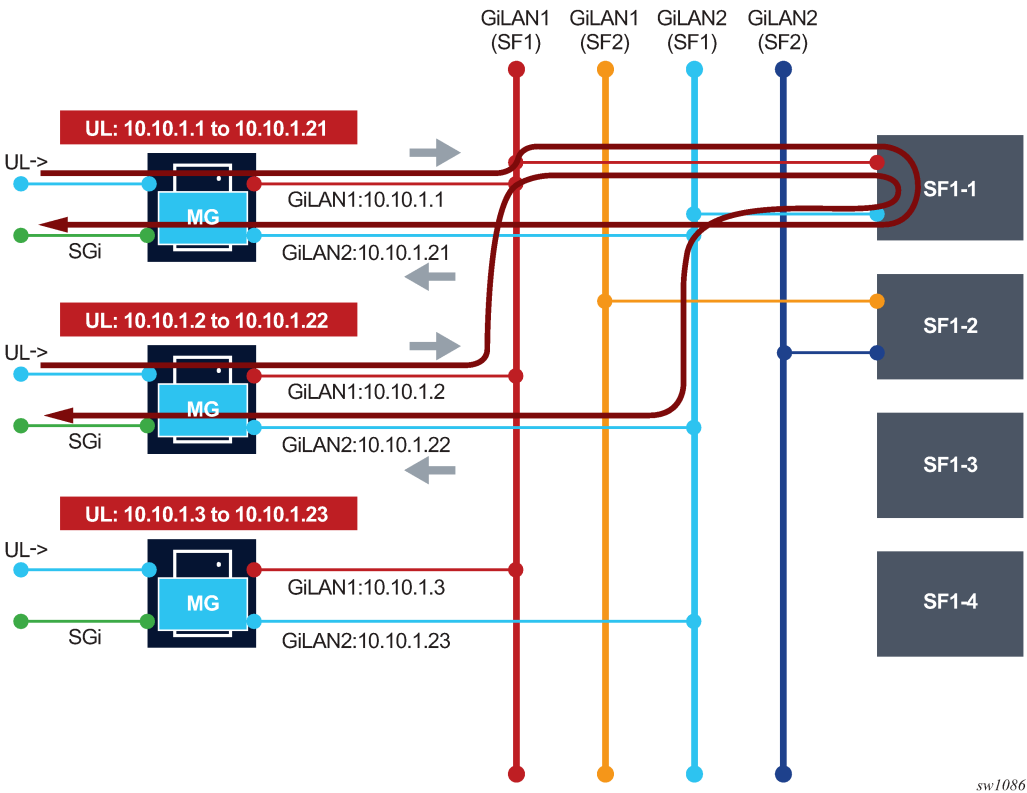
Figure 162: Network function as an SFC classifier: VLAN



Traffic flow with integrated SFF

Classification criteria is provided by PCC rules. The steering profile uses the SFC identifier for SFC path selection and steering. The steering profile and the traffic forwarding policy are configured on gateway such that traffic is forwarded to service function instance and received back on gateway after enforcement of function available on service function instance.

Figure 163: Network function as an SFC classifier with integrated SFF



11.4 Steering policy profiles

Steering policy profiles can be configured and applied at the PDN APN or Sd levels. Steering policy profiles include configuration of the following functions:

- service chaining
- next hop
- encapsulation
- subscriber ID
- application information
- RAT
- traffic redirection

11.4.1 Dynamic ADC rules for traffic steering

Dynamic provisioning of the steering function using ADC rules is supported. With the use of ADC rule support for traffic steering, the operator can:

- enable a steering profile that is predefined on the node
- use dynamic mapping of the flow information to a steering profile over Gx/Sd interfaces

With dynamic ADC rules, provisioning a service chain is rapid and adds flexibility as it does not require complex policy rules to be predefined on the node. The PCRF can download a steering policy and the associated flow information over the Gx/Sd interfaces.

11.5 Traffic steering configuration guidelines

Steering configuration examples.

11.5.1 Traffic redirection and steering profile configuration

A traffic redirection profile configures the uplink and downlink next-hop information that is used within a steering profile configuration.

The traffic redirection profile must be defined in the policy options profile context, prior to association with a steering profile.

Use the following syntax to configure a traffic redirection profile.

```
configure>mobile-gateway# profile policy-options
  traffic-redirect-profile traffic-redirection-profile-name
    mg-group group-id
      router ul router-name dl router-name
```

```

traffic-redirect-target traffic-redirect-target-id
    downlink-next-hop-ipv4 ip-address
    downlink-next-hop-ipv6 ipv6-address
    uplink-next-hop-ipv4 ip-address
    uplink-next-hop-ipv6 ipv6-address
router router-instance
traffic-redirect-target traffic-redirect-target-id
    downlink-next-hop-ipv4 ip-address
    downlink-next-hop-ipv6 ipv6-address
    uplink-next-hop-ipv4 ip-address
    uplink-next-hop-ipv6 ipv6-address

```

11.5.1.1 Traffic redirection and steering profile configuration for LB mode deployment

A traffic redirection profile configures the uplink and downlink next-hop information for all traffic on the CMG instance.



NOTE:

Applies to CMG deployed with LB mode for SGI traffic.

Example: Traffic redirection profile configuration

```

config>mobile>profile>policy-options# traffic-redirect-profile
traffic-redirect-profile traffic-profile1
    router "vprn1"
    traffic-redirect-target 1
        uplink-next-hop-ipv4 34.0.0.2
        uplink-next-hop-ipv6 3ffe::2200:2
        downlink-next-hop-ipv4 35.0.0.2
        downlink-next-hop-ipv6 3ffe::2300:2
    exit
commit

```

11.5.1.2 Traffic redirection and steering profile configuration for LB-less mode deployment

A traffic redirection profile configures the uplink and downlink next-hop information for traffic per MG-VM basis for CMG instance deployed with LB-less mode.



NOTE:

Applies to CMG deployed with LB-less mode for SGI traffic.

Example: Traffic redirection profile per MG group basis configuration

```

config>mobile>profile>policy-options# traffic-redirect-profile
traffic-redirect-profile traffic-profile1
  mg-group 1
    router ul vprn1001 dl vprn 1021
    traffic-redirect-target 1
      uplink-next-hop-ipv4 10.10.1.21
      downlink-next-hop-ipv4 10.10.1.1
    exit
    traffic-redirect-target 2
      uplink-next-hop-ipv4 10.10.2.21
      downlink-next-hop-ipv4 10.10.2.1
    exit
    traffic-redirect-target 3
      uplink-next-hop-ipv4 10.10.3.21
      downlink-next-hop-ipv4 10.10.3.1
    exit
  mg-group 2
    router ul vprn1002 dl vprn 1022
    traffic-redirect-target 41
      uplink-next-hop-ipv4 10.10.1.22
      downlink-next-hop-ipv4 10.10.1.2
    exit
    traffic-redirect-target 42
      uplink-next-hop-ipv4 10.10.2.22
      downlink-next-hop-ipv4 10.10.2.2
    exit
    traffic-redirect-target 43
      uplink-next-hop-ipv4 10.10.3.22
      downlink-next-hop-ipv4 10.10.3.2
    exit
  exit
exit

```

11.5.2 Steering profile configuration

The traffic redirection profile can be associated with a steering profile in the policy options context. The steering profile can then be used for service steering configuration.

Example: Steering profile configuration

```

configure>mobile-gateway# profile policy-options
begin

```

```
steering-profile steering-profile-name traffic-redirect traffic-
redirect-profile
commit
```

Example: Steering profile configuration

```
config>mobile>profile>policy-options>steering-profile# info
-----
traffic-redirect traffic-profile1
-----
```

11.5.3 Steering for the SDF and APN configuration

A steering policy profile can be configured and associated within a policy rule unit for SDF-level steering. It can also be associated at the APN level as a default steering profile for an APN.

When a profile is enabled for the SDF level, it overrides a profile specified at the APN level. If no profile is specified at the SDF level, the one specified at the APN level is inherited.

To configure a steering profile, use the following syntax.

```
configure>mobile-gateway# profile policy-options
begin
  policy-rule-unit policy-name
  svc-steering [steering-profile profile-name]
commit
```

Example: Policy rule unit configuration to enable service steering for an SDF

The service steering is associated with the steering profile. If only service steering is configured for the policy rule unit (no steering profile), the default steering is applied for this policy rule unit.

```
config>mobile>profile>policy-options*
begin
  policy-rule-unit
    flow-description 1
      match
        remote-ip 15.1.1.0/24
      exit
    exit
    svc-steering steering-profile ssg-profile1
  commit
```

The steering profile can also be associated at the APN level as a default steering profile for an APN. This profile is inherited at the SDF level if no profile is specified at the SDF level.

```
configure>mobile-gateway>pdn>apn
svc-steering
    default-steering-profile profile-name
no shutdown
```

Example: Application of a default steering profile for the APN

```
config>mobile>pdn>apn#
    agr-rate-dl 6000
    agr-rate-ul 6000 pcrf-selection
        dynamic-pcc fh-session disable fh-3xxx disable fh-4xxx disable exit
    router "vprn1"
    exit
    svc-steering
        default-steering-profile ssg-profile1 exit
    no shutdown
```

11.5.4 Incoming IP interface configuration for traffic from service chain

The steering function can be enabled using the steering command under the VPRN service or router interface configuration with the dl-from-sfc option. Enabling this CLI option allows policy and charging enforcement functions for downlink traffic to be applied appropriately.

To enable the steering function for the VPRN or router, use the following commands:

```
configure>service# vprn
    interface
    steering dl-from-sfc
```

```
configure# router
    interface
    steering dl-from-sfc
```

Example: VPRN interface configuration

This example shows a VPRN interface configuration for the steering function using the steering command with the dl-from-sfc option, specified for IPv4 and IPv6 traffic.

```
config>service>vprn>if# info
-----
                address 34.0.0.1/24
                ipv6
```

```
        address 3ffe::2200:1/120
    exit
    sap 1/1/3:1 create
    exit
    steering dl-from-sfc
```

11.5.5 IP interface configuration for receiving uplink traffic from the mobile gateway on SSG



NOTE:

This configuration is applicable to SSG independent of the traffic steering function.

Related information

Interface configuration to the PGW/GGSN for uplink traffic

The interface between a standalone SSG and PGW/GGSN can be configured for the SSG TDF.

11.5.6 Traffic forwarding configuration for SFF function with CMG in LB-less mode

CMG deployed in LB-less mode with the SFF function requires traffic to be steered back to the CMG it is processed by the service function.

To send uplink or downlink traffic via a set of service function instances, each MG-group must be configured with the list of next-hop addresses to use to send traffic. An MG-group specific forwarding policy is defined using a traffic profile.

Each next-hop represents an interface that exists locally on the same MG-group. The interface is not configured on the MG-group, but in an independent VPRN (for example, GiLAN). As traffic is forwarded back to the CMG after it is processed on the service function, it must be forwarded back to the VPRN associated with SGi. An IP filter is configured to forward traffic between the GiLAN VPRN and the SGi VPRN.

Uplink traffic forwarding for SFF with CMG in LB-less mode and Downlink traffic forwarding for SFF with CMG in LB-less mode show examples of the uplink and downlink traffic with configured MG-groups.

Figure 164: Uplink traffic forwarding for SFF with CMG in LB-less mode

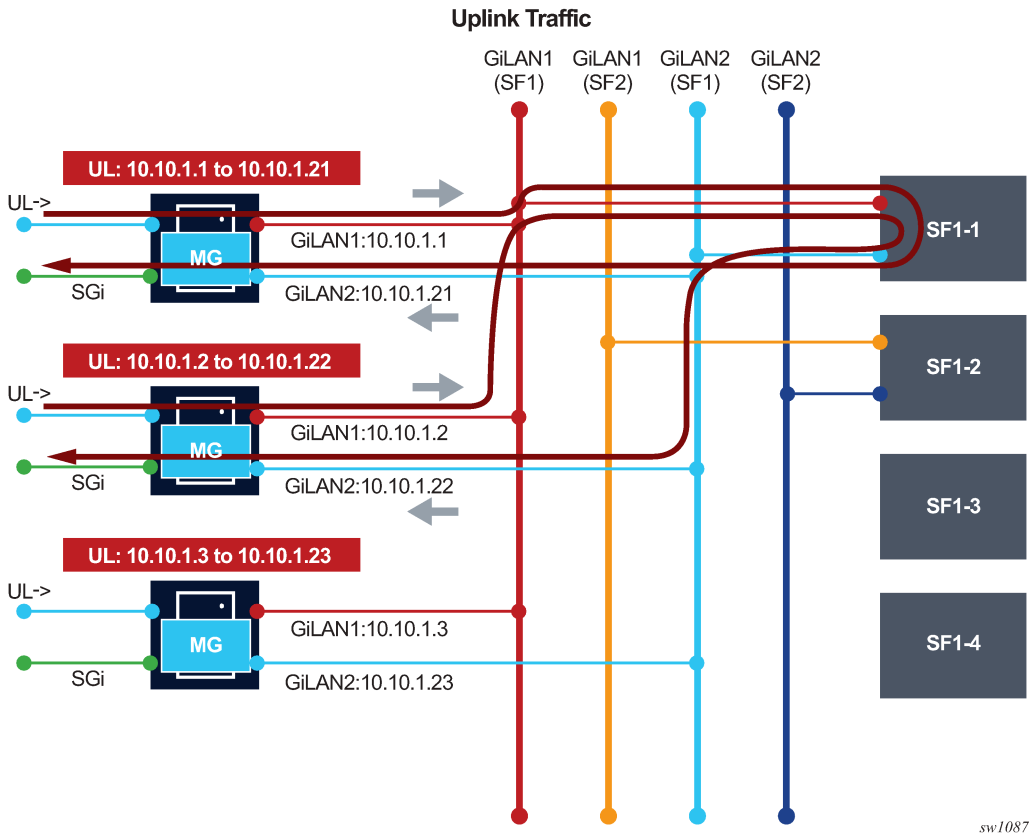
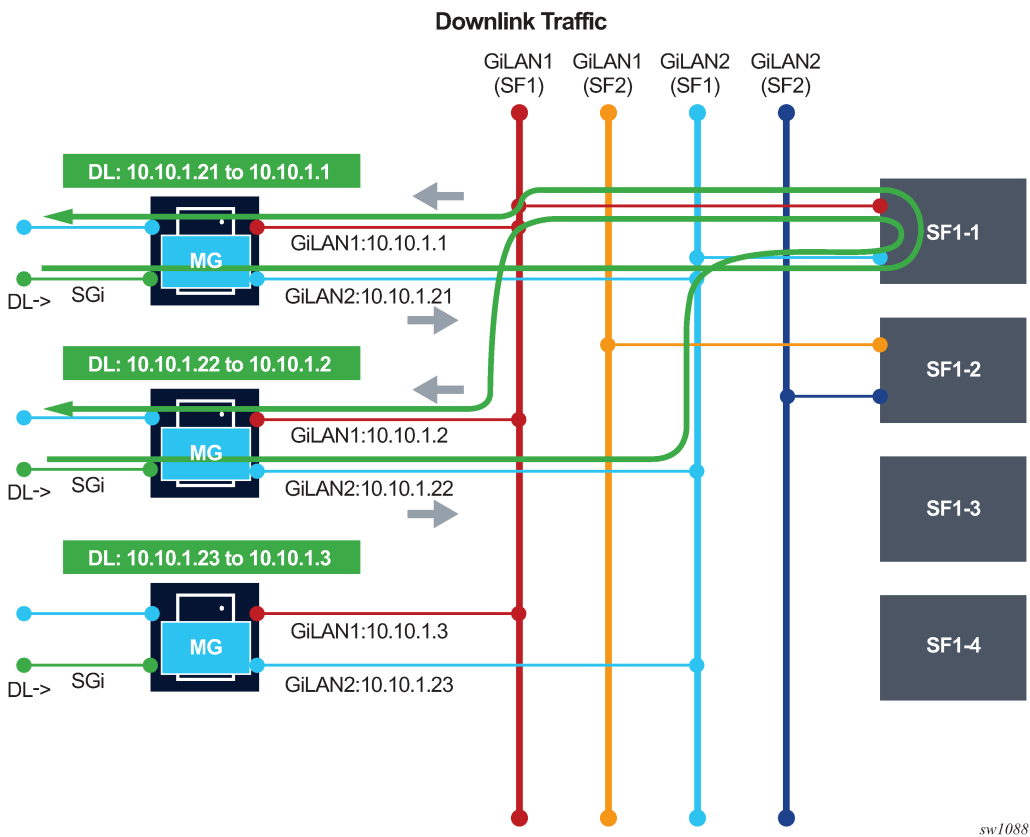


Figure 165: Downlink traffic forwarding for SFF with CMG in LB-less mode



Example: Forwarding uplink and downlink traffic received from the service chain to the VPRN associated with SGi

This example shows an MG-group with VPRN 101 associated with SGi.

For uplink traffic, VPRN 1001 is used for egress traffic, which is received back in VPRN 1021 and forwarded to SGi VPRN.

For downlink traffic, VPRN 1021 is used for egress traffic, which is received back in VPRN 1001 and forwarded to SGi VPRN.

```
vprn 1001 name "gilan1-ul-mg-group1" customer 1 create
  interface "interface-sf1-1" create
    address 10.10.1.1/24
    sap 1/1/1:1 create
      ingress
      filter ip 101
    exit
  exit
exit
interface "interface-sf1-2" create
  address 10.10.2.1/24
  sap 1/1/1:2 create
    ingress
    filter ip 101
  exit
exit
exit
vprn 1021 name "gilan2-dl-mg-group1" customer 1 create
  interface "interface-sf1-1" create
    address 10.10.1.21/24
    sap 1/1/1:41 create
      ingress
      filter ip 101
    exit
  exit
  steering dl-from-sfc
exit
interface "interface-sf1-2" create
  address 10.10.2.21/24
  sap 1/1/1:42 create
    ingress
    filter ip 101
  exit
exit
  steering dl-from-sfc
exit
```

12 Charging configuration

This section describes the charging principles and provides information and configuration guidelines for the charging implementation.

12.1 Supported interfaces

Table 35: Supported charging interfaces

Interface	Description	Gateway support
Bp (ftp/sftp)	Supports non-volatile storage of CDR files	SGW, PGW, TDF (SSG), ePDG
Rf	Supports sending of call data/billing records from IP network elements to the billing servers and to issue offline charging events to the charging system	SGW, PGW, TDF (SSG), ePDG
Ga (GTP prime)	Supports GTP prime to communicate with the charging gateway	SGW, PGW, TDF (SSG), ePDG
Gy	Supports online charging between the PCEF/GW and the Online Charging System	PGW, TDF (SSG)

12.2 Supported 3GPP release

The baseline for all charging interfaces is based on 3GPP Release 16, June 2020.

The following specifications are applicable:

- LTE

32.240 v16.1.0

Charging architecture and principles

32.251 v16.0.0	Packet Switched (PS) domain charging
32.295 v16.0.0	Charging Data Record (CDR) transfer
32.297 v16.1.0	Charging Data Record (CDR) file format and transfer
32.298 v16.5.0	Charging Data Record (CDR) parameter description
• SMF	
32.255 v16.5.0	5G data connectivity domain charging
32.290 v16.4.0	5G system; Services, operations and procedures of charging using Service Based Interface (SBI)
32.291 v16.4.2	5G system, charging service; Stage 3

The following functionality is supported by this feature:

- RAN-NAS Cause feature
- NB-IoT feature
- CSG Information Reporting feature
- Presence Reporting Area Information feature
- Sponsored Data Connectivity
- ePDG charging
- TDF charging

To control the inclusion and exclusion of UWAN user location information for online charging, use the following command:

```
configure mobile-gateway profile charging gy-avp-options uwan-user-location-info
```

To view the UWAN user location information, if configured, use the following commands:

```
show mobile-gateway profile charging cdr-avp-options
```

```
show mobile-gateway profile charging gy-avp-options
```


12.3 OFCS summary

The following key OFCS functionality is supported.

- Diameter ACR (start/interim/stop) and ACA commands
- Diameter over TCP and IPv6 or IPv4 transport
- Diameter charging sessions are set up from the ePDG, SGW, PGW, or TDF to the CDF on a per PDN Session basis. Each IP-CAN session has a corresponding Diameter Rf charging session to the CDF.
- Each bearer within the PDN session has a corresponding traffic data volumes (TDV) container (ePDG, PGW, SGW) or service data container (SD) (TDF) for the Diameter charging session.
- Charging-information addition triggers and partial record event triggers
- Default charging profiles for home subscribers, roamers (home-routed) as well as visiting (local break out) scenarios.
- CCF connectivity outage and recovery
- Gy Rf synchronization events including Rf charging information addition triggers related to offline/online charging at the PGW

12.4 Home, visitor, and roamer charging profile configuration

Configure the charging profiles to be applied for home, visitors, and roamers on the gateway.

The following configuration ignores the charging characteristics sent in S11 signaling and applies the charging profiles to homers, visitors, roamers as configured.

```
charging
  charging-prof home 10
  charging-prof visiting 10
  charging-prof roaming 15
  cc-ignore any
exit
ga
  gtp-prime-group gtp_server_group1
exit
```

12.5 Charging characteristics and charging profile

The charging characteristics (CC) IE received from MME/HSS is used to pick a locally configured charging profile. The selected charging profile is applicable for the IP-CAN session.

The charging method (disable offline; enable online) provisioned within the charging profile is the Default Charging Method, for the subscriber's IP-CAN session. When Policy and Charging Control (PCC) is enabled

(Gx), the PCRF has the option to override the default charging method, at the start of the IP-CAN session (Gx session establishment), if needed.

A charging profile supports an option to enable or disable online charging. Primary and Secondary OCS peers can be configured as part of the charging profile, along with the associated DCCA profile. The same DCCA profile is applicable to both the primary and secondary OCS peers.

12.5.1 PDN/APN level charging profile selection

The charging characteristics are supplied by the HSS to the MME as part of the subscription information and upon activation of the PDN connection. The MME forwards the CC field to the PGW (through the SGW) for selecting a locally configured charging profile to be used for that PDN Connection.

The charging profile selection is based on a mapping table configured either on the PDN or the APN level.

To enable charging profile selection on the PDN level for a specified PLMN, use the following command:

```
configure mobile-gateway pdn charging plmn-chrg-profile-map
```

To enable charging profile selection on the APN level for a specified PLMN, use the following command:

```
configure mobile-gateway pdn apn charging plmn-chrg-profile-map
```

When configuring APN level charging profile selection, the operator can differentiate charging for the same CC field value received when the activation requests point to different APNs.

Charging profile selection is made based on the following rule priorities:

- If APN level mapping is configured and the related APN entry exists in the configured mapping table, the charging profile is selected based on that entry.
If APN level mapping is not configured, the PDN level mapping is used.
- if no entry exists in the mapping table for the specific APN and the `config>mobile>pdn>apn>charging>cc-reject` command is enabled, the received session activation is rejected
- if the `config>mobile>pdn>apn>charging>cc-reject` command is disabled, the default charging profile is used

12.6 Bill Shock Prevention

The Bill Shock Prevention (BSP) feature enables operators to control session data costs in real time.

When this feature is enabled, the roaming status is used to pick up a locally configured charging profile to be used for the IP-CAN session. The charging method will be set to enable online for real time control of the session data usage. The Online Charging System (OCS) will be configured to notify roaming subscribers when their network charges have reached a certain threshold and redirect them to an operator portal for account top-up. Customers can feel more in control when charging schemes are transparent and known

up front. If the roaming status changes during the PDN connection lifetime, the PDN connection will be terminated. The subscriber/UE must re-activate (that is, refresh the browser) the PDN in the new PLMN. For backward compatibility, this feature provides the operator with the option to revert to the serving node supplied Charging Characteristic (CC) value that will be reported toward the charging interfaces (Ga, Gy, and Rf), regardless of the one used for selecting the charging profile.

To enable the BSP feature, use the following command:

```
configure mobile-gateway pdn charging bill-shock-prevention
```

To define which CC value will be reported toward the charging interfaces (Ga, Gy, and Rf), use the following command:

```
configure mobile-gateway profile charging pgw-charging report-serving-node-supplied-cc
```

12.7 Gy Rf synchronization events

This feature provides additional Rf charging information addition triggers related to online/offline charging at the PGW.

CLI commands can be used to enable information addition triggers. When a CCR-U is sent to the OCS for a Rating Group with one of the Reporting Reason codes shown in *Change condition values populated at the SDC level*, the corresponding service data containers (SDCs) for the Rating Group are cached and the maximum number of change conditions partial record trigger count is incremented. An ACR interim is sent out to the OFCS if the configured maximum number of change conditions (using the max-change-condition command) is met.

Table 36: Change condition values populated at the SDC level

Change condition	Change condition value	Reporting-Reason
Final	522	Reporting-Reason=Final
Validity Time	523	Reporting-Reason=Validity-Time
Rating Condition Change	524	Reporting-Reason=Rating-Condition-Change
Forced Reauthorization	525	Reporting-Reason=Forced-Reauthorization
Quota Hold Time	526	Reporting-Reason=Quota-Hold-Time

Use the following commands to enable the information addition triggers in the context indicated:

```
configure mobile-gateway profile charging pgw-charging off-line charging-info-trigger
  gy-final
  gy-forced-reauthorization
  gy-qht
  gy-quota-exhausted
  gy-rating-condition-change
  gy-tariff-time-change
  gy-threshold-vqt
  gy-validity-time
```

12.8 CSG reporting

A Closed Subscriber Group (CSG) identifies a group of subscribers who are permitted to access a group of cells that have restricted access.

The CSG can consist of a group of users who are allowed to connect to a single small cell hosted by a residential customer, or to a group of small cells hosted by a business. The CSG ID is used to identify a unique group of subscribers. A CSG list is created based on input from the host of either a single or group of small cells, and the operator. The CSG access list is referenced by the MME to validate subscription at a CSG cell, where it performs access or handover when the UE connects to a small cell. Each Home eNodeB (HeNB) provides a list of supported CSG IDs to the MME, which is used for CSG paging optimization during S1 setup.

Small cells can operate in either closed, hybrid, or open access mode.

- Closed access mode provides services only to its associated CSG group members.
- Hybrid access mode provides services to its associated CSG group members and to non-CSG members.
- Open access mode operates as a normal eNB, but with a small cell.

The UE is provided access to the CSG enabled cell by the MME. Once access is granted, the PGW requests the MME to enable CSG-ID reporting for policy enforcement and charging functions. For example, a QCI update if the UE moves between the macro eNB and the CSG-enabled small cell.

PGW supports, including Change Reporting Action IE in GTP message toward MME, processing CSG information received in GTP message from MME, event triggers on Gx interface for policy enforcement and event triggers for CSG update to offline and online charging systems.

For information about CLI configuration of CSG reporting, see the `charging-info-trigger` and related CLI commands.

Example: CSG charging information commands

```
config>mobile>profile>charging>pgw>offline># info
-----
enable-offline
pri-cdf-dia-peer "Peer-Profile2"
charging-info-trigger
  qos-change
  user-loc-change
  change-csg-cell
  change-csg-subscribed-hybrid-cell
  change-csg-unsubscribed-hybrid-cell
exit
partial-record-closure-trigger
  ms-time-zone-change
  rat-change
exit
-----
```

To view information about CSG reporting updates, use the following commands:

```
show mobile-gateway pdn ref-point-stat>s11
```

```
show mobile pdn ref-point-stat s5
```

```
show mobile-gateway pdn ref-point-stat s8
```

To view CSG event trigger configuration information, use the following commands:

```
show mobile-gateway profile charging pgw-charging
```

```
show mobile-gateway profile charging sgw-charging
```

12.8.1 CSG Reporting Action IE handling

The GTP-C profile configuration provides an option to exclude CSG reporting actions IEs in CS Response messages. Including CSG Reporting Action IEs in the CS Response messages from the PGW informs the MME to report when the user CSG information changes.

CSG change reporting can be avoided by configuring the GTP options profile to exclude CSG Reporting Action IE in CSResp message.

The CSG reporting action IE can be enabled or disabled using the `exclude-csg-reporting-action` command in the `config>mobile>profile>gtp>gtpv2-options` context.

12.9 Location-based charging in the OFCS

The OFCS location-based charging feature supports the configuration of user location change charging information triggers within a charging profile (SGW, PGW, TDF). The user - loc - change trigger can be enabled to populate the service change condition with corresponding specific (CGI-SAI, eCGI, TAI, RAI) change condition values, as applicable. The user - loc - change trigger also supports an optional parameter cc - user - loc - change, which populates the service change condition with the generic userLocationChange.

The ability to configure the following granular (specific) charging information triggers within the charging profile is also supported:

- CGI-SAI change
- eCGI change
- RAI change
- TAI change

The specific granular triggers and the cc - user - loc - change option can coexist for the charging information trigger. If one of the specific granular change conditions is reported while the specific granular trigger is also armed, the granular value is populated in the service change condition, instead of the generic userLocationChange value indicated by cc - user - loc - change.

The following examples describe how the service change condition is populated when the serving node is an MME:

- If user - loc - change is enabled without the cc - user - loc - change option specified, the PGW/SGW/TDF asks the serving node for both the eCGI and TAI (START REPORTING ECGI AND TAI). The service change condition is then populated by whichever particular (granular) change is notified.
- If user - loc - change is enabled with the cc - user - loc - change option specified, and the tai - change trigger is also enabled, the following occurs:
 - If only a TAI change condition is reported by the MME, the service change condition will be populated as TAISChange.
 - If only an eCGI change is reported by the MME, the generic userLocationChange will be populated.



NOTE:

The change condition is a bit-map within a PGW-CDR and therefore more than one change condition can be populated. For the SGW, only a single change condition can be included. Levels of granularity are supported in this case. For example, if both the tai - change and ecgi - change triggers are enabled then the more granular eCGI change will be populated.

The following commands can be used to configure the user location change and specific granular charging information triggers for the PGW, SGW, or TDF charging profiles:

```
configure mobile-gateway profile charging {pgw-charging | sgw-charging | tdf-  
charging}  
    off-line charging-info-trigger  
        ecgi-change  
        tai-change  
        cgi-sai-change  
        rai-change  
        user-loc-change
```

To view the charging profile configuration, use the following commands:

```
show mobile-gateway profile charging  
    pgw-charging  
    sgw-charging  
    tdf-charging
```

12.10 Location-based charging in the OCS

For online charging, similar location based charging is supported using DCCA profile event triggers. The ability to configure the following granular (specific) location triggers within the DCCA profile is supported:

- change in location RAC (Routing Area Code)
- change in location LAC (Location Area Code)
- change in location ECGI (EUTRAN Cell Global Identifier)
- change in location CellId (Cell Identity)
- change in location TAC (Tracking Area Code)

The following usage notes apply for this functionality:

- If a generic CHANGE_IN_LOCATION event trigger is received in conjunction with any of these specific event triggers, the CHANGE_IN_LOCATION trigger will be ignored, and the specific trigger will be enforced.
- If both change in location RAC and change in location LAC triggers are enabled, then both triggers are reported to the OCS.

To enable the Gy event triggers locally on the gateway, including the change in location TAC (Tracking Area Code), use the following commands:

```
configure mobile-gateway profile dcca event-triggers  
    user-loc-change-rac  
    user-loc-change-lac
```

```
user-loc-change-ecgi
user-loc-change-cellid
user-loc-change-tac
```

A `trigger-override-option` command is also supported in the `config>mobile>profile>dcca>event-triggers` context, to provide the following options for enforcing the event triggers at the gateway:

locally configured	ignores OCS-subscribed triggers
OCS subscribed	ignores locally-configured triggers (default)
locally configured and OCS subscribed	enforces both OCS-subscribed and locally-configured triggers

12.11 User location change notification over Gn/Gp

The GGSN supports user location reporting functionality as per 3GPP TS 29.060, section 7.5B.1.1 & 7.5B.1.2 and section 7.7.80.

Population of the MS Info Change Reporting Action IE in Create PDP Context Response and Update PDP Context Response messages is supported over the Gn/Gp interface. The MS Info Change Reporting mechanism is also supported, including MS Info Change Notification Request and Response messages. Starting and stopping of MS Info Change Reporting mechanism is facilitated by including the MS Info MS Info Change Reporting Action IE in the following messages:

- Create PDP context response
- Update PDP context response
- GGSN-Initiated Update PDP Context Request

MS Info Change Reporting Action IE values are populated based on PCRF/OCS subscription to corresponding event triggers and/or location reporting event triggers that are enabled within the charging profile for offline charging. Per GTPv1 standard, the MS Info Change Reporting Action IE value over Gn/Gp can be one of the following:

0	stop reporting
1	start reporting cgi/sai
2	start reporting rai

When the `user-loc-change` event trigger and the more specific `rai-change` and `cgi-sai-change` are enabled (subscribed to), the more granular option (start reporting cgi/sai) is subscribed toward the SGSN.

The GGSN supports the following KPI/KCIs for Change Notification Request and Response message counts over Gn/Gp:

- MS Info Change Notification Request messages received
- Successful MS Info Change Notification Response messages sent
- Unsuccessful MS Info Change Notification Response messages sent

To view statistics for change reporting action messages for the Gn/Gp interface, use the command `show>mobile>pdn>ref-point-stats gn`.

12.12 Presence Reporting Area information reporting

The Core Network Overload - User Location Information (CNO-ULI) reporting feature helps operators deploy differentiated charging and other business logic functions based on UE location without incurring massive network signaling loads. The CNO-ULI reporting feature includes support for Presence Reporting Area (PRA) information reporting.

In some use cases, policy control and charging decisions, such as QoS modification or charging rate change, depend on whether the UE is located inside or outside a specific area of interest (a PRA), and especially on whether the UE enters or leaves that specific area of interest.

PRA is an area defined within the 3GPP Packet Domain for the purpose of reporting UE presence within that area for policy control and/or charging reasons. A PRA for E-UTRAN can consist of a set of neighbor or non-neighbor tracking areas, or eNBs, and/or cells. A PRA for GERAN/UTRAN can consist of a set of neighbor or non-neighbor cells/service areas or routing areas. There are two types of PRAs. UE-dedicated PRAs and core network pre-configured PRAs that apply to an MME pool.

The UE presence in PRA charging information, as defined in TS 23.401 and TS 23.203, is collected for each UE by the SGWs, PGWs, and TDFs which are serving that UE.

During IP-CAN session establishment, a PRA for the PDN connection is determined. The PCRF, when deployed, may provide a PRA to the PGW. The PCRF-selected PRA is then delivered to the TDF and OCS. When the PCRF does not provide a PRA, the OCS may provide the PRA to the PGW. Only one PRA can be selected per PDN connection. The PCEF will ignore any Presence-Reporting-Area-Identifier AVPs provided during the lifetime of the IP-CAN session after it is established.

During IP-CAN session establishment, the PCRF, TDF (via PCRF), and OCS may independently subscribe to notifications of whether the UE is entering or leaving a PRA. At any time during the IP-CAN session, the PCRF, TDF (via PCRF), and OCS may independently subscribe or unsubscribe to these notifications. When subscribed, the UE presence status in the PRA at the time of the subscription, and subsequently, whether the UE enters or leaves the PRA, will be reported by the PGW or TDF to the OCS for online charging when subscribed by OCS, and by the SGW, PGW, and TDF for offline charging.

For offline charging, the initial status of the UE presence in the PRA is captured in current counts and reported on the first charging event. For online charging, the initial status is reported when received by the PCEF if a quota has already been requested for the service usage, otherwise the initial status is reported on the first quota request.

12.12.1 Gx PRA functionality

The PGW indicates support for PRA information reporting in the Supported-Features AVP. The `config>mobile>profile>gx>supported-features>cno-uli` command is used to enable or disable this feature indication.

The following event trigger is supported over Gx:

CHANGE_OF_UE_PRESENCE_IN_PRESENCE_REPORTING_AREA_REPORT (48)

Table 37: PRA information reporting AVPs in Gx messages

AVP	CCR-I	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA
Presence-Reporting-Area-Information		√	√					
Presence-Reporting-Area-Identifier		√	√					
Presence-Reporting-Area-Status			√					
Presence-Reporting-Area-Elements-List		√						
Event-Trigger		√	√	√			√	
Change-of-UE-Presence-in-Presence-Reporting-Area-Report		√	√	√			√	

12.12.2 S5 / S8 PRA functionality

Table 38: PRA information reporting IEs supported in GTP-C messages

IE	Create-Session-Req	Create-Session-Resp	Create-Bearer-Req	Create-Bearer-Resp	Update-Bearer-Req	Update-Bearer-Req	Modify-Bearer-Req	Modify-Bearer-Resp
Presence-Reporting-Area-Information	√			√		√	√	
Presence Reporting Area Action		√	√		√			√

12.12.3 Offline charging PRA functionality

The SGW, PGW, and TDF support the following charging information addition trigger:

Change of UE Presence in Presence Reporting Area

The following Change-Condition value is defined:

24 (Change of UE Presence in Presence Reporting Area)

Table 39: PRA Information Reporting AVPs in Rf Messages Supported by SGW and PGW

AVP	ACR(Start)	ACR(Interim)	ACR(Stop)
Presence-Reporting-Area-Information	√	√	√
Presence-Reporting-Area-Identifier	√	√	√
Presence-Reporting-Area-Status	√	√	√
Presence-Reporting-Area-Elements-List			

At the SDC level, the Presence-Reporting-Area-Status contains the status of the UE presence (that is, whether the UE is inside or outside) in the PRA identified by the Presence-Reporting-Area-Identifier contained in the PGW-CDR or TDF-CDR main level. This is included in the SDC when the initial status is reported after the PGW-CDR or TDF-CDR is opened, or if the previous container's change condition is "Presence in Presence Reporting Area Change".

At the TDV level - the Presence-Reporting-Area-Status contains the status of the UE presence (that is, whether the UE is inside or outside) in the PRA identified by Presence-Reporting-Area-Identifier contained in the SGW-CDR main level. This is included in the traffic data container when the initial status is reported after the SGW-CDR is opened, or if a previous container's change condition is "Presence in Presence Reporting Area Change".

12.12.4 Online charging PRA functionality

Table 40: PRA Information Reporting AVPs in Gy Messages Supported by PGW

AVP	CCR-1	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA	ASR	ASA
Presence-Reporting-Area-Information	√	√	√		√					
Presence-Reporting-Area-Identifier	√	√	√		√					
Presence-Reporting-Area-Status	√	√	√		√					
Presence-Reporting-Area-Elements-List										
Event-Trigger		√	√	√	√					
Change-of-UE-Presence-in-Presence-Reporting-Area-Report		√	√	√	√					

The PGW and SSG/TDF support the Event-Trigger Trigger-Type:

73 CHANGE_OF_UE_PRESENCE_IN_PRESENCE_REPORTING_AREA

12.12.5 Sd-related PRA functionality

Table 41: PRA information reporting AVPs in Sd messages supported by SSG and TDF

AVP	TSR	TSA	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA
Presence-Reporting-Area-Information	√						√	
Presence-Reporting-Area-Identifier	√						√	
Presence-Reporting-Area-Status	√						√	
Presence-Reporting-Area-Elements-List								
Event-Trigger		√	√				√	√
Change-of-UE-Presence-in-Presence-Reporting-Area-Report		√	√				√	√

12.12.6 PRA configuration restriction and usage notes

- Event-Report-Indication:
 - From the TDF PCRF: used to subscribe to the required event triggers (per Gzn and Gyn event trigger needs)
 - From the PCRF TDF: used to report an event coming from the PCEF (PGW) and relevant (event specific) information to TDF
- Feature compliance is as per R12 of 3GPP specifications.
- This feature is not applicable to Gn and Gp interfaces.
- Only one PRA may be selected per PDN connection.

- When the UE is at non-EPS access, only the PRA ID is reported to the peer elements and not the status.

12.12.7 PRA CLI commands

To configure the PRA, use the following commands:

```
configure mobile-gateway profile gx supported-features cno-uli
```

```
configure mobile-gateway profile sd supported-features cno-uli
```

```
configure mobile-gateway profile charging pgw-charging off-line charging-info-trigger change-ue-presence-pra
```

```
configure mobile-gateway profile charging sgw-charging off-line charging-info-trigger change-ue-presence-pra
```

```
configure mobile-gateway profile charging tdf-charging off-line charging-info-trigger change-ue-presence-pra
```

```
cconfigure mobile-gateway profile dcca event-triggers change-ue-presence-pra
```

```
configure mobile-gateway profile charging gy-avp-options include-pra-info
```

```
configure mobile-gateway profile charging cdr-avp-options  
    include-pra-info  
    include-sdc-pra-status  
    include-tdv-pra-status
```

12.13 Multiple PRA reporting

3GPP Release 12 supported a single PRA for a PDN connection. 3GPP Release 14 enhances this functionality to support multiple PRAs for a PDN connection. With this feature, the PCRF and OCS can subscribe to reporting of UE presence in multiple PRAs. The PCRF and OCS subscribed PRAs can be different.



NOTE:

Multiple PRA for offline charging is currently not supported.

During an IP-CAN session establishment or modification, the PCRF determines, based on the local configuration, if the UE is located in an access type that supports reporting of changes of a UE presence in a PRA. If the access type supports it, the PCRF may subscribe to a Change of UE Presence in Presence Reporting Area reporting at any time during the IP-CAN session.

When a Change of UE Presence in Presence Reporting Area reporting is activated, the PCRF provides all of the PRA identifiers to be activated for Core Network pre-configured PRAs. The PCRF can change (activate/modify/remove) these PRAs by providing the updated PRA identifiers to the PCEF. PRA identifiers define the PRAs to monitor for the UE. The PCRF may unsubscribe to the Change of UE Presence in Presence Reporting Area reporting at any time during the IP-CAN session.

When the PCRF or OCS subscribes to reporting change of UE presence in a particular PRA and reporting in that PRA has not been activated before, the PCEF activates the IP CAN procedure that reports when the UE enters or leaves a PRA. An initial report is sent when the IP CAN procedure is activated.

The PCEF combines requests from the PCRF and the OCS, and reports the following:

- PRA identifiers
- indications of whether the UE is inside or outside the PRAs
- indications of whether the corresponding PRAs is set to inactive by the serving node to the PCRF and/or OCS

The PCEF stores the PCRF or OCS subscriptions to Change of UE Presence in Presence Reporting Area reporting and forwards the PRA identifiers and indications received from the serving node according to the corresponding subscription.

For a PRA set subscribed to by the PCRF or OCS, the PCEF receives from the serving node, an identifier for the PRA set, along with individual identifiers belonging to the PRA set and indications of whether the UE is inside or outside the individual PRAs.

12.13.1 Gx AVPs and messages

Table 42: Gx AVPs and messages for multiple PRA

Attribute name	AVP code	Clause defined (29.212)	Value type	AVP Flag Rules				May Encr.	Acc. type	Applicability
				Must	May	Should not	Must not			
PRA-Install	2845	5.3.130	Grouped	V	P	–	M	–	3GPP-EPS	Multiple-PRA

Table 42: Gx AVPs and messages for multiple PRA (continued)

Attribute name	AVP code	Clause defined (29.212)	Value type	AVP Flag Rules					Acc. type	Applicability
				Must	May	Should not	Must not	May Encr.		
PRA-Remove	2846	5.3.131	Grouped	V	P	–	M	–	3GPP-EPS	Multiple-PRA
Presence-Reporting-Area-Elements-List	2820	5.3.107	OctetString	V	P	–	M	Y	3GPP-EPS	CNO-ULI Multiple-PRA
Presence-Reporting-Area-Identifier	2821	5.3.108	OctetString	V	P	–	M	Y	3GPP-EPS	CNO-ULI Multiple-PRA
Presence-Reporting-Area-Information	2822	5.3.109	Grouped	V	P	–	M	Y	3GPP-EPS	CNO-ULI Multiple-PRA
Presence-Reporting-Area-Status	2823	5.3.110	Unsigned32	V	P	–	M	Y	3GPP-EPS	CNO-ULI Multiple-PRA

Table 43: PRA-Install AVP in messages supported by PGW

AVP	CCR-I	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA
PRA-Install	N/A	√	N/A	√	N/A	N/A	√	N/A

Table 43: PRA-Install AVP in messages supported by PGW (continued)

AVP	CCR-I	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA
Presence-Reporting-Area-Information	N/A	√	N/A	√	N/A	N/A	√	N/A

Table 44: PRA-Remove AVP in messages supported by PGW

AVP	CCR-I	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA
PRA-Remove	N/A	N/A	N/A	√	N/A	N/A	√	N/A
Presence-Reporting-Area-Identifier	N/A	N/A	N/A	√	N/A	N/A	√	N/A

Table 45: Event-Trigger AVP in messages supported by PGW

AVP	CCR-I	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA
Event-Trigger	N/A	√	√	√	√	N/A	√	N/A
Change-of-UE-Presence-in-Presence-Reporting-Area-Report	N/A	√	√	√	√	N/A	√	N/A

CHANGE_OF_UE_PRESENCE_IN_PRESENCE_REPORTING_AREA_REPORT (48) — This value will be used in CCA or RAR commands by the PCRF to indicate that the PCEF will report the event when the user enters or leaves the area as indicated in the Presence-Reporting-Area-Information AVP. This includes reporting the initial status at the time the request for reports is initiated.

When used in a CCR command, this value indicates whether the UE is within or outside the requested PRA. The Presence-Reporting-Area-Identifier AVP and Presence-Reporting-Area-Status AVP will be provided in the same request included in the Presence-Reporting-Area-Information AVP. This is applicable only to 3GPP-EPS access type and to functionality introduced with the CNO-ULI or Multiple-PRA feature.

Table 46: Supported Features of Feature-List-ID 1 used in Gx

Feature bit	Feature	M/O	Description
23	CNO-ULI	O	This feature indicates support for Presence Reporting Area Information reporting. If the PCEF supports this feature, the PCRF will behave as described in 29.212 Annex B.3.16.

Table 47: Supported Features of Feature-List-ID 2 used in Gx

Feature bit	Feature	M/O	Description
3	Multiple-PRA	O	This feature indicates support for Multiple Presence Reporting Area Information reporting. If the PCEF supports this feature, the PCRF shall behave as described in 29.212 Annex B.3.17.

**NOTE:**

CNO-ULI will only be used when the PCEF and/or the PCRF does not support Multiple-PRA and both PCEF and PCRF support CNO-ULI.

12.13.2 Support for multiple PRAs on Gy

During a credit management session establishment or modification, the OCS determines, based on the local configuration, if the UE is located in an access type that supports reporting changes of UE presence in PRA. If the access type supports it, the OCS may subscribe to a Change of UE Presence in Presence Reporting Area reporting at any time during the credit management session. During IP-CAN session establishment or modification, and independently from the PCRF, the OCS may provide to the PGW, a list of:

- Presence Reporting Area (PRA) Identifiers to be activated for Core Network pre-configured Presence Reporting Areas
- Presence Reporting Area (PRA) Identifiers and their elements for UE-dedicated Presence Reporting Areas

By providing the corresponding lists in Presence-Reporting-Area-Information AVPs, the OCS subscribes to the notifications of whether the UE is entering or leaving a PRA through the Trigger-Type AVP.

The OCS may change (activate/modify/remove) the PRAs to be reported by providing the updated PRA identifiers to the PCEF.

The OCS may unsubscribe to a Change of UE Presence in Presence Reporting Area at any time during the credit management session.

Table 48: Event-Trigger AVP in messages supported by PGW

AVP	CCR-1	CCA-I	CCR-U	CCA-U	CCR-T	CCA-T	RAR	RAA	ASR	ASA
Event-Trigger		✓	✓	✓	✓					
Change-of-UE-Presence-in-Presence-Reporting-Area		✓	✓	✓	✓					

73 CHANGE_OF_UE_PRESENCE_IN_PRESENCE_REPORTING_AREA — This value is used to indicate a request to report the event of the user entering/leaving the area as indicated in the Presence-Reporting-Area-Information AVP. This includes reporting the initial status at the time the request is initiated. When used in a CCR, the Presence-Reporting-Area-Identifier AVP and Presence-Reporting-Area-Status AVP will be provided in the Presence-Reporting-Area-Information AVP with the event report. For the UE-dedicated presence area, the Presence-Reporting-Area-Elements-List AVP will be provided in the Presence-Reporting-Area-Information AVP. When used in a CCA with a list of PRAs, the new list of PRAs will supersede the previous provided list. In case this trigger is not included in the Trigger-Type AVP, it indicates all PRAs are unsubscribed.

The PGW supports Presence-Reporting-Area-Information AVPs under the PS-Information grouped AVP in the following messages:

- CCR-U
- CCR-T
- CCA-I
- CCA-U

12.13.3 GTP IEs and messages

With the introduction of Multiple PRA support in Release 14 feature Improvements of Awareness of User Location Change (AULC), the Presence Reporting Area Action and Presence Reporting Area Information will be included in the relevant messages with multiple instances. Because these IEs are used for the same purpose semantically, they can use the same instance number.

Table 49: PRA information reporting IEs for multiple PRAs supported in GTP-C messages

IE	Create-Session-Req	Create-Session-Resp	Create-Bearer-Req	Create-Bearer-Resp	Update-Bearer-Req	Update-Bearer-Resp	Modify-Bearer-Req	Modify-Bearer-Resp	Change-Notification-Req	Change-Notification-Resp
Presence-Reporting-Area-Information	√			√		√	√		√	
Presence Reporting Area Action		√	√		√			√		√

Refer to 29.274 section 8.109 *Presence Reporting Area Information* and 29.274 section 8.108 *Presence Reporting Area Action* for the definition of these information elements.

12.13.4 Configuration restriction and usage notes

The following are restrictions and usage notes for the Multiple PRA feature.

- Support for Multiple PRAs for offline charging is currently not supported.
- When enabled, the Multiple PRA feature needs to be supported uniformly across the relevant network elements and interfaces. (In other words, support for Release 12 Single PRA and Release 14 Multiple PRA cannot be mixed across network elements).

12.13.5 CLI commands

The following commands are used to enable the multiple PRA feature:

```
configure mobile-gateway pdn charging multiple-pra
```

```
configure mobile-gateway profile gx supported-features multiple-pra
```

12.14 NarrowBand-IoT charging

The following NB-IoT-Charging functionalities have been implemented according to 3GPP Release 13, December 2016 charging specifications:

- Non-IP traffic

- PLMN rate control
- APN rate control
- Mobile Originated (MO) exception data counter

This feature adds support for the following Offline Charging requirements for NB-IoT, as defined in 3GPP Release 13, December 2016:

- 32.240 Charging architecture and principles
- 32.251 Packet Switched (PS) domain charging
- 32.295 Charging Data Record (CDR) transfer
- 32.297 Charging Data Record (CDR) file format and transfer
- 32.298 Charging Data Record (CDR) parameter description

**NOTE:**

NB-IoT-Charging is not supported in the Online Charging interface.

The metering of non-IP traffic is implemented on the S11-U and S1-U interfaces (MO and Mobile Terminated (MT) directions) affecting SGW and PGW CDRs. A single PCC rule is used with the service data flow descriptor to match the whole non-IP data traffic.

The NB-IoT Offline Charging triggers are the following:

- a change in the UE's user plane (a switch between the S11-U and S1-U interface) is a charging trigger for charging information addition (closes containers and opens new ones), affecting SGW-CDRs
- the serving PLMN rate control change is a charging trigger for charging information addition (closes containers and opens new ones), affecting SGW and PGW CDRs
- the APN rate control change is a charging trigger for charging information addition, affecting PGW-CDRs
- the reception of the MO exception data counter is a charging trigger for CDR closure (it closes a CDR and opens a new one), affecting SGW and PGW CDRs

To configure the NB-IoT Offline charging triggers, use the following commands:

```
configure mobile-gateway profile charging pgw-charging off-line charging-info-trigger apn-rate-control-change
```

```
configure mobile-gateway profile charging pgw-charging off-line charging-info-trigger plmn-rate-control-change
```

```
configure mobile-gateway profile charging sgw-charging off-line charging-info-trigger plmn-rate-control-change
```

```
configure mobile-gateway profile charging pgw-charging off-line partial-record-closure-trigger mo-exception-data-receipt
```

```
configure mobile-gateway profile charging sgw-charging off-line partial-record-closure-trigger mo-exception-data-receipt
```

The NB-IoT-related CDR fields are as follows:

- SGi PtP tunneling method, applicable to PGW-CDRs
- UNI PDU CP-only flag, applicable to SGW and PGW CDRs
- PDP/PDN type extension, applicable to SGW and PGW CDRs
- serving PLMN rate control, applicable to SGW and PGW CDRs
- APN rate control, applicable to PGW-CDRs
- MO exception data counter, applicable to PGW-CDRs

To configure the NB-IoT-related CDR fields, use the following commands:

```
configure mobile-gateway profile charging cdr-avp-options sgi-ptp-tunnelling-method
```

```
configure mobile-gateway profile charging cdr-avp-options uni-pdu-cp-only-flag
```

```
configure mobile-gateway profile charging cdr-avp-options pdp-pdn-type-extension
```

```
configure mobile-gateway profile charging cdr-avp-options serving-plmn-rate-control
```

```
configure mobile-gateway profile charging cdr-avp-options apn-rate-control
```

```
configure mobile-gateway profile charging cdr-avp-options mo-exception-data-counter
```

**NOTE:**

- In case of simultaneous triggers for Charging Information Addition (closed containers but not CDRs) or for CDR closure (closed containers and CDRs), the NB-IoT triggers will take a lower precedence than the already implemented triggers.
- The NB-IoT RAT types will be visible to the SGW and PGW CDRs. The NB-IoT RAT type is received from GTPv2 and copied to the charging interfaces. This affects PGW and SGW CDRs.

To view the include/exclude configuration about the NB-IoT CDR fields, use the following command:

```
show mobile-gateway profile charging cdr-avp-options
```

12.15 Rating group level charging accuracy for delayed accounting

This section describes how charging accuracy is improved at Flow Based Charging level.

If the UE traffic is generated, the session may end before this traffic is mapped to a policy rule. With the `disable-delayed-accounting` approach, these packets are accounted to the correct bearer, but the Rating Group/Service ID identified by Flow Based Charging is usually incorrect. This results in a mismatch between the received traffic and the reported Rating Group/Service ID in CDRs.

A new Rating Group for the unidentified flows can be configured for this traffic to be consistently reported.

To configure a Rating Group for the incomplete-signaling functionality, use the following command.

```
configure mobile-gateway pdn apn charging incomplete-signaling rating-group
```

To configure the charging method for the incomplete-signaling functionality, use the following command.

```
configure mobile-gateway pdn apn charging incomplete-signaling charging-method
```

To configure the service identifier for the incomplete-signaling functionality, use the following command.

```
configure mobile-gateway pdn apn charging incomplete-signaling service-id
```

To configure the Usage-Reporting-Rule identifier for the incomplete-signaling functionality, use the following command.

```
configure mobile-gateway pdn apn charging incomplete-signaling urr-id
```

**NOTE:**

If the rating-group and service-id used for the incomplete-signaling are also used in a charging rule, the charging-method must match that of the charging rule. The urr-id must have a unique value that is not used in any of the policy rules.

12.16 OAM switchover in case of lost access to storage nodes

The OAM node provides access to two different storage nodes, CF-1 and CF-2. If access to a storage node is lost, you can configure CMG to trigger an OAM switchover, so that after the switchover, the secondary OAM node continues to write to the disk.

To enable the OAM switchover, set the `cffsw` SMBIOS attribute to 1. For more information about the `cffsw` attribute, refer to the *CMG Installation Guide*.

**NOTE:**

This feature only applies to CMG.

When the OAM switchover is enabled, the switchover takes place when a CF node fails. The OAM node waits for 30 seconds for the CF storage node to recover. If the standby OAM is not available at that time, the switchover is not performed, and the check is repeated every 15 seconds, until either the standby OAM becomes available or the CF storage node recovers.

**NOTE:**

- Nokia recommends that the storage nodes are always mapped to different physical disks.

- Check and repair, if needed, the file system on the failed CF drive before placing it back into service.
- For information about limitations in the current release, refer to the applicable *7750 MG and CMG Release Notes*.

12.17 Offline charging

This key offline charging system (OFCS) functionality is supported by the CMG.

For more information about OFCS functionality related to gateway deployments, see the gateway-specific sections of this guide.

12.17.1 CDR generation

MG allows generation of SGW-CDRs, PGW-CDRs, TDF-CDRs, and EPDG-CDRs.

12.17.1.1 Configurable CDR AVP options

A CDR AVP options profile can be created to control which values are included and excluded in records. This profile can be referenced at the PDN and APN levels. The CDR AVP options profile is used for both Ga CDRs and Rf ACRs.

The profile can be applied to the PDN using the following CLI commands:

- `config>mobile>pdn>charging>cdr-avp-options-profile`
- `config>mobile>pdn>apn>charging>cdr-avp-options-profile`

12.17.1.2 Disabling event triggers on Gx

The mobile gateway allows the operator to disable certain event triggers on the Gx interface which according to 3GPP specifications are enabled without the PCRF subscription to them. This reduces Gx interface signaling when notification for certain events, is not required.

If this functionality is not configured, the specific event triggers are enabled by default, and the PCRF is informed about those events even if the PCRF has not subscribed to them.

To disable the event triggers that are enabled by default, use the following commands:

```
configure mobile-gateway profile gx forced-triggers disable-ambr-modification-failure
```

```
configure mobile-gateway profile gx forced-triggers disable-default-eps-bearer-qos-change
```

```
configure mobile-gateway profile gx forced-triggers disable-default-eps-bearer-qos-modification-failure
```

```
configure mobile-gateway profile gx forced-triggers disable-qos-change
```

12.17.1.3 Minimum data threshold for CDR generation

This feature reduces the number of generated CDRs by inhibiting the generation of time-based partial CDRs which contain a small amount of total volume.

Although there are several charging triggers which can close partial CDRs, few happen frequently. A large portion of CDRs is generated because of a set time limit (for example, every hour a CDR is generated). When there is no traffic, those CDRs carry none or little chargeable information, but require the same resources as CDRs reporting higher volumes.

With this feature, if the total volume is less than a configurable threshold, the time-based partial CDR is not generated. The volume is stored so that it can be reported in the next CDR. If the total volume is higher than the configured threshold, then the CDR is generated and the volume is reported in that CDR.

This feature does not prevent the generation of CDRs for triggers other than the configured time limit. The reason is that other charging triggers are used by operators to select the tariff for billing the UE. For example, if there is a change in the RAT, the operator may want to charge differently according to the RAT type. In this case, the CDR needs to be closed because of the change in RAT, even if the volume is less than the configured threshold. If different charging is not required (for example, for RAT type), then operators can disable charging triggers in the gateway configuration.



NOTE:

This feature does not apply to the Rf interface because it only affects time-limit reporting (that is, time-limit CDRs). Suppressing an Rf time-based interim update due to the volume being less than the minimum data threshold may cause problems at the CDF server. The CDF server may rely on the time-based interim updates to assume that the Rf session is still alive at the client.

To set the minimum data threshold value, use the following commands:

```
configure mobile-gateway profile charging epdg-charging off-line part-rec-  
closure-trig min-data-threshold
```

```
configure mobile-gateway profile charging pgw-charging off-line part-rec-  
closure-trig min-data-threshold
```

```
configure mobile-gateway profile charging sgw-charging off-line part-rec-  
closure-trig min-data-threshold
```

```
configure mobile-gateway profile charging tdf-charging off-line part-rec-  
closure-trig min-data-threshold
```

12.17.1.4 Partial record trigger configuration

Single volume limit and time limit triggers are supported for partial record closure for charging profiles, irrespective of 2G, 3G or LTE. This applies for PGW, SGW and ePDG charging profiles.

LTE has a higher traffic rate than 2G/3G. For subscribers with online charging disabled, too many CDRs can be generated for LTE, resulting in a higher processing load on the combined SGW/PGW. If a larger volume limit is configured based on the LTE traffic rate, the total actual usage might exceed the volume limit on service plans, but the PGW/GGSN does not close the PGW-CDR and sends it to the CGF until the time limit closure trigger that is specified.

To avoid over charging situations, volume limit partial record trigger can also be configured on a per RAT basis on the PGW Ga interface. This enables operators to specify a separate volume limit (partial record trigger) for supported Radio Access Technologies with their network. Time limit partial record trigger on a per RAT basis is also supported on the PGW Ga interface.

To configure the partial record trigger, use the following commands:

```
configure mobile-gateway profile charging pgw-charging off-line partial-  
record-closure-trigger  
    max-change-condition  
    min-data-threshold  
    mo-exception-data-receipt  
    ms-time-zone-change  
    plmn-change  
    rat-change  
    serving-node-change-limit  
    time-limit  
    time-limit-eutran  
    time-limit-eutran-nb-iot  
    time-limit-utran  
    volume-limit
```

```
volume-limit-eutran  
volume-limit-eutran-nb-iot  
volume-limit-utran
```

12.171.5 Generation of partial CDRs with zero volume

The mobile gateway allows operators to configure whether partial CDRs with zero volume are generated or not. This functionality allows the reduction of the number of CDRs generated when only CDRs with chargeable volume are selected for processing.

Before enabling the exclude zero volume CDR feature, you must take into account that partial CDRs with zero volume may include valuable information, such as change in RAT, MS Timezone change, PLMN change, and so on.

When the exclude zero volume CDR feature is enabled and there is a charging trigger to close the partial CDR, but the CDR contains zero volume, then the partial CDR will not be closed but will be kept open. The last CDR for the bearer will always be generated, even if it has zero volume since it marks the end of the bearer.



NOTE:

In the case of the Rf interface, ACR messages due to a time limit will be reported even if they carry zero charging data.

To enable this functionality for each CDR type, use the following commands:

```
configure mobile-gateway profile charging epdg-charging offline exclude-zero-  
vol-cdr
```

```
configure mobile-gateway profile charging pgw-charging offline exclude-zero-  
vol-cdr
```

```
configure mobile-gateway profile charging sgw-charging offline exclude-zero-  
vol-cdr
```

```
configure mobile-gateway profile charging tdf-charging offline exclude-zero-  
vol-cdr
```

To view the charging profile configuration, use the following commands:

```
show mobile-gateway profile charging epdg-charging
```

```
show mobile-gateway profile charging pgw-charging
```

```
show mobile-gateway profile charging sgw-charging
```

```
show mobile-gateway profile charging tdf-charging
```

12.17.1.6 Generation of session CDRs with zero volume

The mobile gateway allows you to configure whether CDRs, which report zero volume and are at the same time the only CDRs for the session, are generated or not.

This functionality applies when a UE attaches to the network, generates no traffic and then detaches. These sessions generate a single CDR which reports zero volume (and zero or non-zero duration).

To configure zero volume CDR generation (with zero or non-zero duration) when the last CDR is also the first CDR of the session for each CDR type, use the following commands:

```
configure mobile-gateway profile charging epdg-charging offline exclude-zero-vol-complete-cdr
```

```
configure mobile-gateway profile charging pgw-charging offline exclude-zero-vol-complete-cdr
```

```
configure mobile-gateway profile charging sgw-charging offline exclude-zero-vol-complete-cdr
```

```
configure mobile-gateway profile charging tdf-charging offline exclude-zero-vol-complete-cdr
```

To view the charging profile configuration, use the following commands:

```
show mobile-gateway profile charging epdg-charging
```

```
show mobile-gateway profile charging pgw-charging
```

```
show mobile-gateway profile charging sgw-charging
```

```
show mobile-gateway profile charging tdf-charging
```

12.17.1.7 Bearer metering in PGW CDRs

It is possible that both IP CAN Bearer Charging and Flow Based Charging (FBC) can be reported in PGW CDRs (or eG-CDRs), with the use of rating group 0 in a List-Of-Service-Data container.

If this functionality is enabled, rating group 0 is added to the PGW CDRs (or eG-CDRs) for IP CAN Bearer Charging data report, while other rating groups are used for FBC data report. If this functionality is not enabled, only FBC is reported in CDRs. For RG-0 the following conditions apply:

- Rating group 0 is reserved for IP CAN Bearer Charging. No other policy rules with this rating group should be installed.
- Partial CDRs will include rating group 0 even if there is no traffic which matches any FBC flow. Partial CDRs will report only time in this case.
- When a partial CDR is closed due to a time limit, RG-0 and all rating groups are closed.
- The volume limit that is reported in RG-0 is used for closing the partial CDR.

To enable IP CAN bearer charging through rating group 0, use the following command:

```
configure mobile-gateway profile charging pgw-charging off-line include-  
bearer-charging
```

12.17.1.8 Directory structure configuration for local CDR storage

When CDR files are closed, they are written to permanent storage.

The CMG provides the ability to configure the number of files that will be stored in a single subdirectory. When the subdirectory is full, the CMG stores the remaining CDR files in the next subdirectory.

To configure the maximum number of files per subdirectory, use the following command:

```
configure mobile-gateway profile gtp-prime-group local-cdr-storage file-params  
max-files-per-subdirectory
```

To configure the maximum number of subdirectories, use the following command:

```
configure mobile-gateway profile gtp-prime-group local-cdr-storage file-params  
max-number-subdirectories
```



NOTE:

There might be a performance degradation depending on the number of files per subdirectory. It is the operator's responsibility to choose a value for *max-files-per-subdirectory* which offers an acceptable performance according to the operator's needs.

If the configured subdirectory structure is full (all the configured subdirectories have the maximum number of allowed files per subdirectory), CDR files are not discarded. An alarm is triggered and the configured *max-files-per-subdirectory* value is ignored.

12.17.2 Offline charging record transfer

The mobile gateway supports the Rf, Ga and Bp interfaces for transferring offline charging records.

12.17.2.1 RF interface

The following Rf interface configurations are supported:

- *Configure Rf reference-point, primary, and secondary peers*
- *Configure the flash storage location*
- *Include Release Identifier Extension field in ACR file and record header*

12.17.2.1.1 Configure Rf reference-point, primary, and secondary peers

Associate a primary Diameter peer profile to the Rf reference-point.

The following configuration applies to accounting at the PDN level instead of the bearer level.

```
configure mobile-gateway pdn 1
  rf
    acct-interim-interval 44
    acct-level pdn-level
    charging-group-id
    retry-count 2
    transaction-timer 15

    interface
      diameter "rf"
    exit
    operator-string "test"
    primary-diameter-peer "Peer Profile 6"
    secondary-diameter-peer "Peer Profile 7"
  exit
```

12.17.2.1.2 Configure the flash storage location

Configure the flash storage location as: Rf server failure.

CF1 is the default primary storage location. CF2 is used if CF1 is full, or if CF2 is specified as the primary storage location using the `config>mobile>pdn >rf outage>location` command.

The following example shows the configuration of the primary flash storage location to be CF2 instead of the default CF1.

```
rf
  acct-interim-interval 5
  acct-level pdn-level
  outage
    location cf1
    no primary
  exit
  location cf2
    primary
  exit
exit
no shutdown
```

**NOTE:**

Flash storage may also be configured for the Ga interface under the config>mobile>profile>gtp-prime-group context.

12.17.2.1.3 Include Release Identifier Extension field in ACR file and record header

The Release Identifier Extension field can be included in the ACR file header and record header (similar to how it is included in the CDR file header and header format as specified by 3GPP TS 32.297 - CDR File Format and Transfer).

This functionality is applicable to the Diameter Rf interface and can be enabled using the following command:

```
configure mobile-gateway pdn rf outage acr-file-and-record-hdr-rel-id-ext
```

12.17.2.2 Ga interface

The following Ga interface configurations are supported:

- *GTP prime group configuration*
- *Duplicate CDR prevention algorithm for Ga charging*
- *CDR file transfer*
- *Ga Interface (GTP') messages*

12.17.2.2.1 GTP prime group configuration

The following CLI can be used to configure the GTP prime group for any gateway configuration that supports Ga.

```
configure mobile-gateway profile  
gtp-prime-group
```

```
configure mobile-gateway pdn ga  
gtp-prime-group gtp-prime-group  
interface-gtp-prime [router router-instance] interface-name
```

The GTP prime server which is no longer used (in a shutdown state), is deleted without PDN shutdown. There are two implications when the GTP prime server is deleted or modified on the fly and RR (round-robin) is enabled:

1. The show>mobile>pdn>bearer-context command displays no charging server address when the GTP prime server is deleted
2. If the Ga Peer is deleted or modified on the fly, the CFG address, which is sent to the SGW/SGSN in future CSResp and MBResp messages for sessions assigned to that GTP prime server, is not updated

12.17.2.2.2 Duplicate CDR prevention algorithm for Ga charging

This feature implements the duplicate prevention algorithm defined in 3GPP TS 32.295, sections 5.2.2.2, 5.2.2.3, and 5.2.2.4.

When the feature is disabled (default), the legacy behavior is used in which a DRT-REQUEST that is not acknowledged by the primary CGF is sent to the next CGF (marked as possibly duplicate). The CGF/ billing system is then responsible for implementing a duplicate prevention mechanism to check if the CDRs marked as possibly duplicate were processed by another CGF.

When this feature is enabled, the new behavior is activated in which the CDF implements the duplicate detection algorithm on top of GTP Prime signaling, as defined in 3GPP section 5.2.2.

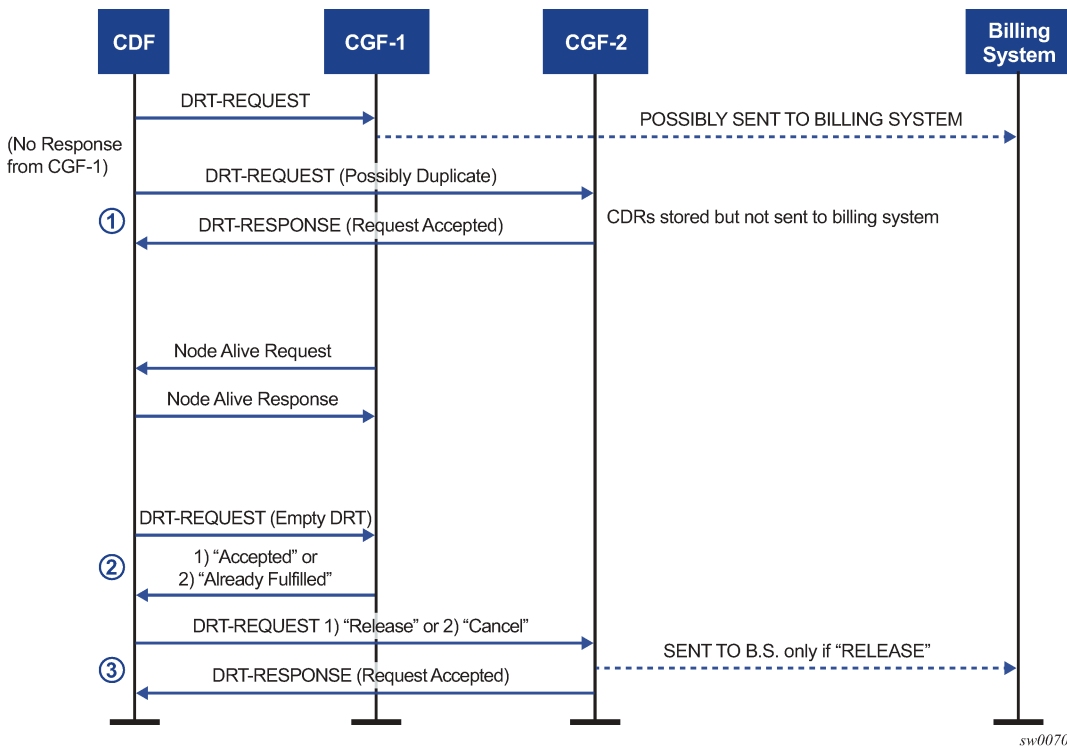
In addition to the basic duplicate detection algorithm, if there is a switchover or shutdown at the CDF while DRT-REQUESTS are waiting for acknowledgement, the operator can configure the option to discard or store possible duplicate CDRs in the flash memory (permanent storage).

The following CLI can be used to configure the duplicate CDR prevention algorithm.

```
configure mobile-gateway profile gtp-prime-group duplicate-prevention
```

```
configure mobile-gateway profile gtp-prime-group duplicate-prevention-storage
```

Figure 166: Duplicate CDR prevention algorithm



12.17.2.2.3 CDR file transfer

The mobile gateway allows you to write CDR files in a permanent storage for later processing.

By default, CDR files are transferred over the Ga interface and the Bp interface is used as backup if the Ga interface fails. You can also select the Bp interface for the main method for CDR file delivery, in which case the Ga interface is used as backup.

The following command is used to change the file transfer interface:

```
configure mobile-gateway pdn charging cdr-transfer {ga | bp}
```



NOTE:

In CMG CDR files are written to the operator's provided storage node. It is expected that operator allocates a storage node of maximum size 200 GB. Replication of the allocated storage node lies in the operator's responsibility (for example by using CEPH or other cloud replication services).

12.17.2.2.4 Ga Interface (GTP') messages

The SGW, PGW, ePDG and TDF support the GTP protocol based on GTPv0, GTPv1 and GTPv2. GTP may use UDP or TCP transport. The SGW and PGW support UDP only as the transport for Ga.

The following message types are supported:

- Echo Request
- Echo Response
- Version Not Supported
- Node Alive Request
- Node Alive Response
- Redirection Request
- Redirection Response
- Data Record Transfer Request
- Data Record Transfer Response

By default the gateway uses GTP'v1 to send Ga messages to a charging server peer. However, if GTP'v2 is enabled using the `enable-gtpv2` command in the `config>mobile>profile>gtp-prime-group>gtp-prime-server` context, the gateway will use GTP'v2 to send initial messages (such as an Echo Request). If the charging server peer does not support GTP'v2, the peer responds with a Version Not Supported message and the gateway will automatically revert to the version indicated by the message (GTP'v1 or GTP'v0).

The ability to configure the Data Record Format version used for a CDR is also supported. The Data Record Format version defaults to the baseline release of TS 32.298 that is supported by the mobile gateway. However, the operator can also override the version using the command `config>mobile>pdn>ga>data-record-format-version` for the PDN Ga interface.

**NOTE:**

The content of the CDRs is not affected by setting the Data Record Format version. The individual IEs included in the CDRs can be managed using the CLI command `config>mobile>pdn>charging>cdr-avp-options-profile`. Operators should ensure that the Data Record Format version and the CDR AVP options profile are configured to be consistent with each other.

Redirection Request messages are used by the CGF to inform the gateway that it should start sending CDRs to a different CGF or the current CGF is about to go down or may have lost connectivity with the billing server. When a Redirection Request is received, the gateway picks a locally configured CGF to start sending the CDRs.

The SGW generates SGW-CDRs and transmits them to a Charging Gateway Function using the Ga interface (GTP'). The PGW generates PGW-CDRs and transmits them to a Charging Gateway Function using the Ga interface (GTP'). This allows the PGW to support flow-based charging for EPS bearers. When configured for PGW-CDRs, the PGW supports charging for EPS bearers and PDP contexts, if the UE is supported via the Gn/Gp interface.

When configured to use the Ga interface, the ePDG generates the ePDG-CDR. The ePDG supports charging for IP-CAN bearers. Upon encountering this event, a new ePDG-CDR for this IP-CAN bearer is created and

the data volume is captured for the IPCAN bearer. The ePDG-CDR is closed when the ePDG encounters the "end of IP-CAN bearer" event.

Configure GTP options for the Ga in PDN context, using the GTP prime group profile created in the mobile gateway profile context. The GTP prime group also includes options for CDR storage management.

12.17.2.3 Bp interface

The mobile gateway allows operators to write CDR files in permanent storage for later processing. The Bp interface (FTP and SFTP) can be used for collecting CDR files.

**NOTE:**

In CMG CDR files are written to the operator's provided storage node. The maximum size of storage that the operator can allocate is 200 GB. Replication of the allocated storage node lies in the operator's responsibility (for example by using CEPH or other cloud replication services).

By default the Ga interface is used for CDR file transfers. If the CDR transfer method is Ga, the Bp interface is used as backup if the Ga interface fails. The operator also has the option to select Bp as the main method for CDR delivery.

To enable the Bp interface as the main method for CDR delivery, use the following command:

```
configure mobile-gateway pdn charging cdr-transfer bp
```

12.17.3 CDR containers report to OCS

This section describes the functionality supported for the CDR containers that have not reported charging data to the OCS when Failure Handling starts.

When a CDR is closed at the moment the Failure Handling starts, it includes containers that have been reported to OCS and containers that have not been reported to OCS. The containers that have not been sent to the OCS are marked with the value of the PS-FCI field set to 11, while the containers that have been sent to the OCS are marked with the value of the PS-FCI field set to 11.

Thus, only CDR containers with PS-FCI value set to 11 at container level are processed for billing.

To configure the PS-FCI field reporting at the container level, use command in the following context.

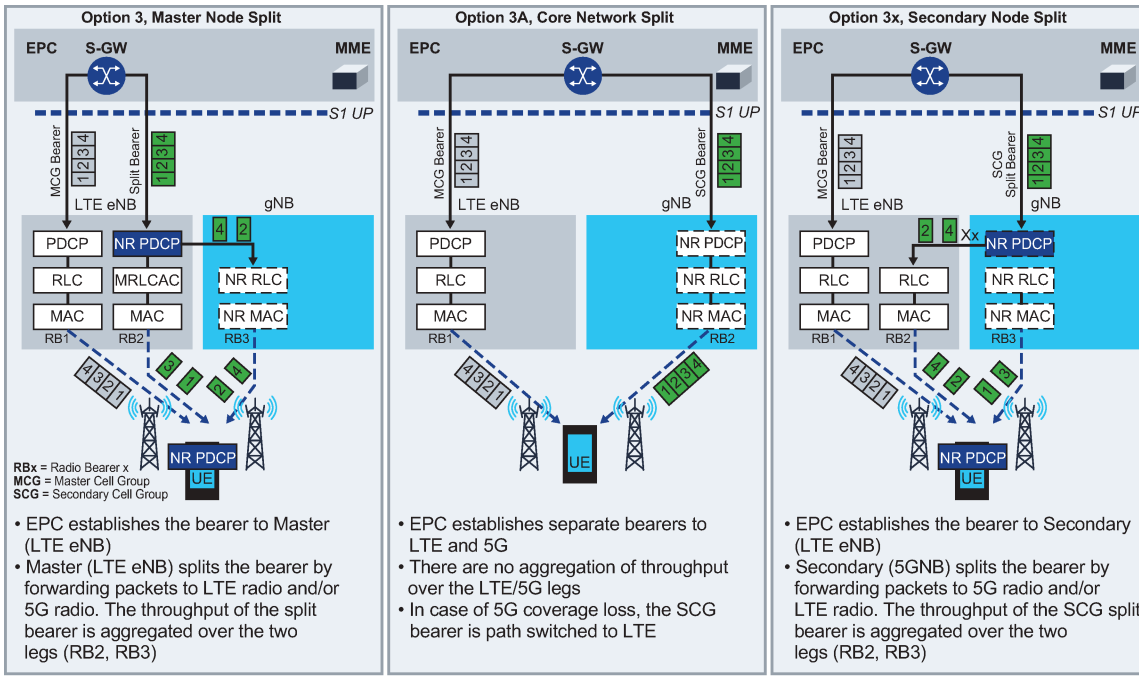
```
configure mobile-gateway profile dcca offline-interwork ps-fci-fh-container
```

12.17.4 Secondary RAT usage data reporting for 5G NR (Option 3x)

Secondary RAT usage data reporting is part of 5G NR Option 3x functionality and helps operators to monitor 5G usage.

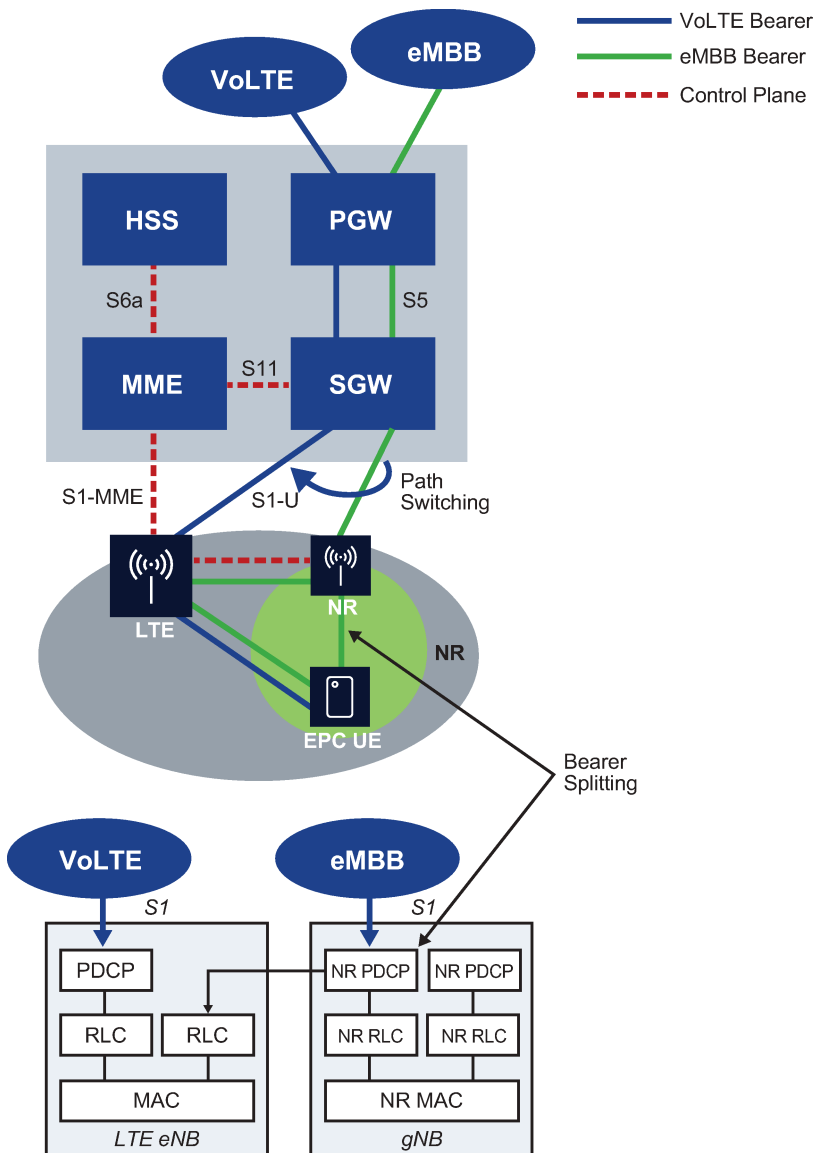
The data that is reported can be used for statistical purposes to facilitate capacity planning, or for charging purposes. The core network does not know when 5G coverage is lost and the master eNodeB has decided to split bearers. As a result, the RAN reports the secondary RAT usage to the core network per EPS bearer.

Figure 167: NSA Option 3 family — Options 3, 3A, and 3X data flows



sw0676

Figure 168: Option 3X functionality



sw0675

The following are details for Option 3x using an example scenario depicting a VoLTE bearer and an enhanced Mobile Broadband bearer (eMBB).

- When 5G coverage is lost, the gNodeB in 3X may continue to receive eMBB/5G traffic from the core and forward it to the LTE eNodeB to reach the UE.
- In the scenario shown in *Option 3X functionality* uplink traffic follows the same path; that is, from the UE to the LTE eNodeB, from the LTE eNodeB to NR, and then to the core. The core does not realize that 5G coverage has been lost and the LTE eNodeB is being used.
- Alternatively, if 5G coverage is lost, the LTE eNodeB may decide to switch the eMBB/5G traffic to it.
- For the VoLTE APN, the LTE eNodeB may decide to split the video bearer and forward it via the NR using combined Options 3 and 3X scenarios. Option 3 is used for the VoLTE APN for the video bearer and 3X is used for the eMBB APN. This decision is made at the bearer level.

- Idle/connected state applies at the UE level, that is, either the UE becomes idle and all PDN connections become idle or the UE is connected and all PDN connections are also active.
- Using secondary RAT usage reporting allows operators to charge for 5G usage separately, to monetize 5G investments. The RAN reports, via signaling, RAT usage and the core adjusts and existing messages are enhanced accordingly. The core alone cannot make the decision due to split bearers; for example, if 5G coverage is lost, the LTE eNodeB does not switch eMBB/5G traffic back to it, so although the core believes that 5G is used, this not the case.

12.17.4.1 CDR parameter, IE, and AVP bindings for EPC offline charging

Table 50: RAN Secondary RAT Usage Report Bindings

Information element	CDR parameter	AVP
RAN Secondary RAT Usage Reports; see <i>RAN Secondary RAT Usage Reports in GTP messages</i> and <i>Secondary RAT Usage Reports in Diameter Rf charging</i>	List of RAN Secondary RAT Usage Reports; see <i>List of RAN Secondary RAT Usage Reports CDR parameter</i>	RAN-Secondary-RAT-Usage Report; see <i>RAN Secondary RAT Usage Report AVP</i>

12.17.4.2 RAN Secondary RAT Usage Reports in GTP messages

When a Modify Bearer Request message that includes the RAN Secondary RAT Usage Reports IE is received from the MME, the SGW will send a Modify Bearer Request message on the S5/S8 interface, if the IRPGW flag is set to 1 in the RAN Secondary RAT Usage Reports IE.

The RAN Secondary RAT Usage Reports IE is specified in [29.274: section 8.132].

Table 51: GTP messages for RAN Secondary RAT Usage Reports

AVP	3GPP standard
Create Session Request	329.274, section 7.2.1
Modify Bearer Request	29.274, section 7.2.7
Delete Session Request	29.274, section 7.2.9.1

Table 51: GTP messages for RAN Secondary RAT Usage Reports (continued)

AVP	3GPP standard
Delete Bearer Response	29.274, section 7.2.10.2
Delete Bearer Command	29.274, section 7.2.17.1
Modify Access Bearers Request	29.274, section 7.2.24
Change Notification Request	29.274, section 7.3.14
Release Access Bearers Request	29.274, section 7.2.21

12.17.4.3 List of RAN Secondary RAT Usage Reports CDR parameter

The SGW supports the List of RAN Secondary RAT Usage Reports field in the SGW-CDR as per [32.251: 6.1.2] IP-CAN bearer charging data in the SGW (SGW-CDR). If CDR data collection is enabled, the List of RAN Secondary RAT Usage Reports will be available for each IP-CAN bearer.

The PGW supports the List of RAN Secondary RAT Usage Reports field in the PGW-CDR as per [32.251: 6.1.3] FBC IP-CAN bearer charging data in PGW (PGW-CDR). If FBC and CDR data collection are enabled, the List of RAN Secondary RAT Usage Reports will be available for each IP-CAN bearer when charging per IP-CAN session is not active, or for each IP-CAN session when charging per IP-CAN session is active.

For more information about IP-CAN Bearer Data for the SGW-CDR and PGW-CDR, refer to the *CMG CDR Description*.

12.17.4.4 Secondary RAT Usage Reports in Diameter Rf charging

The SGW and PGW support RAN Secondary RAT Usage Reports (within PS-Information) in ACR(Interim) and ACR(Stop) messages for Diameter Rf charging as per [32.251: 6.3.2] Detailed message format for offline charging.

The RAN Secondary RAT Usage Reports IE is supported in the Charging Data Request message, for the SGW and PGW. The SGW and PGW communicate the accounting information via the Diameter ACR messages and the OFCS (Offline Charging Server) generates the CDRs.

12.17.4.5 RAN Secondary RAT Usage Report AVP

The RAN-Secondary-RAT-Usage-Report AVP (code 1302) is of type Grouped. It contains the volume count reported by the RAN for the secondary RAT (separated for uplink and downlink), including the time of the report.

Secondary RAT Usage Report AVP lists the supported AVP fields. For more detailed descriptions, see *SGW Rf attributes SGW Rf attributes* and *PGW Rf attributes PGW Rf attributes* in *Appendix A: Messages and information element reference*.

Table 52: Secondary RAT Usage Report AVP

AVP	3GPP standard
RAN-Secondary-RAT-Usage-Report	32.299, 7.2.160Db
Secondary-RAT-Type	32.299, 7.2.189aA
RAN-Start-Timestamp	32.299, 7.2.160Dc
RAN-End-Timestamp	32.299, 7.2.160Da
Accounting-Input-Octets	32.299, 7.1.1
Accounting-Input-Octets	32.299, 7.1.3

12.17.4.6 CLI commands

The RAN secondary RAT usage report functionality is disabled by default. To enable the functionality, use the following command.

```
configure mobile-gateway profile charging
cdr-avp-options
  ran-secondary-rat-usage reports {include | exclude}
```

To view configuration information for RAN secondary RAT usage reporting, use the following command.

```
show mobile-gateway profile charging cdr-avp-options profile-id
```

12.18 Online charging

The online charging functionality that is supported for the CMG.

For more information about specific supported OCS standards, see *Online charging standards summary*.

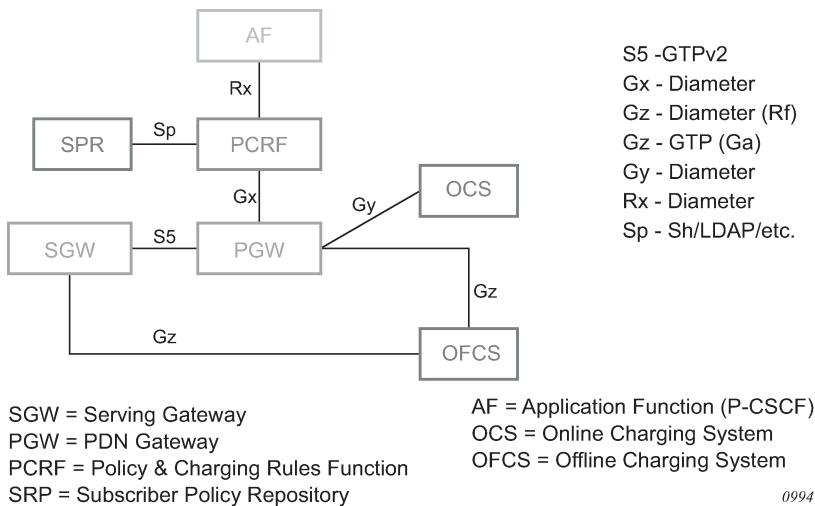
12.18.1 Online charging deployment scenarios

The following are examples of the online charging deployment scenarios:

- *Non-roaming scenario: PGW/GGSN and OCS deployed in the H-PLMN*
- *Roaming scenario: Home-routed (PGW/GGSN and OCS seployed in the H-PLMN)*
- *Roaming Scenario: Local-breakout (PGW/GGSN deployed in the V-PLMN and OCS deployed in the H-PLMN)*

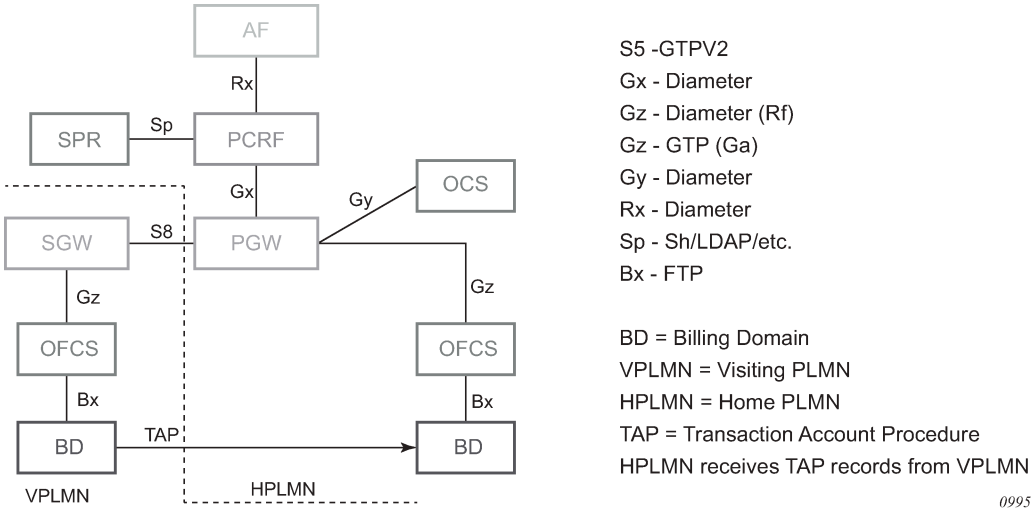
12.18.1.1 Non-roaming scenario: PGW/GGSN and OCS deployed in the H-PLMN

Figure 169: Policy and charging architecture (non-roaming)



12.18.1.2 Roaming scenario: Home-routed (PGW/GGSN and OCS seployed in the H-PLMN)

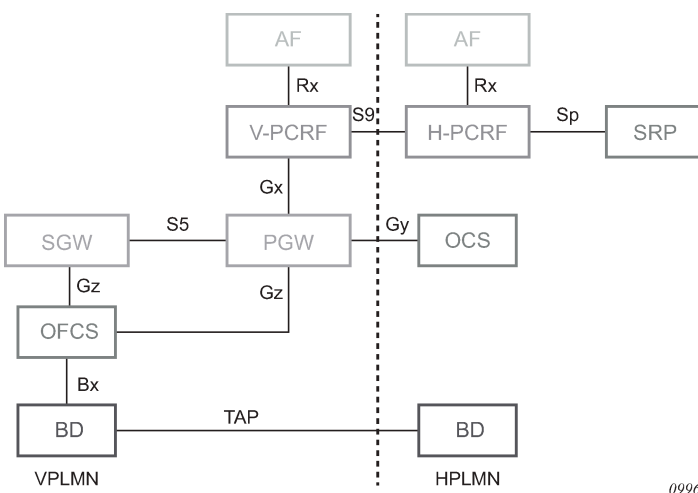
Figure 170: Policy and charging architecture (roaming home-routed)



12.18.1.3 Roaming Scenario: Local-breakout (PGW/GGSN deployed in the V-PLMN and OCS deployed in the H-PLMN)

Because the subscriber's online account information exists at the OCS in the H-PLMN, the PGW in the visiting PLMN needs to communicate with the OCS in the H-PLMN. Communicating with an OCS in a different operator's network will likely involve going through Diameter Proxies and Relay Agents.

Figure 171: Policy and charging architecture (roaming - LBO)



12.18.1.4 OCS provisioning options

Some of the options for specifying the primary and secondary OCS addresses that a PGW interacts with include:

- Local provisioning on the PGW/GGSN at the gateway level.

Configuration context: `config>mobile>pdn>gy`

- Local provisioning on the PGW/GGSN at the APN-level. This overrides the configuration specified at the gateway level.

Configuration context: `config>mobile>pdn>apn>charging`

- Local provisioning on the PGW/GGSN as part of the Charging Profile. This is the preferred option to support the Local Breakout scenario mentioned above. Note that the Charging Profile is picked using the Octet 1 from the Charging Characteristics IE received from the HSS/MME at the time of IP-CAN session establishment.

Configuration context: `config>mobile>profile>charging>pgw-charging`

12.18.1.5 DCCA profile provisioning

The Diameter Credit Control Application (DCCA) profile associated with each primary and secondary OCS must be configured. The specified DCCA profiles are associated with the OCS peers (primary and secondary) during OCS provisioning.

The configuration context for a DCCA profile is: `config>mobile>profile>dcca`.

For information about the supported DCCA profile configuration options, refer to the *DCCA Profile Commands* in the *CMG CLI Reference Guide*.

12.18.2 Diameter messages for online charging

The CTF function in PGW supports the following message pairs:

Credit-Control-Request (CCR) (CTF to OCS)

Credit-Control-Answer (CCA) (OCS to CTF)

- These two messages are mandatory for Diameter Credit Control Application (DCCA).
- The formats of these messages can be found in RFC-4006, section 3 and 3GPP TS 32.299, sections 6.4.2 and 6.4.3.

The following AVPs are not supported in a CCR command:

- Requested-Action
- AoC-Request-Type
- within MSCC Requested-Service-Unit
 - CC-Time

- CC-Total-Octets
- CC-Input-Octets
- CC-Output-Octets
- CC-Service-Specific-Units
- Event-Charging-Timestamp
- within Reporting-Reason, the POOL-EXHAUSTED value
- Refund-Information
- within Envelope
 - CC-Total-Octets
 - CC-Input-Octets
 - CC-Output-Octets
 - CC-Service-Specific-Units
 - Envelope-Start-Time
 - Envelope-End-Time
- within Service-Specific-Info
 - Service-Specific-Type
 - Service-Specific-Data
- within PS-Information
 - User-CSG-Information
 - Offline-Charging
- within Qos-Profile
 - CG-Address
 - Charging-Rule-Base-Name
 - User-CSG-Information
 - Offline-Charging

The following AVPs are not supported in a CCA command:

- within MSCC Granted-Service-Unit
 - CC-Total-Octets
 - CC-Input-Octets
 - CC-Output-Octets
 - CC-Service-Specific-Units

- G-S-U-Pool-Reference
 - G-S-U-Pool-Identifier
 - CC-Unit-Type
 - Unit-Value
 - Value-Digits
 - Exponent
 - Restriction-Filter-Rule
 - Filter-Id
- Unit-Quota-Threshold
- Refund-Information
- Envelope-Reporting
- Time-Quota-Mechanism
 - Time-Quota-Type
 - Base-Time-Interval
- Cost-Information
 - Unit-Value
 - Value-Digits
 - Exponent
 - Currency-Code
 - Cost-Unit
- Low-Balance-Indication
- Remaining-Balance
 - Unit-Value
 - Value-Digits
 - Exponent
 - Currency-Code
- Direct-Debiting-Failure-Handling
- Redirect-Host
- Redirect-Host-Usage
- Redirect-Max-Cache-Time
- within Service information

– Offline-Charging

The PGW also supports the following message pairs:

Re-Auth-Request (RAR) (OCS to CTF)

Re-Auth-Answer (RAA) (CTF to OCS)

- A Diameter server may initiate a re-authentication/re-authorization service for a session by sending RAR.
- The formats of these messages can be found in RFC 6733, section 8.3 and 3GPP TS 32.299, sections 6.4.4 and 6.4.5.

**NOTE:**

The G-S-U-Pool-Reference AVPs are not supported in a RAR command. The following AVPs are not supported in a RAA command:

- Redirect-Host/Redirect-Host-Usage/Redirect-Max-Cache-Time
 - Abort-Session-Request (ASR) (OCS to CTF)
 - Abort-Session-Answer (ASA) (CTF to OCS)
- A Diameter server may request that the client stop providing service for a particular session by sending an ASR message.
 - The formats of these messages can be found in RFC 6733, section 8.5.

**NOTE:**

The Redirect-Host/Redirect-Host-Usage/Redirect-Max-Cache-Time AVPs are not supported in an ASA command.

For more information about the Diameter messages, see [Appendix A: Messages and information element reference](#).

12.18.3 Gateway and OCS re-authorization

The re-authorization may be initiated either by the gateway (CTF) or by the OCS:

- The CTF may initiate mid-session re-authorizations of single or multiple active resource quotas within a DCC session by using a single Diameter Credit Control Request/Answer message sequence.
- The OCS may also re-authorize multiple active resource quotas within a DCC session by using a single Diameter Re-Auth-Request/Answer message sequence.

When re-authorization is triggered, gateway reports quota usage along with the reason why the quota is being reported through the use of the Reporting-Reason AVP in the CCR, as described in section 6.5.1.3 of 3GPP TS 32.299.

Supported Rating-Group level trigger conditions include:

- Threshold Reached
- QHT (Quota-Holding-Time) expiration
- Quota exhaustion
- Validity Timer expiration
- Forced re-authorization (RAR from OCS)

Supported Bearer/Session level trigger conditions include:

- RAT-Change
- User Location Change
- QoS-Change
- Serving-Node Change (SGW/SGSN)

The following event triggers are supported over the Gy interface to OCS:

- CHANGEINQOS_MAXIMUM_BIT_RATE_FOR_UPLINK

This value is used to indicate that a change in the end user negotiated uplink maximum bit rate will cause the Credit-Control client to ask for a re-authorization of the associated quota.

- CHANGEINQOS_MAXIMUM_BIT_RATE_FOR_DOWNLINK

This value is used to indicate that a change in the end user negotiated downlink maximum bit rate will cause the Credit-Control client to ask for a re-authorization of the associated quota

- CHANGEINQOS_APN_AGGREGATE_MAXIMUM_BIT_RATE

This value is used to indicate that a change in the APN aggregate maximum bit rate for the IP-CAN session will cause the Credit-Control client to ask for a re-authorization of the associated quota. This value is only applicable when charging per IP-CAN session is active.

12.18.4 Time quota consumption

In context of time-based credit, the distinction between active-usage and active-connection has to be made. The best way to do so is by defining an activity-threshold, an object which defines an average data rate under which the subscriber-host is considered silent.

As long as the effective rate of the application-usage will not exceed the rate defined by activity-threshold (configurable), given subscriber host will be considered silent and its corresponding credit will not be used. As long as the application-usage will exceed the rate, the application-credit will be consumed (in terms of time). The system will sample queue counters at the fixed rate corresponding to sampling interval (Ts).